



US Army Corps
of Engineers®

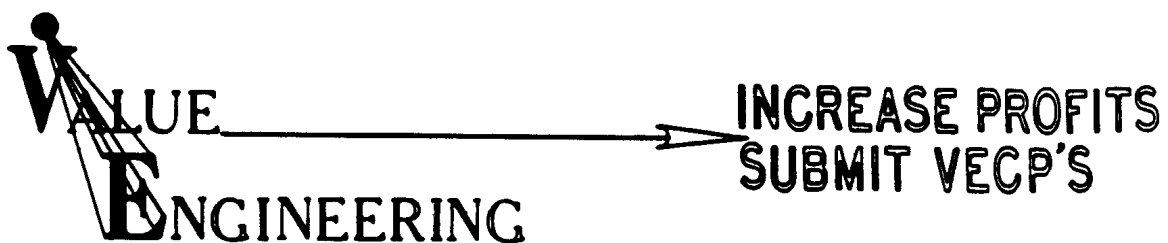
Caribbean District

Rio Puerto Nuevo Flood Control Project

**Supplemental Contract 3:
Channel Bottom and Wall Improvements
Sta. 88+33 to Sta. 118+50 and
Sta 123+50 to Sta. 147+40**

San Juan, Puerto Rico

**Construction Solicitation and Specifications
Volume 2 of 3**



See Volume 1 of 3 for issue date

RPN FLOOD CONTROL PROJECT
SUPP. CONTRACT 3: CHANNEL BOTTOM AND WALL IMPROVEMENTS

SPECIFICATIONS

SUPERVISING ENGINEER CERTIFICATION DOCUMENT

DATE: May 28, 2024

DISCIPLINE	SPECIFICATION SECTIONS	
	PUERTO RICO PROFESSIONAL	SUPERVISING PROFESSIONAL
PROJECT ENGINEER OF RECORD Sergio Gaitan, PE 	00 31 32 31 23 19 00 33 50 31 32 23 01 11 00 31 36 00 01 22 00 31 41 16 02 22 13 31 52 10 02 41 00 31 63 29 03 11 13 32 15 00 03 15 00 32 31 13 03 20 00 33 05 07 03 31 01 33 30 00 03 35 00 33 40 00 03 39 00 35 20 23 05 51 00 35 20 26 06 73 01 35 20 27 06 82 14 35 31 19 09 97 02 35 31 19.20 31 00 00 35 31 19.26 31 05 19 35 42 35 31 11 00	31 23 19 33 05 07
STRUCTURAL-BRIDGE Garrett Masiulis, PE 	03 41 00 05 50 13 05 52 00	

RPN FLOOD CONTROL PROJECT
SUPP. CONTRACT 3: CHANNEL BOTTOM AND WALL IMPROVEMENTS

SPECIFICATIONS

SUPERVISING ENGINEER CERTIFICATION DOCUMENT

DATE: May 28, 2024

DISCIPLINE	SPECIFICATION SECTIONS	
	PUERTO RICO PROFESSIONAL	SUPERVISING PROFESSIONAL
STRUCTURAL-BRIDGE Patrick Pecot, PE 		03 41 00
STRUCTURAL-CHANNEL Benjamin Curole, PE 		03 11 13 03 15 00 03 20 00 03 31 01 03 35 00 03 39 00 05 50 13 05 52 00
STRUCTURAL-CHANNEL Tanvir Hossain, PE 		09 97 02

RPN FLOOD CONTROL PROJECT
SUPP. CONTRACT 3: CHANNEL BOTTOM AND WALL IMPROVEMENTS

SPECIFICATIONS

SUPERVISING ENGINEER CERTIFICATION DOCUMENT

DATE: May 28, 2024

DISCIPLINE	SPECIFICATION SECTIONS	
	PUERTO RICO PROFESSIONAL	SUPERVISING PROFESSIONAL
GEOTECHNICAL Scott Breen, PE 		00 31 32 31 41 16 31 52 10
GEOTECHNICAL Timothy Newton, PG 		02 22 13 31 00 00 31 05 19 31 32 23 31 63 29
ROADWAY DESIGN Gary Sowell, PE 	01 55 26 32 01 16 32 11 23 32 12 16 32 16 19 32 17 23 32 92 19 32 92 23 34 71 13	

RPN FLOOD CONTROL PROJECT
SUPP. CONTRACT 3: CHANNEL BOTTOM AND WALL IMPROVEMENTS

SPECIFICATIONS

SUPERVISING ENGINEER CERTIFICATION DOCUMENT

DATE: May 28, 2024

DISCIPLINE	SPECIFICATION SECTIONS	
	PUERTO RICO PROFESSIONAL	SUPERVISING PROFESSIONAL
CIVIL Garret Keller, PE 		00 33 50 32 15 00 01 11 00 32 16 19 01 22 00 32 17 23 01 55 26 32 31 00 02 41 00 32 92 19 05 51 00 32 92 23 06 73 01 33 30 00 06 82 14 34 71 13 31 11 00 35 20 23 32 01 16 35 20 26 32 11 23 35 20 27 32 12 16
HYDRAULICS Brian Bandy, PE 		31 36 00 33 40 00 35 31 19 35 31 19.20 35 31 19.26 35 42 35
ELECTRICAL Eric Battle, PE 	26 00 00 26 08 00 26 56 00 33 71 01	

RPN FLOOD CONTROL PROJECT
SUPP. CONTRACT 3: CHANNEL BOTTOM AND WALL IMPROVEMENTS

SPECIFICATIONS

SUPERVISING ENGINEER CERTIFICATION DOCUMENT

DATE: May 28, 2024

DISCIPLINE	SPECIFICATION SECTIONS	
	PUERTO RICO PROFESSIONAL	SUPERVISING PROFESSIONAL
ELECTRICAL Bobby Sharpe, PE 		26 00 00 26 08 00 26 56 00 33 71 01
WATER UTILITIES Francisco Sola, PE 	33 11 00	33 11 00
The Specification Sections listed here have been provided by the Government and are not under the purview of the designers of the project.	01 30 00 01 32 01 01 33 00 01 35 25 01 35 26 01 45 00 01 45 04 01 45 08 01 50 02	01 52 10 01 57 20 01 57 25 01 74 19 01 78 02 01 78 23 02 42 00

PROJECT TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

00 31 32	GEOTECHNICAL DATA REPORT (SEE SEPARATE FILE)
00 33 50	WEATHER, WATER STAGE AND TIDE DATA

DIVISION 01 - GENERAL REQUIREMENTS

01 11 00	SUMMARY OF WORK
01 22 00	MEASUREMENT AND PAYMENT
01 30 00	ADMINISTRATIVE PROCEDURES
01 32 01	PROJECT SCHEDULE
01 33 00	SUBMITTAL PROCEDURES
01 35 25	OWNER SAFETY REQUIREMENTS - DIVING
01 35 26	GOVERNMENTAL SAFETY REQUIREMENTS
01 45 00	RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)
01 45 04	CONTRACTOR QUALITY CONTROL
01 50 02	TEMPORARY CONSTRUCTION FACILITIES
01 52 10	CONTRACTING OFFICER'S FIELD OFFICE
01 55 26	TRAFFIC CONTROL
01 57 20	ENVIRONMENTAL PROTECTION
01 57 25	TURBIDITY AND DISPOSAL MONITORING
01 74 19	CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL
01 78 02	CLOSEOUT SUBMITTALS
01 78 23	OPERATION AND MAINTENANCE DATA

DIVISION 02 - EXISTING CONDITIONS

02 22 13	VIBRATION AND NOISE MONITORING REQUIREMENTS
02 41 00	DEMOLITION
02 42 00	DECONSTRUCTION OF ELECTRIC TRANSMISSION AND DISTRIBUTION UTILITY

DIVISION 03 - CONCRETE

03 11 13	STRUCTURAL CAST-IN-PLACE CONCRETE FORMING
03 15 00	CONCRETE ACCESSORIES
03 20 00	CONCRETE REINFORCING
03 31 01	CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS
03 35 00	CONCRETE FINISHING
03 39 00	CONCRETE CURING
03 41 00	PRECAST, PRESTRESSED CONCRETE BRIDGE GIRDERS

DIVISION 05 - METALS

05 50 13	MISCELLANEOUS METAL FABRICATIONS
05 51 00	METAL STAIRS
05 52 00	METAL HAND RAILINGS

DIVISION 06 - WOOD, PLASTICS, AND COMPOSITES

06 73 01	FIBERGLASS REINFORCED PLASTIC (FRP) GRATING
06 82 14	FIBERGLASS REINFORCED PLASTIC (FRP) PIPE AND TUBE RAILINGS

DIVISION 09 - FINISHES

09 97 02	PAINTING: COAL TAR EPOXY
----------	--------------------------

DIVISION 26 - ELECTRICAL

26 00 00	BASIC ELECTRICAL MATERIALS AND METHODS
26 08 00	APPARATUS INSPECTION AND TESTING
26 56 00	EXTERIOR LIGHTING

DIVISION 31 - EARTHWORK

31 00 00	EARTHWORK
31 05 19	GEOTEXTILE
31 11 00	CLEARING AND GRUBBING
31 23 19	UNWATERING
31 32 23	JET GROUTING
31 36 00	GABION MATTRESS
31 41 16	METAL SHEET PILING
31 52 10	TEMPORARY RETAINING STRUCTURES
31 63 29	DRILLED CONCRETE PIERS AND SHAFTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

32 01 16	COLD MILLING ASPHALT PAVING
32 11 23	AGGREGATE BASE COURSES
32 12 16	HOT-MIX ASPHALT (HMA) FOR ROADS
32 15 00	AGGREGATE SURFACING
32 16 19	CONCRETE CURBS, GUTTERS AND SIDEWALKS
32 17 23	PAVEMENT MARKINGS
32 31 00	FENCES AND GATES
32 92 19	SEEDING
32 92 23	SODDING

DIVISION 33 - UTILITIES

33 05 07	PILOT TUBE GUIDED BORING AND JACKING
33 11 00	WATER UTILITY DISTRIBUTION PIPING
33 30 00	SANITARY SEWERAGE
33 40 00	STORM DRAINAGE UTILITIES
33 71 01	OVERHEAD TRANSMISSION AND DISTRIBUTION

DIVISION 34 - TRANSPORTATION

34 71 13	VEHICLE CRASH BARRIERS
----------	------------------------

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

35 20 23	DREDGING
35 20 26	NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SCOW - ULLAGE PROFILE
35 20 27	NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM PIPELINE HYDRAULIC DREDGE
35 31 19	STONE AND CHANNEL PROTECTION FOR STRUCTURES
35 31 19.20	ARTICULATING CONCRETE BLOCK REVETMENTS
35 31 19.26	CONCRETE ARMOR UNITS
35 42 35	MARINE MATTRESS

-- End of Project Table of Contents --

SECTION TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

SECTION 00 31 32

GEOTECHNICAL DATA REPORT (SEE SEPARATE FILE)

PART 1 GENERAL

PART 2 PRODUCTS (NOT APPLICABLE)

PART 3 EXECUTION (NOT APPLICABLE)

-- End of Section Table of Contents --

SECTION TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

SECTION 00 33 50

WEATHER, WATER STAGE AND TIDE DATA

PART 1 GENERAL

PART 2 PRODUCTS (NOT APPLICABLE)

PART 3 EXECUTION (NOT APPLICABLE)

-- End of Section Table of Contents --

SECTION 00 33 50 WEATHER AND WATER STAGE DATA

1. The below stated water levels are for informational purposes only and are not to be utilized in conjunction with any contract-related hydrographic surveying purposes as noted on the control drawing(s) of the contract plans.
2. The Rio Puerto Nuevo Flood Control Project (RPN Project) Contract 3, which encompasses Channel Improvements over Rio Piedras between the Margarita Channel near the Jose De Diego Bridge (PR-22) and south of the Franklin Delano Roosevelt Bridge, as shown in the plans and specifications of the RPN Project.
3. Rio Piedras water levels at the Channel Improvements are affected by rainfall, evaporation, local runoff and groundwater levels, and are subject to changes resulting from rainfall runoff from steep terrain and impermeable areas.
4. In general, historic water levels are not necessarily indicative of future water levels. The contractor should not expect that water levels/flows will be manipulated for the contractor. It shall be the contractor's responsibility to obtain information concerning water levels and flows. The United States Geological Survey (USGS) has historical observations (water level and flow data) available for a stream gage located at Latitude: 18°24'34.38" N, Longitude: 66°04'09.60" W known as Rio Piedras at Hato Rey, Puerto Rico (USGS Gage No. 50049100) that can be accessed at the USGS website: <https://waterdata.usgs.gov/pr/nwis/nwis>. In addition, the National Weather Service (NWS) Forecast Office in San Juan, Puerto Rico has an Advanced Hydrologic Prediction Service for the USGS Gage No. 50049100 that can be accessed at the NWS website: <https://water.weather.gov/ahps2/hydrograph.php?gage=rpop4&wfo=sju>.
5. It shall be the contractor's responsibility to obtain information concerning rain and wind. Reference is made to the National Oceanic and Atmospheric Administration (NOAA) publication entitled Local Climatological Data - Monthly Summary which contains climatological and meteorological observations and data, including wind, rain, and temperature data. Subscription price and ordering information is available from the National Centers for Environmental Information, Federal Plaza, 151 Patton Avenue, Asheville, North Carolina 28801-5001. NOAA also has an internet information page at: <https://www.ncdc.noaa.gov/>.
6. The climate is essentially tropical and humid. Temperatures below freezing are rare. The wet season is from May through November. The hurricane season is from June through November. In general, the winter months constitute the dry season and the summer months comprise the wet season. Occasionally, daily rainfall in the dry season can be quite heavy. The project area is subject to tropical storms and hurricanes from June through November and to windy and/or rainy weather throughout the year. Weather forecasts as well as actual rainfall information from the NWS Forecast Office in San Juan, Puerto Rico can be found at: <https://www.weather.gov/sju/>.
7. The following data are provided:

MONTHLY ANTICIPATED ADVERSE WEATHER DELAY WORK DAYS
BASED ON A 5-DAY WORK WEEK

SECTION 00 33 50 WEATHER AND WATER STAGE DATA

The average number of work days in each calendar month with precipitation greater than or equal to 0.50 inches (1.27 cm) is provided for the area in the below table. This information is based on data obtained from Station Rio Piedras EXP STN located at Latitude: 18.3906 N, Longitude: 66.0542 W.

Average Number of Days per Month with
Precipitation $\geq$ 0.50 Inches (1.27 cm)

Station	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Rio Piedras EXP STN	2	2	1	2	3	2	3	3	3	3	4	2

The data in the above table, which have been rounded to the nearest whole number, was obtained from the Summary of Monthly Normals 1981-2010 internet publication of the National Centers for Environmental Information, NOAA for Station Rio Piedras EXP STN via the internet page:

<http://www.ncdc.noaa.gov/cdo-web/datatools/normals>

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 11 00

SUMMARY OF WORK

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 WORK COVERED BY CONTRACT DOCUMENTS
 - 1.3.1 Project Description
 - 1.3.2 Location
 - 1.3.3 Channel Bypass Requirements
 - 1.3.4 Safety Requirements
 - 1.3.5 Work Performed by Others
- 1.4 PRECONSTRUCTION LOGISTICS PLAN
 - 1.4.1 PCL Conference
 - 1.4.2 Related Specification Sections
- 1.5 SCHEDULE MILESTONES
- 1.6 COMMENCEMENT, PROSECUTION AND COMPLETION OF WORK
- 1.7 LIQUIDATED DAMAGES - CONSTRUCTION
- 1.8 PHYSICAL DATA
 - 1.8.1 Physical Conditions
 - 1.8.2 Weather, Water Stage and Tide Data
 - 1.8.3 Contractor Investigation
 - 1.8.4 Maritime Traffic
 - 1.8.5 Obstruction of Channel
- 1.9 LAYOUT OF WORK
 - 1.9.1 Established Project Control
 - 1.9.2 Layout
 - 1.9.3 Survey
- 1.10 DAMAGE TO WORK
- 1.11 DAMAGE TO WORK -- BYPASS SHEET PILES
- 1.12 EXISTING WORK
- 1.13 LOCATION OF UNDERGROUND FACILITIES
 - 1.13.1 Notification Prior to Excavation
- 1.14 TIME EXTENSIONS FOR UNUSUALLY SEVERE WEATHER (31 OCT 1989)
 - 1.14.1 Schedule of Monthly Anticipated Adverse Weather Delays
 - 1.14.2 Contractor Responsibility
- 1.15 UNITS OF MEASURE

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section Table of Contents --

SECTION 01 11 00

SUMMARY OF WORK

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

GDR Section 00 31 32, Geotechnical Data Report
Rio Puerto Nuevo Supplemental Contract 3

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Transmit submittal items in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Preconstruction Logistics Plan

1.3 WORK COVERED BY CONTRACT DOCUMENTS

1.3.1 Project Description

The work for the Rio Puerto Nuevo Supplemental Contract 3 includes the widening and deepening of the existing channel for flood control purposes by constructing two drilled shaft walls and dredging to the required depth. Each wall is approximately 5200 feet long with exposed heights ranging from 24 feet to 31 feet and will be tied together with a cap beam. Construction of the walls will require temporary sheet piling to maintain a minimum flow area for flood events. This contract also includes a tie-in structure at the Josefina Confluence, a transition the existing river upstream, and a transition to an existing project at the downstream outlet. Other work includes sanitary sewer, water, and electrical utility relocations, as well as, implementation of new stormwater inlets for drainage across the wall into the channel.

1.3.2 Location

The work is located along the Rio Piedras between the Margarita Channel near the Jose De Diego Bridge (PR-22) and approximately 2740 feet south of the Franklin Delano Roosevelt Bridge in San Juan, Puerto Rico.

1.3.3 Channel Bypass Requirements

Contractor shall maintain the required channel conveyance capacity as described within drawings GC101 through GC104 and the C0300 drawing series. Refer to Channel Bypass Requirements and provisions stated on drawing G-005.

1.3.4 Safety Requirements

See Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and EM 385-1-1.

1.3.5 Work Performed by Others

See paragraph PROJECT COORDINATION of Section 01 30 00 ADMINISTRATIVE PROCEDURES.

1.4 PRECONSTRUCTION LOGISTICS PLAN

Submit a project-wide plan demonstrating the ability to complete the construction activities within the required project schedule and to the quality described in the contract documents. The Preconstruction Logistics (PCL) Plan is to be an integrated plan and narrative to demonstrate the means and methods and sequence of construction for the timely completion of the work while maintaining the bypass flows as required within the Contract Plans.

Submit the PCL Plan with the Initial Project Schedule which is required 42 days after issuance of notice to proceed. The PCL Plan is to be consistent with and comply with the durations within the Initial Project Schedule including the required milestones listed within the paragraph SCHEDULE MILESTONES.

The PCL Plan may also include modifications or deviations to the overall project sequencing as allowed within the GC series of the contract drawings. Submit changes to the construction sequencing which may affect the features depicted on drawings or specifications as a variation to the contract. Any proposed modifications or deviations must be consistent with the Channel Bypass Requirements noted in paragraph "Channel Bypass Requirements". The deviations to the construction sequencing must be signed and sealed by an engineer licensed in the Commonwealth of Puerto Rico.

As part of the PCL Plan, provide certificates or letters of commitment from suppliers for materials and equipment, as well as, a narrative describing the sequence and means and methods required to complete the activities below. These items include:

- A. Existing Construction Remnant Demolition
 - 1. Debris Disposal Site
 - 2. Method for Removing Existing Piles, Sheetpiles, Structural Members, and Other Debris at the De Diego Bridge
- B. Drilled Shaft Construction
 - 1. Crane Supplier or Source
 - 2. Delivery Logistics
 - 3. Methods for Construction
- C. Concrete Ready Mix Plants
 - 1. Supplier List
 - 2. Concrete Plant Capacities
 - 3. Delivery Logistics
- D. Reinforcing Steel For Drilled Shafts

1. Supplier List
2. Delivery Logistics
3. Methods for Assembling Rebar Cages
4. Methods for Placing Rebar Cages
- E. Metal Sheet Pile
 1. Supplier List
 2. Delivery Logistics
 3. Driving Method
 4. Extraction Method for Temporary Sheet Piles
- F. Precast Girder Construction
 1. Supplier List
 2. Delivery Logistics
 3. Casting Yard Capabilities and FIB Girder Formwork Procurement
 3. Methods for Placing Girders
- G. Dredging And Disposal
 1. Equipment Supplier or Source
 2. Delivery Logistics
 3. Method of Dredging
 4. Method of Disposal
- H. In-Channel Material Mechanical Excavation And Disposal
 1. Equipment Supplier or Source
 2. Delivery Logistics
 3. Method of In-Channel Excavation
 4. Method and Logistics of Material Disposal
- I. Articulating Concrete Blocks and Concrete Armor Units
 1. Supplier List
 2. Delivery Logistics
 3. Method of Placement
- J. Water Utility Modifications
 1. Pipe Supplier List
 2. Valve Supplier List
 3. Delivery Logistics
 4. Phasing of Installation
- K. Sanitary Sewer Utility Modifications
 1. Pipe Supplier List
 2. Delivery Logistics
 3. Method of Installation
 4. Phasing of Installation
- L. Traffic Management
 1. Temporary Signal Supplier
 2. Temporary Barricade Supplier

1.4.1 PCL Conference

Schedule a conference 30 days after issuance of notice to proceed to review and discuss progress of the PCL plan with the Government. Contractor should be prepared to discuss general schedule and outline the major aspects of the PCL Plan. Allow for a 4-hour meeting duration.

Do not perform work at the site, with the exception of site inspections and surveys, until the PCL Review Conference has been completed and the PCL has been submitted.

1.4.2 Related Specification Sections

Review the following sections prior to developing the PCL Plan for specific requirements that may require consideration within the plan:

01 32 01 PROJECT SCHEDULE
01 55 26 TRAFFIC CONTROL
02 41 00 DEMOLITION
03 41 00 PRECAST, PRESTRESSED CONCRETE BRIDGE GIRDERS
26 00 00 BASIC ELECTRICAL MATERIALS AND METHODS
31 00 00 EARTHWORK
31 41 16 METAL SHEET PILING
31 63 29 DRILLED CONCRETE PIERS AND SHAFTS
33 11 00 WATER UTILITY DISTRIBUTION PIPING
33 30 00 SANITARY SEWERAGE
33 40 00 STORM DRAIN UTILITIES
35 20 23 DREDGING
35 31 19.20 ARTICULATING CONCRETE BLOCK REVETMENTS
35 31 19.26 CONCRETE ARMOR UNITS

1.5 SCHEDULE MILESTONES

Include the following schedule milestones in the Project Schedule for use by the Government in evaluating construction progress. Complete construction phasing is shown on the Contract Documents listed in Table 2. Additional requirements of the schedule are located in 01 32 01 PROJECT SCHEDULE.

Table 1 - Required Schedule Milestones		
<u>Milestone</u>	<u>Activity</u>	<u>Reference Contract Document</u>
I.	Installation of the Demonstration Wall and Approval to Proceed with Production Shafts	31 63 29 C0302
II.	Dredging Downstream and Below De Diego Bridge and Place Bridge Pier Scour Protection	GC101
III.	Completion of Subreaches R1, R2A, R2C R3, L1A, L1B, L6A, L6C and LC1 Including Drilled Shaft Construction, Pump Station Outlet, and Channel Dredging	GC101
IV.	Completion of Subreaches R4, R5, RC-1A, RC-1B L7A, L7B, and LC-2 Including Drilled Shaft Construction, 48" Waterline Relocation, and Channel Dredging	GC101

<u>Milestone</u>	<u>Activity</u>	<u>Reference Contract Document</u>
V.	Completion of Subreaches RC-2, R6, L8, and LC-3 Including Drilled Shaft Construction, 10" Waterline Relocation, and Channel Dredging	GC102
VI.	Completion of Subreaches J-R1, J-R2, J-L1, J-L2, J-C1, J-C2, and J-M1 Including Drilled Shaft Construction, Channel Dredging, and Josefina Transition	GC102
VII.	Completion of Subreaches R7, RC-3, and L9 Including Drilled Shaft Construction and Channel Dredging	GC102
VIII.	Completion of Subreaches R8, R9, LC-4, L10, C1 and C2 Including Drilled Shaft Construction, Channel Dredging, and RPN Transition	GC102
IX.	Completion of Subreaches R2B, L6B, and Relocation of the Sanitary Sewer Siphon	GC101

1.6 COMMENCEMENT, PROSECUTION AND COMPLETION OF WORK

Read this paragraph in conjunction with the Clause COMMENCEMENT, PROSECUTION, AND COMPLETION OF WORK (FAR 52.211-10) of Section 00700 CONTRACT CLAUSES in Volume 1. Work will not be considered complete until turf has been established as defined in 32 92 19 SEEDING and 32 92 23 SODDING

1.7 LIQUIDATED DAMAGES - CONSTRUCTION

Refer to the Clause LIQUIDATED DAMAGES - CONSTRUCTION (FAR 52.211-12) of Section 00700 CONTRACT CLAUSES in Volume 1.

1.8 PHYSICAL DATA

Read this paragraph in conjunction with the Clause PHYSICAL DATA (FAR 52.236-4) of Section 00700 CONTRACT CLAUSES in Volume 1.

1.8.1 Physical Conditions

The indications of physical conditions on the drawings and in the specifications are the result of site investigations by surveys and/or by core borings. When the indicated physical conditions are the result of site investigations by core borings, the core boring logs and laboratory data are in Section 00 31 32 GEOTECHNICAL DATA REPORT (GDR) and the core

boring locations are shown on the drawings. Also, the GDR for instructions concerning availability of core borings for inspection.

1.8.2 Weather, Water Stage and Tide Data

See Section 00 33 50 WEATHER, WATER STAGE AND TIDE DATA.

1.8.3 Contractor Investigation

Refer to the Clause SITE INVESTIGATION AND CONDITIONS AFFECTING THE WORK (FAR 52.236-3) of Section 00700 CONTRACT CLAUSES in Volume 1.

1.8.4 Maritime Traffic

Marine Traffic on the dredging disposal route consists of Naval, cruise ships, commercial, pleasure, and small recreational vessels of all types and sizes which can be accommodated by existing depths.

1.8.5 Obstruction of Channel

The Government will not undertake to keep the channel free from vessels or other obstructions, except to the extent of such regulations, if any, as may be prescribed by the Secretary of the Army, in accordance with the provisions of Section 7 of the River and Harbor Act approved 8 August 1917. Conduct the work in such manner as to obstruct navigation as little as possible, and in case the Contractor's plant so obstructs the channel as to make difficult or endanger the passage of any vessels, promptly move said plant on the approach of any vessel to such an extent as may be necessary to afford a practicable passage. Upon completion of the work promptly remove his plant, including ranges, buoys, piles, and other marks placed under the contract in navigable waters or on shore.

1.9 LAYOUT OF WORK

Read this paragraph in conjunction with the Clause LAYOUT OF WORK (FAR 52.236-17) of Section 00700 CONTRACT CLAUSES in Volume 1.

1.9.1 Established Project Control

The Government has established monuments along the work site(s). Each monument is represented in the contract drawings by a Designation and in most cases, an alphanumeric Permanent Identifier (PID). Information concerning the location and elevations can be found within the Survey Notes on the plan sheets. The Survey Notes for each included field survey will cite a particular survey number, and specify the control used to conduct that survey. Additional information is included in the Survey Notes that must be duplicated if an attempt to recreate the subject survey is desired.

For navigation channels, the Government has established tidal calibration sites for projects constructed and maintained to a tidal datum. Each tidal calibration site is represented in the plan sheet drawings by Station ID, and is cited in the Survey Notes. All projects constructed and maintained to a tidal datum are to be calibrated to the Station ID specified in the plan sheet drawings. Information regarding the location and associated benchmarks of the referenced Station ID is included in the Monument Description section of these specifications.

See the Contract Drawings for Benchmark location and information.

1.9.2 Layout

From the monuments, control data and elevations established by the Government, complete the layout of the work and obtain all measurements that may be required for the execution of the work to the location and limit marks prescribed in the specifications or on the contract drawings, subject to such modifications as the Contracting Officer may require to meet changed conditions or as a result of necessary modifications to the contract work.

1.9.3 Survey

Provide tools, material, and labor for laying out any part of the work from the monuments, control data and elevations established by the Government. Maintain and preserve all monuments, stakes and other marks established by the Contracting Officer until authorized to remove them, and if such marks or monuments are destroyed by the Contractor or through negligence, prior to their authorized removal, they may be replaced by the Contracting Officer, at their discretion, and the expense of replacement will be deducted from any amounts due or to become due the Contractor. The Contracting Officer may require that work be suspended at any time when location and limit marks established by the Contractor are not reasonably adequate to permit checking of the work.

1.10 DAMAGE TO WORK

The responsibility for damage to any part of the permanent work will be as set forth in Clause PERMITS AND RESPONSIBILITIES of Section 00700 CONTRACT CLAUSES in Volume 1. However, if, in the judgment of the Contracting Officer, any part of the permanent work performed by the Contractor is damaged by flood, earthquake, hurricane, or tornado, which damage is not due to the failure of the Contractor to take reasonable precautions or to exercise sound engineering and construction practices in the conduct of the work, make the repairs as ordered by the Contracting Officer and full compensation for such repairs will be made at the applicable contract unit or job prices as fixed and established in the contract. If, in the opinion of the Contracting Officer, there are no contract unit or job prices applicable to any part of such work, an equitable adjustment pursuant to Clause CHANGES of Section 00700 CONTRACT CLAUSES in Volume 1 will be made as full compensation for the repairs of that part of the permanent work for which there are no applicable contract unit or job prices. Except as herein provided, repair damage to all work (including temporary construction), utilities, materials, equipment and plant to the satisfaction of the Contracting Officer at the Contractor's expense, regardless of the cause of such damage.

1.11 DAMAGE TO WORK -- BYPASS SHEET PILES

- a. The responsibility for damage to any part of the work to be performed under this contract is set forth in Clause PERMITS AND RESPONSIBILITIES of Section 00700 CONTRACT CLAUSES in Volume 1. However, if the bypass sheet piles are constructed in accordance with plans and progress schedules approved by the Contracting Officer, but are overtopped by flood and such flood causes damage to the bypass sheet piles or if any part of the permanent work is damaged by flood, earthquake, hurricane, or tornado, which damage is not due to the failure of the Contractor to take reasonable precaution or to exercise sound engineering and construction practices in the conduct of the

work, make the repairs ordered by the Contracting Officer. Compensation will be made for any such repairs in accordance with the Clause CHANGES of Section 00700 CONTRACT CLAUSES in Volume 1. The equitable adjustment shall be based on the contract unit or job prices applicable to any part of the damaged work.

- b. The Contracting Officer may order the Contractor to, flood or breach the bypass sheet piles during a rise prior to, and in anticipation of, natural flooding due to overtopping. Such flooding or breach will be considered the same as though the bypass sheet piles, if constructed in accordance with plans and progress schedules approved by the Contracting Officer, had been overtopped, in which event an equitable adjustment will be made for damages to the bypass sheet piles and/or any part of the permanent work, as provided in subparagraph a. above.

1.12 EXISTING WORK

In addition to FAR 52.236-9, Protection of Existing Vegetation, Structures, Equipment, Utilities, and Improvements:

- a. Remove or alter existing work in such a manner as to prevent injury or damage to any portions of the existing work which remain.
- b. Repair or replace portions of existing work which have been altered during construction operations to match existing or adjoining work, as approved by the Contracting Officer. At the completion of operations, existing work must be in a condition equal to or better than that which existed before new work started.

1.13 LOCATION OF UNDERGROUND FACILITIES

Obtain digging permits 15 calendar prior to start of excavation, and comply with installation requirements for locating and marking underground utilities. Contact local utility locating service a minimum of 48 hours prior to excavating, to mark utilities, and within sufficient time required if work occurs on a Monday or after a Holiday. Verify existing utility locations indicated on contract drawings, within area of work.

Identify and mark all other utilities not managed and located by the local utility companies. Scan the construction site with Ground Penetrating Radar (GPR), electromagnetic, or sonic equipment, and mark the surface of the ground or paved surface where existing underground utilities are discovered. Verify the elevations of existing piping, utilities, and any type of underground obstruction not indicated, or specified to be removed, that is indicated or discovered during scanning, in locations to be traversed by piping, ducts, and other work to be conducted or installed. Verify elevations before installing new work closer than nearest manhole or other structure at which an adjustment in grade can be made.

1.13.1 Notification Prior to Excavation

Notify the Contracting Officer at least 14 days prior to starting excavation work. Obtain excavation authorization from DTOP "Dirección de Excavaciones, Demoliciones y Tuberías". Contractor is responsible for marking all utilities not marked by utility companies after completing DTOP excavation authorization.

1.14 TIME EXTENSIONS FOR UNUSUALLY SEVERE WEATHER (31 OCT 1989)

This provision specifies the procedure for the determination of time extensions for unusually severe weather in accordance with the Clause DEFAULT (FIXED-PRICE CONSTRUCTION) of Section 00700 CONTRACT CLAUSES in Volume 1. In order for the Contracting Officer to award a time extension under this clause, the following conditions must be satisfied:

- a. The weather experienced at the project site during the contract period must be found to be unusually severe; that is, more severe than the adverse weather anticipated for the project location during any given month.
- b. The unusually severe weather must actually cause a delay to the completion of the project. The delay must be beyond the control and without the fault or negligence of the Contractor.

1.14.1 Schedule of Monthly Anticipated Adverse Weather Delays

The schedule of monthly anticipated adverse weather delays is based upon National Oceanic and Atmospheric Administration (NOAA) or similar data for the project location and will constitute the base line for monthly weather time evaluations. The Contractor's progress schedule must reflect these anticipated adverse weather delays in all weather dependent activities. See Section 00 33 50 WEATHER, WATER STAGE AND TIDE DATA to obtain MONTHLY ANTICIPATED ADVERSE WEATHER DELAY WORK DAYS BASED ON 5-DAY WORK WEEK. A reasonable number of anticipated weather delay work days is obtained from the average number of days per month (rounded to the nearest whole number) with precipitation greater than that indicated in Section 00 33 50 WEATHER, WATER STAGE AND TIDE DATA.

1.14.2 Contractor Responsibility

Upon acknowledgment of the Notice to Proceed (NTP) and continuing throughout the Contract, record on the daily CQC report the occurrence of adverse weather and resultant impact to normally scheduled work. Actual adverse weather delay days must prevent work on critical activities for 50 percent or more of the Contractor's scheduled work day. Include the number of actual adverse weather delay days impacted by actual adverse weather (even if adverse weather occurred in previous month), be calculated chronologically from the first to the last day of each month, and be recorded as full days. If the number of actual adverse weather delay days exceeds the number of days anticipated in Section 00 33 50 WEATHER, WATER STAGE AND TIDE DATA (rounded to the nearest whole number), the Contracting Officer will convert any qualifying delays to calendar days, giving full consideration for equivalent fair weather work days, and issue a modification in accordance with the Clause DEFAULT (FIXED PRICE CONSTRUCTION) of Section 00700 CONTRACT CLAUSES in Volume 1.

1.15 UNITS OF MEASURE

The standard U.S. system is to be utilized by the Contractor in the performance of work. The unit of measure used in Contract Drawings and Specifications is the standard U.S. system. Refer to 01 78 02 CLOSEOUT SUBMITTALS for units required for record and as-built drawings.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 22 00

MEASUREMENT AND PAYMENT

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 Reference
- 1.3 RESIDENT MANAGEMENT SYSTEM (RMS)
 - 1.3.1 Definition
 - 1.3.2 Instructions
- 1.4 SUBMITTALS
- 1.5 JOB PAYMENT ITEMS
 - 1.5.1 Utility Invoices
 - 1.5.2 Mobilization and Demobilization (Line Items 0001)
 - 1.5.3 Erosion and Sediment Control and Turbidity Monitoring (Line Item 0002)
 - 1.5.4 Environmental Species Monitoring (Line Item 0003)
 - 1.5.5 Vibration Control and Monitoring (Line Item 0004)
 - 1.5.6 Contracting Officer's Field Office (Fixed Costs) (Line Item 0005)
 - 1.5.7 As-Built Drawings & Close Out Submittals (Line Item 0008)
 - 1.5.8 Demolition (Line Item 0010)
 - 1.5.9 Demolition - Remnant Removal at De Diego Bridge (Line Item 0011)
 - 1.5.10 Demonstration Wall (Line Item 0016)
 - 1.5.11 Drilled Shaft Cap Beam (Line Items 0021)
 - 1.5.12 Jet Grouting - Ground Improvement (Line Item 0023)
 - 1.5.13 Concrete Double Box Culvert - 5feetx6.1feet (CIP) (Line Item 0052)
 - 1.5.14 Stormwater Bypass (Line Item 0059)
 - 1.5.15 Utility Bridge - Middle Pier Cap (CIP) (Line Item 0063)
 - 1.5.16 Utility Bridge - Bridge Slab (CIP) (Line Item 0064)
 - 1.5.17 48" DIP Waterline Crossing (Line Item 0065)
 - 1.5.18 Traffic Control (Line Item 0066)
 - 1.5.19 Parking Lot Striping (Line Item 0070)
 - 1.5.20 Metal Steps (Line Item 0074)
 - 1.5.21 Restoration (Line Item 0079)
 - 1.5.22 LUMA Electrical Power Charges (Line Items 0082)
 - 1.5.23 AT&T/Liberty Telecommunications Charges (Line Item 0083)
 - 1.5.24 Claro Telecommunications Charges (Line Item 0084)
 - 1.5.25 Sanitary Siphon - Temporary Retaining Structures (Line Item 0085)
 - 1.5.26 Sanitary Siphon - Headwalls (Line Item 0086)
 - 1.5.27 Sanitary Siphon - Influent Box, Effluent Box, and Piping (Line Item 0087)
 - 1.5.28 Sanitary Siphon - Boring and Jacking (Line Item 0088)
 - 1.5.29 Concrete Privacy Wall Replacement (Line Item 0089)
 - 1.5.30 Municipal Taxes (Line Item 0091)
- 1.6 UNIT PRICE PAYMENT ITEMS

- 1.6.1 Contracting Officer's Field Office (Recurring Costs)(Estimated Quantity) (Line Item 0006)
 - 1.6.1.1 Payment
 - 1.6.1.2 Measurement
 - 1.6.1.3 Unit of Measure
- 1.6.2 Partnering Regroups (Estimated Quantity) (Line Item 0007)
 - 1.6.2.1 Payment
 - 1.6.2.2 Measurement
 - 1.6.2.3 Unit
- 1.6.3 Clearing and Grubbing (Estimated Quantity) (Line Item 0009)
 - 1.6.3.1 Payment
 - 1.6.3.2 Measurement
 - 1.6.3.3 Unit of Measure
- 1.6.4 Temporary Bypass Sheet Pile (Estimated Quantity) (Line Item 0012)
 - 1.6.4.1 Measurement
 - 1.6.4.2 Unit of Measure
- 1.6.5 Permanent and Transition Sheet Piles (Estimated Quantity) (Line Items 0013, 0014 and 0015)
 - 1.6.5.1 Measurement
 - 1.6.5.2 Unit of Measure
- 1.6.6 Drilled Shaft (Estimated Quantity) (Line Items 0017, 0018, 0019, 0020 & 0061)
 - 1.6.6.1 Payment
 - 1.6.6.2 Measurement
 - 1.6.6.3 Unit of Measure
- 1.6.7 Jet Grout Seals (Estimated Quantity) (Line Items 0022)
 - 1.6.7.1 Payment
 - 1.6.7.2 Measurement
 - 1.6.7.3 Unit of Measure
- 1.6.8 Guardrail (Estimated Quantity) (Line Item 0024)
 - 1.6.8.1 Payment
 - 1.6.8.2 Measurement
 - 1.6.8.3 Unit of Measure
- 1.6.9 Handrail (Estimated Quantity) (Line Item 0025)
 - 1.6.9.1 Payment
 - 1.6.9.2 Measurement
 - 1.6.9.3 Unit of Measure
- 1.6.10 Excavation (Estimated Quantity) (Line Items 0026, 0027 & 0028)
 - 1.6.10.1 Payment
 - 1.6.10.2 Measurement
 - 1.6.10.3 Unit of Measure
- 1.6.11 Backfill (Estimated Quantity) (Line Items 0029 & 0030)
 - 1.6.11.1 Payment
 - 1.6.11.2 Measurement
 - 1.6.11.3 Unit of Measure
- 1.6.12 Wall Groundwater Drains and Filter (Estimated Quantity) (Line Item 0031)
 - 1.6.12.1 Payment
 - 1.6.12.2 Measurement
 - 1.6.12.3 Unit of Measure
- 1.6.13 Finish Grading (Estimated Quantity) (Line Item 0032)
 - 1.6.13.1 Payment
 - 1.6.13.2 Measurement
 - 1.6.13.3 Unit of Measure
- 1.6.14 Gabion Baskets for Bedding (Estimated Quantity) (Line Item 0033)
 - 1.6.14.1 Payment
 - 1.6.14.2 Measurement

- 1.6.14.3 Unit of Measure
- 1.6.15 Marine Mattress (Estimated Quantity) (Line Item 0034)
 - 1.6.15.1 Measurement
 - 1.6.15.2 Unit of Measure
- 1.6.16 Riprap (Estimated Quantity) (Line Items 0035, 0036, 0037 & 0038)
 - 1.6.16.1 Payment
 - 1.6.16.2 Measurement
 - 1.6.16.3 Unit of Measure
- 1.6.17 Articulating Concrete Block Revetment (Estimated Quantity) (Line Item 0039)
 - 1.6.17.1 Payment
 - 1.6.17.2 Measurement
 - 1.6.17.3 Unit of Measure
- 1.6.18 Concrete Armor Units (Estimated Quantity) (Line Item 0040)
 - 1.6.18.1 Measurement
 - 1.6.18.2 Unit of Measure
- 1.6.19 Storm Drainage Headwalls (Line Items 0041, 0042, & 0043)
 - 1.6.19.1 Payment
 - 1.6.19.2 Measurement
 - 1.6.19.3 Unit of Measure
- 1.6.20 Storm Drainage Piping (Estimated Quantity) (Line Items 0044 through 0051)
 - 1.6.20.1 Payment
 - 1.6.20.2 Measurement
 - 1.6.20.3 Unit of Measure
- 1.6.21 Flap Gates (Line Items 0053 through 0056)
 - 1.6.21.1 Payment
 - 1.6.21.2 Measurement
 - 1.6.21.3 Unit of Measure
- 1.6.22 Storm Drainage Structures (Line Items 0057 & 0058)
 - 1.6.22.1 Payment
 - 1.6.22.2 Measurement
 - 1.6.22.3 Unit of Measure
- 1.6.23 Concrete Slab on Grade (Estimated Quantity) (Line Items 0060)
 - 1.6.23.1 Payment
 - 1.6.23.2 Measurement
 - 1.6.23.3 Unit of Measure
- 1.6.24 Utility Bridge - Precast Beams (Line Item 0062)
 - 1.6.24.1 Payment
 - 1.6.24.2 Measurement
 - 1.6.24.3 Unit of Measure
- 1.6.25 Granular Subbase Course (Estimated Quantity) (Line Item 0067)
 - 1.6.25.1 Payment
 - 1.6.25.2 Measurement
 - 1.6.25.3 Unit of Measure
- 1.6.26 Asphaltic Concrete Pavement (Estimated Quantity) (Line Item 0068)
 - 1.6.26.1 Payment
 - 1.6.26.2 Measurement
 - 1.6.26.3 Unit of Measure
- 1.6.27 Milling Asphalt (Estimated Quantity) (Line Item 0069)
 - 1.6.27.1 Payment
 - 1.6.27.2 Measurement
 - 1.6.27.3 Unit of Measure
- 1.6.28 Permanent Access Road (Estimated Quantity) (Line Item 0071)
 - 1.6.28.1 Payment
 - 1.6.28.2 Measurement
 - 1.6.28.3 Unit of Measure

- 1.6.29 Chain Link Fencing (Estimated Quantity) (Line Items 0072)
 - 1.6.29.1 Payment
 - 1.6.29.2 Measurement
 - 1.6.29.3 Unit of Measure
- 1.6.30 Ornamental Fencing (Estimated Quantity) (Line Item 0073)
 - 1.6.30.1 Payment
 - 1.6.30.2 Measurement
 - 1.6.30.3 Unit of Measure
- 1.6.31 Vehicular Gates (Line Item 0075)
 - 1.6.31.1 Payment
 - 1.6.31.2 Measurement
 - 1.6.31.3 Unit of Measure
- 1.6.32 Bollards (Line Item 0076)
 - 1.6.32.1 Payment
 - 1.6.32.2 Measurement
 - 1.6.32.3 Unit of Measure
- 1.6.33 Sodding (Estimated Quantity) (Line Item 0077)
 - 1.6.33.1 Payment
 - 1.6.33.2 Measurement
 - 1.6.33.3 Unit of Measure
- 1.6.34 Seeding (Estimated Quantity) (Line Item 0078)
 - 1.6.34.1 Payment
 - 1.6.34.2 Measurement
 - 1.6.34.3 Unit of Measure
- 1.6.35 TRS Overtopping (Estimated Quantity) (Line Item 0080)
 - 1.6.35.1 Payment
 - 1.6.35.2 Measurement
 - 1.6.35.3 Unit of Measure
- 1.6.36 TRS Overtopping - Dewatering Equipment Standby (Estimated Quantity) (Line Item 0081)
 - 1.6.36.1 Payment
 - 1.6.36.2 Measurement
 - 1.6.36.3 Unit of Measure
- 1.6.37 Debris Removal for Dredging and Excavation (Estimated Quantity) (Line Item 0090)
 - 1.6.37.1 Payment
 - 1.6.37.2 Measurement
 - 1.6.37.3 Unit of Measure

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

- 3.1 PAYMENT PROCEDURES
 - 3.1.1 Requesting Progress Payment
 - 3.1.2 Options and Modification CLINS

-- End of Section Table of Contents --

SECTION 01 22 00

MEASUREMENT AND PAYMENT

PART 1 GENERAL

1.1 SUMMARY

This section describes how Line Items will be measured and paid for when making progress payments. Work to be measured is described in specification sections listed for each Line Item. Measurement procedures for payment, required quantity survey or procurement documentation and payment restrictions are described in applicable specification sections. Allocate costs for work not specifically mentioned to the Line Item most closely associated with work involved. Unless there is a specific Line Item for administrative costs, such as Quality Control and Safety, allocate such costs proportionally across all Line Items.

1.2 Reference

The publication listed below forms a part of this specification to the extent referenced. This publication is referred to in the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 1110-2-1003

(2013) Hydrographic Surveying

1.3 RESIDENT MANAGEMENT SYSTEM (RMS)

1.3.1 Definition

The terms Contract Line Item Number (CLIN) and Line Item are interchangeable herein (e.g.: CLIN 0001 is Line Item 0001). The term CLIN is a contracting term used in the Resident Management System (RMS) payment data base.

1.3.2 Instructions

See Section 01 45 00 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM) for more information on CLINs, schedule of values of pay activities and construction schedule, and payment procedures. The Contractor mode user manual with in-depth instructions can be downloaded from the RMS web site indicated in Section 01 45 00 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM).

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Transmit submittal items in accordance with Section 01 33 00 SUBMITTAL PROCEDURES. Submit the following preconstruction submittal items no later than 30 calendar

days after Notice to Proceed:

SD-01 Preconstruction Submittals

Schedule of Values; G, R0

SD-07 Certificates

Utility Invoices

Request For Progress Payment; G, R0

1.5 JOB PAYMENT ITEMS

Payment items for the work of this contract for which contract lump sum payments will be made are listed in the LINE ITEMS AND PRICING SCHEDULE with the unit JOB and described below. All costs for items of work, which are not specifically mentioned to be included in a particular job or unit price payment item, will be included in the listed job item most closely associated with the work involved. The job price and payment made for each item listed will constitute full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, and for performing all work required for which separate payment is not otherwise provided.

Progress payments for job CLINs will be made in accordance with the Payments Under Fixed-Price Construction Contracts clause of Section 00700 CONTRACT CLAUSES in Volume 1. Submit **Schedule of Values** for Government approval with the baseline schedule submittal (see Section **01 32 01** PROJECT SCHEDULE). Include a list of pay activities to breakdown bid for each job CLIN as part of the RMS Schedule. Schedule of Values will become basis for CLIN and Pay Activity data in the RMS payment data base. An unbalanced Schedule of Values and Pay Activity Schedule will be returned for revision. If this contract contains either the Continuing Contracts clause or the Continuing Contracts (Alternate) clause, take into account the amount reserved for contract payments when preparing the construction schedule.

1.5.1 Utility Invoices

Submit invoices from the utility company indicating payment for services covered in Line Items 0082, 0083, and 0084. Payment under this line item will not be made prior to submission of valid invoices to the Government. The Contractor will be reimbursed for costs indicated on these invoices only.

1.5.2 Mobilization and Demobilization (Line Items 0001)

Payment will be made for costs associated with or incidental to mobilization and demobilization for all activities associated with the project including those associated with the establishment of initial project management, initial partnering meeting, and coordination. See Clause PAYMENT FOR MOBILIZATION AND DEMOBILIZATION of Section 00700 CONTRACT CLAUSES in Volume 1.

1.5.3 Erosion and Sediment Control and Turbidity Monitoring
(Line Item 0002)

Payment will be made for costs associated with or incidental to installing and maintenance of erosion and sediment controls for construction and obtaining, analyzing, and reporting the results of monitoring for turbidity. See Section 01 57 25 TURBIDITY AND DISPOSAL MONITORING.

1.5.4 Environmental Species Monitoring (Line Item 0003)

Payment will be made for costs associated with Environmental Species Monitoring as specified in 01 57 20 ENVIRONMENTAL PROTECTION.

1.5.5 Vibration Control and Monitoring (Line Item 0004)

Payment will be made for costs associated with or incidental to the protection of existing structures from construction activities and monitoring. See Section 02 22 13 VIBRATION MONITORING REQUIREMENTS FOR EXISTING STRUCTURES.

1.5.6 Contracting Officer's Field Office (Fixed Costs) (Line Item 0005)

Payment will be made for costs associated with or incidental to furnish, deliver, construct and any utility connection required to establish the Contracting Officer's field office as specified in 01 52 10 CONTRACTING OFFICER'S FIELD OFFICE .

1.5.7 As-Built Drawings & Close Out Submittals (Line Item 0008)

Payment will be made for costs associated with or incidental to preparing as-built drawings and closeout submittals. See Section 01 78 02 CLOSEOUT SUBMITTALS.

1.5.8 Demolition (Line Item 0010)

Payment will be made for costs associated with or incidental to demolition, permanent removal, hauling, and offsite disposal of existing features as identified on the contract drawings. See Section 02 41 00 DEMOLITION and Section 02 42 00 DECONSTRUCTION OF ELECTRIC TRANSMISSION AND DISTRIBUTION UTILITY. Photos are included as an attachment to 02 41 00 DEMOLITION for select critical existing structures which record drawing information is not available and/or are to be salvaged as part of the project.

1.5.9 Demolition - Remnant Removal at De Diego Bridge (Line Item 0011)

Payment will be made for costs associated with or incidental to demolition of remnant construction debris and infrastructure located beneath the De Diego Bridge. Existing features include helical anchors, structural steel wide-flange members, steel sheet piles, and concrete debris. Excavation to the final channel depth and excavation required for riprap installation is also included in this item. See Section 02 41 00 DEMOLITION. Record drawings showing the existing known remnants are included as part of the drawings.

1.5.10 Demonstration Wall (Line Item 0016)

Payment will be made for costs associated with or incidental to placement of three demonstration wall drilled shafts, including a correlation

boring, drilling and placement of temporary casing and slurry, removal of material and testing of slurry, placement of steel reinforcement cage, placement of concrete and removal of temporary casing, installation of the cap beam, installation of required jet grout seals, installation of wall drainage material, and final testing and acceptance and demolition to required elevation after acceptance in accordance with Section 31 63 29 DRILLED CONCRETE PIERS AND SHAFTS. A demonstration wall is required for each drilled shaft crew, and when there is a change in personnel (including subcontractors), or change in methods or equipment.

1.5.11 Drilled Shaft Cap Beam (Line Items 0021)

Payment will be made for costs associated with or incidental to placement of the cap beam including formwork, preformed joint filler, steel reinforcing, dowels, concrete, finishing, and curing. See Sections 03 11 13 STRUCTURAL CAST-IN-PLACE CONCRETE FORMING, 03 20 00 CONCRETE REINFORCING, 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS, 03 35 00 CONCRETE FINISHING, AND 03 39 00 CONCRETE CURING.

1.5.12 Jet Grouting - Ground Improvement (Line Item 0023)

Payment will be made for costs associated with or incidental to the installation of the required jet grout columns for ground improvement including, but not limited to, plant, tools, material, and equipment to the locations and depths shown on the plans. The costs for core sampling, testing, logging, reporting, and Contractor quality control procedures shall be incidental. See section 31 32 23 JET GROUTING.

1.5.13 Concrete Double Box Culvert - 5feetx6.1feet (CIP) (Line Item 0052)

Payment will be made for costs associated with or incidental to construction of the cast-in-place double box culvert formwork, preformed joint filler, steel reinforcing, dowels, concrete, finishing, and curing. This item also includes payment for excavation, bedding material, connection to headwalls, and backfilling. See Sections 03 11 13 STRUCTURAL CAST-IN-PLACE CONCRETE FORMING, 03 20 00 CONCRETE REINFORCING, 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS, 03 35 00 CONCRETE FINISHING, AND 03 39 00 CONCRETE CURING.

1.5.14 Stormwater Bypass (Line Item 0059)

Payment will be made for costs associated with or incidental to establishing and maintaining the required stormwater bypass at existing outfalls while construction of the new outfall is being performed. This includes, but is not limited to, temporary piping, installation and removal of temporary blockouts, and pipe stubs and temporary flap gate at channel bypass sheet piling. See contract drawing CU510 for stormwater bypass requirements and sequencing.

1.5.15 Utility Bridge - Middle Pier Cap (CIP) (Line Item 0063)

Payment will be made for costs associated with or incidental to placement of the cast-in-place middle pier cap formwork, preformed joint filler, steel reinforcing, dowels, concrete, finishing, and curing. See Sections 03 11 13 STRUCTURAL CAST-IN-PLACE CONCRETE FORMING, 03 20 00 CONCRETE REINFORCING, 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS, 03 35 00 CONCRETE FINISHING, AND 03 39 00 CONCRETE CURING.

1.5.16 Utility Bridge - Bridge Slab (CIP) (Line Item 0064)

Payment will be made for costs associated with or incidental to placement of the cast-in-place bridge slab including formwork, preformed joint filler, steel reinforcing, dowels, concrete, finishing, and curing. See Sections 03 11 13 STRUCTURAL CAST-IN-PLACE CONCRETE FORMING, 03 20 00 CONCRETE REINFORCING, 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS, 03 35 00 CONCRETE FINISHING, AND 03 39 00 CONCRETE CURING.

1.5.17 48" DIP Waterline Crossing (Line Item 0065)

Payment will be made for costs associated with or incidental to the removal of service, and required excavation of the existing 48-inch waterline and replacement with a new 48-inch crossing of the channel over the new bridge including pipe, fittings, valves, valve box, supports, and connection to the existing system.

1.5.18 Traffic Control (Line Item 0066)

Payment will be made for costs associated with or incidental to traffic control including coordination with all agencies/entities having jurisdiction, signs, temporary pavement markings, and temporary concrete barriers as applicable.

1.5.19 Parking Lot Striping (Line Item 0070)

Payment will be made for costs associated with or incidental to furnish, deliver, and install permanent lane striping on the permanent bridge and approaches. Temporary striping and markings are excluded from this item. See section 32 17 23 PAVEMENT MARKINGS.

1.5.20 Metal Steps (Line Item 0074)

Payment will be made for costs associated with or incidental to furnishing and installing metal steps for access to the utility bridge. See section 05 50 13 MISCELLANEOUS METAL FABRICATIONS, 05 51 00 METAL STAIRS and the Contract Drawings.

1.5.21 Restoration (Line Item 0079)

Payment will be made for costs associated with or incidental to returning staging areas, parking areas, temporary work areas, and temporary access routes to preconstruction conditions in accordance with the Contract Drawings.

1.5.22 LUMA Electrical Power Charges (Line Items 0082)

Payment will include full compensation for the charges paid to LUMA by the Contractor for furnishing all materials, equipment, and labor required to remove, relocate, and install power lines in accordance with 26 00 00 BASIC ELECTRICAL MATERIALS AND METHODS and Section 33 71 01 OVERHEAD TRANSMISSION AND DISTRIBUTION and detailed work as shown in the drawings. Payment will be made for actual cost incurred for utility relocation, installation, coordination, inspection, and outages. Actual cost incurred will be based on utility invoices indicated in paragraph SUBMITTALS above. Payment under these line items will not be made prior to submission of valid invoices to the Government. The Contractor will be reimbursed for costs indicated on these invoices only.

1.5.23 AT&T/Liberty Telecommunications Charges (Line Item 0083)

Payment will include full compensation for the charges paid to AT&T and Liberty by the Contractor for furnishing all materials, equipment, and labor required to remove, relocate, and install telecommunication lines and detailed work as shown in the drawings. Payment will be made for actual cost incurred for utility relocation and installation. Actual cost incurred will be based on utility invoices indicated in paragraph SUBMITTALS above. Payment under these line items will not be made prior to submission of valid invoices to the Government. The Contractor will be reimbursed for costs indicated on these invoices only.

1.5.24 Claro Telecommunications Charges (Line Item 0084)

Payment will include full compensation for the charges paid to Claro by the Contractor for furnishing all materials, equipment, and labor required to remove, relocate, and install telecommunication lines and detailed work as shown in the drawings. Payment will be made for actual cost incurred for utility relocation and installation. Actual cost incurred will be based on utility invoices indicated in paragraph SUBMITTALS above. Payment under these line items will not be made prior to submission of valid invoices to the Government. The Contractor will be reimbursed for costs indicated on these invoices only.

1.5.25 Sanitary Siphon - Temporary Retaining Structures (Line Item 0085)

Payment will be made for costs associated with or incidental to the installation, procurement, and any required design of the temporary retaining structures required for excavating to the bottom of the jacking and receiving pits. This item includes costs for driving and removal of sheetpiles, excavation, backfill, tremie concrete, unwatering, exit rings, and unwatering of the coffercell. See sections 31 23 19 UNWATERING, 31 52 10 TEMPORARY RETAINING STRUCTURES, and 31 41 16 METAL SHEET PILING.

1.5.26 Sanitary Siphon - Headwalls (Line Item 0086)

Payment will be made for costs associated with or incidental to construction of the headwalls including formwork, preformed joint filler, embedded sleeves, sleeve boxes, anchors, steel reinforcing, dowels, concrete, finishing, and curing. See Sections 03 11 13 STRUCTURAL CAST-IN-PLACE CONCRETE FORMING, 03 20 00 CONCRETE REINFORCING, 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS, 03 35 00 CONCRETE FINISHING, AND 03 39 00 CONCRETE CURING.

1.5.27 Sanitary Siphon - Influent Box, Effluent Box, and Piping (Line Item 0087)

Payment will be made for costs associated with or incidental to furnishing and installing the required influent box, effluent box, and piping. Influent box and effluent box construction includes but is not limited to excavation, earthwork, reinforced concrete walls, lean concrete fill, and roof hatches, as well as appurtenances including fiberglass pipe railing, fiberglass grating, ladder rungs, epoxy coatings, and sluice gates. This item also includes furnishing and installing ductile iron pipe located outside of the channel limits, trenching, backfill, sanitary sewer manholes, and connections to structures. Operation and testing of the system shall also be included in this item. See Section 33 30 00 SANITARY SEWERAGE.

1.5.28 Sanitary Siphon - Boring and Jacking (Line Item 0088)

Payment will be made for costs associated with or incidental to the boring and jacking operation required to install the ductile iron sewer pipe below the channel. This item includes the pilot tube installation, boring, casing installation, furnishing and installing the ductile iron pipe within the casing, annular sand fill, end seals, modular seals, and soil improvement such as jet grouting or horizontal drains to prevent water ingress into the jacking or receiving pits. See Section 33 05 07 PILOT TUBE GUIDED BORING AND JACKING.

1.5.29 Concrete Privacy Wall Replacement (Line Item 0089)

Payment will be made for costs associated with or incidental to designing, furnishing, and installing the required replacement of the existing concrete privacy wall at Nemesio Canales. This item includes but is not limited to the design, excavation, earthwork, reinforcement, concrete, concrete blocks, connections, foundations, grout, finishing, and work incidental thereto. For bidding purposes, assume the wall is 8-inches thick, 8-feet tall, and of length as shown on the Drawings. See Section 32 31 00 FENCES AND GATES for additional requirements.

1.5.30 Municipal Taxes (Line Item 0091)

Pursuant to San Juan Municipality local regulations, taxes may apply with each award of this contract. See Paragraph 999.229-1 in Section 00100 of this solicitation. The contract price will be adjusted to compensate the Contractor for the amount of construction taxes. No mark-ups will be allowed in this line item. The Contractor will be paid on the basis of proof of payment from the appropriate Commonwealth of Puerto Rico or Municipal authorities.

1.6 UNIT PRICE PAYMENT ITEMS

Payment items for the work of this contract on which the contract unit price payments will be made are listed in the LINE ITEMS AND PRICING SCHEDULE and described below. The unit price and payment made for each item listed will constitute full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, and for performing all work required for each of the unit price items. Break down of unit price activities into activities with varying units is not allowed

Each Unit Price CLIN may be a single pay activity item or may be broken down into pay activities with smaller quantities equal to CLIN total. Contract unit price multiplied by agreed quantity is full compensation.

1.6.1 Contracting Officer's Field Office (Recurring Costs)(Estimated Quantity) (Line Item 0006)

1.6.1.1 Payment

Payment will be made for costs associated with or incidental to operating and maintaining the Contracting Officer's field office including all costs for rent and provision of all services indicated in Section 01 52 10 CONTRACTING OFFICER'S FIELD OFFICE. Costs for such provision and maintenance of the Contractor's own field office and other facilities will not be paid for under this line item.

1.6.1.2 Measurement

The basis for payment will be the amount of time that the Contracting Officer's field office is available to the Government within the duration of the contract.

1.6.1.3 Unit of Measure

Month.

1.6.2 Partnering Regroups (Estimated Quantity) (Line Item 0007)

1.6.2.1 Payment

Payment will be made for costs associated with or incidental to providing facilitator, venue, and associated follow-up deliverables as indicated in Section 01 30 00 ADMINISTRATIVE PROCEDURES. Partnering Regroups shall be assumed to occur every 180 days throughout the project duration.

1.6.2.2 Measurement

The basis for payment will be for each meeting completed and associated deliverables submitted.

1.6.2.3 Unit

Each.

1.6.3 Clearing and Grubbing (Estimated Quantity) (Line Item 0009)

1.6.3.1 Payment

Payment will be made for costs associated with or incidental to clearing and grubbing of existing vegetation, trees, etc., as specified in Section 31 11 00 CLEARING AND GRUBBING. Payment constitutes full compensation for all plant, labor, materials, equipment, tools, and incidentals necessary to complete the work.

1.6.3.2 Measurement

Clearing and grubbing will be measured for payment based upon the actual acreage cleared and grubbed in accordance with the contract drawings and specifications.

1.6.3.3 Unit of Measure

Acre.

1.6.4 Temporary Bypass Sheet Pile (Estimated Quantity) (Line Item 0012)

Payment will be made for costs associated with or incidental to the installation, procurement, and any required design of the temporary bypass sheet piles required for working platform establishment and drilled shaft installation purposes. Costs for removal and reuse of the temporary bypass sheet piles shall also be included in this item. See section 31 41 16 METAL SHEET PILING.

1.6.4.1 Measurement

Temporary Bypass Sheet Pile measurement will be based on design quantities; that is the length from tip elevation to top elevation times the length of temporary retaining wall; as documented in the driving records. In areas where the design tip elevation is not reached, the square footage is based upon the actual achieved tip elevation.

1.6.4.2 Unit of Measure

Square foot.

1.6.5 Permanent and Transition Sheet Piles (Estimated Quantity) (Line Items 0013, 0014 and 0015)

Payment will be made for costs associated with or incidental to the installation of the new permanent sheetpiles at the existing pump station site (R10, L3, and L4) and the transition sheet piles at the upstream transition of the project (C1, C2, R9, and L10) and Josefinia (J-C1, J-C2, J-M1, J-R2, and J-L2). See section 31 41 16 METAL SHEET PILING and section 09 97 02 PAINTING: COAL TAR EPOXY.

1.6.5.1 Measurement

Permanent and Transition Sheet Piles measurement will be based on design quantities; that is the length from tip elevation to top elevation times the length of wall; as documented in the driving records. In areas where the design tip elevation is not reached, the square footage is based upon the actual achieved tip elevation.

1.6.5.2 Unit of Measure

Square foot.

1.6.6 Drilled Shaft (Estimated Quantity) (Line Items 0017, 0018, 0019, 0020 & 0061)

1.6.6.1 Payment

Payment will be made for costs associated with or incidental to placement of drilled shafts, including establishment of a work platform, crane matting design and placement, drilling and placement of temporary casing and slurry, removal and disposal of excavated material and testing of slurry, placement of steel reinforcement cage, placement of concrete and removal of temporary casing, and testing and acceptance of each drilled shaft. See Section 31 63 29 DRILLED CONCRETE SHAFTS AND PIERS.

1.6.6.2 Measurement

Drilled shafts shall be measured by the actual vertical length taken from the bottom of the cap beam, as shown on the contract drawings, to the final tip elevation which have been satisfactorily installed. Tip elevations shown on the contract drawings are based on the best information available and are for bidding purposes only. Final tip elevations are to be field-adjusted and approved by the Contracting Officer based on varying geotechnical conditions.

1.6.6.3 Unit of Measure

Vertical linear foot.

1.6.7 Jet Grout Seals (Estimated Quantity) (Line Items 0022)

1.6.7.1 Payment

Payment will be made for costs associated with or incidental to the installation of the required jet grout columns including, but not limited to, the costs for core sampling, testing, logging, reporting, acceptance, and Contractor quality control procedures. Jet grouting for correction of verticality and plumbness during drilled shaft construction is specifically excluded from this item. See section 31 32 23 JET GROUTING.

1.6.7.2 Measurement

Measurement of jet grout seals will be by the vertical length acceptably installed. The length of each column installed will be measured to the nearest tenth of a linear foot. The portion of any column installed below the elevation specified on the drawings will not be measured for payment unless the over-depth is directed by the Contracting Officer.

1.6.7.3 Unit of Measure

Vertical linear foot.

1.6.8 Guardrail (Estimated Quantity) (Line Item 0024)

1.6.8.1 Payment

Payment will be made for costs associated with or incidental to placement of the guardrails. See Section 34 71 13 VEHICLE CRASH BARRIERS.

1.6.8.2 Measurement

Guardrail will be measured for payment by the length satisfactorily placed and accepted.

1.6.8.3 Unit of Measure

Linear foot.

1.6.9 Handrail (Estimated Quantity) (Line Item 0025)

1.6.9.1 Payment

Payment will be made for costs associated with or incidental to placement of the handrails. See Section 05 52 00 METAL HAND RAILINGS.

1.6.9.2 Measurement

Handrail will be measured for payment by the length satisfactorily placed and accepted.

1.6.9.3 Unit of Measure

Linear foot.

1.6.10 Excavation (Estimated Quantity) (Line Items 0026, 0027 & 0028)

1.6.10.1 Payment

Payment will be made for costs associated with or incidental to excavation, transportation, and disposal of materials; excavation, hauling and stockpiling of topsoil; providing and maintaining access to the work site(s) and disposal area(s) and debris removal. Dry excavated material above approximately El. +1.0 may be used on-site as fill if meeting requirements in 31 00 00. All excess material, dry material not meeting backfill requirements, and wet material below approximately El. +1.0 shall be disposed of offsite. See Section 31 00 00 EARTHWORK and 35 20 23 DREDGING.

Excavation within the channel banks is designated for offshore disposal and shall be paid for under the CLINS "Channel Excavation (Sta. 83+40 to Sta. 118+50)" and "Channel Excavation (Sta. 118+50 to Sta. 147+40)". See the Contract Drawings for the pay limit lines. Dewatering for dry excavation is prohibited within the channel banks.

Excavation outside of the channel banks, not incidental to other aspects of the work, and not eligible for offshore disposal, shall be paid for under the CLIN "Embankment Excavation" and disposed of at La Chuleta or an approved disposal site.

The channel bank is defined by a vertical plane that is located where the mean higher high water (MHHW) meets the ascending bank. The MHHW is defined as El. 4.5 from station 83+40 to 118+50 and El. 6.0 from station 123+50 to 147+40 for project purposes.

1.6.10.2 Measurement

The total amount of material removed, and to be paid for under this contract, will be measured by the volume in-place prior to excavation with quantities determined by digital terrain model (DTM) surface to surface computations. The quantity of material measured for Channel Excavation versus Embankment Excavation will be determined by theoretical computation based on the pay limit lines shown on the Contract Drawings. The Government will perform initial and final surveys in accordance with the clause QUANTITY SURVEYS of Section 00700 CONTRACT CLAUSES in Volume 1. The initial and final DTM surfaces used for calculation of final quantities will be determined from the original (initial surface) survey to be performed by the Government after clearing and before excavation, and the finished grades (final surface) as shown on the drawings. Interim surveys for Contractor quality control or to support payment requests shall be the responsibility of the Contractor. The final survey performed by the Government will be used to verify that excavation is complete to the required finished grades, and that tolerances have not been exceeded. DTM surfaces used for calculation of quantities for progress payments will be determined from the original survey performed by the Government, and elevations above finished grade obtained from progress surveys performed by the Contractor in accordance with the clause QUANTITY SURVEYS of Section 00700 CONTRACT CLAUSES in Volume 1.

The final quantity shall include excavation to the final channel invert as shown on the Contract Drawings. No payment will be made for excavation below this depth. Estimated bid quantities shown within Section 00010A LINE ITEMS AND PRICING SCHEDULE are calculated to the final channel invert as shown on the Contract Drawings and do not include any overdepth or

tolerances.

1.6.10.3 Unit of Measure

Cubic yard.

1.6.11 Backfill (Estimated Quantity) (Line Items 0029 & 0030)

1.6.11.1 Payment

Payment will be made for costs associated with or incidental to borrow, processing (removal of unsuitable fill), transportation, and placement of suitable materials; providing and maintaining access to the work site(s) and borrow area(s). See Section 31 00 00 EARTHWORK.

1.6.11.2 Measurement

The total amount of material placed, and to be paid for under this contract, will be measured by the volume in-place with quantities determined by digital terrain model (DTM) surface to surface computations. The Government will perform initial and final surveys in accordance with the clause QUANTITY SURVEYS of Section 00700 CONTRACT CLAUSES in Volume 1. The initial and final DTM surfaces used for calculation of final quantities will be determined from the original (initial surface) survey performed by the Government after clearing and before fill placement, and the finished grades (final surface) as shown on the drawings. The final survey performed by the Government will be used to verify that fill placement is complete to the required finished grades, and that tolerances have not been exceeded. Fill above required finished grades will not be included in quantities calculated for final payment. The estimated quantity for this line item does not include tolerances. The Contractor is responsible for considering the cost of required tolerances, and including this cost in the unit price for this line item (See Section 00010A LINE ITEMS AND PRICING SCHEDULE). DTM surfaces used for calculation of quantities for progress payments will be determined from the original survey performed by the Government, and elevations below finished grade obtained from progress surveys performed by the Contractor in accordance with the clause QUANTITY SURVEYS of Section 00700 CONTRACT CLAUSES in Volume 1. See Section 31 00 00 EARTHWORK.

1.6.11.3 Unit of Measure

Cubic yard.

1.6.12 Wall Groundwater Drains and Filter (Estimated Quantity) (Line Item 0031)

1.6.12.1 Payment

Payment will be made for costs associated with or incidental to placement of the wall groundwater drains including required excavation, wall penetrations, piping, geotextile fabric, and stone filter.

1.6.12.2 Measurement

Wall groundwater drains and filter will be measured for payment by the length satisfactorily placed and accepted.

1.6.12.3 Unit of Measure

Linear Foot.

1.6.13 Finish Grading (Estimated Quantity) (Line Item 0032)

1.6.13.1 Payment

Payment will be made for costs associated with or incidental to establishing the final grades as shown on the contract drawings for areas outside of the channel. See Section 31 00 00 EARTHWORK.

1.6.13.2 Measurement

Finish grading will be measured for payment by the area shown on the drawings.

1.6.13.3 Unit of Measure

Square yard.

1.6.14 Gabion Baskets for Bedding (Estimated Quantity) (Line Item 0033)

1.6.14.1 Payment

Payment will be made for costs associated with or incidental to furnishing, transportation, assembling, filling, and placing gabion baskets including material costs for the bedding stone and geotextile. See Section 31 36 00 GABION MATTRESS

1.6.14.2 Measurement

Gabion Baskets for Bedding will be measured for payment by each basket in the arrangement and grades shown on the drawings.

1.6.14.3 Unit of Measure

Each.

1.6.15 Marine Mattress (Estimated Quantity) (Line Item 0034)

Payment will be made for costs associated with or incidental to furnishing geogrid, stone fill material, and placement. This includes, but not limited to, costs associated with assembling, filling, lifting, and placing mattresses as shown on the plans. Costs for Contractor quality control procedures shall be incidental. See section 35 42 35 MARINE MATTRESS.

1.6.15.1 Measurement

Marine mattresses will be measure for payment by the area of marine mattress satisfactorily placed and accepted.

1.6.15.2 Unit of Measure

Square foot.

1.6.16 Riprap (Estimated Quantity) (Line Items 0035, 0036, 0037 & 0038)

1.6.16.1 Payment

Payment will be made for costs associated with or incidental for furnishing, hauling, handling, placing, and maintaining the riprap until final acceptance by the Government. No separate payment will be made for the stockpiling of riprap, and all cost in connection with stockpiling shall be included in the contract unit price for riprap. See Section 35 31 19 STONE AND CHANNEL PROTECTION FOR STRUCTURES.

1.6.16.2 Measurement

Riprap will be measured for payment by the ton. Quantities will be computed to the nearest whole ton. Riprap will be measured for payment, in the presence of the Contracting Officer, by weighing on approved, accurately calibrated scales furnished by and at the expense of the Contractor. The scales shall be capable of printing a weight ticket including time, date, truck number, and weight.

1.6.16.3 Unit of Measure

Ton.

1.6.17 Articulating Concrete Block Revetment (Estimated Quantity) (Line Item 0039)

1.6.17.1 Payment

Payment will be made for costs associated with or incidental to furnishing and installing articulating concrete block revetments. Work includes incidental grading and preparatory work; furnishing and installing the geotextile, geogrid, and ACB, filling the voids, securing cable fasteners, and seeding (where specified). Engineering services and product testing shall be incidental, if required. Placing cast-in-place concrete joints and cutting blocks shall be incidental, if required. Payment will be full compensation for all material, labor and equipment to complete the work.

1.6.17.2 Measurement

Measurement of ACB revetment for payment will be made on the basis of the face area. The pay lines of ACB revetment will be neat lines taken off the approved shop drawings; and will include embedded blocks and anchor trenches.

1.6.17.3 Unit of Measure

Square Foot.

1.6.18 Concrete Armor Units (Estimated Quantity) (Line Item 0040)

Payment will be made for costs associated with or incidental to furnishing and installing concrete armor units. This includes, but not limited to, costs associated with assembling individual concrete units, assembling concrete armor unit bundles, cables for securing bundles, and placement of units as shown on the plans. Costs for Contractor quality control procedures shall be incidental. See Section 35 31 19.26 CONCRETE ARMOR UNITS.

1.6.18.1 Measurement

Marine mattresses will be measure for payment by the area of concrete armor units satisfactorily placed and accepted.

1.6.18.2 Unit of Measure

Square foot.

1.6.19 Storm Drainage Headwalls (Line Items 0041, 0042, & 0043)

1.6.19.1 Payment

Payment will be made for costs associated with or incidental to construction of the headwalls including formwork, preformed joint filler, joint ring, steel reinforcing, dowels, concrete, finishing, and curing. See Sections 03 11 13 STRUCTURAL CAST-IN-PLACE CONCRETE FORMING, 03 20 00 CONCRETE REINFORCING, 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS, 03 35 00 CONCRETE FINISHING, AND 03 39 00 CONCRETE CURING.

1.6.19.2 Measurement

Storm drainage headwalls will be measured for payment for each of the headwalls satisfactorily placed and accepted.

1.6.19.3 Unit of Measure

Each.

1.6.20 Storm Drainage Piping (Estimated Quantity) (Line Items 0044 through 0051)

1.6.20.1 Payment

Payment will be made for costs associated with or incidental to placement of the storm drain piping including required excavation, bedding material, piping, connection to headwalls, and backfilling. Removal of existing pipe shall also be included in this item. See Section 33 40 00 STORM DRAINAGE UTILITIES.

1.6.20.2 Measurement

Storm drainage piping will be measured for payment for the length of pipes satisfactorily placed and accepted

1.6.20.3 Unit of Measure

Linear foot.

1.6.21 Flap Gates (Line Items 0053 through 0056)

1.6.21.1 Payment

Payment will be made for costs associated with or incidental to furnishing and installing flapgates including anchors, metal ladder rungs, fall protection grab hooks, and grab bars. See Section 33 40 00 STORM DRAINAGE UTILITIES.

1.6.21.2 Measurement

Flap gates will be measured for payment for each of the flap gates satisfactorily placed and accepted.

1.6.21.3 Unit of Measure

Each.

1.6.22 Storm Drainage Structures (Line Items 0057 & 0058)

1.6.22.1 Payment

Payment will be made for costs associated with or incidental to the placement of the drainage structures including manholes, inlets, frames, covers, grating, and ladders as indicated on the contract drawings, as well as, any required excavation, bedding material, foundation preparation, and backfilling. See Section 33 40 00 STORM DRAINAGE UTILITIES.

1.6.22.2 Measurement

Storm drainage structures will be measured for payment by each drainage structure satisfactorily placed and accepted

1.6.22.3 Unit of Measure

Each.

1.6.23 Concrete Slab on Grade (Estimated Quantity) (Line Items 0060)

1.6.23.1 Payment

Payment will be made for costs associated with or incidental to the placement of the concrete slab at the pump station outlet including formwork, concrete material, placing of concrete, curing, finishing, and any required Contractor QC requirements. See Section 03 11 13 STRUCTURAL CAST-IN-PLACE CONCRETE FORMING, 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS, 03 20 00 CONCRETE REINFORCING, 03 35 00 CONCRETE FINISHING, and 03 39 00 CONCRETE CURING.

1.6.23.2 Measurement

Concrete slab on grade will be measured for payment by the area satisfactorily placed and accepted

1.6.23.3 Unit of Measure

Square Yard.

1.6.24 Utility Bridge - Precast Beams (Line Item 0062)

1.6.24.1 Payment

Payment will be made for costs associated with or incidental to furnish, deliver, and install precast beams including bearing pads, anchors, shear studs, utility blockouts/supports, and bearing pads in accordance with Section 03 41 00 PRECAST, PRESTRESSED CONCRETE BRIDGE GIRDERS.

1.6.24.2 Measurement

Precast beams will be measured for payment by each precast beam completed and accepted.

1.6.24.3 Unit of Measure

Each.

1.6.25 Granular Subbase Course (Estimated Quantity) (Line Item 0067)

1.6.25.1 Payment

Payment will be made for costs associated with or incidental to furnishing all materials, equipment, plant, and tools required to place, grade, and compact granular subbase course layers.

1.6.25.2 Measurement

Granular Subbase Course will be measured for payment by area of thicknesses as indicated on the contract drawings completed and accepted by the Contracting Officer.

1.6.25.3 Unit of Measure

Square yard.

1.6.26 Asphaltic Concrete Pavement (Estimated Quantity) (Line Item 0068)

1.6.26.1 Payment

Payment will be made for costs associated with or incidental to furnishing all materials, equipment, plant, tools, testing, and certificates required to satisfactorily complete the work. Cost for adjustments to handholes, manholes, and utility boxes to grade shall be included herein. See Section 32 12 16 HOT-MIX ASPHALT (HMA) FOR ROADS.

1.6.26.2 Measurement

Asphaltic Concrete will be measured for payment by weight of asphaltic concrete completed and accepted.

1.6.26.3 Unit of Measure

Ton.

1.6.27 Milling Asphalt (Estimated Quantity) (Line Item 0069)

1.6.27.1 Payment

Payment will be made for costs associated with or incidental to furnishing all equipment, plant, and tools required to mill asphalt pavement areas indicated on the plans. No payment will be made for milling outside the specified area of work. See Section 32 01 16 COLD MILLING ASPHALT PAVEMENT.

1.6.27.2 Measurement

The quantity of milled pavement will be the number of square yards completed and accepted as determined by the Contracting Officer.

Determine the number of square yards of milled pavement by measuring the length and width of the milled surface within the specified work area. Measure the width of the area to the closest inch and measure the length of the area to the closest foot.

1.6.27.3 Unit of Measure

Square yard.

1.6.28 Permanent Access Road (Estimated Quantity) (Line Item 0071)

1.6.28.1 Payment

Payment will be made for costs associated with or incidental to constructing the permanent access roads on both sides of the channel. This includes grading and compaction of the subgrade, stone, stone placement, and stone compaction. See Section 32 15 00 AGGREGATE SURFACING

1.6.28.2 Measurement

Permanent access road will be measured for payment by the area shown on the drawings.

1.6.28.3 Unit of Measure

Square yard.

1.6.29 Chain Link Fencing (Estimated Quantity) (Line Items 0072)

1.6.29.1 Payment

Payment will be made for costs associated with or incidental to placement of the chain link fencing and gates. See Section 32 31 00 FENCES AND GATES.

1.6.29.2 Measurement

Chain link fencing will be measured for payment by the length satisfactorily placed and accepted.

1.6.29.3 Unit of Measure

Linear foot.

1.6.30 Ornamental Fencing (Estimated Quantity) (Line Item 0073)

1.6.30.1 Payment

Payment will be made for costs associated with or incidental to placement of the ornamental fencing.

1.6.30.2 Measurement

Ornamental fencing will be measured for payment by the length satisfactorily placed and accepted.

1.6.30.3 Unit of Measure

Linear foot.

1.6.31 Vehicular Gates (Line Item 0075)

1.6.31.1 Payment

Payment will be made for costs associated with or incidental to placement of the vehicular gates including footings, gate members, latches, hinges, and supports.

1.6.31.2 Measurement

Vehicular gates will be measured for payment by the number of gates satisfactorily placed and accepted.

1.6.31.3 Unit of Measure

Each.

1.6.32 Bollards (Line Item 0076)

1.6.32.1 Payment

Payment will be made for costs associated with or incidental to furnishing and installing bollards including excavation, concrete fill, pipe, and backfill.

1.6.32.2 Measurement

Bollards will be measured for payment by the number of bollards satisfactorily placed and accepted.

1.6.32.3 Unit of Measure

Each.

1.6.33 Sodding (Estimated Quantity) (Line Item 0077)

1.6.33.1 Payment

Payment will be made for costs associated with or incidental to placement and establishment of sod. See Section 32 92 23 SODDING.

1.6.33.2 Measurement

Measurement will be made by the area of sod established.

1.6.33.3 Unit of Measure

Square yard.

1.6.34 Seeding (Estimated Quantity) (Line Item 0078)

1.6.34.1 Payment

Payment will be made for costs associated with or incidental to placement and establishment of seed. See Section 32 92 19 SEEDING.

1.6.34.2 Measurement

Measurement will be made by the area of seeding established.

1.6.34.3 Unit of Measure

Square yard.

1.6.35 TRS Overtopping (Estimated Quantity) (Line Item 0080)

1.6.35.1 Payment

Payment will be made for costs associated with or incidental to installation, operation, and fueling of the dewatering pumps and equipment which are required to dewater the work site after a temporary retaining structure (TRS) overtopping event including labor, material and equipment used to reposition, operate, and secure dewatering equipment after operation. Enough dewatering pumps and equipment are to be operated to have the excavated work area dewatered within 48 hours. Dewatering of the excavated work area is to start once the channel water surface elevation recedes below the top of sheet pile wall within the TRS currently under construction. Pumps and dewatering equipment needed for normal daily construction activities operations are not to be included in this line item.

1.6.35.2 Measurement

Measurement will be made by each event that occurs.

1.6.35.3 Unit of Measure

Each.

1.6.36 TRS Overtopping - Dewatering Equipment Standby (Estimated Quantity)
(Line Item 0081)

1.6.36.1 Payment

Unit price payment will be made for costs associated with or incidental to installation, lease, maintenance, periodic testing, and removal of dewatering pumps and appurtenant equipment on standby which are required to dewater the TRS after an overtopping event. Pumps and dewatering equipment needed for normal daily construction activities shall not be included in this line item.

Perform testing of standby pumps on a 2-month cycle to ensure operability condition when needed.

1.6.36.2 Measurement

Measurement will be made by time on standby.

1.6.36.3 Unit of Measure

Month.

1.6.37 Debris Removal for Dredging and Excavation (Estimated Quantity)
(Line Item 0090)

1.6.37.1 Payment

Payment will be made for costs associated with or incidental to debris

removal and disposal from the channel bottom and side slopes in areas to be dredged or excavated only. Debris removal and disposal from other areas will not be paid for under this line item.

1.6.37.2 Measurement

Debris removed from areas to be dredged or excavated will be paid for by weight. Weight of debris to be disposed offsite from the jobsite will be measured by the delivery ticket from the landfill.

1.6.37.3 Unit of Measure

Ton.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 PAYMENT PROCEDURES

Upon receiving initial Resident Management System import file, go to PAY ACTIVITIES and establish a link between bid breakdown schedule of values of PAY ACTIVITIES to contract CLINs using SCHEDULE ACTIVITIES data entry page.

3.1.1 Requesting Progress Payment

For progress payments, ensure ACTIVITY SCHEDULE, FEATURE SCHEDULE, submittal register, and punchlists are all up to date. Use PROGRESS PAYMENTS to REQUEST ACTIVITY EARNINGS for both ACTIVITY EARNINGS data entry page and OTHER EARNING. Provide hard copies of supporting invoices and quantity measurements to support all requested earnings. Ensure that sums of payment activities do not exceed contract award CLIN funding amounts, or unbalanced CLINs error will prevent processing the payment. Submit a [request for progress payment](#) monthly in accordance with Clause 52.232-5 PAYMENTS UNDER FIXED-PRICE CONSTRUCTION CONTRACTS of Section 00700 CONTRACT CLAUSES. In addition to the items listed in Clause 52.232-5(b)(1), submit the following in the request for progress payment:

- a. Approved construction schedule.
- b. Up-to-date payroll records.
- c. Current submittal register.
- d. Approved submittals on installed property.
- e. QA/QC deficiency tracking list.
- f. Environmental reports.
- g. Progress photographs.
- h. Working/current as-built drawings.

Once the complete pay application package is received to include the Contractor's signed ENG 93, the 14 calendar days will commence.

3.1.2 Options and Modification CLINs

When additional work is added by modification, existing CLINs funding amounts must be updated, or new CLINs for modification will be created. If contract has option CLINs not yet awarded, option CLINs will appear as zero dollar CLINs until option is awarded by modification. No payment may be requested for Options or Modification CLINs until contract modification has been funded and signed.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 30 00

ADMINISTRATIVE PROCEDURES

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 SUBMITTALS
- 1.3 PROJECT COORDINATION
 - 1.3.1 Resident Management System (RMS)
 - 1.3.2 Access by Others
 - 1.3.3 Inspections by Others
- 1.4 PROJECT MEETINGS
- 1.5 PROJECT MEETINGS FORMAT
- 1.6 PRECONSTRUCTION CONFERENCE
 - 1.6.1 Preconstruction Conference Minutes and Phone
 - 1.6.2 Preconstruction Conference Submittals
 - 1.6.2.1 Other Division 01 Submittals
 - 1.6.2.2 Divisions 02 through 48 Submittals
 - 1.6.3 Joint Risk Register
 - 1.6.4 List of Subcontractors
- 1.7 PARTNERING
 - 1.7.1 Initial Partnering Meeting
 - 1.7.1.1 Contractor Responsibility
 - 1.7.1.2 Tasks and Deliverables
 - 1.7.2 Partnering Regroups
 - 1.7.2.1 Contractor Responsibility
 - 1.7.2.2 Tasks and Deliverables
 - 1.7.3 Cancellation of Partnering Meeting or Regroup
- 1.8 NOTICE TO PROCEED (NTP)
- 1.9 COORDINATION MEETING
- 1.10 PROGRESS MEETINGS
 - 1.10.1 Progress Meeting Participants
- 1.11 CONTRACTOR PERFORMANCE EVALUATIONS
 - 1.11.1 Training

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

- 3.1 GENERAL MEETING REQUIREMENTS
- 3.2 PROGRESS MEETING AGENDA
- 3.3 SIGNATURE AUTHORITY
- 3.4 DRUG-FREE WORK PLACE RECORD

-- End of Section Table of Contents --

SECTION 01 30 00

ADMINISTRATIVE PROCEDURES

PART 1 GENERAL

1.1 SUMMARY

Manage project and coordinate activities of own employees, subcontractors, suppliers and offsite fabricators. Use computers, e-mail, and internet resources for administrative work. Notify Contracting Officer of important meetings, scheduled events, and activities. Furnish labor and materials and equipment required to plan and execute project management functions and coordination. Coordinate activities and manage resources to construct project conforming to Contract, on time and within budget. Related Sections are 01 32 01 PROJECT SCHEDULE, and 01 45 04 CONTRACTOR QUALITY CONTROL

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Transmit submittal items in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Draft Joint Risk Register

List of Subcontractors

SD-07 Certificates

Partnering Meeting(s) Written Synopsis; G, R0

Signature Authority

Drug-Free Work Place Record

1.3 PROJECT COORDINATION

1.3.1 Resident Management System (RMS)

Use the U.S. Army Corps of Engineers' Resident Management System for construction information management. See Section 01 45 00 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM).

1.3.2 Access by Others

Maintain access to power poles and other items owned or operated by utility companies. Coordinate with utility companies as required for line marking, hookups, and relocations as needed to perform work. Incoming electrical utilities become the responsibility of Contractor at transformers. Coordinate emergency incident response with local law

enforcement and fire rescue authorities. Notify Contracting Officer in writing of coordination problems encountered affecting work.

1.3.3 Inspections by Others

During construction period, inspection, or maintenance, work will be performed by others within the construction limits. The Contractor is responsible for neatly mowing all grass and vegetation (not to include established trees or other protected vegetation designated as wetland areas) within the construction limits for the duration of the Contract. Neatly edge or trim areas not accessible to mowers. Perform mowing or trimming every 3 weeks or when vegetation exceeds 12 inches in height.

1.4 PROJECT MEETINGS

Contracting Officer requires following types of project meetings:

- Preconstruction Conference
- Partnering Meeting
- Partnering Regroup Meetings
- Coordination Meeting
- Preparatory and Initial Phase meetings for each feature of work
- Project Progress Meetings

Project meetings are described in detail in subparts below.

1.5 PROJECT MEETINGS FORMAT

All project meetings will occur face to face or virtually; the default format will be face to face. The Contracting Officer will approve or request a virtual meeting when a face-to-face meeting format cannot be used. Meetings at the Contractor's facility will have system capabilities for both audio and video. The audio and video system must provide for full coverage of the conference room.

1.6 PRECONSTRUCTION CONFERENCE

Contracting Officer will conduct a Preconstruction Conference for this project in accordance with Clause PRECONSTRUCTION CONFERENCE of Section 00700 CONTRACT CLAUSES in Volume 1. The Preconstruction Conference will be held within 30 to 60 calendar days after Notice to Proceed (NTP). (Refer to subparagraph PRECONSTRUCTION CONFERENCE SUBMITTALS below.) Contracting Officer will notify the Contractor of the time, place, and agenda. Notify key subcontractors and suppliers to attend. Contracting Officer will discuss Contract ground rules and general issues including:

- Lines of Contracting Officer authority
- Lines of Contractor authority
- 00700 Contract Clauses
- 00800 Special Contract requirements
- Contract administration
- Progress payment
- Correspondence procedures
- Project Schedule
- Submittal Register
- Labor requirements
- General Site Safety
- FDEP Permit requirements
- Joint Risk Management Processes

1.6.1 Preconstruction Conference Minutes and Phone

Contracting Officer will take detailed minutes of Preconstruction Conference discussions and may record the meeting. Copies of typed minutes will be provided to the Contractor to review for accuracy, sign, and return. Signed minutes become part of the Contract file. Recordings, if used, will be made available for the Contractor to review or copy.

1.6.2 Preconstruction Conference Submittals

The timing of submission of submittals and completion of the Preconstruction Conference is intended to allow the Contractor and the Government adequate time to prepare for commencement of work. However, if the Contractor fails to submit required items and receive acceptance and/or approval within the times stated, the Contractor will not be permitted to commence work until these requirements have been satisfied. Any delays attributable to the Contractor's failure to comply with these pre-work requirements must be at the Contractor's expense and may be cause for remedial action by the Contracting Officer. Submittals required by this Section are described in paragraph SUBMITTALS above.

1.6.2.1 Other Division 01 Submittals

Contracting Officer reviews submitted draft plans to prepare the Coordination Meeting agenda and to understand Contractor procedures. Submit the following preconstruction submittal items no later than 30 calendar days after NTP unless specifically identified otherwise in the following specifications:

- Project Schedule - See Section 01 32 01 PROJECT SCHEDULE
- Accident Prevention Plan - See Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS
- Quality Control Plan - See Section 01 45 04 CONTRACTOR QUALITY CONTROL
- Temporary Facilities Plan - See Section 01 50 02 TEMPORARY CONSTRUCTION FACILITIES
- Traffic Control Plan - See Section 01 55 26 TRAFFIC CONTROL
- Environmental Protection Plan - See Section 01 57 20 ENVIRONMENTAL PROTECTION
- Dewatering Work Plan - See Section 31 00 00 EARTHWORK

1.6.2.2 Divisions 02 through 48 Submittals

In addition to the above, bring submittal items for materials, workmanship, plans, or events required early in project schedule, which are ready for transmittal to Contracting Officer. Prepare transmittal of submittal items in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

1.6.3 Joint Risk Register

Request a sample template from Contracting Officer or receive Government approval for a Contractor's template form prior to submission. Submit the [draft joint risk register](#) no later than 30 calendar days after NTP.

1.6.4 List of Subcontractors

Documentation of compliance with Clauses SUBCONTRACTS (LABOR STANDARDS) and LIMITATIONS ON SUBCONTRACTING of Section 00700 CONTRACT CLAUSES in

Volume 1, and if included, in paragraph LIMITATIONS ON SUBSTITUTIONS FOR CERTAIN POSITIONS AND/OR SUBCONTRACTORS of Section 00800 SPECIAL REQUIREMENTS in Volume 1. As part of the RMS Schedule, submit a [list of subcontractors](#) with company name, person to contact, street address, mailing address, phone number, type of specialty, and estimated subcontract quote no later than 30 calendar days after NTP.

1.7 PARTNERING

Contracting Officer wants to have a bilateral project partnership with the Contractor that draws on strengths of both organizations to identify and achieve common goals. Typical partnering goals include:

- Effective and efficient Contract performance
- Project completion on time and within budget
- Construction completed according to plans and specifications
- Development of cooperative management teams
- Project success with customer (stakeholders) satisfaction
- Improved Project communication
- Joint Risk Management Plan and Register

Provide a facilitator from an approved list of vendors located on the website

<https://www.saj.usace.army.mil/About/Divisions-Offices/Construction/> or propose a vendor as indicated below.

1.7.1 Initial Partnering Meeting

Contractor, key subcontractors, and suppliers must attend a 1-day project Partnering Meeting. Attendants will be determined by the Contracting Officer. At a minimum, the Contractor's Project Manager, Quality Control Manager, and Site Superintendent must attend the Partnering Meeting. The Partnering Meeting is a project team building meeting facilitated by qualified persons. Contracting Officer attendees and invited persons may include:

- Administrative Contracting Officer (Area Engineer/Resident Engineer)
- Construction Quality Assurance Representatives
- Caribbean District - Construction Branch persons
- Caribbean District - Engineering Division Designers
- Local Sponsor (Owner) - Puerto Rico Department of Natural and Environmental Resources

The goal of the Initial Partnering Meeting is to build trust, identify common goals, develop an initial Joint Risk Plan and Register, and understand individual project members' expectations and organizational values. Usual results include better communication between Contract parties, a shortened project learning curve, and a cohesive two-party Contract partnership. Participants usually sign a voluntary partnering agreement generally describing a mutual obligation to cooperate to achieve project goals and maintain effective communication. The Initial Partnering Meeting is mandatory. The expected duration for the initial partnering session is 1 day (at least 8 hours).

1.7.1.1 Contractor Responsibility

Provide a facilitator for all Partnering Meetings from an approved list of vendors on the web site

<https://www.saj.usace.army.mil/About/Divisions-Offices/Construction/> under

the BUSINESS WITH US column to the bottom right of the page. The Contractor may propose a vendor not included on the approved vendor list for Contracting Officer approval. Pay for all associated facilitator costs including travel. Provide a meeting venue adequate to comfortably accommodate a minimum of 40 attendees. Provide all office supplies (i.e., easels, projectors, markers) necessary to facilitate the meeting. The Contractor will be responsible for all costs associated with and incidental to the Partnering Meeting.

1.7.1.2 Tasks and Deliverables

The facilitator must provide a written agenda and appropriate work products for the Initial Partnering Meeting. Upon completion of the meeting, submit a [partnering meeting\(s\) written synopsis](#) of the meeting to the Government. At a minimum, the written synopsis must include the partnering charter, goals, attendance register, lines of communication, working lists, items of discussion, action items, risk register, and any other information addressed in the meeting as requested by the Government in agreement with the partnering team. Provide all deliverables in editable Microsoft Office Suite (i.e., Word, Excel) and PDF formats to the Government for review within 10 business days after completion of the meeting.

1.7.2 Partnering Regroups

After the Initial Partnering Meeting, participation will be voluntary in accordance with a signed Partnering Agreement. Partnering Agreements include provisions for Contracting Officer and Contractor to request Partnering Regroups when needed. A Partnering Regroup is usually a partnering session of 1 day (at least 4 hours) where senior-level Contractor and Contracting Officer project members and other interested parties review project issues of concern to include an update to the Joint Risk Register. Issues of conflict are reviewed with attempts to affirm common project goals and understand the concerns of other partners.

1.7.2.1 Contractor Responsibility

Provide a facilitator for all Partnering Meetings from an approved list of vendors on the web site indicated in paragraph CONTRACTOR RESPONSIBILITY above. The Contractor may propose a vendor not included on the approved vendor list for Contracting Officer approval. The Contractor must pay for all associated facilitator costs including travel. Provide a meeting venue adequate to comfortably accommodate a minimum of 25 attendees. Provide all office supplies (i.e., easels, projectors, markers) necessary to facilitate the meeting. The Contractor will be responsible for all costs associated with and incidental to the Partnering Meeting.

1.7.2.2 Tasks and Deliverables

The facilitator must provide a written agenda and appropriate work products for the Regroup Partnering Meeting. Upon completion of the meeting, a written synopsis of the meeting must be provided to the Contractor and the Government. At a minimum the written synopsis must include updates of the partnering charter, goals, attendance register, lines of communication, working lists, items of discussion, action items, risk register, and any other information addressed in the meeting as requested by the Government in agreement with the partnering team. Provide all deliverables in editable Microsoft Office Suite (i.e., Word, Excel) and PDF formats to the Government for review within 10 business days after

completion of the meeting.

1.7.3 Cancellation of Partnering Meeting or Regroup

Partnering Meetings may be postponed or cancelled by the Contracting Officer at no cost to the Government when the Government notifies the Contractor in writing with at least 72 hours notice prior to the scheduled meeting. No payment will be made for a postponed or cancelled Partnering Meeting.

1.8 NOTICE TO PROCEED (NTP)

NTP will be issued 15 to 30 calendar days after award or after all required bonds are in place, whichever is later. If the Contractor failed to submit and receive acceptance and/or approval of specified plans including, but not limited to, Accident Prevention Plan - Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS, Quality Control Plan - Section 01 45 04 CONTRACTOR QUALITY CONTROL, and Environmental Protection Plan - Section 01 57 20 ENVIRONMENTAL PROTECTION or has not yet received the Contracting Officer's conditional approval to work under an Interim Plan, the Contractor cannot proceed with the work, and the work will be suspended in accordance with the Suspension of Work clause within the Contract. While the Contractor is working under a conditionally accepted Interim Plan, funds may be retained from progress payments in accordance with the Payments under Fixed-Price Construction Contracts clause of this Contract until such time as the Contractor submits an acceptable plan. If the Contractor does not submit an acceptable plan within a reasonable time as determined by the Contracting Officer, the Contracting Officer may order the Contractor to suspend work. Any suspension order issued for the Contractor's failure to submit an acceptable plan will not constitute an unreasonable delay under the Suspension of Work clause, and the Contractor will not be entitled to an equitable adjustment of either performance period or Contract price.

1.9 COORDINATION MEETING

A Coordination Meeting is scheduled, convened, and conducted by Contracting Officer after a Preconstruction Conference and prior to starting physical construction. Plans submitted after NTP (i.e., Quality Control Plan, Environmental Protection Plan, and Accident Prevention Plan) will have been reviewed. The Coordination Meeting is primarily for on-site Contractor Quality Control staff, including subcontractor and supplier employees performing quality control, to meet and discuss the project in detail with Contracting Officer's Quality Assurance Representatives. Attendants will be determined by the Administrative Contracting Officer. At a minimum, the Contractor's Quality Control Manager, Site Safety and Health Officer, Site Superintendent, and major subcontractor's foreman must attend the Coordination Meeting. Purposes of the Coordination Meeting are:

- Achieve a mutual understanding with Contractor of required Quality Control
- Jointly review submitted draft plans; resolve issues of concern
- Discuss project plans and specifications, schedule, and documentation
- Discuss project Joint Risk Management Plan and Register
- Establish a good working relationship between the Contractor's Quality Control Staff and Quality Assurance Representatives
- Discuss the number of hard-copy submittals requirements

1.10 PROGRESS MEETINGS

Schedule, convene, and preside over progress meetings as required. As project activities increase, a minimum of one progress meeting per week is typical of a project of this scope. Convene additional meetings as required or when requested by Contracting Officer. Notify persons needed to be present to discuss agenda issues. Contracting Officer may direct attendance by key Contractor suppliers or fabricators as needed. A sample meeting agenda is provided in paragraph GENERAL MEETING REQUIREMENTS below. Take minutes of the meetings and distribute copies of the minutes within 3 business days after the meetings. Provide a professional-grade teleconference system with a conference phone with speakers for adequate audio coverage during the meeting. Provide a conference phone line to accommodate up to 25 individuals. Provide a meeting room to accommodate up to 15 people for weekly progress meetings.

1.10.1 Progress Meeting Participants

Typical participants include:

- Contracting Officer
- Construction Quality Assurance Representatives
- Contractor's Site Superintendent
- Contractor's Quality Control Manager
- Contractor's Safety Coordinator
- Subcontractors as appropriate to the agenda
- Suppliers as appropriate to the agenda
- Others as appropriate to the agenda

1.11 CONTRACTOR PERFORMANCE EVALUATIONS

Contractors are given an opportunity to provide and/or enter performance evaluation comments directly into the Contractor Performance Assessment Reporting System (CPARS). The system is a web-enabled application that supports the completion, distribution, and retrieval of Contract performance evaluations. Provide to the responsible area office the name and email address of the Contractor user who will be accessing CPARS.

1.11.1 Training

Contractor online training is available at the following web site: <https://www.cpars.gov/lc.htm>. Training gives the student an understanding of the Contractor Representative and Contractor Corporate Senior Management levels of access, the policies and regulations governing CPARS, and the electronic workflow with particular attention to the Contractor comment function.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 GENERAL MEETING REQUIREMENTS

See Section 01 45 04 CONTRACTOR QUALITY CONTROL for Preparatory and Initial Phase meeting checklists. The Contractor is responsible for Phase and Progress Meetings to include:

- sending a meeting notification to participants

- preparing an agenda for meetings
- using phase checklists for Preparatory and Initial Phase meetings
- setting the physical arrangements for meetings
- presiding at meetings
- taking minutes and recording proceedings and decisions
- copying and sending minutes to:
 - meeting participants
 - project parties affected by decisions
 - Contracting Officer (no later than 3 working days)

3.2 PROGRESS MEETING AGENDA

Modify agenda as needed for on-going work.

- Review minutes from previous progress meetings.
- Review RMS Contractor Action Item Report.
- Review work progress since previous meeting.
- Review current definable features of work:
 - Identify phases of current features of work.
 - Identify pending phase changes.
 - Identify features for discussion in next scheduled meeting.
- Discuss problem prevention:
 - Field observations
 - Deficiencies and tracking
 - Procedures working well
 - Problems or conflicts
 - Methods to improve
- Review construction schedule:
 - Identify delays
 - Discuss proposed corrective actions to regain schedule.
- Submittals and Requests for Information (design interpretation):
 - Review Submittal Register.
 - Identify submittals to expedite as required.
- Review off-site activities:
 - Fabrications
 - Material and equipment delivery schedule
- Review Testing:
 - Type and schedule
 - Received results
- Review changes to construction schedule:
 - Planned progress during succeeding work period
 - Coordination of various schedules
 - Effect of changes on construction and completion date
- Review site safety.
- Review and update Joint Risk Management Register.
- Discuss maintaining Contract quality for materials and workmanship.
- Discuss pending modifications, changes, and substitutions.
- Discuss other business as appropriate.

3.3 SIGNATURE AUTHORITY

Submit a Power of Attorney or a notarized letter of authority identifying local representatives authorized to sign Contract documents.

3.4 DRUG-FREE WORK PLACE RECORD

Submit documentation of compliance with Clause DRUG-FREE WORKPLACE of Section 00700 CONTRACT CLAUSES in Volume 1 (see subparagraphs (b)(1) through (b)(7)).

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 32 01

PROJECT SCHEDULE

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 PROJECT SCHEDULER QUALIFICATIONS
- 1.4 CORRESPONDENCE

PART 2 PRODUCTS

- 2.1 SOFTWARE
 - 2.1.1 Government Default Software
 - 2.1.2 Contractor Software
 - 2.1.2.1 Primavera
 - 2.1.2.2 Other Than Primavera

PART 3 EXECUTION

- 3.1 GENERAL REQUIREMENTS
 - 3.1.1 Scheduling Expectations Kickoff Meeting (SEK0)
- 3.2 BASIS FOR PAYMENT AND COST LOADING
 - 3.2.1 Activity Cost Loading
 - 3.2.2 Withholdings / Payment Rejection
- 3.3 PROJECT SCHEDULE DETAILED REQUIREMENTS
 - 3.3.1 Level of Detail Required
 - 3.3.2 Activity Durations
 - 3.3.3 Procurement Activities
 - 3.3.4 Mandatory Activities
 - 3.3.5 Government Activities
 - 3.3.6 Standard Activity Coding Dictionary
 - 3.3.6.1 Workers Per Day (WRKP)
 - 3.3.6.2 Responsible Party Coding (RESP)
 - 3.3.6.3 Area of Work Coding (AREA)
 - 3.3.6.4 Modification Number (MODF)
 - 3.3.6.5 Bid Item Coding (BIDI)
 - 3.3.6.6 Phase of Work Coding (PHAS)
 - 3.3.6.7 Category of Work Coding (CATW)
 - 3.3.6.8 Feature of Work Coding (FOW)
 - 3.3.7 Contract Milestones and Constraints
 - 3.3.7.1 Project Start Date Milestone and Constraint
 - 3.3.7.2 End Project Finish Milestone and Constraint
 - 3.3.7.3 Interim Completion Dates and Constraints
 - 3.3.7.3.1 Start Phase
 - 3.3.7.3.2 End Phase
 - 3.3.8 Adverse Weather
 - 3.3.9 Calendars
 - 3.3.10 Open Ended Logic

- 3.3.11 Default Progress Data Disallowed
- 3.3.12 Out-of-Sequence Progress
- 3.3.13 Added and Deleted Activities
- 3.3.14 Original Durations
- 3.3.15 Leads, Lags, and Start to Finish Relationships
- 3.3.16 Retained Logic
- 3.3.17 Percent Complete
- 3.3.18 Remaining Duration
- 3.3.19 Cost Loading of Closeout Activities
 - 3.3.19.1 As-Built Documents
 - 3.3.19.2 O & M Manuals
- 3.3.20 Early Completion Schedule and the Right to Finish Early
- 3.4 PROJECT SCHEDULE SUBMISSIONS
 - 3.4.1 Preliminary Project Schedule Submission
 - 3.4.2 Initial Project Schedule Submission
 - 3.4.3 Periodic Schedule Updates
- 3.5 SUBMISSION REQUIREMENTS
 - 3.5.1 Electronic Scheduling Data
 - 3.5.2 Narrative Report
 - 3.5.3 Schedule Reports
 - 3.5.3.1 Activity Report
 - 3.5.3.2 Logic Report
 - 3.5.3.3 Total Float Report
 - 3.5.3.4 Earnings Report by CLIN
 - 3.5.3.5 Scheduling and Leveling Report
 - 3.5.4 Network Diagram
 - 3.5.4.1 Continuous Flow
 - 3.5.4.2 Project Milestone Dates
 - 3.5.4.3 Critical Path
 - 3.5.4.4 Grouping and Sorting
 - 3.5.4.5 Cash Flow / Schedule Variance Control (SVC) Diagram
 - 3.5.5 Half Step Schedule
- 3.6 PERIODIC SCHEDULE UPDATE
 - 3.6.1 Periodic Schedule Update Meetings
 - 3.6.2 Update Submission Following Progress Meeting
- 3.7 WEEKLY PROGRESS MEETINGS
- 3.8 REQUESTS FOR TIME EXTENSIONS
 - 3.8.1 Justification of Delay
 - 3.8.2 Changes That Do Not Cause Delay
 - 3.8.3 Time Impact Analysis (Prospective Analysis)
 - 3.8.4 Forensic Schedule Analysis (Retrospective Analysis)
 - 3.8.5 Fragmentary Network (Fragnet)
 - 3.8.6 Time Extension
 - 3.8.7 Impact to Early Completion Schedule
- 3.9 FAILURE TO ACHIEVE PROGRESS
 - 3.9.1 Artificially Improving Progress
 - 3.9.2 Failure to Perform
 - 3.9.3 Recovery Schedule
- 3.10 OWNERSHIP OF FLOAT
- 3.11 TRANSFER OF SCHEDULE DATA INTO RESIDENT MANAGEMENT SYSTEM
- 3.12 PRIMAVERA P6 MANDATORY REQUIREMENTS

-- End of Section Table of Contents --

SECTION 01 32 01

PROJECT SCHEDULE

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AACE INTERNATIONAL (AACE)

- | | |
|-------------|--|
| AACE 29R-03 | (2011) Forensic Schedule Analysis |
| AACE 52R-06 | (2006) Time Impact Analysis - As Applied in Construction |
| AACE 84R-13 | (2015) Planning and Accounting for Adverse Weather |

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

- | | |
|------------|--------------------------------|
| ASCE 67-17 | (2017) Schedule Delay Analysis |
|------------|--------------------------------|

U.S. ARMY CORPS OF ENGINEERS (USACE)

- | | |
|-----------|--|
| ER 1-1-11 | (2017) Administration -- Project Schedules |
|-----------|--|

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Project Scheduler Qualifications; G, R0

Preliminary Project Schedule; G, R0

Initial Project Schedule; G, R0

SD-07 Certificates

Periodic Schedule Update; G, R0

1.3 PROJECT SCHEDULER QUALIFICATIONS

Designate an authorized representative to be responsible for the preparation of the schedule and all required updating and production of

reports. The Designated Project Scheduler must have a minimum of 2 years experience with and have prepared and maintained at least three previous construction schedules for projects of similar size and complexity to this contract, using Primavera P6. Representative must have a comprehensive knowledge of CPM scheduling principles and application.

1.4 CORRESPONDENCE

Identify or otherwise reference the applicable existing schedule activity(ies) in other project correspondence (e.g. serial letters, Requests for Information, QC Daily Reports, e-mails).

PART 2 PRODUCTS

2.1 SOFTWARE

The scheduling software utilized to produce and update the schedules required herein must be capable of meeting all requirements of this specification.

2.1.1 Government Default Software

The Government uses Primavera P6. Ensure exported schedule files are compatible with the version of P6 used by the Government.

2.1.2 Contractor Software

Scheduling software used by the contractor must be commercially available from the software vendor for purchase with vendor software support agreements available. The software routine used to create the required sdef file must be created and supported by the software manufacturer.

2.1.2.1 Primavera

If Primavera P6 is selected for use, provide the "xer" export file in a version of P6 importable by the Government system. Verify at the Scheduling Expectations Kickoff Meeting (SEKO) meeting which version of P6 is in use by the Government. Export the schedule in a version of P6 no newer than that used by the Government.

2.1.2.2 Other Than Primavera

If the Contractor chooses software other than Primavera P6, that is compliant with this specification, provide for the Government's use two licenses, two computers, and training for two Government employees in the use of the software. These computers will be stand-alone and not connected to Government or any other network. Computers and licenses will be returned at project completion.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

Prepare for approval a Project Schedule, as specified herein, pursuant to FAR Clause 52.236-15 Schedules for Construction Contracts. Show in the schedule the proposed sequence to perform the work and dates contemplated for starting and completing all schedule activities. The scheduling of the entire project is required. The scheduling of construction is the responsibility of the Contractor. Contractor management personnel must

actively participate in its development. Subcontractors, and suppliers working on the project must also contribute in developing and maintaining an accurate Project Schedule. Provide a schedule that is a forward planning as well as a project monitoring tool. Use the Critical Path Method (CPM) of network calculation to generate all Project Schedules. Prepare each Project Schedule using the Precedence Diagram Method (PDM).

3.1.1 Scheduling Expectations Kickoff Meeting (SEK0)

In conjunction with the post award conference or pre construction conference the Government and the Contractor will hold a separate and distinct Scheduling Expectations Kickoff Meeting (SEK0).

The intent of this meeting is for the Government to review with the contractor the requirements of the scheduling specification, communicate the Government's expectations regarding schedule development and management, and answer any questions the Contractor may have regarding the schedule requirements.

3.2 BASIS FOR PAYMENT AND COST LOADING

The schedule is the basis for determining contract earnings during each update period and therefore the amount of each progress payment. The aggregate value of all activities coded to a contract CLIN must equal the value of the CLIN. Match the progress payment "Pay Period Thru" date in RMS to the schedule data date.

3.2.1 Activity Cost Loading

Activity cost loading must be reasonable and without front-end loading. Activities with a negative cost loading is not allowed. Provide additional documentation to demonstrate reasonableness if requested by the Contracting Officer. Only tangible deliverables to the Government are cost loaded. Unless a specific line item is for administrative costs, such as preparation of preconstruction submittals and quality control, allocate such costs proportionally across all line items or to the closest related tangible deliverable.

3.2.2 Withholdings / Payment Rejection

Failure to meet the requirements of this specification may result in the disapproval of the preliminary, initial, or periodic schedule updates and subsequent rejection of payment requests until compliance is met.

In the event that the Contracting Officer directs schedule revisions and those revisions have not been included in subsequent Project Schedule revisions or updates, the Contracting Officer may also withhold 10 percent of pay request amount from each payment period until such revisions to the project schedule have been made.

3.3 PROJECT SCHEDULE DETAILED REQUIREMENTS

3.3.1 Level of Detail Required

Develop the Project Schedule to the appropriate level of detail to address major milestones and to allow for satisfactory project planning and execution. Failure to develop the Project Schedule to an appropriate level of detail will result in its disapproval. The Contracting Officer will consider, but is not limited to, the following characteristics and

requirements to determine appropriate level of detail:

3.3.2 Activity Durations

Reasonable activity durations are those that allow the progress of ongoing activities to be accurately determined between update periods. Non-procurement activities Original Durations (OD) are not to exceed 20 workdays or 30 calendar days.

3.3.3 Procurement Activities

Include in the schedule activities associated with the critical submittals and their approvals, procurement, fabrication, and delivery of long lead materials, equipment, fabricated assemblies, and supplies. Long lead procurement activities are those with an anticipated procurement sequence of over 90 calendar days.

3.3.4 Mandatory Activities

Include the following activities in the initial project schedule and all updates.

- a. Submission, review and acceptance of SD-01 Preconstruction Submittals (individual activity for each).
- b. Submission, review and acceptance of features requiring design completion
- c. Submission of mechanical/electrical/information systems layout drawings.
- d. Submission and approval of O & M manuals.
- e. Submission and approval of as-built drawings.
- f. Submission and approval of DD1354 data and installed equipment lists.
- g. Submission and approval of testing and air balance (TAB).
- h. Submission of TAB specialist design review report.
- i. Submission and approval of fire protection specialist.
- j. Submission and approval of Building Commissioning Plan, Controls testing plan, test data, and reports: Develop the schedule logic associated with testing and commissioning of mechanical systems to a level of detail consistent with the contract commissioning requirements. All tasks associated with building testing and commissioning will be completed prior to submission of building commissioning report and subsequent contract completion.

These activities may include but are not limited to: air and water balancing, building commissioning -functional performance testing, controls testing, and performance verification testing. Include other systems testing, if required. Include additional commissioning requirements noted elsewhere in the Contract as required in the schedule and discuss during the SEKO meeting, noted in paragraph SCHEDULING EXPECTATIONS KICKOFF MEETING (SEKO).

- k. Contractor's pre-final inspection.
- m. Correction of punch list from Contractor's pre-final inspection.
- n. Government's pre-final inspection.
- o. Correction of punch list from Government's pre-final inspection.
- p. Final inspection.

3.3.5 Government Activities

Show in schedule, Government and other agency activities that could impact progress. These activities include, but are not limited to [approvals](#), environmental permit approvals by State regulators, inspections, utility tie-in, Government Furnished Equipment (GFE) and Notice to Proceed (NTP) for phasing requirements.

3.3.6 Standard Activity Coding Dictionary

Use the activity coding structure defined in the Standard Data Exchange Format (SDEF) in [ER 1-1-11](#). This exact structure is mandatory. Develop and assign all Activity Codes to activities as detailed herein.

The SDEF format is as follows:

Field	Activity Code	Length	Description
1	WRKP	3	Workers per day
2	RESP	4	Responsible party
3	AREA	4	Area of work
4	MODF	6	Modification Number
5	BIDI	6	Bid Item (CLIN)
6	PHAS	2	Phase of work
7	CATW	1	Category of work
8	FOW	20	Feature of work*
*Some systems require that FEATURE OF WORK values be placed in several activity code fields. The notation shown is for Primavera P6. Refer to the specific software guidelines with respect to the FEATURE OF WORK field requirements.			

3.3.6.1 Workers Per Day (WRKP)

Assign Workers per Day for all field construction or direct work activities, if directed by the Contracting Officer. Workers per day is based on the average number of workers expected each day to perform a task for the duration of that activity.

3.3.6.2 Responsible Party Coding (RESP)

Assign responsibility code for all activities to the Prime Contractor, Subcontractor(s) or Government agency(ies) responsible for performing the activity.

- a. Activities coded with a Government Responsibility code include, but are not limited to: Government approvals, Government design reviews, environmental permit approvals by State regulators, Government Furnished Property/Equipment (GFP) and Notice to Proceed (NTP) for phasing requirements.
- b. Activities cannot have more than one Responsibility Code. Examples of acceptable activity code values are: DOR (for the designer of record); ELEC (for the electrical subcontractor); MECH (for the mechanical subcontractor); and GOVT (for USACE).

3.3.6.3 Area of Work Coding (AREA)

Assign Work Area code to activities based upon the work area in which the activity occurs. Define work areas based on resource constraints or space constraints that would preclude a resource, such as a particular trade or craft work crew from working in more than one work area at a time due to restraints on resources or space. Examples of Work Area Coding include different areas within a floor of a building, different floors within a building, and different buildings within a complex of buildings. Activities cannot have more than one Work Area Code.

3.3.6.4 Modification Number (MODF)

Assign a Modification Number Code to any activity or sequence of activities added to the schedule as a result of a Contract Modification, when approved by Contracting Officer. Key all Code values to the Government's modification numbering system. An activity can have only one Modification Number Code.

3.3.6.5 Bid Item Coding (BIDI)

Assign a Bid Item Code to all activities using the Contract Line Item Number (CLIN) to which the activity belongs, even when an activity is not cost loaded. An activity can have only one BIDI Code.

3.3.6.6 Phase of Work Coding (PHAS)

Assign Phase of Work Code to all activities. Examples of phase of work are procurement phase and construction phase. Each activity can have only one Phase of Work code.

- a. Code proposed fast track design and construction phases proposed to allow filtering and organizing the schedule by fast track design and construction packages.
- b. If the contract specifies phasing with separately defined performance periods, identify a Phase Code to allow filtering and organizing the schedule accordingly.

3.3.6.7 Category of Work Coding (CATW)

Assign a Category of Work Code to all activities. Category of Work Codes

include, but are not limited to construction submittal, procurement, fabrication, weather sensitive installation, non-weather sensitive installation, start-up, and testing activities. Each activity can have no more than one Category of Work Code.

3.3.6.8 Feature of Work Coding (FOW)

Assign a Feature of Work Code to appropriate activities based on the Definable Feature of Work to which the activity belongs based on the approved QC plan.

Definable Feature of Work is defined in Section 01 45 00 QUALITY CONTROL. An activity can have only one Feature of Work Code.

3.3.7 Contract Milestones and Constraints

Milestone activities are to be used for significant project events including, but not limited to, project phasing, project start and end activities, or interim completion dates. The use of artificial float constraints such as "zero free float" or "zero total float" are prohibited.

Mandatory constraints that ignore or effect network logic are prohibited. No constrained dates are allowed in the schedule other than those specified herein. Submit additional constraints to the Contracting Officer for approval on a case by case basis.

3.3.7.1 Project Start Date Milestone and Constraint

The first activity in the project schedule must be a start milestone titled "NTP Issued," which must have a "Start On" constraint date equal to the date that the NTP is issued.

3.3.7.2 End Project Finish Milestone and Constraint

The last activity in the schedule must be a finish milestone titled "End Project."

Constrain the project schedule to the Contract Completion Date in such a way that if the schedule calculates an early finish, then the float calculation for "End Project" milestone reflects positive float on the longest path. If the project schedule calculates a late finish, then the "End Project" milestone float calculation reflects negative float on the longest path.

3.3.7.3 Interim Completion Dates and Constraints

Constrain contractually specified interim completion dates to show negative float when the calculated late finish date of the last activity in that phase is later than the specified interim completion date.

3.3.7.3.1 Start Phase

Use a start milestone as the first activity for a project phase. Call the start milestone "Start Phase X" where "X" refers to the phase of work.

3.3.7.3.2 End Phase

Use a finish milestone as the last activity for a project phase. Call the

finish milestone "End Phase X" where "X" refers to the phase of work.

3.3.8 Adverse Weather

Ensure anticipated adverse weather is accounted for in the Project Schedule. [AACE 84R-13](#) provides methodologies for techniques in planning for adverse weather. The schedule will incorporate a weather calendar. If a method other than a weather calendar is proposed, the reason will be discussed at the SEKO meeting and the method must have approval from the Contracting Officer before it can be incorporated into the Project Schedule.

3.3.9 Calendars

Schedule activities on a Calendar to which the activity logically belongs. Develop calendars to accommodate any contract defined work period such as a 7-day calendar for Government Acceptance activities, concrete cure times, etc. Develop the default Calendar to match the physical work plan with non-work periods identified including weekends and holidays. Develop Seasonal Calendar(s) and assign to seasonally affected activities as applicable.

If a weather calendar is used its implementation and execution is to be in accordance with [AACE 84R-13](#)

3.3.10 Open Ended Logic

Only two open ended activities are allowed: the first activity "NTP Acknowledged" is to have no predecessor logic, and the last activity "-End Project" is to have no successor logic. Except as noted above all activities are to have at least a start-to-start or finish-to-start relationship with its predecessor(s) and a finish-to-finish or finish-to-start relationship with its successor(s).

Predecessor open ended logic may be allowed in a time impact analyses upon the Contracting Officer's approval.

3.3.11 Default Progress Data Disallowed

Actual Start and Finish dates must not automatically update with default mechanisms included in the scheduling software. Updating of the percent complete and the remaining duration of any activity must be independent functions. Disable program features that calculate one of these parameters from the other. Activity Actual Start (AS) and Actual Finish (AF) dates assigned during the updating process must match those dates provided in the Contractor Quality Control Reports. Failure to document the AS and AF dates in the Daily Quality Control report will result in disapproval of the Contractor's schedule.

3.3.12 Out-of-Sequence Progress

Activities that have progressed before all preceding logic has been satisfied (Out-of-Sequence Progress) will be allowed only on a case-by-case basis subject to approval by the Contracting Officer. Propose logic corrections to eliminate out of sequence progress or justify not changing the sequencing for approval prior to submitting an updated project schedule. Address out of sequence progress or logic changes in the Narrative Report and in the periodic schedule update meetings.

3.3.13 Added and Deleted Activities

Do not delete activities from the project schedule or add new activities to the schedule without approval from the Contracting Officer. Activity ID and description changes are considered new activities and cannot be changed without Contracting Officer approval.

3.3.14 Original Durations

Activity Original Durations (OD) must be reasonable to perform the work item. OD changes are prohibited unless justification is provided and approved by the Contracting Officer.

3.3.15 Leads, Lags, and Start to Finish Relationships

Lags must be reasonable as determined by the Government and not used in place of realistic original durations, must not be in place to artificially absorb or create float, or to replace proper schedule logic.

- a. Leads (negative lags) are prohibited.
- b. Start to Finish (SF) relationships are prohibited.

3.3.16 Retained Logic

Schedule calculations must retain the logic between predecessors and successors ("retained logic" mode) even when the successor activity(s) starts and the predecessor activity(s) has not finished (out-of-sequence progress). Software features that in effect sever the tie between predecessor and successor activities when the successor has started and the predecessor logic is not satisfied ("progress override") are not be allowed.

3.3.17 Percent Complete

Update the percent complete for each activity started, based on the realistic assessment of earned value. Activities which are complete but for remaining minor punch list work and which do not restrain the initiation of successor activities may be declared 100 percent complete to allow for proper schedule management.

3.3.18 Remaining Duration

Update the remaining duration for each activity based on the number of estimated workdays it will take to complete the activity. Remaining duration is not required to mathematically correlate with percentage found under paragraph entitled Percent Complete. The remaining duration for unstarted activities are not to be less than its original duration.

3.3.19 Cost Loading of Closeout Activities

Cost load the "Correction of punch list from Government pre-final inspection" activity(ies) not less than 1 percent of the present contract value. Activity(ies) may be declared 100 percent complete upon the Government's verification of completion and correction of all punch list work identified during Government pre-final inspection(s).

3.3.19.1 As-Built Documents

If there is no separate contract line item (CLIN) for as-built drawings, cost load the "Submission and approval of as-built documents" activity not less than \$35,000 or 1 percent of the present contract value, whichever is greater, up to \$200,000. Activity will be declared 100 percent complete upon the Government's approval of all as-built documents.

3.3.19.2 O & M Manuals

Cost load the "Submission and approval of O & M manuals" activity not less than \$20,000. Activity will be declared 100 percent complete upon the Government's approval of all O & M manuals.

3.3.20 Early Completion Schedule and the Right to Finish Early

An Early Completion Schedule is an Initial Project Schedule (IPS) that indicates all scope of the required contract work will be completed before the contractually required completion date.

- a. No IPS indicating an Early Completion will be accepted without being fully resource-loaded (including crew sizes and manhours) and the Government agreeing that the schedule is reasonable and achievable.
- b. The Government is under no obligation to accelerate work items it is responsible for to ensure that the early completion is met nor is it responsible to modify incremental funding (if applicable) for the project to meet the contractor's accelerated work.
- c. If an early completion schedule is accepted, the Contract completion date will be revised through a bilateral modification which must be agreed to and signed by both the Contracting Officer and the Contractor.

3.4 PROJECT SCHEDULE SUBMISSIONS

Provide the submissions as described below. The data reports and network diagrams required for each submission are contained in paragraph SUBMISSION REQUIREMENTS. If the Contractor fails or refuses to furnish the information and schedule updates as set forth herein, the Contractor will be deemed not to have provided an estimate upon which a progress payment can be made.

Review comments made by the Government on the schedule(s) do not relieve the Contractor from compliance with requirements of the Contract Documents.

3.4.1 Preliminary Project Schedule Submission

Within 15 calendar days after the NTP is acknowledged, submit the [Preliminary Project Schedule](#) defining the planned operations detailed for the first 90 calendar days for approval. The approved Preliminary Project Schedule will be used for payment purposes not to exceed 90 calendar days after NTP. Completely cost load the Preliminary Project Schedule to balance the contract award CLINS shown on the Price Schedule. The Preliminary Project Schedule may be summary in nature for the remaining performance period. It must be early start and late finish constrained and logically tied as specified. The Preliminary Project Schedule forms the basis for the Initial Project Schedule specified herein and must include all of the required plan and program preparations, submissions and

approvals identified in the contract (for example, Quality Control Plan, Safety Plan, and Environmental Protection Plan) as well as design activities, planned submissions of all early design packages, permitting activities, design review conference activities, and other non-construction activities intended to occur within the first 90 calendar days. Government acceptance of the associated design package(s) and all other specified Program and Plan approvals must occur prior to any planned construction activities. Activity code any activities that are summary in nature after the first 90 calendar days with Bid Item (CLIN) code (BIDI), Responsibility Code (RESP) and Feature of Work code (FOW).

3.4.2 Initial Project Schedule Submission

Submit the [Initial Project Schedule](#) for approval within 42 calendar days after notice to proceed is issued. The schedule must demonstrate a reasonable and realistic sequence of activities which represent all work through the entire contract performance period. No payment will be made for work items not fully detailed in the Project Schedule.

3.4.3 Periodic Schedule Updates

Update the Project Schedule on a regular basis, monthly at a minimum. Provide a draft Periodic Schedule Update for review at the schedule update meetings as prescribed in the paragraph PERIODIC SCHEDULE UPDATE MEETINGS. These updates will enable the Government to assess Contractor's progress.

- a. Update information including Actual Start Dates (AS), Actual Finish Dates (AF), Remaining Durations (RD), and Percent Complete. Updated information is subject to the approval of the Government at the meeting.
- b. AS and AF dates must match the date(s) reported on the Contractor's Quality Control Report for an activity start or finish.

3.5 SUBMISSION REQUIREMENTS

Submit the following items for the Preliminary Schedule, Initial Schedule, and every Periodic Schedule Update throughout the life of the project:

3.5.1 Electronic Scheduling Data

Provide electronic scheduling data containing the current project schedule and all previously submitted schedules in the format of the scheduling software (e.g. .xer). Also include the Narrative Report and all required Schedule Reports. The electronic file is to identify the type of schedule (Preliminary, Initial, Update), full contract number, Data Date and schedule file name. Each schedule must have a unique file name and use project specific settings. The file naming convention is to be determined at SEKO meeting.

3.5.2 Narrative Report

Provide a Narrative Report with each schedule submission. The Schedule Narrative Report is a "stand alone" report in PDF or Word format. Title the report "Schedule Narrative Report". Show the Report date, schedule name, and contract number on the first page. Indicate who is responsible for the schedule update and narrative report. Provide a brief description of the Project Scope. The Narrative Report is expected to communicate to

the Government the thorough analysis of the schedule output and the plans to compensate for problems, either current or potential, which are revealed through that analysis. Include the following information at a minimum in the Narrative Report:

- a. Identify and discuss the work scheduled to start in the next update period.
- b. Describe activities along the critical path in addition to activities on the next float path after the critical path.
- c. Describe current and anticipated problem areas, delaying factors and their impact, and an explanation of corrective actions taken or required to be taken. Identify the party responsible for delay, if applicable.
- d. Identify and explain why activities, based on their calculated late dates, should have either started or finished during the update period but did not.
- e. Identify and discuss all schedule changes by activity ID and activity name including what specifically was changed and why the change was needed. Include at a minimum new and deleted activities, logic changes, duration changes, calendar changes, lag changes, resource changes, and actual start and finish date changes.
- f. Identify and discuss out-of-sequence work.
- g. If the longest path changes from a monthly update to the next, provide an explanation of the factors driving the change.
- h. Respond to USACE Schedule Review Comments from the previous update(s).

3.5.3 Schedule Reports

The format, filtering, organizing, and sorting for each schedule report will be as directed by the Contracting Officer. Typically, reports contain Activity Numbers, Activity Description, Original Duration, Remaining Duration, Early Start Date, Early Finish Date, Late Start Date, Late Finish Date, Total Float, Actual Start Date, Actual Finish Date, and Percent Complete. Provide the reports electronically in .pdf format. Provide one set of hardcopy reports. The following lists typical reports that will be requested:

3.5.3.1 Activity Report

List of all activities sorted according to activity number.

3.5.3.2 Logic Report

List of detailed predecessor and successor activities for every activity in ascending order by activity number.

3.5.3.3 Total Float Report

A list of all incomplete activities sorted in ascending order of total float. List activities which have the same amount of total float in ascending order of Early Start Dates. Do not show completed activities on this report.

3.5.3.4 Earnings Report by CLIN

A compilation of the Total Earnings on the project from the NTP to the data date, which reflects the earnings of activities based on the agreements made in the schedule update meeting defined herein. Provided a complete schedule update has been furnished, this report serves as the basis of determining progress payments. Group activities by CLIN number and sort by activity number. Provide a total CLIN percent earned value, CLIN percent complete, and project percent complete. The printed report must contain the following for each activity: the Activity Number, Activity Description, Original Budgeted Amount, Earnings to Date, Earnings this period, Total Quantity, Quantity to Date, and Percent Complete (based on cost).

3.5.3.5 Scheduling and Leveling Report

Provide a Scheduling/Leveling Report (Schedule Log Report) generated from the current project schedule being submitted.

3.5.4 Network Diagram

The Network Diagram is required for the Preliminary, Initial and Periodic Updates. Depict and display the order and interdependence of activities and the sequence in which the work is to be accomplished. The Contracting Officer will use, but is not limited to, the following conditions to review compliance with this paragraph:

3.5.4.1 Continuous Flow

Show a continuous flow from left to right with no arrows from right to left. Show the activity number, description, duration, and estimated earned value on the diagram.

3.5.4.2 Project Milestone Dates

Show dates on the diagram for start of project, any contract required interim completion dates, and contract completion dates.

3.5.4.3 Critical Path

Show all activities on the critical path. The critical path is defined as the longest path.

3.5.4.4 Grouping and Sorting

Group and sort activities as directed to assist in the understanding of the activity sequence. Typically, this flow will group activities by major elements of work, category of work, work area and/or responsibility.

3.5.4.5 Cash Flow / Schedule Variance Control (SVC) Diagram

With each schedule submission, provide a SVC diagram showing 1) Cash Flow S-Curves indicating planned project cost based on projected early and late activity finish dates, and 2) Earned Value to-date.

3.5.5 Half Step Schedule

Should the Contracting Officer find it necessary, submit separate schedule

updates segregating progress reporting from non-progress revisions.

3.6 PERIODIC SCHEDULE UPDATE

3.6.1 Periodic Schedule Update Meetings

Conduct periodic schedule update meetings for the purpose of reviewing the proposed Periodic Schedule Update, Narrative Report, Schedule Reports, and progress payment. Conduct meetings at least monthly within five days of the proposed schedule data date. Provide a computer with the scheduling software loaded and a projector which allows all meeting participants to view the proposed schedule during the meeting. The Contractor's authorized scheduler must organize, group, sort, filter, perform schedule revisions as needed and review functions as requested by the Contractor and/or Government. The Contractor scheduler must attend in person unless virtual attendance is approved by Contracting Officer. The meeting is a working interactive exchange which allows the Government and Contractor the opportunity to review the updated schedule on a real time and interactive basis. The meeting will last no longer than 8 hours. Submit schedule file and narrative to the Government a minimum of two workdays in advance of the meeting. The Contractor's Project Manager must attend the meeting with the authorized representative of the Contracting Officer. Superintendents, foremen and major subcontractors must attend the meeting as required to discuss the project schedule and work. Following the schedule update meeting, make corrections to the schedule and resubmit, following the required schedule naming convention. Include only those changes approved by the Government in the schedule review meeting. Only approved schedules may be used to generate invoices for payment.

3.6.2 Update Submission Following Progress Meeting

Submit the complete [Periodic Schedule Update](#) of the Project Schedule containing all approved progress, revisions, and adjustments, pursuant to paragraph SUBMISSION REQUIREMENTS not later than 4 workdays after the periodic schedule update meeting.

3.7 WEEKLY PROGRESS MEETINGS

Conduct a weekly meeting with the Government (or as otherwise mutually agreed to) between the meetings described in paragraph entitled PERIODIC SCHEDULE UPDATE MEETINGS for the purpose of jointly reviewing the actual progress of the project as compared to the as planned progress and to review planned activities for the upcoming two weeks. Use the current approved schedule update for the purposes of this meeting and for the production and review of reports. At the weekly progress meeting, address the status of RFIs, RFPs and Submittals.

3.8 REQUESTS FOR TIME EXTENSIONS

[ASCE 67-17](#) provides delay analysis guidelines. If there is a conflict between the contract and guidance provided in [ASCE 67-17](#), the contract will govern. Provide justification of delay to the Contracting Officer in accordance with the contract provisions and clauses for approval, within 10 days of a delay occurring. Also prepare a time impact analysis for each Government request for proposal (RFP).

3.8.1 Justification of Delay

Provide a description of the event(s) that caused the delay and/or impact

to the work. As part of the description, identify all schedule activities impacted. Show that the event that caused the delay/impact was the responsibility of the Government. Provide a time impact analysis that demonstrates the effects of the delay or impact on the project completion date, or interim completion date(s). Evaluate multiple impacts chronologically; each with its own justification of delay. With multiple impacts consider any concurrency of delay. A time extension and the schedule fragnet becomes part of the project schedule and all future schedule updates upon approval by the Contracting Officer.

3.8.2 Changes That Do Not Cause Delay

If it is determined that a change does not constitute a schedule time extension, a fragnet as described in paragraph FRAGMENTARY NETWORK, is to be created and inserted into the Project schedule.

3.8.3 Time Impact Analysis (Prospective Analysis)

Prepare a time impact analysis for approval by the Contracting Officer based on industry standard [AACE 52R-06](#). Utilize a copy of the last approved schedule prior to the first day of the impact or delay for the time impact analysis. If Contracting Officer determines the time frame between the last approved schedule and the first day of impact is too great, prepare an interim updated schedule to perform the time impact analysis. Unless approved by the Contracting Officer, no other changes may be incorporated into the schedule being used to justify the time impact.

3.8.4 Forensic Schedule Analysis (Retrospective Analysis)

Prepare an analysis for approval by the Contracting Officer based on methodologies identified in industry standard [AACE 29R-03](#). Identify which methodology was chosen and explain why it is more appropriate than other methodologies available.

3.8.5 Fragmentary Network (Fragnet)

Prepare a proposed fragnet for time impact analysis consisting of a sequence of new activities that are proposed to be added to the project schedule to demonstrate the influence of the delay or impact to the project's contractual dates. Clearly show how the proposed fragnet is to be tied into the project schedule including all predecessors and successors to the fragnet activities. The proposed fragnet must be approved by the Contracting Officer prior to incorporation into the project schedule.

3.8.6 Time Extension

The Contracting Officer must approve the Justification of Delay including the time impact analysis before a time extension will be granted. No time extension will be granted unless the delay consumes all available Project Float and extends the projected finish date ("End Project" milestone) beyond the Contract Completion Date. The time extension will be in calendar days.

Actual delays that are found to be caused by the Contractor's own actions, which result in a calculated schedule delay, are not a cause for an extension to the performance period, completion date, or any interim milestone date.

3.8.7 Impact to Early Completion Schedule

No extended overhead will be paid for delay prior to the original Contract Completion Date for an Early Completion IPS unless a bilateral modification was issued revising the Contract Completion Date.

3.9 FAILURE TO ACHIEVE PROGRESS

Should the progress fall behind the approved project schedule for reasons other than those that are excusable within the terms of the contract, the Contracting Officer may require provision of a written recovery plan for approval. The plan must detail how progress will be made-up to include which activities will be accelerated by adding additional crews, longer work hours, extra workdays, etc.

3.9.1 Artificially Improving Progress

Artificially improving progress by means such as, but not limited to, revising the schedule logic, modifying, or adding constraints, shortening activity durations, or changing calendars in the project schedule is prohibited. Indicate assumptions made and the basis for any logic, constraint, duration, and calendar changes used in the creation of the recovery plan. Any additional resources, manpower, or daily and weekly work hour changes proposed in the recovery plan must be evident at the work site and documented in the daily report along with the Schedule Narrative Report.

3.9.2 Failure to Perform

Failure to perform work and maintain progress in accordance with the supplemental recovery plan may result in an interim and final unsatisfactory performance rating and may result in corrective action directed by the Contracting Officer pursuant to FAR 52.236-15 Schedules for Construction Contracts, FAR 52.249-10 Default (Fixed-Price Construction), and other contract provisions.

3.9.3 Recovery Schedule

Should the Contracting Officer find it necessary, submit a recovery schedule pursuant to FAR 52.236-15 Schedules for Construction Contracts.

3.10 OWNERSHIP OF FLOAT

Except for the provision given in the paragraph IMPACT TO EARLY COMPLETION SCHEDULE, float available in the schedule, at any time, belongs to the Project and is available for Contractor and Government use. This includes activity and project float. Activity float is the number of workdays that an activity can be delayed without causing a delay to the "End Project" finish milestone. Project float (if applicable) is the number of work days between the projected early finish and the "End Project" finish.

3.11 TRANSFER OF SCHEDULE DATA INTO RESIDENT MANAGEMENT SYSTEM

Ensure schedule data is uploaded to RMS. This data is additional supporting data in a form and detail required by the Contracting Officer pursuant to FAR 52.232-5 Payments under Fixed-Price Construction Contracts. The receipt of a proper payment request pursuant to FAR 52.232-27 Prompt Payment for Construction Contracts is contingent upon the

Government receiving both acceptable and approvable hard copies and matching electronic versions of the application for progress payment.

3.12 PRIMAVERA P6 MANDATORY REQUIREMENTS

Ensure Primavera P6 settings provide a schedule capable of fulfilling the requirements of the contract. The following settings are mandatory and required in all schedule submissions to the Government:

- a. Activity Codes must be Project Level, not Global or EPS level.
- b. Calendars must be Project Level, not Global or Resource level.
- c. Activity Duration Types must be set to "Fixed Duration & Units".
- d. Percent Complete Types must be set to "Physical".
- e. Time Period Admin Preferences must remain the default "8.0 hr/day, 40 hr/week, 172 hr/month, 2000 hr/year". Set Calendar Work Hours/Day to 8.0 Hour days.
- f. Set Schedule Option for defining Critical Activities to "Longest Path".
- g. Set Schedule Option for defining progressed activities to "Retained Logic".
- h. Set up cost loading using a single lump sum non-labor resource. The Price/Unit must be \$1/hr, Default Units/Time must be "8h/d", and settings "Auto Compute Actuals" and "Calculate costs from units" un-selected.
- i. Activity ID's must not exceed 10 characters.
- j. Activity Names must have a verb-noun structure and contain the most defining and detailed description within the first 30 characters.
- k. The daily ending hour for all work calendars must be the same.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 33 00

SUBMITTAL PROCEDURES

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 DEFINITIONS
 - 1.2.1 Submittal
 - 1.2.2 Transmittal
 - 1.2.3 Submittal Descriptions (SD)
 - 1.2.4 Approving Authority
 - 1.2.5 Work
 - 1.2.6 Contractor Reviewer
 - 1.2.7 Primary Reviewer
 - 1.2.8 Secondary Reviewer
 - 1.2.9 Reviewer Codes
 - 1.2.10 Review and Action Codes
 - 1.2.11 Review Comments to Contractor
- 1.3 SUBMITTALS
- 1.4 SUBMITTAL CLASSIFICATION
 - 1.4.1 Government Approved (G)
 - 1.4.2 Information Only
- 1.5 APPROVED SUBMITTALS
- 1.6 DISAPPROVED SUBMITTALS
- 1.7 WITHHOLDING OF PAYMENT
- 1.8 GENERAL
- 1.9 PROJECT SUBMITTAL REGISTER
- 1.10 SCHEDULING
 - 1.10.1 Avoiding Delays
- 1.11 TRANSMITTAL FORM (ENG FORM 4025)
 - 1.11.1 Approving Authority Review Codes
- 1.12 SUBMITTAL PROCEDURES
 - 1.12.1 Reviewing, Certifying, Approving Authority
 - 1.12.2 Procedures
 - 1.12.3 Number of Copies or Samples
 - 1.12.4 Identifying Submittal Items and Transmittals
 - 1.12.5 Submittal Format
 - 1.12.5.1 Format for Shop Drawings
 - 1.12.5.2 Format for Construction Drawings
 - 1.12.5.3 Format for Computer Aided Design (CAD) Drawings
 - 1.12.5.4 Administrative Submittal Items Format
 - 1.12.5.5 Operation and Maintenance Data
 - 1.12.6 Deviations
 - 1.12.7 Considering Variations
 - 1.12.8 Proposing Variations
 - 1.12.9 Warranty Variations
 - 1.12.10 Electronic Correspondence
- 1.13 CONTROL OF SUBMITTALS
- 1.14 GOVERNMENT APPROVED SUBMITTALS

- 1.15 INFORMATION-ONLY SUBMITTALS
- 1.16 REQUEST FOR INFORMATION (RFI)
- 1.17 SUBMITTAL REGISTER
- 1.18 STAMPS

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section Table of Contents --

SECTION 01 33 00
SUBMITTAL PROCEDURES

PART 1 GENERAL

1.1 SUMMARY

This section describes submittal procedures in Caribbean District contracts using Quality Control (QC) System submittal procedures in accordance with Section 01 45 00 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM). It covers submittal item identification, number of copies, review periods, QC review, QC/QA reviewer coding, and transmittal tracking procedures using the Resident Management System (RMS) shared data base. Use RMS software to identify, track, and transmit submittal items.

1.2 DEFINITIONS

1.2.1 Submittal

Contract Clauses FAR 52.236-5, Material and Workmanship, paragraph (b) and FAR 52.236-21, Specifications and Drawings for Construction, paragraphs (d), (e), and (f) apply to all submittals.

1.2.2 Transmittal

This is a routine contract correspondence action using a transmittal form (ENG Form 4025) to send submittal items to the Government. The RMS Transmittal Log module uses database entry to create a hard copy of ENG Form 4025.

1.2.3 Submittal Descriptions (SD)

Submittals requirements are specified in the technical sections. Submittals are identified by SD numbers and titles as follows.

SD-01 Preconstruction Submittals

Submittals that are required prior to start of construction (work) or the start of the next major phase of the construction on a multi-phase contract, includes schedules, tabular list of data, or tabular list including location, features, or other pertinent information regarding products, materials, equipment, or components to be used in the work. Submit preconstruction submittals no later than 30 calendar days after Notice to Proceed.

SD-02 Shop Drawings

Drawings, diagrams, and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project.

Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions, and brochures illustrating size, physical appearance, and other characteristics of materials, systems, or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-04 Samples

Fabricated or unfabricated physical examples of materials, equipment, or workmanship that illustrate functional and aesthetic characteristics of a material or product and establish standards by which the work can be judged.

Color samples from the manufacturer's standard line (or custom color samples if specified) to be used in selecting or approving colors for the project.

Field samples and mock-ups constructed on the project site establish standards ensuring work can be judged. Includes assemblies or portions of assemblies that are to be incorporated into the project and those that will be removed at conclusion of the work.

SD-05 Design Data

Design calculations, mix designs, analyses, or other data pertaining to a part of work.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product, or system identical to the material, product, or system to be provided has been tested in accordance with specified requirements. Unless specified in another section, testing must have been within 3 years of date of contract award for the project.

Report that includes findings of a test required to be performed on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report that includes finding of a test made at the job site or on a sample taken from the job site or on a portion of work during or after installation.

Investigation reports.

Daily checklists.

Final acceptance test and operational test procedure.

SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by

responsible officials of manufacturer of product, system, or material attesting that the product, system, or material meets specification requirements. Date after award of project contract and clearly name the project.

Document required of Contractor or of a manufacturer, supplier, installer, or subcontractor through the Contractor. The document's purpose is to further promote the orderly progression of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications.

Confined space entry permits.

SD-08 Manufacturer's Instructions

Preprinted material describing installation of a product, system, or material including special notices and Material Safety Data sheets concerning impedances, hazards, and safety precautions.

SD-09 Manufacturer's Field Reports

Documentation of the testing and verification actions taken by manufacturer's representative to confirm compliance with manufacturer's standards or instructions.

Factory test reports.

SD-10 Operation and Maintenance Data

Data that is furnished by the manufacturer or the system provider to the equipment operating and maintenance (O&M) personnel. This data is needed by O&M personnel for the safe and efficient operation, maintenance, and repair of the item.

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

1.2.4 Approving Authority

Office authorized to approve submittal.

1.2.5 Work

As used in this section, on-site and off-site construction required by Contract documents including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction.

1.2.6 Contractor Reviewer

Contractor employee performing QC review on submittal items.

1.2.7 Primary Reviewer

Initial Government reviewer is usually among construction field personnel who perform Quality Assurance (QA) review on submittal items.

1.2.8 Secondary Reviewer

Secondary Government reviewer is usually an in-house designer or Architect/Engineer firm hired to perform engineering during construction. This is a submittal reviewer for design critical "G" submittal items. Secondary reviewers may also be specialized district staff such as safety or environmental specialists, users, or O&M reviewers. Secondary reviewers are advisors to Approving Authority.

1.2.9 Reviewer Codes

Two- and three-letter codes used to identify Government primary and secondary reviewer on Submittal Register are as follows:

PO -- Construction Project Office
RO -- Construction Resident Office
AO -- Construction Area Office
DO -- Caribbean District Office
AE -- Architect Engineer Contractor providing engineering during construction

1.2.10 Review and Action Codes

Contractor QC Staff and Government Primary and Secondary Reviewers use reviewer code letters A, B, C, E, F, G, and X to indicate approval, acceptance, or rejection of a submittal item.

1.2.11 Review Comments to Contractor

ENG Form 4025, Section III is generated in RMS and contains consolidated remarks for Primary and Secondary Reviewers. When RMS exports to Contractor Mode, Review Comments to Contractor will electronically update. A hard copy of ENG Form 4025, Part III is attached to a submittal item when returned.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.

1.4 SUBMITTAL CLASSIFICATION

Submittals are classified as follows:

1.4.1 Government Approved (G)

Government approval is required for extensions of design, critical materials, deviations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Contracting Officer. Within the terms of the Contract Clause entitled SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are considered to be shop drawings.

1.4.2 Information Only

All submittals not requiring Government approval will be for information only. The Contractor must review, code, and approve Information-Only

submittal items. They are not considered to be shop drawings within the terms of the Contract Clause referred to above.

1.5 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals must not be construed as a complete check but will indicate only that the general method of construction, materials, detailing, and other information are satisfactory. Approval will not relieve the Contractor of the responsibility for any error that may exist as the Contractor, under the Contractor Quality Control (CQC) requirements of this Contract, is responsible for dimensions, the design of adequate connections and details, and the satisfactory construction of all work. After submittals have been approved by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.6 DISAPPROVED SUBMITTALS

Make all corrections required by the Contracting Officer and promptly furnish a corrected submittal in the form and number of copies specified for the initial submittal. If the Contractor considers any correction indicated on the submittals to constitute a change to the Contract, a notice in accordance with the Contract Clause CHANGES must be given promptly to the Contracting Officer.

1.7 WITHHOLDING OF PAYMENT

No payment for materials incorporated in the work will be made if required approvals have not been obtained until the Contracting Officer determines that materials meet Contract requirements.

1.8 GENERAL

Make submittals as required by the specifications using the RMS software Transmittal Log. The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections. Units of weights and measures used on all submittals must be the same as those used in the Contract drawings. Each submittal must be complete and in sufficient detail to allow ready determination of compliance with Contract requirements. Prior to transmittal, the CQC System Manager must check and approval code all items, and the CQC System Manager must stamp, sign, and date each item indicating action taken. Proposed deviations from the Contract requirements must be clearly identified. Submittals must include items such as: Contractor's, manufacturer's, or fabricator's drawings, descriptive literature including, but not limited to, catalog cuts, diagrams, operating charts, or curves, test reports, test cylinders, samples, O&M manuals (including parts list), certifications, warranties, and other such required submittals. Submittals requiring Government approval must be scheduled and made prior to the acquisition of the material or equipment covered thereby. Pick up and dispose of samples remaining upon completion of the work in accordance with manufacturer's Safety Data Sheets (SDS) and in compliance with existing laws and regulations.

1.9 PROJECT SUBMITTAL REGISTER

As part of the Initial Project Schedule, review and correct the Submittal

Register. Provide the revised Submittal Register to the Government for consideration. If Contract sections are added or revised via Contract modifications, update the Submittal Register with the corresponding transmittals for each Contract section and submit to the Government for review.

At the end of this section is a Submittal Register showing items of equipment and materials for which submittals are required by the specifications; this list may not be all inclusive, and additional submittals may be required. The Submittal Register using RMS is a shared electronic database constantly updated by both Government and Contractor. Examine Contract drawings and specifications, verify RMS data, and add required submittal items. The Contractor's initial update to the Submittal Register must be made prior to submission of the Initial Project Schedule. The Contractor can print or e-mail RMS-generated current Submittal Register ENG Form 4288 and Transmittal Log as RMS reports.

1.10 SCHEDULING

Make submittals as required by the specifications using the RMS software Transmittal Log. The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections. Units of weights and measures used on all submittals must be the same as those used in the contract drawings. Each submittal must be complete and in sufficient detail to allow ready determination of compliance with contract requirements. Prior to transmittal, all items must be checked and approval coded by the CQC System Manager, and each item must be stamped, signed, and dated by the CQC System Manager indicating action taken. Clearly identify proposed deviations from the contract requirements. Submittals must include items such as: Contractor's, manufacturer's, or fabricator's drawings, descriptive literature including, but not limited to, catalog cuts, diagrams, operating charts or curves, test reports, test cylinders, samples, O&M manuals (including parts list), certifications, warranties, and other such required submittals. Submittals requiring Government approval must be scheduled and made prior to the acquisition of the material or equipment covered thereby. Pick up and dispose of samples remaining upon completion of the work in accordance with manufacturer's SDSs and in compliance with existing laws and regulations.

Information Only - Contractor QC Review - 10 Calendar Days
G - Government QA Review - 30 Calendar Days

The Government has the number of calendar days indicated above from the date a transmitted G submittal item is received to review, code and return comments. Resubmit disapproved submittal items (codes C, D, E, G, or X) within 10 calendar days after return. No delay damages or time extensions will be allowed for time lost caused by late submittals.

1.10.1 Avoiding Delays

Notify the Contracting Officer immediately when procurement or construction delays due to submittal review are possible.

1.11 TRANSMITTAL FORM (ENG FORM 4025)

These forms are included in the RMS software. Use the RMS transmittal log to generate ENG Form 4025 - Transmittal of Shop Drawings, Equipment Data, Material Samples, or Manufacturer's Certificates of Compliance to transmit

submittal items to the Government. EN Form 4025R has three sections as follows:

Section I - Request for approval of the following items is initiated by the Contractor.

Section II - Approval Action combined with a review code in column For CE Use Code indicates submittal item acceptance or rejection by Approving Authority.

Section III - Government Review Remarks.

Properly complete this form using RMS software to fill out all the heading blank spaces and identify each item submitted. Exercise special care to ensure proper listing of the specification paragraph or sheet number of the contract drawings pertinent to the data submitted for each item.

1.11.1 Approving Authority Review Codes

Use RMS to act as Approving Authority to code FIO submittals. The Government will use RMS to prepare reviewer comments and code G submittal items. These comments and coding will update RMS when the .RFX file update is imported by the Contractor from the RMS SFTP repository. The action codes are as follows (Resubmit in parentheses means the Contractor is required to resubmit):

- A -- Approved as submitted (Complete)
- B -- Approved, except as noted on drawings (Complete)
- C -- Approved, except as noted on drawings (Resubmit)
- D -- Returned by correspondence (Complete)
- E -- Disapproved (See Attached and Resubmit)
- F -- Receipt Acknowledged (Completed)
- G -- Other (Specify Action in Detail; Resubmittal may be required)
- S -- Sustainability Reporting
- X -- Receipt Acknowledged, does not comply (Resubmit)

1.12 SUBMITTAL PROCEDURES

Make submittals as follows:

1.12.1 Reviewing, Certifying, Approving Authority

Perform QC review of all submittal items and certify for compliance or note submittal item is a variation. Approving authorities are as follows:

G - Government will approve G submittal items.

Information Only - Contractor must approve Information-Only submittal items.

The Government reserves the right to perform QA on an Information-Only submittal even after it has been approved by the Contractor's Approving Authority. See paragraph INFORMATION-ONLY SUBMITTALS below.

1.12.2 Procedures

Provide submittals required by Contract and transmit them to the Contracting Officer using ENG Form 4025 as contract transmittal form.

Transmit all submittals to the address provided by the Resident Engineer prior to the Preconstruction Conference. The address may also be provided by contacting the area office indicated in Page 00010-2 of the Standard Form 1442.

1.12.3 Number of Copies or Samples

Transmit one hard copy and one soft copy of G submittal items and Information-Only submittals to Contracting Officer. Each transmittal of a set of hard-copy submittal items must have a coded and signed hard copy of an RMS-generated ENG Form 4025. Transmit soft-copy submittal items on CD-ROM in PDF format unless otherwise approved by the Contracting Officer. Only one transmittal must be transmitted per CD-ROM. Attach a signed copy of ENG Form 4025 on each CD-ROM disk case, mark the contract and transmittal numbers on each CD-ROM disk, and include a PDF of the signed ENG Form 4025 on each CD-ROM.

1.12.4 Identifying Submittal Items and Transmittals

RMS creates a unique number using section number and generated number (01 30 00-X) for each submittal item. Also, RMS creates a unique number for each transmittal using section number and a number assigned by the Contractor (01 33 00-XXXXX). The Contractor-assigned numbers are not consecutive and may be an activity number from the procurement schedule. Attach an RMS-generated ENG Form 4025 to each set and copy of submittal items. Identify each sample submittal item with project title, contract number, section/submittal item number, name, address, and phone number of Contractor's point of contact.

1.12.5 Submittal Format

1.12.5.1 Format for Shop Drawings

Paper copies of shop drawings must be ANSI standard sizes A, B, C or E (8.5-inch x 11-inch, 11-inch x 17-inch, 17-inch x 22-inch, or 34-inch x 44-inch, respectively). Each drawing must have a right side margin information block similar to contract drawings with project title, contract number, contractor/supplier names, section/submittal item number, drawing title, number and date, revision date, and approval box. Units must be the same as those in contract drawings. Dimensions must be to the scale shown or noted as Not to Scale.

1.12.5.2 Format for Construction Drawings

Construction drawings are large sized shop drawings for complex structures and systems, layout of site temporary construction (dewatering), mapping access, and land-use descriptions. When required, prepare construction drawings on ANSI standard sizes C (17-inch x 22-inch) or E (34-inch x 44-inch) as approved by the Contracting Officer. The layout of construction drawings must be as indicated in subparagraph FORMAT FOR SHOP DRAWINGS above.

1.12.5.3 Format for Computer Aided Design (CAD) Drawings

Transmit CAD drawings electronically as approved by the Contracting Officer via RMS. Transmit the same number of CD-ROM disks as the number of hard copies required. Attach a copy of ENG Form 4025 on each CD-ROM disk case, and mark the contract and transmittal numbers on each CD-ROM disk. Include one paper copy of drawings with each set of CD-ROM disks

transmitted.

1.12.5.4 Administrative Submittal Items Format

When an administrative submittal becomes part of the contract or project record, use 8.5-inch x 11-inch pages for text and 11-inch x 17-inch sheets for drawings. Use a table of contents with tabs for large submittal items. Use a loose-leaf binder or binder fastener to allow for disassembly and the addition of documents.

1.12.5.5 Operation and Maintenance Data

See Section 01 78 23 OPERATION AND MAINTENANCE DATA.

1.12.6 Deviations

Deviations are also known as variations from the Contract requirements. Variations require Government approval and will be considered when beneficial to the Government. An Information-Only submittal item becomes a G submittal item requiring Government review and approval when a variation is proposed. When the RMS variation block is checked, explain why the variation should be accepted in place of the specified item in Remarks to Government. The Remarks to Government will populate the ENG Form 4025R REMARKS block. The variation will become a modification to contract requirements if it is approved by the Contracting Officer. Approved variations may require adjustments to contract price for increased (basic change document) or decreased (Government's credit) costs, and construction time. The Government reserves the right to rescind inadvertent approval of submittals containing unnoted deviations.

1.12.7 Considering Variations

Discuss proposed variations with the Contracting Officer prior to submission. Ensure function and quality requirements are met. Submit proposed variation as a Value Engineering Change Proposal if it is likely to result in lower cost. See the clause VALUE ENGINEERING - CONSTRUCTION of Section 00700 CONTRACT CLAUSES of Volume 1.

1.12.8 Proposing Variations

Use the RMS Transmittal Log to identify submittal items as variations. In Remarks to Government, describe what the proposed variation is, how it differs from the contract, and why it is desirable and beneficial to the Government. Include an estimate of cost savings if lower cost is a benefit.

1.12.9 Warranty Variations

Use the RMS transmittal log Remarks to Government to warrant that the variation submitted for approval will be compatible with other work if incorporated.

1.12.10 Electronic Correspondence

Submit all correspondence and contract-related documents in searchable Adobe Portable File format (PDF) with the only exceptions being those stated at the end of this paragraph. Submit any documents that require signature in PDF format with signature. Minimum scan quality must be 150 dpi. The Contracting Officer will discuss electronic correspondence procedures at Preconstruction and Coordination Meetings. These procedures

may include sending files by e-mail or submitting on electronic media such as compact disks, a Contractor-provided FTP site, a website indicated by the Contracting Officer at the Preconstruction Conference, or other methods approved by the Contracting Officer. The Contractor will be responsible for the cost of the electronic media. Examples of correspondence and contract-related documents that must be submitted in PDF format include, but are not limited to, Contractor serial letters, notifications to Contracting Officer, requests for equitable adjustment, change requests, claims, Contract Modifications (SF 30), QC Reports, RFIs, pay requests, meeting minutes, insurance certificates, schedules, shop drawings, submittals, and ENG 4025 transmittal forms and supporting data. QC Reports will include all required attachments including QC Test Reports. Submit both a hard copy and a PDF file of the signed QC Report including required attachments daily. In addition, 1 day after the end of each month, combine all QC Reports for the month into one PDF and submit. Submit copies of correspondence to other businesses, entities, or agencies such as Notices to Mariners, permit-required reports, and other contract-related documents in PDF format. Include signatures if applicable for final versions of scanned contracts. Also submit all correspondence and contract-related documents in hard copy form in the quantities and with the requirements listed in these documents unless the Contractor obtains a waiver from the Contracting Officer. E-mail correspondence and payroll documentation do not have to be provided in PDF format.

1.13 CONTROL OF SUBMITTALS

Carefully control procurement operations to ensure that each individual submittal is made on or before the Contractor scheduled submittal date shown on the approved Submittal Register.

1.14 GOVERNMENT APPROVED SUBMITTALS

Upon completion of review of submittals requiring Government approval, the submittals will be identified as having received approval by being so stamped and dated. Two copies of the submittal will be retained by the Contracting Officer and one copy of the submittal will be returned to the Contractor.

1.15 INFORMATION-ONLY SUBMITTALS

Normally submittals for information only are approved by the Contractor's approving authority. The Government will notify the Contractor if the submittal has received an F action code, and the submittal will not be returned. Approval of the Contracting Officer is not required on Information-Only submittals. Submittal items found not to comply with contract requirements will be given an X action code, returned to the Contractor, and will require resubmittal by the Contractor as a G submittal item. The Government reserves the right to require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications, will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work, and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.

1.16 REQUEST FOR INFORMATION (RFI)

RFIs are not submittal items. They are time-sensitive, QC-related correspondence. See Section 01 45 04 CONTRACTOR QUALITY CONTROL.

1.17 SUBMITTAL REGISTER

The Submittal Register is appended after the end of this section.

1.18 STAMPS

Stamps used by the Contractor on the submittal data to certify that the submittal meets Contract requirements must be similar to the following (see next page):

CONTRACTOR
(Firm Name)
_____ Approved
_____ Approved with corrections as noted on submittal data and/or attached sheets(s).
SIGNATURE: _____
TITLE: _____
DATE: _____

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 11 00	SD-01 Preconstruction Submittals														
			Preconstruction Logistics Plan	1.4													
		01 22 00	SD-01 Preconstruction Submittals														
			Schedule of Values	1.5	G RO												
			SD-07 Certificates														
			Utility Invoices	1.5.1													
			Request For Progress Payment	3.1.1	G RO												
		01 30 00	SD-01 Preconstruction Submittals														
			Draft Joint Risk Register	1.6.3													
			List of Subcontractors	1.6.4													
			SD-07 Certificates														
			Partnering Meeting(s) Written Synopsis	1.7.1.2	G RO												
			Signature Authority	3.3													
			Drug-Free Work Place Record	3.4													
		01 32 01	SD-01 Preconstruction Submittals														
			Project Scheduler Qualifications	1.3	G RO												
			Preliminary Project Schedule	3.4.1	G RO												
			Initial Project Schedule	3.4.2	G RO												
			SD-07 Certificates														
			Periodic Schedule Update	3.6.2	G RO												
		01 35 25	SD-01 Preconstruction Submittals														
			Dive Operations Plan	3.2	G DO												
			SD-07 Certificates														
			Dive Personnel Qualifications	1.6.2	G DO												
			Daily Dive Logs	3.2.1													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 35 26	SD-01 Preconstruction Submittals														
			Dive Operations Plan	1.17	G DO												
			Accident Prevention Plan	1.8.1	G DO												
			SD-06 Test Reports														
			Monthly Exposure Reports	1.5													
			Notifications and Reports	1.13													
			Reports	1.13.2	G DO												
			LHE Inspection Reports	1.13.3													
			SD-07 Certificates														
			Crane Operators, Signal Persons and Riggers	1.7.1.3	G DO												
			Standard Lift Plan	1.8.3.2	G DO												
			Critical Lift Plan	1.8.3.3	G DO												
			Naval Architecture Analysis	1.8.3.4	G DO												
			Activity Hazard Analysis (AHA)	1.9	G DO												
			Confined Space Entry Permit	1.10.1													
			Hot Work Permit	1.10.1													
			Certificate of Compliance	1.13.4	G DO												
			License Certificates	1.15	G DO												
			Radiography Operation Planning Work Sheet	1.15.1	G DO												
			Portable Gauge Operations Planning Worksheet	1.15.1	G DO												
		01 45 04	SD-01 Preconstruction Submittals														
			Contractor Quality Control (CQC) Plan	3.2	G RO												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 45 04	Work Schedule	1.5	G RO												
			SD-07 Certificates														
			Laboratory Qualifications	3.7.2	G RO												
			Letter of Authority	3.2.2													
		01 50 02	SD-01 Preconstruction Submittals														
			Mobilization/Demobilization Plan	1.4.1	G RO												
			Security Plan	1.4.2	G RO												
			Vegetation Protection and Restoration Plan	1.4.3	G DO												
			Documentation Of Existing Conditions and Post-construction Restoration	1.4.4	G RO												
			SD-02 Shop Drawings														
			Contractor's Temporary Facilities	3.5													
			Temporary Electric Drawings	3.2.6													
		01 52 10	SD-02 Shop Drawings														
			Contracting Officer's Field Office	2.1													
		01 55 26	SD-01 Preconstruction Submittals														
			Traffic Control Plan	1.3	G RO												
		01 57 20	SD-01 Preconstruction Submittals														
			Environmental Protection Plan	1.6	G DO												
			SD-03 Product Data														
			Meteorological Monitoring	1.11	G RO												
			SD-07 Certificates														
			Manatee Observer Qualifications	1.5.1.1	G DO												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 20	Puerto Rican Boa Monitor Qualifications	1.5.1.2	G DO												
			SD-11 Closeout Submittals														
			Logs/Final Summary Report	3.2.2													
			Project Environmental Summary Sheet	3.2.3	G DO												
		01 57 25	SD-03 Product Data														
			Calibration Standard	3.1.1.1	G RO												
			SD-06 Test Reports														
			Turbidity Monitoring	3.1.2	G RO												
		01 74 19	SD-01 Preconstruction Submittals														
			Construction Waste Management Plan	1.7	G DO												
			SD-06 Test Reports														
			Quarterly Reports	1.9.2													
			Annual Report	1.9.3													
			SD-11 Closeout Submittals														
			Final Construction Waste Diversion Report	1.10	G DO												
		01 78 02	SD-02 Shop Drawings														
			As-Builts	1.3	G RO												
			SD-03 Product Data														
			As-Built Record of Equipment and Materials	1.3.3	G RO												
			Warranty Management Plan	1.4.1	G DO												
			Warranty Tags	1.4.5	G DO												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	CLASSIFICATION GOVT OR A/E REVIEWER	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 78 02	Final Cleaning	1.6	G DO												
			SD-11 Closeout Submittals														
			Load Rating Analysis Report	1.3	G DO												
		02 22 13	SD-01 Preconstruction Submittals														
			Vibration Control Specialist	1.2.1	G DO												
			Vibration Control Specialist	1.2.1	G DO												
			Alternate														
			Vibration Monitoring	1.2.2	G DO												
			Technician(s)														
			Structural Inspection/Evaluation	1.2.3	G DO												
			Engineer														
			Pre-Construction Structural	3.2	G DO												
			Survey														
			Vibration Control Plan	3.3	G DO												
			Noise Control Plan	3.4	G DO												
			SD-06 Test Reports														
			Vibration Monitoring Report	3.6													
			Noise Monitoring Report	3.7													
			Daily Instrument Logs	3.5													
			Post-Construction Structural	3.8	G DO												
			Survey														
		02 41 00	SD-01 Preconstruction Submittals														
			Demolition Plan	1.2.2	G RO												
			Existing Conditions	1.9													
			Gondola Cables	1.3.1	G RO												
			Luma Equipment Inventory	1.9.1	G RO												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		02 41 00	SD-07 Certificates														
			Notification	1.6	G RO												
			SD-11 Closeout Submittals														
			Receipts	3.3.3													
		02 42 00	SD-01 Preconstruction Submittals														
			Demolition Plan		G												
			Deconstruction Plan		G												
			Existing Conditions														
			SD-07 Certificates														
			Notification		G												
			Notification of Demolition and														
			Renovation Form														
			SD-11 Closeout Submittals														
			Receipts														
		03 11 13	SD-02 Shop Drawings														
			Formwork	3.1.1	G DO												
			Stay-In-Place Metal Bridge Deck	3.1.2	G RO												
			Forms														
			SD-03 Product Data														
			Form Materials	2.2													
			Form Releasing Agents	2.2.5													
			SD-05 Design Data														
			Calculations	2.1	G RO												
			SD-06 Test Reports														
			Inspection	3.5													
		03 15 00	SD-03 Product Data														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION			APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		03 15 00	Preformed Joint Filler	2.1	G RO												
			Splicing	3.1.2	G RO												
			Waterstops	2.2	G RO												
			Bridge Bearing Pads	2.3	G RO												
			Bridge Joint Seal	2.4	G RO												
			Curing Compound Or Bituminous	3.1.1	G RO												
			Paint														
			Waterproofing Membrane	2.5	G RO												
			SD-04 Samples														
			Splicing Waterstops	2.2.2	G RO												
		03 20 00	SD-02 Shop Drawings														
			Reinforcement	3.1	G DO												
			SD-03 Product Data														
			Materials	2.8	G DO												
			Mechanical Butt-Splices	2.4.1	G DO												
			Mechanical Butt-Splices	3.1.3.2	G DO												
			SD-06 Test Reports														
			Tests, Inspections, and	2.8	G DO												
			Verifications														
			Test Butt-Splices	1.3.2	G DO												
			SD-07 Certificates														
			Reinforcing Steel	2.4	G DO												
			Qualification of Steel Bar	1.3.1	G DO												
			Butt-Splicers														
		03 31 01	SD-01 Preconstruction Submittals														
			Random Sampling Plan	3.5.2.6.6	G DO												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		03 31 01	Concrete Placement Plan	3.1.1	G DO												
			Deck Concrete Placement Procedure Plan	3.4	G DO												
			Quality Control Plan	2.1.3.3.2													
			Quality Control Personnel Certifications	1.4.1	G DO												
			Quality Control Organizational Chart	1.4.1	G DO												
			Laboratory Accreditation	1.4.3	G DO												
			SD-03 Product Data														
			Concrete Mixture Proportions	1.2	G RO												
			Batch Plant	2.2	G RO												
			Concrete Mixers	2.2.5	G RO												
			Conveying Equipment	2.2.6	G RO												
			Placing Equipment	2.2	G RO												
			Deck Screeds	2.2.8	G RO												
			Construction Joint Treatment	3.1.4	G DO												
			Hot-Weather Placing	3.2.4	G DO												
			Cementitious Materials	2.1.1	G DO												
			Admixtures	2.1.3	G DO												
			Pumping Concrete Materials and Methods	3.2.2	G DO												
			Nonshrink Grout	2.1.5	G DO												
			SD-04 Samples														
			Aggregates	2.1.2	G RO												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		03 31 01	Cementitious Materials, Admixtures, and Curing Compound	1.4.6	G RO												
			SD-05 Design Data														
			Concrete Mix	1.2	G RO												
			SD-06 Test Reports														
			Quality of Aggregates	3.5.2.3	G DO												
			Mixer Uniformity	3.5.2.10	G RO												
			Tests and Inspections	2.1	G DO												
			Tests and Inspections	3.5	G DO												
			Water	2.1.4	G DO												
			Slump Loss	3.5.2.6.3	G DO												
			Cumulative Mass Loss	2.1.3.4	G DO												
			Concrete Mix	1.2													
			Fly Ash	2.1.1.2	G DO												
			Pozzolan	2.1.1.3	G DO												
			Slag	2.1.1.5	G DO												
			Tolerance	2.2.8	G DO												
			Compressive Strength Tests	3.5.2.6.6	G DO												
			Air Content	3.5.2.6.1	G DO												
			Slump Test	3.5.2.6.3	G DO												
			Permeability	3.5.2.6.10	G DO												
			SD-07 Certificates														
			Concrete Field Testing	1.4	G RO												
			Technicians														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		03 31 01	Concrete Construction Inspector (CCI)	1.4	G RO												
			Cementitious Materials	2.1.1	G DO												
			Blended Hydraulic Cement	2.1.1.6	G DO												
			Air-Entraining Admixture	2.1.3.1	G DO												
			Other Chemical Admixtures	2.1.3.6	G DO												
			Epoxy Resin	2.1.7	G DO												
			Latex Bonding Compound	2.1.6	G DO												
			Nonshrink Grout	2.1.5	G DO												
		03 35 00	SD-01 Preconstruction Submittals														
			Finishing Plan	3.1.3	G DO												
			SD-04 Samples														
			Sample Wall Panels	1.3.1													
			Latex Bonding Compound	2.1													
			Epoxy Resin	2.2													
		03 39 00	SD-01 Preconstruction Submittals														
			Concrete Curing Plan	3.1	G DO												
			SD-03 Product Data														
			Concrete Curing Materials	2.1	G DO												
			SD-06 Test Reports														
			Testing and Inspection for CQC	3.2	G RO												
			SD-08 Manufacturer's Instructions														
			Curing Compound	2.1	G DO												
		03 41 00	SD-02 Shop Drawings														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION						CONTRACTOR											
RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3																	
ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		03 41 00	Precast, Prestressed Florida I-Beam 36 Concrete Bridge Girders	2.2.1	G RO												
			Precast, Prestressed Florida I-Beam 36 Concrete Bridge Girders	3.1.1	G RO												
			Placement	1.5.1													
			Tolerances	1.5.1	G RO												
			SD-03 Product Data														
			Mixture	2.2.4	G RO												
			Disposition	1.5.1	G RO												
			Equipment	2.2.6	G RO												
			Tensioning	2.2.6													
			Detensioning	2.2.7	G RO												
			Delivery, Storage, and Handling	1.4	G RO												
			Curing of Girders	2.2.8	G RO												
			SD-06 Test Reports														
			Test Reports	3.1.4	G RO												
			Concrete Test Cylinders	2.2.8	G RO												
			SD-07 Certificates														
			Certificates of Compliance	2.3.2													
			Records	1.5.1	G RO												
			Records	3.1.4	G RO												
			Manufacturer Qualifications	1.6	G RO												
		05 50 13	SD-03 Product Data														
			Corner Guards	2.3													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		05 50 13	Filler Metal	3.5.2.1													
			Access Hatches	2.1.8													
			SD-06 Test Reports														
			Bolt Testing Reports	3.3													
			SD-07 Certificates														
			Welder Qualifications	1.3													
		05 51 00	SD-02 Shop Drawings														
			Iron and Steel Hardware	2.1	G RO												
			Steel Shapes and Plates	2.1	G RO												
			Metal Stair System	2.2.1	G RO												
			SD-03 Product Data														
			Structural Steel Plates and Shapes	2.4.1	G RO												
			Concrete Anchors	2.3.2	G RO												
			Protective Coating	2.2.3	G RO												
			Steel Stair Components	2.3.1	G RO												
			SD-08 Manufacturer's Instructions														
			Structural Steel Plates and Shapes	2.4.1	G RO												
			Protective Coating	2.2.3	G RO												
		05 52 00	SD-02 Shop Drawings														
			Fabrication Drawings	1.2.1	G RO												
			Iron and Steel Hardware	3.2	G RO												
			Steel Shapes, Plates, Bars and Strips	3.2	G RO												
			SD-03 Product Data														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		05 52 00	Structural-Steel Plates, Shapes, and Bars	2.2.1	G RO												
			Structural-Steel Tubing	1.2.1	G RO												
			Cold-Drawn Steel Tubing	1.2.1	G RO												
			Concrete Inserts	1.2.1	G RO												
			Anchorage Devices	3.2	G RO												
			Protective Coating	2.1.2	G RO												
			Steel Railings and Handrails	2.2.4	G RO												
			Anchorage and Fastening Systems	1.2.1	G RO												
			SD-07 Certificates														
			Welding Procedures	1.4.2	G DO												
			Welder Qualification	1.4.3	G DO												
			SD-08 Manufacturer's Instructions														
			Installation Instructions	3.2													
		06 73 01	SD-02 Shop Drawings														
			Installation Drawings, Templates, and Directions	1.3	G DO												
			SD-03 Product Data														
			FRP Grating	1.3	G DO												
			Clips and Anchorage	1.3	G DO												
			SD-06 Test Reports														
			Bearing Strength Testing	2.1.1													
			Flexural Properties	2.1.1													
			Ultraviolet Testing	2.1.1													
			Shear Strength	2.1.1													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION						CONTRACTOR											
RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3																	
ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		06 73 01	Tensile Properties	2.1.1													
			Toxicity Testing	2.1.1													
			Coefficient of Lineal Thermal Expansion	2.1.1													
			Flame Spread Testing	2.1.1													
			SD-07 Certificates														
			Manufacturer's Sample Warranty	1.3													
			Certification of Anchorage	1.3													
			System compliance with ASCE 7-16														
			Proof Of Certification From A Minimum Of Two Quality Assurance Programs For Its Facilities Or Products (Ul, Dnv, Abs, Uscg, Aarr)	1.3													
			SD-08 Manufacturer's Instructions Shipping, Handling, And Erection Procedures	1.4													
			Care and Maintenance Instructions	1.4													
			SD-09 Manufacturer's Field Reports														
			Manufacturer's Certification of Installation	3.1													
			SD-11 Closeout Submittals														
			Manufacturer's Warranty	3.2.1													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		06 82 14	SD-01 Preconstruction Submittals														
			Qualifications of Manufacturer	1.3.1	G DO												
			Qualifications of Engineer of Record	1.3.2	G DO												
			SD-02 Shop Drawings														
			Installation Drawings	1.3.2	G DO												
			SD-03 Product Data														
			FRP Pipe and Tube	2.1.2	G DO												
			Railings/Guards	2.1.2	G DO												
			Anchorage Materials	2.1.2	G DO												
			Adhesives	2.1.2	G DO												
			Resins	2.1.2	G DO												
			Hardeners	2.1.2	G DO												
			Manufacturer's Sample Warranty	1.3.1													
			SD-06 Test Reports														
			Ultraviolet Test Reports	2.1.3													
			Thermal Expansion Test Reports	2.1.3													
			Flame Spread Test Reports	2.1.3													
			SD-07 Certificates														
			Proof of Certification from a minimum of two quality assurance programs for its facilities or products (UL, DNV, ABS, USCG, AARR)	1.3.1													
			SD-08 Manufacturer's Instructions														
			Manufacturer's Instructions	3.2.2													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		06 82 14	SD-09 Manufacturer's Field Reports														
			Manufacturer's Certification of Installation	3.1													
			SD-11 Closeout Submittals														
			Manufacturer's Warranty	3.2.1													
		09 97 02	SD-01 Preconstruction Submittals														
			Safety, Health, and Environmental Requirements	1.2	G DO												
			Safety Plan	1.2.1	G DO												
			Respiratory Protection Plan	1.2.1.5	G DO												
			Airborne Sampling Plan	1.2.2.1	G DO												
			Medical Surveillance Plan	1.2.2	G DO												
			Worker Protection Plan	1.2.2.2	G DO												
			Environmental Protection Plan	1.2.3	G DO												
			Waste Manifest	1.2.3.1													
			Waste Disposal Plan	1.2.3.1	G DO												
			Water Quality Plan	1.2.3.2	G DO												
			Soil Quality Plan	1.2.3.3	G DO												
			SD-03 Product Data														
			Manufacturer's Product Data Sheet	Part 2	G DO												
			SD-04 Samples														
			Product Samples	Part 2	G DO												
			Special Paint Formulas	2.1	G DO												
			SD-06 Test Reports														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	CLASSIFICATION GOVT OR A/E REVIEWER	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		09 97 02	Soil Quality Test Report	1.2.3.3													
			Inspection Reports	3.4													
			Medical Status Report	1.2.2.2													
			Change in Medical Status Report	1.2.2.3													
			Air Monitoring Test Plan	1.2.2.1	G DO												
			Air Monitoring Test Report	1.2.2.1													
			SD-07 Certificates														
			Certified EHS Professional	1.4.1													
			Certified Lead Laboratory	1.4.2													
			SSPC QP 1 Certificate	1.4.3													
			SSPC QP 2 Certificate	1.4.4													
			Qualified Hazardous Paint	1.4.4	G DO												
			Removal Contractor														
			Coating Thickness Gage	1.4.5													
			Qualification														
		26 08 00	SD-06 Test Reports														
			Acceptance Tests and	3.1	G DO												
			Inspections														
			SD-07 Certificates														
			Qualifications	1.4.1	G DO												
			Acceptance Test and Inspections	1.4.3	G DO												
			Procedure														
		26 56 00	SD-02 Shop Drawings														
			Luminaire Drawings	1.5.1.1	G DO												
			SD-03 Product Data														
			Luminaires	2.2	G DO												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		26 56 00	Light Sources	2.3	G DO												
			LED Drivers	2.4	G DO												
			Luminaire Warranty	1.6.1	G DO												
			Photosensors	2.5.2.1	G DO												
			Brackets	2.6	G DO												
			SD-05 Design Data														
			Luminaire Design Data	1.5.2	G DO												
			Photometric Plan	1.5.6	G DO												
			SD-06 Test Reports														
			ANSI/IES LM-79 Test Report	1.5.3	G DO												
			ANSI/IES LM-80 Test Report	1.5.4	G DO												
			ANSI/IES TM-21 Test Report	1.5.5	G DO												
			SD-10 Operation and Maintenance Data														
			Lighting System	1.7.1	G												
			Exterior Lighting Control System	1.7.2	G												
			Maintenance Staff Training Plan	3.3.1.1	G DO												
			End-User Training Plan	3.3.1.2	G DO												
		31 00 00	SD-01 Preconstruction Submittals														
			Shoring and Sheeting	3.6	G DO												
			Water Control Plan	1.4.3													
			In-Channel Excavation Plan	3.3	G DO												
			Geotechnical Engineer	3.6.2	G DO												
			Borrow Site	2.1	G DO												
			Disposal Site	1.7	G DO												
			SD-03 Product Data														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		31 00 00	Utilization of Excavated Materials	3.9	G DO												
			Opening of any Excavation or Borrow Pit	3.5													
			SD-06 Test Reports														
			Testing	3.17													
			Borrow Site Testing	2.1													
			SD-07 Certificates														
			Testing	3.17													
			Laboratory Qualifications	3.17	G DO												
		31 05 19	SD-02 Shop Drawings														
			Layout	3.2	G DO												
			Seams	3.3.2	G DO												
			SD-03 Product Data														
			Thread	2.1.2	G DO												
			Manufacturing Quality Control Sampling and Testing	2.2	G DO												
			SD-04 Samples														
			Quality Assurance Samples	3.1													
			SD-06 Test Reports														
			Quality Assurance Tests	3.1.2	G DO												
			SD-07 Certificates														
			Geotextile	2.1.1	G DO												
		31 11 00	SD-03 Product Data														
			Tree Wound Paint	2.1.1													
		31 23 19	SD-01 Preconstruction Submittals														
			Unwatering Design Package	1.8.1	G DO												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		31 23 19	Work Plan	1.4	G DO												
			SD-06 Test Reports														
			Fabrication And Workmanship Records	1.9.1													
			Installation, Operation, And Removal Records	1.9.1													
			Inspection	1.9.2													
			Piezometer Installation Records	1.9.1													
			Relief Well Installation Records	1.9.1													
		31 32 23	SD-01 Preconstruction Submittals														
			Experience	1.6.1.1	G DO												
			Grouting Plans	1.11	G DO												
			Grout Mix Proportions	2.2	G DO												
			Jet Grouting Containment Plan	3.1.3	G DO												
			Drilling Deviation Measurement	1.4	G DO												
			Sample Log	1.6.5	G DO												
			SD-02 Shop Drawings														
			Jet Grouting	3.1	G DO												
			SD-03 Product Data														
			Drilling and Grouting Equipment	3.1.1	G DO												
			Grout Plant	3.1.1	G DO												
			Borehole Camera	3.2.5	G DO												
			SD-06 Test Reports														
			Field and Laboratory Testing	3.2.2	G DO												
			Metering Equipment	1.6.6													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		31 32 23	Instrument For Drilling Deviation Measurements	1.6.2	G DO												
			Factory Calibrations On Instruments Used For Control Of Jet Grouting	1.6.6	G DO												
			Factory Calibration On Drilling Deviation And Borehole Surveying Instruments	1.6.6	G DO												
			SD-11 Closeout Submittals														
			Final Demonstration Test Report	3.1.2	G DO												
			Final Jet Grouting Reports	3.3.1	G DO												
			Record Drawings Of Jet Grout Installation	3.3.2	G DO												
		31 36 00	SD-01 Preconstruction Submittals														
			Installation Plan	3.4	G DO												
			SD-03 Product Data														
			Product Data	1.4.3	G RO												
			SD-04 Samples														
			Gabions	1.4.1													
			Wire Mesh	1.4.3													
			SD-06 Test Reports														
			Test Reports	1.4.3	G RO												
			SD-07 Certificates														
			Stone Fill	2.1.3	G RO												
			Certificate Of Compliance	1.4.3	G RO												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		31 36 00	SD-09 Manufacturer's Field Reports														
			Mill Certifications	1.4.3	G RO												
		31 41 16	SD-01 Preconstruction Submittals														
			Work Plan	1.5.1	G DO												
			Monitoring Plan	1.5.3	G DO												
			SD-02 Shop Drawings														
			Steel Sheet Piling	1.5.2	G DO												
			Sheet Pile Splicing	1.5.2	G DO												
			SD-03 Product Data														
			Driving	3.3.2.2													
			Pile Driving Equipment	3.3.1	G DO												
			Pulling and Redriving	3.3.6	G DO												
			SD-05 Design Data														
			Design Calculations	1.5.4	G DO												
			SD-06 Test Reports														
			Materials Tests	2.3.1													
			SD-07 Certificates														
			Sheet Pile Contractor Experience	1.5.4	G DO												
			Sheet Pile Design Experience	1.5.4	G DO												
			Surveyor Experience	1.5.5	G DO												
			Supervisor Experience	1.5.6	G DO												
			Rig Operator Experience	1.5.7	G DO												
			SD-11 Closeout Submittals														
			Pile Driving Record	3.6													
		31 52 10	SD-01 Preconstruction Submittals														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION						CONTRACTOR											
RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3																	
ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		31 52 10	TRS Work Plan	1.8	G DO												
			Tremie Concrete Placement Plan	3.2	G DO												
			SD-02 Shop Drawings														
			Shop Drawings	1.4.1	G DO												
			SD-05 Design Data														
			Temporary Retaining Structure	1.4	G DO												
			Design Calculations														
			Tremie Mix Design	2.2	G DO												
			SD-07 Certificates														
			Concrete Tremie Slab Contractor	1.7.2	G DO												
			Experience														
			Temporary Retaining Structure	1.7.1	G DO												
			(TRS) Contractor Experience														
		31 63 29	SD-01 Preconstruction Submittals														
			Confirmation Boring Drilling	3.1	G DO												
			Correlation Boring Drilling	3.1	G DO												
			Demonstration Wall	3.2.3	G DO												
			Survey	1.6.10	G DO												
			SD-02 Shop Drawings														
			Shaft Installation Plan	3.2.4	G DO												
			Casing Installation	2.6.2	G DO												
			Excavation	2.6.3	G DO												
			Concreting	3.10.1	G DO												
			Sheeting and Shoring	3.2.2	G DO												
			SD-05 Design Data														
			Correlation Boring Report	3.1													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		31 63 29	Confirmation Boring Report	3.1													
			Demonstration Wall Report	3.2.3	G DO												
			Shaft Installation Record Form	3.18	G DO												
			Slurry Data	3.18	G DO												
			Slurry	3.5	G DO												
			Cleanup	3.4	G DO												
			Concreting Data	3.18	G DO												
			SD-06 Test Reports														
			Slurry Test Reports	3.5.2.3	G DO												
			Slump Loss Test	3.14.4	G DO												
			CSL Test Report	3.13.1	G DO												
			Condition of Existing Structures	3.2.2	G DO												
			Individual Shaft Installation	3.9	G DO												
			Records														
			SD-07 Certificates														
			Drilled Shaft Contractor	1.6.7	G DO												
			Experience														
			Site Superintendent	1.6.7.1	G DO												
			Supervisor Experience	1.6.7.2	G DO												
			Drill Operator Experience	1.6.7.3	G DO												
			Cross Hole Sonic Logging (CSL)	1.6.7.4	G DO												
			Testing Experience														
			Ultrasonic or Sonar Shaft	1.6.7.5	G DO												
			Excavation Measuring Device														
			Testing System Experience														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		31 63 29	Shaft Inspection Device Experience	1.6.7.6	G DO												
			Soil Testing Laboratory	3.5.2.3	G DO												
			Quality Control Personnel Qualifications	1.6.9	G DO												
			Geotechnical Engineer Qualifications	1.6.11	G DO												
			Structural Engineer Qualifications	1.6.12	G DO												
			Welding Qualifications	1.6.8	G DO												
			SD-08 Manufacturer's Instructions														
			Mechanical Butt	3.11.4	G DO												
		32 11 23	SD-03 Product Data														
			Plant, Equipment, and Tools	1.4	G RO												
			SD-06 Test Reports														
			In-Place Tests	3.11.1	G RO												
		32 12 16	SD-03 Product Data														
			Mix Design	2.4	G RO												
			SD-04 Samples														
			Asphalt Cement Binder	2.3													
			Aggregates	2.2													
			SD-06 Test Reports														
			Aggregates	2.2	G RO												
			SD-07 Certificates														
			Asphalt Cement Binder	2.3	G RO												
		32 15 00	SD-03 Product Data														
			Plant, Equipment, and Tools	1.4	G RO												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		32 15 00	SD-06 Test Reports														
			Initial Tests	2.3.1	G RO												
			In-Place Tests	3.11.1	G RO												
		32 16 19	SD-03 Product Data														
			Concrete	2.1													
			Biodegradable Form Release Agent	2.6.5													
			Biodegradable Form Release Agent	3.2													
			SD-06 Test Reports														
			Field Quality Control	3.8													
		32 17 23	SD-03 Product Data														
			Surface Preparation Equipment List	2.1.1.1	G RO												
			Application Equipment List	2.1.2	G RO												
			Exterior Surface Preparation	3.2													
			Safety Data Sheets	1.3.1	G DO												
			Materials	2.2	G RO												
			Thermoplastic compound	3.3.2	G RO												
			SD-06 Test Reports														
			Materials	2.2	G RO												
			Thermoplastic Compound	3.3.2	G RO												
			Test Reports	3.4.1													
			SD-07 Certificates														
			Qualifications	1.3.2	G DO												
			Materials	2.2	G DO												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS SIF ACTION OR REVIEW N	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		32 17 23	Volatile Organic Compound	1.3.1	G DO												
			Thermoplastic Compound	3.3.2													
			SD-08 Manufacturer's Instructions														
			Thermoplastic Compound	3.3.2	G RO												
		32 31 00	SD-02 Shop Drawings														
			Fence Assembly	2.1	G RO												
			Location of Gate, Corner, End, and Pull Posts	3.2.1.1	G RO												
			Gate Assembly	2.1	G RO												
			Gate Hardware and Accessories	2.2.12	G RO												
			Erection/Installation Drawings	Part 3	G RO												
			SD-03 Product Data														
			Fence Assembly	2.1	G RO												
			Gate Assembly	2.1	G RO												
			Gate Hardware and Accessories	2.2.12	G RO												
			Zinc Coating	2.3.1	G RO												
			Fabric	2.2.1	G RO												
			Stretcher Bars	2.2.7	G RO												
			Concrete	2.3.3	G RO												
			Barbed Wire	2.2.15	G RO												
			Barbed Wire Supporting Arms	2.2.16	G RO												
			Privacy Screen	2.3.5	G RO												
			SD-04 Samples														
			Fabric	2.2.1	G RO												
			Braces	2.2.6	G RO												
			Line Posts	2.2.2	G RO												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		32 31 00	Top Rail	2.2.4	G RO												
			Tension Wire	2.3.2	G RO												
			Stretcher Bars	2.2.7	G RO												
			Gate Posts	2.2.10	G RO												
			Gate Hardware and Accessories	2.2.12	G RO												
			Wire Ties	2.2.14	G RO												
			Privacy Screen	2.3.5	G RO												
			SD-05 Design Data														
			Reinforced Concrete Or	1.5	G DO												
			Reinforced Concrete Block Privacy														
			Wall														
			SD-07 Certificates														
			Certificates of Compliance	1.3.1													
			SD-08 Manufacturer's Instructions														
			Fence Assembly	2.1													
			Gate Assembly	2.1													
			Hardware Assembly	2.1													
			Accessories	2.1													
			SD-11 Closeout Submittals														
			Recycled Material Content	3.3													
		32 92 19	SD-03 Product Data														
			Fertilizer	2.1													
			SD-06 Test Reports														
			Topsoil	2.1													
			SD-07 Certificates														
			Seed	2.1													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		32 92 23	SD-03 Product Data														
			Fertilizer	2.3													
			Topsoil composition	2.2.3													
			SD-07 Certificates														
			sods	2.1													
		33 05 07	SD-01 Preconstruction Submittals														
			Guided Boring and Jacking Plan	1.5.1	G DO												
			SD-06 Test Reports														
			Pilot Guide And Driven Casing	3.1.3	G DO	P											
			Test														
			SD-07 Certificates														
			Certified Of Compliance	1.5.2													
			SD-11 Closeout Submittals														
			Driving Records	3.1													
		33 11 00	SD-01 Preconstruction Submittals														
			Disinfection Method	3.3.4	G DO												
			48' Existing Watermain Work	3.1.1	G DO												
			Plan														
			SD-02 Shop Drawings														
			Pipe, Fittings, Joints and	2.1.1	G RO												
			Couplings														
			Butterfly Valve	2.1.2.1	G RO												
			Combination Air Valve	2.1.2.2	G RO												
			Extension Stems	2.1.2	G RO												
			SD-03 Product Data														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		33 11 00	Pipe, Fittings, Joints and Couplings	2.1.1	G RO												
			Valves	2.1.2	G RO												
			Valve Boxes	3.2.2	G RO												
			Pipe Restraint	2.2.1	G RO												
			Tapping Sleeves	2.2.2	G RO												
			Corporation Stops	3.2.2	G RO												
			Detectable Marking Tape	2.2.4	G RO												
			SD-06 Test Reports														
			Leakage Test	3.3.3													
			Hydrostatic Pressure Test	3.3.3													
			SD-07 Certificates														
			Pipe, Fittings, Joints and Couplings	2.1.1													
			Valves	2.1.2													
			SD-08 Manufacturer's Instructions														
			Ductile Iron Piping	2.1.1.1													
		33 30 00	SD-01 Preconstruction Submittals														
			Contractor's License	1.3.1	G DO												
			Bypass Pumping Plan	1.5	G DO												
			SD-02 Shop Drawings														
			Installation Drawings	3.1.1	G DO												
			SD-03 Product Data														
			Precast Concrete Manholes	2.2.7													
			Frames, Covers, and Gratings	2.2.10													
			Gravity Pipe	2.2.1													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		33 30 00	Resilient Manhole Connector	2.2.8													
			HDPE Lining For Concrete Structures	2.2.13	G DO												
			Sluice Gates	2.2.12	G DO												
			Polyethylene Encasement	2.2.4													
			Ceramic Epoxy Lining	2.2.2	G DO												
			SD-06 Test Reports														
			Infiltration Test	3.3.1.1	G DO												
			Exfiltration Test	3.3.1.2	G DO												
			Low-Pressure Air Test	3.3.2	G DO												
			Sluice Gate Operation	3.2.5.1	G DO												
			Acceptance Trial Operation And Test	3.4.2													
			SD-07 Certificates														
			Portland Cement	2.2.5													
			Welder Certification	2.2.12.7	G DO												
		33 40 00	SD-01 Preconstruction Submittals														
			Outfall Diversion Plan	3.1.5	G RO												
			SD-04 Samples														
			Pipe for Culverts and Storm Drains	2.1	G RO												
			SD-07 Certificates														
			Oil Resistant Gasket	2.3.10.1													
			Leakage Test	3.8.1.2	G RO												
			Hydrostatic Test on Watertight Joints	3.8.1.1	G RO												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		33 40 00	Flap Gates	2.3.9	G RO												
			Determination of Density	3.8.1.3													
			Frame and Cover for Gratings	2.3.7	G RO												
			Post-Installation Inspection Report	3.8.2.1.2	G RO												
			SD-08 Manufacturer's Instructions														
			Placing Pipe	3.3													
		33 71 01	SD-02 Shop Drawings														
			Connection Diagrams	1.3.1	G DO												
			Fabrication Drawings	1.3.1	G DO												
			Installation Drawings	1.3.1	G DO												
			SD-03 Product Data														
			Conductors	2.4.1	G DO												
			Insulators	2.3.3	G DO												
			Steel Poles	2.3.1	G DO												
			Nameplates	2.3.8	G DO												
			Pole Top Switch	3.1.7	G DO												
			Cutouts	2.3.6	G DO												
			Transformer	2.3.7	G DO												
			Surge Arresters	2.3.5	G DO												
			SD-05 Design Data														
			Steel Pole Design	2.3.1.1	G DO												
			Drilled Shaft Foundations	2.3.1.3	G DO												
			Direct Embedment Foundation	2.3.1.1	G DO												
			SD-06 Test Reports														
			Field Quality Control	3.2	G DO												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION
RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		33 71 01	Ground Resistance Test Reports	1.6.3	G DO												
			SD-07 Certificates														
			Transformer Losses	1.3.1	G DO												
			Qualifications of Licensed Engineer	1.6.4	G DO												
			Certification	2.3.1.2	G DO												
			SD-09 Manufacturer's Field Reports														
			Routine and Other Tests	2.5.2	G DO												
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance Manuals	1.5.1	G DO												
			SD-11 Closeout Submittals														
			Transformer Test Schedule	2.5.1	G DO												
		35 20 23	SD-01 Preconstruction Submittals														
			Notice Of Intent To Dredge	1.5.1													
			Relocation Of Navigation Aids	1.5.2													
			Relocation Of Navigation Aids	3.1.1													
			Drag Bar	1.5.3													
			Dredging Plan	1.6	G DO												
			SD-07 Certificates														
			Acceptance Sections Plans	3.4.4	G DO												
			Historical Period Shipwreck Sites	1.8.1													
			Surveys	3.5													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		35 20 23	Daily/Monthly Report Of Operations	1.5.4													
			Misplaced Materials	3.3.4													
			Declaration Of Inspection	1.11													
			Surveillance Of Ocean Disposal	3.1.3													
		35 20 26	SD-01 Preconstruction Submittals														
			Revisions Or Addendum To The Dredge Plant Instrumentation Plan	1.5	G												
			SD-07 Certificates														
			Letter of National Dredging Quality Management Program Certification	1.4	G DO												
		35 20 27	SD-01 Preconstruction Submittals														
			Dredge Plant Instrumentation Plan	1.5	G												
			SD-07 Certificates														
			National Dredging Quality Management Program Certification	1.4	G RO												
		35 31 19	SD-01 Preconstruction Submittals														
			Stone Source	1.7.2	G DO												
			Construction Methods	3.1	G DO												
			SD-04 Samples														
			Riprap	2.5	G DO												
			SD-05 Design Data														
			Surveys	3.7	G DO												
			SD-06 Test Reports														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		35 31 19	Initial Stone Test	1.7.3	G DO												
			Gradation	1.7.4	G DO												
			SD-07 Certificates														
			Testing Laboratory	1.7.5	G DO												
			Weekly Certificates	1.7.1	G DO												
			Weekly Certificates	1.7.1	G DO												
			Weight Scale Certificate	1.7.6	G DO												
		35 31 19.20	SD-02 Shop Drawings														
			Block Installation	3.4	G RO												
			Microgrid Installation	3.3	G RO												
			SD-03 Product Data														
			Articulating Concrete Block	2.1	G RO												
			Microgrid	2.3	G RO												
			Stabilized Stone Drainage Layer	2.6	G RO												
			SD-04 Samples														
			Articulating Concrete Block	2.1													
			Microgrid	2.3													
			SD-06 Test Reports														
			Flume Test	2.1.1.1													
			SD-07 Certificates														
			Certificates Of Compliance	2.3													
		35 31 19.26	SD-02 Shop Drawings														
			Unit Installation and Layout Plan	3.2	G RO												
			SD-03 Product Data														
			Concrete Armor Unit	2.1	G RO												
			SD-04 Samples														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3

CONTRACTOR

[illegible]

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 35 25

OWNER SAFETY REQUIREMENTS - DIVING

PART 1 GENERAL

- 1.1 SUMMARY
 - 1.1.1 Related Section
- 1.2 REFERENCES
- 1.3 SAFETY AND HEALTH REQUIREMENTS MANUAL
- 1.4 DEFINITIONS
- 1.5 SUBMITTALS
- 1.6 DIVE PLAN
 - 1.6.1 Dive Plan Reviewer
 - 1.6.2 Dive Personnel Qualifications
- 1.7 MEETINGS
 - 1.7.1 Phase Meetings

PART 2 PRODUCTS (NOT APPLICABLE)

PART 3 EXECUTION

- 3.1 EMERGENCY MEDICAL TREATMENT
- 3.2 DIVING OPERATIONS
 - 3.2.1 Dive Operations Execution

-- End of Section Table of Contents --

SECTION 01 35 25

OWNER SAFETY REQUIREMENTS - DIVING

PART 1 GENERAL

1.1 SUMMARY

This section covers job-specific underwater diving requirements and dive safety.

1.1.1 Related Section

Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1030 Bloodborne pathogens

1.3 SAFETY AND HEALTH REQUIREMENTS MANUAL

EM 385-1-1 is available at the USACE publications web site (<https://www.publications.usace.army.mil>). Download the most recent edition for information and comply with all the regulatory requirements of the EM 385-1-1.

1.4 DEFINITIONS

Use definitions found in EM 385-1-1 in transmitted Dive Plan submittal items.

Dive Supervisor - A person with specialized training or experience in Dive safety.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Dive Operations Plan; G, D0

SD-07 Certificates

Dive Personnel Qualifications; G, D0

Daily Dive Logs

1.6 DIVE PLAN

The Dive Plan must address all requirements of Section DIVING OPERATIONS of EM 385-1-1. See paragraph DIVE OPERATIONS below. An approved Dive Plan and Safe Practices Manual is required on all projects with work on or over water; see paragraph DIVING OPERATIONS below.

1.6.1 Dive Plan Reviewer

Submit all Dive Plans to the Contracting Officer in accordance with Section 01 33 00 SUBMITTAL PROCEDURES. Dive Operations Plans are District Office (D0) review with secondary review by Caribbean District Diving Coordinator or Alternate District Diving Coordinator. The names and contact information of the Diving Coordinator and Alternate(s) will be provided at the Preconstruction Conference. When a transmittal with a Dive Plan submittal item is delivered to Resident Engineer, send a copy directly to the Diving Coordinator.

1.6.2 Dive Personnel Qualifications

Submit qualifications and training certificates of dive personnel. Annual diver physicals must conform to the requirements of EM 385-1-1. Submit only a statement from a licensed physician that states the diver is Fit To Dive. Submitted physician statement must contain at a minimum the diver name, examination date, physician's name, physician's signature, physician's medical license number, and physician's telephone number. The signing physician must be an MD or an OD. Submitted statement must not contain divers Personal Health Information (PHI) or Personal Identification Information (PII).

First Aid (FA), Cardiopulmonary Resuscitation (CPR), and Emergency Oxygen System (E02) training must be in accordance with the EM 385-1-1, Chapter 3 and all applicable OSHA regulations. FA, CPR, and E02 training are required to be hands-on training. Online training will not be accepted. Provide proof of Hands-on training within this submittal.

Blood-Borne Pathogen (BBP) Program and Annual Training must comply with the EM 385-1-1 and OSHA 29 CFR 1910.1030, as well as all other applicable regulations. Provide proof of BBP Program and Annual Training within this submittal.

1.7 MEETINGS

1.7.1 Phase Meetings

Treat each dive as a definable feature of work in accordance with accepted Quality Control Plan. Perform a Preparatory and Initial Phase meeting for all dives.

PART 2 PRODUCTS (NOT APPLICABLE)

PART 3 EXECUTION

3.1 EMERGENCY MEDICAL TREATMENT

Arrange with local authorities for emergency dive medical response, treatment and evacuation to medical facility and nearest hyperbaric chamber. Provide a map with route directions to the nearest hospital and hyperbaric chamber in the Dive Plan's Emergency Management Plan required by EM 385-1-1.

3.2 DIVING OPERATIONS

Submit a Dive Operations Plan when the work is performed on or over water at least 30 days before the planned drive. No matter if a dive is actually planned or only required as a contingency (i.e., most dredging projects) submit a Diving Operations Plan for Contracting Officer's approval. Dive Operations Plan must cover all requirements in Chapter 30 DIVING OPERATIONS of EM 385-1-1. The Dive Operations Plan consists of a Safe Practices Manual describing Contractor's diving program and a Dive Plan describing site specific information of proposed dive or contingency dive. Safe Practices Manual, Dive Plan, and revisions must have cover sheets signed and dated by Contractor. When diving is subcontracted, cover sheets must also be signed and dated by diving contractor's principal or authorized representative.

3.2.1 Dive Operations Execution

Execute dives in accordance with approved Dive Operations Plan submittal; Section DIVING OPERATIONS of EM 385-1-1. Submit completed daily dive logs at the end of each dive day. Submit daily dive logs with the daily Quality Control report and provide to the District Dive Coordinator or authorized representative upon request. Use USACE form ENG 4615 to record daily diving activities. The ENG form can be found in the USACE publication website
<https://www.publications.usace.army.mil/USACE-Publications/Engineer-Forms>.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Competent Person (CP)
 - 1.2.2 Competent Person, Confined Spaces in Ships and Vessels (CPCSSV)
 - 1.2.3 Competent Person, Cranes and Rigging
 - 1.2.4 Competent Person, Excavation/Trenching
 - 1.2.5 Competent Person, Fall Protection
 - 1.2.6 Competent Person, Scaffolding
 - 1.2.7 High Hazard Activities
 - 1.2.8 High Visibility Accident
 - 1.2.9 Load Handling Equipment (LHE)
 - 1.2.10 Medical Treatment
 - 1.2.11 Accident
 - 1.2.12 Mishap
 - 1.2.13 Near Miss
 - 1.2.14 Operating Envelope
 - 1.2.15 Qualified Person (QP)
 - 1.2.16 Qualified Person, Fall Protection (QP for FP)
 - 1.2.17 Recordable Injuries or Illnesses
 - 1.2.18 Government Property and Equipment
 - 1.2.19 Load Handling Equipment (LHE) Accident or Load Handling Equipment Mishap
- 1.3 SUBMITTALS
- 1.4 PUBLIC HEALTH EMERGENCIES
- 1.5 MONTHLY EXPOSURE REPORTS
- 1.6 REGULATORY REQUIREMENTS
- 1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS
 - 1.7.1 Personnel Qualifications
 - 1.7.1.1 Site Safety and Health Officer (SSHO)
 - 1.7.1.1.1 Additional Site Safety and Health Officer (SSHO) Requirements and Duties
 - 1.7.1.2 Competent Person Qualifications
 - 1.7.1.2.1 Entry Supervisor for Confined Space Entry
 - 1.7.1.2.2 Competent Person for Scaffolding
 - 1.7.1.2.3 Competent Person for Fall Protection
 - 1.7.1.3 Crane Operators, Signal Persons and Riggers
 - 1.7.2 Personnel Duties
 - 1.7.2.1 Duties of the Site Safety and Health Officer (SSHO)
 - 1.7.3 Meetings
 - 1.7.3.1 Preconstruction Conference
 - 1.7.3.2 Safety Meetings
- 1.8 ACCIDENT PREVENTION PLAN (APP)
 - 1.8.1 Accident Prevention Plan
 - 1.8.2 Names and Qualifications

- 1.8.3 Plans
 - 1.8.3.1 Confined Space Entry Plan
 - 1.8.3.2 Standard Lift Plan (SLP)
 - 1.8.3.3 Critical Lift Plan - Crane or Load Handling Equipment
 - 1.8.3.3.1 Critical Lift Plan Planning and Schedule
 - 1.8.3.3.2 Lifts of Personnel
 - 1.8.3.4 Floating Cranes/Derricks, Crane Barges, and Auxiliary Shipboard-Mounted Crane
 - 1.8.3.5 Multi-Purpose Machines, Material Handling Equipment, and Construction Equipment Lift Plan
 - 1.8.3.6 Fall Protection and Prevention (FP&P) Plan
 - 1.8.3.7 Rescue Plan
 - 1.8.3.8 Hazardous Energy Control Program (HECP)
 - 1.8.3.9 Excavation Plan
 - 1.8.3.10 Site Demolition Plan
- 1.9 ACTIVITY HAZARD ANALYSIS (AHA)
 - 1.9.1 AHA Management
 - 1.9.2 AHA Signature Log
- 1.10 DISPLAY OF SAFETY INFORMATION
 - 1.10.1 Safety Bulletin Board
 - 1.10.2 Safety and Occupational Health (SOH) Deficiency Tracking System
- 1.11 SITE SAFETY REFERENCE MATERIALS
- 1.12 EMERGENCY MEDICAL TREATMENT
- 1.13 NOTIFICATIONS AND REPORTS
 - 1.13.1 Accident and Near Miss Notification
 - 1.13.2 Reports
 - 1.13.3 LHE Inspection Reports
 - 1.13.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging
- 1.14 HOT WORK
 - 1.14.1 Permit and Personnel Requirements
 - 1.14.2 Work Around Flammable Materials
- 1.15 RADIATION SAFETY REQUIREMENTS
 - 1.15.1 Radiography Operation Planning Work Sheet
 - 1.15.2 Site Access and Security
 - 1.15.3 Loss or Release and Unplanned Personnel Exposure
 - 1.15.4 Site Demarcation and Barricade
 - 1.15.5 Security of Material and Equipment
 - 1.15.6 Transportation of Material
 - 1.15.7 Schedule for Exposure or Unshielding
 - 1.15.8 Transmitter Requirements
- 1.16 CONFINED SPACE ENTRY REQUIREMENTS
 - 1.16.1 Entry Procedures
 - 1.16.2 Ventilation
 - 1.16.3 Sewer Wet Wells
 - 1.16.4 Rescue Procedures and Coordination with Local Emergency Responders
- 1.17 DIVE SAFETY REQUIREMENTS
- 1.18 SEVERE STORM PLAN

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

- 3.1 CONSTRUCTION AND OTHER WORK
 - 3.1.1 Worksite Communication
 - 3.1.2 Hazardous Material Exclusions

- 3.1.3 Unforeseen Hazardous Material
- 3.2 UTILITY OUTAGE REQUIREMENTS
- 3.3 OUTAGE COORDINATION MEETING
- 3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)
 - 3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility
 - 3.4.2 Lockout/Tagout Isolation
 - 3.4.3 Lockout/Tagout Removal
- 3.5 FALL PROTECTION PROGRAM
 - 3.5.1 Training
 - 3.5.2 Fall Protection Equipment and Systems
 - 3.5.2.1 Additional Personal Fall Protection Measures
 - 3.5.2.2 Personal Fall Protection Equipment
 - 3.5.3 Fall Protection for Roofing Work
 - 3.5.4 Flexible Horizontal Lifelines (HLL)
 - 3.5.5 Guardrail and Safety Net Systems
 - 3.5.6 Rescue Plan and Procedures
- 3.6 WORK PLATFORMS
 - 3.6.1 Scaffolding
 - 3.6.2 Elevated Aerial Work Platforms (AWPs)
- 3.7 EQUIPMENT
 - 3.7.1 Material Handling Equipment (MHE)
 - 3.7.2 Load Handling Equipment (LHE)
 - 3.7.3 Machinery and Mechanized Equipment
 - 3.7.4 Base Mounted Drum Hoists
 - 3.7.5 Use of Explosives
- 3.8 EXCAVATIONS
 - 3.8.1 Utility Locations
 - 3.8.2 Utility Location Verification
 - 3.8.3 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces
- 3.9 ELECTRICAL
 - 3.9.1 Conduct of Electrical Work
 - 3.9.2 Qualifications
 - 3.9.3 Arc Flash
 - 3.9.4 Grounding
 - 3.9.5 Testing

-- End of Section Table of Contents --

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B30.3	(2020) Tower Cranes
ASME B30.5	(2021) Mobile and Locomotive Cranes
ASME B30.7	(2021) Winches
ASME B30.8	(2020) Floating Cranes and Floating Derricks
ASME B30.9	(2018) Slings
ASME B30.20	(2018) Below-the-Hook Lifting Devices
ASME B30.22	(2016) Articulating Boom Cranes
ASME B30.23	(2022) Personnel Lifting Systems Safety Standard for Cableways, Cranes, Derricks, Hoists, Hooks, Jacks, and Slings
ASME B30.26	(2015; R 2020) Rigging Hardware

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ANSI/ASSP A10.22	(2007; R 2017) Safety Requirements for Rope-Guided and Non-Guided Workers' Hoists
ANSI/ASSP A10.34	(2021) Protection of the Public on or Adjacent to Construction Sites
ANSI/ASSP A10.44	(2020) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations
ANSI/ASSP Z244.1	(2016) The Control of Hazardous Energy Lockout, Tagout and Alternative Methods
ANSI/ASSP Z359.0	(2018) Definitions and Nomenclature Used for Fall Protection and Fall Arrest
ANSI/ASSP Z359.1	(2020) The Fall Protection Code

ANSI/ASSP Z359.2	(2017) Minimum Requirements for a Comprehensive Managed Fall Protection Program
ANSI/ASSP Z359.3	(2019) Safety Requirements for Lanyards and Positioning Lanyards
ANSI/ASSP Z359.4	(2013) Safety Requirements for Assisted-Rescue and Self-Rescue Systems, Subsystems and Components
ANSI/ASSP Z359.6	(2016) Specifications and Design Requirements for Active Fall Protection Systems
ANSI/ASSP Z359.7	(2019) Qualification and Verification Testing of Fall Protection Products
ANSI/ASSP Z359.11	(2014) Safety Requirements for Full Body Harnesses
ANSI/ASSP Z359.12	(2019) Connecting Components for Personal Fall Arrest Systems
ANSI/ASSP Z359.13	(2013) Personal Energy Absorbers and Energy Absorbing Lanyards
ANSI/ASSP Z359.14	(2014) Safety Requirements for Self-Retracting Devices for Personal Fall Arrest and Rescue Systems
ANSI/ASSP Z359.15	(2014) Safety Requirements for Single Anchor Lifelines and Fall Arresters for Personal Fall Arrest Systems
ANSI/ASSP Z359.16	(2016) Safety Requirements for Climbing Ladder Fall Arrest Systems
ANSI/ASSP Z359.18	(2017) Safety Requirements for Anchorage Connectors for Active Fall Protection Systems

ASTM INTERNATIONAL (ASTM)

ASTM F855	(2019) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment
-----------	---

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048	(2016) Guide for Protective Grounding of Power Lines
IEEE C2	(2023) National Electrical Safety Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10	(2022; ERTA 1 2021) Standard for Portable
---------	---

Fire Extinguishers

NFPA 51B	(2019; TIA 20-1) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work
NFPA 70	(2023; ERTA 4 2023; ERTA 5 2023; ERTA 6 2023) National Electrical Code
NFPA 70E	(2021) Standard for Electrical Safety in the Workplace
NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations
NFPA 306	(2019) Standard for the Control of Gas Hazards on Vessels

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-222	(2018H; Add 1 2019) Structural Standard for Antenna Supporting Structures and Antennas and Small Wind Turbine Support Structures
TIA-1019	(2012; R 2016) Standard for Installation, Alteration and Maintenance of Antenna Supporting Structures and Antennas

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety and Occupational Health (SOH) Requirements
------------	--

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 20	Standards for Protection Against Radiation
29 CFR 1910	Occupational Safety and Health Standards
29 CFR 1910.146	Permit-required Confined Spaces
29 CFR 1910.147	The Control of Hazardous Energy (Lock Out/Tag Out)
29 CFR 1910.333	Selection and Use of Work Practices
29 CFR 1910.1000	Air Contaminants
29 CFR 1915	Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment
29 CFR 1915.7	Competent person
29 CFR 1915.89	Control of Hazardous Energy (Lockout/Tags-Plus)

29 CFR 1926	Safety and Health Regulations for Construction
29 CFR 1926.16	Rules of Construction
29 CFR 1926.500	Fall Protection
29 CFR 1926.552	Material Hoists, Personal Hoists, and Elevators
29 CFR 1926.553	Base-Mounted Drum Hoists
29 CFR 1926.1431	Hoisting personnel
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
CPL 02-01-056	(2014) Inspection Procedures for Accessing Communication Towers by Hoist
CPL 2.100	(1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146

1.2 DEFINITIONS

1.2.1 Competent Person (CP)

A Competent Person is one who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees, and who has the authorization to take prompt corrective measures to eliminate them.

1.2.2 Competent Person, Confined Spaces in Ships and Vessels (CPCSSV)

An individual, designated in writing by the employer, responsible for testing, inspecting, and evaluating spaces and meets the requirements of 29 CFR 1915.7.

1.2.3 Competent Person, Cranes and Rigging

A person designated in writing by the employer to be responsible for the immediate supervision, implementation, and monitoring of the Crane and Rigging Program. Must have sufficient training, knowledge, and experience in crane(s) and rigging and be capable of identifying, evaluating, and addressing existing and potential hazards and must have the authority to take prompt corrective measures with regard to such hazards.

1.2.4 Competent Person, Excavation/Trenching

A person designated in writing by the employer to be responsible for the immediate supervision, implementation, and monitoring of the Excavation and Trenching Plan. Must have training, knowledge, and experience in excavation and trenching to be capable of identifying, evaluating, and addressing existing and potential hazards. This individual must have the authority to take prompt corrective measures with regard to such hazards.

1.2.5 Competent Person, Fall Protection

A person designated in writing by the employer to be responsible for the immediate supervision, implementation, and monitoring of the Fall Protection Program. The CP for Fall Protection, who through training, knowledge, and experience in fall protection and rescue systems and equipment, is capable of identifying, evaluating, and addressing existing and potential fall hazards. The CP for Fall Protection has the authority to take prompt corrective measures with regard to such hazards.

1.2.6 Competent Person, Scaffolding

A person designated in writing by the employer to be responsible for immediate supervision, implementation, and monitoring of the Scaffold Work Plan. Must have training, knowledge, and experience in scaffolding to be capable of identifying, evaluating, and addressing existing and potential hazards. This individual must have the authority to take prompt corrective measures with regard to such hazards. CP qualifications must meet the requirements of [EM 385-1-1](#), Section 22-3.b.

1.2.7 High Hazard Activities

Example of High Hazard Activities are activities that involve work at heights, crane and rigging, excavations and trenching, scaffolding, electrical work, and confined space entry.

1.2.8 High Visibility Accident

A High Visibility Accident is any mishap which may generate publicity or high visibility.

1.2.9 Load Handling Equipment (LHE)

LHE is a term used to describe cranes, hoists and all other hoisting equipment (hoisting equipment means equipment, including crane, derricks, hoists and power operated equipment used with rigging to raise, lower or horizontally move a load).

1.2.10 Medical Treatment

Medical Treatment is treatment administered by a physician or by registered professional personnel under the standing orders of a physician. Medical treatment does not include first aid treatment even when provided by a physician or registered personnel.

1.2.11 Accident

Any unplanned event or series of events that results in death, injury, or illness to personnel, or damage to or loss of equipment or property. Within the context of this manual, accident is synonymous with mishap.

1.2.12 Mishap

A mishap is synonymous with accident.

1.2.13 Near Miss

A potentially serious accident that could have resulted in personal injury, death, or property damage, damage to the environment, or illness

but did not occur due to one or more factors.

1.2.14 Operating Envelope

The Operating Envelope is the area surrounding any crane or load handling equipment. Inside this "envelope" is the crane, the operator, riggers and crane walkers, other personnel involved in the operation, rigging gear between the hook, the load, the crane's supporting structure (i.e. ground or rail), the load's rigging path, the lift and rigging procedure.

1.2.15 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.2.16 Qualified Person, Fall Protection (QP for FP)

A person with a recognized degree or professional certificate with extensive knowledge, training, and experience in the fall protection and rescue field. The QP for Fall Protection must be capable of designing, analyzing, evaluating, and specifying fall protection and rescue systems.

1.2.17 Recordable Injuries or Illnesses

Recordable Injuries or Illnesses are any work-related injury or illness that results in:

- a. Death, regardless of the time between the injury and death, or the length of the illness;
- b. Days away from work (any time lost after day of injury/illness onset);
- c. Restricted work;
- d. Transfer to another job;
- e. Medical treatment beyond first aid;
- f. Loss of consciousness; or
- g. A significant injury or illness diagnosed by a physician or other licensed health care professional, even if it did not result in (a) through (f) above

1.2.18 Government Property and Equipment

Interpret "USACE" property and equipment specified in USACE EM 385-1-1 as Government property and equipment.

1.2.19 Load Handling Equipment (LHE) Accident or Load Handling Equipment Mishap

A LHE accident occurs when any one or more of the eight elements in the operating envelope fails to perform correctly during operation, including operation during maintenance or testing resulting in personnel injury or

death; material or equipment damage; dropped load; derailment; two-blocking; overload; or collision, including unplanned contact between the load, crane, or other objects. A dropped load, derailment, two-blocking, overload and collision are considered accidents, even though no material damage or injury occurs. A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, or roll over). Document all LHE mishaps using the reporting form, ENG 3394, and a mandatory LHE Operator Drug Test.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" designation. Submittals not having a "G" or "S" designation are for information only. When used, a code following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Dive Operations Plan; G, D0

Accident Prevention Plan; G, D0

SD-06 Test Reports

Monthly Exposure Reports

Notifications and Reports

Reports; G, D0

LHE Inspection Reports

SD-07 Certificates

Crane Operators, Signal Persons and Riggers; G, D0

Standard Lift Plan; G, D0

Critical Lift Plan; G, D0

Naval Architecture Analysis; G, D0

Activity Hazard Analysis (AHA); G, D0

Confined Space Entry Permit

Hot Work Permit

Certificate of Compliance; G, D0

License Certificates; G, D0

Radiography Operation Planning Work Sheet; G, D0

Portable Gauge Operations Planning Worksheet; G, D0

1.4 PUBLIC HEALTH EMERGENCIES

In the event of a declared public health emergency, follow safety precautions as required by the Occupational Safety and Health Administration (OSHA) www.osha.gov, the Centers for Disease Control and Prevention (CDC) www.cdc.gov, and as required by federal, state and local requirements.

1.5 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report by the fifth of each month. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the voucher.

1.6 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE [EM 385-1-1](#), and the following federal, state, and local laws, ordinances, criteria, rules and regulations. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.7.1 Personnel Qualifications

1.7.1.1 Site Safety and Health Officer (SSHO)

Provide a minimum of one Level 1 SSHO that meets the requirements of [EM 385-1-1](#) Section 2-3. The SSHO must ensure that the requirements of [29 CFR 1926.16](#) are met for the project and must be present at the site of work and located so they have full mobility and reasonable access to all major work operations occurring during the shift. The SSHO or an equally-qualified Alternate SSHO must be at the work site at all times to implement and administer the Contractor's safety program and Government-accepted Accident Prevention Plan. The SSHO and Alternate SSHO must have the required training, experience, and qualifications in accordance with [EM 385-1-1](#) Section 2-3, and all associated sub-paragraphs.

The utilization of a Level 2 SSHO as identified and outlined in [EM 385-1-1](#), Section 2-3 does not apply in this contract. A Competent/Qualified Person (Level 3 SSHO) will be provided for all of the hazards identified in the Contractor's Safety and Health Program in accordance with the accepted Accident Prevention Plan, and will be on-site at all times when the work that presents the hazards associated with their professional expertise is being performed. The credentials of the Competent/Qualified Persons(s) will be approved by the Contracting Officer in consultation with the Safety Office. The Activity Hazard Analysis (AHA) is the document for this requirement.

1.7.1.1.1 Additional Site Safety and Health Officer (SSHO) Requirements and Duties

The Level 1 SSHO may not serve as the Quality Control Manager. The Level

1 SSHO may not serve as the Superintendent.

1.7.1.2 Competent Person Qualifications

Provide Competent Persons in accordance with EM 385-1-1 and herein. Competent Persons for high risk activities include confined space, cranes and rigging, excavation/trenching, fall protection, and electrical work. The CP for these activities must be designated in writing, and meet the requirements for the specific activity (i.e. competent person, fall protection).

The Competent Person identified in the Contractor's Safety and Health Program and accepted Accident Prevention Plan, must be on-site at all times when the work that presents the hazards associated with their professional expertise is being performed. Provide the credentials of the Competent Persons(s) to the Contracting Officer for information in consultation with the Safety Office.

1.7.1.2.1 Entry Supervisor for Confined Space Entry

Provide a Confined Space Entry Supervisor who meets the requirements of EM 385-1-1 Section 34-3 and herein. The Entry Supervisor must supervise the entry into each confined space in accordance with EM 385-1-1, Section 34.

Since this work involves operations that handle combustible or hazardous materials, this person must have the ability to understand and follow through on the air sampling, Personal Protective Equipment (PPE), and instructions of a Marine Chemist, Coast Guard authorized persons, or Certified Industrial Hygienist. Confined space and enclosed space work must comply with NFPA 306, OSHA 29 CFR 1915, Subpart B, "Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment," or as applicable, 29 CFR 1910.146 for general industry.

1.7.1.2.2 Competent Person for Scaffolding

Provide a Competent Person for Scaffolding who meets the requirements of EM 385-1-1, Section 22-3 and herein.

1.7.1.2.3 Competent Person for Fall Protection

Provide a Competent Person for Fall Protection who meets the requirements of EM 385-1-1, Section 21-3 and herein.

1.7.1.3 Crane Operators, Signal Persons and Riggers

Provide Operators, Signal Persons, and Riggers meeting the requirements in EM 385-1-1, Section 15-3 for Riggers and Section 16-3 for Crane Operators and Signal Persons.

1.7.2 Personnel Duties

1.7.2.1 Duties of the Site Safety and Health Officer (SSHO)

The SSHO must:

- a. Conduct daily safety and health inspections and maintain a written log which includes area/operation inspected, date of inspection, identified hazards, recommended corrective actions, estimated and

actual dates of corrections. Attach safety inspection logs to the Contractors' daily production report.

- b. Report and investigate all accidents and near misses and complete required accident report ENG Form 3394.
- c. Use and maintain OSHA's Form 300 to log work-related injuries and illnesses occurring on the project site for Prime Contractors and subcontractors, and make available to the Contracting Officer upon request. Post and maintain the Form 300A on the site Safety Bulletin Board.
- d. Maintain applicable safety reference material at the site of work including the EM 385-1-1, Safety Data Sheets (SDSs), APP, manufacturer instructions, etc.
- e. Attend all pre-construction conferences, tail-gate meetings, and pre-work meetings, to include preparatory meetings and periodic in-progress meetings.
- f. Review the APP and AHAs for compliance with EM 385-1-1, and approve, sign, implement and enforce them.
- g. Establish a Safety and Occupational Health (SOH) Deficiency Tracking System that lists and monitors outstanding deficiencies until resolution.
- h. Ensure all project personnel comply with SOH requirements (including subcontractors and visitors).
- i. Maintain a list of hazardous chemicals on site and their material Safety Data Sheets (SDS).
- j. Maintain a weekly list of high hazard activities involving energy, equipment, excavation, entry into confined space, and elevation, and be prepared to discuss details during QC Meetings.
- k. Provide and keep a record of site safety orientation and indoctrination training for all project personnel, including subcontractors
- l. Conduct on-going evaluations of the APP throughout the life of the project to ensure it is current and site-specific. Submit changes, revisions, and updates to the APP for review and approval by the prime contractor.
- m. Ensure visitors are escorted, properly protected, are wearing or provided the appropriate PPE, and receive a safety brief on the hazards to be expected onsite and the SOH controls required.

Superintendent, QC Manager, and SSHO are subject to dismissal if the above or any other required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

1.7.3 Meetings

1.7.3.1 Preconstruction Conference

- a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction conference. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).
- b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays.
- c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

1.7.3.2 Safety Meetings

Conduct safety meetings on project sites to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHAs, establish safe working procedures for anticipated hazards, and provide pertinent SOH training and motivation). Meetings must be conducted at least once a month for all supervisors on the project location and at least once a week for all workers. Document the meetings. Include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Include meeting minutes in the Daily QC report and furnish copies to the Contracting Officer. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.8 ACCIDENT PREVENTION PLAN (APP)

1.8.1 Accident Prevention Plan

A competent person must prepare the written site-specific APP. The APP must be job-specific and document all project, contract, and job-specific potential hazards in the workplace, and the company policies, controls, and work practices that will be used to minimize those hazards. They are an integral part of the planning and risk management process. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling authority" for all work site safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting

subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed by a corporate safety officer of the firm, the individual preparing the APP, the on-site superintendent, the designated SSHO, the Contractor Quality Control Manager, and any designated Certified Safety Professional (CSP) or Certified Health Physicist (CIH). The SSHO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Submit the APP to the Contracting Officer within 30 calendar days of Contract award and not less than 10 calendar days prior to the date of the preconstruction conference for acceptance. Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and update the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e. imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ANSI/ASSP A10.34), and the environment. At the time of APP submission, there may be unknown portions of the work. Include all known information in the APP prior to the start of work. Project-specific information that is yet to be known (for example, subcontractors to be used, type of fall protection to be used at a certain point in time, specific cranes or other load handling equipment (LHE) to be brought at the site of work, must be added and submitted prior to each preparatory phase/meeting, along with the appropriate AHAs.

1.8.2 Names and Qualifications

Provide plans in accordance with the requirements outlined in ENG Form 6293, including the following:

- a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.
- b. Qualifications of competent and of qualified persons. As a minimum, designate and submit qualifications of competent persons for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance.

1.8.3 Plans

Provide plans in the APP in accordance with ENG Form 6293, Accident Prevention Plan (APP) Worksheet. Provide the completed ENG Form 6293 with the submission of the project Accident Prevention Plan., including the following:

1.8.3.1 Confined Space Entry Plan

Develop a confined or enclosed space entry plan in accordance with EM 385-1-1, applicable OSHA standards 29 CFR 1910, 29 CFR 1915, and 29 CFR 1926, OSHA Directive CPL 2.100, and any other federal, state and local regulatory requirements identified in this Contract. Identify the qualified person's name and qualifications, training, and experience. Delineate the qualified person's authority to direct work stoppage in the event of hazardous conditions. Include procedure for rescue by Contractor personnel and the coordination with emergency responders. (If there is no confined space work, include a statement that no confined space work exists and none will be created.)

1.8.3.2 Standard Lift Plan (SLP)

Plan all lifts to avoid situations where the operator cannot maintain safe control of the lift. Prepare a written SLP for every lift or series of lifts (if duty cycle or routine lifts are being performed). The SLP must be developed, reviewed, and accepted by all personnel involved in the lift. Keep a copy of the SLP on the LHE for the specific approved lift(s). Keep a copy of every SLP for a minimum of three months. Include in the plan the items identified in EM 385-1-1 Section 16-7.b or use ENG Form 6203 (Crane Standard Lift Plan)

1.8.3.3 Critical Lift Plan - Crane or Load Handling Equipment

Provide a Critical Lift Plan prepared by a Qualified Person or Competent Person. The plan must include at a minimum, all requirements of the EM385-1-1 Sections 16-7.a, 16-7.b, and 16-7.d. ENG Form 6213 (Load-Handling Equipment Crane Operation Critical Lift Plan) may be utilized. In addition, Critical Lift Plans are required for the following:

- a. Lifts involving HAZMAT (for example, explosives, highly volatile substances).
- b. Hoisting personnel with LHE.
- c. Lifts made with more than one LHE.
- d. Lifts where the center of gravity (CG) could change.
- e. Lifts that exceed 75 percent of the load chart of the LHE during construction work.
- f. Lifts without the use of outriggers using rubber tire load charts.
- g. Lifts using more than one hoist on the same LHE.
- h. Lifts involving MLR assemblies or other non-routine or technically difficult rigging arrangements.
- i. Lifts involving submerged loads.

- j. Lifts out of the operator's view.
- k. Load Tests.
- l. When land based LHE mounted on barges, pontoons, or other means of flotation are required to travel while lifting the load.
- m. When two or more tower cranes are set up in the same operating envelope.
- n. Any lift the operator, lift director, supervisor, or employer believes should be considered critical.

1.8.3.3.1 Critical Lift Plan Planning and Schedule

Critical lifts require detailed planning and additional or unusual safety precautions. Develop and submit a critical lift plan to the Contracting Officer 30 calendar days prior to critical lift. Comply with load testing requirements in accordance with EM 385-1-1, Section 16-8.u.

1.8.3.3.2 Lifts of Personnel

In addition to the requirements of EM 385-1-1, Section 16-8.hh, for lifts of personnel, demonstrate compliance with the requirements of 29 CFR 1926.1431.

1.8.3.4 Floating Cranes/Derricks, Crane Barges, and Auxiliary Shipboard-Mounted Crane

Plan all lifts to avoid procedures that could result in configurations where the operator cannot maintain safe control of the lift. Prepare a written Standard Lift Plan (SLP) - Floating Plant meeting the requirements and criteria provided in EM 385-1-1 Sections 16-7.b and 16-7.c. Provide a Naval Architecture Analysis and include an LHE Manufacturer's Floating Service Load Chart in accordance with EM 385-1-1, Chapter 16.

1.8.3.5 Multi-Purpose Machines, Material Handling Equipment, and Construction Equipment Lift Plan

Multi-purpose machines, material handling equipment, and construction equipment used to lift loads that are suspended by rigging gear, require proof of authorization from the machine OEM that the machine is capable of making lifts of loads suspended by rigging equipment. Written approval from a qualified registered professional engineer, after a safety analysis is performed, is allowed in lieu of the OEM's approval. Demonstrate that the operator is properly trained and that the equipment is properly configured to make such lifts and is equipped with a load chart.

1.8.3.6 Fall Protection and Prevention (FP&P) Plan

The plan must be in accordance with the requirements of EM 385-1-1, Section 21-7.a and ANSI/ASSP Z359.2, be site specific, and address all fall hazards in the work place and during different phases of construction. Address how to protect and prevent workers from falling to lower levels when they are at or above 6 feet or as exposed to hazards below.. A competent person or qualified person for fall protection must prepare and sign the plan documentation. Include fall protection and prevention systems, equipment and methods employed for every phase of

work, roles and responsibilities, assisted rescue, self-rescue and evacuation procedures, training requirements, and monitoring methods. Review and revise, as necessary, the Fall Protection and Prevention Plan documentation as conditions change, but at least annually, for lengthy projects, reflecting any changes during the course of construction due to changes in personnel, equipment, systems or work habits. Keep and maintain the accepted Fall Protection and Prevention Plan documentation at the job site for the duration of the project. Include the Fall Protection and Prevention Plan documentation in the Accident Prevention Plan (APP).

1.8.3.7 Rescue Plan

Provide a Rescue Plan in accordance with EM 385-1-1 Section 21-7.b and ANSI/ASSP Z359.2, and include in the FP&P Plan and as part of the APP. Include a detailed discussion of the following: methods of rescue; methods of self-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility.

1.8.3.8 Hazardous Energy Control Program (HECP)

Develop a HECP in accordance with EM 385-1-1 Chapter 12, 29 CFR 1910.147, 29 CFR 1910.333, 29 CFR 1915.89, ANSI/ASSP Z244.1, and ANSI/ASSP A10.44. Submit this HECP as part of the Accident Prevention Plan (APP). Conduct a preparatory meeting and inspection with all effected personnel to coordinate all HECP activities. Document this meeting and inspection in accordance with EM 385-1-1, Section 12-5.a. Ensure that each employee is familiar with and complies with these procedures.

1.8.3.9 Excavation Plan

An Excavation and Trenching Plan must be developed by a CP and is required when excavations or trenches are 5 feet (1.5 m) or more in depth. Identify the safety and health aspects of excavation, and provide and prepare the plan in accordance with EM 385-1-1, Section 25-7 and Section 31 00 00 EARTHWORK.

1.8.3.10 Site Demolition Plan

Demolition Plans are required for all structural or soft demolition activities. The Demolition Plan must be prepared in accordance with EM 385-1-1 Section 17-7 and approved prior to work. The plan will be based on the engineering, lead, and asbestos surveys and must detail procedures for safe dismantling and removal of all building components and debris. Refer to Section 02 41 00 DEMOLITION and other reference sources.

1.9 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with ENG Form 6206 (Activity Hazard Analysis (AHA)) or as directed by the Contracting Officer. Submit the AHA for review at least

30 calendar days prior to the start of each activity task, or DFOV. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.9.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.9.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFOV must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.10 DISPLAY OF SAFETY INFORMATION

1.10.1 Safety Bulletin Board

Within seven calendar days after commencement of contract work, the contractor must erect a safety bulletin board at the site of work. Where site, duration, or logistics of a project do not facilitate a bulletin board, an alternative Contracting Officer accepted method (for example, binder, digital) may be deemed as meeting the requirement for a bulletin board. Include and maintain information on safety bulletin board as required by [EM 385-1-1](#), Section 2-8.f(2). Additional items required to be posted include:

- a. [Confined space entry permit](#).
- b. [Hot work permit](#).

1.10.2 Safety and Occupational Health (SOH) Deficiency Tracking System

Establish a SOH deficiency tracking system that lists and monitors the status of SOH deficiencies in chronological order. Use the tracking system to evaluate the effectiveness of the APP. A monthly evaluation of the data must be discussed in the QC or SOH meeting with everyone on the project. The list must be posted on the project bulletin board and updated daily, and provide the following information:

- a. Date deficiency identified;
- b. Description of deficiency;
- c. Name of person responsible for correcting deficiency;

- d. Projected resolution date;
- e. Date actually resolved.

1.11 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.12 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. Government has no responsibility to provide emergency medical treatment.

1.13 NOTIFICATIONS AND REPORTS

1.13.1 Accident and Near Miss Notification

Notify the Contracting Officer as soon as practical, but no later than the stated timelines included in EM 385-1-1 Section 2-9, Table 2-1 Accident Reporting Required Timeline. The Contractor is responsible for obtaining appropriate medical and emergency assistance and for notifying fire, law enforcement, and regulatory agencies. Immediate reporting is required for electrical accidents, to include Arc Flash; shock; uncontrolled release of hazardous energy (includes electrical and non-electrical); load handling equipment or rigging; fall from height (any level other than same surface); and underwater diving. All accidents and near misses must be investigated, analyzed, and reported in accordance with EM 385-1-1 Section 2-8.d.

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any accident or near miss.

1.13.2 Reports

The reporting and associated investigation of accidents and near misses is considered a leading indicator. Utilize the ENG Form 3394 (U.S. Army Corps of Engineers Mishap Notification and Investigation) to report accidents and near misses.

1.13.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.13.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

Provide a Certificate of Compliance (CoC) for Load Handling Equipment (LHE) and Rigging (ENG Form 6209) for LHE entering an activity under this

Contract and in accordance with EM 385-1-1. Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1, Section 16-7.b and paragraph STANDARD LIFT PLAN (SLP) above for each lift planned. Submit SLP to the Contracting Officer for approval within 30 calendar days in advance of planned lift.

1.14 HOT WORK

1.14.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e. welding or cutting) or operating other flame-producing/spark producing devices, from the local fire department. A permit is required from the Explosives Safety Office for work in and around where explosives are processed, stored, or handled. CONTRACTORS ARE REQUIRED TO MEET ALL CRITERIA BEFORE A PERMIT IS ISSUED. Provide at least two 20 pound 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must be current inspection tagged, and contain an approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch must be trained in accordance with NFPA 51B and remain on-site for a minimum of thirty (30) minutes after the exposure has ended or as determined by the Contractor and documented in the AHA.

When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency fire department phone number. REPORT ANY FIRE, NO MATTER HOW SMALL, TO THE RESPONSIBLE FIRE DEPARTMENT IMMEDIATELY.

1.14.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or Certified Industrial Hygienist for "HOT WORK" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in EM 385-1-1, Section 6-8.j.

1.15 RADIATION SAFETY REQUIREMENTS

Submit License Certificates, employee training records, and Leak Test Reports for radiation materials and equipment to the Contracting Officer and Radiation Safety Office (RSO) for all specialized and licensed material and equipment proposed for use on the construction project (excludes portable machine sources of ionizing radiation including moisture density and X-Ray Fluorescence (XRF)). Maintain on-site records whenever licensed radiological materials or ionizing equipment are on Government property.

Protect workers from radiation exposure in accordance with 10 CFR 20, ensuring any personnel exposures are maintained As Low As Reasonably

Achievable.

1.15.1 Radiography Operation Planning Work Sheet

Submit a Gamma and X-Ray [Radiography Operation Planning Work Sheet](#) to Contracting Officer 14 days prior to commencement of operations involving radioactive materials or radiation generating devices. For portable machine sources of ionizing radiation, including moisture density and XRF, use and submit the [Portable Gauge Operations Planning Worksheet](#) instead. The Contracting Officer will review the submitted worksheet and provide questions and comments.

Contractors must use primary dosimeters process by a National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratory.

1.15.2 Site Access and Security

Coordinate site access and security requirements with the Contracting Officer for all radiological materials and equipment containing ionizing radiation that are proposed for use on a government facility. For gamma radiography materials and equipment, a Government escort is required for any travels on the Installation. The Government authorized representative will meet the Contractor at a designated location outside the Installation, ensure safety of the materials being transported, and will escort the Contractor for gamma sources onto the Installation, to the job site, and off the Installation. For portable machine sources of ionizing radiation, including moisture density and XRF, the Government authorized representative will meet the Contractor at the job site.

Provide a copy of all calibration records, and utilization records for radiological operations performed on the site.

1.15.3 Loss or Release and Unplanned Personnel Exposure

Loss or release of radioactive materials, and unplanned personnel exposures must be reported immediately to the Contracting Officer, RSO, and Base Security Department Emergency Number.

1.15.4 Site Demarcation and Barricade

Properly demark and barricade an area surrounding radiological operations to preclude personnel entrance, in accordance with [EM 385-1-1](#), Nuclear Regulatory Commission, and Applicable State regulations and license requirements, and in accordance with requirements established in the accepted Radiography Operation Planning Work Sheet.

Do not close or obstruct streets, walks, and other facilities occupied and used by the Government without written permission from the Contracting Officer.

1.15.5 Security of Material and Equipment

Properly secure the radiological material and ionizing radiation equipment at all times, including keeping the devices in a properly marked and locked container, and secondarily locking the container to a secure point in the Contractor's vehicle or other approved storage location during transportation and while not in use. While in use, maintain a continuous visual observation on the radiological material and ionizing radiation equipment. In instances where radiography is scheduled near or adjacent

to buildings or areas having limited access or one-way doors, make no assumptions as to building occupancy. Where necessary, the Contracting Officer will direct the Contractor to conduct an actual building entry, search, and alert. Where removal of personnel from such a building cannot be accomplished and it is otherwise safe to proceed with the radiography, position a fully instructed employee inside the building or area to prevent exiting while external radiographic operations are in process.

1.15.6 Transportation of Material

Comply with 49 CFR 173 for Transportation of Regulated Amounts of Radioactive Material. Notify Local Fire authorities and the site Radiation Safety Officer (RSO) of any Radioactive Material use.

1.15.7 Schedule for Exposure or Unshielding

Actual exposure of the radiographic film or unshielding the source must not be initiated until after 5 p.m. on weekdays.

1.15.8 Transmitter Requirements

Adhere to the base policy concerning the use of transmitters, such as radios and cell phones. Obey Emissions control (EMCON) restrictions.

1.16 CONFINED SPACE ENTRY REQUIREMENTS

Confined space entry must comply with Chapter 34 of EM 385-1-1, OSHA 29 CFR 1926, OSHA 29 CFR 1910, OSHA 29 CFR 1910.146, and OSHA Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used.

1.16.1 Entry Procedures

Prohibit entry into a permit required confined space by personnel for any purpose, including hot work, until the qualified person has conducted appropriate tests to ensure the confined or enclosed space is safe for the work intended and that all potential hazards are controlled or eliminated and documented. Comply with EM 385-1-1, Chapter 34 for entry procedures. Hazards pertaining to the space must be reviewed with each employee during review of the AHA.

1.16.2 Ventilation

Use natural or mechanical ventilation controls to provide continuous fresh air into the confined space based on the work that will be performed in accordance with EM 385-1-1, Section 34-8.g

1.16.3 Sewer Wet Wells

Sewer wet wells require continuous atmosphere monitoring with audible alarm for toxic gas detection.

1.16.4 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue from a confined space.

1.17 DIVE SAFETY REQUIREMENTS

Develop a Dive Operations Plan, AHA, emergency management plan, and personnel list that includes qualifications, for each separate diving operation. Submit these documents to the District Dive Coordinator (DDC) via the Contracting Officer for review and approval at least 30 working days prior to commencement of diving operations. These documents must be at the diving location at all times. Provide each of these documents as a part of the project file.

1.18 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

- a. Secure outside equipment and materials and place materials that could be damaged in protected areas.
- b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.
- c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Minimum requirements for PPE are provided in EM 385-1-1.

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.1.2 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with

radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. Notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

3.1.3 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e. 29 CFR 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during construction operations, stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification pursuant to FAR 52.243-4 Changes and FAR 52.236-2 Differing Site Conditions.

3.2 UTILITY OUTAGE REQUIREMENTS

Apply for utility outages in sufficient time as to not cause schedule delays or impacts. At a minimum, the written request must include the location of the outage, utilities being affected, duration of outage, any necessary sketches, and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1, Chapter 11 (Isolation). Some examples of energy isolation devices and procedures are highlighted in EM 385-1-1, Chapter 12. Where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy.

These activities include, but are not limited to, a review of HEC and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1, Chapter 11 and NFPA 70E, work on energized electrical circuits must not be performed without prior Government authorization. Government permission is considered through the permit process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is infeasible.

3.3 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage coordination meeting in accordance with EM 385-1-1, Section 12-4. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the Contracting Officer, the Installation representative, and the Public Utilities representative. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HEC training in accordance with EM 385-1-1, Section 12-3. Clarify when personal

protective equipment is required during switching operations, inspection, and verification.

3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1 Chapter 12, 29 CFR 1910.147, 29 CFR 1915.89, ANSI/ASSP A10.44, NFPA 70E, and paragraph HAZARDOUS ENERGY CONTROL PROGRAM (HECP).

3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1, Section 12-4. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECP. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1, Sections 5-8 and 11-8. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1, Chapter 12.

3.4.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECP. In accordance with EM 385-1-1, Chapter 12, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal notification to the Government (by completing the Government form if provided by Contracting Officer's Representative), confirming that steps of de-energization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment and rescue and evacuation procedures in accordance with ANSI/ASSP Z359.2 and EM 385-1-1, Chapter 21.

3.5.1 Training

Institute a fall protection training program. As part of the Fall Protection Program, provide training for each employee who might be exposed to fall hazards and using personal fall protection equipment. Provide training by a competent person for fall protection in accordance with EM 385-1-1, Section 21-3. Document training and practical application of the competent person in accordance with EM 385-1-1, Section 21-3.c and ANSI/ASSP Z359.2 in the AHA.

3.5.2 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1, Chapter 21.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 Chapter 21, 29 CFR 1910, 29 CFR 1926.500 Subpart M, ANSI/ASSP Z359.0, ANSI/ASSP Z359.1, ANSI/ASSP Z359.2, ANSI/ASSP Z359.3, ANSI/ASSP Z359.4, ANSI/ASSP Z359.6, ANSI/ASSP Z359.7, ANSI/ASSP Z359.11, ANSI/ASSP Z359.12, ANSI/ASSP Z359.13, ANSI/ASSP Z359.14, ANSI/ASSP Z359.15, ANSI/ASSP Z359.16 and ANSI/ASSP Z359.18.

3.5.2.1 Additional Personal Fall Protection Measures

In addition to the required fall protection systems, other protective measures such as safety skiffs, personal floatation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1, Section 21-8.t. Personal fall protection systems and equipment are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

3.5.2.2 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabineers must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 3,600 lbs in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 6 feet, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1, Section 21-8.

3.5.3 Fall Protection for Roofing Work

Implement fall protection controls based on the type of roof being constructed and work being performed. Evaluate the roof area to be accessed for its structural integrity including weight-bearing capabilities for the projected loading.

- a. Low Sloped Roofs: Construction roofing activities on low-sloped roofs (less than or equal to 4:12 slope) with unprotected sides and edges 6 feet (1.8 m) or more above lower levels can utilize a Warning Line System (WLS). At minimum, the WLS must be used in combination with a safety monitor system in accordance with [EM 385-1-1](#), Section 21-8.i.
- b. Steep-Sloped Roofs: Work on a roof having a slope greater than 4:12 (vertical to horizontal) requires use of conventional fall protection system (that is, guardrail system, safety net system, or personal fall protection system). This requirement also applies to residential or housing type construction.

3.5.4 Flexible Horizontal Lifelines (HLL)

Provide Flexible HLL in accordance with [EM 385-1-1](#), Section 21-8.c(10). Commercially manufactured horizontal lifelines (HLL) must be designed, installed, certified and used, under the supervision of a qualified person, for fall protection as part of a complete fall arrest system which maintains a safety factor of 2 ([29 CFR 1926.500](#)). The competent person for fall protection may (if deemed appropriate by the qualified person) supervise the assembly, disassembly, use and inspection of the HLL system under the direction of the qualified person. Locally manufactured HLLs are not acceptable unless they are custom designed for limited or site specific applications by a Registered Professional Engineer who is qualified in designing HLL systems.

3.5.5 Guardrail and Safety Net Systems

Design, install and use guardrail and safety net systems in accordance with [EM 385-1-1](#), Sections 21-8.d, 21-8.g and [29 CFR 1926](#) Subpart M.

3.5.6 Rescue Plan and Procedures

When personal fall arrest systems are used, ensure that the mishap victim can self-rescue or can be rescued promptly should a fall occur. Prepare a Rescue Plan and include a detailed discussion of the following: methods of rescue; methods of self-rescue or assisted-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility. Include the Rescue Plan within the Activity Hazard Analysis (AHA) for the phase of work, in the Fall Protection and Prevention (FP&P) Plan, and the Accident Prevention Plan (APP). The plan must be in accordance with the requirements of [EM 385-1-1](#), [ANSI/ASSP Z359.2](#), and [ANSI/ASSP Z359.4](#).

3.6 WORK PLATFORMS

3.6.1 Scaffolding

Provide employees with a safe means of access to the work area on the scaffold. Climbing of any scaffold braces or supports not specifically

designed for access is prohibited. Comply with the following requirements:

- a. Scaffold platforms greater than 20 feet in height must be accessed by use of a scaffold stair system.
- b. Ladders commonly provided by scaffold system manufacturers are prohibited for accessing scaffold platforms greater than 20 feet maximum in height.
- c. An adequate gate is required.
- d. Employees performing scaffold erection and dismantling must be qualified.
- e. Scaffold must be capable of supporting at least four times the maximum intended load, and provide appropriate fall protection as delineated in the accepted fall protection and prevention plan.
- f. Stationary scaffolds must be attached to structural building components to safeguard against tipping forward or backward.
- g. Special care must be given to ensure scaffold systems are not overloaded.
- h. Side brackets used to extend scaffold platforms on self-supported scaffold systems for the storage of material are prohibited. The first tie-in must be at the height equal to 4 times the width of the smallest dimension of the scaffold base.
- i. Scaffolding other than suspended types must bear on base plates upon wood mudsills (2 in x 10 in x 8 in minimum) or other adequate firm foundation.
- j. Scaffold or work platform erectors must have fall protection during the erection and dismantling of scaffolding or work platforms that are more than 6 feet.
- k. Delineate fall protection requirements when working above 6 feet or above dangerous operations in the Fall Protection and Prevention (FP&P) Plan and Activity Hazard Analysis (AHA) for the phase of work.

3.6.2 Elevated Aerial Work Platforms (AWPs)

Use restraint systems or PFAS on all AWPs. Workers must be anchored to the basket or bucket in accordance with manufacturer's specifications and instructions (anchoring to the boom may only be used when allowed by the manufacturer and permitted by the CP). Lanyards, to include adjustable energy absorbing lanyards, used in the restraint system must be sufficiently short to provide fall restraint and prohibit workers from climbing out of or being ejected from the platform. The climbing of rails is prohibited. Lanyards with built-in shock absorbers are acceptable. Self-retracting devices are not acceptable. Tying off to an adjacent pole or structure is not permitted unless a safe device for 100 percent tie-off is used for the transfer.

Use of AWPs must be operated, inspected, and maintained as specified in the operating manual for the equipment and delineated in the AHA. Operators of AWPs must be designated as qualified operators by the Prime Contractor. Maintain proof of qualifications on site for review and

include in the AHA.

3.7 EQUIPMENT

3.7.1 Material Handling Equipment (MHE)

- a. Material handling equipment such as forklifts must not be modified with work platform attachments for supporting employees unless specifically delineated in the manufacturer's printed operating instructions. Material handling equipment fitted with personnel work platform attachments are prohibited from traveling or positioning while personnel are working on the platform.
- b. The use of hooks on equipment for lifting of material must be in accordance with manufacturer's printed instructions. Material Handling Equipment Operators must be trained in accordance with OSHA 29 CFR 1910, Subpart N.
- c. Operators of forklifts or power industrial trucks must be licensed in accordance with OSHA.

3.7.2 Load Handling Equipment (LHE)

The following requirements apply. In exception, these requirements do not apply to commercial truck mounted and articulating boom cranes used solely to deliver material and supplies (not prefabricated components, structural steel, or components of a systems-engineered metal building) where the lift consists of moving materials and supplies from a truck or trailer to the ground; to cranes installed on mechanics trucks that are used solely in the repair of shore-based equipment; to crane that enter the activity but are not used for lifting; nor to other machines not used to lift loads suspended by rigging equipment. However, LHE accidents occurring during such operations must be reported.

- a. Equip cranes and derricks as specified in EM 385-1-1, Chapter 16.
- b. Notify the Contracting Officer 15 working days in advance of any LHE entering the activity so that necessary quality assurance spot checks can be coordinated. Contractor's operator must remain with the crane during the spot check. Rigging gear must be in accordance with OSHA, ASME B30.9 Standards safety standards.
- c. Comply with the LHE manufacturer's specifications and limitations for erection and operation of cranes and hoists used in support of the work. Perform erection under the supervision of a designated person (as defined in ASME B30.5). Perform all testing in accordance with the manufacturer's recommended procedures.
- d. Comply with ASME B30.5 for mobile and locomotive cranes, ASME B30.22 for articulating boom cranes, ASME B30.3 for construction tower cranes, ASME B30.8 for floating cranes and floating derricks, ASME B30.9 for slings, ASME B30.20 for below the hook lifting devices and ASME B30.26 for rigging hardware.
- e. When operating in the vicinity of overhead transmission lines, operators and riggers must be alert to this special hazard and follow the requirements of EM 385-1-1 Chapter 11, and ASME B30.5 or ASME B30.22 as applicable.

- f. Do not use crane suspended personnel work platforms (baskets) unless the Contractor proves that using any other access to the work location would provide a greater hazard to the workers or is impossible. Do not lift personnel with a line hoist or friction crane. Additionally, submit a specific AHA for this work to the Contracting Officer. Ensure the activity and AHA are thoroughly reviewed by all involved personnel.
- g. Inspect, maintain, and recharge portable fire extinguishers as specified in NFPA 10, Standard for Portable Fire Extinguishers.
- h. All employees must keep clear of loads about to be lifted and of suspended loads, except for employees required to handle the load.
- i. Use cribbing when performing lifts on outriggers.
- j. The crane hook/block must be positioned directly over the load. Side loading of the crane is prohibited.
- k. A physical barricade must be positioned to prevent personnel access where accessible areas of the LHE's rotating superstructure poses a risk of striking, pinching or crushing personnel.
- l. Maintain inspection records in accordance by EM 385-1-1, Section 16-5, including shift, monthly, and annual inspections, the signature of the person performing the inspection, and the serial number or other identifier of the LHE that was inspected. Records must be available for review by the Contracting Officer.
- m. Maintain written reports of operational and load testing in accordance with EM 385-1-1, Section 16-5, listing the load test procedures used along with any repairs or alterations performed on the LHE. Reports must be available for review by the Contracting Officer.
- n. Certify that all LHE operators have been trained in proper use of all safety devices (e.g. anti-two block devices).
- o. Take steps to ensure that wind speed does not contribute to loss of control of the load during lifting operations. At wind speeds greater than 20 mph, the operator, rigger and lift director must cease all crane operations, evaluate conditions and determine if the lift may proceed. Base the determination to proceed or not on wind calculations per the manufacturer and a reduction in LHE rated capacity if applicable. Include this maximum wind speed determination as part of the activity hazard analysis plan for that operation.
- q. Follow FAA guidelines when required based on project location.

3.7.3 Machinery and Mechanized Equipment

- a. Proof of qualifications for operator must be kept on the project site for review.
- b. Manufacture specifications or owner's manual for the equipment must be on-site and reviewed for additional safety precautions or requirements that are sometimes not identified by OSHA or USACE EM 385-1-1. Incorporate such additional safety precautions or requirements into the AHAs.

3.7.4 Base Mounted Drum Hoists

- a. Operation of base mounted drum hoists must be in accordance with [EM 385-1-1](#) and [ANSI/ASSP A10.22](#).
- b. Rigging gear must be in accordance with applicable ASME/OSHA standards.
- c. When used on telecommunication towers, base mounted drum hoists must be in accordance with [TIA-1019](#), [TIA-222](#), [ASME B30.7](#), [29 CFR 1926.552](#), and [29 CFR 1926.553](#).
- d. When used to hoist personnel, the AHA must include a written standard operating procedure. Operators must have a physical examination in accordance with [EM 385-1-1](#) Section 16-3.f and trained, at a minimum, in accordance with [EM 385-1-1](#) Section 16-3. The base mounted drum hoist must also comply with OSHA Instruction [CPL 02-01-056](#) and [ASME B30.23](#).
- e. Material and personnel must not be hoisted simultaneously.
- f. Personnel cage must be marked with the capacity (in number of persons) and load limit in [pounds](#).
- g. Construction equipment must not be used for hoisting material or personnel or with trolley/tag lines. Construction equipment may be used for towing and assisting with anchoring guy lines.

3.7.5 Use of Explosives

Explosives must not be used or brought to the project site without prior written approval from the Contracting Officer. Such approval does not relieve the Contractor of responsibility for injury to persons or for damage to property due to blasting operations.

Storage of explosives, when permitted on Government property, must be only where directed and in approved storage facilities. These facilities must be kept locked at all times except for inspection, delivery, and withdrawal of explosives.

3.8 EXCAVATIONS

Soil classification must be performed by a competent person in accordance with [29 CFR 1926](#) and [EM 385-1-1](#).

3.8.1 Utility Locations

Provide a third party, independent, private utility locating company to positively identify underground utilities in the work area in addition to any station locating service and coordinated with the station utility department.

3.8.2 Utility Location Verification

Physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within [3 feet](#) of the underground system.

3.8.3 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces

Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. Whenever Contract work involves chipping, saw cutting, or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with station utility departments in addition to location and depth verification by a third party, independent, private locating company. The third party, independent, private locating company must locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings does not alleviate the Contractor from meeting this requirement.

3.9 ELECTRICAL

Perform electrical work in accordance with [EM 385-1-1](#), Chapters 11 and 12.

3.9.1 Conduct of Electrical Work

As delineated in [EM 385-1-1](#), electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with [ASTM F855](#) and [IEEE 1048](#). Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by [NFPA 70](#), high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety shoes, insulating gloves and electrical arc flash protection for personnel as required by [NFPA 70E](#). Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures, [29 CFR 1910](#) Subpart S, and [29 CFR 1910.147](#).

3.9.2 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with State, Local requirements applicable to where work is being performed.

3.9.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

3.9.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

Check grounding circuits to ensure that the circuit between the ground and a grounded power conductor has a resistance low enough to permit sufficient current flow to allow the fuse or circuit breaker to interrupt the current.

3.9.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 45 00

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 CONTRACT ADMINISTRATION
 - 1.2.1 Electronic Communications
 - 1.2.2 Other Factors
- 1.3 RMS SOFTWARE
 - 1.3.1 RMS CONTRACTOR'S MODE (CM)
- 1.4 SYSTEM REQUIREMENTS
- 1.5 RELATED INFORMATION
 - 1.5.1 RMS User Guide
- 1.6 CONTRACT DATABASE
- 1.7 DATABASE MAINTENANCE
 - 1.7.1 Administration
 - 1.7.1.1 Contractor Information
 - 1.7.1.2 Subcontractor Information
 - 1.7.1.3 Correspondence
 - 1.7.1.4 Equipment
 - 1.7.1.5 Management Reporting
 - 1.7.1.6 Request For Information (RFI)
 - 1.7.2 Finances
 - 1.7.2.1 Pay Activity Data
 - 1.7.2.2 Payment Requests
 - 1.7.3 Quality Control (QC)
 - 1.7.3.1 Daily Contractor Quality Control (CQC) Reports
 - 1.7.3.2 Deficiency Tracking
 - 1.7.3.3 QC Requirements
 - 1.7.3.4 Three-Phase Control Meetings
 - 1.7.3.5 Labor and Equipment Hours
 - 1.7.3.6 Accident/Safety Reporting
 - 1.7.3.7 Features of Work
 - 1.7.3.8 Hazard Analysis
 - 1.7.4 Submittal Management
 - 1.7.5 Schedule
 - 1.7.6 Import/Export of Data
- 1.8 IMPLEMENTATION
- 1.9 MONTHLY COORDINATION MEETING
- 1.10 NOTIFICATION OF NONCOMPLIANCE

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section Table of Contents --

SECTION 01 45 00

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

1.2 CONTRACT ADMINISTRATION

The Government will use the Resident Management System (RMS) to assist in its monitoring and administration of this contract. The Contractor uses the Government-furnished Construction Contractor Mode of RMS, referred to as RMS CM, to record, maintain, and submit various information throughout the contract period. The Contractor mode user manuals, updates, and training information can be downloaded from the RMS web site (<https://rms.usace.army.mil/datafiles/rmsdocwebsite/default.html>). The joint Government-Contractor use of RMS facilitates electronic exchange of information and overall management of the contract. RMS provides the means for the Contractor to input, track, and electronically share information with the Government in the following areas:

- Administration
- Finances
- Quality Control
- Submittal Monitoring
- Scheduling
- Import/Export of Data

1.2.1 Electronic Communications

For ease and speed of communications, exchange correspondence and other documents (i.e., Correspondence and Payment Requests) in electronic format via RMS unless otherwise approved by the Contracting Officer. Correspondence, pay requests, and other documents comprising the official contract record are also to be provided in paper format with signatures and dates where necessary. Paper documents will govern in the event of discrepancy with the electronic version.

1.2.2 Other Factors

Particular attention is directed to Contract Clause, SCHEDULES FOR CONSTRUCTION CONTRACTS, Contract Clause, PAYMENTS, Section 01 32 01 PROJECT SCHEDULE, Section 01 33 00 SUBMITTAL PROCEDURES, and Section 01 45 04 CONTRACTOR QUALITY CONTROL, which have a direct relationship to the reporting to be accomplished through RMS. Also, there is no separate

payment for establishing and maintaining the RMS database; costs associated will be included in the contract pricing for the work.

1.3 RMS SOFTWARE

RMS is a Windows-based program that can be run on a Windows-based PC meeting the requirements as specified in Section 1.4. The Government will make available the RMS software to the Contractor after award of the construction contract. Prior to the Preconstruction Conference, download, install, and use the latest version of the RMS software from the Government's RMS Internet Website. Any program updates of RMS will be made available to the Contractor via the Government RMS Website as the updates become available.

1.3.1 RMS CONTRACTOR'S MODE (CM)

RMS Contractor's Mode, or RMS CM, is the replacement for Quality Control System. The database remains the same. References to RMS in this specification include RMS CM.

1.4 SYSTEM REQUIREMENTS

The following is the minimum system configuration required to run RMS and Contractor Mode:

Minimum RMS System Requirements	
Hardware	
Windows-based PC	1.5 GHz 2 core or higher processor
RAM	8 GB
Hard drive disk	200 GB space for sole use by the RMS system
Monitor	Screen resolution 1366 x 768
Mouse or other pointing device	
Windows compatible printer	Laser printer must have 4 MB+ of RAM
Connection to the Internet	minimum 10 Mbs per user
Software	
MS Windows	Windows 7 x 64 bit (RMS requires 64 bit O/S) or newer
Word Processing software	Viewer for MS Word 2013, MS Excel 2013, or newer
Microsoft.NET Framework	Coordinate with the Contracting Officer for free version required
Email	MAPI compatible

Minimum RMS System Requirements	
Virus protection software	Regularly upgraded with all issued manufacturer's updates and is able to detect most zero day viruses.

1.5 RELATED INFORMATION

1.5.1 RMS User Guide

After contract award, download instructions for the installation and use of RMS from the Government RMS Internet Website.

1.6 CONTRACT DATABASE

Prior to the preconstruction conference, the Government will provide the Contractor with basic contract award data to use for RMS. The Government will provide data updates to the Contractor as needed. These updates will generally consist of submittal reviews, correspondence status, Quality Assurance (QA) comments, and other administrative and QA data.

1.7 DATABASE MAINTENANCE

Establish, maintain, and update data in the RMS database throughout the duration of the contract at the Contractor's site office. Submit data updates to the Government (e.g., daily reports, submittals, RFIs, schedule updates, payment requests) using RMS. The RMS database typically includes current data on the following items:

1.7.1 Administration

1.7.1.1 Contractor Information

Contain within the database the Contractor's name, address, telephone numbers, management staff, and other required items. Within 7 calendar days of receipt of RMS software from the Government, deliver Contractor administrative data in electronic format in RMS.

1.7.1.2 Subcontractor Information

Contain within the database the name, trade, address, phone numbers, and other required information for all subcontractors. A subcontractor is listed separately for each trade to be performed. Assign each subcontractor/trade a unique Responsibility Code provided in RMS. Within 7 calendar days of receipt of RMS software from the Government, deliver subcontractor administrative data in electronic format.

1.7.1.3 Correspondence

Identify all Contractor correspondence to the Government with a serial number. Prefix correspondence initiated by the Contractor's site office with S. Prefix letters initiated by the Contractor's home (main) office with H. Letters are numbered starting from 0001. (e.g., H-0001 or S-0001). The Government's letters to the Contractor will be prefixed with C.

1.7.1.4 Equipment

Contain within the Contractor's RMS database a current list of equipment

planned for use or being used on the jobsite including the most recent and planned equipment inspection dates.

1.7.1.5 Management Reporting

RMS includes a number of reports that Contractor management can use to track the status of the project. The value of these reports is reflective of the quality of the data input and is maintained in the various sections of RMS. Among these reports are Progress Payment Request worksheet, Quality Assurance/Quality Control (QA/QC) comments, Submittal Register Status, Three-Phase Control checklists.

1.7.1.6 Request For Information (RFI)

Exchange all RFIs using the built-in RFI generator and tracker in RMS.

1.7.2 Finances

1.7.2.1 Pay Activity Data

Include within the RMS database a list of pay activities that the Contractor develops in conjunction with the construction schedule. The sum of pay activities equals the total contract amount including modifications. Each pay activity must be assigned to a Contract Line Item Number (CLIN). The sum of the activities equals the amount of each CLIN. The sum of all CLINs equals the contract amount.

1.7.2.2 Payment Requests

Prepare all progress payment requests using RMS. Complete the payment request worksheet, prompt payment certification, and payment invoice in RMS. Update the work completed under the contract, measured as percent or as specific quantities, at least monthly. After the update, generate a payment request report using RMS. Submit the payment request, prompt payment certification, and payment invoice with supporting data using RMS CM. A signed paper copy of the approved payment request is also required and will govern in the event of discrepancy with the electronic version.

1.7.3 Quality Control (QC)

RMS provides a means to track implementation of the three-phase QC System, prepare daily reports, identify and track deficiencies, document progress of work, and support other Contractor QC requirements. Maintain this data on a daily basis. Entered data will automatically output to the RMS-generated daily report. Provide the Government a Contractor Quality Control (CQC) Plan within the time required in Section 01 45 04 CONTRACTOR QUALITY CONTROL. Within 7 calendar days of Government acceptance, submit an RMS update reflecting the information contained in the accepted CQC Plan: schedule, pay activities, features of work, submittal register, QC requirements, and equipment list.

1.7.3.1 Daily Contractor Quality Control (CQC) Reports

RMS includes the means to produce the Daily CQC Report. The Contractor can use other formats to record basic QC data. However, the Daily CQC Report generated by RMS must be the Contractor's official report. Summarize data from any supplemental reports by the Contractor and consolidate onto the RMS-generated Daily CQC Report. Submit daily CQC Reports as required by Section 01 45 04 CONTRACTOR QUALITY CONTROL. Electronically submit reports

to the Government within 24 hours after the date covered by the report.
Also provide the Government a signed, printed copy of the Daily CQC Report.

1.7.3.2 Deficiency Tracking

Use RMS to track deficiencies. Deficiencies identified by the Contractor will be numerically tracked using its QC punch list items. Maintain a current log of its QC punch list items in the RMS database. The Government will log the deficiencies it has identified using its QA punch list items. The Government's QA punch list items will be included in its export file to the Contractor. Regularly update the correction status of both QC and QA punch list items.

1.7.3.3 QC Requirements

Develop and maintain a complete list of QC testing and required structural and life-safety special inspections required by the International Code Council, transferred and installed property, and user training requirements in RMS. Update data on these QC requirements as work progresses and promptly provide the information to the Government via RMS.

1.7.3.4 Three-Phase Control Meetings

Maintain scheduled and actual dates and times of preparatory and initial control meetings in RMS.

1.7.3.5 Labor and Equipment Hours

Log labor and equipment exposure hours on a daily basis. The labor and equipment exposure data will be rolled up into a monthly exposure report.

1.7.3.6 Accident/Safety Reporting

The Government will issue safety comments, directions, or guidance whenever safety deficiencies are observed. The Government's safety comments will be provided via RMS CM. Regularly update the correction status of the safety comments. In addition, use RMS to advise the Government of any accidents occurring on the jobsite. A brief supplemental entry of an accident is not to be considered as a substitute for completion of mandatory reports, e.g., ENG Form 3394 and OSHA Form 300.

1.7.3.7 Features of Work

Include a complete list of the features of work in the RMS database. A feature of work is associated with multiple pay activities. However, each pay activity (see subparagraph PAYMENT ACTIVITY DATA of paragraph FINANCES) will only be linked to a single feature of work.

1.7.3.8 Hazard Analysis

Use RMS CM to develop a hazard analysis for each feature of work included in the CQC Plan. The Activity Hazard Analysis will include information required by [EM 385-1-1](#) Section 1-6.

1.7.4 Submittal Management

For Design-Bid-Build contracts, the Government will provide the initial submittal register in electronic format. For Design-Build contracts, the Designer of Record is responsible for developing the submittal register in

electronic format. Thereafter, maintain a complete list of submittals including completion of data columns. Dates when submittals are received and returned by the Government will be included. Use RMS CM to track and transmit submittals. ENG Form 4025, submittal transmittal form, and the submittal register update is produced using RMS. RMS will be used to update, store, and exchange submittal registers and transmittals. In addition to requirements stated in section 01 33 00 SUBMITTAL PROCEDURES, actual submittals are to be stored in RMS CM with hard copies also provided. Exception will be where the Contracting Officer specifies only hard copies required, where size of document cannot be saved in RMS CM, and where samples, spare parts, color boards, and full-size drawings are to be provided.

1.7.5 Schedule

Develop a construction schedule consisting of pay activities in accordance with Section 01 32 01 PROJECT SCHEDULE. Input and maintain in the RMS database the schedule either manually or by using the Standard Data Exchange Format (SDEF) (see Section 01 32 01 PROJECT SCHEDULE). Include with each pay request the updated schedule. Provide electronic copies of transmittals.

1.7.6 Import/Export of Data

RMS includes the ability to import schedule data using SDEF.

1.8 IMPLEMENTATION

Use of RMS CM as described in the preceding paragraphs is mandatory. Ensure that sufficient resources are available to maintain contract data within the RMS CM system. RMS CM is an integral part of the Contractor's management of quality control.

1.9 MONTHLY COORDINATION MEETING

Update the RMS CM database each workday. At least monthly, generate and submit a schedule update. At least 1 week prior to submittal, meet with the Contracting Officer to review the planned progress payment data submission for errors and omissions.

Make required corrections prior to Government acceptance of the export file and progress payment request. Payment requests accompanied by incomplete or incorrect data submittals will not be accepted. The Government will not process progress payments until all required corrections are processed.

1.10 NOTIFICATION OF NONCOMPLIANCE

The Contracting Officer will notify the Contractor of any detected noncompliance with the requirements of this specification. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, will be deemed sufficient

for the purpose of notification.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 45 04

CONTRACTOR QUALITY CONTROL

PART 1 GENERAL

- 1.1 RESIDENT MANAGEMENT SYSTEM (RMS)
 - 1.1.1 Help Support
- 1.2 DEFINITIONS
 - 1.2.1 Contractor's Representative (Site Superintendent or Project Manager)
 - 1.2.2 Definable Features of Work
 - 1.2.3 Quality Control (QC) System Manager
 - 1.2.4 Quality Control (QC) Staff
 - 1.2.5 Safety Officer
 - 1.2.6 Qualified Scheduler
- 1.3 REFERENCES
- 1.4 SUBMITTALS
- 1.5 WORK SCHEDULE

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

- 3.1 GENERAL
- 3.2 CONTRACTOR QUALITY CONTROL (CQC) PLAN
 - 3.2.1 General
 - 3.2.1.1 CQC Plan Resubmittal
 - 3.2.1.2 Failure
 - 3.2.2 Content of CQC Plan
 - 3.2.3 Acceptance of Plan
 - 3.2.4 Notification of Changes
- 3.3 COORDINATION MEETING
- 3.4 QUALITY CONTROL (QC) ORGANIZATION
 - 3.4.1 General
 - 3.4.2 CQC System Manager
 - 3.4.3 Qualified Scheduler
 - 3.4.4 CQC Personnel
 - 3.4.5 Construction Quality Management for Contractors (CQM-C) Training Requirement
 - 3.4.6 Registered Land Surveyor (RLS)
 - 3.4.7 Organizational Changes
- 3.5 SUBMITTALS AND DELIVERABLES
- 3.6 CONTROL
 - 3.6.1 Preparatory Phase
 - 3.6.2 Initial Phase
 - 3.6.3 Follow-up Phase
 - 3.6.4 Additional Preparatory and Initial Phases
- 3.7 TESTS

- 3.7.1 Testing Procedure
- 3.7.2 Testing Laboratories
 - 3.7.2.1 Laboratory Requirements and Validations
- 3.7.3 Samples for Contracting Officer Testing
- 3.8 COMPLETION INSPECTIONS
 - 3.8.1 Punch-out Inspection
 - 3.8.2 Pre-final Inspection
 - 3.8.3 Final Acceptance Inspection
- 3.9 DOCUMENTATION
- 3.10 NOTIFICATION OF NONCOMPLIANCE
- 3.11 SAMPLE FORMS

-- End of Section Table of Contents --

SECTION 01 45 04

CONTRACTOR QUALITY CONTROL

PART 1 GENERAL

1.1 RESIDENT MANAGEMENT SYSTEM (RMS)

Use the most currently approved version of Government-furnished RMS Contractor Mode software as indicated in Section 01 45 00 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM).

1.1.1 Help Support

Contractor should contact the following help phone numbers for assistance.

Step 1 - Call your Area Office RMS Administrator:

San Juan - 787-729-6901/787-729-6833

Step 2 - Submit a ticket via the RMS support website. Tickets can be submitted at <http://rms.usace.army.mil>.

1.2 DEFINITIONS

1.2.1 Contractor's Representative (Site Superintendent or Project Manager)

Highest level manager located onsite and responsible for site construction and related activities including quality, safety, environmental protection, and production.

1.2.2 Definable Features of Work

Construction task separate and distinct from other tasks and having separate control requirements. A definable feature of work may be identified by different trades or disciplines, or it may be work by the same trade in a different environment. An example for concrete would be formwork, placement, finish, curing.

1.2.3 Quality Control (QC) System Manager

A person assigned duties to manage Contractor's Quality Control (CQC) system. CQC System Manager must have written, delegated authority sufficient to stop non-conforming work.

1.2.4 Quality Control (QC) Staff

Persons assigned CQC functions and performing QC activities. CQC Staff members may be employees of Contractor, subcontractors, testing laboratories, product representatives; however, CQC Staff are working under direction of the CQC System Manager.

1.2.5 Safety Officer

Person assigned responsibility for site safety management. The Safety

Officer must have no other duties. See Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS.

1.2.6 Qualified Scheduler

Person assigned responsibility for the preparation of the schedule, all required schedule updates, and the preparation of all required schedule reports.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C1077	(2017) Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation
ASTM D3666	(2016) Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials
ASTM D3740	(2012a) Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
ASTM E329	(2018) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection

U.S. ARMY CORPS OF ENGINEERS (USACE)

ER 1110-1-261	(1999) Quality Assurance of Laboratory Testing Procedures
ER 1110-1-263	(1998) Chemical Data Quality Management for Hazardous, Toxic, Radioactive Waste Remedial Activities
ER 1180-1-6	(1995) Construction Quality Management

USACE publications Internet location site is:
<http://www.publications.usace.army.mil/>.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittal the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES. Submit the following preconstruction submittal items no later than 30 calendar days after Notice to Proceed:

SD-01 Preconstruction Submittals

Contractor Quality Control (CQC) Plan; G, R0

Work Schedule; G, R0

SD-07 Certificates

Laboratory Qualifications; G, R0

Letter of Authority

1.5 WORK SCHEDULE

Submit Contractor's work schedule (shifts, days, hours) to the Contracting Officer for approval. Notify the Contracting Officer 48 hours prior to any temporary changes in this schedule. Resubmit permanent changes to Initial Work Schedule.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 GENERAL

Contractor is responsible to plan and execute QC in accordance with **ER 1180-1-6**. Establish and maintain an effective QC system in compliance with the Clause INSPECTION OF CONSTRUCTION of Section 00700 CONTRACT CLAUSES in Volume 1. The QC system consists of plans, procedures, and organization necessary to produce a quality end product that complies with the contract. The QC system must cover all construction operations, both onsite and offsite activities, and be keyed to a definable features of work, construction sequence, and schedule. Project Manager/Superintendent is responsible for the quality of work and is subject to removal by the Contracting Officer for non-compliance with contract quality requirements. Project Manager/Superintendent must be onsite at all times except as otherwise approved by the Contracting Officer.

3.2 CONTRACTOR QUALITY CONTROL (CQC) PLAN

Upon receiving Notice to Proceed, prepare a CQC Plan specific to project organization, site, and features of work. Describe proposed procedures to implement requirements of Clause INSPECTION OF CONSTRUCTION of Section 00700 CONTRACT CLAUSES in Volume 1. After a Preconstruction Conference, the Contracting Officer will schedule a Coordination Meeting (see paragraph COORDINATION MEETING below). A CQC Plan will be discussed in detail by onsite personnel from Contractor and Contracting Officer. Contracting Officer may accept an Interim CQC Plan under a conditional acceptance for the first 30 days of operation when CQC Plan first applicable definable features of work are acceptable. Submit no later than 30 calendar days after commencement of work an acceptable overall CQC Plan.

3.2.1 General

No later than 30 calendar days after Notice to Proceed, submit a written CQC Plan for review and acceptance by Contracting Officer. CQC Plan submittal will be reviewed by the Contracting Officer and discussed in detail at the Coordination Meeting. See paragraph COORDINATION MEETING below. Fully describe proposed procedures to implement requirements of

Clause INSPECTION OF CONSTRUCTION of Section 00700 CONTRACT CLAUSES in Volume 1. A CQC Plan must identify personnel, procedures, control, instructions, tests, records, and forms to be used. Contracting Officer may accept an Interim CQC Plan under a conditional acceptance for the first 30 calendar days of operation when construction QC planning for first definable features of work are acceptable. Submit no later than 30 calendar days after commencement of work an acceptable overall CQC Plan.

3.2.1.1 CQC Plan Resubmittal

No construction will be allowed to start until an Interim CQC Plan is conditionally accepted. When an Interim CQC Plan is conditionally accepted, revise and resubmit an overall project CQC Plan for Contracting Officer's acceptance. When the Contractor is working under an Interim CQC Plan, until the Contractor submits an acceptable final CQC Plan, the Contracting Officer will retain funds from progress payments in accordance with Clause PAYMENTS UNDER FIXED-PRICE CONSTRUCTION CONTRACTS of Section 00700 CONTRACT CLAUSES in Volume 1. When no acceptable CQC Plan is resubmitted within a reasonable time as determined by the Contracting Officer, the Contracting Officer may order the Contractor to stop work until such time as a CQC Plan is accepted. Such a directed stop work order must not be considered a suspension of work under Clause SUSPENSION OF WORK of Section 00700 CONTRACT CLAUSES in Volume 1. No pay or construction period adjustments will be allowed as a result of a directed stop work order based on the Contractor's inability to plan QC in a manner acceptable to the Contracting Officer.

3.2.1.2 Failure

Failure to comply with above requirements within the time prescribed will be considered a condition endangering contract performance and may be considered grounds for termination of the contract in accordance with Clause DEFAULT (FIXED-PRICE CONSTRUCTION) of Section 00700 CONTRACT CLAUSES in Volume 1.

3.2.2 Content of CQC Plan

The CQC Plan must cover all construction operations, both onsite and offsite, including work by subcontractors, fabricators, suppliers, and purchasing agents:

- a. Describe QC Organization: Include an Organization Chart with lines of authority and reporting. CQC Staff must include a CQC System Manager who must perform their duties. For CQC matters, the CQC System Manager must directly report to the Contractor other than the Project Manager/Site Superintendent.
- b. Definable Features of Work: Although each section of specifications may generally be considered as a definable feature of work, there are frequently more than one definable feature under a particular section. A definable feature of work list will be agreed upon during the Coordination Meeting.
- c. CQC Staff Qualifications: Names, qualifications (in resume format), duties, responsibilities, and authorities of each person assigned a CQC function. If included, see paragraph LIMITATIONS ON SUBSTITUTIONS FOR CERTAIN POSITIONS AND/OR SUBCONTRACTORS of Section 00800 SPECIAL REQUIREMENTS in Volume 1.

- d. **Letter of Authority:** Letter to CQC System Manager signed by an authorized Contractor official that describes responsibilities and delegates sufficient authorities to perform functions of the CQC System Manager including the authority to stop work not in compliance with contract.
- e. **Submittal Control:** Procedures for scheduling, reviewing, certifying, and managing submittals including submittal items from subcontractors, offsite fabricators, suppliers, and purchasing agents. Procedures must be in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.
- f. **Testing:** Control, verification, and acceptance testing procedures. Provide a list of specific tests. Provide information including test type or testing standard, specification paragraph requiring test, feature of work being tested, test frequency, and identify who (i.e., Contractor, subcontractor, Testing Laboratory) is responsible for each test (Laboratory facilities must be approved by the Contracting Officer).
- g. **Three-Phase Control:** Procedures to implement a three-phase QC system. Procedures to plan and document preparatory, initial, and follow-up control phases.
- h. **Deficiency Tracking:** Procedures for tracking construction deficiencies from identification through acceptable corrective action. Establish procedures that verify deficiencies have been corrected and document correction.
- i. **Reports and Forms:** Reporting procedures including proposed reporting formats and sample forms.
- j. **Table of Contents:** Number pages of the QC Plan including appendices. Provide a table of contents. Index sections of the plan and all appendices in the table of contents.

3.2.3 Acceptance of Plan

Conditional acceptance of a Contractor's Interim CQC Plan is required prior to the start of construction. Within a specified period after commencement of work, Contractor's overall CQC Plan requires Contracting Officer's acceptance, or work is subject to a stop work directive. Contracting Officer's acceptance is conditional and is contingent on satisfactory performance during the construction. Contracting Officer reserves the right to require the Contractor to make changes in their CQC Plan and construction operations, including the removal of personnel, in order to obtain required contract quality.

3.2.4 Notification of Changes

Submit to the Contracting Officer in writing a minimum of 30 calendar days prior to CQC Plan procedure changes. Proposed changes are subject to Contracting Officer acceptance.

3.3 COORDINATION MEETING

Refer to Section 01 30 00 ADMINISTRATIVE PROCEDURES. Contracting Officer will schedule a Coordination Meeting where the Contractor, CQC Staff, and Contracting Officer will develop a mutual understanding of the Contractor's CQC Plan with Contracting Officer's Quality Assurance. The

CQC Plan will be discussed in detail including forms for recording CQC operations, control activities, testing, and administration of the system for both onsite and offsite work. CQC both onsite and offsite, safety and environmental protection, supervision by QC personnel, and the joint Risk Management plan and register will be discussed. Meeting minutes will be prepared by Contracting Officer and signed by both parties. Minutes will become part of contract files. There may be occasions when subsequent conferences will be called by either party to reconfirm mutual understandings or address deficiencies in the CQC system or procedures requiring corrective action.

3.4 QUALITY CONTROL (QC) ORGANIZATION

3.4.1 General

CQC Organization must have a CQC System Manager and a sufficient number of additional qualified personnel to ensure contract QC for workmanship and materials including safety and environmental protection compliance. Designate a Safety Officer who must serve as a member of CQC Staff. Personnel identified in other contract sections as requiring specialized skills to assure work is performed properly must also be included as part of CQC Organization. CQC Staff must be present onsite during work and have authority and responsibility to ensure contract compliance. CQC Staff must be subject to acceptance by Contracting Officer. Provide office space, filing systems, and other resources as necessary to maintain an effective and functional CQC Organization. Promptly give complete records of letters, transmitted submittal items, shop drawings, progress schedules, changes to contract drawings, test results, and other project documentation to CQC Organization. CQC Organization must maintain these documents and records onsite unless otherwise approved by Contracting Officer.

3.4.2 CQC System Manager

Appoint, as CQC System Manager, an individual within the onsite work organization who is responsible for overall management of CQC and has the authority to act in all CQC matters for the Contractor. The CQC System Manager is required to have a degree in engineering and be a Registered Professional Engineer licensed in the Commonwealth of Puerto Rico with a minimum of 5 years construction experience on construction similar to this contract as approved by the Contracting Officer. This CQC System Manager or Alternate must be on site at all times during active construction activities and is employed by the prime Contractor. The CQC System Manager must NOT be assigned other duties. Identify in the QC Plan an Alternate CQC System Manager to serve in the event of the CQC System Manager's absence. The Alternate must not perform other duties when performing CQC activities. The requirements for the Alternate are the same as the CQC System Manager. CQC System Manager and Alternate must meet the Construction Quality Management for Contractors (CQM-C) training listed below.

3.4.3 Qualified Scheduler

See Section 01 32 01 PROJECT SCHEDULE.

3.4.4 CQC Personnel

In addition to CQC personnel specified elsewhere in the contract, provide as part of CQC Organization specialized personnel to assist the CQC System

Manager for the following areas: Safety, including confined space, electrical, mechanical, civil, structural, geotechnical, hazardous materials, including lead paint or asbestos abatement site competent persons, environmental, architectural, materials technician, and submittals clerk. Individuals may be employees of the Contractor, Supplier Product Representative, or subcontractor employee as needed, be responsible to the CQC System Manager, be physically present at the construction site during work on their areas of responsibility, and have the necessary education and experience in accordance with the experience matrix listed herein. These individuals may perform other duties but must be allowed sufficient time to perform their assigned QC duties as described in the CQC Plan. In addition to any other qualification or requirement, all additional CQC personnel must take the CQM-C training within 6 months of appointment.

Experience Matrix

	Area	Qualifications
a.	Surveyor	Refer to paragraph 3.4.6 for Surveyor requirements.
b.	Civil	Graduate Civil Engineer with 2 yrs related experience.
c.	Mechanical	Graduate Mechanical Engineer with 5 yrs experience.
d.	Electrical	Graduate Electrical Engineer with 5 yrs related experience.
e.	Structural	Graduate Structural Engineer with 10 yrs experience and 5 yrs related experience.
f.	Geotechnical Engineer	Graduate Geotechnical Engineer with 10 yrs experience and 5 yrs related experience.
g.	Architectural	Graduate Architect with 2 yrs experience.
h.	Environmental	Graduate Environmental Engineer with 2 yrs experience.
i.	Submittals	Submittal Clerk with 1 yr experience.
f.	Concrete, Pavements, and Soils	Materials Technician with 2 yrs experience for the appropriate area.
k.	Testing, Adjusting, and Balancing Personnel	Specialist must be a member of AABC or an experienced

technician of the firm
certified by the NEBB.

3.4.5 Construction Quality Management for Contractors (CQM-C) Training Requirement

The CQC System Manager and Alternate must have completed a U.S. Army Corps of Engineers (USACE) course CQM-C within the previous 5 years. A completion certificate from any USACE District or Naval Facilities Command is acceptable. In the event the proposed CQC System Manager has not completed CQM-C training, they must complete this course prior to commencement of work. Do not commence work until these training requirements have been satisfied. Any delays attributable to the Contractor's failure to comply with these pre-work requirements must be at the Contractor's expense and may be cause for remedial action by the Contracting Officer. CQM-C training is periodically offered by Caribbean District. Information regarding the course can be obtained by contacting the Caribbean District Construction, Technical Support Branch at jorge.l.ayalaburgos@usace.army.mil. Supporting CQC personnel must complete CQM-C training within 6 months after appointment.

3.4.6 Registered Land Surveyor (RLS)

Contract requires survey QC performed by an RLS registered in the Commonwealth of Puerto Rico with a minimum of 3 years of experience. This person must directly supervise a layout and post-construction survey. RLS must stamp field notes, computations, and other records relating to surveys and layout of work.

3.4.7 Organizational Changes

When CQC staff changes are needed, revise the CQC Organization Chart in the CQC Plan to reflect changes and submit changes to Contracting Officer for acceptance.

3.5 SUBMITTALS AND DELIVERABLES

Prepare and transmit submittals as specified in Section 01 33 00 SUBMITTAL PROCEDURES. The CQC Organization must certify submittals comply with contract requirements. Control, package, transport, and store items delivered to the Contracting Officer in a manner to prevent damage or loss. Control deliverables including QC documentation, invoices, and correspondence to prevent loss or delays.

3.6 CONTROL

CQC is the means by which the Contractor ensures that construction, including that of subcontractors and suppliers, complies with the contract. Conduct Preparatory Phase and Initial Phase meetings for each definable feature of work (refer to Section 01 30 00 ADMINISTRATIVE PROCEDURES). At least three phases of control must be conducted by the CQC System Manager for each definable feature of the construction work as follows:

3.6.1 Preparatory Phase

Preparatory Phase must be performed prior to beginning work on each definable feature of work. Notify the Contracting Officer at least 24 hours in advance of beginning preparatory control phase. Ensure proposed

plans, activity hazard analyses, permits, and submittals are approved, and copies are onsite. Conduct a Preparatory Phase meeting headed by the CQC System Manager and attended by the Superintendent, other CQC personnel, and foremen responsible for supervising workmanship for definable feature of work. Document Preparatory Phase actions using Preparatory Phase Checklist and meeting minutes prepared by the CQC System Manager. Attach the checklist and minutes to the CQC Report (a sample CQC form is on any of the web sites indicated at the end of this Section). Preparatory Phase actions include:

- a. Review each paragraph of specifications, reference codes, and standards. Review copies of referenced codes and standards applicable to work to be accomplished. Make copies available for use by Contracting Officer personnel and CQC staff at the Preparatory Phase meeting. Maintain copies available until final acceptance of work.
- b. Review of contract drawings.
- c. Check to assure that all materials and equipment have been tested, submitted, and approved.
- d. Review provisions that have been made to provide required control inspection and testing.
- e. Examine the work area to assure required preliminary work is complete and in compliance with contract.
- f. Inspect materials, equipment, and sample work to assure that they are on hand, conform to approved or information only submittal items, and are properly stored.
- g. Review of activity hazard analysis to assure safety requirements are met.
- h. Discuss procedures for controlling quality of the work including preventing repetitive deficiencies. Document construction tolerances and workmanship standards for that feature of work.
- i. Check to ensure that portion of plan for work to be performed has been accepted by Contracting Officer.
- j. Discuss initial control phase.

3.6.2 Initial Phase

Notify the Contracting Officer at least 24 hours in advance of beginning the Initial Phase. The Initial Phase is workmanship oriented and must be accomplished at the beginning of physical work on a definable feature of work. CQC Staff, testing personnel, foremen, and workers must attend an Initial Phase Meeting conducted by CQC System Manager. Indicate the exact location of Initial Phase actions for future reference and comparison with follow-up phases. Document the Initial Phase Meeting using an Initial Phase Checklist and minutes prepared by the CQC System Manager. Attach checklist and minutes to CQC Report (sample CQC form is on any of the web sites indicated at the end of this Section). Initial Phase actions include:

- a. Check preliminary work to ensure that it complies with contract. Review minutes of the preparatory meeting.

- b. Verify adequacy of controls to ensure full contract compliance. Verify required control inspection and testing.
- c. Establish level of workmanship and verify that it meets minimum acceptable contract workmanship standards. Compare with required sample panels as appropriate.
- d. Resolve all differences.
- e. Check safety to include compliance with and upgrading of the safety plan and activity hazard analysis. Review activity hazard analysis with each worker.
- f. The Initial Phase must be repeated for new crews working onsite and when contract workmanship quality standards are not being met.

3.6.3 Follow-up Phase

The Follow-up Phase consists of daily checks performed to assure QC activities, including testing, to provide continued compliance with contract requirements until the feature of work is complete. Record inspection and check results in CQC documentation. Complete follow-up checks and inspections and correct deficiencies prior to starting features of work that may be affected by deficient work. No nonconforming work must be concealed to build upon.

3.6.4 Additional Preparatory and Initial Phases

Conduct additional Preparatory and Initial Phases on the same definable features of work when quality of ongoing work is unacceptable, there are changes in applicable CQC Staff, production supervision, or work crews, work on a definable feature is resumed after a period of inactivity, or when other problems develop. The Contracting Officer may direct additional phase meetings as needed to assure contract compliance.

3.7 TESTS

3.7.1 Testing Procedure

Perform specified or required tests to verify that control measures are adequate and provide an end product conforming to contract. When requested, furnish the Contracting Officer duplicate samples of test specimens for possible testing by the Contracting Officer. Testing includes operation and acceptance tests when specified. Procure the services of a USACE-approved testing laboratory or establish an approved testing laboratory at the project site. Perform the following activities and record and provide the following data:

- a. Verify that testing procedures comply with contract requirements.
- b. Verify that facilities and testing equipment are available and comply with testing standards.
- c. Check test instrument calibration data against certified standards.
- d. Verify that recording forms and test identification control number system, including all of the test documentation requirements, have been prepared.

- e. Record results of all tests taken, both passing and failing tests, on the CQC Report for the date taken. Results of tests must include specification paragraph reference, location where tests were taken, and the sequential control number identifying the test. For soil density tests, test results must include dry density found by test, moisture content found by test, proctor optimum water content, proctor maximum density, and proctor reference number. Provide an information copy of tests performed by an offsite or commercial test facility directly to the Contracting Officer. Failure to submit timely test reports as stated may result in nonpayment for related work performed and disapproval of the test facility for this contract.

3.7.2 Testing Laboratories

The Contractor's proposed testing labs require Contracting Officer approval. Submit [laboratory qualifications](#) including:

- a. Laboratory name, address, point of contact, and phone number.
- b. Experience and background of technical personnel in resume format.
- c. The most recent external inspection report.

3.7.2.1 Laboratory Requirements and Validations

- a. **Materials Testing Labs:** Provide material testing laboratories validated by the USACE Materials Testing Center in accordance with [ER 1110-1-261](#). Labs used for testing soils, concrete, asphalt, and steel must meet criteria detailed in [ASTM C1077](#), [ASTM D3666](#), [ASTM D3740](#), and [ASTM E329](#). A list of currently USACE-validated material laboratories is available by accessing the Internet website <https://mtc.erdc.dren.mil/searchvalidation.aspx> and clicking the link Validated Laboratories. If the Contractor proposes a material testing lab not currently validated by USACE, the Contractor must pay costs for USACE lab inspection and validation. Cost will be deducted from contract amount due to the Contractor. Inspection and validation for material testing labs, required onsite inspection, and audit of QC records costs is \$6,500 per lab. The validation process requires labs to describe deficiency correction and respond within 30 days after receiving inspection report. Reinspection costs will be deducted from the contract.
- b. **Onsite Laboratory:** Onsite laboratory must meet inspection validation requirements outlined above. The Contracting Officer reserves the right to use the Contractor's onsite testing lab and equipment to make quality assurance tests, check the Contractor's testing procedures and techniques, and obtain test results at no additional cost to the Contracting Officer.
- c. **Water Quality Labs:** Water testing labs usually only require QC record inspection to assure capability to perform contract tests in accordance with testing standards listed in Appendix A of [ER 1110-1-261](#). The Contracting Officer reserves right to inspect the lab, equipment, and personnel qualifications in proposed water quality labs for compliance with standards specified in contract specifications. An inspection will include checking lab technician's testing procedures and techniques. The Contracting Officer will pay for the first inspection; reinspection costs will be deducted from contract.

- d. Environmental Testing Labs: Environmental Testing Labs for HTRW projects are required to have National Environmental Laboratory Accreditation Conference/Program and will be inspected for Chemical Data Quality Management for HTRW remedial activities as described in ER 1110-1-263. Omaha District - HTRW Technical Center of Expertise inspections are only done by Contracting Officer request on active contracts. The Contracting Officer will pay for the first inspection; reinspection costs will deducted from contract.

3.7.3 Samples for Contracting Officer Testing

Costs incidental to obtaining and transporting samples and materials for Contracting Officer testing are the Contractor's responsibility. Samples of materials for test verification and acceptance testing by the Contracting Officer will be delivered to a lab designated by the Contracting Officer. Coordinate obtaining specific samples and delivery location with the Contracting Officer. Routine soil, concrete, and asphalt testing is usually done within Puerto Rico. Special construction products or components are usually sent to a USACE lab located in Vicksburg, Mississippi or Champaign, Illinois. Caribbean District chemical analysis, rock petrographic, and metallurgy samples are usually sent to Atlanta, Georgia or Jacksonville, Florida for testing.

3.8 COMPLETION INSPECTIONS

3.8.1 Punch-out Inspection

Near end of project or phase of work established for beneficial occupancy, conduct a Punch-out Inspection jointly inspecting completed work with subcontractors. Contractors shall develop a punch list of work that does not conform to contract. Provide the punch list to the Contracting Officer. The punch list must include estimated date by which deficiencies will be corrected. Make a second inspection to ensure deficiencies have been corrected. Once this is accomplished, notify the Contracting Officer that work is ready for Contracting Officer Pre-final Inspection.

3.8.2 Pre-final Inspection

The Contracting Officer will jointly perform a Pre-final Inspection with the Contractor to verify work is complete and ready for acceptance or occupancy. Notify the Contracting Officer in writing when ready for Pre-Final Inspection. The Contracting Officer will schedule a Pre-Final Inspection within 14 working days of Contractor's notification. New punch list items may be developed as a result of Pre-final Inspection. Ensure items on this punch list have been corrected before notifying the Contracting Officer to schedule a Final Inspection with the Owner. Correct items on a Pre-final punch list in a timely manner. Complete inspections and correct any deficiencies within the construction period for completion of work or for a particular phase of work when the contract has separate completion dates.

3.8.3 Final Acceptance Inspection

The Contractor, Project Manager/Superintendent, and CQC System Manager must attend the Final Inspection. The Contracting Officer and additional persons including, but not limited to, those from Civil Facility Engineer user groups, Sponsor, User, Customer or Owner, and other agencies may also attend. The Final Inspection will be scheduled by the Contracting Officer based upon results of the Pre-Final Inspection. Notify the Contracting

Officer in writing when ready for the Final Inspection and propose a date for the Final Inspection. Include a statement in the notification assuring the Contracting Officer that deficiencies will be corrected and work will be acceptable by the proposed Final Inspection date. The Contracting Officer will schedule a Final Inspection within 14 working days of the Contractor's notification. Contractor's failure to correct deficiencies and have work complete for the Final Inspection will be cause for the Contracting Officer to deduct from the Contractor for Contracting Officer's additional reinspection cost from contract payment in accordance with Clause INSPECTION OF CONSTRUCTION of Section 00700 CONTRACT CLAUSES.

3.9 DOCUMENTATION

Maintain records for each construction day documenting QC activities that have been performed. Records must include testing record and work of subcontractors and suppliers. Prepare a CQC Report with supporting attachments daily on an acceptable form that includes the following information:

- a. Contractor/subcontractor and their area of responsibility.
- b. Equipment with hours worked, idle, or down for repair.
- c. Work performed each day giving location, description, and by whom. When Network Analysis (NAS) is used, identify each phase of work performed each day by NAS activity number (refer to Section 01 32 01 PROJECT SCHEDULE).
- d. Testing and control activities performed with results and references to specifications or drawings requirements. Identify control phase (Preparatory, Initial, and Follow-up). List deficiencies noted along with corrective action.
- e. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications or drawings requirements.
- f. Submittals and deliverables reviewed with contract reference, by whom, and action taken.
- g. Offsite surveillance activities including actions taken.
- h. Job safety evaluations stating what was checked, the results, and instructions or corrective actions.
- i. Instructions given or received and conflicts in plans or specifications.
- j. Contractor's verification statement.

Describe trades working on the project, the number of personnel working, weather conditions encountered, and delays encountered. Records must cover both conforming and deficient work and include a statement that equipment and materials incorporated into work and workmanship comply with the contract. Provide the original records to the Contracting Officer daily within 24 hours after the date covered by the report. A daily CQC Report is required for each contract day including days on which no work is performed. On days when no work is performed, the daily report must document weather and note possible effects on restarting work. Account for

all contract calendar days. The CQC System Manager or Alternate must sign and date reports. The CQC System Manager's daily reports must include copies of test reports, meeting minutes, safety inspections and issues, inspector notes, copies of reports prepared by other QC personnel, minutes to Preparatory Phase meetings held, minutes to Initial Phase meetings held, Final Follow-up inspection results, and any other documentation of construction progress or issues. The daily reports must also include a record of the Follow-up Phase checks performed on each definable feature of work each day, problems discovered, and corrective actions taken. The Contracting Officer may stipulate the narrative titles in the daily CQC Reports.

3.10 NOTIFICATION OF NONCOMPLIANCE

The Contracting Officer will notify the Contractor of noncompliance with contract requirements. Take corrective action immediately after receipt of noncompliance notification. Any Contractor personnel notified at the work site is sufficient for the purpose of Contractor notification. If the Contractor fails to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. Such stop orders must not be made a basis of a Contractor's claim for time extension or other damages.

3.11 SAMPLE FORMS

See paragraph RESIDENT MANAGEMENT SYSTEM (RMS) above for construction forms to be completed in the RMS program. Forms are available for Contractor use at the following web address:

<http://www.publications.usace.army.mil/>

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 50 02

TEMPORARY CONSTRUCTION FACILITIES

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 REFERENCES
- 1.3 SUBMITTALS
- 1.4 PRECONSTRUCTION REQUIREMENTS
 - 1.4.1 Mobilization/Demobilization Plan
 - 1.4.2 Security Plan
 - 1.4.3 Vegetation Protection and Restoration Plan
 - 1.4.4 Documentation of Existing Conditions and Post-construction Restoration
- 1.5 EXISTING UTILITIES
 - 1.5.1 Water
 - 1.5.2 Electricity
- 1.6 WORK AND ACCESS DATA
 - 1.6.1 Staging and Access Areas
 - 1.6.2 Contractor Responsibilities
- 1.7 ADJACENT PROPERTY AND STRUCTURES

PART 2 PRODUCTS

- 2.1 Contractor's Temporary Facilities
- 2.2 TEMPORARY SIGNAGE
 - 2.2.1 Bulletin Board
 - 2.2.2 Project Identification and Safety Signs
 - 2.2.3 Construction Forms and Details
 - 2.2.4 Warning Signs
 - 2.2.5 Haul Road Traffic and Construction Warning Signs
- 2.3 TEMPORARY TRAFFIC CONTROL
 - 2.3.1 Haul Roads
 - 2.3.2 Barricades
- 2.4 TEMPORARY FENCING
 - 2.4.1 Chain Link Panel Fencing
 - 2.4.2 Post-Driven Chain Link Fencing
- 2.5 TEMPORARY WIRING
- 2.6 STORAGE CONTAINERS

PART 3 EXECUTION

- 3.1 GENERAL REQUIREMENTS
 - 3.1.1 Identification of Employees
 - 3.1.2 Employee Parking
 - 3.1.3 Onsite Information
 - 3.1.4 Janitorial Services
- 3.2 AVAILABILITY AND USE OF UTILITY SERVICES
 - 3.2.1 Lighting

- 3.2.2 Heating and Cooling
- 3.2.3 Water
- 3.2.4 Fire Extinguisher
- 3.2.5 Utility Lines
- 3.2.6 Electric Power
- 3.3 PROTECTION AND MAINTENANCE OF TRAFFIC
- 3.4 MAINTENANCE OF HAUL AND ACCESS ROADS
- 3.5 CONTRACTOR'S TEMPORARY FACILITIES
 - 3.5.1 Contractor Field Office
 - 3.5.2 Appearance of Trailer(s)
 - 3.5.3 Storage Area
 - 3.5.3.1 Maintenance of Storage Area
 - 3.5.4 Waste Storage
 - 3.5.5 Fuel Storage and Fueling Operations
- 3.6 SECURITY PLAN
 - 3.6.1 Visitors
- 3.7 CLEANUP
- 3.8 RESTORATION OF STORAGE AREA

-- End of Section Table of Contents --

SECTION 01 50 02

TEMPORARY CONSTRUCTION FACILITIES

PART 1 GENERAL

1.1 SUMMARY

In addition to Construction Facilities this Section covers:

Temporary Utilities
Construction Aids
Vehicular Access and Parking
Project Identification

See Section 01 57 20 ENVIRONMENTAL PROTECTION for requirements including silt control, trailer placement, fueling restrictions, dust control, solid waste, and cleanup. Upon completion of project, clean up and restore the area in accordance with Clause CLEANING UP of Section 00700 CONTRACT CLAUSES in Volume 1.

a. Construction facilities include, but are not limited to, the following:

- (1) Contractor Offices
- (2) Information Bulletin Board
- (3) Material and Equipment Storage Area
- (4) Fueling Area
- (5) Secured Storage Area
- (6) Employee Parking Area
- (7) Debris Container (dumpster)
- (8) Construction Signage to include Project Sign, Safety Sign, and Construction Warning Signs

b. Temporary Utilities include, but are not limited to, the following:

- (1) Water
- (2) Electric
- (3) Sewage
- (4) Communications
 - (i) Telephone
 - (ii) Internet (10 Mbps or higher speed)

(5) Lighting

1.2 REFERENCES

The publications listed below form a part of this specification to extent referenced. The publications are referred to in text by basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C2 (2023) National Electrical Safety Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 4 2023; ERTA 5 2023; ERTA 6 2023) National Electrical Code

NFPA 241 (2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health (SOH) Requirements

PUERTO RICO HIGHWAY AND TRANSPORTATION AUTHORITY (PRHTA)

PRHTA Specifications (2005) Standard Specifications for Road and Bridge Construction

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Mobilization/Demobilization Plan; G, R0

Security Plan; G, R0

Vegetation Protection and Restoration Plan; G, D0

Documentation Of Existing Conditions and Post-construction Restoration; G, R0

SD-02 Shop Drawings

Contractor's Temporary Facilities

Temporary Electric Drawings

1.4 PRECONSTRUCTION REQUIREMENTS

Submit the following preconstruction submittal items no later than 30 calendar days after Notice to Proceed (NTP).

1.4.1 Mobilization/Demobilization Plan

Plan must include, but not be limited to, the following:

- a. Mobilization Requirements:
 - (1) Methods, equipment, and materials
 - (2) Connection of utilities
 - (3) Placement of site facilities and temporary controls
 - (4) Construction of facilities
- b. Demobilization Requirements (methods, equipment, and materials required to clean-up and restore site at project conclusion):
 - (1) Collection, recycling, and disposal of solid waste
 - (2) Contract-generated material
 - (3) Utility disconnection
 - (4) Removal of Contractor facilities
 - (5) Repair and restoration of site (i.e., fences, roads, or permanent facilities)

1.4.2 Security Plan

Prepare a Security Plan for the Contracting Officer describing site security as follows:

- a. Day and night security
- b. Weekend and holiday security
- c. General security duties

1.4.3 Vegetation Protection and Restoration Plan

Prior to clearing for access, submit a Vegetation Protection and Restoration Plan describing the methods to be used for the protection of vegetation that is not to be disturbed or otherwise impacted. Include restoration measures for vegetation cleared from access areas and in the event of the accidental damage to, or destruction of, said vegetation.

1.4.4 Documentation of Existing Conditions and Post-construction Restoration

Before beginning any mobilization or work, survey the site and photo document the work area including all staging areas, haul routes, parking areas, and vegetation to show existing limits that could potentially be impacted. Include in the record a description of each picture taken as well as the conditions of roadways, utilities, and other site features and the location and extent of existing damage prior to starting work. It is the Contractor's responsibility to document all pertinent existing conditions that may affect the subject property.

Provide the following at all access sites and staging areas to be used:

- a. Photodocument the condition of all such areas prior to disrupting the site. Include a description of each picture taken.
- b. Limit access width through existing vegetation to 20 feet or less.
- c. Replace any fencing, signage, curbing, or any other structural features disturbed by the Contractor's activities.
- d. Restore and revegetate the access route with native dune plants to the preexisting conditions, subject to the approval of the Contracting Officer. Provide revegetation of access and staging areas with sod (non-dune areas) or viable plant units (dune areas) at 18-inch maximum spacing with species and diversity equivalent to preexisting conditions. Include a survival warranty of 90 percent of the plant material for 90 days for revegetation. Install vegetation with fertilization and irrigation or with initial irrigation, fertilization, and approved water-absorbent polymeric gels at no additional expense to the Government. Replace shrubs and trees to preexisting conditions per the requirements of Section 01 57 20 ENVIRONMENTAL PROTECTION.

1.5 EXISTING UTILITIES

Investigate the availability of utilities at the project site. Provide potable water, sanitary sewer, electricity from transformer, communications, and optional utilities required at the project site. Provide internet services with 10 Mbps minimum speed and wireless capabilities to the Contractor's Field Office complex to include Government office space.

1.5.1 Water

In addition to the above, provide and maintain an adequate supply of water for construction use at no additional cost to the Government. Install and maintain necessary supply connections and piping for same but only at such locations and in such a manner as may be approved by the Contracting Officer. Provide and maintain temporary toilet and washing facilities. Install and maintain toilet and washing facilities in a location approved by the Contracting Officer. Refer to paragraph AVAILABILITY AND USE OF UTILITY SERVICES below.

1.5.2 Electricity

In addition to the above, furnish all electric current at no additional cost to the Government. Install, connect, and maintain all temporary lines in a workmanlike manner satisfactory to the Contracting Officer. Remove temporary lines in like manner at no additional cost to the Government prior to completion of the construction. Refer to paragraph AVAILABILITY AND USE OF UTILITY SERVICES below.

1.6 WORK AND ACCESS DATA

See paragraph WORK RESTRICTIONS in Section 01 11 00 SUMMARY OF WORK.

Staging and Access Area Point of Contact:
To be determined after award.

Right-of-Way Permit Point of Contact:
To be determined after award.

1.6.1 Staging and Access Areas

Obtain and coordinate staging areas and submit for Contracting Officer's approval. Access areas are shown on the contract drawings for the Contractor's use. The Contracting Officer will field determine the final limits of the access areas indicated on the drawings in coordination with the local sponsor and the Contractor. Investigate and obtain any additional areas that may be necessary for construction operations at the Contractor's expense. The additional areas will be subject to the approval of the Contracting Officer.

1.6.2 Contractor Responsibilities

Exclude the public from the work area in the immediate vicinity of operations including, but not limited to, fill placement, grading, and transport. Minimize areas closed to the public. Install warning signs to warn the public and all commercial recreational boats of all construction activities. Provide and maintain all water and land access routes necessary for equipment and plant to and from the work sites. Ascertain the environmental conditions that can affect access such as climate, terrain, and winds.

1.7 ADJACENT PROPERTY AND STRUCTURES

Immediately repair any damage to private or public property within the project boundaries including staging site(s) and work and access areas/roads. Repair any damage as a result of the Contractor's operations at no cost to the Government.

PART 2 PRODUCTS

2.1 Contractor's Temporary Facilities

General layout sketch of temporary site facilities must include, but not be limited to, the following:

- a. Trailer locations
- b. Parking areas
- c. Material storage
- d. Equipment lay-down area
- e. Areas for gravel
- f. Fuel areas
- g. Supplemental or other staging area
- h. Temporary well, water supply
- i. Septic field or holding tanks, port-a-lets
- j. Contaminated water handling

- k. Concrete cleaning area and methods of disposal
- l. Fences -- location and dimensions, entrance and exit points, and details of installation
- m. Utilities (poles, connections, and services)

2.2 TEMPORARY SIGNAGE

2.2.1 Bulletin Board

Prior to commencement of work activities, provide a clear, weatherproof-covered bulletin board not less than 915 by 1,220 mm (36 by 48 inches) in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the contract, Wage Rate Information poster, WH1090 Paid Sick Leave for Contractors poster (<https://www.dol.gov/whd/regs/compliance/wh1090.pdf>), Safety and Health Information as required by EM 385-1-1 Chapter 1 and other information approved by the Contracting Officer. Display legible copies of the aforementioned data until work is completed. Remove the bulletin board upon completion of work. The bulletin board will remain the property of the Contractor. Coordinate requirements herein with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS.

2.2.2 Project Identification and Safety Signs

Provide signs, their content, and location as indicated on the web site provided in paragraph "Construction Forms and Details" below. Erect the signs within 15 days after receipt of the NTP. Correct the data required by the safety sign with light-colored metallic or non-metallic numerals. Upon completion of the project, remove the signs from the site.

2.2.3 Construction Forms and Details

See the CONSTRUCTION FORMS & DETAILS module at the following web address: <https://www.saj.usace.army.mil/About/Divisions-Offices/Engineering/Design-Branch/Specifications-Section/>

2.2.4 Warning Signs

Post temporary signs, tags, and labels to give workers and the public adequate warning and caution of construction hazards according to the EM 385-1-1 Chapter 4. Attach signs to the perimeter fencing every 150 feet or as determined by the Contracting Officer warning the public of the presence of construction hazards. Install signs requiring unauthorized persons to keep out of the construction site. Update the data required by the safety signs daily.

2.2.5 Haul Road Traffic and Construction Warning Signs

Provide 1/2-inch HDO plywood or aluminum sheet conforming to PRHTA Specifications Series 613, Traffic Signs, non-standard signage high visibility orange with black lettering. Orange fabric mesh acceptable for temporary survey crew use.

2.3 TEMPORARY TRAFFIC CONTROL

During construction, provide access and temporarily relocate roads as necessary to maintain traffic. Maintain and protect traffic on all

affected roads during the construction period. Install measures for the protection and diversion of traffic including the provision of watchmen and flagmen, erection of barricades, placement of lights around and in front of equipment and the work, and the erection and maintenance of adequate warning, danger, and direction signs as required by the Commonwealth and local authorities having jurisdiction. Protect the traveling public from damage to person and property. Limit the interface between the Contractor's traffic on roads selected for hauling material to and from the site with public traffic. Investigate the adequacy of existing roads and the allowable load limit on these roads. Repair any damage to roads caused by construction operations.

2.3.1 Haul Roads

Construct access and haul roads necessary for proper prosecution of the work under this contract in accordance with [EM 385-1-1](#) Chapter 4. Construct with suitable grades and widths; avoid sharp curves, blind corners, and dangerous cross traffic. Submit haul road plan for approval. Provide necessary lighting, signs, barricades, and distinctive markings for the safe movement of traffic. Provide a method of dust control, although optional, to ensure safe operation at all times. Location, grade, width, alignment of construction, and hauling roads area subject to approval by the Contracting Officer. Provide adequate lighting to assure full and clear visibility for full width of haul road and work areas during any night work operations. [Upon completion of the work, remove and dispose of portions of any Contractor-constructed access and haul roads. Remove road segments if they impede proper function or maintenance of the permanent project features, adversely impact local stormwater drainage, or adversely impact environmentally sensitive areas. Grade the access and haul road portions removed to approximately match the original ground contours.](#)

[Where haul roads traverse an environmentally sensitive area or if located on the crest of an earthen embankment, one-way haul roads are preferred. Haul roads built and maintained for this work must comply with the following:](#)

- a. [One-way haul roads for off-the-road equipment; e.g., belly dumps, scrapers, and off-the-road trucks must have a minimum usable width of 25 feet. One-way haul roads for over-the-road haulage equipment only \(e.g., dump trucks\) may be reduced to a usable width of 15 feet. When the Contracting Officer determines that it is impractical to obtain the required width for one-way haul roads \(e.g., a road on top of a levee\), a usable width of not less than 10 feet may be approved by the Contracting Officer provided a positive means of traffic control is implemented. Such positive means must be signs, signals, and/or a signalman and an effective means of speed control.](#)
- b. [Two-way haul roads for off-the-road haulage equipment must have a usable width of 60 feet. Two-way haul roads for over-the-road haulage equipment only may be reduced to a usable width of 30 feet.](#)
- c. [Design vertical and horizontal curvature and the sight distance in accordance with AASHTO design criteria. When conditions make it impractical to obtain the required design criteria, implement a positive means of traffic control. All requirements of \[EM 385-1-1\]\(#\) concerning access and haul roads must apply.](#)
- d. [Dust abatement must permit observation of objects on the roadway at a](#)

minimum distance of 300 feet.

- e. Haul roads must have the edges of the usable portion marked with posts at intervals of 50 feet on curves and 200 feet maximum elsewhere. Such markers must extend 6 feet above the road surface, and for nighttime haulage provide markers with reflectors in both directions.
- f. If the required haul road dimensions are not achievable due to the physical limitation of the placement area, provide alternate widths with applicable safety measures to ensure safe transit along the haul routes for review and approval by the Contracting Officer.

2.3.2 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Whenever construction activities prevent safe public access to paved areas such as roads, parking areas, or sidewalks or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic, barricades will be required. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night.

2.4 TEMPORARY FENCING

Provide fencing along the construction limits to control access by unauthorized personnel. Provide highly visible safety fencing to be seen by pedestrians and vehicular traffic. Specific fencing requirements are as described herein. All fencing will meet the requirements of **EM 385-1-1**.

2.4.1 Chain Link Panel Fencing

Temporary panel fencing must be galvanized steel chain link panels minimum 8 feet high. Multiple fencing panels may be linked together at the bases to form long spans as needed. Weigh down each panel base using sand bags or other suitable materials in order for the fencing to withstand anticipated winds while remaining upright. Fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection. Provide privacy screen along fencing areas adjacent to communities, state and public properties, and areas exposed to general public traffic.

2.4.2 Post-Driven Chain Link Fencing

Temporary post-driven fencing must be galvanized chain link fencing minimum 8 feet high supported by tightly secured galvanized steel posts driven below grade. Locate fence posts on minimum 10-foot centers. Posts may be set in various surfaces such as sand, soil, asphalt, or concrete as necessary. Chain link fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection. Completely remove fencing and posts at the completion of construction. Restore any surfaces disturbed or damaged to its original condition. Locate and identify underground utilities prior to setting fence posts. Equip fence with a lockable gate. Keep gate locked when construction personnel are not present. Provide privacy screen along fencing areas adjacent to communities, state and public properties, and areas exposed to general public traffic.

2.5 TEMPORARY WIRING

Provide temporary wiring in accordance with EM 385-1-1 Chapter 11, NFPA 241, and NFPA 70. Include monthly inspection and testing of all equipment and apparatus.

2.6 STORAGE CONTAINERS

Welded steel construction, locking, shipping containers, or equal.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

3.1.1 Identification of Employees

Furnish identification as approved by the Contracting Officer to each employee and require each employee engaged on the work to display them. Immediately deliver prescribed identification to the Contracting Officer for cancellation upon release of any employee. When required, obtain and provide fingerprints of persons employed on the project. Wear identifying markings on hard hats clearly identifying the company for whom the employee works.

3.1.2 Employee Parking

- a. Park employees' vehicles in areas designated by Contractor, away from construction traffic, within reasonable walking distance of site. Maintain area free of ruts, mud holes, and puddles. Place gravel where required by deteriorated conditions.
- b. Protect unattended equipment as it may be subject to vandalism.
- c. Storage trailers and storage area with Government material should be locking type with lighting.

3.1.3 Onsite Information

Keep copy of contract drawings, specifications, and other contract documents at the Contractor's office onsite available for use at all times.

3.1.4 Janitorial Services

- a. Daily -- Sweep and mop floors, empty wastebaskets, service and sanitize bathrooms and other common areas, restock cleaning supplies and toiletries such as, but not limited to, hand soap, hand sanitizer, paper towels, and toilet paper.
- b. Weekly -- Wash floors and windows (inside and out).
- c. Cut grass as needed to keep below 3 inches high.

3.2 AVAILABILITY AND USE OF UTILITY SERVICES

Install temporary facilities and utilities in accordance with IEEE C2, EM 385-1-1, and NFPA 70. Obtain necessary construction, building, zoning, or soil erosion and sediment control approvals required by local authorities and utility companies. Equip trailer(s) with wind tie downs in accordance with local wind and building code requirements.

3.2.1 Lighting

Electric light, non-glare type luminaries to provide a minimum illumination, level of 30-foot candles at desk height level. Comply with temporary lighting, wiring, and Ground Fault Circuit Interrupter requirements found in [EM 385-1-1](#).

3.2.2 Heating and Cooling

Adequate equipment to maintain internal ambient air temperature of 75 degrees F in 105 degrees F and 70 degrees F in 30 degrees F. Equip trailer(s) with smoke alarms. If bottled gas is used to heat, equip trailer(s) with carbon monoxide alarms.

3.2.3 Water

Non-potable well water may be used to flush toilets and for dish and hand washing. Provide bulk potable water in storage tanks or bottled water for drinking.

3.2.4 Fire Extinguisher

Refer to [EM 385-1-1](#). Non-toxic, dry chemical, fire extinguisher meeting Underwriters Laboratories, Inc. approval for Class A, Class B, and Class C fires with a minimum rating of 2A, 10B, and 10C.

3.2.5 Utility Lines

Install, connect, and modify temporary lines as coordinated with owning utility. Conform requirements in accordance with [IEEE C2](#) and [NFPA 70](#) for Temporary Electric Lines. Remove temporary lines at completion of project.

3.2.6 Electric Power

Submit [temporary electric drawings](#) including electricity supply and lighting, signed by licensed electrician, from transformer source to layout locations, fixtures, and materials to include outdoor lighting. Provide electrical service and transformers sufficient to operate heating/air conditioning units. Directly purchase electrical power.

3.3 PROTECTION AND MAINTENANCE OF TRAFFIC

During construction, provide access and temporary relocated roads as necessary to maintain traffic. Maintain and protect traffic on all affected roads during the construction period except as otherwise approved by the Contracting Officer. Provide measures for the protection and diversion of traffic including the provision of watchmen and flagmen, the erection of barricades, placement of lights around and in front of equipment and the work, and the erection and maintenance of adequate warning, danger, and direction signs as required by the Commonwealth and local authorities having jurisdiction. Protect the traveling public from damage to person and property. Ensure limited interference between the Contractor's traffic on roads selected for hauling material to and from the site with public traffic. Investigate the adequacy of existing roads and the allowable load limit on these roads. Repair any damage to roads caused by construction operations.

3.4 MAINTENANCE OF HAUL AND ACCESS ROADS

Provide and maintain safe and passable means of ingress and egress to the site and to all project features on the site for Contractor, Government, and third-party entities. Requirements herein are applicable to both new and existing haul and access roads. Repair and regrade haul and access roads as necessary. Maintain roadway drainage including maintaining roadway crown and profile and keeping drainage ditches clear. Repair potholes, mud holes, ruts, and/or depressions of 6 inches deep or more within 24 hours of discovery. Check condition of haul and access roads at least weekly and after any severe weather event. All requirements of [EM 385-1-1](#) concerning access and haul roads also apply; see Chapter 4.

3.5 CONTRACTOR'S TEMPORARY FACILITIES

Place the field offices, staging areas, stockpile storage, and temporary buildings in areas designated on the drawings and in accordance with Section [01 57 20](#) ENVIRONMENTAL PROTECTION. Make temporary movement or relocation of Contractor facilities when approved by the Contracting Officer.

3.5.1 Contractor Field Office

Provide on-site field office of sufficient size and staff capability to manage project activities. Provide communications and computer capabilities to manage the work including implementing RMS, facsimile, electronic mail, and reproduction to track schedule and cost. Include a conference area with table and chairs to accommodate a minimum of 8 people. Provide handicap access with a 1 on 12 ramp and gravel parking spot marked with a handicap parking sign. Provide high-speed internet services (10 Mbps or higher speed rate) and wireless capabilities to the field office facilities/compound to [include Government office space to](#) facilitate RMS, SharePoint (file transfer), and visitor usage.

3.5.2 Appearance of Trailer(s)

Maintain trailer(s) used for both office and material storage purposes in clean, neat exterior appearance and in good repair. Trailer(s) requiring exterior painting or maintenance will not be allowed on site until determined satisfactory by Contracting Officer.

3.5.3 Storage Area

Provide temporary 6-foot-high chain link fence for storage containers, trailers, or sheds containing Government property. Fence posts may be driven. Ground chain link fence against lightning. Store Contracting Officer's property, owned materials, and equipment within fenced storage area. Store small-size Government property, equipment, tools, and materials in locked steel containers. Do not stockpile Government property and materials outside fence in preparation for the next day's work. Park mobile equipment, tractors, wheeled lifting equipment, cranes, trucks, and like equipment within fenced areas on weekends.

3.5.3.1 Maintenance of Storage Area

Keep fencing in a state of good repair and proper alignment. Should the Contractor elect to traverse with construction equipment or other vehicles, cover grassed or unpaved areas, which are not established roadways, with a layer of gravel as necessary to prevent rutting and the

tracking of mud onto paved or established roadways; gravel gradation will be at the Contractor's discretion. Mow grass within the boundaries of the construction site for the duration of the project. Edge or trim neatly, grass and vegetation along fences, buildings, under trailers, and in areas not accessible to mowers.

3.5.4 Waste Storage

Provide dumpsters or suitable debris containers. Prevent windblown trash; cover as needed. Dispose of offsite when needed. Refer to Section 01 57 20 ENVIRONMENTAL PROTECTION.

3.5.5 Fuel Storage and Fueling Operations

Refer to Section 01 57 20 ENVIRONMENTAL PROTECTION. Provide light when fueling at night.

3.6 SECURITY PLAN

Provide a site-security person at all times when there are no on-site activities. Maintain 24-hour security during weekends and holidays. Site security must include, but not be limited to:

- a. Limiting vehicular access to authorized vehicles and personnel only.
- b. Limiting public access to travel along access road only. Keeping access road closed at all times except for passage of authorized personnel and vehicles.
- c. Maintaining a list of authorized personnel and subcontractors available at site offices.
- d. Maintaining a sign-in log documenting visitors, deliveries, and security incidents. The log must include the date, name, address, company, time in, and time out for each employee and visitor.
- e. Checking fenced areas, equipment, and trailers on a daily basis. If damage is observed or vandalism is found, reporting to the Contracting Officer.

3.6.1 Visitors

No visitors will be allowed on site without the knowledge of the Contractor and permission of the Contracting Officer. Direct visitors to report upon arrival to Contractor's Field Office for site safety and accident prevention briefing. Provide visitors appropriate protective equipment (i.e., ear plugs, safety glasses).

3.7 CLEANUP

Remove construction debris, temporary fencing, waste materials, packaging material, and the like from the work site daily. Clean away any dirt or mud tracked onto paved or surfaced roadways. Store materials resulting from demolition activities, which are salvageable, within the fenced area described above or at the supplemental storage area. Neatly stack stored material not in trailers, whether new or salvaged, when stored. Refer to Section 01 57 20 ENVIRONMENTAL PROTECTION for solid waste and post-construction cleanup.

3.8 RESTORATION OF STORAGE AREA

Remove the fence upon completion of the project. After removal of trailers, materials, and equipment from within the fenced area, the area will become the property of the Contractor. Restore areas used by the Contractor for the storage of equipment or material, or other use, to the original or better condition. Remove gravel used to traverse grassed areas and restore the area to its original condition including top soil and seeding as necessary.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 52 10

CONTRACTING OFFICER'S FIELD OFFICE

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUMMARY
- 1.3 SUBMITTALS
- 1.4 EXISTING UTILITIES
- 1.5 TEMPORARY SERVICES

PART 2 PRODUCTS

- 2.1 CONTRACTING OFFICER'S FIELD OFFICE

PART 3 EXECUTION

- 3.1 GENERAL REQUIREMENTS
 - 3.1.1 Parking
 - 3.1.2 Janitorial Service
- 3.2 AVAILABILITY AND USE OF UTILITY SERVICES
- 3.3 FIELD OFFICE
 - 3.3.1 Portable Office
 - 3.3.2 Office Equipment

-- End of Section Table of Contents --

SECTION 01 52 10

CONTRACTING OFFICER'S FIELD OFFICE

PART 1 GENERAL

1.1 REFERENCES

The publication listed below forms a part of this section to the extent referenced. The publication is referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

1.2 SUMMARY

This section covers the requirements regarding the Contracting Officer's Field Office. See Section 01 50 02 TEMPORARY CONSTRUCTION FACILITIES and Section 01 57 20 ENVIRONMENTAL PROTECTION for requirements including silt control, portable office placement, fueling restrictions, dust control, solid waste, and cleanup. Upon completion of project, clean up and restore area in accordance with Clause CLEANING UP of Section 00700 CONTRACT CLAUSES in Volume 1. Services required in this section will be paid for by separate line item (see Section 01 22 00 MEASUREMENT AND PAYMENT).

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Within 10 calendar days following date of receipt of Notice to Proceed (NTP) and prior to mobilization to site submit following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Contracting Officer's Field Office

1.4 EXISTING UTILITIES

See Section 01 50 02 TEMPORARY CONSTRUCTION FACILITIES

1.5 TEMPORARY SERVICES

Furnish following services for Contracting Officer's Field Office:

- a. Janitorial service
- b. Lawn service
- c. Waste removal

- d. Pest control
- e. Potable water by the most economic means available

PART 2 PRODUCTS

2.1 CONTRACTING OFFICER'S FIELD OFFICE

Submit supplier or manufacturer scale drawings of office floor plans and site plan for Contracting Officer's Field Office with requirement indicated in paragraph CONTRACTING OFFICER'S FIELD OFFICE below. Provide a portable office in good working condition that has a minimum of 1,728 square feet. Refer to suggested trailer floor plan at the end of this section. Portable office must have two lavatory/restroom facilities, a conference room, and a pantry/kitchenette area. The private offices are to have a minimum of 100 square feet to accommodate Government employees. At a minimum, the portable unit must have air conditioning, running water, and restroom facilities and accommodate equipment required below and IT equipment to be installed by the Government. See paragraph FIELD OFFICE below. A suitable office replacement for the portable office may be accepted at the Contracting Officer's discretion.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

Install Contracting Officer's Field Office ready for use no later than 30 calendar days after NTP.

3.1.1 Parking

Provide Contracting Officer 10 gravel parking spaces at the Contracting Officer's Field Office. Place parking signs reserved for use of Contracting Officer and visitors. Maintain these spaces clear of other vehicles. Designate one parking spot with a sign for handicap use only. Comply with all ADA parking requirements for size and accessibility.

3.1.2 Janitorial Service

- a. Daily -- Sweep and mop floors, empty wastebaskets, service and sanitize bathrooms and other common areas, restock cleaning supplies and toiletries such as, but not limited to, hand soap, hand sanitizer, paper towels, and toilet paper.
- b. Weekly -- Wash floors and windows (inside and out).

3.2 AVAILABILITY AND USE OF UTILITY SERVICES

See Section 01 50 02 TEMPORARY CONSTRUCTION FACILITIES. Provide Contracting Officer's Field Office sufficient continuous power and electric supply to assure operation of systems including computer, light, and HVAC. Provide one 20-amp, dedicated circuit to support copier/printer location and communications rack located in a 6-foot by 6-foot IT/storage closet. Install dedicated circuit wall termination at a central location where Government equipment will be installed.

3.3 FIELD OFFICE

Supply and maintain lockable field office separate from the Contractor for

use by the Contracting Officer. Find a suitable location and place the portable office on the project site, perform all required services installation, and furnish the office.

3.3.1 Portable Office

- a. An all-metal exterior, sides and roof, double-insulated walls, floor, and roof, self-contained, built-in heat and air conditioning, and 110-volt wall outlets in the quantity necessary to support equipment and furnishings but no fewer than two double receptacles per space.
- b. Security guard screens, screens, and blinds for all windows.
- c. Toilet facility -- Two separately enclosed rooms properly ventilated and complying with applicable sanitary codes including hot and cold running water and operable flush-type toilets.
- d. Installed during site mobilization.
- e. Make the portable office handicap accessible and comply with all ADA requirements.
- f. Provide lockable individual offices.
- g. Prewire portable office with network and phone drops in each office and the main conference area. Coordinate with the Contracting Officer the quantity and location of drops. Install network drops using Cat6, plenum cable from POTS terminal block in office IT closet to the end user/device locations, to include a Cat6, 24-port patch panel with wall mounting brackets, and any necessary conduit boxes, face plates and jacks for a complete cable installation. Install phone drops standard cabling to support any telephony required (to include POTS, VOIP, Centrix or other similar system) and plenum cable from the office IT closet to the end user/device locations. Include pre-wiring with 2-inch conduit and CAT6 wire from the DMARK (utility pole) to the CSU/DSU location (inside the 6-foot x 6-foot IT/Storage Closet). If the Telco or other telephony/network access provider positions their access point (or DMARK) in a different location, provide all cabling necessary to connect to the trailer cable plant. Locate portable office within 100 feet of utility services (telephony/network communications provider) location. Provide infrastructure to connect to the nearest services (i.e., install additional pole and services) if there are no such utilities within 100 feet. Install trailer to meet all federal, Commonwealth, and local codes, and applicable IT standards. Connect the trailer cabling for phones and network to whatever handoff the telephony/network service provider provides regardless of technology.
- h. Tie down office trailer per local wind and building codes. Refer to USACE Safety and Occupational Health (SOH) Requirements (EM 385-1-1) for additional requirements.
- i. At the completion of the project, remove portable office and associated equipment from the site. Connect and disconnect utilities in accordance with local codes and to the satisfaction of the Contracting Officer.

3.3.2 Office Equipment

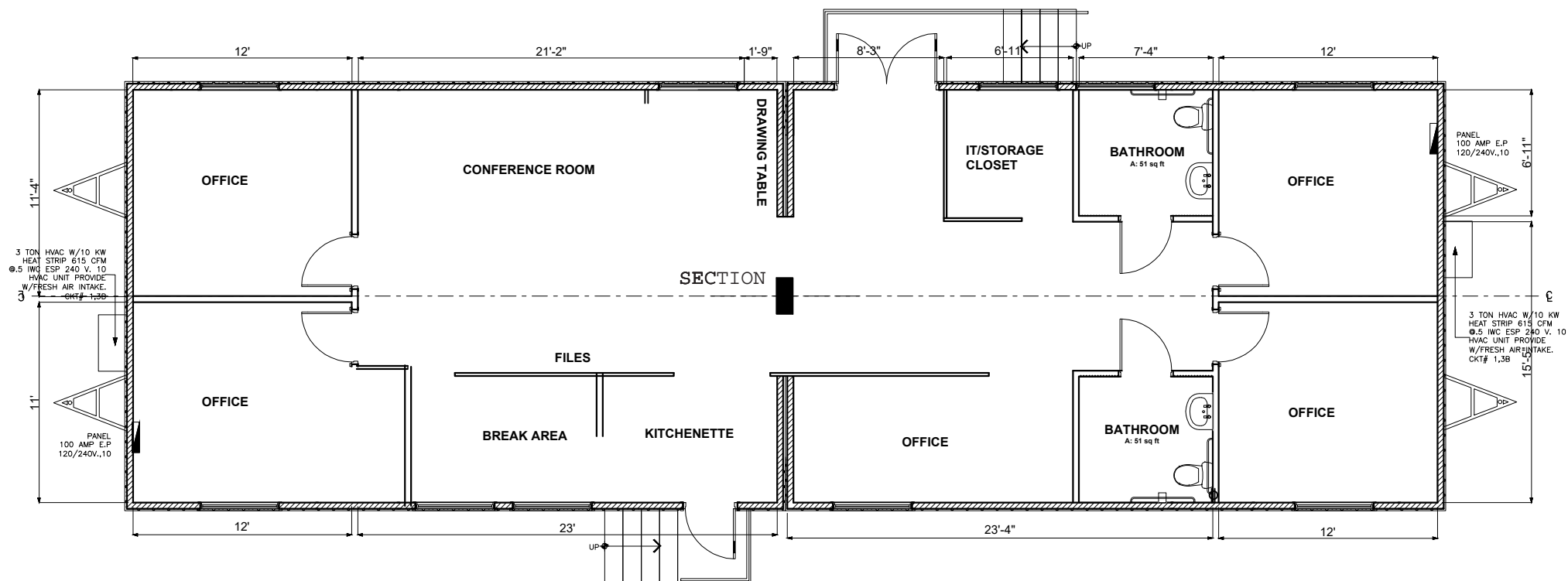
Provide and install the following office equipment. Uninstall and remove at the completion of the project:

- 2 - Office desks with lockable drawers (60 inches by 30 inches, laminated top), ergonomic swivel chair, and power strip for each (desks should be designed for use with PCs).
 - 1 - Conference table with ten chairs.
 - 22 - Folding chairs.
 - 1 - 16-unit first aid kit, wall mounted.
 - 12 - Five-drawer, letter-size lockable filing cabinets and one office step stool.
 - 1 - Kitchenette table.
 - 5 - Two-shelf bookshelves(35-inch x 14-inch x 28-inch Typ).
 - 1 - 3-foot x 6-foot cork bulletin boards.
 - 4 - 3-foot x 6-foot dry erase boards and set of color markers for each board.
 - 12 - 4- to 6-gallon waste baskets, one for each office and bathroom.
 - 1 - 10-gallon waste basket for conference room.
 - 1 - Vertical filing plan rack sufficient for full-size contract drawings.
 - 1 - 20-cubic-foot refrigerator.
 - 1 - 1.5-cubic-foot microwave oven.
 - 1 - 12-cup coffee maker.
 - 1 - Wall-mounted rack enclosure closet, 10 U (Unit), 19 inches deep, locking on all sides (Example: Tripp Lite-SRW10US or equal as approved by the Contracting Officer).
 - 8 - Office desks with lockable drawers (60 inches x 30 inches, laminated top) with ergonomic swivel desk chair and power strip for each (desks should be designed for use with PCs).
 - 1 - 12- to 16-gallon waste basket at the pantry/kitchenette area.
 - 1 - 12- to 16-gallon recycle bin at the pantry/kitchenette area.
- Extinguishers in compliance with applicable requirements.

Confirm and coordinate all quantities and the location of furnishings herein with the Contracting Officer. Refer to suggested trailer floor plan at the end of this section.

-- End of Section --

RIO PUERTO NUEVO PROJECT SUPPLEMENTAL CONTRACT 3
San Juan, Puerto Rico



COR TRAILER SUGGESTED FLOOR PLAN

NOTE: MAKE PORTABLE OFFICE HANDICAP ACCESSIBLE AND COMPLY WITH ALL ADA REQUIREMENTS.

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 55 26

TRAFFIC CONTROL

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 TRAFFIC CONTROL PLAN
- 1.4 GENERAL REQUIREMENTS
 - 1.4.1 Codes
 - 1.4.2 Safety
 - 1.4.3 Responsibility
 - 1.4.4 Cleanup

PART 2 PRODUCTS

- 2.1 MATERIALS
- 2.2 ROADWAY MAINTENANCE

PART 3 EXECUTION

- 3.1 CONSTRUCTION IN GENERAL
 - 3.1.1 Delay to Traffic
 - 3.1.2 Debris
 - 3.1.3 Dust Control
 - 3.1.4 Ingress and Egress
 - 3.1.5 Existing Signage
 - 3.1.6 Parking
 - 3.1.7 Temporary Traffic Control Devices
- 3.2 TRAFFIC REGULATIONS
- 3.3 TRAFFIC CONTROL STANDARDS
- 3.4 CONSTRUCTION SIGNS
- 3.5 BARRICADES, CONES, AND DRUMS
- 3.6 FLAGGERS

-- End of Section Table of Contents --

SECTION 01 55 26

TRAFFIC CONTROL

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to in the text by basic designation only.

PUERTO RICO HIGHWAY AND TRANSPORTATION AUTHORITY (PRHTA)

PRHTA Specifications (2005) Standard Specifications for Road and Bridge Construction

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health (SOH) Requirements

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)

MUTCD, PART 6 (2009; R 2012) Manual on Uniform Traffic Control Devices for Streets and Highways (<https://mutcd.fhwa.dot.gov>) - Temporary Traffic Control

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where

PRHTA Specifications Section and/or Special Provision(s) is cited that contains references to other PRHTA sections, the other PRHTA sections are to also form a part of this specification to the extent referenced. Where reference is made to "Highway Authority" or "Authority's Representative" in PRHTA Standard Specifications, they are to mean "Government" or "Contracting Officer" as defined in the contract documents. Where PRHTA Standard Specifications refer to the "Engineer," "Engineer of Tests," or "Division of Tests," it is understood to mean the "Contracting Officer" as stated in the contract documents.

1.2 SUBMITTALS

Government approval is required for submittals with a G designation; submittals not having a G designation are for information only. When used, a designation following the G designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section **01 33 00** SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Traffic Control Plan; G, R0

1.3 TRAFFIC CONTROL PLAN

The Contractor must develop a [Traffic Control Plan](#) for the project and submit to all concerning agencies having authority for approval. The Plan is to be in written form and indicate the type and location of all signs, lights, barricades, striping, and barriers to be used for safe passage of vehicular traffic through the project and for the protection of construction traffic and workers throughout the project including along the channel and disposal areas. Include specific details addressing ingress and egress at all access points along public roads. Lane and shoulder closures needed to complete the work must be included in the plan. Indicate conditions and setups for each phase of the Contractor's activities. This includes coordination with Contracting Officer and local governing authorities for the closing of public access points during working hours. Do not begin work until the Plan has been approved by all concerning agencies having authority and a letter of approval from all concerning agencies having authority has been submitted to the Government. Submit modifications to the Plan that become necessary in writing for approval to all concerning agencies having authority. Except in an emergency, no modifications to the approved Plan must be implemented on the project until the Contractor submits and receives approval of such modification from all concerning agencies. Also submit the Plan to Metropistas to coordinate proposed lane closures with adjacent road work.

1.4 GENERAL REQUIREMENTS

1.4.1 Codes

Work includes the furnishing, erection, relocation, maintenance, and eventual removal of temporary signs, barricades, pavement markings, lighting, signals, flaggers, and other traffic control devices as may be required for the maintenance and protection of traffic. Traffic control devices and traffic control operations must be in conformance with the [MUTCD, PART 6](#).

1.4.2 Safety

Conduct operations in a manner to ensure the convenience and safety of travelers and abutting property owners as well as the safety of the Contractor's employees following all requirements of [EM 385-1-1](#) and Section [01 35 26](#) GOVERNMENTAL SAFETY REQUIREMENTS.

1.4.3 Responsibility

The Contractor has sole responsibility for providing the necessary traffic control devices and taking other appropriate measures for the protection of the public and their personnel. The fact that provisions may be included in the plans and other contract documents, or may be ordered for the maintenance and protection of traffic during construction, does not relieve the Contractor of their responsibilities. In the absence of specific provisions in the contract plans and specifications or instructions from the Contracting Officer, provide additional devices and operational measures as may be considered necessary for the protection of the public and employees.

1.4.4 Cleanup

Temporary traffic control devices installed under this specification will remain the property of the Contractor and be removed from the project

after release by the Contracting Officer.

PART 2 PRODUCTS

2.1 MATERIALS

Traffic control devices must conform to the design, materials, colors, fabrication, and installation requirements specified by **PRHTA Specifications, EM 385-1-1** Chapter 8, and the applicable specification as may be modified by this Section. Traffic control devices are to be in good working condition and be subject to approval. Use reflective sheeting of the encapsulated lens grade on all devices requiring reflectorization.

2.2 ROADWAY MAINTENANCE

Keep existing and temporary pavements in good repair using the same materials compatible with the pavement. In general, plant-mixed bituminous concrete is suitable for all pavement surfaces. Other suitable material may be used if approved.

PART 3 EXECUTION

3.1 CONSTRUCTION IN GENERAL

Notify the Contracting Officer before starting any work that might inconvenience or endanger traffic or where the installation of signs, barricades, flaggers, and/or other appurtenances for the maintenance and protection of traffic.

3.1.1 Delay to Traffic

Conduct construction operations to ensure a minimum of delay to traffic. Stopping traffic for more than 5 minutes is not permitted unless specifically authorized.

3.1.2 Debris

Maintain the travel way free of foreign objects such as spilled earth, rocks, timber, and other items that may fall from transporting vehicles. Remove materials spilled by or dropped from any vehicle used in the Contractor's hauling operations along or across any public traveled way, both within and outside the project limits, immediately.

3.1.3 Dust Control

Address dusty conditions resulting from the Contractor's operations by the use of calcium chloride and/or water.

3.1.4 Ingress and Egress

Provide and maintain safe and adequate ingress and egress with soil tracking prevention devices to and from intersecting highways, private points of access, and other establishments at existing or new access points, consistent with the work, unless otherwise shown on the plans or authorized.

3.1.5 Existing Signage

Protect existing traffic signs and other traffic control devices within the project limits that are to remain from damage by construction operations. Replace any such items lost or damaged because of negligence on the part of the Contractor at no additional cost to the Government. Should temporary relocation of any such items be required because of construction operations, place as approved by the Contracting Officer.

3.1.6 Parking

Do not permit employees to park their vehicles nor store equipment or materials adjacent the traveled way or the PRHTA and/or municipality rights-of-way where it may be a hazard to traffic. Normally, keep a clear distance of at least 30 feet from the edge of the pavement free of any such obstacles unless otherwise authorized.

3.1.7 Temporary Traffic Control Devices

Maintain temporary devices in operation only as long as they are needed for traffic guidance and safety. Place only those devices that apply to conditions and operations in existence at a particular time. Clean, repaint, reflectorize, or replace traffic control devices as necessary to provide adequate visibility and legibility.

3.2 TRAFFIC REGULATIONS

Keep highway ramps, crossings, approaches to crossings, and PRHTA and/or municipality rights-of-way used by the Contractor's equipment clear of debris and excavated material resulting from the Contractor's operations. Construct ramps approaching and leaving the highway pavement. Do not exceed the commonwealth and local gross weight limits of each loaded vehicle using roadways and crossings. Restore the roadway at the ramp and crossing sites to conditions existing before hauling operations started. If the highway pavement has been damaged as a result of the Contractor's operations, reconstruct the pavement to the proper line and grade in conformance with [PRHTA Specifications](#).

3.3 TRAFFIC CONTROL STANDARDS

The [MUTCD, PART 6](#) sets forth the basic principles and prescribes minimum standards to be followed in the design, application, installation, maintenance, and removal of all traffic control devices and all warning devices and barriers that are necessary to protect the public and workers from hazards within the project limits. The standards established in [MUTCD, PART 6](#) constitute the minimum requirements for normal conditions and additional traffic control devices, warning devices, barriers, or other safety conditions exist. PRHTA requirements are to be met where more stringent.

3.4 CONSTRUCTION SIGNS

Furnish, erect, maintain, move, and remove signs as required to adequately and safely inform and direct the traveling public. The number of signs indicated in the [MUTCD, PART 6](#) are a minimum, and the Contracting Officer may require additional signs. Provide an adequate quantity of these signs on hand prior to starting construction operations for use as required. Sign sizes and details must conform to the [MUTCD, PART 6](#) or standard models. Signs and markers must indicate actual existing conditions and be

moved, removed, relocated, or changed as needed. Keep signs clean and in good condition for the duration of the contract. Mount signs at the required height on the adequate supports and place in a proper position and alignment so as to give maximum visibility both day and night. During nonworking hours and following the completion of a particular construction operation, remove or cover all warning signs except those required to remain for the safety of the public.

3.5 BARRICADES, CONES, AND DRUMS

Furnish, erect, maintain, move, replace, and remove barricades, cones, and drums where indicated in the **MUTCD, PART 6**, Part VI. Supplement construction barricades either by approved flashing or steady burning lights as appropriate where required or needed.

3.6 FLAGGERS

Provide a sufficient number of trained, PRHTA-certified flaggers where vehicles are entering or leaving public roads. Flaggers must wear orange helmets and vests and operate in conformance with the more stringent procedures and requirements of **PRHTA Specifications** and **EM 385-1-1**.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 57 20

ENVIRONMENTAL PROTECTION

PART 1 GENERAL

- 1.1 SCOPE
- 1.2 ENVIRONMENTAL LAWS, REGULATIONS, AND DOCUMENTS
 - 1.2.1 Miscellaneous Environmental Laws and Regulations
 - 1.2.2 NEPA Reference
- 1.3 REFERENCES
- 1.4 SUBMITTALS
- 1.5 QUALITY CONTROL
 - 1.5.1 Personnel Qualifications
 - 1.5.1.1 Manatee Observer Qualifications (Clamshell/Mechanical Only)
 - 1.5.1.2 Puerto Rican Boa Monitor Qualifications
- 1.6 ENVIRONMENTAL PROTECTION PLAN
- 1.7 PERMITS AND AUTHORIZATIONS
- 1.8 SUBCONTRACTORS
- 1.9 NOTIFICATION
- 1.10 CONTRACTOR PERSONNEL QUALIFICATIONS IN POLLUTION CONTROL
- 1.11 Meteorological Monitoring

PART 2 PRODUCTS

PART 3 EXECUTION

- 3.1 PROTECTION OF ENVIRONMENTAL RESOURCES
 - 3.1.1 General Project Environmental Design and Installation Criteria
 - 3.1.1.1 Sewage-Based Systems Environmental Design and Installation Criteria
 - 3.1.2 Protection of Land Resources
 - 3.1.2.1 Work Area Limits
 - 3.1.2.2 Protection of Landscape
 - 3.1.2.3 Unprotected Erodible Soils
 - 3.1.2.4 Disturbed Areas
 - 3.1.2.5 Contractor Facilities and Other Work Areas
 - 3.1.2.6 Solid Wastes
 - 3.1.2.7 Fuel, Oil, and Lubricants
 - 3.1.2.8 Hazardous Waste
 - 3.1.2.9 Hazardous Materials
 - 3.1.2.10 Disposal of Other Materials
 - 3.1.3 Preservation and Recovery of Historic, Archeological, and Cultural Resources
 - 3.1.3.1 Applicable Law
 - 3.1.3.2 Known Resources
 - 3.1.3.3 Inadvertent Discoveries
 - 3.1.3.4 Claims for Downtime due to Inadvertent Discoveries
 - 3.1.4 Protection of Water Resources
 - 3.1.4.1 Washing and Curing Water
 - 3.1.4.2 Temporary Sheet Pile and Diversion Operations

- 3.1.4.3 Stream Crossings
- 3.1.4.4 Monitoring of Water Areas
- 3.1.4.5 Turbidity
- 3.1.4.6 Oil, Fuel, and Hazardous Substance Spill Prevention and Mitigation
- 3.1.4.7 Wetlands Protection
- 3.1.5 Protection of Fish and Wildlife Resources
- 3.1.6 Endangered Species Protection
 - 3.1.6.1 Manatees and Sea Turtle Protection
 - 3.1.6.1.1 Manatee, Sea Turtle and Whale Sighting Reports
 - 3.1.6.1.2 Report Submission
 - 3.1.6.2 Puerto Rican Boa
- 3.1.7 Protection of Air Resources
 - 3.1.7.1 Particulates
 - 3.1.7.2 Burning
 - 3.1.7.3 Odors
- 3.1.8 Protection of Sound Intrusions
- 3.2 POSTCONSTRUCTION
 - 3.2.1 Cleanup
 - 3.2.2 Logs/Final Summary Report
 - 3.2.3 Project Environmental Summary Sheet
- 3.3 PRESERVATION AND RESTORATION OF LANDSCAPE
- 3.4 MAINTENANCE OF POLLUTION CONTROL FACILITIES
- 3.5 CONSTRUCTION FORMS AND DETAILS

-- End of Section Table of Contents --

SECTION 01 57 20

ENVIRONMENTAL PROTECTION

PART 1 GENERAL

1.1 SCOPE

This Section covers prevention of environmental damage as the result of construction operations under this contract and for those measures set forth in other Technical Requirements of these specifications. For the purpose of this specification, environmental damage is defined as the presence of hazardous, physical, chemical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; affect other species, biological communities, or ecosystems; or degrade the quality of the environment for aesthetic, cultural, and/or historical purposes. The control of environmental damage requires consideration of land, water, and air, and includes management of visual aesthetics, noise, solid waste, radiant energy, and radioactive materials, as well as other pollutants.

1.2 ENVIRONMENTAL LAWS, REGULATIONS, AND DOCUMENTS

1.2.1 Miscellaneous Environmental Laws and Regulations

There are numerous environmental laws and regulations. At the federal level, the applicable laws and regulations include compliance with the Clean Water Act; Clean Air Act; Coastal Zone Management Act; Comprehensive Environmental Response, Compensation, and Liability Act; Endangered Species Act (ESA); Fish and Wildlife Coordination Act; Marine Protection, Research, and Sanctuaries Act; Magnuson-Stevens Fishery Conservation and Management Act; National Environmental Policy Act (NEPA); National Historic Preservation Act; National Pollutant Discharge Elimination System; Research and Sanctuaries Act (RSA); Native American Graves Protection and Repatriation Act; Resource Conservation and Recovery Act; Rivers and Harbors Act; Safe Drinking Water Act; Toxic Substance Control Act; Wild and Scenic Rivers Act; Federal Insecticide, Fungicide, and Rodenticide Act; Marine Mammal Protection Act; Migratory Bird Treaty Act; Code of Federal Regulations (CFRs); Executive Orders; and, Environmental Protection Agency (EPA) requirements.

1.2.2 NEPA Reference

Commitments made in the NEPA document have been incorporated into these specifications. The NEPA document can be found at the following website: <https://www.saj.usace.army.mil/About/Divisions-Offices/Planning/Environmental-Branch/Environmental-Documents/>

1.3 REFERENCES

Any publication listed below forms a part of this specification to the extent referenced. Publications are referred to in the text by basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

ER 1110-1-5 (1984) Plant Pest Quarantined Areas and Foreign Soil Samples

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

33 CFR 153 Control of Pollution by Oil and Hazardous Substances, Discharge Removal

40 CFR 109 Criteria for State, Local, and Regional Oil Removal Contingency Plans

40 CFR 113 Liability Limits for Small Onshore Storage Facilities

40 CFR 261 Identification and Listing of Hazardous Waste

40 CFR 279 Standards for the Management of Used Oil

49 CFR 173 Shippers - General Requirements for Shipments and Packagings

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Environmental Protection Plan; G, D0

SD-03 Product Data

Meteorological Monitoring; G, R0

SD-07 Certificates

Manatee Observer Qualifications; G, D0

Puerto Rican Boa Monitor Qualifications; G, D0

SD-11 Closeout Submittals

Logs/Final Summary Report

Project Environmental Summary Sheet; G, D0

1.5 QUALITY CONTROL

Establish and maintain quality control (QC) for environmental protection of all items set forth herein. Record on daily QC reports or attachments thereto any problems in complying with laws, regulations, and ordinances and corrective action taken.

1.5.1 Personnel Qualifications

1.5.1.1 Manatee Observer Qualifications (Clamshell/Mechanical Only)

No later than 15 calendar days after Notice to Proceed (NTP), furnish to the Contracting Officer for approval the qualifications of the manatee observer. Demonstrate appropriate qualifications for the manatee observer through experience monitoring manatees and their behaviors in association with in-water construction projects.

1.5.1.2 Puerto Rican Boa Monitor Qualifications

No later than 15 calendar days after NTP, furnish to the Contracting Officer for approval, the qualifications of the Puerto Rican Boa Monitor. Demonstrate appropriate qualifications for Puerto Rican Boa Monitor through experience monitoring Boa and their behaviors in association with projects.

1.6 ENVIRONMENTAL PROTECTION PLAN

Submit a draft Environmental Protection Plan (EPP) no later than 15 calendar days after NTP for review and acceptance by the Contracting Officer. The Government will consider an interim plan for the first 30 days of operations. However, furnish an acceptable final plan no later than 30 calendar days after receipt of NTP. Acceptance of the plan will not relieve the Contractor of their responsibility for adequate and continuing control of pollutants and other environmental protection measures. Acceptance of the plan is conditional and predicated on satisfactory performance during construction. The Government reserves the right to require the Contractor to make changes to the EPP or operations if the Contracting Officer determines that environmental protection requirements are not being met. No physical work at the site is to begin prior to acceptance of the plan or an interim plan covering the work to be performed. The EPP to include, but not be limited to, the following:

- a. A list of federal, Commonwealth, and local laws, regulations, and permits concerning environmental protection, pollution control, and abatement that are applicable to the proposed operations and the requirements imposed by those laws, regulations, and permits.
- b. Methods for protection of features to be preserved within authorized work areas. Prepare a listing of methods to protect resources needing protection, i.e., trees, shrubs, vines, grasses, and ground cover, landscape features, air, and water quality, fish and wildlife, soil, historical, archeological, and cultural resources.
- c. Procedures to be implemented to provide the required environmental protection and to comply with the applicable laws and regulations. Provide written assurance that immediate corrective action will be taken to correct pollution of the environment due to accident, natural causes, or failure to follow the procedures set out in accordance with the EPP.
- d. An erosion and sediment control plan which identifies the type and location of the erosion and sediment controls to be provided. The plan must include monitoring and reporting requirements to assure that the control measures are in compliance with the erosion and sediment control plan, federal, Commonwealth, and local laws and regulations. A

Storm Water Pollution Prevention Plan may be substituted for this plan.

- e. A permit or license for and the location of the solid waste disposal area.
- f. Drawings showing locations of any proposed temporary excavations or embankments for haul roads, stream crossing, material storage areas, structures, sanitary facilities, and stockpiles of excess or spoil materials.
- g. Environmental monitoring plans for the job site including land, water, air, and noise monitoring.
- h. Traffic control plan.
- i. Methods of protecting surface and groundwater during construction activities.
- j. Spill prevention. Specify all potentially hazardous substances to be used on the job site and intended actions to prevent accidental or intentional introduction of such materials into the air, ground, water, wetlands, or drainage areas. Specify the provisions to be taken to meet federal, Commonwealth, and local laws and regulations regarding labeling, storage, removal, transport, and disposal of potentially hazardous substances.
- k. Spill contingency plan for hazardous, toxic, or petroleum material.
- l. Work area plan showing the proposed activity in each portion of the area and identifying the areas of limited use or nonuse. Include measures for marking the limits of use areas.
- m. Include a statement as to the person who will be responsible for implementation of the EPP. The Contractor personnel responsible is to report directly to the Contractor's top management and are to have the authority to act for the Contractor in all environmental protection matters.
- n. Recycling and Waste Management Plan. Executive Order 12873 of 20 October 1993 requires a number of considerations in planning a project. Fallen trees should not be burned or buried to the maximum extent practical. Mulching, composting, and other uses for trees should be considered. Also, recovery of metals at the job site including aluminum cans, should be considered with proceeds to be retained by the Contractor. Non-federal recycling and waste minimization efforts are to also be incorporated into this plan.
- o. Turbidity Monitoring Plan. Provide a statement with the person responsible for turbidity monitoring and reporting to be performed and the method of turbidity monitoring and reporting that will be performed.
- p. Meteorological Monitoring Plan. Provide a statement with the person responsible for the installation of the meteorological monitoring equipment and reporting to be performed.

1.7 PERMITS AND AUTHORIZATIONS

The Contractor is responsible for obtaining National Pollutant Discharge

Elimination System (NPDES) Stormwater Construction Generic Permit to provide coverage under the NPDES Generic Permit for Stormwater Discharges from Construction Activities.

Offsite Dewatering. The issuance of this permit does not constitute coverage under the NPDES General Permit for the Discharge of Produced Ground Water from any Non-Contaminated Site Activity (if applicable).

Obtain all needed permits or licenses. The Government will not obtain any permits for this project; see Clause PERMITS AND RESPONSIBILITIES of Section 00700 CONTRACT CLAUSES. The Contractor shall be responsible for implementing the terms and requirements of the appropriate permits as needed and for payment of all fees.

Pursuant to the Puerto Rico Department of Natural and Environmental Resources (DNER) Administrative Order 2004-04, dated March 9, 2004, the requirement for permits and franchises for the systems administered by the DNER regarding waters, vegetation and earth crust materials will be inapplicable to flood control projects sponsored by the agency. This exemption will only apply to activities directly related to the construction, operation and maintenance of these projects. A copy of DNER Administrative Order 2004-04 and its working translation is appended after the end of this section.

In addition to the above, comply with all requirements under the terms and conditions set out in the following permit(s) and authorization(s) obtained by USACE listed below.

- a. Puerto Rico Environmental Quality Board, Water Quality Certificate for the Control de Inundaciones Rio Puerto Nuevo, San Juan, Puerto Rico, dated 11 June 1993.

The above permits indicate issuing agency approval of work required by this contract. Permits and environmental compliance documents including Environmental Compliance Matrix defining roles and responsibilities are available at the following web address:
<http://www.saj.usace.army.mil/About/Divisions-Offices/Planning/Environmental-Branch/Environmental-Compliance/>

Dispose of excavated material off-site at approved disposal sites or approved landfill.

Obtain and comply with the requirements of all permits or licenses required for construction of this project in accordance with the Clause PERMITS AND RESPONSIBILITIES of Section 00700 CONTRACT CLAUSES in Volume 1.

1.8 SUBCONTRACTORS

Assurance of compliance with this section by subcontractors will be the responsibility of the Contractor.

1.9 NOTIFICATION

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with the aforementioned federal, Commonwealth, or local laws or regulations, permits and other elements of the environmental protection specifications. After receipt of such notice, inform the Contracting Officer of proposed corrective action and take such action as may be approved. If the Contractor fails to comply promptly, the

Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No time extensions are to be granted or costs or damages allowed to the Contractor for any such suspension.

Additionally, notify the Contracting Officer, in writing, of the absence or occurrence of environmental incidents, as required on the Project Environmental Summary Sheet, copy on the first web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. Refer to paragraph SUBMITTALS above.

1.10 CONTRACTOR PERSONNEL QUALIFICATIONS IN POLLUTION CONTROL

Provide qualified personnel to perform all phases of environmental protection including methods of detecting and avoiding pollution, familiarization with pollution standards, both statutory and contractual, and careful installation and monitoring of the project to ensure adequate and continuous environmental pollution control. QC and supervisory personnel are to be thoroughly knowledgeable of federal, Commonwealth, and local laws, regulations, and permits as listed in the EPP submitted by the Contractor.

1.11 Meteorological Monitoring

Provide a portable meteorological station for the continuous observation and recording of wind speed, wind direction, ambient air temperature, atmospheric pressure, atmospheric humidity, solar insolation, and atmospheric precipitation. Ensure the equipment and its placement conform to EPA's Ambient Monitoring Guidelines for PSD. Include a continuous readout temperature gauge and a rainfall gauge at the station.

Position the meteorological station to provide representative data on overall atmospheric diffusion conditions at the site. Establish visual wind direction indicators in a central location at each active work area. Install the meteorological station in an area relatively free of trees and houses, either at the office trailer or in the site's SZ. Ensure the station can record all measured parameters every 5 minutes and produce a 24-hour average for each parameter to characterize weather influences on the VOC meters. Enable the station to transmit real-time data to a central location to correlate real-time VOC monitoring with wind direction.

Maintain written or electronic records of the meteorological monitoring results for the duration of the project. Document meteorological monitoring results in the Quality Control report.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 PROTECTION OF ENVIRONMENTAL RESOURCES

For contract work, comply with all applicable federal, Commonwealth, or local laws and regulations. Protect the environmental resources within the project boundaries and those affected outside the limits of permanent work under this Contract at least during the entire period of this Contract. Confine activities to areas defined by the drawings and specifications. Environmental protection is to be as stated in the following subparagraphs.

3.1.1 General Project Environmental Design and Installation Criteria

Some project sites have features that must not be impacted in any way including cultural, historic, or archeological features. At all sites, minimize disturbance to existing features at the site to the extent possible including vegetative, topographic, and drainage pattern features. Wetland, seagrass or hardbottoms impacts (temporary access, detours, staging areas, and other work area impacts) to project sites should be avoided and may require separate permitting action. Any wetlands temporarily impacted are to have its soil restored upon project completion. Expansion of previously permitted project footprints may likewise require separate permitting action.

In all cases, the installation of temporary systems to aid construction will provide for protection of the environment during handling, installing, storing, utilizing, transporting, servicing, testing, refilling, transferring, pumping, processing, removing waste products, repairing, and maintaining systems and their components. Necessary protection is to also be considered that would prevent contamination of the environment from impacts to the systems caused by storm water runoff and flooding. Retrofit of temporary systems on project sites to modern environmental protection design standards will also be considered.

In the event environmental protection measures fail, implement procedures to control and correct environmental damage. Coordinate with the Contracting Officer if environmental protection measures fail.

3.1.1.1 Sewage-Based Systems Environmental Design and Installation Criteria

In general, there will be no waste or debris discharges of any kind for a project unless authorized by the Contracting Officer. This is to include the Contractor's providing sufficient temporary sanitary equipment and facilities for the project. The design and/or installation of temporary or permanent sewage systems are to ensure that waters will be free of effects of sewage discharges. Adhere to federal, Commonwealth, or local codes and requirements regarding sewage in the design, such as those of the EPA. Adhere to Best Management Practices from the applicable agencies in the design.

3.1.2 Protection of Land Resources

Prior to the beginning of any construction, identify all land resources to be preserved or avoided within the work area. Remove materials displaced into not cleared areas. Do not remove, cut, deface, injure, or destroy land resources including trees, shrubs, vines, grasses, topsoil, and land forms without special permission from the Contracting Officer. Engage a qualified tree surgeon to perform all tree surgery. Repair injuries to bark, trunk, branches, and roots of protected trees by dressing, cutting, and painting as specified for Class I Fine Pruning, of the National Arborist Association Pruning Standards for Shade Tree or as per the Commonwealth's Agricultural Extension Agency Guidelines, immediately as occurrences arise. Do not attach ropes, cables, or guys to any trees for anchorage unless specifically authorized. Where such special emergency use is permitted, provide effective protection for land and vegetation resources at all times as defined in the following subparagraphs.

Maintain all construction equipment and tools free of fluid leaks (coolant, refrigerant, oil, and lubricants). In the event of equipment

breakage, remove the equipment from the construction area and away from wetlands or sensitive areas. Where the equipment is park/store, install metallic pans to collect any leakage of fluid and tag with the date of breakage, reason of why is park/store, and if waiting for parts, indicate so. Repair the equipment within 30 days or completely remove from the project area.

3.1.2.1 Work Area Limits

Prior to any construction, mark the areas that are not required to accomplish all work to be performed under this Contract. Mark or fence isolated areas within the general work area to be saved and protected. Protect existing trees designated to remain. Provide protection of tree roots against noxious materials in solution caused by run-off or spillage. Fires are to be located outside the canopy of protected trees. Do not store materials, trailers, or equipment within the drip line of any protected tree. Protect monuments and markers before construction operations commence. Where construction operations are to be conducted during darkness, the markers will be visible. Convey to personnel the purpose of marking and/or protection of all necessary objects.

Thoroughly clean all construction equipment and tools at the prior job site in a manner that ensures all residual soil is removed and that egg deposits from plant pests are not present. Consult with the U.S. Department of Agriculture regarding additional cleaning requirements that may be necessary. In addition, if this contract involves the identification, shipping, storage, testing, or disposal of soils from such a quarantined area, the Contractor agrees to comply with the provisions of [ER 1110-1-5](#) and attachments. The Contractor agrees to assure compliance with this obligation by all subcontractors.

3.1.2.2 Protection of Landscape

Clearly identify and protect trees and their roots, shrubs, vines, grasses, land forms, and other landscape features, indicated and defined on the drawings to be preserved, by fencing or any other approved techniques. Illustrate the protection of trees in the Tree Protection Plan Detail on the first web site indicated in the paragraph CONSTRUCTION FORMS AND DETAILS below. Place tree protection fencing before excavation or grading is begun and maintain in place until construction is complete. Remove branches of protected trees, if required, to clear for construction and then prune to subsequently restore the natural shape of the entire tree. Cut branches or roots, if required, with sharp pruning instruments; do not break or chop. Fertilize protected trees to compensate for root loss with 6-6-6 as per manufacturer's application direction. Replace any damage to tree crowns or roots promptly after damage occurs.

- a. Trench or Bore Under Trees: Where trenching for utilities is required within tree driplines, hand dig under and around roots or bore under them. Protect roots from drying and cover exposed roots within an hour as specified in subparagraph EXCAVATION FOR STRUCTURES below. Do not cut lateral roots that interfere with new construction. Boring is permitted.
- b. Excavation for Structures: Where excavating for new construction is required within tree drip lines, hand excavate to minimize damage to root systems. Use narrow tine pitchforks and comb soil to expose roots. Relocate roots in backfill areas. If large, main lateral roots are encountered that are exposed beyond the excavation limits, bend

and relocate these roots without breaking or girdling. If roots are encountered immediately adjacent to new construction such that relocation is not practical, saw roots approximately 3 inches back from the new construction, seal with tree wound dressing, and protect any exposed embankment of roots from drying by covering with straw and black plastic. Irrigate affected areas daily until final grade conditions are established and the exposed roots are backfilled properly for continued plant growth.

- c. Replacement: Remove dead or damaged protected trees determined, by the Government, to be incapable of restoration to normal health growth. Replace each removed tree up to a 4-inch caliper with tree of equal specie and size. For each tree removed larger than a 4-inch caliper, the replace the tree with one 4-inch caliper tree per 4-inch caliper increment or fraction thereof.
- d. Grade Change: See Section 31 00 00 EARTHWORK for method of handling grade changes at existing trees to be protected during construction.

3.1.2.3 Unprotected Erodible Soils

Finish earthwork brought to final grade as indicated in the contract drawings. Protect side slopes and back slopes as soon as practicable upon completion of rough grading. All earthwork is to be planned and conducted to minimize the duration of exposure of unprotected soils. Except in instances where the constructed feature obscures borrow areas, quarries, and waste material areas, these areas are to not initially be totally cleared. Clearing of such areas will progress in reasonably sized increments as needed to use the areas developed as approved by the Contracting Officer.

3.1.2.4 Disturbed Areas

Prevent erosion and control sedimentation through approved methods including, but not limited to, the following:

- a. Retardation and Control of Runoff: Runoff from the construction site or from storms are to be controlled, retarded, and diverted to protected drainage courses by means of diversion ditches, benches, and by any measures required by area wide plans approved under paragraph 208 of the Clean Water Act.
- b. Erosion and Sedimentation Control Devices: Construct or install temporary and permanent erosion and sedimentation control features as required to comply with contract requirements. Place temporary velocity dissipation devices along drainage courses so as to provide for non-erosive flows. Maintain temporary erosion and sediment control measures such as berms, dikes, drains, sediment traps, sedimentation basins, grassing, mulching, baled hay or straw, and silt fences until permanent drainage and erosion control facilities are completed and operative. For silt fences, the filter fabric is to be of nylon, polyester, propylene, or ethylene yarn of at least 50 lb/in strength and able to withstand a flow rate of at least 0.3 gal/ft sq/minute. The fabric is to contain ultraviolet ray inhibitors and stabilizers and be a minimum of 36 inches in width. Bury the toe of the fence at least 8 inches deep to prevent undercutting and secure to posts by suitable staples, tie wire, or hog rings. Posts are to have a minimum of 4 feet in length. If work is performed in area with karst/limestone soils, metal stakes or rebar may be used. Overlap fence to the next

post if fabric joints are necessary.

- c. Sediment Basins: Trap sediment from construction areas in temporary or permanent sediment basins. The basins are to accommodate the runoff of a local 24-hour storm. After each storm, pump the basins dry and accumulated sediment and remove as necessary to maintain basin effectiveness. Control overflow by paved weir or by vertical overflow pipe, draining from the surface. Reuse the collected topsoil sediment for fill on the construction site, and/or conserved (stockpiled) for use at another site(s). Institute effluent quality monitoring programs as required by Commonwealth and local environmental agencies.

3.1.2.5 Contractor Facilities and Other Work Areas

Locate facilities, staging, and other work areas as indicated in Section 01 50 02 TEMPORARY CONSTRUCTION FACILITIES, and in compliance with permits listed in the paragraph PERMITS AND AUTHORIZATIONS above and obtained for performance of work. Manage borrow or dredge material management areas to minimize erosion and to prevent sediment from entering nearby watercourses, wetlands, or lakes. Manage and control dredged material areas to limit spoil intrusion into areas designated on the drawings and to prevent erosion of soil or sediment from entering nearby watercourses, wetlands, or lakes. Develop dredged material areas in accordance with the grading plan indicated on the drawings. Control temporary excavation and embankments for plant and/or work areas to protect adjacent areas from despoilment. If there is suspicion that sediment may be unsuitable for disposal at a specified location, immediately take measures to contain the suspect sediment and notify the Contracting Officer.

3.1.2.6 Solid Wastes

Place solid wastes (excluding clearing debris) in containers which are emptied on a regular schedule. Conduct handling and disposal to prevent contamination. Haul solid waste materials to a solid waste disposal site approved by the Contracting Officer. Comply with federal, Commonwealth, and local regulations pertaining to the use of the solid waste disposal site.

3.1.2.7 Fuel, Oil, and Lubricants

Manage fuel, oil, and lubricants in order to prevent spills and evaporation. Provide a 4-foot square, 16-gauge metal pan with borders banded up and welded at corners right below the bibb for fuel dispensers to prevent spills. Edges of the pans are to be 8-inch minimum in depth to ascertain that no contamination of the ground takes place. Clean pans by an approved method immediately after every dispensing of fuel and wastes disposed of offsite in an approved area. Should any spilling of fuel occur, immediately recover the contaminated ground and dispose of it offsite in an approved area. Store petroleum waste generated in marked corrosion-resistant containers and recycled or disposed of in accordance with 40 CFR 279, Commonwealth, and local regulations.

3.1.2.8 Hazardous Waste

Hazardous wastes are defined in 40 CFR 261. Ensure that hazardous wastes are stored and disposed of in accordance with 40 CFR 261 and Commonwealth and local regulations. Ensure that hazardous wastes are packed, labeled, and transported in accordance with 49 CFR 173 and Commonwealth and local regulations.

3.1.2.9 Hazardous Materials

Ensure that hazardous materials are labeled, stored, and transported in accordance with 49 CFR 173 and Commonwealth and local regulations.

3.1.2.10 Disposal of Other Materials

Handle other materials than previously discussed (construction and demolition, vegetative waste, etc.) as approved by the Contracting Officer.

3.1.3 Preservation and Recovery of Historic, Archeological, and Cultural Resources

3.1.3.1 Applicable Law

A number of federal laws require protection of cultural resources. Two laws, in particular, can be potentially involved with construction activities: (1) the National Historic Preservation Act, as amended; and, (2) the Abandoned Shipwreck Act.

3.1.3.2 Known Resources

If known historic, archeological, and cultural resources within the Contractor's work area(s) are present, it will be designated as an avoidance area on the contract drawings or other documents. If so designated, install protection for these resources and preserve during the Contract's duration. Do not distribute maps or other information on these resource locations except for distribution among the Contractor's staff with a need to know technical responsibility for protecting the resources.

3.1.3.3 Inadvertent Discoveries

If, during construction activities, the Contractor observes items that may have historic or archeological origin, such observations are to be reported immediately to the Contracting Officer, so the Project Archaeologist may be notified and a determination for what, if any, additional action is needed. Examples of historic, archeological, and cultural resources are bones, human remains, brick, pottery, metal, artifacts, shell, midden, modified rocks or coral, constructed features, charcoal, or other deposits. Cease all activities that may result in the destruction of these resources and prevent employees from further removing, or otherwise damaging, such resources.

Once reported, the Project Archaeologist will initiate coordination with the appropriate federal and Commonwealth agencies to determine if archaeological investigation is required. Additional work in the area of the discovery will be suspended at the site until all federal and Commonwealth regulations have been successfully completed and the Contracting Officer will provide further directive. Project activities in the vicinity of the discovery may not resume until the Contracting Officer approves work to proceed. No interview or other contact with the media or federal or Commonwealth agencies are to occur without clear authorization from the Contracting Officer.

In the unlikely event that unmarked human remains are identified, all work and ground disturbing activities within a 50-meter diameter of the unmarked human remains are to immediately cease and, and the Contracting Officer or the appropriate Corps representative is to be immediately

notified. The Project Archaeologist will notify the local medical examiner, and the appropriate State Historic Preservation Office and Tribal Historic Preservation Office(s). Based on the circumstances of the discovery, equity to all parties, and considerations of the public interest, the Contracting Officer may modify, suspend, or terminate work activities. Such activity is not to resume without written authorization from the Commonwealth Archeologist and from the Contracting Officer.

3.1.3.4 Claims for Downtime due to Inadvertent Discoveries

Upon discovery and subsequent reporting of a possible inadvertent discovery of cultural resources, continue work well away from, or otherwise protectively avoiding, the area of interest, or in some other manner that strives to continue productive activities in keeping with the Contract. Should an inadvertent discovery be of the nature that substantial impact(s) to the work schedule are evident, such delays are to be coordinated with the Contracting Officer. Contract adjustments resulting from compliance with this paragraph are to be determined in accordance with Clause DIFFERING SITE CONDITIONS of Section 00700 CONTRACT CLAUSES in Volume 1.

3.1.4 Protection of Water Resources

Keep construction activities under surveillance, management, and control to avoid pollution of surface, ground waters, and wetlands. Implement special management techniques as set out below to control water pollution by the listed construction activities which are included in this Contract. Use construction methods that protect wetland and surface water areas from damage due to mechanical grading, erosion, sedimentation and turbid discharges. Storing or stockpiling equipment, tools, or materials within wetlands or along the shoreline within the littoral zone is not allowed unless specifically authorized.

3.1.4.1 Washing and Curing Water

Waste waters directly derived from construction activities are not allowed to enter water areas. These waste waters are to be collected and placed in retention ponds where suspended materials can be settled out or the water evaporates so that pollutants are separated from the water. Analysis is to be performed and results reviewed and approved by USACE staff before water in retention ponds is discharged.

3.1.4.2 Temporary Sheet Pile and Diversion Operations

Control construction for dewatering, removal of temporary sheet pile, tailrace excavation, and tunnel closure at all times to limit the impact of water turbidity on the habitat for wildlife and impacts on water quality for downstream use.

3.1.4.3 Stream Crossings

Control stream crossings during construction. Crossings are to provide movement of materials or equipment without violating water pollution control standards of the federal, Commonwealth, or local government.

3.1.4.4 Monitoring of Water Areas

Monitor all water areas affected by construction activities.

3.1.4.5 Turbidity

Conduct operations in a manner to minimize turbidity. Refer to Section 01 57 25 TURBIDITY AND DISPOSAL MONITORING for further instructions.

3.1.4.6 Oil, Fuel, and Hazardous Substance Spill Prevention and Mitigation

Prevent oil, fuel, or other hazardous substances from entering the air, ground, drainage, local bodies of water, or wetlands. This is to be accomplished by design and procedural controls. In the event that a spill occurs, despite the design and procedural controls, the following is to occur:

- (1) Take immediate action to contain and cleanup any spill of oil, fuel or other hazardous substance.
 - (2) Immediately report spills to the Contracting Officer.
 - (3) Spill contingency planning is to be strictly in accordance with the criteria of 40 CFR 109.
 - (4) To control the spread of any potential spill, absorbent materials are to be readily available and capable of absorbing the contents of the single largest tank.
 - (5) To control the spread of any potential spill, provide a written certification of commitment of manpower, equipment, and materials required to expeditiously cleanup and dispose of spill materials.
- a. Spill Preventive Systems: System design and installation requirements have been discussed at the beginning of this Section. Temporary or portable tanks will conform to applicable federal, Commonwealth, and local codes and requirements and is not to be placed where they may be affected by storm, flooding, or washout. Diversionary structures for spills are to be put in place in advance where practical. The Contracting Officer will approve both spill preventive systems and any deviations from associated requirements prior to implementation.
 - b. Liabilities: Liable in the amounts established in 40 CFR 113 when it can be shown that oil was discharged as a result of willful negligence or willful misconduct. The penalty for failure to report the discharge of oil will be in accordance with the provision of 33 CFR 153.

3.1.4.7 Wetlands Protection

- a. Determine the location of wetlands within the project area and adjacent to the project area from the information provided in the contract documents. Instruct all personnel associated with the project of the presence of wetlands if the wetlands are located within 1,000 feet of staging areas, access roads or any other areas used during construction activities.
- b. Advise construction personnel that there are civil and criminal penalties for harming or destroying wetlands beyond actions specifically identified, anticipated, and authorized in these specifications and associated plans and environmental documents. Erect a silt fence at least 30 feet upland and along the entire length of all wetland delineation lines adjacent the work site and staging

areas, prohibit all access into the wetland, and ensure compliance with the paragraph PROTECTION OF WATER RESOURCES above.

- c. Do not anchor, place pipelines, or stage equipment in a manner that will cause any damage to wetlands beyond those specifically identified, anticipated, and authorized in these specifications and associated drawings and environmental documents. Avoid anchoring, placing pipeline, or staging equipment in these sensitive wetland areas. If such activities cannot be done without affecting sensitive areas outside the construction area identified in the Contract, cease the activities notify the Contracting Officer and Chief, Environmental Branch (904-232-2336) immediately (no later than the morning following the working day if the incident occurs after normal working hours). Report any actual incident involving damage to, or disturbance of wetlands.

3.1.5 Protection of Fish and Wildlife Resources

Keep construction activities under surveillance, management, and control to minimize interference with, disturbance to, and damage of fish and wildlife. List species that require specific attention along with measures for their protection in the EPP prior to the beginning of construction operation.

In the event that a threatened or endangered species is harmed as a result of construction activities, cease all work and notify the Contracting Officer. The order of contact within USACE will be as follows:

Order of Contact of USACE Personnel

<u>Title</u>	<u>Telephone Number</u>
USACE, Inspector	Onsite
Antilles Engineer	787-729-6880
Chief, Planning Technical Section (CESAJ-TFP-MIP)	904-699-1567
Chief, Construction Branch Construction Division	904-232-3278
Chief, Construction Division	904-232-1118

Information provided above will be updated during the Preconstruction Conference.

3.1.6 Endangered Species Protection

3.1.6.1 Manatees and Sea Turtle Protection

Instruct personnel associated with the project of the potential presence of manatees and sea turtles in the area and the need to avoid collisions with and/or harming these animals. Advise all construction personnel that there are civil and criminal penalties for harming, harassing, or killing, or attempting to harm, harass or kill manatees in the area, which are protected under the Marine Turtle Protection Act and the Marine Mammal Protection Act of 1972, the Endangered Species Act of 1973. The Contractor will be held responsible for any manatee harmed, harassed, or killed as a

result of construction activities.

- a. Siltation Barriers: Made of material in which manatees and sea turtles cannot become entangled, properly secured, and regularly monitored to avoid manatee entrapment. Barriers must not block manatee or sea turtle entry to or exit from essential habitat.
- b. Special Operating Conditions:
 - (1) All vessels associated with the project will operate at no wake/idle speeds at all times while in waters where the draft of the vessel provides less than a 4-foot clearance from the bottom, and vessels will follow routes of deep water whenever possible. Boats used to transport personnel must be shallow-draft vessels, preferably of the light-displacement category, where navigational safety permits. Place mooring bumpers on all barges, tugs, and similar large vessels wherever and whenever there is a potential for manatees to be crushed between two moored vessels. The bumpers are to provide a minimum stand-off distance of 4 feet.
 - (2) If a manatee(s) or sea turtles(s) is sighted within 100 yards of the project area, implement all appropriate precautions to ensure protection of the manatee or marine turtle. These precautions will include the operation of all moving equipment including vessels, no closer than 50 feet of a manatee or sea turtle. If a manatee or sea turtle is closer than 50 feet to moving equipment or the project area, shut down the equipment and cease all construction activities within the waterway to ensure protection of the manatee or sea turtle. Do not resume construction activities until the manatee or sea turtle has departed the project area. If construction activity shall cease, notify the Contracting Officer.
- c. Manatee Signs: Prior to commencement of construction, each vessel involved in construction activities shall display at the vessel control station or in a prominent location, visible to all employees operating the vessel, a temporary sign at least 8-1/2 inches x 11 inches reading, "CAUTION: MANATEE HABITAT/IDLE SPEED IS REQUIRED IN CONSTRUCTION AREA." In the absence of a vessel, a temporary 3-foot x 4-foot sign reading "CAUTION: MANATEE AREA" will be posted adjacent the issued construction permit. Post a second temporary sign measuring 8-1/2 inches x 11 inches reading "CAUTION: MANATEE HABITAT. EQUIPMENT MUST BE SHUTDOWN IMMEDIATELY IF A MANATEE COMES WITHIN 50 FEET OF OPERATION" at the dredge operator control station and at a location prominently adjacent the issued construction permit. Remove the signs upon completion of construction. Sample Manatee Caution Signs are on the first web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below.
- d. Manatee Monitoring (Clamshell/Mechanical Only): During clamshell dredging operations, a qualified dedicated observer is to monitor for the presence of manatees. The dedicated observer must be in a location with direct line of sight for all in-water construction activity (i.e., direct line of sight of clamshell bucket or excavator arm entering water). The dedicated observer is to have experience in manatee observation and be equipped with polarized sunglasses to aid in observing. Nighttime lighting of waters within and adjacent to the work area is to be illuminated, using shielded or low-pressure sodium-type lights, to a degree that allows the dedicated observer to sight any manatee on the surface within 200 feet of the operation. The

dredge operator must gravity-release the clamshell bucket only at the water surface, and only after confirmation that there are no manatees within the safety distance identified in the standard construction conditions.

3.1.6.1.1 Manatee, Sea Turtle and Whale Sighting Reports

Any take concerning a manatee, sea turtle, or whale; or sighting of any injured or incapacitated manatees is to be reported immediately to the Corps of Engineers by notifying the personnel indicated in the table "Order of Contact of Corps Personnel" above.

Provide a copy of the incidental take report in PDF format with photographs to the Contracting Officer within 24 hours of the incident. Also immediately report any collision with and/or injury to a manatee to the Boqueron Field Office 787-851-7273 for Puerto Rico and U.S. Fish and Wildlife Service 787-724-5700. If a sea turtle is taken by construction activities (live or dead), email a PDF version of the incidental take report to NOAA-Fisheries Southeast Region at the following email address within 24 hours of the take - takereport.nmfsser@noaa.gov also providing a cc copy to the Contracting officer.

3.1.6.1.2 Report Submission

Maintain a log detailing all incidents including sightings, collisions with, injuries, or killing of manatees, sea turtles, or whales occurring during the contract period. Record the data on forms provided by the Contracting Officer (sample forms are on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below). Forward all data in original form directly to Caribbean District Construction, Technical Support Branch, 383 Franklin Delano Roosevelt Avenue, San Juan, Puerto Rico 00918 within 10 days of collection and supply copies of the data to the Contracting Officer. Following project completion, submit a Project Environmental Summary Sheet summarizing the above incidents and sightings to the following:

Caribbean District Construction, Technical Support Branch
U.S. Army Corps of Engineers
383 Franklin Delano Roosevelt Avenue
San Juan, Puerto Rico 00918

Area Engineer for Area Office indicated
on Standard Form 1442

U.S. Fish and Wildlife Service
P.O. Box 491
Boqueron, Puerto Rico 00622-0491

Wildlife Bureau, Puerto Rico
Department of Natural and Environmental Resources
P.O. Box 9066600
San Juan, Puerto Rico

3.1.6.2 Puerto Rican Boa

Puerto Rican boa, *Epicrates inornatus*. Instruct all personnel associated with the project of the potential presence of Puerto Rico Boas. Advise all construction personnel that there are civil and criminal penalties for harming, harassing, or killing Puerto Rico Boas which are protected under

the Endangered Species Act of 1973. The Contractor may be held responsible for any Puerto Rico Boa harmed, harassed, or killed as a result of construction activities. The Puerto Rico Boa, is an endangered species under federal and Commonwealth of Puerto Rico Law. For detailed information on Puerto Rican Boa, the Contractor is referred to either:

The Puerto Rico Department of Natural Resources (787-999-2200), Wildlife Bureau, or U.S. Fish and Wildlife Service, Boquerón Office, Boquerón, Puerto Rico (787-851-7297).

a. Boa Monitor. The Puerto Rico Boa is a large snake found only on the island of Puerto Rico. It is an endangered species under federal and Commonwealth of Puerto Rico law. The Federal Endangered Species Act prescribes severe penalties for killing, injuring or harming endangered species including the Puerto Rico Boa. Contractor is to engage the services of a boa monitor acceptable to the Puerto Rico Department of Natural Resources in accordance with regulation 6766. Boas may initially be present along the margins of the work site, in trees or bushes, under stored materials or inactive equipment stored in shady locations. The Monitor's job will be to provide orientation to the Contractor and his employees regarding boa identification and avoidance; to lead a daily search for snakes in construction equipment prior to work start-up each and every day of operation; and to secure and relocate, if necessary, any boas that are sighted during work at the subject extraction site.

The monitor will determine prior to commencement of extraction or removal of earth's crust materials at the site, in cooperation with the Puerto Rico DNER, what relocation sites are to be used for any boas encountered during quarrying. Communicate this information to DNER and to the USACE Contracting Officer. Include the name of the Boa monitor, his/her qualifications, and information about the pre-selected relocation site(s) in the Contractor's EPP.

b. Pre-briefing for Contractor's employees. Prior to beginning of construction, the Contractor's employees are to receive a briefing on the Puerto Rican Boa, the snake's characteristics, types of material in which it may hide, actions to take if a boa is spotted, and penalties for failure to comply with Commonwealth and federal law regarding this species. This briefing is to be prepared by the Boa Monitor.

c. Daily Search. Boas are active mostly during the night. Therefore, complete a daily search around and in machinery at the beginning of each working day, prior to start-up of engines of quarry machinery, bulldozers, trucks, etc. Pay particular attention to motors and other warm areas that may be entered at night by the animals in an attempt to warm themselves. If search of machinery does not discover any specimens, inspect areas that are about to be cleared of vegetation next, especially piles of brush, leaf litter and rotting vegetation. These areas may be prodded gently with a blunt stick.

d. Capture and relocation supplies and equipment: Provide at least three items to the boa monitor, and maintain these items available on-site to handle and carry snakes if snakes are spotted: These are: a blunt snake hook, netting or burlap bags with closing ties, and a 6 x 6 or 8 x 8 foot tarpaulin. These items, except the snake hook, can be procured locally in agricultural or hardware stores. The snake hook, if the monitor does not possess one, may be obtained from "Valentine

Herpetological Supply" or "Bush Herpetological Supply."

e. Capture and/or Relocation Actions. If a snake is discovered, cease all work within a 50-foot radius of the snake's location. One person keeps watch on the snake while another contacts the Project Inspector and the boa monitor. The boa monitor will determine if the snake is to be captured or allowed to leave the site naturally. If capture is required, the monitor will perform the capture, notify the USACE project inspector and the Puerto Rico DNER endangered species coordinator, and proceed to capture and relocate the snake. The captured animal is to be maintained in a cool, shady place (not inside a parked car) until relocation is completed. It is the responsibility of the boa monitor to relocate the animal and make appropriate entry into the boa log regarding the capture/relocation.

f. Reports. Make a log entry using the supplied form for every boa discovery, whether the snake is observed on or adjacent the work site, leaves the work site voluntarily, or is relocated.

The log includes, but is not limited to, date, time of day, general weather, activity where the snake was found, action from first sighting until relocation, behavior of the snake, and personnel involved. Note the time and date of notification of USACE and DNER personnel also on the form. The boa monitor or his representative signs the report. Submit boa monitoring reports to the Contract Inspector monthly.

g. Boa injury and/or mortality. If a boa is killed or injured on the project, all work stops immediately. Contact both the U.S. Fish and Wildlife Service, Boqueron Puerto Rico, at 787-851-7297, and the Puerto Rico Department of Natural Resources, Wildlife Bureau at 787-999-2200 x2652, as well as the USACE Project Inspector and follow all instructions regarding disposition of the specimen.

h. Work Delay. Delays in work due to the fault or negligence of the Contractor or the Contractor's failure to comply with this specification is not compensable. Make any adjustments to the contract performance period or price that are required as a result of compliance with this section in accordance with the provisions of the Clause SUSPENSION OF WORK of Section 00700 CONTRACT CLAUSES.

Also, immediately report any death of and/or injury to a boa to the U.S. Fish and Wildlife Service, Boqueron Field Office at 787-851-7297 for Puerto Rico.

i. Report Submission. Maintain a log detailing sightings, injuries, relocations of or deaths of Puerto Rico Boas occurring during the contract period. Record the data on forms provided by the Contracting Officer. Forward all data in original form directly to Caribbean District Construction, Technical Support Branch, 383 Franklin Delano Roosevelt Avenue, San Juan, Puerto Rico 00918 within 10 days of collection and supply copies of the data to the Contracting Officer. Following project completion, submit a report summarizing the above incidents and sightings to the following:

Wildlife Bureau
Puerto Rico Department of Natural and Environmental Resources
P.O. Box 906600
San Juan, Puerto Rico 00906-6600

Caribbean District Construction, Technical Support Branch
U.S. Army Corps of Engineers (PD-E)
Caribbean District
383 Franklin Delano Roosevelt Avenue
San Juan, Puerto Rico 00918

Resident Engineer, San Juan Resident Office
U.S. Army Corps of Engineers (DS-CO-S)

U.S. Fish and Wildlife Service
P.O. Box 491
Boqueron, Puerto Rico 00622-0491

3.1.7 Protection of Air Resources

Keep construction activities under surveillance, management, and control to minimize pollution of air resources. All activities, equipment, processes and work operated or performed by the Contractor in accomplishing the specified construction is to be in strict accordance with the applicable air pollution standards of the Commonwealth and all federal emission and performance laws and standards including the U.S. Environmental Protection Agency's Ambient Air Quality Standards.

3.1.7.1 Particulates

Control particulates, such as dust, at all times including weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and work areas within or outside the project boundaries free from particulates that would cause air pollution standards to be exceeded or that would cause a hazard or nuisance. Provide equipment and approved methods to control particulates as the work proceeds and before a problem develops.

3.1.7.2 Burning

Burning is not allowed in the Commonwealth of Puerto Rico.

3.1.7.3 Odors

Control odors at all times for all construction activities.

3.1.8 Protection of Sound Intrusions

Keep construction activities under surveillance and control to minimize damage to the environment by noise.

3.2 POSTCONSTRUCTION

3.2.1 Cleanup

Clean up any area(s) used for construction.

3.2.2 [Logs/Final Summary Report](#)

Submit, as specified, logs and final summary report of sightings and incidents with listed species.

3.2.3 Project Environmental Summary Sheet

Submit within 30 calendar days following completion of the project, a written report of the absence or occurrence of environmental incidents. In addition, for construction activities whose anticipated duration is more than one calendar year, complete a sheet each 31 May (plus/minus 14 days).

3.3 PRESERVATION AND RESTORATION OF LANDSCAPE

Restore all landscape features and aquatic vegetation damaged or destroyed during construction operations outside the limits of the approved work areas. Such restoration is to be a part of the EPP as defined in subparagraph ENVIRONMENTAL PROTECTION PLAN of paragraph SUBMITTALS above. This work is to be accomplished at the Contractor's expense.

3.4 MAINTENANCE OF POLLUTION CONTROL FACILITIES

Maintain all constructed facilities and pollution control facilities and devices for the duration of the contract or for that length of time construction activities create the particular pollutant.

3.5 CONSTRUCTION FORMS AND DETAILS

See the Construction Forms and Details module at the following web address:
<https://www.saj.usace.army.mil/About/Divisions-Offices/Engineering/Design-Branch/Specifications-Section/>

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 57 25

TURBIDITY AND DISPOSAL MONITORING

PART 1 GENERAL

- 1.1 SCOPE
- 1.2 SUBMITTALS

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

- 3.1 MONITORING REQUIREMENTS
 - 3.1.1 General
 - 3.1.1.1 Turbidity Monitoring Equipment
 - 3.1.2 Dredging and Disposal Locations
 - 3.1.2.1 Sampling Stations
 - 3.1.2.2 Turbidity
- 3.2 TURBIDITY TESTS
 - 3.2.1 Testing
 - 3.2.2 Reporting
 - 3.2.2.1 Report Contents
- 3.3 Notification
- 3.4 WORK DELAY
- 3.5 CONSTRUCTION FORMS AND DETAILS

-- End of Section Table of Contents --

SECTION 01 57 25

TURBIDITY AND DISPOSAL MONITORING

PART 1 GENERAL

1.1 SCOPE

The work covered by this section consists of furnishing all labor, materials, and equipment and performing all work required to obtain, analyze, and report the results of turbidity and disposal monitoring.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Calibration Standard; G, R0

SD-06 Test Reports

Turbidity Monitoring; G, R0

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 MONITORING REQUIREMENTS

3.1.1 General

Obtain and analyze nearshore or inland water samples for turbidity. Conduct sampling in accordance with techniques described in the latest edition of Standard Methods published by the American Public Health Association, American Water Works Association, and Water Pollution Control Federation, and other current techniques recognized by the scientific community and approved by the U.S. Army Corps of Engineers, Caribbean District. Analyze samples obtained for turbidity analysis within 30 minutes of collection. Take samples with a sampler obtaining samples uncontaminated by water from any other depth.

3.1.1.1 Turbidity Monitoring Equipment

Measure monitoring required for turbidity in Nephelometric Turbidity Units (NTU) using a standard Nephelometer. Global Positioning System (GPS) is also required to record sampling stations. Provide the Government with a certification attesting to the accuracy of testing equipment and procedure. Provide a duplicate of the calibration standard used to calibrate the testing instrument as well as a complete set of operating instructions for the turbidity testing equipment. Use this standard

throughout the project to maintain the calibration of the equipment.

Furnish to the Contracting Officer a copy of the operating instructions and calibration standard used in calibrating equipment used in collecting samples for turbidity.

3.1.2 Dredging and Disposal Locations

Perform monitoring at dredging and disposal locations provided sea, wind, visibility, marine traffic, and other conditions allow for safe performance of the activity. Document all instances where [turbidity monitoring](#) within the bounds of the above parameters could not be performed. Perform routine monitoring at the following locations:

3.1.2.1 Sampling Stations

Perform turbidity sampling at locations meeting the following criteria. Adjust the sampling station locations based on the following criteria for the work being performed at the time of sampling.

- a. Compliance Turbidity Station: No more than 150 meters downcurrent of the dredge or point of discharge and in the direction of any visible plume.
- b. Background Turbidity Station: At least 150 meters upcurrent from the dredge or point of discharge and outside of any turbidity generated by the project.

3.1.2.2 Turbidity

Begin samples to be analyzed for turbidity after 30 minutes of commencement of operations and continue thereafter at least every 4 hours during daylight hours only at surface and mid depth at the determined stations/locations. Perform additional sampling when the Contracting Officer determines that there may be non compliance with water quality standards.

3.2 TURBIDITY TESTS

3.2.1 Testing

Whenever there is doubt as to the adequacy of the testing or validity of the results, the Contracting Officer may direct that additional tests be performed at no additional cost to the Government.

3.2.2 Reporting

Submit all required daily turbidity test reports (by electronic mail in Excel format) to the Contracting Officer within 24 hours after completion of each test. Record the turbidity monitoring data on summary forms and as a daily quality control test in the RMS system that contain the pertinent information in the following paragraphs. Example forms are on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. Submit other data in the form supplied by the laboratory chosen to do the analysis. Forward all data (electronically) to the Contracting Officer within 7 days of collection. Electronic mail addresses of the Corps personnel to receive these reports are provided below. Provide reports in Excel Spreadsheet files.

NAME	ORGANIZATION	E-MAIL ADDRESS
To be determined	USACE ACO	To be determined
To be determined	PD-E POC	To be determined

Information to be determined will be provided in the Preconstruction Conference. See Section 01 57 20 ENVIRONMENTAL PROTECTION.

3.2.2.1 Report Contents

- a. Permit application number.
- b. A statement describing the methods used in collection, handling, storage, and quality control methods used in the analysis of the samples.
- c. A map indicating the sampling location and plume configuration, if any.
- d. A map plotting the dredge location. This map can be combined with the map indicating the sampling location.
- e. A statement by the individual responsible for implementation of the sampling program concerning the authenticity, precision, limits of detection, and accuracy of the data.
- f. Monitoring Reports. Monitoring reports to also include the following information for each day that samples are taken:
 - (1) Time of commencement and completion of in-water work for that day.
 - (2) Time of day and date samples were taken.
 - (3) Depth of water body.
 - (4) Depth of sample.
 - (5) Antecedent weather conditions.
 - (6) Dredge or disposal location (station location and map).
 - (7) Water sample location. GPS coordinates.

3.3 Notification

If turbidity exceeds background levels by more than 10 NTU, immediately notify Caribbean District Construction, Technical Support Branch and the Contracting Officer or on the morning of the following work day if it occurs after normal work hours. In addition, cease all dredging or disposal activity immediately and take all measures to reduce turbidity. Do not resume dredging or disposal until corrective measures have been taken and turbidity has returned to acceptable levels as determined by proper testing described in subparagraph "Dredging and Disposal Locations" above.

3.4 WORK DELAY

Delays in work due to the fault or negligence of the Contractor or the Contractor's failure to comply with this specification shall not be compensable. Make any adjustments to the contract performance period or

price that are required as a result of compliance with this section in accordance with the provisions of the Clause SUSPENSION OF WORK of Section 00700 CONTRACT CLAUSES in Volume 1.

3.5 CONSTRUCTION FORMS AND DETAILS

See the Construction Forms and Details module at the following web address:

<http://www.saj.usace.army.mil/About/Divisions-Offices/Engineering/Design-Branch/Specifications-Section/>

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Co-mingle
 - 1.2.2 Construction Waste
 - 1.2.3 Demolition Debris/Waste
 - 1.2.4 Disposal
 - 1.2.5 Diversion
 - 1.2.6 Final Construction Waste Diversion Report
 - 1.2.7 Recycling
 - 1.2.8 Reuse
 - 1.2.9 Salvage
 - 1.2.10 Source Separation
- 1.3 CONSTRUCTION WASTE (INCLUDES DEMOLITION DEBRIS/WASTE)
- 1.4 CONSTRUCTION WASTE MANAGEMENT
 - 1.4.1 Implementation of Construction Waste Management Program
 - 1.4.2 Special Programs
 - 1.4.3 Special Instructions
 - 1.4.4 Waste Streams
- 1.5 SUBMITTALS
- 1.6 MEETINGS
- 1.7 CONSTRUCTION WASTE MANAGEMENT PLAN
- 1.8 RECORDS (DOCUMENTATION)
 - 1.8.1 General
 - 1.8.2 Accumulated
- 1.9 REPORTS
 - 1.9.1 General
 - 1.9.2 Quarterly Reporting
 - 1.9.3 Annual Reporting
- 1.10 FINAL CONSTRUCTION WASTE DIVERSION REPORT
- 1.11 COLLECTION
 - 1.11.1 Source Separation Method
 - 1.11.2 Co-Mingled Method
 - 1.11.3 Other Methods
- 1.12 DISPOSAL
 - 1.12.1 Reuse
 - 1.12.2 Recycle
 - 1.12.3 Waste

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 273	Standards for Universal Waste Management
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
49 CFR 178	Specifications for Packagings

1.2 DEFINITIONS

1.2.1 Co-mingle

The practice of placing unrelated materials together in a single container usually for benefits of convenience and speed.

1.2.2 Construction Waste

Waste generated by construction activities, such as scrap materials, damaged or spoiled materials, temporary and expendable construction materials, and other waste generated by the workforce during construction activities.

1.2.3 Demolition Debris/Waste

Waste generated from demolition activities including minor incidental demolition waste materials generated as a result of intentional dismantling of all or portions of a building to include clearing of building contents that have been destroyed or damaged.

1.2.4 Disposal

Depositing waste in a solid waste disposal facility, usually a managed landfill or incinerator, regulated in the U.S. under the Resource Conservation and Recovery Act.

1.2.5 Diversion

The practice of diverting waste from disposal in a landfill or incinerator by means of eliminating or minimizing waste or reuse of materials.

1.2.6 Final Construction Waste Diversion Report

A written assertion by a material recovery facility operator identifying

constituent materials diverted from disposal, usually including summary tabulations of materials, weight in short-ton.

1.2.7 Recycling

The series of activities including collection, separation, and processing by which products or other materials are diverted from the solid waste stream for use in the form of raw materials in the manufacture of new products sold or distributed in commerce or the reuse of such materials as substitutes for goods made of virgin materials other than fuel.

1.2.8 Reuse

The use of a product or materials again for the same purpose in its original form or with little enhancement or change.

1.2.9 Salvage

Usable, salable items derived from buildings undergoing demolition or deconstruction, parts from vehicles, machinery, other equipment, or other components.

1.2.10 Source Separation

The practice of administering and implementing a management strategy to identify and segregate unrelated waste at the first opportunity.

1.3 CONSTRUCTION WASTE (INCLUDES DEMOLITION DEBRIS/WASTE)

Divert a minimum of 60 percent by weight of the project construction waste and demolition debris/waste from the landfill or incinerator. Follow applicable industry standards in the management of waste. Apply sound environmental principles in the management of waste. Practice efficient waste management when sizing, cutting, and installing products and materials and use all reasonable means to divert construction waste and demolition debris/waste from landfills and incinerators and to facilitate the recycling or reuse of excess construction materials.

1.4 CONSTRUCTION WASTE MANAGEMENT

Implement a Construction Waste Management Program for the project. Take a proactive, responsible role in the management of construction waste, recycling process, and disposal of demolition debris/waste and require all subcontractors, vendors, and suppliers to participate in the Construction Waste Management Program. Establish a process for clear tracking and documentation of construction waste and demolition debris/waste.

1.4.1 Implementation of Construction Waste Management Program

Develop and document how the Construction Waste Management Program will be implemented in a Construction Waste Management Plan. Submit a Construction Waste Management Plan to the Contracting Officer for approval. Construction waste and demolition debris/waste materials include unused construction materials not incorporated in the final work as well as demolition debris/waste materials from demolition activities or deconstruction activities. In the management of waste, consider the availability of viable markets, the condition of materials, the ability to provide material in suitable condition and in a quantity acceptable to available markets, and time constraints imposed by internal project

completion mandates.

1.4.2 Special Programs

Implement special programs involving rebates or similar incentives related to recycling of construction waste and demolition debris/waste materials. Retain revenue or savings from salvaged or recycling unless otherwise directed. Ensure firms and facilities used for recycling, reuse, and disposal are permitted for the intended use to the extent required by federal, commonwealth, and local regulations.

1.4.3 Special Instructions

Provide on-site instruction of appropriate separation, handling, recycling, salvage, reuse, and return methods to be used by all parties at the appropriate stages of the projects. Clearly mark designation of single-source separating or commingling on the containers.

1.4.4 Waste Streams

Delineate waste streams and characterization, including estimated material types and quantities of waste, in the Construction Waste Management Plan. Manage all waste streams associated with the project. Typical waste streams are listed below. Include additional waste streams not listed:

- a. Land Clearing Debris
- b. Asphalt
- c. Masonry and CMU
- d. Concrete
- e. Metals (includes, but is not limited to, banding, stud trim, ductwork, piping, rebar, roofing, other trim, steel, iron, galvanized, stainless steel, aluminum, copper, zinc, and bronze.)
- f. Wood (nails and staples allowed)
- g. Vegetative Waste (trees and grass)

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Waste Management Plan; G, D0

SD-06 Test Reports

Quarterly Reports

Annual Report

SD-11 Closeout Submittals

Final Construction Waste Diversion Report; G, D0

1.6 MEETINGS

Conduct Construction Waste Management meetings. After award of the Contract and prior to commencement of work, schedule and conduct a meeting with the Contracting Officer to discuss the proposed Construction Waste Management Plan and to develop a mutual understanding relative to the management of the Construction Waste Management Program and how waste diversion requirements will be met.

The requirements of this meeting may be fulfilled during the coordination and mutual understanding meeting outlined in Section 01 45 04 CONTRACTOR QUALITY CONTROL. At a minimum, discuss and document waste management goals at following meetings:

- a. Preconstruction meeting.
- b. Regular site meetings.
- c. Work safety meeting (if applicable).

1.7 CONSTRUCTION WASTE MANAGEMENT PLAN

Submit Construction Waste Management Plan within 15 days after Notice to Proceed. Revise and resubmit Construction Waste Management Plan until it receives final approval from the Contracting Officer in order for construction to begin. Execute demolition or deconstruction activities in accordance with Section 02 41 00 DEMOLITION AND DECONSTRUCTION. Manage demolition debris/waste or deconstruction materials in accordance with the approved Construction Waste Management Plan.

An approved Construction Waste Management Plan will not relieve the Contractor of responsibility for compliance with applicable environmental regulations or meeting project cumulative waste diversion requirement. Ensure all subcontractors receive a copy of the approved Construction Waste Management Plan. The plan demonstrates how to meet the project waste diversion requirement. Also, include the following in the plan:

- a. Identify the names of individuals responsible for waste management and waste management tracking along with roles and responsibilities on the project.
- b. Actions that will be taken to reduce solid waste generation including coordination with subcontractors to ensure awareness and participation.
- c. Description of the regular meetings to be held to address waste management.
- d. Description of the specific approaches to be used in recycling/reuse of the various materials generated including the areas on site and equipment to be used for processing, sorting, and temporary storage of materials.
- e. Name of landfill and incinerator to be used.
- f. Identification of local and regional re-use programs including

non-profit organizations such as schools, local housing agencies, and organization that accept used materials such as material exchange networks and resale stores. Include the name, location, phone number for each re-use facility identified and provide a copy of the permit or license for each facility.

- g. List of specific materials, by type and quantity, that will be salvaged for resale, salvaged, and reused on the current project, salvaged and stored for reuse on a future project, or recycled. Identify the recycling facilities by name, address, and phone number.
- h. Identification of materials that cannot be recycled or reused with an explanation or justification, to be approved by the Contracting Officer.
- i. Description of the means by which materials identified in item (g) above will be protected from contamination.
- j. Description of the means of transportation of the recyclable materials (whether materials will be site separated and self hauled to designated centers or whether mixed materials will be collected by a waste hauler and removed from the site).
- k. Copy of training plan for subcontractors and other services to prevent contamination by comingling materials identified for diversion and waste materials.
- l. Facilities or subcontractors offering construction waste transport on site or off site must ensure that proper shipping orders, bills of lading, manifests, or other shipping documents containing waste diversion information meet requirements of [40 CFR 273](#) Universal Waste Management, [49 CFR 173](#) Shippers - General Requirements for Shipments and Packagings, and [49 CFR 178](#) Specifications for Packaging. Individuals signing manifests or other shipping documents should meet the minimum training requirements.
- m. Identify local jurisdiction requirements for waste management. Include local requirements and points of contact.

Distribute copies of the Construction Waste Management Plan to each subcontractor, [Quality Control Manager](#), and the Contracting Officer.

1.8 RECORDS (DOCUMENTATION)

1.8.1 General

Maintain records to document the types and quantities of waste generated and diverted through reuse, recycling, and sale to third parties and through disposal to a landfill or incinerator facility. Provide explanations for materials not recycled, reused, or sold. Collect and retain manifests, weight tickets, sales receipts, and invoices specifically identifying diverted project waste materials or disposed materials.

1.8.2 Accumulated

Maintain a running record of materials generated and diverted from landfill disposal including accumulated diversion rates for the project. Make records available to the Contracting Officer during construction or

incidental demolition activities. Provide a copy of the diversion records to the Contracting Officer upon completion of the construction, incidental demolitions, or minor deconstruction activities.

1.9 REPORTS

1.9.1 General

Maintain current construction waste diversion information on site for periodic inspection by the Contracting Officer. Include in the quarterly reports, annual reports, and final reports: the project name, contract information, information for waste generated, diverted, and disposed of for the current reporting period and show cumulative totals for the project. Reports must identify quantities of waste by type and disposal method. Also include in each report supporting documentation to include manifests, weigh tickets, receipts, and invoices specifically identifying the project and waste material type and weighted sum.

1.9.2 Quarterly Reporting

Provide cumulative reports at the end of each quarter (December, March, June, and September, corresponding with the federal fiscal year for reporting purposes). Submit **quarterly reports** not later than 15 calendar days after the preceding quarter has ended.

1.9.3 Annual Reporting

Provide a cumulative construction waste diversion report annually. Submit **annual report** not later than 30 calendar days after the preceding fourth quarter has ended.

1.10 FINAL CONSTRUCTION WASTE DIVERSION REPORT

A Final Construction Waste Diversion Report is required at the end of the project. Provide **Final Construction Waste Diversion Report** 60 days prior to the anticipated Final Inspection.

1.11 COLLECTION

Collect, store, protect, and handle reusable and recyclable materials at the site in a manner that prevents contamination and provides protection from the elements to preserve their usefulness and monetary value. Provide receptacles and storage areas designated specifically for recyclable and reusable materials and label them clearly and appropriately to prevent contamination from other waste materials. Keep receptacles or storage areas neat and clean.

Train subcontractors and other service providers to either separate waste streams or use the co-mingling method as described in the Construction Waste Management Plan. Handle hazardous waste and hazardous materials in accordance with applicable regulations and coordinate with Section 01 57 20 ENVIRONMENTAL PROTECTION. Separate materials by one of the following methods described herein:

1.11.1 Source Separation Method

Separate waste products and materials that are recyclable from trash and sort as described below into appropriately marked separate containers and then transport to the respective recycling facility for further

processing. Deliver materials in accordance with recycling or reuse facility requirements (e.g., free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process). Separate materials into the category types as defined in the Construction Waste Management Plan.

1.11.2 Co-Mingled Method

Place waste products and recyclable materials into a single container and then transport to an authorized recycling facility, which meets all applicable requirements to accept and dispose of recyclable materials in accordance with all applicable local, commonwealth, and federal regulations. Sort and process the co-mingled materials in accordance with the approved Construction Waste Management Plan.

1.11.3 Other Methods

Other methods proposed by the Contractor may be used when approved by the Contracting Officer.

1.12 DISPOSAL

Control accumulation of waste materials and trash. Recycle or dispose of collected materials off site at intervals approved by the Contracting Officer and in compliance with waste management procedures as described in the Construction Waste Management Plan. Except as otherwise specified in other sections of the specifications, dispose of in accordance with the following:

1.12.1 Reuse

Give first consideration to reusing construction and demolition materials as a disposition strategy. Recover for reuse materials, products, and components as described in the approved Construction Waste Management Plan. Coordinate with the Contracting Officer to identify onsite reuse opportunities or material sales or donation available through Government resale or donation programs. Sale of recovered materials is not allowed on the site.

1.12.2 Recycle

Recycle non-hazardous construction and demolition/debris materials that are not suitable for reuse. Track rejection of contaminated recyclable materials by the recycling facility. Do not count rejected recyclables materials as a percentage of diversion calculation. Legally dispose of all fluorescent lamps, HID lamps, and PCB-containing ballasts, and electrical components as needed. Do not crush lamps on site as this creates a hazardous waste stream with additional handling requirements.

1.12.3 Waste

Dispose by landfill or incineration only those waste materials with no practical use, economic benefit, or recycling opportunity.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 78 02

CLOSEOUT SUBMITTALS

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 PROJECT RECORD DOCUMENTS
 - 1.3.1 Definitions
 - 1.3.1.1 Design Files
 - 1.3.1.2 Working As-Built Drawings
 - 1.3.1.3 Completed As-Built Drawings
 - 1.3.1.4 Record Drawings
 - 1.3.1.5 As-Builts
 - 1.3.1.6 As-Built Shop Drawings
 - 1.3.2 Preparation of As-Builts and Record Drawings
 - 1.3.2.1 Government-Furnished Materials
 - 1.3.2.2 Working As-Built and Completed As-Built Drawings
 - 1.3.2.3 Record Drawing Preparation
 - 1.3.2.4 Electronic Record Drawings
 - 1.3.2.5 Completion of As-Builts
 - 1.3.3 As-Built Record of Equipment and Materials
 - 1.3.4 Record Drawings
 - 1.3.5 Real Property Equipment
- 1.4 WARRANTY MANAGEMENT
 - 1.4.1 Warranty Management Plan
 - 1.4.2 Performance Bond
 - 1.4.3 Pre-Warranty Conference
 - 1.4.4 Contractor's Response to Construction Warranty Service Requirements
 - 1.4.5 Warranty Tags
- 1.5 OPERATION AND MAINTENANCE MANUALS
- 1.6 FINAL CLEANING

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section Table of Contents --

SECTION 01 78 02

CLOSEOUT SUBMITTALS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

ERDC/ITL TR-12-1 (2015) A/E/C CAD Graphics Standard -
Release 2.0

ERDC/ITL TR-12-6 (2015) A/E/C CAD Standard - Release 6.0

<http://www.saj.usace.army.mil/About/Divisions-Offices/Engineering/Design-Branch/CADD-Management-Support/>

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where a

PRHTA Specifications Section and/or Special Provision(s) is cited that contains references to other PRHTA sections, they shall also form a part of this specification to the extent referenced. Where reference is made to "Highway Authority" or "Authority's Representative" in PRHTA Specifications shall mean "Government" or "Contracting Officer" as defined in the Contract Documents. Where PRHTA Specifications refer to the "Engineer," "Engineer of Tests," or "Division of Tests," it shall be understood to mean the "Contracting Officer" as stated in the Contract Documents. In case of conflict between the PRHTA Specifications and the Specifications in this Section that follows, the Specifications in this Section shall govern.

Paragraphs title "Method of Measurement" and "Basis of Payment" of referenced PRHTA Specifications shall not apply.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. For each submittal identified in this section an equal number of electronic copies must be submitted on CD-ROM media in PDF format. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

As-Builts; G, R0

SD-03 Product Data

As-Built Record of Equipment and Materials; G, R0

Warranty Management Plan; G, D0

Warranty Tags; G, D0

Final Cleaning; G, D0

SD-11 Closeout Submittals

Load Rating Analysis Report; G, D0

1.3 PROJECT RECORD DOCUMENTS

Submit [As-Builts](#) as indicated below and a [Load Rating Analysis Report](#). Check all drawing revisions and as-builts for compliance with [ERDC/ITL TR-12-6](#), [ERDC/ITL TR-12-1](#), and District PPM.

1.3.1 Definitions

1.3.1.1 Design Files

Design files are the latest version electronic CADD files.

1.3.1.2 Working As-Built Drawings

Working as-built drawings are contract drawings in PDF format marked up in red to indicate as-built conditions.

1.3.1.3 Completed As-Built Drawings

Completed as-built drawings are the final contract drawings in PDF format prepared by the Contractor. They show, in red, on-site changes to the original construction documents.

1.3.1.4 Record Drawings

Record drawings are corrected design files (CADD) showing the as-constructed condition and reflect on-site changes the Contractor noted in the completed as-built drawings. Record drawings are produced by making changes to the design files using CADD.

1.3.1.5 As-Builts

As-builts are the complete set, as indicated below, including record drawings (CADD) and completed as-built drawings in electronic format. As-builts in both English and Metric (soft conversion) units must be submitted.

1.3.1.6 As-Built Shop Drawings

As-built shop drawings are detailed shop drawings of components incorporated into the completed work indicating as-built conditions.

1.3.2 Preparation of As-Builts and Record Drawings

This paragraph covers obtaining completed as-built drawings and record

drawings (CADD) as a requirement of the contract. These as-builts are part of the permanent records of this project and will be retained by the Contracting Officer after approval by the Government. Satisfactorily replace any drawings lost by the Contractor at no expense to the Government.

1.3.2.1 Government-Furnished Materials

One set of electronic CADD files (design files) in Bentley Microstation Format software and format revised to reflect all bid amendments will be provided by the Government at the Preconstruction Conference for projects requiring CADD file record drawings. One set of electronic design files from any contract modification resulting in a change to the original design files will be provided by the Government as necessary.

1.3.2.2 Working As-Built and Completed As-Built Drawings

Revise drawings in PDF format by red-line process to show the as-built conditions during the prosecution of the project. Working as-built drawings must be kept current on a weekly basis and must be available on the jobsite at all times. Accurately and neatly record changes from the contract plans that are made in the work or additional information that might be uncovered in the course of construction as they occur by means of details and notes. Incorporate all mechanical and electrical changes due to field Request for Information process, equipment shop drawings reflecting modified data due to submittal and approval process, and contract field and design modifications in the working as-built drawings. Prepare completed as-built drawings after the completion of each definable feature of work as listed in the Contractor Quality Control Plan (foundations, utilities, structural steel, etc., as appropriate for the project). The working as-built drawings and completed as-built drawings will be jointly reviewed for accuracy and completeness by the Contracting Officer and the Contractor prior to submission of each monthly pay estimate. The working as-built drawings and completed as-built drawings must show, but are not limited to, the following information:

- a. The actual location, kinds, and sizes of all subsurface utility lines.
- b. Drawings to depict actual locations of all underground and aboveground water, wastewater, and reclaimed water improvements and related appurtenances.
- c. Drawings to clearly depict all details not on the original contract drawings but constructed in the field. Accurately depict the location of all equipment and piping relocation to scale.
- d. Depict the location of existing utilities where crossed or uncovered during the course of the work.
- e. Field verify and depict dimensions between all manholes. Depict the inverts and grade elevations of all lines entering and leaving manholes. Accurately locate and depict sewer laterals.
- f. Indicate the depth from finish grade to the top of potable water mains, reclaimed water mains, and sewer force mains at all valve and fitting locations.
- g. In order that the location of these lines and appurtenances may

be determined in the event the surface openings or indicators become covered over or obscured, these drawings must show, by offset dimensions to two permanently fixed surface features, the end of each run including each change in direction. Locate valves, splice boxes, and similar appurtenances by dimensioning along the utility run from a reference point.

h. The location, dimensions, and finished floor elevations of the building structures installed or affected as part of the project construction.

i. As-built grade, elevations, cross sections, or alignment of canals, levees, roads, earthwork, structures, or utilities including any changes that were made from contract plans. Provide cross sections at increments of 100 feet.

j. Prepare and/or furnish changes in details of design or additional information obtained from shop drawings specified including, but not limited to, fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment foundations, etc.

k. The topography, invert elevations, pipe sizes, and grades of drainage installed or affected as part of the project construction.

l. The invert elevations for all water control structures (including weir crest elevations, culverts, etc.). The completed as-built drawings to depict all elevations at [PRVD02](#).

m. Changes or modifications that result from the final inspection.

n. Where contract drawings or specifications present options, only show the option selected for construction on the completed working as-built drawings, or as an alternative, box, draw over with an "X" (corner to corner), and label options not constructed with a note stating OPTION NOT EXERCISED.

o. Systems designed or enhanced by the Contractor, such as, but not limited to, HVAC controls, fire alarm, fire sprinkler, irrigation systems, pumping equipment, hydraulic operating system, fuel system, electrical controls, one-line diagram telemetry and SCADA system, and conduit and piping layout, etc.

p. Modifications (change order price must include the Contractor's cost to change working as-built drawings and completed as-built drawings to reflect modifications) and compliance with the following procedures.

(1) Follow directions in the modification for posting descriptive changes.

(2) Place a revision symbol at the location of each deletion.

(3) For new details or sections that are added to a drawing, place a revision symbol by the detail or section title.

(4) For minor changes, place a revision symbol by the area changed on the drawing (each location).

(5) For major changes to a drawing, place a revision symbol by the title of the affected plan, section, or detail at each location.

(6) For changes to schedules or drawings, place a revision symbol either by the schedule heading or by the change in the schedule.

(7) Make the revision symbol shown on plan 1.5 times larger than the corresponding revision symbol shown in the revision block.

1.3.2.3 Record Drawing Preparation

The record drawings (CADD) must correctly show the features of the project as it has been constructed by bringing the contract design files into agreement with approved completed as-built drawings and adding such additional drawings as may be necessary. Ensure the completed record drawings are neat, legible, and accurate. Record drawings (CADD) must remain in the originally provided design file digital file format.

1.3.2.4 Electronic Record Drawings

Only personnel proficient in the preparation of CADD drawings must be employed to modify the design files and prepare additional new drawings files when preparing the record drawings.

a. Perform all drawing revisions to conform to **ERDC/ITL TR-12-6**, **ERDC/ITL TR-12-1**, and District PPM.

b. Ensure additions and corrections to the contract design files are equal in quality and detail to that of the original files. Line colors, line weights, lettering, and symbols must be the same as the original line colors, line weights, lettering, and symbols. If additional drawings are required in the record drawings, prepare them using the specified electronic file format applying the same graphic standards specified for original drawings. The title block and drawing border to be used for any new drawings in the record drawings must be identical to that used on the design file drawings. Accomplish additions and corrections to the record drawings using CADD program/system to prepare the design files. The Contractor will be furnished Microstation design files. The electronic files will be supplied on compact disc, read-only memory (CD-ROM). Use the electronic design files provided by the Government at the Preconstruction Conference to prepare changes and additions to the electronic record drawings. Prepare new drawings added to the original set of drawings in CADD format. Provide all programs and hardware necessary to prepare the record drawings. Provide a digital to-scale version of the record drawings in PDF format to the Contracting Officer for review. Make required corrections, changes, additions, and deletions as indicated by the Contracting Officer.

c. Create record drawings by appending an uppercase R to the design file name in the first user definable character. For a previously amended or modified design file, replace an existing A identifier or M identifier in the design file name with an uppercase R in the first user definable character. The renamed files corrected to reflect the "as-built" condition are hereafter referred to as "record drawings."

1.3.2.5 Completion of As-Builts

The as-built drawings in PDF format and record drawings (CADD) to be

completed as follows:

a. When final revisions have been completed, the cover sheet drawing must show the wording COMPLETED AS-BUILT followed by the name of the Contractor in letters at least 3/16 inch high relative to full size. Mark all other completed as-built drawings and record drawings either AS-BUILT drawing denoting no revisions on the sheet or REVISED AS-BUILT denoting one or more revisions. All completed as-built drawings and record drawings requiring revisions on the sheet must have an entry entered into the revision block. Sign the cover sheet of the completed as-built drawings in the following manner: I CERTIFY THAT THESE CORRECTED AS-BUILT DRAWINGS INDICATE CONSTRUCTION AS ACTUALLY PERFORMED AND ARE AN ACCURATE REPRESENTATION OF THE SPECIFIED WORK. THESE COMPLETED AS-BUILT DRAWINGS ARE APPROVED FOR PREPARATION OF RECORD CONSTRUCTION DRAWINGS.

b. Within 30 days after Government approval of all of the completed working as-built drawings for a phase of work, prepare the record drawings (CADD) for that phase of work and submit a digital copy in PDF format on a CD for Government review and approval. The Government will promptly return a PDF copy annotated with any necessary corrections. Within 10 days, revise the record drawings accordingly at no additional cost and submit one digital copy in PDF format on a CD for the completed phase of work to the Government. Within 20 days of substantial completion of all phases of work, submit the final record drawings for the entire project. The submittal must consist of four sets of CADD-generated record drawings in PDF format on compact disc(s), four sets of record drawing CADD files on compact disc(s), and two sets of the completed, approved, working as-built drawings in PDF format on compact disc(s). Compact discs must be read-only memory (CD-ROM). Complete files in all details and identical in form and function to the design files supplied by the Government. The PDF files produced from the approved completed working as-built drawings must be of high resolution and quality and easily readable. Any transactions or adjustments necessary to accomplish this are the responsibility of the Contractor. The Government reserves the right to reject any drawing files it deems incompatible. Storage media submitted will become the property of the Government upon final approval. Failure to submit as-builts and record drawings as specified must be cause for withholding any payment due the Contractor as retainage under the clause PAYMENTS UNDER FIXED-PRICE CONSTRUCTION CONTRACTS of Section 00700 CONTRACT CLAUSES in Volume 1. Approval and acceptance of as-builts and record drawings must be accomplished before final payment is made to the Contractor.

1.3.3 As-Built Record of Equipment and Materials

Furnish two copies of preliminary record of equipment and materials as approved during shop drawing submittal, review, and approval process used on the project 15 days prior to final inspection. This preliminary submittal will be reviewed and returned 2 days after final inspection with Government comments. Submit six sets of final record of equipment and materials 10 days after final inspection. Key the designations to the related area depicted on the contract drawings. The record must list the following data:

RECORD OF DESIGNATED EQUIPMENT AND MATERIALS DATA

Description	Specification	Manufacturer	Composition	Where
-------------	---------------	--------------	-------------	-------

RECORD OF DESIGNATED EQUIPMENT AND MATERIALS DATA			
Section	and Catalog, Model, and Serial Number	and Size	Used

1.3.4 Record Drawings

Furnish final approved as-built shop drawings 30 days after transfer of the completed facility.

1.3.5 Real Property Equipment

Furnish a list of installed equipment furnished under this contract. The list must include all information usually listed on manufacturer's name plate. The EQUIPMENT-IN-PLACE LIST must include, as applicable, the following for each piece of equipment installed: description of item, location (by room number), model number, serial number, capacity, name and address of manufacturer, name and address of equipment supplier, condition, spare parts list, manufacturer's catalog, and warranty. Furnish a draft list at time of transfer. Furnish the final list 30 days after transfer of the completed facility.

1.4 WARRANTY MANAGEMENT

1.4.1 Warranty Management Plan

Develop a Warranty Management Plan, which must contain information relevant to the Clause WARRANTY OF CONSTRUCTION of Section 00700 CONTRACT CLAUSES in Volume 1. At least 30 days before the planned pre-warranty conference, submit six sets of the Warranty Management Plan for Government approval. The Warranty Management Plan must include all required actions and documents to assure that the Government receives all warranties to which it is entitled. The Warranty Management Plan must be in narrative form and contain sufficient detail to render it suitable for use by future maintenance and repair personnel, whether tradesmen or of engineering background, not necessarily familiar with this contract. The term status as indicated below must include due date and whether item has been submitted or was accomplished. Warranty information made available during the construction phase must be submitted to the Contracting Officer for approval prior to each monthly pay estimate. Assemble approved information in a binder and turn over to the Government upon acceptance of the work. The construction warranty period must begin on the date of project acceptance and must continue for the full product warranty period. Conduct a joint 4-month and 9-month warranty inspection, measured from time of acceptance, with the Contractor, Contracting Officer, and the Customer Representative. Information contained in the Warranty Management Plan must include, but is not limited to, the following:

- a. Roles and responsibilities of all personnel associated with the warranty process including points of contact and telephone numbers within the organizations of the Contractors, subcontractors, manufacturers, or suppliers involved.
- b. Listing and status of delivery of all Certificates of Warranty for extended warranty items to include roofs, HVAC balancing, pumps, motors, transformers, and for all commissioned systems such as fire protection and alarm systems, sprinkler systems, lightning protection systems, etc.

c. A list for each warranted equipment, item, feature of construction, or system indicating:

- (1) Name of item.
- (2) Model and serial numbers.
- (3) Location where installed.
- (4) Name and phone numbers of manufacturers or suppliers.
- (5) Names, addresses, and telephone numbers of sources of spare parts.
- (6) Warranties and terms of warranty. Must include 1-year overall warranty of construction. Indicate items that have extended warranties with separate warranty expiration dates.
- (7) Cross-reference to warranty certificates as applicable.
- (8) Starting point and duration of warranty period.
- (9) Summary of maintenance procedures required to continue the warranty in force.
- (10) Cross-reference to specific pertinent Operation and Maintenance Manuals.
- (11) Organization, names, and phone numbers of persons to call for warranty service.
- (12) Typical response time and repair time expected for various warranted equipment.

d. The Contractor's plans for attendance at the 4- and 9-month post-construction warranty inspections conducted by the Government.

e. Procedure and status of tagging of all equipment covered by extended warranties.

f. Post copies of instructions near selected pieces of equipment where operation is critical for warranty and/or safety reasons.

g. Maintain RMS capabilities during the warranty period to correspond with the Government regarding the correction of warranty items.

1.4.2 Performance Bond

The Contractor's Performance Bond must remain effective throughout the construction period.

a. In the event the Contractor fails to commence and diligently pursue any construction warranty work required, the Contracting Officer will have the work performed by others, and after completion of the work, will charge the remaining construction warranty funds of expenses incurred by the Government while performing the work including, but not limited to, administrative expenses.

b. In the event sufficient funds are not available to cover the

construction warranty work performed by the Government at the Contractor's expense, the Contracting Officer will have the right to recoup expenses from the bonding company.

c. Following oral or written notification of required construction warranty repair work, respond in a timely manner. Written verification will follow oral instructions. Failure of the Contractor to respond will be cause for the Contracting Officer to proceed against the Contractor.

1.4.3 Pre-Warranty Conference

Prior to contract completion, and at a time designated by the Contracting Officer, meet with the Contracting Officer to develop a mutual understanding with respect to the requirements of this section. Communication procedures for Contractor notification of construction warranty defects, priorities with respect to the type of defect, reasonable time required for Contractor response, and other details deemed necessary by the Contracting Officer for the execution of the construction warranty must be established/reviewed at this meeting. In connection with these requirements and at the time of the Contractor's quality control completion inspection, furnish the name, telephone number, and address of a licensed and bonded company that is authorized to initiate and pursue construction warranty work action on behalf of the Contractor. This point of contact will be located within the local service area of the warranted construction, must be continuously available, and must be responsive to Government inquiry on warranty work action and status. This requirement does not relieve the Contractor of any of its responsibilities in connection with other portions of this provision.

1.4.4 Contractor's Response to Construction Warranty Service Requirements

Following oral or written notification by the Contracting Officer, the Contractor must respond to construction warranty service requirements in accordance with the Construction Warranty Service Priority List and the three categories of priorities listed below. Submit a report on any warranty item that has been repaired during the warranty period. The report is to include the cause of the problem, date reported, corrective action taken, and when the repair was completed. If the Contractor does not perform the construction warranty within the time frames specified, the Government will perform the work and back charge the construction warranty payment item established.

a. First Priority Code 1. Perform onsite inspection to evaluate situation, determine course of action within 4 hours, initiate work within 6 hours, and work continuously to completion or relief.

b. Second Priority Code 2. Perform onsite inspection to evaluate situation, determine course of action within 8 hours, initiate work within 24 hours, and work continuously to completion or relief.

c. Third Priority Code 3. All other work to be initiated within 3 work days and work continuously to completion or relief.

d. The Construction Warranty Service Priority List is as follows:

Code 2 - Electrical

Code 3 - All other work not listed above.

1.4.5 Warranty Tags

Submit six record copies of the warranty tags showing the layout and design. At the time of installation, tag each warranted item with a durable, oil-and-water-resistant tag approved by the Contracting Officer. Attach each tag with a copper wire and spray with a silicone waterproof coating. The date of acceptance and the quality control signature must remain blank until project is accepted for beneficial occupancy. The tag must show the following information.

- a. Type of product/material_____.
- b. Model number_____.
- c. Serial number_____.
- d. Contract number_____.
- e. Warranty period_____ from_____ to_____.
- f. Inspector's signature_____.
- g. Construction Contractor_____.
- Address_____.
- Telephone number_____.
- h. Warranty contact_____.
- Address_____.
- Telephone number_____.
- i. Warranty response time priority code_____.
- j. WARNING - PROJECT PERSONNEL TO PERFORM ONLY OPERATIONAL MAINTENANCE DURING THE WARRANTY PERIOD.

1.5 OPERATION AND MAINTENANCE MANUALS

Submit operation manuals and maintenance manuals as specified. Clearly differentiate and separately index operation manuals and maintenance manuals provided in a common volume. Refer to Section 01 78 23 OPERATION AND MAINTENANCE DATA.

1.6 FINAL CLEANING

Submit six copies of the listing of completed final clean-up items on the day of final inspection. Leave the premises broom clean. Remove stains, foreign substances, and temporary labels from surfaces. Vacuum carpet and soft surfaces. Clean equipment and fixtures to a sanitary condition. Replace filters of operating equipment. Remove debris from roofs, drainage systems, gutters, and downspouts. Sweep paved areas and rake clean landscaped areas. Remove waste, surplus materials, and rubbish from the site. Remove temporary structures, barricades, project signs, and construction facilities from the project area.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA

PART 1 GENERAL

- 1.1 SUBMISSION OF OPERATION AND MAINTENANCE DATA
 - 1.1.1 Quantity
 - 1.1.2 Package Quality
 - 1.1.3 Package Content
 - 1.1.4 Delivery
 - 1.1.5 Changes to Submittals
- 1.2 TYPES OF INFORMATION REQUIRED IN O&M DATA PACKAGES
 - 1.2.1 Operating Instructions
 - 1.2.1.1 Safety Precautions
 - 1.2.1.2 Operator Prestart
 - 1.2.1.3 Startup, Shutdown, and Post-shutdown Procedures
 - 1.2.1.4 Normal Operations
 - 1.2.1.5 Emergency Operations
 - 1.2.1.6 Operator Service Requirements
 - 1.2.1.7 Environmental Conditions
 - 1.2.2 Preventive Maintenance
 - 1.2.2.1 Lubrication Data
 - 1.2.2.2 Preventive Maintenance Plan and Schedule
 - 1.2.3 Corrective Maintenance (Repair)
 - 1.2.3.1 Troubleshooting Guides and Diagnostic Techniques
 - 1.2.3.2 Wiring Diagrams and Control Diagrams
 - 1.2.3.3 Maintenance and Repair Procedures
 - 1.2.3.4 Removal and Replacement Instructions
 - 1.2.3.5 Spare Parts and Supply Lists
 - 1.2.3.6 Corrective Maintenance Work Hours
 - 1.2.4 Appendices
 - 1.2.4.1 Parts Identification
 - 1.2.4.2 Warranty Information
 - 1.2.4.3 Personnel Training Requirements
 - 1.2.4.4 Testing Equipment and Special Tool Information
 - 1.2.4.5 Contractor Information
- 1.3 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES
 - 1.3.1 Data Package 5

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

- 3.1 DATA PACKAGE SUBMITTALS
- 3.2 PACKAGE FORMAT REQUIREMENTS

-- End of Section Table of Contents --

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA

PART 1 GENERAL

1.1 SUBMISSION OF OPERATION AND MAINTENANCE DATA

Submit Operation and Maintenance (O&M) Data and Manuals specifically applicable to this contract and a complete and concise depiction of the provided equipment, product, or system. Organize and present information in sufficient detail to clearly explain O&M requirements at the system, equipment, component, and subassembly level. Include an index preceding each submittal. Submit in accordance with this Section and Section 01 33 00 SUBMITTAL PROCEDURES.

1.1.1 Quantity

Submit five copies and one reproducible original for a total of six copies of the supplier/manufacturers' O&M information specified herein for the components, assemblies, subassemblies, attachments, and accessories. Submit six electronic copies of the above submittal in PDF format on CD-ROM media. The items for which O&M Data and Manuals are required are listed in the Technical Sections, which specifies those particular items.

1.1.2 Package Quality

Documents must be fully legible. Poor quality copies and material with hole punches obliterating the text or drawings will not be accepted.

1.1.3 Package Content

Data package content must be as shown in the subpart SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES below. Comply with the data package requirements specified in the individual Technical Sections including the content of the packages and addressing each product, component, and system designated for data package submission.

1.1.4 Delivery

Submit O&M Data and Manuals to the Contracting Officer for review and acceptance; submit data specified for a given item within 30 calendar days after the item is delivered to the contract site.

- a. In the event the Contractor fails to deliver O&M Data/Manuals within the time limits set forth above, the Contracting Officer may withhold from progress payments the retained percentage for the contract in accordance with Clause PAYMENTS UNDER FIXED-PRICE CONSTRUCTION CONTRACTS of Section 00700 CONTRACT CLAUSES in Volume 1.

1.1.5 Changes to Submittals

Furnish manufacturer-originated changes or revisions to submitted data if a component of an item is so affected subsequent to acceptance of the O&M Data. Submit changes, additions, or revisions required by the Contracting

Officer for final acceptance of submitted data within 30 calendar days of the notification of this change requirement.

1.2 TYPES OF INFORMATION REQUIRED IN O&M DATA PACKAGES

1.2.1 Operating Instructions

Include specific instructions, procedures, and illustrations for the following phases of operation:

1.2.1.1 Safety Precautions

List personnel hazards and equipment or product safety precautions for all operating conditions.

1.2.1.2 Operator Prestart

Include procedures required to set up and prepare each system for use.

1.2.1.3 Startup, Shutdown, and Post-shutdown Procedures

Provide narrative description for startup, shutdown, and post-shutdown operating procedures including the control sequence for each.

1.2.1.4 Normal Operations

Provide narrative description of Normal Operating Procedures. Include Control Diagrams with data to explain operation and control of systems and specific equipment.

1.2.1.5 Emergency Operations

Include Emergency Procedures for equipment malfunctions to permit a short period of continued operation or to shutdown the equipment to prevent further damage to system and equipment. Include Emergency Shutdown Instructions for fire, an explosion, spills, or other foreseeable contingencies. Provide guidance and procedures for emergency operation of all utility systems including required valve positions, valve locations and zones, or portions of systems controlled.

1.2.1.6 Operator Service Requirements

Include instructions for services to be performed by the operator such as lubrication, adjustment, inspection, and the recording of gage readings.

1.2.1.7 Environmental Conditions

Include a list of Environmental Conditions (temperature, humidity, and other relevant data) that are best suited for the operation of each product, component, or system. Describe conditions under which the item of equipment should not be allowed to run.

1.2.2 Preventive Maintenance

Include the following information for preventative and scheduled maintenance to minimize corrective maintenance and repair.

1.2.2.1 Lubrication Data

Include preventative maintenance lubrication data in addition to instructions for lubrication in accordance with subpart OPERATOR SERVICE REQUIREMENTS above:

- a. A table showing recommended lubricants for specific temperature ranges and applications.
- b. Charts with a schematic diagram of the equipment showing lubrication points, recommended types and grades of lubricants, and capacities.
- c. A Lubrication Schedule showing service interval frequency.

1.2.2.2 Preventive Maintenance Plan and Schedule

Include manufacturer's schedule for routine preventive maintenance inspections, tests, and adjustments required to ensure proper and economical operation and to minimize corrective maintenance. Provide manufacturer's projection of preventive maintenance work hours on a daily, weekly, monthly, and annual basis including craft requirements by type of craft. For periodic calibrations, provide manufacturer's specified frequency and procedures for each separate operation.

1.2.3 Corrective Maintenance (Repair)

Include manufacturer's recommended procedures and instructions for correcting problems and making repairs.

1.2.3.1 Troubleshooting Guides and Diagnostic Techniques

Include step-by-step procedures to promptly isolate the cause of typical malfunctions. Describe clearly why the checkout is performed and what conditions are to be sought. Identify tests or inspections and test equipment required to determine whether parts and equipment may be reused or require replacement.

1.2.3.2 Wiring Diagrams and Control Diagrams

Wiring diagrams and control diagrams must be point-to-point drawings of wiring and control circuits including factory-field interfaces. Provide a complete and accurate depiction of the actual job-specific wiring and control work. On diagrams, number electrical and electronic wiring and pneumatic control tubing and the terminals for each type identically to actual installation numbering.

1.2.3.3 Maintenance and Repair Procedures

Include instructions and a list tools required to repair or restore the product or equipment to proper condition or operating standards.

1.2.3.4 Removal and Replacement Instructions

Include step-by-step procedures and a list of required tools and supplies for removal, replacement, disassembly, and assembly of components, assemblies, subassemblies, accessories, and attachments. Provide tolerances, dimensions, settings, and adjustments required. Instructions must include a combination of text and illustrations.

1.2.3.5 Spare Parts and Supply Lists

Include lists of spare parts and supplies required for maintenance and repair to ensure continued service or operation without unreasonable delays. Parts lists must include the identifications, nomenclature, part numbers, required number of parts, recommended list of spare parts to be stocked at the project, and actual spare parts supplied. All data must match the actual equipment furnished, and standard catalog sheets, cuts, and diagrams will not be acceptable unless all irrelevant parts are marked out and relevant parts are identified by heavy arrows at each side of the applicable data. Special consideration is required for facilities at remote locations. List spare parts and supplies that have a long lead time to obtain.

1.2.3.6 Corrective Maintenance Work Hours

Include manufacturer's projection of corrective maintenance work hours including requirements by type of craft. Identify and tabulate separately corrective maintenance that requires completion or participation of the equipment manufacturer.

1.2.4 Appendices

Provide information required below and information not specified in the preceding subparts but pertinent to the maintenance or operation of the product or equipment. Include the following:

1.2.4.1 Parts Identification

Provide identification and coverage for all parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Include special hardware requirements such as a requirement to use high-strength bolts and nuts. Identify parts by make, model, serial number, and source of supply to allow reordering without further identification. Provide clear and legible illustrations, drawings, and exploded views to enable easy identification of the items. When illustrations omit the part numbers and description, both the illustrations and separate listing must show the index, reference, or key number that will cross-reference the illustrated part to the listed part. Group parts shown in the listings by components, assemblies, and subassemblies in accordance with the manufacturer's standard practice. Parts data may cover more than one model or series of equipment, components, assemblies, subassemblies, attachments, or accessories such as is typically shown in a master parts catalog.

1.2.4.2 Warranty Information

List and explain the various warranties and include the servicing and technical precautions prescribed by the manufacturers or contract documents in order to keep warranties in force. Include warranty information for primary components such as the compressor of an air conditioning system.

1.2.4.3 Personnel Training Requirements

Provide information available from the manufacturers that is needed for use in training designated personnel to properly operate and maintain the equipment and systems.

1.2.4.4 Testing Equipment and Special Tool Information

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components.

1.2.4.5 Contractor Information

Provide a list that includes the name, address, and telephone number of the General Contractor and each Subcontractor who installed the product, or equipment, or system. For each item, also provide the name, address, and telephone number of the manufacturer's representatives and service organization most convenient to the project site. Provide the name, address, and telephone number of the product, equipment, and system manufacturers.

1.3 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES

Furnish the O&M Data Package specified in individual Technical Sections. The required information for each O&M Data Package is as follows:

1.3.1 Data Package 5

- a. Safety precautions and hazards
- b. Operator prestart
- c. Startup, shutdown, and post-shutdown procedures
- d. Normal operations
- e. Environmental conditions
- f. Preventive maintenance plan, schedule, and procedures
- g. Troubleshooting guides and diagnostic techniques
- h. Wiring and control diagrams
- i. Maintenance and repair procedures
- j. Removal and replacement instructions
- k. Spare parts and supply list
- l. Product submittal data
- m. Manufacturer's instructions
- n. O&M submittal data
- o. Parts identification
- p. Testing equipments and special tool information
- q. Warranty information
- r. Extended warranty information

- s. Testing and performance data
- t. Contractor information
- u. Field test reports

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 DATA PACKAGE SUBMITTALS

Furnish the following O&M Manuals for the identified Data Packages in addition to the requirements specified in Technical Section of the specifications. Furnish the O&M Manuals not listed here as required under each Technical Section of the specifications.

Item	Data Package
Electrical Systems (Per Sections 26 56 00 EXTERIOR LIGHTING)	Data Package 5
Electrical Distribution System (Per Section 33 71 01 OVERHEAD TRANSMISSION AND DISTRIBUTION;	Data Package 5

3.2 PACKAGE FORMAT REQUIREMENTS

In addition to the above requirements, assemble the above data for each group or type of equipment in a durable plastic or leatherette binder. Binders must be for 8-1/2 by 11-inch sheets with slide binding or fastening with screwposts (with sufficient length for future expansion to add additional sheets) enabling ready replacement of sheets. Provide an identifiable title, visible from the front cover, giving the name of project, equipment or system title, contract number, and bid item number. Ring-type, loose lead binders will not be acceptable. Shop drawings, assembly drawings, or specially prepared drawings for these manuals or parts catalogs must be of a size that requires folding only in left-to-right coordinates as the manual or catalog is opened. Furnish a permanent film or cloth reproducible for all drawings specially prepared and of special sizes as specified in the preceding sentences. This permanent reproducible must be in addition to the drawings required to provide the five complete copies specified above. Number each sheet in the binder and provide an index for ready reference to the data. All standard catalog cuts, manufacturer's printed data or descriptive literature parts sheets, or illustrations must be either original manufacturer sheet or reproduced copies equal in clarity and durability to the original copies. At least one copy of such manual must contain all original copies of such data. Thermofax and similar nonpermanent copies are not acceptable. All non-applicable data, such as descriptive literature for other drawings, sketches, or data, must be included or the reference marked out. The final section of each manual must consist of a list of equipment covered by the manual, the equipment manufacturer, and the name, address, and telephone number of the local manufacturer's representative or equipment supplier. List the procurement order number of the equipment for this contract. Submit electronic copies of the above package on CD-ROM media in PDF format.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 22 13

VIBRATION AND NOISE MONITORING REQUIREMENTS

PART 1 GENERAL

- 1.1 SUBMITTALS
- 1.2 PERSONNEL QUALIFICATIONS
 - 1.2.1 Vibration Control Specialist
 - 1.2.2 Vibration Monitoring Technician(s)
 - 1.2.3 Structural Inspection/Evaluation Engineer
 - 1.2.4 Organizational Changes

PART 2 PRODUCTS (NOT APPLICABLE)

PART 3 EXECUTION

- 3.1 VIBRATION CONTROL PROGRAM
- 3.2 PRE-CONSTRUCTION STRUCTURAL SURVEY
- 3.3 VIBRATION CONTROL PLAN
- 3.4 NOISE CONTROL PLAN
- 3.5 DAILY INSTRUMENT LOGS
- 3.6 VIBRATION MONITORING REPORT
- 3.7 NOISE MONITORING REPORT
- 3.8 POST-CONSTRUCTION STRUCTURAL SURVEY
- 3.9 RESTITUTION OF DAMAGED PROPERTY

-- End of Section Table of Contents --

SECTION 02 22 13

VIBRATION AND NOISE MONITORING REQUIREMENTS

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Vibration Control Specialist; G, D0
Vibration Control Specialist Alternate; G, D0
Vibration Monitoring Technician(s); G, D0
Structural Inspection/Evaluation Engineer; G, D0

Within 30 calendar days after Notice to Proceed, submit qualifications for the Vibration Control Specialist and his Alternate; Vibration Monitoring Technician(s); and, Structural Inspection/Evaluation Engineer. Refer to qualifications in paragraph PERSONNEL QUALIFICATIONS below for each person.

Pre-Construction Structural Survey; G, D0

Within 30 calendar days after Notice to Proceed and prior to mobilization of equipment, submit 3 copies of the Pre-Construction Structural Survey. Refer to requirements in paragraph PRE-CONSTRUCTION STRUCTURAL SURVEY below.

Vibration Control Plan; G, D0

Within 30 calendar days after Notice to Proceed and prior to mobilization of equipment, submit 3 copies of a Vibration Control Plan (including Monitoring Plan). Submit the Vibration Control Plan for compliance with the requirements in paragraph VIBRATION CONTROL PLAN below. The Plan shall include, but not be limited to, the following:

- a. Scaled drawings of the construction site with vibration sensitive structures identified.
- b. List of structures that are susceptible to vibration damage from construction activity.
- c. Number of monitors (including seismographs and other instruments subject to approval such as strain and movement gages at cracks) required for the project and location of monitors that will operate simultaneously in each work area.
- d. Calibration data for each instrument that will be used for

the project. Calibrations shall be current, not older than one year, and follow the manufacturer's recommended procedures.

e. List of methods and procedures to reduce ground vibrations induced by construction activities to below the pre-determined maximum allowable vibration level for the designated vibration sensitive structure(s); i.e., reducing equipment speed, changing fill placement method, reducing equipment size, and using manual labor.

f. Plan for each work area showing the proposed construction equipment in the area, the description of vibration susceptible structure(s) in the work area, monitors in the work area, and the list of methods and procedures in subparagraph e. above.

g. The minimum safe working distance that vibration producing equipment may operate from each vibration sensitive structure(s).

h. The maximum allowable ground vibration level that is permissible without causing threshold damage to each vibration sensitive structure(s).

i. Frequency of instrument maintenance/inspection by the Vibration Control Specialist. Frequency shall not be greater than every 48 hours.

Noise Control Plan; G, D0

Within 30 calendar days after Notice to Proceed and prior to mobilization of equipment, submit 3 copies of a Noise Control Plan (including Monitoring Plan). Submit the Noise Control Plan for compliance with the requirements in paragraph NOISE CONTROL PLAN below. The Plan shall include, but not be limited to, the following:

a. Scaled drawings of the construction site with noise sensitive locations identified.

b. Number of monitors (including sound level analyzers) required for the project and location of monitors that will operate simultaneously in each work area.

c. Calibration data for each instrument that will be used for the project. Calibrations shall be current, not older than one year, and follow the manufacturer's recommended procedures.

d. List of methods and procedures for noise abatement induced by construction activities to below the pre-determined maximum noise levels for noise sensitive areas.

e. Plan for each work area showing the proposed construction equipment in the area, the description of noise sensitive commercial and residential locations in the work area, monitors in the work area, and the list of methods and procedures in subparagraph d. above.

f. Frequency of instrument maintenance/inspection by the Vibration Control Specialist. Frequency shall not be greater than every 48 hours.

SD-06 Test Reports

Vibration Monitoring Report

Submit a Vibration Monitoring Report every 2 weeks for compliance with the requirements in paragraph VIBRATION MONITORING REPORT below. Submit the Vibration Monitoring Report and the Daily Instrument Logs concurrently. Reports shall indicate if vibration limits were exceeded and describe mitigation measures.

Noise Monitoring Report

Submit a Noise Monitoring Report every 2 weeks for compliance with the requirements in paragraph NOISE MONITORING REPORT below. Submit the Noise Monitoring Report and the Daily Instrument Logs concurrently. Reports shall indicate if noise levels were exceeded and describe noise abatement measures.

Daily Instrument Logs

Submit the Daily Instrument Logs every 2 week for compliance with the requirements in paragraph DAILY INSTRUMENT LOGS below. Submit the Daily Instrument Logs and the Vibration Monitoring Report concurrently.

Post-Construction Structural Survey; G|D0

Submit 3 copies of the Post-Construction Structural Survey within 15 calendar days after completion of the inspection. Submit the Post-Construction Structural Survey for compliance with the requirements in paragraph POST-CONSTRUCTION STRUCTURAL SURVEY below.

1.2 PERSONNEL QUALIFICATIONS

1.2.1 Vibration Control Specialist

The Vibration Control Specialist and his alternate shall have a background in structural and/or geotechnical engineering, have a minimum of five years of demonstrated experience in vibration monitoring and related work, and be a registered professional engineer or registered professional geologist in the Commonwealth of Puerto Rico. The alternate may serve in the event of the Vibration Control Specialist's absence. Periods of absence with the alternate substituting shall not exceed one week at any one time and not more than fifteen workdays during a calendar year. The Vibration Control Specialist shall be designated in the Vibration Control Plan. [Vibration Control Specialist Alternate](#) shall meet the same qualifications listed herein.

1.2.2 Vibration Monitoring Technician(s)

Personnel responsible for operation, maintenance and calibration of sound level analyzers and seismographs and other instruments used on site shall have a minimum of five years of demonstrated experience in vibration and noise monitoring and related work.

1.2.3 Structural Inspection/Evaluation Engineer

Structural inspections shall be performed by professional structural engineers registered in the Commonwealth of Puerto Rico with a minimum of five years of demonstrated experience in structural condition inspections. The Structural Inspection Engineer shall have responsibility for pre-construction and post-construction structural surveys. The Structural Inspection Engineer shall be designated in the Vibration Control Plan.

1.2.4 Organizational Changes

When it is necessary to make changes to personnel, resubmit qualifications in accordance with paragraph SUBMITTALS above.

PART 2 PRODUCTS (NOT APPLICABLE)

PART 3 EXECUTION

3.1 VIBRATION CONTROL PROGRAM

Implement a program that protects existing structures from damages that result from construction equipment operations and vibrations. The purpose of the program is to avoid damages and potential claims that allege damages were caused by construction activities. The Contractor is responsible to document pre-existing conditions, to avoid damaging existing structures that were determined to be susceptible to vibration damage, and to avoid damaging existing structures that were not determined to be susceptible to vibration damage. Related responsibilities include inspection, damage claims, and work stoppage that results from monitoring.

The Contractor's attention is called to Clauses PERMITS AND RESPONSIBILITIES and PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES AND IMPROVEMENTS of Section 00700 CONTRACT CLAUSES in Volume 1 of this contract that define the Contractor's responsibilities. The Contractor shall process any claim arising from his operation; in particular, all property damage claims shall be acknowledged by the Contractor (or his agent) immediately, and the claimed damage inspected within 30 calendar days following initial notification, and processed to a conclusion (honored, denied, or compromised) within 90 calendar days after completion of the contract; but, in no case shall the claim(s) remain unresolved for a period exceeding six months.

3.2 PRE-CONSTRUCTION STRUCTURAL SURVEY

Inspect existing structures within 400 feet from the channel walls as to their potential susceptibility to vibration damage from construction equipment induced ground vibration. Include below grade structures such as utilities and vaults. Visible structural and/or cosmetic damage to buildings, exterior walls, foundations, decks, and other ancillary structures shall be documented by photographs, sketches, and field notes.

Factors to consider in determining potential susceptibility shall include but not be limited to: foundation design; foundation conditions; soils testing data; changes in structural loads and local water levels ; structural condition including construction materials; past damage history and existing stresses, magnitude, frequency, and duration of predicted vibrations from construction equipment and distance from fill placement or sheet pile driving.

Inspect all existing structures that are determined to be vibration sensitive. Any damage found shall be documented thoroughly by date-stamped photographs (supplemented with video as necessary), sketches of visible structural and/or cosmetic damage, and field notes. Photographs shall be at least 3- by 5-inch (or digital) and shall provide a detailed visual explanation of the damage. Include a reference scale in each close-up photograph. Sketches shall show the general damage location and extent. All inspection items shall be indexed and cross-referenced and shall use the stationing and locations shown on the contract drawings. Include building names and addresses where applicable. Structural damage shall be additionally documented by measuring crack or damage size, width, and length. Every effort shall be made to inspect and document the condition of the building's interior where the building has been determined to be extremely susceptible to vibration damage. Structure(s) determined not to be susceptible to vibration damage shall be noted as such.

3.3 VIBRATION CONTROL PLAN

Use the results of the Pre-Construction Structural Survey to develop the Vibration Control Plan. The Plan shall include procedures to monitor vibration sensitive structures, adjust daily mobilization and demobilization activities, and alter fill placement and pile driving operations. The Plan shall include the appropriate tolerable ground vibration level for each structure that has been determined to be susceptible to vibration damage. Should ground vibrations equal or exceed the pre-determined maximum vibration level(s), construction operations shall be halted and corrective measures taken in accordance with the approved Plan.

The minimum safe working distance that vibration producing equipment may operate from each vibration sensitive structure shall be documented in the Plan.

The maximum allowable ground vibration level that is permissible without causing threshold damage to each vibration sensitive structure shall be documented in the Plan. Threshold damage is defined as the occurrence of cosmetic damage.

Each seismograph shall have the capability to measure peak particle velocity and frequency and shall be equipped with an alarm system to alert the Vibration Control Specialist that ground vibrations are approaching the maximum tolerable ground vibration level. Seismographs shall record measurements 24 hours a day, 7 days a week. Seismographs shall be installed and shall be properly working prior to any construction activities, including mobilization and demobilization of all construction equipment, and fill placement and pile driving operations.

Strain and movement gages shall be implemented and monitored at pre-existing critical stress cracks located on vibration monitored structures. Strain and moment gage activity shall be monitored and reported under the same conditions as the vibration monitors.

The Contractor's personnel responsible for implementation of the Plan is hereinafter called Vibration Control Specialist. The Vibration Control Specialist shall be on the site during mobilization and demobilization of all construction equipment; and during the fill placement and/or pile driving operations that can affect the vibration sensitive structures (as

defined in paragraph PRE-CONSTRUCTION STRUCTURAL SURVEY above). When the construction activities do not involve any vibration sensitive structure, the Vibration Control Specialist may choose not to be on site. However, seismographs used for the 24/7 monitoring shall be capable of 24-hour real time notifications capable of being sent directly to the Vibration Control Specialist and Vibration Control Technicians. Other approved instruments shall be recorded and reported as outlined in the approved monitoring program. Ground vibrations shall be kept within allowable levels to avoid damaging existing structures that were determined to be susceptible to vibration damage, and to avoid damaging existing structures that were not determined to be susceptible to vibration damage.

3.4 NOISE CONTROL PLAN

Noise levels shall be kept below the maximum permissible sound levels as authorized by the Reglamento para el Control de la Contaminación por Ruidos. Notify the Contracting Officer and coordinate with Junta de Calidad Ambiental (JCA) for any activities that may potentially not meet regulatory requirements and cannot be mitigated. Failure to comply with the regulations or the JCA may result in required corrective action and monitoring in accordance with the JCA. Corrective action or schedule delays associated with non-compliance shall be at the Contractor's expense. Use identified commercial and residential locations that may be sensitive to noise annoyance to develop the Noise Control Plan. The Plan shall include procedures to monitor sensitive areas, adjust daily mobilization and demobilization activities, and alter fill placement and pile driving operations. The Plan shall include the permissible noise levels, project-specific permits listed in the Contract Documents, and applicable federal, commonwealth, and local codes, regulations and standards. Should noise levels equal or exceed the maximum level(s), construction operations shall be halted, and corrective measures taken in accordance with the approved Plan.

Corrective measures include the use of equipment with special exhaust silencers or enclosures and construction of temporary enclosures or noise barriers around activities.

Sound monitoring stations shall be equipped with sound level analyzers capable of measuring the A-Weighted and C-Weighted scales required by regulatory criteria and Noise Level Limits. Analyzers shall be capable of sound recording and have sufficient internal memory for two (2) weeks of logged data and sound recordings. The free-field microphone shall be housed in an environmental shroud providing protection from rain and wind conditions. Sound monitoring stations shall be installed and be properly working prior to any construction activities, including mobilization and demobilization of all construction equipment, and fill placement and pile driving operations.

3.5 DAILY INSTRUMENT LOGS

Daily instrument logs shall document satisfactory performance of the equipment during monitoring periods. Strip charts shall be documented daily with monitoring station number, date, technician signature, and instrument serial number.

3.6 VIBRATION MONITORING REPORT

The vibration monitoring report shall detail the daily activities of the vibration monitoring program. This report shall include, but not be

limited to, locations of monitoring equipment; instrument serial number; date and times of readings; magnitude of vibration levels; a sketch and photograph (3- by 5-inch) of each monitoring station showing the relationship of the monitor to vibration sensitive structures; daily instrument logs; instructions transmitted to the Contractor's personnel regarding the modification or stoppage of work operations to keep vibrations below the allowable levels; description of complaint(s); and, any other information pertinent to the vibration monitoring program.

3.7 NOISE MONITORING REPORT

The noise monitoring report shall detail the daily activities of the noise monitoring program. This report shall include, but not be limited to, locations of monitoring equipment; instrument serial number; date and times of readings; magnitude of noise levels; a sketch and photograph (3- by 5-inch) of each monitoring station showing the relationship of the monitor to residential or commercial areas; daily instrument logs; instructions transmitted to the Contractor's personnel regarding the modification or stoppage of work operations to keep noise levels below the allowable levels; description of complaint(s); and, any other information pertinent to the monitoring program.

3.8 POST-CONSTRUCTION STRUCTURAL SURVEY

Immediately after completion of work, conduct a post-construction inspection of the structures previously inspected under the Pre-Construction Structural Survey. Documentation procedures shall be identical to those performed under the Pre-Construction Structural Survey. Changes or deviations from the pre-construction inspection conditions in any structure shall be identified and described in the inspection documentation. All citizen complaints and remedies that occurred during fill placement or pile driving operations shall be summarized and included in the report.

3.9 RESTITUTION OF DAMAGED PROPERTY

Repair or replace structures damaged as a result from construction activities at no additional cost to the Government.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 41 00

DEMOLITION

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 PROJECT DESCRIPTION
 - 1.2.1 Definitions
 - 1.2.1.1 Demolition
 - 1.2.1.2 Demolition Plan
 - 1.2.2 Demolition Plan
 - 1.2.3 General Requirements
- 1.3 ITEMS TO REMAIN IN PLACE
 - 1.3.1 Gondola Cables
 - 1.3.2 Existing Construction Limits and Protection
 - 1.3.3 Trees
 - 1.3.4 Utility Service
 - 1.3.5 Facilities
- 1.4 BURNING
- 1.5 SUBMITTALS
- 1.6 QUALITY ASSURANCE
 - 1.6.1 Dust and Debris Control
- 1.7 PROTECTION
 - 1.7.1 Traffic Control Signs
 - 1.7.2 Protection of Personnel
- 1.8 RELOCATIONS
- 1.9 EXISTING CONDITIONS
 - 1.9.1 LUMA EQUIPMENT INVENTORY

PART 2 PRODUCTS

- 2.1 FILL MATERIAL

PART 3 EXECUTION

- 3.1 EXISTING FACILITIES TO BE REMOVED
 - 3.1.1 Structures
 - 3.1.2 Utilities and Related Equipment
 - 3.1.2.1 General Requirements
 - 3.1.2.2 Disconnecting Existing Utilities
 - 3.1.3 Chain Link Fencing
 - 3.1.4 Paving and Slabs
 - 3.1.5 Concrete
 - 3.1.6 Structural Steel
 - 3.1.7 Miscellaneous Metal
 - 3.1.8 Carpentry
 - 3.1.9 Acoustic Ceiling Tile
 - 3.1.10 Patching
 - 3.1.11 Air Conditioning Equipment

- 3.1.12 Cylinders and Canisters
- 3.1.13 Locksets on Swinging Doors
- 3.1.14 Mechanical Equipment and Fixtures
 - 3.1.14.1 Preparation for Storage
 - 3.1.14.2 Piping
 - 3.1.14.3 Ducts
 - 3.1.14.4 Fixtures, Motors and Machines
- 3.1.15 Electrical Equipment and Fixtures
 - 3.1.15.1 Fixtures
 - 3.1.15.2 Electrical Devices
 - 3.1.15.3 Wiring Ducts or Troughs
 - 3.1.15.4 Conduit and Miscellaneous Items
- 3.2 CONCURRENT EARTH-MOVING OPERATIONS
- 3.3 DISPOSITION OF MATERIAL
 - 3.3.1 Title to Materials
 - 3.3.1.1 MATERIALS OWNED BY LUMA
 - 3.3.2 Reuse of Materials and Equipment
 - 3.3.3 Disposal of Ozone Depleting Substance (ODS)
 - 3.3.3.1 Special Instructions
 - 3.3.3.2 Fire Suppression Containers
 - 3.3.4 Transportation Guidance
 - 3.3.5 Non-Recyclable Material
- 3.4 CLEANUP
- 3.5 DISPOSAL OF REMOVED MATERIALS
 - 3.5.1 Regulation of Removed Materials
 - 3.5.2 Removal from Government Property

ATTACHMENTS:

Kiosk

Garbage Facility

Gondola Cables

-- End of Section Table of Contents --

SECTION 02 41 00

DEMOLITION

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI Guideline K (2009) Guideline for Containers for Recovered Non-Flammable Fluorocarbon Refrigerants

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.6 (2006) Safety & Health Program Requirements for Demolition Operations - American National Standard for Construction and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health (SOH) Requirements

U.S. DEFENSE LOGISTICS AGENCY (DLA)

DLA 4145.25 (Jun 2000; Reaffirmed Oct 2010) Storage and Handling of Liquefied and Gaseous Compressed Gases and Their Full and Empty Cylinders
<http://www.aviation.dla.mil/UserWeb/aviationengineerir>

U.S. DEPARTMENT OF DEFENSE (DOD)

DOD 4000.25-1-M (2006) MILSTRIP - Military Standard Requisitioning and Issue Procedures

MIL-STD-129 (2014; Rev R) Military Marking for Shipment and Storage

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 61 National Emission Standards for Hazardous Air Pollutants

40 CFR 82 Protection of Stratospheric Ozone

49 CFR 173.301 Shipment of Compressed Gases in Cylinders and Spherical Pressure Vessels

1.2 PROJECT DESCRIPTION

1.2.1 Definitions

1.2.1.1 Demolition

Demolition is the process of wrecking or taking out any load-supporting structural member of a facility together with any related handling and disposal operations.

1.2.1.2 Demolition Plan

Demolition Plan is the planned steps and processes for managing demolition activities and identifying the required sequencing activities and disposal mechanisms.

1.2.2 Demolition Plan

Prepare a [Demolition Plan](#) and submit proposed demolition, and removal procedures for approval before work is started. Include in the plan procedures for careful removal and disposition of materials, coordination with other work in progress, a disconnection schedule of utility services, a detailed description of methods and equipment to be used for each operation and of the sequence of operations. Coordinate with Waste Management Plan in accordance with Section [01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL](#). Provide procedures for safe conduct of the work in accordance with [EM 385-1-1](#). Plan shall be approved by Contracting Officer prior to work beginning.

1.2.3 General Requirements

Do not begin demolition or deconstruction until authorization is received from the Contracting Officer. The work includes demolition and removal of resulting rubbish and debris. Remove rubbish and debris from Government property daily. Store materials that cannot be removed daily in areas specified by the Contracting Officer. In the interest of occupational safety and health, perform the work in accordance with [EM 385-1-1](#), Section 23, Demolition, and other applicable Sections.

1.3 ITEMS TO REMAIN IN PLACE

Take necessary precautions to avoid damage to existing items to remain in place, to be reused, or to remain the property of the Government. Repair or replace damaged items at no cost to the Government. Coordinate the work of this section with all other work indicated. Construct and maintain shoring, bracing, and supports as required. Ensure that structural elements are not overloaded. Increase structural supports or add new supports as may be required as a result of any cutting, removal, deconstruction, or demolition work performed under this contract. Do not overload structural elements. Provide new supports and reinforcement for existing construction weakened by demolition, deconstruction, or removal work. Repairs, reinforcement, or structural replacement require approval by the Contracting Officer prior to performing such work.

1.3.1 Gondola Cables

Provide a plan for temporarily displacing or removing, storing, and reinstalling the existing gondola cables. Include documentation of

coordination with the Municipality of San Juan. Damaged cables or equipment shall be replaced at no cost to the Government.

1.3.2 Existing Construction Limits and Protection

Do not disturb existing construction beyond the extent indicated or necessary for installation of new construction and for staging areas as indicated on the Drawings. Provide temporary shoring and bracing for support of building components to prevent settlement or other movement. Provide protective measures to control accumulation and migration of dust and dirt in all work areas. Remove dust, dirt, and debris from work areas daily.

1.3.3 Trees

Protect trees within the project site which might be damaged during demolition or deconstruction, and which are indicated to be left in place, by a 6 foot high construction safety fence. Erect and secure fence a minimum of 5 feet from the trunk of individual trees or follow the outer perimeter of branches or clumps of trees. Replace any tree designated to remain that is damaged during the work under this contract with like-kind or as approved by the Contracting Officer.

1.3.4 Utility Service

Maintain existing utilities indicated to stay in service and protect against damage during demolition and deconstruction operations. Prior to start of work, utilities serving each area of alteration or removal will be shut off by the Government and disconnected and sealed by the Contractor .

1.3.5 Facilities

Protect electrical and mechanical services and utilities. Where removal of existing utilities and pavement is specified or indicated, provide approved barricades, temporary covering of exposed areas, and temporary services or connections for electrical and mechanical utilities. Floors, roofs, walls, columns, pilasters, and other structural components that are designed and constructed to stand without lateral support or shoring, and are determined to be in stable condition, must remain standing without additional bracing, shoring, or lateral support until demolished or deconstructed, unless approved otherwise by the Contracting Officer. Ensure that no elements determined to be unstable are left unsupported and place and secure bracing, shoring, or lateral supports as may be required as a result of any cutting, removal, deconstruction, or demolition work performed under this contract.

1.4 BURNING

The use of burning at the project site for the disposal of refuse and debris will not be permitted.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Designation R0 stands for Resident Office and designation D0 stands for District Office. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Demolition Plan; G, R0

Existing Conditions

Gondola Cables; G, R0

Luma Equipment Inventory; G, R0

SD-07 Certificates

Notification; G, R0

SD-11 Closeout Submittals

Receipts

1.6 QUALITY ASSURANCE

Submit timely [notification](#) of demolition projects to Federal, Commonwealth, regional, and local authorities in accordance with [40 CFR 61](#), Subpart M. Notify the local air pollution control district/agency and the Contracting Officer in writing 10 working days prior to the commencement of work in accordance with [40 CFR 61](#), Subpart M. Comply with federal, commonwealth, and local hauling and disposal regulations. In addition to the requirements of the "Contract Clauses," conform to the safety requirements contained in [ASSP A10.6](#). Comply with the Environmental Protection Agency requirements specified. Use of explosives will not be permitted.

1.6.1 Dust and Debris Control

Prevent the spread of dust and debris and avoid the creation of a nuisance in the surrounding area. Do not use water if it results in hazardous or objectionable conditions such as, but not limited to, ice, flooding, or pollution.

1.7 PROTECTION

1.7.1 Traffic Control Signs

a. Where pedestrian and driver safety is endangered in the area of removal work, use traffic barricades with flashing lights. Notify the Contracting Officer prior to beginning such work.

1.7.2 Protection of Personnel

Before, during and after the demolition work continuously evaluate the condition of the structure being demolished and take immediate action to protect all personnel working in and around the project site. No area, section, or component of floors, roofs, walls, columns, pilasters, or other structural element will be allowed to be left standing without sufficient bracing, shoring, or lateral support to prevent collapse or failure while workmen remove debris or perform other work in the immediate area.

1.8 RELOCATIONS

Perform the removal and reinstallation of relocated items as indicated with workmen skilled in the trades involved. Repair or replace items to be relocated which are damaged by the Contractor at no cost to the Government with new undamaged items as approved by the Contracting Officer.

1.9 EXISTING CONDITIONS

Before beginning any demolition or deconstruction work, survey the site and examine the drawings and specifications to determine the extent of the work. Record existing conditions in the presence of the Contracting Officer showing the condition of structures and other facilities adjacent to areas of alteration or removal. Digital photographs reproduced at a size of 4 inches x 6 inches and a minimum of 300 dpi resolution will be acceptable as a record of existing conditions. Include in the record the elevation of the top of foundation walls, finish floor elevations, possible conflicting electrical conduits, plumbing lines, alarms systems, the location and extent of existing cracks and other damage and description of surface conditions that exist prior to before starting work. It is the Contractor's responsibility to verify and document all required outages which will be required during the course of work, and to note these outages on the record document. Submit site condition survey results.

1.9.1 LUMA EQUIPMENT INVENTORY

Submit an inventory of material or equipment owned by LUMA within the project limits which is to be demolished or salvaged. Provide equipment type, material information, quantity, and location of all equipment within the inventory. Notify LUMA and the Contracting Officer a minimum of 15 calendar days prior to the demolition or deconstruction of materials or equipment owned by LUMA.

PART 2 PRODUCTS

2.1 FILL MATERIAL

- a. Comply with excavating, backfilling, and compacting procedures for soils used as backfill material to fill basements, voids, depressions or excavations resulting from demolition or deconstruction of structures.
- b. Fill material shall conform to Section 31 00 00 EARTHWORK.

PART 3 EXECUTION

3.1 EXISTING FACILITIES TO BE REMOVED

3.1.1 Structures

- a. Existing bridge components including but not limited to, entire structure and channel lining shall be completely removed. Existing concrete piles shall be demolished to a minimum of 2 feet below the final channel invert.

- b. Remove existing structures indicated to be removed to the ground surface, including supporting slabs. Interior walls, other than retaining walls and partitions, shall be removed to 2 feet below grade. Break up basement slabs to permit drainage. Remove sidewalks, curbs, gutters and street light bases as indicated on the Drawings.
- c. Demolish structures in a systematic manner from the top of the structure to the ground. Complete demolition work above each tier or floor before the supporting members on the lower level are disturbed. Demolish concrete and masonry walls in small sections. Remove structural framing members and lower to ground by means of derricks, platforms hoists, or other suitable methods as approved by the Contracting Officer.
- d. Locate demolition and deconstruction equipment throughout the structure and remove materials so as to not impose excessive loads to supporting walls, floors, or framing.
- e. Building, or the remaining portions thereof, not exceeding 80 feet in height may be demolished by the mechanical method of demolition.

3.1.2 Utilities and Related Equipment

3.1.2.1 General Requirements

Do not interrupt existing utilities serving occupied or used facilities, except when authorized in writing by the Contracting Officer. Do not interrupt existing utilities serving facilities occupied and used by the Government except when approved in writing and then only after temporary utility services have been approved and provided. Do not begin demolition or deconstruction work until all utility disconnections have been made. Shut off and cap utilities for future use, as indicated.

3.1.2.2 Disconnecting Existing Utilities

Remove existing utilities , as indicated and terminate in a manner conforming to the nationally recognized code covering the specific utility and approved by the Contracting Officer. When utility lines are encountered but are not indicated on the drawings, notify the Contracting Officer prior to further work in that area. Remove meters and related equipment and deliver to a location in accordance with instructions of the Contracting Officer.

3.1.3 Chain Link Fencing

Remove chain link fencing, gates and other related items scheduled for removal and transport to designated areas. Remove gates as whole units. Cut chain link fabric to 25 foot lengths and store in rolls off the ground.

3.1.4 Paving and Slabs

Remove concrete and asphaltic concrete paving and slabs including aggregate base as indicated to a depth of 12 inches below existing adjacent new finish grade. Provide neat sawcuts at limits of pavement removal as indicated. Pavement and slabs designated to be recycled and utilized in this project shall be moved, ground and stored as approved by the Contracting Officer. Pavement and slabs not to be used in this project shall be removed from the Installation at Contractor's expense.

3.1.5 Concrete

Saw concrete along straight lines to a depth of a minimum 2 inch. Make each cut in walls perpendicular to the face and in alignment with the cut in the opposite face. Break out the remainder of the concrete provided that the broken area is concealed in the finished work, and the remaining concrete is sound. At locations where the broken face cannot be concealed, grind smooth or saw cut entirely through the concrete. Removed concrete debris, which meets project material requirements, designated to be recycled and utilized in this project shall be moved, ground and stored as approved by the Contracting Officer.

3.1.6 Structural Steel

Dismantle structural steel at field connections and in a manner that will prevent bending or damage. Flame-cutting torches are permitted when other methods of dismantling are not practical. Transport steel joists and girders as whole units and not dismantled. Transport structural steel shapes to a designated area as directed by the Contracting Officer, stacked according to size, type of member and length, and stored off the ground, protected from the weather.

3.1.7 Miscellaneous Metal

Recycle scrap metal as part of demolition and deconstruction operations. Provide separate containers to collect scrap metal and transport to a scrap metal collection or recycling facility, in accordance with the Waste Management Plan.

3.1.8 Carpentry

Remove windows, doors, frames, and cabinets, and similar items as whole units, complete with trim and accessories. Do not remove hardware attached to units, except for door closers. Brace the open end of door frames to prevent damage.

3.1.9 Acoustic Ceiling Tile

Remove, neatly stack, and recycle acoustic ceiling tiles. Recycling may be available with manufacturer. Otherwise, priority shall be given to a local recycling organization. Recycling is not required if the tiles contain or may have been exposed to asbestos material.

3.1.10 Patching

Where removals leave holes and damaged surfaces exposed in the finished work, patch and repair these holes and damaged surfaces to match adjacent finished surfaces, using on-site materials when available. Where new work is to be applied to existing surfaces, perform removals and patching in a manner to produce surfaces suitable for receiving new work. Finished surfaces of patched area shall be flush with the adjacent existing surface and shall match the existing adjacent surface as closely as possible as to texture and finish. Patching shall be as specified and indicated, and shall include:

- a. Concrete and Masonry: Completely fill holes and depressions, left as a result of removals in existing masonry walls to remain, with an approved masonry patching material, applied in accordance with the

manufacturer's printed instructions.

- b. Where existing partitions have been removed leaving damaged or missing resilient tile flooring, patch to match the existing floor tile.
- c. Patch acoustic lay-in ceiling where partitions have been removed. The transition between the different ceiling heights shall be effected by continuing the higher ceiling level over to the first runner on the lower ceiling and closing the vertical opening with a painted sheet metal strip.

3.1.11 Air Conditioning Equipment

Remove air conditioning, refrigeration, and other equipment containing refrigerants without releasing chlorofluorocarbon refrigerants to the atmosphere in accordance with the Clean Air Act Amendment of 1990.

3.1.12 Cylinders and Canisters

Remove all fire suppression system cylinders and canisters and dispose of in accordance with the paragraph entitled "Disposal of Ozone Depleting Substance (ODS)."

3.1.13 Locksets on Swinging Doors

Remove all locksets from all swinging doors indicated to be removed and disposed of. Deliver the locksets and related items to a designated location for receipt by the Contracting Officer after removal.

3.1.14 Mechanical Equipment and Fixtures

Disconnect mechanical hardware at the nearest connection to existing services to remain, unless otherwise noted. Disconnect mechanical equipment and fixtures at fittings. Remove service valves attached to the unit. Do not remove equipment until approved. Do not offer low-efficiency equipment for reuse.

3.1.14.1 Preparation for Storage

Remove water, dirt, dust, and foreign matter from units; tanks, piping and fixtures shall be drained; interiors, if previously used to store flammable, explosive, or other dangerous liquids, shall be steam cleaned. Seal openings with caps, plates, or plugs. Secure motors attached by flexible connections to the unit. Change lubricating systems with the proper oil or grease.

3.1.14.2 Piping

Disconnect piping at unions, flanges and valves, and fittings as required to reduce the pipe into straight lengths for practical storage. If the piping that remains can become pressurized due to upstream valve failure, end caps, blind flanges, or other types of plugs or fittings with a pressure gage and bleed valve shall be attached to the open end of the pipe to ensure positive leak control. Carefully dismantle piping that previously contained gas, gasoline, oil, or other dangerous fluids, with precautions taken to prevent injury to persons and property. Store piping outdoors until all fumes and residues are removed. Box prefabricated supports, hangers, plates, valves, and specialty items according to size and type. Wrap sprinkler heads individually in plastic bags before

boxing. Classify piping as scrap metal where applicable.

3.1.14.3 Ducts

Classify removed duct work as scrap metal.

3.1.14.4 Fixtures, Motors and Machines

Remove fixtures, motors and machines associated with plumbing, heating, air conditioning, refrigeration, and other mechanical system installations. Box and store auxiliary units and accessories with the main motor and machines. Classify broken, damaged, or otherwise unserviceable units and not caused to be broken, damaged, or otherwise unserviceable as debris to be disposed of by the Contractor.

3.1.15 Electrical Equipment and Fixtures

Remove motors, motor controllers, and operating and control equipment that are attached to the driven equipment. Remove wiring systems and components. Box loose items and tag for identification. Disconnect primary, secondary, control, communication, and signal circuits at the point of attachment to their distribution system.

3.1.15.1 Fixtures

Remove electrical fixtures. Remove unprotected glassware from the fixture separately. Remove incandescent, mercury-vapor, and fluorescent lamps and fluorescent ballasts manufactured prior to 1978, boxed and tagged for identification, and protected from breakage.

3.1.15.2 Electrical Devices

Remove switches, switchgear, transformers, conductors including wire and nonmetallic sheathed and flexible armored cable, regulators, meters, instruments, plates, circuit breakers, panelboards, outlet boxes, and similar items. Box and tag these items for identification according to type and size.

3.1.15.3 Wiring Ducts or Troughs

Remove wiring ducts or troughs. Dismantle plug-in ducts and wiring troughs into unit lengths. Remove plug-in or disconnecting devices from the busway and store separately.

3.1.15.4 Conduit and Miscellaneous Items

Remove conduit except where embedded in concrete or masonry. Consider corroded, bent, or damaged conduit as scrap metal. Sort straight and undamaged lengths of conduit according to size and type. Classify supports, knobs, tubes, cleats, and straps as debris to be removed and disposed.

3.2 CONCURRENT EARTH-MOVING OPERATIONS

Do not begin excavation, filling, and other earth-moving operations that are sequential to demolition or deconstruction work in areas occupied by structures to be demolished or deconstructed until all demolition and deconstruction in the area has been completed and debris removed. Fill holes, open basements and other hazardous openings.

3.3 DISPOSITION OF MATERIAL

3.3.1 Title to Materials

All materials and equipment removed, except as noted in the paragraph MATERIALS OWNED BY LUMA, shall become the property of the Contractor and shall be removed from Government property. Title to materials resulting from demolition and deconstruction, and materials and equipment to be removed, is vested in the Contractor upon approval by the Contracting Officer of the Contractor's demolition, deconstruction, and removal procedures, and authorization by the Contracting Officer to begin demolition and deconstruction. The Government will not be responsible for the condition or loss of, or damage to, such property after contract award. Showing for sale or selling materials and equipment on site is prohibited.

3.3.1.1 MATERIALS OWNED BY LUMA

All demolished or salvaged materials and equipment owned by LUMA shall be stockpiled onsite for material disposition to be done by LUMA. LUMA may elect to salvage material for reuse or recycling and will remain the property of LUMA. Removal of this specific material and equipment from the site will be completed by LUMA. Any remaining material and equipment shall become the property of the Contractor and shall be removed from Government property. For bidding purposes, assume all material will be disposed of by the Contractor. Refer to Section 02 42 00 DECONSTRUCTION OF ELECTRIC TRANSMISSION AND DISTRIBUTION UTILITY for additional information related to demolished or salvaged materials and equipment owned by LUMA.

3.3.2 Reuse of Materials and Equipment

Remove and store materials and equipment listed in the Demolition Plan to be reused or relocated to prevent damage, and reinstall as the work progresses. Coordinate the re-use of materials and equipment with the re-use requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture re-use of materials in the diversion calculations for the project. Items designated for reuse include:

1. Kiosk - Sta. 110+90, Right Channel Bank (Photo Attached)
2. Garbage Facility - Sta. 113+75, Left Channel Bank (Photo Attached)
3. Gondola Cables - Sta. 114+00, Crossing Channel (Photo Attached)

3.3.3 Disposal of Ozone Depleting Substance (ODS)

Class I and Class II ODS are defined in Section, 602(a) and (b), of The Clean Air Act. Prevent discharge of Class I and Class II ODS to the atmosphere. Place recovered ODS in cylinders meeting AHRI Guideline K suitable for the type ODS (filled to no more than 80 percent capacity) and provide appropriate labeling. Recovered ODS shall be removed from Government property and disposed of in accordance with 40 CFR 82. Products, equipment and appliances containing ODS in a sealed, self-contained system (e.g. residential refrigerators and window air conditioners) shall be disposed of in accordance with 40 CFR 82. Submit Receipts or bills of lading, as specified. Submit a shipping receipt or bill of lading for all containers of ozone depleting substance (ODS) shipped to the Defense Depot, Richmond, Virginia.

3.3.3.1 Special Instructions

No more than one type of ODS is permitted in each container. A warning/hazardous label shall be applied to the containers in accordance with Department of Transportation regulations. All cylinders including but not limited to fire extinguishers, spheres, or canisters containing an ODS shall have a tag with the following information:

- a. Activity name and unit identification code
- b. Activity point of contact and phone number
- c. Type of ODS and pounds of ODS contained
- d. Date of shipment
- e. National stock number (for information, call (804) 279-4525).

3.3.3.2 Fire Suppression Containers

Deactivate fire suppression system cylinders and canisters with electrical charges or initiators prior to shipment. Also, safety caps must be used to cover exposed actuation mechanisms and discharge ports on these special cylinders.

3.3.4 Transportation Guidance

Ship all ODS containers in accordance with MIL-STD-129, DLA 4145.25 (also referenced one of the following: Army Regulation 700-68, Naval Supply Instruction 4440.128C, Marine Corps Order 10330.2C, and Air Force Regulation 67-12), 49 CFR 173.301, and DOD 4000.25-1-M.

3.3.5 Non-Recyclable Material

Dispose of non-recyclable combustible material off the site.

3.4 CLEANUP

Remove debris and rubbish from basement and similar excavations. Remove and transport the debris in a manner that prevents spillage on streets or adjacent areas. Apply local regulations regarding hauling and disposal.

3.5 DISPOSAL OF REMOVED MATERIALS

3.5.1 Regulation of Removed Materials

Dispose of debris, rubbish, scrap, and other materials resulting from removal operations with all applicable federal, commonwealth and local regulations as contractually specified in the Waste Management Plan in Section 01 74 19.

3.5.2 Removal from Government Property

Transport waste materials removed from demolished structures, except waste soil, from Government property for legal disposal. Dispose of waste soil and excess excavated material at approved disposal sites or approved landfill.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 42 00

DECONSTRUCTION OF ELECTRIC TRANSMISSION AND DISTRIBUTION UTILITY

PART 1 GENERAL

- 1.1 DESCRIPTION
- 1.2 SUBMITTALS
- 1.3 RELATED DOCUMENTS
- 1.4 DIVISION 02 41 00 - DEMOLITION AND DECONSTRUCTION

PART 2 PRODUCTS

- 2.1 EQUIPMENT DISASSEMBLY
- 2.2 EQUIPMENT SEGREGATION
- 2.3 INVENTORY MAINTENANCE
- 2.4 RECYCLABLE MATERIALS
- 2.5 TRANSFORMER RECYCLING
- 2.6 CONCRETE OR METAL POLES

PART 3 EXECUTION

- 3.1 COORDINATION

-- End of Section Table of Contents --

SECTION 02 42 00

DECONSTRUCTION OF ELECTRIC TRANSMISSION AND DISTRIBUTION UTILITY

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies requirements for the disassembly, recycling, and reuse of electric power transmission and distribution equipment including both underground and overhead systems. The scope encompasses poles, transformers, conductors, underground components, overhead lines, and associated materials.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Demolition Plan; G

Deconstruction Plan; G

Existing Conditions

SD-07 Certificates

Notification; G

Notification of Demolition and Renovation Form

SD-11 Closeout Submittals

Receipts

Submit the following to the Contracting Officer for approval:

- a. Inventory of removed equipment for underground or overhead systems.
- b. Coordination plan for the pickup of concrete or steel poles, transformers, and underground and overhead components according to LUMA's Assets Recovery SOP.

1.3 RELATED DOCUMENTS

Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this section. Refer also to LUMA's Assets Recovery SOP for additional guidance.

1.4 DIVISION 02 41 00 - DEMOLITION AND DECONSTRUCTION

Comply with the requirements of Section 02 41 00 for demolition and deconstruction activities related to the removal of electric power transmission and distribution equipment.

PART 2 PRODUCTS

2.1 EQUIPMENT DISASSEMBLY

Completely disassemble all poles including, but not limited to, cross arms, bolts, fuse cutouts, insulators, transformers, overhead lines, and underground components.

2.2 EQUIPMENT SEGREGATION

Segregate equipment for disposal, recycling, or reuse categorizing them by material type, such as aluminum, metal, concrete, wood, and differentiate between overhead and underground components. As the wood materials are neither recyclable nor reusable, dispose of them.

2.3 INVENTORY MAINTENANCE

Maintain an inventory of all removed equipment including, but not limited to, poles, transformers, conductors, and underground components.

2.4 RECYCLABLE MATERIALS

Designate a sorting area for recyclable materials such as aluminum, copper, and metals.

Process recyclable materials through LUMA's recycling program.

LUMA will provide a dumpster for recyclable materials and equipment collection once all materials are segregated by material type and ready for collection. Coordinate with LUMA's assets recovery office for the dumpster.

2.5 TRANSFORMER RECYCLING

Place transformers inside a special transformer containment bag and a small containment area for recycling. Transformer bags should safely contain both non-leaking and leaking transformers for transport. The contractor is responsible of acquiring the special transformer containment bags. LUMA will not provide special transformer containment bags.

2.6 CONCRETE OR METAL POLES

Coordinate with LUMA for the pickup of concrete or metal poles.

PART 3 EXECUTION

3.1 COORDINATION

- a. Coordinate with LUMA for the pickup of recyclable materials and equipment.

LUMA Asset Recovery

Contact: Giovanni Taffanelli, Supervisor
Email: AR@lumapr.com
Phone: (787)-349-6931

- b. Ensure that all materials are disassembled and ready for pickup before LUMA's scheduled collection.
- c. Comply with LUMA's Assets Recovery SOP for recycling transformers and coordinating pickup.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 03 - CONCRETE

SECTION 03 11 13

STRUCTURAL CAST-IN-PLACE CONCRETE FORMING

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
- 2.2 FORM MATERIALS
 - 2.2.1 Forms For Class B Finish
 - 2.2.2 Forms For Class D Finish
 - 2.2.3 Stay-In-Place Metal Bridge Deck Forms
 - 2.2.4 Form Ties
 - 2.2.5 Form Releasing Agents

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Formwork
 - 3.1.2 Stay-In-Place Metal Bridge Deck Forms
- 3.2 CHAMFERING
- 3.3 COATING
- 3.4 FORM REMOVAL
 - 3.4.1 Formwork Not Supporting Weight of Concrete
 - 3.4.2 Formwork Supporting Weight of Concrete
- 3.5 INSPECTION
 - 3.5.1 Formwork
 - 3.5.2 Stay-In-Place Metal Bridge Deck Forms

-- End of Section Table of Contents --

SECTION 03 11 13

STRUCTURAL CAST-IN-PLACE CONCRETE FORMING

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE INTERNATIONAL (ACI)

ACI 347 (2004; Errata 2008; Errata 2012) Guide to Formwork for Concrete

APA - THE ENGINEERED WOOD ASSOCIATION (APA)

APA L870 (2010) Voluntary Product Standard, PS 1-09, Structural Plywood

ASTM INTERNATIONAL (ASTM)

ASTM C272/C272M (2016) Standard Test Method for Water Absorption of Core Materials for Sandwich Constructions

ASTM C1077 (2017) Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation

ASTM A653/A653M (2020) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A742/A742M (2018) Standard Specification for Steel Sheet, Metallic Coated and Polymer Precoated for Corrugated Steel Pipe

ASTM A762/A762M (20015) Standard Specification for Corrugated Steel Pipe, Polymer Precoated for Sewers and Drains

ASTM C31/C31M (2019a) Standard Practice for Making and Curing Concrete Test Specimens in the Field

ASTM C39/C39M (2020) Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens

ASTM D520 (2000; R 2011) Zinc Dust Pigment

ASTM D3895

(2014) Oxidative-Induction Time of
Polyolefins by Differential Scanning
Calorimetry

COMMONWEALTH OF PUERTO RICO, DEPARTMENT OF TRANSPORTATION AND
PUBLIC WORKS (PRHTA)

PRHTA Specification

(2005) Standard Specifications for Road
and Bridge Construction

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC Paint 20

(2002; E 2004) Zinc-Rich Primers (Type I,
Inorganic, and Type II, Organic)

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Designation R0 stands for Resident Office and designation D0 stands for District Office. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Formwork; G, D0

Drawings showing details of formwork, including dimensions of fiber voids, joints, supports, studding and shoring, formwork materials, and sequence of form and shoring removal. At least 30 days either before fabrication on site or before delivery of prefabricated forms.

Stay-In-Place Metal Bridge Deck Forms; G, R0, D0

Drawings and calculations signed and sealed by a licensed professional engineer in Puerto Rico showing the deck form configuration, material, section properties, specifications, and method of support and attachment to bridge beams.

SD-03 Product Data

Form Materials

Manufacturer's data including literature describing form materials, accessories, and form releasing agents.

Form Releasing Agents

Manufacturer's recommendation on method and rate of application of form releasing agents.

SD-05 Design Data

Calculations; G, R0, D0

Design analysis and calculations, signed and sealed by a licensed professional engineer in Puerto Rico, for form design and

methodology used in the design. At least 30 days either before fabrication on site or before delivery of prefabricated forms.

SD-06 Test Reports

Inspection

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

The design, engineering, and construction of the formwork is the responsibility of the Contractor. Design formwork in accordance with methodology of [ACI 347](#) for anticipated loads, lateral pressures, and stresses. Formwork shall be capable of withstanding the pressures resulting from placement and vibration of concrete. Comply with the tolerances specified in Section [03 31 01](#) CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS, paragraph CONSTRUCTION TOLERANCES. Design the formwork as a complete system with consideration given to the effects of cementitious materials and mixture additives such as fly ash, cement type, plasticizers, accelerators, retarders, air entrainment, and others. Monitor the adequacy of formwork design and construction prior to and during concrete placement as part of the Contractor's approved Quality Control Plan. Submit design analysis and [calculations](#), signed and sealed by a licensed professional engineer in Puerto Rico, for form design and methodology used in the design. Submit at least 30 days either before fabrication on site or before delivery of prefabricated forms.

2.2 FORM MATERIALS

Submit manufacturer's data, including literature describing form materials, accessories, and form releasing agents.

2.2.1 Forms For Class B Finish

This class of finish shall apply to all formed surfaces except those specified to receive Class D. Forms for Class B finished surfaces shall be plywood panels conforming to [APA L870](#), Grade B-B concrete form panels, Class I or II. Other form materials or liners may be used provided the smoothness and appearance of concrete produced will be equivalent to that produced by the plywood concrete form panels. Forms for round columns shall be the prefabricated seamless type. Steel lining on wood sheathing will not be permitted.

2.2.2 Forms For Class D Finish

Forms for Class D finished surfaces shall be wood or steel or other approved concrete form material.

2.2.3 Stay-In-Place Metal Bridge Deck Forms

Use of stay-in-place metal forms is permitted in lieu of removable forms to form concrete bridge decks between beams when designated in the Plans. Stay-in-place metal forms must meet or exceed the requirements of [PRHTA Specification](#) Sections 601 and 715. Stay-in-place metal forms may be of the cellular, non-cellular or non-cellular with top cover sheet type. The flutes of non-cellular stay-in-place metal forms may be filled with polystyrene foam or concrete. When polystyrene foam is used to fill the forms, fill form flutes completely; do not allow any portion of the

polystyrene foam to extend beyond the limits of the flutes. Ensure that the polystyrene foam remains in its required position within flutes during the entire concrete placement process. Do not use reinforcing supports or other accessories in such a manner as to cause damage to the polystyrene foam. Replace all damaged polystyrene foam to the satisfaction of the Contracting Officer.

Apply polymer sheeting to stay-in-place metal forms in accordance with the requirements in the following table. Apply polymer sheeting to all faces and edges (including sheared edges) of support angles used on bridges with Moderately and Extremely Aggressive Superstructure Environmental Classifications (as shown in the Plans). No polymer sheeting is required for beam attachment straps or clips partially embedded in concrete, and for support angles used on bridges with a Slightly Aggressive Superstructure Environmental Classification. Use polymer sheeting materials and application methods as described herein.

Polymer Sheeting Usage Requirements				
Form Type		Superstructure Environmental Classification (as shown in Plans)		
		Slightly Aggressive	Moderately Aggressive	Extremely Aggressive
Non-cellular form with concrete filled flutes		No polymer sheeting required	Polymer sheeting required on bottom side	Polymer sheeting required on bottom side
Non-cellular form with polystyrene foam filled flutes		Polymer sheeting required on top side ¹	Polymer sheeting required on both sides ¹	Polymer sheeting required on both sides ¹
Non-cellular form with Top Cover Sheet	Top Cover Sheet	Polymer sheeting required on bottom side	Polymer sheeting required on bottom side	Polymer sheeting required on bottom side
	Non-cellular form	Polymer sheeting required on top side	Polymer sheeting required on both sides	Polymer sheeting required on both sides
Cellular form		No polymer sheeting allowed or required	Cellular form not permitted	Cellular form not permitted
¹ Polymer sheeting not required on top side of form when foam filled flutes are used only at interior supports on continuous decks and the remainder of the flutes are concrete filled.				

Prior to using stay-in-place metal forms, submit detailed plans for approval of the forming system, including method of support and attachment. Submit design calculations for the forming system, which have been signed and sealed by the Specialty Engineer. Detail stay-in-place metal forms such that they in no way infringe upon the concrete outline of the slab shown on the Plans. Use stay-in-place metal forms that provide and maintain the dimensions and configuration of the original slab regarding thickness and slope.

Do not connect polymer coated angles or other hardware that support polymer coated metal forms to the beam attachment straps or clips by welding. Electrical grounding to steel reinforcing or fiber reinforced polymer (FRP) reinforcing is prohibited.

Do not use stay-in-place metal forms until the forming system has been approved by the Contracting Officer. The Contractor is responsible for the performance of the stay-in-place forms.

Prior to using stay-in-place metal forms, submit detailed plans for approval of the forming system (including method of support and attachment) together with design calculations. Include an analysis of the actual unit weight of the proposed forming system over the projected plan area of the metal forms. If the weight thus calculated exceeds the weight allowance for stay-in-place metal forms and concrete required to fill the forms shown on the Plans, then modify the supporting components to support the excess weight as specified by the Contractor's Specialty Engineer.

For non-polymer sheeting form surfaces, use zinc paint coating to all accessories cut from galvanized sheets, which are not embedded in concrete. For areas less than or equal to 100 square inches, use a cold galvanizing compound containing at least 92% zinc dust in the dry film. For areas greater than 100 square inches, use a galvanizing compound with zinc dust pigment meeting the minimum requirements of Type II in accordance with [ASTM D520](#) and organic zinc rich primers shall meet the requirements of [SSPC Paint 20](#), Type II, Level 2.

Meet the following criteria for the design of stay-in-place bridge deck forms:

- a. The maximum self-weight of the stay in place metal forms, plus the weight of the concrete or expanded polystyrene required to fill the form flutes (where used), shall not exceed [20 psf](#).
- b. Design the forms on the basis of dead load of form, reinforcement, and plastic concrete plus [50 pounds per square foot](#) for construction loads. Use a unit working stress in the steel sheet of not more than 0.725 of the specified minimum yield strength of the material furnished, but not to exceed [36,000 psi](#).
- c. Do not allow deflection under the weight of the forms, reinforcement, and plastic concrete to exceed 1/180 of the form span or 1/2 inch, whichever is less, for form spans of 10 feet or less, or 1/240 of the form span or 3/4 inch, whichever is less, for form spans greater than 10 feet. In all cases, do not use a total loading (psf) that is less than 20 plus the product of the deck thickness measured in inches times 12.5.
- d. Use a design span of the form equal to the clear span of the form plus 2 inches. Measure the span parallel to the form flutes.
- e. Compute physical design properties in accordance with requirements of the AISI Specifications for the Design of Cold Formed Steel Structural Members, latest published edition.
- f. For all reinforcement, maintain the design concrete cover required by the Plans.

- g. Maintain the plan dimensions of both layers of primary deck reinforcement from the top surface of the concrete deck.
- h. Do not consider the permanent bridge deck form as lateral bracing for compression flanges of supporting structural members.
- i. Do not use permanent steel bridge deck forms in panels where longitudinal deck construction joints are located between stringers.
- j. Secure forms to the supporting members by means other than welding directly to the member.

Fabricate stay-in-place metal forms and supports from steel meeting the requirements of [ASTM A653/A653M](#) having a coating designation G165. Do not use form materials that are less than 0.03 inch uncoated thickness.

Use polymer sheeting comprised of at least 85% ethylene acrylic acid copolymer capable of being applied to both G165 and G210 steel sheet as described in [ASTM A742/A742M](#). Ensure that the polymer sheeting has a nominal thickness of 12 mils as manufactured and a minimum thickness of 10 mils after lamination to the steel sheet.

Ensure that the polymer sheeting remains free of holes, tears and discontinuities and sufficiently flexible to withstand the forming process without any detrimental effects to bond, durability or performance. Ensure that the polymer sheeting is UV stabilized and contains antioxidants.

Ensure that the as-manufactured polymer sheeting (prior to application) has an Oxidative Induction Time (OIT) of 60 to 75 minutes at 170°C in air when tested according to [ASTM D3895](#). Perform additional OIT tests on samples taken from the finished product (polymer sheeting applied to forms) resulting in a minimum OIT according to [ASTM D3895](#) of 32 minutes at 170°C in air. Ensure that the polymer sheeting adheres to galvanized metal sufficient to prevent undercutting at penetrations made through the polymer sheeting or metal forms to the satisfaction of the Contracting Officer. Ensure that edges subjected to shear cutting are coated by the form manufacturer with two coats of a compatible liquid coating repair material before delivery to the site. Ensure that steel used to produce polymer laminated metal forms is appropriately cleaned and prepared per NCCA (National Coil Coating Association) standard continuous coil coating practices. Ensure that pretreatment for use in conjunction with the manufacturer's polymer sheeting material is approved as compatible by the polymer sheeting manufacturer. Apply pretreatment in accordance with the polymer sheeting manufacturer's procedures. Apply polymer sheeting in accordance with the manufacturer's recommendations and procedures. Ensure that all steel has the polymer sheeting applied prior to fabrication of the stay-in-place forms and accessories.

Ensure that the screws to be used in the fastening of the stay-in-place laminated metal forms have a corrosion resistant cladding that will not have an adverse effect to the system due to the contact of dissimilar metals.

Submit a written certification from the manufacturer stating the product meets the requirements of this specification along with the delivery of the coated forms to the jobsite.

Use polystyrene foam comprised of expanded polystyrene manufactured from virgin resin of sufficient density to support the weight of concrete

without deformation. Extrude the polystyrene foam to match the geometry of the flutes and provide a snug fit. Use polystyrene foam that has a density of not less than 0.8 pounds per cubic foot. Use polystyrene foam that has water absorption of less than 2.6% when tested according to ASTM C272/C272M. Submit a written certification from the manufacturer stating the product meets the requirements of this Specification along with the delivery of the product.

2.2.4 Form Ties

Form ties shall be factory-fabricated metal ties, shall be of the removable or internal disconnecting or snap-off type, and shall be of a design that will not permit form deflection and will not spall concrete upon removal. Provide solid backing for each tie. Except where removable tie rods are used, ties shall not leave holes in the concrete surface less than 1/4 inch nor more than 1 inch deep and not more than 1 inch in diameter. Terminate the embedded portion of metal ties not less than 2 inches from any concrete surface exposed to water. Removable tie rods shall be not more than 1-1/2 inches in diameter. Plastic snap ties may be used in locations where the surface will not be exposed to view.

2.2.5 Form Releasing Agents

Form releasing agents shall be commercial formulations that will not bond with, stain or adversely affect concrete surfaces. Agents shall not impair subsequent treatment of concrete surfaces depending upon bond or adhesion nor impede the wetting of surfaces to be cured with water or curing compounds. If special form liners are to be used, follow the recommendation of the form coating manufacturer. Submit manufacturer's recommendation on method and rate of application of form releasing agents.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Formwork

Forms shall be constructed true to the structural design and required alignment. Forms shall be mortar tight, properly aligned and adequately supported to produce concrete surfaces meeting the surface requirements specified in Section 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS and conforming to construction tolerance given in TABLE 1 of this specification. Continuously monitor the alignment and stability of the forms during all phases to assure the finished product will meet the required surface class specified. Failure of any supporting surface either due to surface texture, deflection or form collapse shall be the responsibility of the Contractor as will the replacement or correction of unsatisfactory surfaces. Where concrete surfaces are to have a Class B finish, joints in form panels shall be arranged as approved. When forms for continuous surfaces are placed in successive units, care shall be taken to fit the forms over the completed surface to obtain accurate alignment of the surface and to prevent leakage of mortar. Forms shall not be re-used if there is any evidence of defects which would impair the quality of the resulting concrete surface. All surfaces of used forms shall be cleaned of mortar and any other foreign material before reuse. Form ties that are to be completely withdrawn shall be coated with a nonstaining bond breaker. Submit drawings showing details of formwork, including dimensions of fiber voids, joints, supports, studding and shoring, and sequence of form and shoring removal.

3.1.2 Stay-In-Place Metal Bridge Deck Forms

Install all forms in accordance with approved fabrication and erection plans.

Do not rest form sheets directly on the top of the stringer or floor beam flanges. Fasten sheets securely to form supports and maintain a minimum bearing length of 1 inch at each end for metal forms. Place form supports in direct contact with the flange of the stringer or floor beam. Make all attachments for coated metal forms by bolts, clips, screws, or other approved means.

For any permanent exposed steel where the galvanized coating has been damaged, thoroughly clean, wire brush, and paint it with two coats of galvanizing compound to the satisfaction of the Contracting Officer. For areas less than or equal to 100 square inches, use a cold galvanizing compound containing at least 92% zinc dust in the dry film. For areas greater than 100 square inches, use a galvanizing compound with zinc dust pigment meeting the minimum requirements of Type II in accordance with [ASTM D520](#) and organic zinc rich primers shall meet the requirements of [SSPC Paint 20](#), Type II, Level 2. Do not touch up minor heat discoloration in areas of welds.

Inspect and identify areas for damage to the polymer sheeting and repair with liquid polymer coating similar and compatible with respect to durability, adhesion and appearance in accordance with [ASTM A762/A762M](#), as furnished by the stay-in-place form manufacturer. Ensure that the inspection includes checking the polymer sheeting for cuts, tears, cracking, surface pits, peeling, dirt, grease, oil, stains, rust or bare areas. Reject any panels that show coating blistering, peeling or cracking. Repair all polymer sheeting damage according to the following:

- a. Surface Preparation: Ensure that all surfaces to be repaired are clean and free of any deleterious substances. Remove all traces of dirt, soil, oil deposits, greases, and other surface contaminants in accordance with the polymer sheeting and coating manufacturer's written specifications prior to touch-up and recoating.
- b. Application Procedures: Ensure that the liquid polymer repair coating is applied to a clean dry surface and in accordance with the manufacturer's written specifications. Apply the repair coating using a suitable paintbrush or other means acceptable to the Contracting Officer. Apply a first coat of product to the surface at 2-4 mils in thickness. Let the first coat air dry. Apply a second coat to form a complete layer and increase the thickness, immediately after verifying the first coat is dry to the touch (15 - 25 minutes depending on the local air-drying temperature and atmospheric conditions). Apply the second coat at the same coating thickness as the first at 2-4 mils. Ensure that the total dry film thickness of the two coats is not less than 6 mils. Apply additional coats in this same manner until desired coating thickness is achieved.

Vibrate concrete to avoid honeycomb and voids, especially at construction joints, expansion joints, valleys and ends of form sheets. Use approved pouring sequences. Do not use calcium chloride or any other admixture containing chloride salts in the concrete.

3.2 CHAMFERING

All exposed joints, edges and external corners shall be chamfered by molding placed in the forms unless the drawings specifically state that chamfering is to be omitted or as otherwise specified. Chamfered joints shall not be permitted where earth or rockfill is placed in contact with concrete surfaces. Chamfered joints shall be terminated **twelve inches** outside the limit of the earth or rockfill so that the end of the chamfers will be clearly visible.

3.3 COATING

Forms for Class B finished surfaces shall be coated with a form releasing agent before the form or reinforcement is placed in final position. The coating shall be used as recommended in the manufacturer's printed or written instructions. Forms for Class D finished surfaces may be wet with water in lieu of coating immediately before placing concrete. Surplus coating on form surfaces and coating on reinforcing steel and construction joints shall be removed before placing concrete.

3.4 FORM REMOVAL

Forms shall not be removed without approval. The minimal time required for concrete to reach a strength adequate for removal of formwork without risking the safety of workers or the quality of the concrete depends on a number of factors including, but not limited to, ambient temperature, concrete lift heights, type and amount of concrete admixture, and type and amount of cementitious material in the concrete. It is the responsibility of the Contractor to consider all applicable factors and leave the forms in place until it is safe to remove them. In any case forms shall not be removed unless the minimum time, requirements below are met. When conditions dictate, forms shall remain in place for a longer period. All removal shall be accomplished in a manner which will prevent damage to the concrete and ensure the complete safety of the structure. Where forms support more than one element, the forms shall not be removed until the form removal criteria are met by all supported elements. Form removal shall be scheduled so that any necessary repairs can be performed. Evidence that concrete has gained sufficient strength to permit removal of forms shall be determined by tests on control cylinders. All control cylinders shall be stored in the structure or as near the structure as possible so they receive the same curing conditions and protection methods as given those portions of the structure they represent. Control cylinders shall be removed from the molds at an age of no more than 24 hours. All control cylinders shall be prepared and tested in accordance with **ASTM C31/C31M** and **ASTM C39/C39M** at the expense of the Contractor by an independent laboratory that complies with **ASTM C1077** and shall be tested within 4 hours after removal from the site.

3.4.1 Formwork Not Supporting Weight of Concrete

Formwork for sides of cap beam and other vertical type formwork not supporting the weight of concrete shall not be removed in less than 24 hours after concrete placement is completed.

3.4.2 Formwork Supporting Weight of Concrete

Formwork supporting weight of concrete and shoring shall not be removed until structural members have acquired sufficient strength to safely support their own weight and any construction or other superimposed loads

to which the supported concrete may be subjected. As a minimum, forms shall be left in place until control concrete test cylinders or maturity instrumentation indicate evidence the concrete has attained at least 70 percent of the compressive strength required for the structure in accordance with the quality and location requirements. Such evidence shall be submitted to the contracting officer for approval prior to removing formwork.

3.5 INSPECTION

3.5.1 Formwork

Inspect forms and embedded items in sufficient time prior to each concrete placement to certify to the Contracting Officer that they are ready to receive concrete. Report the results of each inspection in writing. Submit field inspection reports for concrete forms and embedded items.

TABLE 1 TOLERANCES FOR FORMED SURFACES	
1. Variations from the plumb:	
a. In the lines and surfaces of columns, walls and in arises	1/4 inch in any 10 feet of length Maximum for entire length -- 1 inch
b. For exposed corner columns, control-joint grooves, and other conspicuous lines	1/4 inch in any 20 feet of length Maximum for entire length -- 1/2 inch
2. Variation from the level or from the grades indicated on the drawings:	
a. In exposed lintels, sills, cap beams, parapets, horizontal grooves, slabs, and other conspicuous lines	1/4 inch in any bay or in any 20 feet of length Maximum for entire length -- 1/2 inch
3. Variation of distance between walls, columns and partitions	1/4 inch per 10 feet of distance, but not more than one bay, and not more than 1 inch total variation
4. Variation in the sizes and locations of sleeves, cap beam opening and scuppers	Minus 1/4 inch, Plus 1/2 inch
5. Variation in cross-sectional dimensions of columns, cap beams, drilled shafts and in the thickness of slabs, walls and parapets	Minus 1/4 inch, Plus 1/4 inch
6. Footings:	

TABLE 1 TOLERANCES FOR FORMED SURFACES	
a. Variation of dimensions in plan	Minus 1/2 inch, plus 2 inches when formed or plus 3 inches when placed against unformed excavation
b. Misplacement of eccentricity	2 percent of the footing width in the direction of misplacement but not more than 2 inches
c. Reduction in thickness	Minus 5 percent of the specified thickness

3.5.2 Stay-In-Place Metal Bridge Deck Forms

The Contracting Officer will observe the Contractor's method of construction during all phases of the construction of the bridge deck slab, including the installation of the metal form system; location and fastening of the reinforcement; composition of concrete items; mixing procedures, concrete placement, and vibration; and finishing of the bridge deck. Should the Contracting Officer determine that the procedures used during the placement of the concrete warrant inspection of the underside of the deck, remove at least one section of the metal forms in each span for this purpose. Do this as soon after placing the concrete as practicable in order to provide visual evidence that the concrete mix and the procedures are obtaining the desired results. Remove an additional section in any span if the Contracting Officer determines that there has been any change in the concrete mix or in the procedures warranting additional inspection.

If, in the Contracting Officer's judgment, inspection is needed to check for defects in the bottom of the deck or to verify soundness, sound the metal forms with a hammer after the deck concrete has been in place a minimum of two days. If sounding discloses areas of doubtful soundness to the Contracting Officer, remove the metal forms from such areas for visual inspection after the concrete has attained adequate strength. Remove metal bridge deck forms at the expense of the Contractor.

At locations where sections of the metal forms have been removed, the Contracting Officer will not require the Contractor to replace the metal forms. Repair the adjacent metal forms and supports to present a neat appearance and to ensure their satisfactory retention and where they are polymer sheeted, coat all exposed surfaces of stay-in-place metal form system elements that are not coated or are damaged with a field applied liquid polymer coating as specified in 3.1.2. As soon as the form is removed, the Contracting Officer will examine the concrete surfaces for cavities, honeycombing, and other defects. If irregularities are found, and the Contracting Officer determines that these irregularities do not justify rejection of the work, repair the concrete as approved, and provide a Class B surface finish in accordance with Section 03 35 00 CONCRETE FINISHING Paragraph 3.1.1. If the Contracting Officer determines that the concrete where the form is removed is unsatisfactory, remove additional metal forms as necessary to inspect and repair the slab, and modify the method of construction as required to obtain satisfactory concrete in the slab. Remove and replace all unsatisfactory concrete as approved, at the expense of the Contractor.

Corrosion of assembly screws will not be considered a structural or aesthetic problem and is considered acceptable.

Provide the facilities for the safe and convenient conduct of the inspection Procedures.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 03 - CONCRETE

SECTION 03 15 00

CONCRETE ACCESSORIES

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 DELIVERY, STORAGE, AND HANDLING

PART 2 PRODUCTS

- 2.1 PREFORMED JOINT FILLER
- 2.2 WATERSTOPS
 - 2.2.1 Hydrophilic Waterstop
 - 2.2.2 Splicing Waterstops
 - 2.2.2.1 Procedure and Performance Qualifications
 - 2.2.2.2 Hydrophilic Waterstops
- 2.3 BRIDGE BEARING PADS
- 2.4 BRIDGE JOINT SEAL
- 2.5 BRIDGE WATERPROOFING

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Contraction Joints
 - 3.1.2 WATERSTOP INSTALLATION AND SPLICING
 - 3.1.2.1 Hydrophilic Waterstop
 - 3.1.3 Expansion Joints
 - 3.1.4 Bridge Joint Seal
 - 3.1.5 Bridge Bearing
 - 3.1.6 Bridge Waterproofing
- 3.2 CONSTRUCTION JOINTS

-- End of Section Table of Contents --

SECTION 03 15 00
CONCRETE ACCESSORIES

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A1011/A1011M	(2018a) Standard Specification for Steel Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength
ASTM D545	(2019) Standard Test Methods for Preformed Expansion Joint Fillers for Concrete Construction (Nonextruding and Resilient Types)
ASTM D1751	(2004; E 2013; R 2013) Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types)
ASTM D1752	(2018) Standard Specification for Preformed Sponge Rubber, Cork and Recycled PVC Expansion Joint Fillers for Concrete Paving and Structural Construction
ASTM D146/D146M	(2004; E 2012; R 2012) Sampling and Testing Bitumen-Saturated Felts and Woven Fabrics for Roofing and Waterproofing
ASTM D412	(2016) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension
ASTM D3574	(2017) Standard Test Methods for Flexible Cellular Materials—Slab, Bonded, and Molded Urethane Foams
ASTM E96/E96M	(2016) Standard Test Methods for Water Vapor Transmission of Materials
ASTM E154/E154M	(2008a; R 2013; E 2013) Water Vapor

Retarders Used in Contact with Earth Under
Concrete Slabs, on Walls, or as Ground
Cover

ASTM E1399/E1399M

(1997; R 2017) Standard Test Method for
Cyclic Movement and Measuring the Minimum
and Maximum Joint Widths of Architectural
Joint Systems

ASTM G155

(2013) Standard Practice for Operating
Xenon Arc Light Apparatus for Exposure of
Non-Metallic Materials

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 6/NACE No.3

(2007) Commercial Blast Cleaning

COMMONWEALTH OF PUERTO RICO, DEPARTMENT OF TRANSPORTATION AND
PUBLIC WORKS (PRHTA)

PRHTA Specifications

(2005) Standard Specifications for Road
and Bridge Construction

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where PRHTA Specifications Section, and/or Special Provision(s) is cited that contains references to other PRHTA sections, they shall also form a part of this specification to the extent referenced. Where reference is made to "Highway Authority" or "Authority's Representative" in PRHTA Specifications shall mean "Government" or "Contracting Officer" as defined in the Contract Documents. Where PRHTA Specifications refer to the "Engineer", "Engineer of Tests", or "Division of Tests" it shall be understood to mean the "Contracting Officer" as stated in the Contract Documents. In case of conflict between the PRHTA specifications and the Specifications in this Section that follows, the Specifications in this Section shall govern.

Paragraphs title "Method of Measurement" and "Basis of Payment" of referenced PRHTA Specifications shall not apply.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Designation R0 stands for Resident Office and DO stands for District Office. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Preformed Joint Filler; G, R0

Splicing Waterstops; G, R0

Procedures for splicing waterstops shall be submitted for approval.

Waterstops; G, R0

Bridge Bearing Pads; G, R0

Submit a manufacturer's certification that all components of the pads furnished conform to the applicable requirements of the contract documents and this specification. The certification shall be supported by a certified copy of the tests performed upon components representative of those actually used on the pads furnished.

Bridge Joint Seal; G, R0

Submit product datasheets, performance test reports from an independent laboratory showing the product meets the requirements of this Section, PRHTA Specifications Section 601, and Contract Plans, and method of installation of the selected product.

Curing Compound Or Bituminous Paint; G, R0

Waterproofing Membrane Material; G, R0

Submit product datasheets, performance test reports from an independent laboratory showing the product meets the requirements of this Section, PRHTA Specifications Section 601, and Contract Plans, and method of installation of the selected product.

SD-04 Samples

Splicing Waterstops; G, R0

1.3 DELIVERY, STORAGE, AND HANDLING

Protect material delivered and placed in storage off the ground from moisture, dirt, and other contaminants.

PART 2 PRODUCTS

2.1 PREFORMED JOINT FILLER

Use preformed joint filler material conforming to ASTM D1751 or ASTM D1752, Type I, or resin impregnated fiberboard conforming to the physical requirements of ASTM D1752.

Submit manufacturer's data sheet on preformed joint filler.

2.2 WATERSTOPS

Submit manufacturer's data sheet on waterstop materials and splices.

2.2.1 Hydrophilic Waterstop

Swellable strip type compound of polymer modified chloroprene rubber that swells upon contact with water must conform to the following requirements when tested in accordance to ASTM D412: Tensile strength 420 psi minimum; ultimate elongation 600 percent minimum. Hardness must be 50 minimum on the type A durometer and the volumetric expansion ratio in distilled water at 70 degrees F must be 3 to 1 minimum.

2.2.2 Splicing Waterstops

2.2.2.1 Procedure and Performance Qualifications

Demonstrate procedure and performance qualifications for splicing waterstops in the presence of the Contracting Officer. Submit procedures for splicing waterstops for approval.

2.2.2.2 Hydrophilic Waterstops

Procedure and performance qualifications for splicing hydrophilic waterstops shall be demonstrated by the manufacturer at the factory and the Contractor at the job site by each making three spliced samples of each type and size of finished waterstop specified in the plans.

2.3 BRIDGE BEARING PADS

Bridge bearing pads shall meet the material and sampling requirements of **PRHTA Specifications** Section 717 ELASTOMERIC BEARINGS with the following modifications:

- a. The steel layers of the laminated pads shall utilize 10 gauge steel sheet (**0.1345 inches** thick). The steel utilized for the steel layers and for external load bearing plates (if specified) shall meet the requirements of **ASTM A36/A36M** or **ASTM A1011/A1011M** Grade 36 Type I steel sheet. External load bearing plates shall be finished or machined flat to within 0.01 inches. The bottom surfaces of external load plates (masonry plates) designed to rest on bearing pads shall not exceed an out of flatness value of **0.0625 inches**. External load bearing plates shall be protected from rust until all exposed surfaces can be field painted.
- b. The dimensional tolerance for the variation from a plane parallel to the theoretical surface (as determined by measurements at the edge of the bearings) must adhere to a 0.005 radians or less top slope relative to the bottom of the bearing.
- c. Bearings with steel laminates shall be cast as a unit in a mold and bonded and vulcanized under heat and pressure. Bearings with steel laminates which are designed to act as a single unit with a given shape factor must be manufactured as a single unit. The mold shall have a standard shop practice mold finish. The internal steel laminates shall be blast cleaned to a cleanliness that conforms to **SSPC SP 6/NACE No.3** at the time of bonding. Plates shall be free of sharp edges and burrs and shall have a minimum edge cover of **0.25 inches**. External load plates (sole plates) shall be hot bonded to the bearing during vulcanization. Edges of the embedded steel laminates, including the laminate restraining devices and around holes and slots shall be covered with not less than **3/16 inches** of elastomer. All exposed laminations or imperfections that result in less than the specified elastomer cover of any surface of the steel laminations shall be repaired by the manufacturer at the point of manufacture. The repair shall consist of sealing the imperfections flush on the finished pads with a bonded vulcanized patch material compatible with the elastomeric bearing pad. Repairs employing caulking type material or repairing the bearings in the field shall not be permitted.

2.4 BRIDGE JOINT SEAL

Sealant system shall be comprised of three components: 1) cellular polyurethane foam impregnated with hydrophobic 100% acrylic, water-based emulsion, factory coated with highway-grade, fuel resistant silicone; 2) field-applied epoxy adhesive primer, 3) field-injected silicone sealant bands. Impregnation agent to have proven non-migratory characteristics. Silicone coating to be highway-grade, low-modulus, fuel resistant silicone applied to the impregnated foam sealant at a width greater than maximum allowable joint extension and which when cured and compressed will form a bellows.

Table 1: Typical Physical Properties of Preformed, Precompressed, Foam Supported Silicone Expansion Joint System		
Property	Value	Test Method
Thermal Movement	500 cycles at -60%, +60% movement	ASTM E1399/E1399M
Base Material	Min. 5.65 kg/m3 cellular, high density, polyurethane foam	ASTM D3574
Density	Min. 6 pcf	ASTM D545
Tensile Strength	Min. 18 psi	ASTM D3574
Elongation	Min. 150%	ASTM D3574
UV / Moisture Resistance	No changes - 2000 hours	ASTM G155
Compression Set	Max 3%	ASTM D3574

Joint shall adhere to the requirements shown in Table 1. Dimensions of joint shall be determined per manufacturer's recommendations and contract plans.

Table 2: Requirements for waterproofing membrane		
Property	Value	Test Method
Thickness	Min. 0.060 in	N/A
Thickness of polyethylene film	Min. 0.004 in	N/A
Tensile Strength	Min. 250 psi	ASTM D412 (Die C)
Ultimate Elongation	Min. 250%	ASTM D412 (Die C)
Puncture Resistance	Min. 40 lbs	ASTM E154/E154M
Pliability (180 deg bend over 1 inch mandrel at 25 F)	No cracks	ASTM D146/D146M

Table 2: Requirements for waterproofing membrane		
Permeance-perms grains/ft ² /hr/in Hg	0.1 (5.7) maximum	ASTM E96/E96M Procedure B
Cycling over crack at 15 F	No effect 100 cycles	Apply and roll membrane across two primed concrete blocks with no separation between the blocks. Open and close the crack from 0 to 1/4 in (6 mm)

2.5 BRIDGE WATERPROOFING

Waterproofing materials serve as a barrier between the concrete bridge deck and the overlay of asphaltic concrete. The membranes included herein are for bridge endwall joints. Use these waterproofing barriers for concrete and other masonry surfaces at locations shown on the plans. Use [waterproofing membrane](#) that incorporates a cross-laminated, high density polyethylene film, adhered to a flexible, self-adhesive, rubberized asphalt. Select a primer from the membrane manufacturer or an approved equal compatible with the membrane. Use membranes that meet the following requirements when tested with the required test method:

PART 3 EXECUTION

3.1 INSTALLATION

Provide joint locations and details of expansion, construction, contraction, and isolations joints, including materials and methods of installation of joint fillers, as specified and indicated. In no case shall any fixed metal be continuous through any isolation joint.

3.1.1 Contraction Joints

Coat contraction joints with [curing compound or bituminous paint](#).

3.1.2 WATERSTOP INSTALLATION AND [SPLICING](#)

- a. Provide waterstops in construction joints as indicated.
- b. Install formwork to accommodate waterstop materials. Locate waterstops in joints where indicated in Contract Documents. Minimize number of splices in waterstop. Splice waterstops in accordance with manufacturer's written instructions. Install factory-manufactured premolded mitered corners.
- c. Install waterstops to form a continuous diaphragm in each joint. Make adequate provisions to support and protect waterstops during progress of work. Protect waterstops protruding from joints from damage per manufacturer's recommendations.
- d. Repair or replace any waterstop punctured or damaged at no cost to the Government. Protect exposed waterstops during application of form release agents to avoid being coated. Provide suitable guards to protect exposed projecting edges and ends of partially embedded

waterstops from damage when concrete placement has been discontinued. Accomplish splices with certified trained personnel using approved equipment and procedures. Any waterstop punctured or damaged shall be replaced or repaired at the Contractor's expense.

3.1.2.1 Hydrophilic Waterstop

Miter cut ends to be joined with sharp knife or shears. The ends must be adhered with adhesive. If damaged during installation, report to RO and follow manufacturer's recommendations for repair.

3.1.3 Expansion Joints

Use preformed expansion joint filler in expansion and isolation joints between slabs on grade and vertical surfaces where indicated. Extend the filler to the full slab depth, unless otherwise indicated. Neatly finish the edges of the joint with an edging tool of 1/8 inch radius, except where a resilient floor surface will be applied. Where the joint is to receive a sealant, the filler strips shall be installed at the proper level below the finished floor with a slightly tapered, dressed and oiled wood strip temporarily secured to the top to form a recess to the size shown on the drawings. Remove the wood strip after the concrete has set. Contractor may opt to use a removable expansion filler cap designed and fabricated for this purpose in lieu of the wood strip. Thoroughly clean the groove of laitance, curing compound, foreign materials, protrusions of hardened concrete, and any dust. If blowing out the groove use oil-free compressed air. Install bridge joint seals in accordance with Contract Plans and approved method from manufacturer.

3.1.4 Bridge Joint Seal

Install bridge joint seals in accordance with Contract Plans and approved method from manufacturer.

3.1.5 Bridge Bearing

Install bridge bearing pads in accordance with [PRHTA Specifications](#) Section 717 ELASTOMERIC BEARINGS and the Contract Plans.

3.1.6 Bridge Waterproofing

Install bridge waterproofing in accordance with Contract Plans and approved method from manufacturer.

3.2 CONSTRUCTION JOINTS

Treat construction joints coinciding with expansion and contraction joints as expansion or contraction joints as applicable.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 03 - CONCRETE

SECTION 03 20 00

CONCRETE REINFORCING

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY ASSURANCE
 - 1.3.1 Qualification of Steel Bar Butt-Splacers
 - 1.3.2 Qualification of Butt-Splicing Procedure
- 1.4 DELIVERY, STORAGE, AND HANDLING

PART 2 PRODUCTS

- 2.1 STAINLESS STEEL DOWELS
- 2.2 GALVANIZED DOWELS
- 2.3 FABRICATED BAR MATS
- 2.4 REINFORCING STEEL
 - 2.4.1 Mechanical Butt-Splices
- 2.5 WELDED WIRE REINFORCING
- 2.6 WIRE TIES
- 2.7 SUPPORTS
- 2.8 TESTS, INSPECTIONS, AND VERIFICATIONS
 - 2.8.1 Reinforcement Steel Tests

PART 3 EXECUTION

- 3.1 REINFORCEMENT
 - 3.1.1 Placement
 - 3.1.2 Placing Tolerances
 - 3.1.2.1 Spacing
 - 3.1.2.2 Concrete Cover
 - 3.1.3 Splicing
 - 3.1.3.1 Lap Splices
 - 3.1.3.2 Mechanical Butt-Splices
- 3.2 DOWEL INSTALLATION
- 3.3 FIELD TESTS AND INSPECTIONS
 - 3.3.1 Identification of Splices
 - 3.3.2 EXAMINING, TESTING, AND CORRECTING
 - 3.3.2.1 Tension Tests
 - 3.3.2.2 Test Specimens
 - 3.3.2.3 Correction of Deficiencies
 - 3.3.2.4 Supplemental Examination

-- End of Section Table of Contents --

SECTION 03 20 00
CONCRETE REINFORCING

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318/318M (2019; R 2022) Building Code Requirements for Structural Concrete (ACI 318-19) and Commentary (ACI 318R-19)

ACI SP-66 (2004) ACI Detailing Manual

ASTM INTERNATIONAL (ASTM)

ASTM A1034/A1034M-10a (2015) Standard Test Methods for Testing Mechanical Splices for Steel Reinforcing Bars

ASTM A1064/A1064M (2018a) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete

ASTM A184/A184M (2006; E2011) Standard Specification for Fabricated Deformed Steel Bar Mats for Concrete Reinforcement

ASTM A276 (2010) Standard Specification for Stainless Steel Bars and Shapes

ASTM A370 (2020) Standard Test Methods and Definitions for Mechanical Testing of Steel Products

ASTM A615/A615M (2020) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement

ASTM A706/A706M (2016) Standard Specification for Low-Alloy Steel Deformed and Plain Bars for Concrete Reinforcement

ASTM A767/A767M (2016) Standard Specification for Zinc-Coated (Galvanized) Steel Bars for Concrete Reinforcement

ASTM A780/A780M (2020) Standard Practice for Repair of

Damaged and Uncoated Areas of Hot-Dip
Galvanized Coatings

ASTM A955/A955M

(2012; E 2012) Standard Specification for
Deformed and Plain Stainless-Steel Bars
for Concrete Reinforcement

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Designation RO stands for Resident Office and designation DO stands for District Office. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Reinforcement; G, DO

Detail drawings showing reinforcing steel placement, schedules, sizes, grades, and splicing and bending details. Drawings shall show support details including types, sizes and spacing.

SD-03 Product Data

Materials; G, DO

Mechanical Butt-Splices; G, DO

SD-06 Test Reports

Tests, Inspections, and Verifications; G, DO

Test Butt-Splices; G, DO

SD-07 Certificates

Reinforcing Steel; G, DO

Qualification of Steel Bar Butt-Splicers; G, DO

1.3 QUALITY ASSURANCE

1.3.1 Qualification of Steel Bar Butt-Splicers

Personnel installing steel bar butt-splices shall have satisfactorily completed a course of instruction in the proposed method of butt-splicing or have satisfactorily performed such work within the preceding year. Submit certificates on the Qualifications of Steel Bar Butt-Splicers prior to commencing butt-splicing.

1.3.2 Qualification of Butt-Splicing Procedure

As a condition of approval of the butt-splicing procedure, make three **test butt-splices** of steel bars of each size to be spliced using the proposed butt-splicing method, in the presence of the Contracting Officer. Tension test to ultimate tensile strength these test butt-splices and unspliced bars of the same size, and plot stress-strain curves for each test. Test

results must show that the butt-splices meet the specified strength and deformation requirements in order for the splicing procedure to be approved.

1.4 DELIVERY, STORAGE, AND HANDLING

Store reinforcement and accessories off the ground on platforms, skids, or other supports. Protect mechanical splice threads from rusting.

PART 2 PRODUCTS

2.1 STAINLESS STEEL DOWELS

Conform to [ASTM A955/A955M](#) or [ASTM A276](#); 316 Series UNS S31600, UNS S31603, UNS S31653 or UNS S 31803, as specified or shown, for stainless steel dowels. All stainless steel dowels shall be Grade 60, minimum yield strength of 60,000 psi.

2.2 GALVANIZED DOWELS

Galvanized dowel bars are deformed bars conforming to [ASTM A706/A706M](#) and [ASTM A767/A767M](#). All galvanized dowels shall be Grade 60, minimum yield strength of 60,000 psi. Provide a Class 1 coating and fabricate dowel bars prior to galvanizing.

Repair damaged surfaces with galvanizing repair method and paint conforming to [ASTM A780/A780M](#) or by application of zinc rich paint galvanizing, as approved by Contracting Officer. The length and width of the damaged surface area that shall be allowed is not to exceed 1 in. in only one of the two dimensions. Areas to be repaired shall be cleaned, dry, free of oil, grease, preexisting paint, and corrosion products. Hand clean the surface to the requirements of SSPC SP2. Apply zinc rich paints as soon as possible after surface preparation to avoid the development of any oxide. Follow the paint manufacturer's instructions and be aware of the recommended temperature, humidity conditions, curing time, and paint thickness per application. Spray or brush the zinc paint onto the surface with horizontal passes per the manufacturer's recommendations. Once the repair coating has cured, the product can be inspected and the thickness verified. The final coating thickness for areas repaired with the zinc rich paint should be 50% more than the galvanized coating thickness requirements as specified in [ASTM A767/A767M](#), but not more than 4.0 mils. Multiple applications of zinc rich paint may be necessary to achieve coating thickness repair requirements. Follow the manufacturer's curing instructions between applications.

2.3 FABRICATED BAR MATS

Fabricated bar mats conforming to [ASTM A184/A184M](#).

2.4 REINFORCING STEEL

Reinforcing steel of deformed bars conforming to [ASTM A615/A615M](#) or [ASTM A706/A706M](#) as indicated. Welding of any reinforcement is not permitted. When the grade of steel is not specified in the plans, Grade 60 shall be provided. Cold drawn wire used for spiral reinforcement must conform to [ASTM A1064/A1064M](#).

Submit certified copies of mill reports attesting that the reinforcing steel furnished contains no less than 25 percent recycled scrap steel and

meets the requirements specified herein, prior to the installation of reinforcing steel. Certified mill reports shall not be older than 6 months.

2.4.1 Mechanical Butt-Splices

Mechanical butt splices must be exothermic, threaded coupling, swaged sleeve or other positive connecting type, and shall develop in tension or compression at least 1.25 times the yield strength of the bar as well capable of developing the specified tensile strength of the spliced bar according to [ACI 318/318M](#), Type 2. In addition to this strength requirement, the additional deformation of number 14 and smaller bars due to slippage or other movement within the splice sleeve shall not exceed [0.015 inches](#) (unit strain [0.0015 inches/inch](#)) beyond the elongation of an unspliced bar based upon a [10 inch](#) gage length spanning the extremities of the sleeve at a stress of [30,000 psi](#). The additional deformation of number 18 bars must not exceed [0.03 inches](#) (unit strain [0.003 inches/inch](#)) beyond the elongation of an unspliced bar based upon a [10 inch](#) gage length spanning the extremities of the sleeve at a stress of [30,000 psi](#). Determine the amount of the additional deformation from the stress-strain curves of the unspliced and spliced bars tested as required in paragraph QUALIFICATION OF BUTT-SPLICING PROCEDURE for qualification of the butt-splicing procedure.

2.5 WELDED WIRE REINFORCING

Welded wire reinforcing conforming to [ASTM A1064/A1064M](#). For wire with a specified yield strength (f_y) exceeding [60,000 psi](#), f_y must be the stress corresponding to a strain of 0.35 percent.

2.6 WIRE TIES

Use wire ties that are 16 gauge or heavier black annealed steel wire.

2.7 SUPPORTS

Provide bar supports complying with the requirements of [ACI SP-66](#). Provide plastic-coated wire, stainless steel or precast concrete supports for bars in concrete with formed surfaces exposed to view or to be painted. Use wedge-shaped precast concrete supports, not larger than [3-1/2 by 3-1/2 inches](#), of thickness equal to that indicated for concrete cover and with an embedded hooked tie-wire for anchorage. Bar supports used in precast concrete with formed surfaces exposed to view must be the same quality, texture and color as the finish surfaces.

2.8 TESTS, INSPECTIONS, AND VERIFICATIONS

Perform material tests, specified and required by applicable standards, by an approved laboratory and certified to demonstrate that the [materials](#) are in conformance with the specifications. Perform and certify tests, inspections, and verifications. Submit certified tests reports not older than 6 months of reinforcement steel showing that the steel complies with the applicable specifications for each steel shipment and identified with specific lots prior to placement. Submit three copies of the heat analyses for each lot of steel furnished certifying that the steel conforms to the heat analyses.

2.8.1 Reinforcement Steel Tests

Perform mechanical testing of steel (one test for every 5 tons per lot of

steel) in accordance with **ASTM A370** except as otherwise specified or required by the material specifications. Perform tension tests on full cross-section specimens using a gage length that spans the extremities of specimens with welds or sleeves included. From chemical analyses of steel heats, report the percentages of carbon, phosphorous, manganese, sulphur and silicon present in the steel.

PART 3 EXECUTION

3.1 REINFORCEMENT

Fabricate and place reinforcement steel and accessories as specified, as indicated, and as shown on approved shop drawings. Fabrication and placement details of steel and accessories not specified or shown must be in accordance with **ACI SP-66** and **ACI 318/318M**. Cold bend reinforcement unless otherwise authorized. Bending may be accomplished in the field or at the mill. Do not bend bars after embedment in concrete. Place safety caps on all exposed ends of vertical concrete reinforcement bars that pose a danger to life safety. Face wire tie ends away from the forms. Submit detail drawings showing reinforcing steel placement, schedules, sizes, grades, and splicing and bending details. Show support details including types, sizes and spacing.

3.1.1 Placement

Reinforcement must be free from loose rust and scale, dirt, oil, or other deleterious coating that could reduce bond with the concrete. Place reinforcement at locations indicated plus or minus one bar diameter. Do not continue reinforcement through isolation joints and place as indicated through construction, expansion, or contraction joints. Cover with concrete coverage as indicated in plans. If bars are moved more than one bar diameter to avoid interference with other reinforcement, conduits or embedded items, the resulting arrangement of bars, including additional bars required to meet structural requirements, requires approval before concrete is placed.

3.1.2 Placing Tolerances

3.1.2.1 Spacing

The spacing between adjacent bars and the distance between layers of bars may not vary from the indicated position by more than one bar diameter nor more than **1 inch**.

3.1.2.2 Concrete Cover

The minimum concrete cover of main reinforcement steel bars shall be as shown. The allowable variation for minimum cover shall be as follows:

MINIMUM COVER (inch)	VARIATION (inch)
6	plus 1/2
4	plus 3/8
3	plus 3/8

MINIMUM COVER (inch)	VARIATION (inch)
2	plus 1/4
1-1/2	plus 1/4
1	plus 1/8
3/4	plus 1/8

3.1.3 Splicing

Conform splices of reinforcement to [ACI 318/318M](#) and make only as required or indicated. Bars may be spliced at alternate or additional locations at no additional cost to the Government subject to approval. Splicing must be by lapping or by mechanical splices; except that lap splices must not be used for bars larger than [No. 11](#) unless otherwise indicated.

3.1.3.1 Lap Splices

Place lapped bars in contact and securely tied or spaced transversely apart to permit the embedment of the entire surface of each bar in concrete. Do not space lapped bars farther apart than 1/5 the required length of lap or [6 inches](#). Lap splices shall be staggered so that no more than half of the bars are spliced in any one section unless otherwise indicated.

3.1.3.2 Mechanical Butt-Splices

Fabricate mechanical butt-splices in accordance with the mechanical splicing device manufacturer's recommendations as well as [ASTM A615/A615M](#) or [ASTM A706/A706M](#), Grade 60. Bars to be spliced by a mechanical butt-splicing process may be sawed or sheared, provided the ends of sheared bars are reshaped after shearing. Clean surfaces to be enclosed within a splice sleeve or coupling by wire brushing or other approved method prior to splicing. Make splices using manufacturer's standard jigs, clamps, and other required accessories. Longitudinally stagger tension splices of number [14](#) or smaller bar a minimum of [5 feet](#) or as otherwise indicated so that no more than half of the bars are spliced at any one section. Longitudinally stagger tension splices of number [18](#) bars a minimum of [5 feet](#) so that no more than 1/3 of the bars are spliced at any one section. In addition, splices in adjacent bars shall be staggered at least 30 inches.

3.2 DOWEL INSTALLATION

Install dowels at locations indicated on drawings and at right angles to joint being doweled. Accurately position and align dowels parallel to the finished concrete surface before concrete placement. Rigidly support dowels during concrete placement. Coat one end of dowels with a bond breaker, asphalt, grease or plastic sleeve to prevent a bond with the concrete as noted on drawings.

3.3 FIELD TESTS AND INSPECTIONS

3.3.1 Identification of Splices

Establish and maintain an approved method of identification of all field butt-splices which will indicate the splicer and the number assigned each splice made by the splicer.

3.3.2 EXAMINING, TESTING, AND CORRECTING

Perform the following during the butt-splicing operations as specified:

3.3.2.1 Tension Tests

Prepare butt-splicing test specimens and test in accordance with [ASTM A1034/A1034M-10a](#) to the specified ultimate tensile strength of the spliced bars or to destruction. The number of tests to be performed are as follows:

- Six(6) tension tests for the bridge middle pier shaft splice specimens selected by the Contracting Officer for the approval of the splicing procedure.
- Three(3) tests per each sampling lot for channel wall shafts. Splice specimens shall be selected by the Contracting Officer per each sampling lot for approval of the splicing procedure and 10 additional specimens selected at the discretion of the Contracting Officer during installation.

Each sampling lot is defined as 500 splicers (or fraction thereof) from the same coupler producer and of the same nominal size, length, and grade.

3.3.2.2 Test Specimens

Butt-splicing test specimens shall be assembled with the same type, kind, and rating of equipment or tools intended to be used for the production butt-splices. Provide minimum bar lengths on each side of butt-splice as follows:

- 19-inch for #9 bars and smaller
- 24-inch for #10 bars and larger

3.3.2.3 Correction of Deficiencies

No splice shall be embedded in concrete until satisfactory results of visual examination and the required tests or examinations have been obtained. All splices having visible defects or represented by test specimens which do not satisfy the tests or examinations shall be removed. If any of the tension test specimens fail to meet the strength requirements or deformation limitations two production splices from the same lot represented by the test specimens which failed shall be cut out and tension tested. If both of the retests pass the strength requirements and deformation limitations all of the splices in the lot will be accepted. If one or both of the retests fail to meet the strength requirements or deformation limitations all of the splices in the lot will be rejected. All costs of removal, testing and resplicing of the additional production splices shall be borne by the Contractor. The bars of rejected splices shall be cut off outside the splice zone of coupling or sleeve. The cut

ends shall be finished as specified and the joints shall be respliced and reinspected at no additional cost.

3.3.2.4 Supplemental Examination

The Contracting Officer may require additional or supplemental radiographic examination and/or tension test of any completed splice.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 03 - CONCRETE

SECTION 03 31 01

CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DESIGN REQUIREMENTS
 - 1.2.1 Self-Consolidating Concrete
 - 1.2.2 Tremie Concrete
 - 1.2.3 Air Content
 - 1.2.4 Slump
 - 1.2.5 Concrete Proportioning
 - 1.2.6 Required Average Compressive Strength
 - 1.2.6.1 Average Compressive Strength from Test Records
 - 1.2.6.2 Average Compressive Strength without Previous Test Records
 - 1.2.6.3 Drying Shrinkage
 - 1.2.6.4 Acid Soluble Chloride Ion Content
 - 1.2.7 Concrete Strength
 - 1.2.8 Minimum Cementitious Content
 - 1.2.9 Maximum Water-Cementitious Material (W/C) Ratio
 - 1.2.10 Chloride Ion Penetrability
 - 1.2.11 Construction Tolerances
 - 1.2.11.1 Appearance
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Quality Control Personnel Certifications
 - 1.4.1.1 Quality Manager Qualifications
 - 1.4.1.2 Field Testing Technician and Testing Agency
 - 1.4.2 Laboratory Qualifications For Concrete Qualification Testing
 - 1.4.3 Laboratory Accreditation
 - 1.4.4 Concrete Strength
 - 1.4.5 Aggregates
 - 1.4.6 Cementitious Materials, Admixtures, and Curing Compound
 - 1.4.7 Chloride Penetrability
- 1.5 DELIVERY, STORAGE, AND HANDLING

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Cementitious Materials
 - 2.1.1.1 Portland Cement
 - 2.1.1.2 Fly Ash
 - 2.1.1.3 Raw or Calcined Natural Pozzolan
 - 2.1.1.4 Ultra Fine Fly Ash and Ultra Fine Pozzolan
 - 2.1.1.5 Ground Granulated Blast-Furnace Slag
 - 2.1.1.6 Blended Hydraulic Cement
 - 2.1.2 Aggregates
 - 2.1.2.1 Aggregate Expansion
 - 2.1.2.2 Unfavorable Properties

- 2.1.3 Chemical Admixtures
 - 2.1.3.1 Air-Entraining Admixture
 - 2.1.3.2 Accelerating Admixture (Optional)
 - 2.1.3.3 Water-Reducing or Retarding Admixture
 - 2.1.3.3.1 Water-Reducing or Retarding Admixtures
 - 2.1.3.3.2 High-Range Water Reducing Admixture
 - 2.1.3.4 Anti-Washout or Viscosity Modifying Admixture
 - 2.1.3.5 Corrosion-Inhibiting Admixtures
 - 2.1.3.6 Other Chemical Admixtures
- 2.1.4 Water
- 2.1.5 Nonshrink Grout
- 2.1.6 Latex Bonding Compound
- 2.1.7 Epoxy Resin
- 2.2 EQUIPMENT
 - 2.2.1 Batching Equipment
 - 2.2.2 Scales
 - 2.2.3 Batching Tolerances
 - 2.2.4 Moisture Control
 - 2.2.5 Concrete Mixers
 - 2.2.5.1 Stationary Mixers
 - 2.2.5.2 Truck Mixers
 - 2.2.6 Conveying Equipment
 - 2.2.6.1 Buckets
 - 2.2.6.2 Transfer Hoppers
 - 2.2.6.3 Trucks
 - 2.2.6.4 Chutes
 - 2.2.6.5 Belt Conveyors
 - 2.2.6.6 Concrete Pumps
 - 2.2.7 Vibrators
 - 2.2.8 Bridge Deck Screeds

PART 3 EXECUTION

- 3.1 PREPARATION FOR PLACING
 - 3.1.1 Concrete Placement Plan
 - 3.1.2 Embedded Items
 - 3.1.3 Concrete on Earth Foundations
 - 3.1.4 Construction Joint Treatment
 - 3.1.4.1 Joint Preparation
 - 3.1.4.2 Air-Water Cutting
 - 3.1.4.3 High-Pressure Water Jet
 - 3.1.4.4 Wet Sandblasting
 - 3.1.4.5 Waste Disposal
- 3.2 PLACING CONCRETE
 - 3.2.1 Placing Procedures
 - 3.2.2 Pumping Concrete Materials and Methods
 - 3.2.3 Time Interval Between Mixing and Placing
 - 3.2.4 Hot-Weather Placing
 - 3.2.5 Certification of Batches
 - 3.2.6 Consolidation
 - 3.2.7 Placing Concrete in Congested Areas
 - 3.2.8 Placing Tremie Concrete
 - 3.2.9 Placing Bridge Deck Concrete
 - 3.2.10 Placing Load on Bridge Substructure
- 3.3 PLACING NONSHRINK GROUT
 - 3.3.1 Mixing and Placing of Nonshrink Grout
 - 3.3.2 Treatment of Exposed Surfaces
- 3.4 DECK PLACEMENT
- 3.5 TESTS AND INSPECTIONS

- 3.5.1 General
- 3.5.2 Testing and Inspection Requirements
 - 3.5.2.1 Fine Aggregate
 - 3.5.2.1.1 Grading
 - 3.5.2.1.2 Fineness Modulus Control Chart
 - 3.5.2.1.3 Corrective Action for Fine Aggregate Grading
 - 3.5.2.1.4 Moisture Content Testing
 - 3.5.2.1.5 Moisture Content Corrective Action
 - 3.5.2.2 Coarse Aggregate
 - 3.5.2.2.1 Grading
 - 3.5.2.2.2 Corrective Action for Grading
 - 3.5.2.2.3 Coarse Aggregate Moisture Content
 - 3.5.2.2.4 Coarse Aggregate Moisture Corrective Action
 - 3.5.2.3 Quality of Aggregates
 - 3.5.2.3.1 Frequency of Quality Tests
 - 3.5.2.3.2 Corrective Action for Aggregate Quality
 - 3.5.2.4 Scales
 - 3.5.2.4.1 Weighing Accuracy
 - 3.5.2.4.2 Batching and Recording Accuracy
 - 3.5.2.4.3 Scales Corrective Action
 - 3.5.2.5 Batch-Plant Control
 - 3.5.2.6 Concrete Mixture
 - 3.5.2.6.1 Air Content Testing
 - 3.5.2.6.2 Air Content Corrective Action
 - 3.5.2.6.3 Slump Testing
 - 3.5.2.6.4 Slump Corrective Action
 - 3.5.2.6.5 Temperature
 - 3.5.2.6.6 Compressive Strength Specimens
 - 3.5.2.6.7 Protection from Mechanical Injury
 - 3.5.2.6.8 Strength of Concrete Structure
 - 3.5.2.6.9 Testing Concrete Structure for Strength
 - 3.5.2.6.10 Permeability
 - 3.5.2.6.11 Anti-Washout Testing
 - 3.5.2.6.12 Aggregate for Travel Way
 - 3.5.2.7 Inspection Before Placing
 - 3.5.2.8 Placing
 - 3.5.2.8.1 Placing Inspection
 - 3.5.2.8.2 Placing Corrective Action
 - 3.5.2.9 Vibrators
 - 3.5.2.9.1 Vibrator Testing and Use
 - 3.5.2.9.2 Vibrator Corrective Action
 - 3.5.2.10 Mixer Uniformity
 - 3.5.2.10.1 Stationary Mixers
 - 3.5.2.10.2 Truck Mixers
 - 3.5.2.11 Mixer Uniformity Corrective Action
- 3.5.3 Reports
- 3.6 REPAIR, REHABILITATION AND REMOVAL
 - 3.6.1 Crack Repair
 - 3.6.2 Repair of Weak Surfaces
 - 3.6.3 Failure of Quality Control Test Results

-- End of Section Table of Contents --

SECTION 03 31 01

CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 117	(2010; Errata 2011) Specifications for Tolerances for Concrete Construction and Materials and Commentary
ACI 211.1	(1991; R 2009) Standard Practice for Selecting Proportions for Normal, Heavyweight and Mass Concrete
ACI 214R	(2011) Evaluation of Strength Test Results of Concrete
ACI 304R	(2000; R 2009) Guide for Measuring, Mixing, Transporting, and Placing Concrete
ACI 318/318M	(2019; R 2022) Building Code Requirements for Structural Concrete (ACI 318-19) and Commentary (ACI 318R-19)
ACI SP-2	(2007; Abstract: 10th Edition) ACI Manual of Concrete Inspection
ACI SP-15	(2011) Field Reference Manual: Standard Specifications for Structural Concrete ACI 301-05 with Selected ACI References

ASTM INTERNATIONAL (ASTM)

ASTM C1017/C1017M	(2013; E 2015) Standard Specification for Chemical Admixtures for Use in Producing Flowing Concrete
ASTM C1059/C1059M	(2013) Standard Specification for Latex Agents for Bonding Fresh to Hardened Concrete
ASTM C1064/C1064M	(2017) Standard Test Method for Temperature of Freshly Mixed Hydraulic-Cement Concrete
ASTM C1077	(2017) Standard Practice for Agencies Testing Concrete and Concrete Aggregates

for Use in Construction and Criteria for
Testing Agency Evaluation

ASTM C109/C109M	(2020b) Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (Using 2-in. or (50-mm) Cube Specimens)
ASTM C1107/C1107M	(2020) Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
ASTM C117	(2017) Standard Test Method for Materials Finer than 75-um (No. 200) Sieve in Mineral Aggregates by Washing
ASTM C123/C123M	(2014) Standard Test Method for Lightweight Particles in Aggregate
ASTM C1152/C1152M	(2020) Standard Test Method for Acid-Soluble Chloride in Mortar and Concrete
ASTM C1202	(2019) Standard Test Method for Electrical Indication of Concrete's Ability to Resist Chloride Ion Penetration
ASTM C1260	(2014) Standard Test Method for Potential Alkali Reactivity of Aggregates (Mortar-Bar Method)
ASTM C127	(2015) Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Coarse Aggregate
ASTM C128	(2015) Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Fine Aggregate
ASTM C131/C131M	(2020) Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
ASTM C136/C136M	(2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
ASTM C142/C142M	(2017) Standard Test Method for Clay Lumps and Friable Particles in Aggregates
ASTM C143/C143M	(2020) Standard Test Method for Slump of Hydraulic-Cement Concrete
ASTM C150/C150M	(2020) Standard Specification for Portland Cement
ASTM C1567	(2013) Standard Test Method for Potential Alkali-Silica Reactivity of Combinations of Cementitious Materials and Aggregate (Accelerated Mortar-Bar Method)

ASTM C1602/C1602M	(2018) Standard Specification for Mixing Water Used in Production of Hydraulic Cement Concrete
ASTM C1610/C1610M	(2017) Standard Test Method for Static Segregation of Self-Consolidating Concrete Using Column Technique
ASTM C1611/C1611M	(2014) Standard Test Method for Slump Flow of Self-Consolidating Concrete
ASTM C1621/C1621M	(2017) Standard Test Method for Passing Ability of Self-Consolidating Concrete by J-Ring
ASTM C157/C157M	(2017) Standard Test Method for Length Change of Hardened Hydraulic-Cement Mortar and Concrete
ASTM C172/C172M	(2017) Standard Practice for Sampling Freshly Mixed Concrete
ASTM C173/C173M	(2016) Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method
ASTM C192/C192M	(2019) Standard Practice for Making and Curing Concrete Test Specimens in the Laboratory
ASTM C231/C231M	(2017a) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method
ASTM C260/C260M	(2010a; R 2016) Standard Specification for Air-Entraining Admixtures for Concrete
ASTM C295/C295M	(2019) Standard Guide for Petrographic Examination of Aggregates for Concrete
ASTM C31/C31M	(2019a) Standard Practice for Making and Curing Concrete Test Specimens in the Field
ASTM C33/C33M	(2018) Standard Specification for Concrete Aggregates
ASTM C39/C39M	(2020) Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens
ASTM C40/C40M	(2020) Standard Test Method for Organic Impurities in Fine Aggregates for Concrete
ASTM C42/C42M	(2020) Standard Test Method for Obtaining and Testing Drilled Cores and Sawed Beams of Concrete
ASTM C494/C494M	(2019) Standard Specification for Chemical

Admixtures for Concrete

ASTM C535	(2016) Standard Test Method for Resistance to Degradation of Large-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
ASTM C566	(2013) Standard Test Method for Total Evaporable Moisture Content of Aggregate by Drying
ASTM C595/C595M	(2020) Standard Specification for Blended Hydraulic Cements
ASTM C597	(2009) Pulse Velocity Through Concrete
ASTM C618	(2019) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete
ASTM C666/C666M	(2015) Resistance of Concrete to Rapid Freezing and Thawing
ASTM C803/C803M	(2018) Standard Test Method for Penetration Resistance of Hardened Concrete
ASTM C805/C805M	(2018) Standard Test Method for Rebound Number of Hardened Concrete
ASTM C78/C78M	(2018) Standard Test Method for Flexural Strength of Concrete (Using Simple Beam with Third-Point Loading)
ASTM C87/C87M	(2017) Standard Test Method for Effect of Organic Impurities in Fine Aggregate on Strength of Mortar
ASTM C88	(2018) Standard Test Method for Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate
ASTM C881/C881M	(2020a) Standard Specification for Epoxy-Resin-Base Bonding Systems for Concrete
ASTM C94/C94M	(2020) Standard Specification for Ready-Mixed Concrete
ASTM C989/C989M	(2018a) Standard Specification for Slag Cement for Use in Concrete and Mortars
ASTM D75/D75M	(2019) Standard Practice for Sampling Aggregates
ASTM D3319	(2017) Standard Practice for the Accelerated Polishing of Aggregates Using the British Wheel
ASTM E329	(2021) Standard Specification for Agencies

Engaged in Construction Inspection,
Testing, or Special Inspection

COMMONWEALTH OF PUERTO RICO, DEPARTMENT OF TRANSPORTATION AND
PUBLIC WORKS (PRHTA)

PRHTA SP 934 (2003) Special Provision - Structural
Concrete

CONCRETE REINFORCING STEEL INSTITUTE (CRSI)

CRSI 10MSP (2018) Manual of Standard Practice

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST HB 44 (2018) Specifications, Tolerances, and
Other Technical Requirements for Weighing
and Measuring Devices

NATIONAL READY MIXED CONCRETE ASSOCIATION (NRMCA)

NRMCA CPMB 100 (2000; R 2006) Concrete Plant Standards

U.S. ARMY CORPS OF ENGINEERS (USACE)

COE CRD-C 100 (1975) Method of Sampling Concrete
Aggregate and Aggregate Sources, and
Selection of Material for Testing

COE CRD-C 104 (1980) Method of Calculation of the
Fineness Modulus of Aggregate

COE CRD-C 114 (1997) Test Method for Soundness of
Aggregates by Freezing and Thawing of
Concrete Specimens

COE CRD-C 143 (1962) Specifications for Meters for
Automatic Indication of Moisture in Fine
Aggregate

COE CRD-C 400 (1963) Requirements for Water for Use in
Mixing or Curing Concrete

COE CRD-C 521 (1981) Standard Test Method for Frequency
and Amplitude of Vibrators for Concrete

COE CRD-C 61 (1989A) Test Method for Determining the
Resistance of Freshly Mixed Concrete to
Washing Out in Water

COE CRD-C 94 (1995) Corps of Engineers Specification
for Surface Retarders

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where PRHTA Specifications Section, and/or Special Provision(s) is cited that contains references to other PRHTA sections, they shall also form a part of this specification to the extent referenced. Where reference is made to "Highway Authority" or "Authority's Representative" in

PRHTA Specifications shall mean "Government" or "Contracting Officer" as defined in the Contract Documents. Where PRHTA Specifications refer to the "Engineer", "Engineer of Tests", or "Division of Tests" it shall be understood to mean the "Contracting Officer" as stated in the Contract Documents. In case of conflict between the PRHTA specifications and the Specifications in this Section that follows, the Specifications in this Section shall govern.

Paragraphs titled "Method of Measurement" and "Basis of Payment" of referenced PRHTA Specifications shall not apply.

1.2 DESIGN REQUIREMENTS

For each portion of the structure, select **concrete mixture proportions** so that the strength and water/cementitious materials (W/C) ratio requirements are met. Submit **concrete mixture proportions** as determined by the Contractor and submitted for review. The submission shall be accompanied by test reports from a laboratory complying with **ASTM C1077** showing that proportions thus selected will produce concrete of the qualities indicated. No substitution shall be made in the source or type of materials used in the work without additional tests to show that the quality of the new materials and concrete are satisfactory. The concrete mixture quantities of all ingredients per cubic **yard** and nominal maximum coarse aggregate size that will be used in the manufacture of each quality of concrete shall be stated. Proportions shall indicate the mass of cement, pozzolan and ground granulated blast-furnace (GGBF) slag when used, and water; the mass of aggregates in a saturated surface-dry condition; and the quantities of admixtures. Nominal maximum-size coarse aggregate shall be **1 inch**.

1.2.1 Self-Consolidating Concrete

When self-consolidating concrete is proposed for use, the mixture shall be proportioned and tested for qualification using:

- a. **ASTM C1611/C1611M** slump flow shall not be greater than 24 inches, with visual stability index not greater than 1.
- b. **ASTM C1621/C1621M** passing ability using the J-ring. Spread within 1 inch less than the slump flow.
- c. Passing ability using the L-box between 4 and 8 seconds.
- d. **ASTM C1610/C1610M**, static segregation shall be no less than 4.0 percent.

For process control sampling, the slump flow limit as determined by **ASTM C1611/C1611M** shall be no greater than 24 inches, and the visual stability index limit shall be no greater than 1.0.

1.2.2 Tremie Concrete

When the concrete is intended for placement using the tremie technique, the concrete shall be proportioned to be cohesive and flow with minimal segregation. Viscosity modifying or anti-washout admixtures shall be used for tremie concrete. Proportioning guidance in **ACI 304R** shall be considered. Concrete mixtures shall be qualified for tremie placement methods based on a trial placement approved by the Contracting Officer.

1.2.3 Air Content

Concrete must have an air content between 4 and 7 percent as delivered to the forms and determined by [ASTM C231/C231M](#). For tremie concrete and concrete placed under drilling fluid, the air content must be between 1 and 3 percent as delivered to the forms and as determined by [ASTM C231/C231M](#).

1.2.4 Slump

The slump within the range of 2 to 4 inches as determined in accordance with [ASTM C143/C143M](#).

Where placement by pump is approved, the slump must not exceed 6 inches.

Concrete to be placed by tremie method may contain a chemical admixture for use in producing flowing concrete in accordance with [ASTM C1017/C1017M](#). The slump of the concrete (tremie concrete and concrete under drilling fluid) must be between 7 to 10 inches. Concrete with a slump over 9 inches shall be determined in accordance with [ASTM C1611/C1611M](#).

1.2.5 Concrete Proportioning

Trial batches and testing requirements for various qualities of concrete specified are the responsibility of the Contractor. Obtain samples of aggregates in accordance with the requirements of [ASTM D75/D75M](#). Samples of materials other than aggregate shall be representative of those proposed for the project and shall be accompanied by the manufacturer's test reports indicating compliance with applicable specified requirements. Make trial mixtures having proportions, consistencies, and air content suitable for the work based on methodology described in [ACI 211.1](#), using at least three different water-cement ratios, which will produce a range of strength encompassing those required for the work. The maximum water-cement ratios required in paragraph MAXIMUM WATER-CEMENT (W/C) RATIO below, shall be converted to a weight ratio of water to cement plus pozzolan by mass, silica fume, or GGBF slag by mass equivalency as described in [ACI 211.1](#). The concrete mix must contain Class F fly ash or GGBF slag in the proportions specified in Paragraph 2.1.2.1. In the case where GGBF slag is used, include the weight of the slag in the equations for the term P, which is used to denote the mass of pozzolan. If pozzolan is used in the concrete mixture, the minimum pozzolan content shall be 15 percent of the total cementitious material. Proportion trial mixtures for maximum permitted slump and air content with due consideration to the approved conveying and placement method. The temperature of concrete in each trial batch shall be reported. For each water-cement ratio, at least three test cylinders for each test age shall be made and cured in accordance with [ASTM C192/C192M](#); they shall be tested at 7 days and at the design age specified in accordance with [ASTM C39/C39M](#). From these test results, a curve shall be plotted showing the relationship between water-cement ratio and strength.

1.2.6 Required Average Compressive Strength

In meeting the strength requirements specified below, the selected concrete mix proportion shall produce a required average compressive strength f'_{cr} exceeding the specified strength f'_c by the amount indicated below.

1.2.6.1 Average Compressive Strength from Test Records

Where a concrete production facility has test records, establish a standard deviation in accordance with the applicable provisions of ACI 214R. Test records from which a standard deviation is calculated shall represent materials, quality control procedures, and conditions similar to those expected, shall represent concrete produced to meet a specified strength or strengths (f'_c) within 1,000 psi of that specified for proposed work, and shall consist of at least 30 consecutive tests. A strength test shall be the average of the strengths of two cylinders made from the same sample of concrete and tested at 28 days or at another test age designated for determination of f'_c . Required average compressive strength f'_{cr} used as the basis for selection of concrete proportions shall be the larger of the equations that follow using the standard deviation as determined above:

$$f'_{cr} = f'_c + 1.34S$$

$$f'_{cr} = f'_c + 2.33S - 500$$

Where S = standard deviation

Where a concrete production facility does not have test records meeting the requirements above but does have a record based on 15 to 29 consecutive tests, a standard deviation shall be established as the product of the calculated standard deviation and a modification factor from the following table:

NUMBER OF TESTS*	MODIFICATION FACTOR FOR STANDARD DEVIATION
less than 15	Use tabulation in paragraph REQUIRED AVERAGE COMPRESSIVE STRENGTH
15	1.16
20	1.08
25	1.03
30 or more	1.00
*Interpolate for intermediate numbers of tests.	

1.2.6.2 Average Compressive Strength without Previous Test Records

When a concrete production facility does not have sufficient field strength test records for calculation of the standard deviation, determine the required average strength f_{cr} as follows:

If the specified compressive strength f'_c is 3,000 to 5,000 psi,

$$f'_{cr} = f'_c + 1,200$$

If the specified compressive strength f'_c is over 5,000 psi,

$$f'_{cr} = f'_c + 1,400.$$

1.2.6.3 Drying Shrinkage

Determine drying shrinkage for qualification of concrete mixtures prior to fabrication. No test results shall exceed the limits of 0.05 percent

according to **ASTM C157/C157M**, at 28 days except for high volume fly ash (HVFA) at 56 days. A drying shrinkage test result shall be the mean value from three or more individual specimens constituting a test set. If an individual specimen's measurements deviate from the mean value by more than 0.009 percent length change, the specimen's measurements shall be discarded and a new average established. Casting more than three specimens for each set is permitted. Test procedures and test specimens shall conform to the following:

Drying shrinkage specimens, typically **3 by 3 by 11.25 inch** prisms for **1 inch** maximum size aggregate or smaller, shall be fabricated, cured, dried, and measured at 28 days in the manner delineated in **ASTM C157/C157M**. Mixtures containing 50 percent or more supplementary cementing materials shall meet the shrinkage criteria at 56 days.

1.2.6.4 Acid Soluble Chloride Ion Content

Determine the chloride ion content only for qualification of concrete mixtures. Determine acid soluble chloride ion content in accordance with **ASTM C1152/C1152M**. The limits for allowable acid-soluble chloride ion concentrations in hardened concrete shall not exceed 0.10 percent by weight of cementitious material.

1.2.7 Concrete Strength

Provide specified compressive strength f'_c as follows:

COMPRESSIVE STRENGTH (PSI)	STRUCTURE OR PORTION OF STRUCTURE
5,000 @ 28 days	Drilled shaft, cap beam, valve box, stormwater outfall structure headwall and shelf, bridge deck, middle pier columns, diaphragm, cap, influent/effluent boxes, maintenance box, siphon headwalls, and all other cast-in-place concrete, unless otherwise noted
4,000 @ 28 days	flowable concrete for embedded poles

1.2.8 Minimum Cementitious Content

MINIMUM CEMENTITIOUS CONTENT (LB/CY)	STRUCTURE OR PORTION OF STRUCTURE
650	Drilled shaft, cap beam, valve box, stormwater outfall structure headwall and shelf, middle pier columns, diaphragm, cap, flowable concrete and all other cast-in-place concrete except the utility bridge deck

MINIMUM CEMENTITIOUS CONTENT (LB/CY)	STRUCTURE OR PORTION OF STRUCTURE
600	utility bridge deck

The Minimum Cementitious Content (lbs. /cu. yd.) refers to the total weight of Portland cement, ground granulated blast-furnace slag, microsilica and fly ash added to the concrete expressed in lbs./cu. yd.

1.2.9 Maximum Water-Cementitious Material (W/C) Ratio

Maximum W/C allowed as follows:

WATER-CEMENT RATIO, BY MASS	STRUCTURE OR PORTION OF STRUCTURE
0.36	Drilled shafts, cap beams, pier, and non-shrink grout
0.40	All other cast-in-place concrete

These W/C's may cause higher strengths than that required by paragraph CONCRETE STRENGTH.

1.2.10 Chloride Ion Penetrability

Determine the resistance of concrete to the penetration of chloride ions in accordance with [ASTM C1202](#) during the qualification of concrete mixture. The average of three samples shall meet the limit specified in Table 934-2 of [PRHTA SP 934](#).

1.2.11 Construction Tolerances

Except as specified otherwise, a plus tolerance increases and a minus tolerance decreases the dimension to which it applies. A tolerance without sign means plus or minus. Where only one sign is specified, there is no limit in the other direction. Tolerances are not cumulative. The most restrictive tolerance will control. Tolerances can not extend the structure beyond legal boundaries.

- Make level and grade tolerance measurements of slabs, cap beam, valve box, headwall and concrete shelf as soon as possible after finishing. When forms or shoring are used, make the measurements prior to removal. For drilled shaft construction tolerances refer to Section [31 63 29](#) DRILLED CONCRETE PIERS AND SHAFTS paragraph 1.6.1.2.
- Construction tolerances must meet the requirements of [ACI 117](#) in addition to the following tables:.

TOLERANCES FOR BRIDGES, EROSION-PROTECTION STRUCTURES, AND SMALL HYDRAULIC STRUCTURES		
(1)	Vertical alignment	
	Exposed surfaces	3/4 inch
	Concealed surfaces	1-1/2 inch
(2)	Lateral alignment	
	Centerline alignment	1 inch
(3)	Level alignment	
	Profile grade	1 inch
	Top of other concrete surfaces and horizontal grooves	
	Exposed	3/4 inch
	Concealed	1-1/2 inch
	Mainline pavements in longitudinal direction, the gap below 10 feet unleveled straightedge resting on highspots shall not exceed	1/8 inch
	Mainline pavements in transverse direction, the gap below 10 feet unleveled straightedge resting on highspots shall not exceed	1/4 inch
	Ramps, sidewalks, and intersections, in any direction, the gap below a 10 feet unleveled straightedge resting on highspots shall not exceed	1/4 inch
(4)	Cross-sectional dimensions	
	Bridge slab thickness	+1/4 inch, -1/8 inch
	Members such as columns, beams, piers, walls, and others (slabs--thickness only)	+1/2 inch, -1/4 inch
	Openings through concrete members	1/2 inch

TOLERANCES FOR BRIDGES, EROSION-PROTECTION STRUCTURES, AND SMALL HYDRAULIC STRUCTURES		
(5)	Relative alignment	
	Location of openings through concrete members	1/2 inch
	Formed surface slope with respect to the specified plane	
	Watertight joints	1/8 in. in 10 ft
	Other exposed surfaces	1/2 in. in 10 ft
	Concealed surfaces	1 in. in 10 ft
	Unformed exposed surfaces slopes with respect to the specified plane	1/4 in. in 10 ft
		3/8 in. in 20 ft
TOLERANCES FOR TUNNEL LININGS, CONDUITS, AND FILLING AND EMPTYING CULVERTS		
(1)	Lateral alignment	
	Centerline alignment	
	Water conveying tunnels, conduits, and culverts	1/2 inch
	Other	1 inch
	Inside dimensions	0.005 times inside dimension
(2)	Level alignment	
	Profile grade	
	Water conveying tunnels, conduits, and culverts	1/2 inch
	Other	1 inch
	Surface of invert	1/4 inch
	Surface of side slope	1/2 inch
(3)	Cross-sectional dimension	
	Thickness at any point	
	Tunnel and culvert lining	-0 inch
	Conduits	+5 percent thickness but not less than 1/2 inch
		-2.5 percent thickness but not less than 1/4 inch

1.2.11.1 Appearance

Clean permanently exposed surfaces, if stained or otherwise discolored, by a method that does not harm the concrete and that is approved by the Contracting Officer.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Designation R0 stands for Resident Office and designation D0 stands for District Office. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Random Sampling Plan; G, D0

Concrete Placement Plan; G, D0

Deck Concrete Placement Procedure Plan; G, D0

Quality Control Plan; G, D0

Quality Control Personnel Certifications; G, D0

Quality Control Organizational Chart; G, D0

Laboratory Accreditation; G, D0

SD-03 Product Data

Concrete Mixture Proportions; G, R0, D0

Concrete mixture proportions as determined by the Contractor shall be submitted for review. The submission shall be accompanied by test reports from a laboratory complying with ASTM C1077 showing that proportions thus selected will produce concrete of the qualities indicated. No substitution shall be made in the source or type of materials used in the work without additional tests to show that the quality of the new materials and concrete are satisfactory.

Batch Plant; G, R0

Batch plant data to the Contracting Officer for review for conformance with applicable specifications.

Concrete Mixers; G, R0

Concrete mixer data which includes the make, type, and capacity of concrete mixers proposed for mixing concrete.

Conveying Equipment; G, R0

Data on the conveying equipment and methods for transporting, handling, and depositing the concrete.

Placing Equipment; G, R0

Data on placing equipment and methods.

Statements that the concrete testing technicians and the concrete inspectors meet the specified requirements.

Deck Screeds; G, R0

Data on screed.

Construction Joint Treatment; G, D0

The method and equipment proposed for joint cleanup and waste disposal, for review and approval.

Hot-Weather Placing; G, D0

The proposed materials and methods for review and approval, if concrete is to be placed under hot-weather conditions.

Cementitious Materials; G, D0

Admixtures; G, D0

Pumping Concrete Materials and Methods; G, D0

Nonshrink Grout; G, D0

SD-04 Samples

Aggregates; G, R0

Cementitious Materials, Admixtures, and Curing Compound; G, R0

Samples of materials for Government testing and approval.

SD-05 Design Data

Concrete Mix Design; G, R0, D0

SD-06 Test Reports

Quality of Aggregates; G, D0

Aggregate quality tests at least 30 days prior to start of concrete placement.

Mixer Uniformity; G, R0

The results of the initial mixer uniformity tests prior to the initiation of placing.

Tests and Inspections; G, D0

Test results and inspection reports, as required and specified in this Section.

Water; G, D0

Slump Loss Test; G, D0

Cumulative Mass Loss Test; G, D0

Results of slump loss and cumulative mass loss tests.

Concrete Mix Design; G, D0

Fly Ash; G, D0

Pozzolan; G, D0

Submit test results in accordance with **ASTM C618** for fly ash and pozzolan. Submit test results performed within 6 months of submittal date.

Slag Cement; G, D0

Submit test results in accordance with **ASTM C989/C989M** for slag cement. Submit test results performed within 6 months of submittal date.

Tolerance Report; G, D0

Compressive Strength Tests; G, D0

Air Content; G, D0

Slump Test; G, D0

Permeability; G, D0

SD-07 Certificates

Concrete Field Testing Technicians; G, R0

Concrete Construction Inspector (CCI); G, R0

Cementitious Materials; G, D0

Mill test reports attesting that materials meet the requirements of the specification under which they are furnished. Certification and mill test reports shall be from samples taken from the particular lot furnished and not older than 6 months.

Blended Hydraulic Cement; G, D0

Blended hydraulic cement certified for compliance with all specification requirements dated within 6 months of submittal date.

Air-Entraining Admixture; G, D0

Air-Entraining Admixture certified for compliance with all specification requirements dated within 6 months of submittal date.

Other Chemical Admixtures; G, D0

Other Chemical Admixtures certified for compliance with all

specification requirements dated within 6 months of submittal date.

Epoxy Resin; G, D0

Latex Bonding Compound; G, D0

Epoxy Resin and Latex Bonding Compound certified for compliance with all specification requirements dated within 6 months of submittal date.

Nonshrink Grout; G, D0

Descriptive literature of the Nonshrink Grout proposed for use together with a certificate from the manufacturer stating that it is suitable for the application or exposure for which it is being considered.

1.4 QUALITY ASSURANCE

The Government will sample and test aggregates and concrete to determine compliance with the specifications. The Contractor shall provide facilities and labor as may be necessary for procurement of representative test samples. Samples of aggregates will be obtained at the point of batching in accordance with [ASTM D75/D75M](#). Concrete will be sampled in accordance with [ASTM C172/C172M](#). Do not use a material until the Contracting Officer gives notice that test results are satisfactory. The Contractor shall provide individuals who sample and test concrete or the constituents of concrete as required in this specification that must have demonstrated a knowledge and ability to perform the necessary test procedures equivalent to the ACI minimum guidelines for certification of [Concrete Field Testing Technicians](#), Grade I. The individuals who perform the inspection of concrete construction must have demonstrated a knowledge and ability equivalent to the ACI minimum guidelines for certification of [Concrete Construction Inspector \(CCI\)](#). Submit statements that the concrete testing technicians and the concrete inspectors meet the specified requirements. Maintain a copy of [ACI SP-15](#) and [CRSI 10MSP](#) at project site.

1.4.1 Quality Control Personnel Certifications

The Contractor must submit for approval the responsibilities of the various quality control personnel, including the names and qualifications of the individuals in those positions and a [quality control organizational chart](#) defining the quality control hierarchy and the responsibility of the various positions. Quality control personnel must be employed by the Contractor.

Submit American Concrete Institute certification for the following:

- a. CQC personnel responsible for inspection of concrete operations.
- b. Lead Foreman or Journeyman of the Concrete Placing, Finishing, and Curing Crews.
- c. Field Testing Technicians: ACI Concrete Field Testing Technician, Grade I.

1.4.1.1 Quality Manager Qualifications

The quality manager must hold a current license as a professional engineer in a U.S. state or territory with experience on at least five similar projects. Evidence of extraordinary proven experience may be considered by the Contracting Officer as sufficient to act as the Quality Manager.

1.4.1.2 Field Testing Technician and Testing Agency

Submit data on qualifications of proposed testing agency and technicians for approval by the Contracting Officer prior to performing testing on concrete.

- a. Work on concrete under this contract must be performed by an ACI Concrete Field Testing Technician Grade 1 qualified in accordance with [ACI SP-2](#) or equivalent. Equivalent certification programs must include requirements for written and performance examinations as stipulated in [ACI SP-2](#).
- b. Testing agencies that perform testing services on reinforcing steel must meet the requirements of [ASTM E329](#).
- c. Testing agencies that perform testing services on concrete materials must meet the requirements of [ASTM C1077](#).

1.4.2 Laboratory Qualifications For Concrete Qualification Testing

The concrete testing laboratory must have the necessary equipment and experience to accomplish required testing. The laboratory must meet the requirements of [ASTM C1077](#) and be Cement and Concrete Reference Laboratory (CCRL) inspected.

1.4.3 Laboratory Accreditation

Laboratory and testing facilities must be provided by and at the expense of the Contractor. The laboratories performing the tests must be accredited in accordance with [ASTM C1077](#), including [ASTM C78/C78M](#) and [ASTM C1260](#). The accreditation must be current and must include the required test methods, as specified. Furthermore, the testing must comply with the following requirements:

- a. Aggregate Testing and Mix Proportioning: Aggregate testing and mixture proportioning studies must be performed by an accredited laboratory and under the direction of a registered professional engineer in a U.S. state or territory competent in concrete materials who is competent in concrete materials and must sign all reports and designs.
- b. Acceptance Testing: Furnish all materials, labor, and facilities required for molding, curing, testing, and protecting test specimens at the site and in the laboratory. Furnish and maintain boxes or other facilities suitable for storing and curing the specimens at the site while in the mold within the temperature range stipulated by [ASTM C31/C31M](#).
- c. Contractor Quality Control: All sampling and testing must be performed by an approved, onsite, independent, accredited laboratory.

1.4.4 Concrete Strength

Compressive strength test specimens will be made by the Contractor and cured in accordance with [ASTM C31/C31M](#) and tested in accordance with [ASTM C39/C39M](#). The strength of the concrete will be considered satisfactory so long as the average of all sets of three consecutive test results equals or exceeds the specified compressive strength f'c and no individual test result falls below the specified strength f'c by more than [500 psi](#) (or [200 psi](#) for concrete used in bridge construction). A "test" is defined as the average of two companion cylinders, or if only one cylinder is tested, the results of the single cylinder test. Additional analysis or testing, including nondestructive testing, taking cores and/or load tests may be required at the Contractor's expense when the strength of the concrete in the structure is considered potentially deficient.

- a. Investigation of Low-Strength Test Results - When any strength test of standard-cured test cylinders falls below the specified strength requirement by more than [500 psi](#) (or [200 psi](#) for concrete used in bridge construction) or if tests of field-cured cylinders indicate deficiencies in protection and curing, take steps to ensure that the load-carrying capacity of the structure is not jeopardized. Nondestructive testing in accordance with [ASTM C597](#), [ASTM C803/C803M](#), or [ASTM C805/C805M](#) may be permitted by the Contracting Officer to estimate the relative strengths at various locations in the structure as an aid in evaluating concrete strength in place or for selecting areas to be cored. Such tests shall not be used as a basis for acceptance or rejection.
- b. Testing of Cores - When the strength of concrete in place is considered potentially deficient, obtain cores and test them in accordance with [ASTM C42/C42M](#). At least three representative cores shall be taken from each member or area of concrete in place that is considered potentially deficient. The location of cores will be determined by the Contracting Officer to least impair the performance of the structure. Concrete in the area represented by the core testing will be considered adequate if the average strength of the cores is equal to at least 85 percent of the specified strength requirement and if no single core is less than 75 percent of the specified strength requirement.
- c. Load Tests - If the core tests are inconclusive or impractical to obtain or if structural analysis does not confirm the safety of the structure, load tests may be directed by the Contracting Officer in accordance with the requirements of [ACI 318/318M](#). Concrete work evaluated by structural analysis or by results of a load test shall be corrected in a manner satisfactory to the Contracting Officer. All investigations, testing, load tests, and correction of deficiencies will be performed at the expense of the Contractor and approved by the Contracting Officer.

1.4.5 Aggregates

Provide aggregate from sources capable of producing materials of a quality required for this project and meeting the specified requirements. Samples from any source of coarse aggregate and any source of fine aggregate selected by the Contractor, consisting of not less than [150 pounds](#) of each size coarse aggregate and [75 pounds](#) of fine aggregate taken under the supervision of the Contracting Officer in accordance with [COE CRD-C 100](#) shall be delivered to the Government's Resident Office in San Juan, Puerto Rico.

Rico within 15 days after notice to proceed. Sampling and shipment of samples shall be at the Contractor's expense. 60 days will be required to complete evaluation of the aggregates. Testing will be performed by and at the expense of the Government in accordance with the applicable COE CRD-C or ASTM test methods. The cost of testing one source for each size of aggregate will be borne by the Government. Except if the Contractor selects more than one source for each aggregate size or selects a substitute source for any size aggregate after the original source was tested, the cost of that additional testing will be borne by the Contractor. Tests to which aggregate may be subjected are listed in paragraph 1.4 QUALITY ASSURANCE. The material from the proposed source shall meet the quality requirements of this paragraph. Testing of aggregates by the Government does not relieve the Contractor of the requirements outlined in paragraph TESTS AND INSPECTIONS.

1.4.6 Cementitious Materials, Admixtures, and Curing Compound

At least 60 days in advance of concrete placement, notify the Contracting Officer of the sources for cementitious materials, admixtures, and curing compound, along with sampling location, brand name, type, and quantity to be used in the manufacture and/or curing of the concrete. Cementitious Materials, including Cement and Pozzolan, and Ground Granulated Blast-Furnace Slag will be accepted on the basis of the manufacturer's certification of compliance. Do not use cementitious materials until receiving Contracting Officer notice of acceptance. Cementitious materials will be subject to check testing from samples obtained at the source, at transfer points, or at the project site, as scheduled by the Contracting Officer, and such sampling will be by or under the supervision of the Government at its expense. Promptly remove material not meeting specifications from the work site. Submit samples of materials for Government testing and approval. Submit manufacturer's instructions for placement of curing compound.

1.4.7 Chloride Penetrability

The contractor shall cast specimens, test, and report hardened concrete property tests for chloride ion penetrability in accordance with [ASTM C1202](#) and [PRHTA SP 934](#) during the concrete qualification. If the concrete mix fails, the test shall be repeated at the Contractor's cost. Test concrete for chloride ion penetrability at 7 and 28 days for each concrete design mix placed each day, not less than once a day, nor less than once for each [30 cu yd.](#) of concrete.

1.5 DELIVERY, STORAGE, AND HANDLING

Retest chemical admixtures that have been in storage at the project site for longer than 6 months or that have been subjected to freezing and reject if test results are not satisfactory. Chemical admixtures will be accepted based on compliance with the requirements of paragraph 2.1.3 CHEMICAL ADMIXTURES.

PART 2 PRODUCTS

2.1 MATERIALS

Submit mill test reports ([Tests and Inspections](#)) attesting that materials meet the requirements of the specification under which they are furnished. Certification and mill test reports must be from samples taken from the particular lot furnished. Certify the following for compliance

with all specification requirements: Air-Entraining Admixture, Other Chemical Admixtures, Anti-Washout, Epoxy Resin and Latex Bonding Compound. Submit descriptive literature of the Nonshrink Grout proposed for use together with a certificate from the manufacturer stating that it is suitable for the application or exposure for which it is being considered.

2.1.1.1 Cementitious Materials

Cementitious materials are portland cement, portland-pozzolan cement, portland blast-furnace slag cement, portland cement in combination with pozzolan or GGBF slag conforming to appropriate specifications listed below.

2.1.1.1.1 Portland Cement

ASTM C150/C150M, Type II.

2.1.1.1.2 Fly Ash

ASTM C618, Class F, except that the maximum allowable loss on ignition must not exceed 6 percent. Class F fly ash for use in mitigating Alkali-Silica Reactivity must have a Calcium Oxide (CaO) content of less than 8 percent and a total equivalent alkali content less than 1.5 percent.

When fly ash is used, it shall be incorporated either as a cement replacement, as a mineral admixture added as a replacement of fine aggregate or a combination thereof. All fly ash shall be added on an equal weight basis of the material being replaced. Recommended amounts of fly ash are in the range of up to 40 percent added as a cement replacement and up to 20 percent added as a mineral admixture.

2.1.1.1.3 Raw or Calcined Natural Pozzolan

Natural pozzolan must be raw or calcined and conform to ASTM C618, Class N, including the optional requirements for uniformity and effectiveness in controlling Alkali-Silica reaction and must have an ignition loss not exceeding 3 percent. Class N pozzolan for use in mitigating Alkali-Silica Reactivity must have a Calcium Oxide (CaO) content of less than 13 percent and total equivalent alkali content less than 3 percent.

2.1.1.1.4 Ultra Fine Fly Ash and Ultra Fine Pozzolan

Ultra Fine Fly Ash (UFFA) and Ultra Fine Pozzolan (UFP) must conform to ASTM C618, Class F (UFFA) or N (UFP), and the following additional requirements:

- a. The strength activity index at 28 days of age is at least 95 percent of the control specimens.
- b. The average particle size does not exceed 6 microns.
- c. The sum of SiO₂ + Al₂O₃ + Fe₂O₃ is greater than 77 percent.

2.1.1.1.5 Ground Granulated Blast-Furnace Slag

Conform to ASTM C989/C989M, Grade 120. Ground Granulated Blast-Furnace Slag should be used as a Portland cement replacement on an equal weight basis. Recommended amounts of ground granulated blast-furnace slag are in

the range of 25 to 65 percent added as a cement replacement on an equal weight basis of the cement being replaced. Add with cement.

2.1.1.6 Blended Hydraulic Cement

Conform to [ASTM C595/C595M](#), Type IP or IS.

2.1.2 Aggregates

[ASTM C33/C33M](#), except as modified herein. Furnish aggregates for exposed concrete surfaces from one source. Manufactured sand shall not be used as fine aggregate for concrete which will serve as the travel surface for vehicular traffic such as bridge decks. Provide aggregates that do not contain any substance which may be deleteriously reactive with the alkalis in the cement. Submit test report showing compliance with [ASTM C33/C33M](#). Fine and coarse aggregate shall meet the following requirements:

TEST LIMITS			
PROPERTY	FINE AGGREGATE	COARSE AGGREGATE	TEST
Specific Gravity	> 2.5	> 2.4	ASTM C127 ASTM C128
Absorption	< 2.0%	< 2.0%	ASTM C127 ASTM C128
Durability Factor using (Procedure A)	F10	F10	COE CRD-C 114 ASTM C666/C666M
Clay Lumps and Friable Particles	< 3.0%	< 5.0%	ASTM C142/C142M
Material Finer than No. 200 Sieve	< 5.0%	< 5.0%	ASTM C117
Organic Impurities	Not darker than No. 3 Not less than 95 percent		ASTM C40/C40M ASTM C87/C87M
L.A. Abrasion	-	< 40%	ASTM C131/C131M ASTM C535
Chert, less than 2.40 specific gravity	< 5.0%	< 5.0%	ASTM C123/C123M

TEST LIMITS			
PROPERTY	FINE AGGREGATE	COARSE AGGREGATE	TEST
Coal and Lignite, less than 2.00 specific gravity	< 0.5%	< 0.5%	ASTM C123/C123M
Petrographic Examination	< 5.0%	< 5.0%	ASTM C295/C295M
Magnesium sulfate soundness (5 cycles)	< 15%	< 18%	ASTM C88

For concrete that is to serve as the travel way for vehicular traffic, such as bridge decks, the coarse aggregate shall have a minimum polishing value of 48% as determined by ASTM D3319.

For concrete used in flowable concrete for embedded poles, the nominal maximum size shall be one half inch (1/2").

2.1.2.1 Aggregate Expansion

Provide fine and coarse aggregates with expansions less than 0.08 percent at 28 days after casting when testing in accordance with ASTM C1260. Should the test data indicate an expansion of 0.08 percent or greater, reject the aggregate(s) or perform additional testing using ASTM C1567 and using the proposed mix design. In this case, include the mix design low alkali portland cement and one of the following supplementary cementitious materials:

- a. GGBF slag at a minimum of 40 percent of total cementitious material.
- b. Fly ash or natural pozzolan at a minimum of total cementitious material as stated below:
 - (1) 30 percent if (SiO₂ plus Al₂O₃ plus Fe₂O₃) is 65 percent or more,
 - (2) 25 percent if (SiO₂ plus Al₂O₃ plus Fe₂O₃) is 70 percent or more,
 - (3) 20 percent if (SiO₂ plus Al₂O₃ plus Fe₂O₃) is 80 percent or more,
 - (4) 15 percent if (SiO₂ plus Al₂O₃ plus Fe₂O₃) is 90 percent or more.
- c. Silica fume at a minimum of 7 percent of total cementitious material.

If a combination of these materials is chosen, the minimum amount must be a linear combination of the minimum amounts above. Include these materials in sufficient proportion to show less than 0.08 percent expansion at 16 days after casting when tested in accordance with ASTM C1567.

2.1.2.2 Unfavorable Properties

Do not provide aggregates possessing properties or constituents that are known to have specific unfavorable effects in concrete when tested in

accordance with [ASTM C295/C295M](#). Submit test results of petrographic examination for aggregate quality in accordance with [ASTM C295/C295M](#).

2.1.3 Chemical Admixtures

Provide chemical admixtures, when required or permitted, that conform to the appropriate specification listed.

2.1.3.1 Air-Entraining Admixture

Conform to [ASTM C260/C260M](#) and consistently cause the concrete to have an air content in the specified ranges under field conditions.

2.1.3.2 Accelerating Admixture (Optional)

If an accelerating admixture is used, meet the requirements of [ASTM C494/C494M](#), Type C or E, except that calcium chloride or admixtures containing calcium chloride is not permitted.

2.1.3.3 Water-Reducing or Retarding Admixture

2.1.3.3.1 Water-Reducing or Retarding Admixtures

[ASTM C494/C494M](#), Type A, B, or D, except that the 6-month and 1-year compressive strength tests are waived.

2.1.3.3.2 High-Range Water Reducing Admixture

[ASTM C494/C494M](#), Type F or G except that the 6-month and 1-year strength requirements are waived. The admixture may be used only when approved by the Contracting Officer, such approval being contingent upon particular mixture control as described in the Contractor's [Quality Control Plan](#).

2.1.3.4 Anti-Washout or Viscosity Modifying Admixture

Provide anti-washout or viscosity modifying admixtures for concrete placement by tremie. Provide certification that the admixture is compatible with the cementitious materials and other chemical admixtures in the proposed concrete mixture. The anti-washout or viscosity modifying admixture shall require approval by the Contracting Officer and have a proven record of performance with a minimum of five similar projects. Provide test result per [COE CRD-C 61](#) to determine [cumulative mass loss](#). The acceptable range for mass loss is between 1% and 3%. Tests shall be performed once for each [350 cubic yards](#) of underwater concrete and results submitted to Contracting Officer for approval prior to continued use.

2.1.3.5 Corrosion-Inhibiting Admixtures

- a. General. A calcium nitrite based corrosion inhibitor shall be added at a dosage rate of up to five gallons per cubic yard of concrete to the specified mixture proportions for all concrete to inhibit active corrosion to [15 pounds](#) of chloride per cubic yard of concrete at the rebar level. The product shall meet [ASTM C494/C494M](#) admixture requirements as to their effect on other concrete properties, and shall have a minimum of 10 years of successful field application. The product shall be a minimum 30 percent calcium nitrite solution. A calcium nitrite based corrosion inhibitor acts as an accelerating admixture in

accordance with [ASTM C494/C494M](#) Type C. This shall be considered when addressing the requirements of "Hot-Weather Placing". Prohibited admixtures include calcium chloride, thiocyanates and all admixtures that contribute free chloride ion in excess of 0.1% by weight of cement.

- b. Manufacturer's Field Service. The Contractor shall engage the services of a qualified admixture manufacturer's representative. The admixture manufacturer shall provide a source for incidental consultation through concrete completion of project as herein specified. The representative shall be employed full time by the manufacturer and possess technical admixture training and concrete mix design experience. The mix design shall be reviewed by the admixture supplier prior to trial mixture placement and prior to making any changes in the mix design. A qualified manufacturer's representative shall observe start up of concrete placement and offer recommendations as appropriate.
- c. Trial Mixtures. The required trial mixtures shall include the corrosion inhibiting admixture along with other required admixtures. The water content of the corrosion-inhibiting admixture shall be included in the water-cement ratio. Calcium nitrite treated concrete shall contain a high range water reducer and/or retarder ([ASTM C494/C494M](#), Type F or G) as necessary to provide workability.

2.1.3.6 Other Chemical Admixtures

Provide other chemical admixtures for use in producing flowing concrete for tremie placement in compliance with [ASTM C1017/C1017M](#), Type 1 or 2. Only use these admixture for concrete listed in paragraph 1.2.4 SLUMP. Provide certification that the admixture is compatible with cementitious materials and other chemical admixtures in the proposed concrete mixture.

2.1.4 Water

Provide water for mixing and curing compliance with the requirements of [ASTM C1602/C1602M](#) or [COE CRD-C 400](#); free of injurious amounts of oil, acid, salt, or alkali. Submit test report showing water complies with [ASTM C1602/C1602M](#) or [COE CRD-C 400](#).

2.1.5 Nonshrink Grout

Provide a commercial formulation that conforms to [ASTM C1107/C1107M](#) and is suitable for the proposed application. Nonshrink grout shall be cement-based with a minimum compressive strength of 5,000 psi at 28 days when tested in accordance with [ASTM C109/C109M](#). Grout shall be mixed and placed in strict accordance with the manufacturer's recommendations.

2.1.6 Latex Bonding Compound

Latex bonding compound agents for bonding fresh to hardened concrete shall conform to [ASTM C1059/C1059M](#), Type II.

2.1.7 Epoxy Resin

Epoxy resin for use in repairs shall conform to [ASTM C881/C881M](#), Type III, Grade I or II.

2.2 EQUIPMENT

Submit data on placing equipment and methods. The batching, mixing, conveying, and placing equipment must have a capacity of at least 120 cubic yards per hour. Conform the batch plant to the requirements of NRMCA CPMB 100 and as specified; however, rating plates attached to batch plant equipment are not required. Submit batch plant data for conformance with applicable specifications.

2.2.1 Batching Equipment

Use partially automatic, semiautomatic, or automatic batching controls. Provide the semiautomatic batching system with interlocks such that the discharge device cannot be actuated until the indicated material is within the applicable tolerance. Equip the batching system with an accurate recorder or recorders that meet the requirements of NRMCA CPMB 100. Provide separate bins or compartments for each size group of aggregate and cement, pozzolan, and GGBF slag. Weigh aggregates either in separate weigh batchers with individual scales or cumulatively in one weigh batcher on one scale. Do not weigh aggregate in the same batcher with cement, pozzolan, or GGBF slag. If both cement and pozzolan or GGBF slag are used, they may be batched cumulatively provided that the Portland cement is batched first. If measured by mass, do not weigh the mass of the water cumulatively with another ingredient. Interlock water batcher filling and discharging valves so that the discharge valve cannot be opened before the filling valve is fully closed. Provide an accurate mechanical device for measuring and dispensing each admixture. Interlock each dispenser with the batching and discharging operation of the water so that each admixture is separately batched and discharged automatically in a manner to obtain uniform distribution throughout the batch in the specified mixing period. Do not combine admixtures prior to introduction in water. Arrange the plant to facilitate the inspection of all operations at all times. Provide facilities for obtaining representative samples of aggregates from each bin or compartment. Clearly mark all filling ports for cementitious materials bins or silos with a permanent sign stating the contents.

2.2.2 Scales

Provide equipment for batching by mass conforming to the applicable requirements of NIST HB 44, except that the accuracy shall be plus or minus 0.2 percent of scale capacity. Provide standard test weights and any other auxiliary equipment required for checking the operating performance of each scale or other measuring devices. Perform tests at the frequency required in paragraph 3.4 TESTS AND INSPECTIONS, and in the presence of a Government inspector.

2.2.3 Batching Tolerances

a. Weighing Tolerances

MATERIAL	PERCENT OF REQUIRED MASS
Cementitious materials	0 to plus 2
Aggregate	plus or minus 2

MATERIAL	PERCENT OF REQUIRED MASS
Water	plus or minus 1
Chemical admixture	0 to plus 6

- b. Volumetric Tolerances - For volumetric batching equipment, the following tolerances apply to the required volume of material being batched:

Water	Plus or minus 1 percent
Chemical admixtures	Zero to plus 6 percent

2.2.4 Moisture Control

The plant must be capable of ready adjustment to compensate for the varying moisture content of the aggregates and to change the masses of the materials being batched. Provide an electric moisture meter complying with the provisions of [COE CRD-C 143](#) for measuring moisture in the fine aggregate. Arrange the sensing element so that the measurement is made near the batcher charging gate of the sand bin or in the sand batcher.

2.2.5 Concrete Mixers

Do not charge the concrete mixers in excess of the capacity recommended by the manufacturer. Operate the mixers at the drum or mixing blade speed designated by the manufacturer. Maintain the mixers in satisfactory operating condition, and keep the mixer drums free of hardened concrete. Should any mixer at any time produce unsatisfactory results, promptly discontinue its use until it is repaired. Submit concrete mixer data including the make, type, and capacity of concrete mixers proposed for mixing concrete.

2.2.5.1 Stationary Mixers

Concrete plant mixers must be tilting, nontilting, horizontal-shaft, vertical-shaft, or pugmill and provided with an acceptable device to lock the discharge mechanism until the required mixing time has elapsed. The mixing time and uniformity must conform to all the requirements in [ASTM C94/C94M](#) applicable to central-mixed concrete.

2.2.5.2 Truck Mixers

Truck mixers, the mixing of concrete therein, and concrete uniformity must conform to the requirements of [ASTM C94/C94M](#). A truck mixer may be used either for complete mixing (transit-mixed) or to finish the partial mixing done in a stationary mixer (shrink-mixed). Equip each truck with two counters from which it will be possible to determine the number of revolutions at mixing speed and the number of revolutions at agitating speed.

2.2.6 Conveying Equipment

Submit data on the conveying equipment and methods for transporting, handling, and depositing the concrete. Conform the conveying equipment to the following requirements:

2.2.6.1 Buckets

The interior hopper slope shall be not less than 58 degrees from the horizontal, the minimum dimension of the clear gate opening shall be at least five times the nominal maximum-size aggregate, and the area of the gate opening shall not be less than 2 square feet. The maximum dimension of the gate opening shall not be greater than twice the minimum dimension. The bucket gates shall be essentially grout tight when closed and may be manually, pneumatically, or hydraulically operated except that buckets larger than 2 cubic yards shall not be manually operated. The design of the bucket shall provide means for positive regulation of the amount and rate of deposit of concrete in each dumping position.

2.2.6.2 Transfer Hoppers

Concrete may be charged into nonagitating hoppers for transfer to other conveying devices. Transfer hoppers must be capable of receiving concrete directly from delivery vehicles and have conical-shaped discharge features. Equip the transfer hopper with a hydraulically operated gate and with a means of external vibration to effect complete discharge. Do not hold concrete in nonagitating transfer hoppers more than 30 minutes.

2.2.6.3 Trucks

Truck mixers operating at agitating speed or truck agitators used for transporting plant-mixed concrete shall conform to the requirements of ASTM C94/C94M. Nonagitating equipment may be used for transporting plant-mixed concrete over a smooth road when the hauling time is less than 15 minutes. Bodies of nonagitating equipment shall be smooth, watertight, metal containers specifically designed to transport concrete, shaped with rounded corners to minimize segregation, and equipped with gates that will permit positive control of the discharge of the concrete.

2.2.6.4 Chutes

When concrete is placed directly from a truck mixer, agitator, or nonagitating equipment, the chutes attached to this equipment by the manufacturer may be used. Use a discharge deflector when required by the Contracting Officer. Separate chutes and other similar equipment will not be permitted for conveying concrete.

2.2.6.5 Belt Conveyors

Design and operate to assure a uniform flow of concrete from mixer to final place of deposit without segregation of ingredients or loss of mortar and provide with positive means for preventing segregation of the concrete at the transfer points and the point of placing. Construct belt conveyors such that the idler spacing does not exceed 36 inches. The belt speed must be a minimum of 300 feet per minute and a maximum of 750 feet per minute. If concrete is to be placed through installed horizontal or sloping reinforcing bars, discharge the conveyor concrete into a pipe or elephant trunk that is long enough to extend through the reinforcing bars.

2.2.6.6 Concrete Pumps

Concrete may be conveyed by positive displacement pump when approved. The pumping equipment must be piston or squeeze pressure. The pipeline must be rigid steel pipe or heavy-duty flexible hose. The inside diameter of

the pipe must be at least three times the nominal maximum-size coarse aggregate in the concrete mixture to be pumped but not less than 4 inches. Do not use aluminum pipe.

2.2.7 Vibrators

Use vibrators of the proper size, frequency, and amplitude for the type of work being performed in conformance with the following requirements:

APPLICATION	HEAD DIAMETER (inches)	FREQUENCY (VPM)	AMPLITUDE (inches)
Thin walls, beams, etc.	1-1/4 to 2-1/2	9,000 to 13,500	0.02 to 0.04
General construction	2 to 3-1/2	8,000 to 12,000	0.025 to 0.05

Determine the frequency and amplitude in accordance with COE CRD-C 521.

2.2.8 Bridge Deck Screeds

Do not use vibratory screeds without approval. Use screeds that are mechanically operated, designed and constructed to screed with the strike-off parallel to the center line, readily adjustable, and capable of maintaining proper adjustment throughout the screeding operation. Use transverse screeds for all deck pours.

Support screeds outside the pour area that will receive a surface finish. Do not use intermediate supports or guides.

Adjust screeds to the camber specified on the Plans. Verify the finish grade elevations at the intervals specified in the Contract Plans during operations.

The following information for screeds and their supports shall be submitted for Contracting Officer approval:

- a. Weight
- b. Durability
- c. Adjustability
- d. Accuracy
- e. Mechanical condition
- f. Operational results

Check screed clearances and tolerances before pouring bridge deck.

PART 3 EXECUTION

3.1 PREPARATION FOR PLACING

3.1.1 Concrete Placement Plan

Prior to commencement of work, submit for approval details and descriptions of concrete placement in, above and adjacent to water showing conformance to requirements of the Environmental Protection Plan in

Section 01 57 20, ENVIRONMENTAL PROTECTION, including preventative measures for spillage, containment and cleanup.

3.1.2 Embedded Items

Before placement of concrete, take care to determine that all embedded items are firmly and securely fastened in place as indicated on the drawings, or required. Embedded items must be free of oil and other foreign matter such as loose coatings or rust, paint, and scale. The embedding of wood in concrete will be permitted only when specifically authorized or directed. Temporarily fill voids in sleeves, inserts, and anchor slots with readily removable materials to prevent the entry of concrete into voids. Welding, including tack welding, will not be permitted on embedded metals within 2 feet of the surface of the concrete.

3.1.3 Concrete on Earth Foundations

Earth surfaces upon which concrete is to be placed must be clean, damp, and free from debris, frost, ice, and standing or running water. Prior to placement of concrete, compact the earth foundation in accordance with Section 31 00 00 EARTHWORK.

3.1.4 Construction Joint Treatment

Submit the method and equipment proposed for joint cleanup and waste disposal, for review and approval. Conform construction joint treatment to the following requirements:

3.1.4.1 Joint Preparation

Construction joints shall be roughened to 1/4 inch amplitude minimum roughness per ACI 318/318M. Prepare concrete surfaces to which additional concrete is to be bonded for receiving the next lift or adjacent concrete by cleaning with either air-water cutting, sandblasting, high-pressure water jet, or other approved method. Air-water cutting will not be permitted on formed surfaces or surfaces congested with reinforcing steel. Regardless of the method used, the resulting surfaces must be free from all laitance and inferior concrete so that clean, well bonded coarse aggregate is exposed uniformly throughout the lift surface. Do not undercut the edges of the coarse aggregate. Wash the surface again as the last operation prior to placing the next lift. No standing water is allowed on the surface upon which concrete is placed.

3.1.4.2 Air-Water Cutting

Perform air-water cutting of a construction joint at the proper time and only on horizontal construction joints. The air pressure used in the jet must be 90 to 110 psi, and the water pressure must be just sufficient to bring the water into effective influence of the air pressure. When approved by the Contracting Officer, a retarder complying with the requirements of COE CRD-C 94 may be applied to the surface of the lift to prolong the period of time during which air-water cutting is effective. Prior to receiving approval, furnish samples of the material to be used and demonstrate the method to be used in applications. After cutting, wash and rinse the surface until there is no trace of cloudiness in the wash water. Where necessary to remove accumulated laitance, coatings, stains, debris, and other foreign material, high-pressure water jet or sandblasting will be required as the last operation before placing the next lift.

3.1.4.3 High-Pressure Water Jet

A stream of water under a pressure of not less than 3,000 psi may be used for cleaning. Delay its use until the concrete is sufficiently hard so that only the surface skin or mortar is removed and there is no undercutting of coarse-aggregate particles. If the water jet is incapable of a satisfactory cleaning, clean the surface by sandblasting.

3.1.4.4 Wet Sandblasting

This method may be used when the concrete has reached sufficient strength to prevent undercutting of the coarse aggregate particles. Then thoroughly wash the surface of the concrete to remove all loose materials.

3.1.4.5 Waste Disposal

Dispose of the waste water employed in cutting, washing, and rinsing of concrete surfaces such that the waste water does not stain, discolor, or affect exposed surfaces of the structures, or damage the environment of the project area. The method of disposal is subject to approval.

3.2 PLACING CONCRETE

3.2.1 Placing Procedures

The surfaces of horizontal construction joints shall be kept continuously wet for the first 12 hours during the 24-hour period prior to placing concrete. Surfaces may be dampened immediately before placement if necessary. Concrete placement will not be permitted when weather conditions prevent proper placement and consolidation. Deposit concrete as close as possible to its final position in the forms and, in so depositing, there shall be no vertical drop greater than 5 feet except where suitable equipment is provided to prevent segregation and where specifically authorized. Depositing of the concrete shall be so regulated that it may be effectively consolidated in horizontal layers 2.0 feet or less in thickness with a minimum of lateral movement. The amount deposited in each location shall be that which can be readily and thoroughly consolidated. Sufficient placing capacity shall be provided so that concrete placement can be kept plastic and free of cold joints while concrete is being placed. Concrete shall be placed by methods that will prevent segregation or loss of ingredients. Any concrete transferred from one conveying device to another shall be passed through a hopper that is conical in shape. The concrete shall not be dropped vertically more than 5 feet, except where a properly designed and sized elephant trunk with rigid drop chute bottom section is provided to prevent segregation and where specifically authorized. In no case shall concrete be discharged to free-fall through reinforcing bars. Retempering of the concrete by adding water or other means is not permitted.

3.2.2 Pumping Concrete Materials and Methods

When concrete is to be placed by pump, the nominal maximum-size coarse aggregate shall not be reduced to accommodate the pumps. The distance to be pumped shall not exceed limits recommended by the pump manufacturer. The concrete shall be supplied to the concrete pump continuously. When pumping is completed, concrete remaining in the pipeline shall be ejected without contamination of concrete in place. After each operation, equipment shall be thoroughly cleaned, and flushing water shall be wasted

outside of the forms. Grout used to lubricate the pumping equipment at the beginning of the placement shall not be incorporated into the placement.

3.2.3 Time Interval Between Mixing and Placing

Place concrete within 30 minutes after discharge into nonagitating equipment. When concrete is truck-mixed or when a truck mixer or agitator is used for transporting concrete mixed by a concrete plant mixer, the concrete shall be delivered to the site of the work, and discharge shall be completed within 1-1/2 hours after either introduction of mixing water to the cement and aggregates or addition of cement to aggregates if the air temperature is less than 84 degrees F. Reduce mixing time and place concrete within 60 minutes if the air temperature is greater than 84 degrees F except as follows: if set retarding admixture is used and slump requirements can be met, limit for placing concrete may remain at 1-1/2 hours. Additional water may be added, provided that both the specified maximum slump and submitted water-cementitious material ratio are not exceeded and the required concrete strength is still met. When additional water is added, an additional 30 revolutions of the mixer at mixing speed is required. When the length of haul makes it impossible to deliver truck-mixed concrete within these time limits, batching of cement and a portion of the mixing water shall be delayed until the truck mixer is at or near the construction site.

3.2.4 Hot-Weather Placing

Concrete shall be properly placed and finished with procedures previously submitted. The concrete-placing temperature shall not exceed 90 degrees F when measured in accordance with ASTM C1064/C1064M. Cooling of the mixing water and aggregates, or both, may be required to obtain an adequate placing temperature. A retarder meeting the requirements of paragraph 2.1.3.3.1 WATER-REDUCING OR RETARDING ADMIXTURES may be used to facilitate placing and finishing. Steel forms and reinforcement shall be cooled prior to concrete placement when steel temperatures are greater than 120 degrees F. Conveying and placing equipment shall be cooled, if necessary, to maintain proper concrete-placing temperature. Submit the proposed materials and methods for review and approval if concrete is to be placed under hot-weather conditions.

Concrete shall not be placed in bridge decks and other exposed slabs when any combination of air temperature, relative humidity, concrete temperature and wind speed is expected to result in an evaporation rate greater than 0.2 pound per square foot per hour. Table below may be used as a guide to determine wind speeds and relative humidity combination at which evaporation rates in excess of 0.2 in/ft²/hr, at an ambient temperature of 90 degrees F, may be expected. The Contracting Officer may authorize placement of concrete in bridge decks and other exposed slabs at temperatures in excess of the critical values if the Contractor submits for approval acceptable procedures which will reduce the evaporation rate throughout the placement area to less than 0.2 lb/ft²/hr. Such procedures shall include one or more of the following: shading and cooling aggregates and other components at the batching plant, dampening subgrade and forms, erecting windbreaks to reduce wind speeds throughout placement area, placing concrete at the lowest possible temperature by reducing time between mixing and placing, fog spraying throughout the placement area to increase relative humidity, or placing concrete at lower ambient temperature such as early morning, late afternoon or at night.

Typical Critical Concrete Temperatures for Various Wind Speeds and
Relative Humidities

(Maximum Concrete temperatures at an ambient temperature of 90 degrees F for different wind speeds and relative humidities to limit the rate of evaporation to about 0.2 in/ft²/hr.)

Concrete Temperature (F)	Relative Humidity % at 10 mph	Relative Humidity % at 15 mph	Relative Humidity % at 20 mph	Relative Humidity % at 25 mph
100	80	95	*	*
95	65	80	85	90
90	45	60	70	75
85	30	45	55	60
80	20	35	40	45
75	10	20	30	35

* -> Required relative humidities in excess of 100%.

The Contractor may also use the following website link to calculate evaporation rates during hot-weather placing:
<http://apps.acpa.org/applibrary/EvaporationRate/>

3.2.5 Certification of Batches

Before unloading at the site of delivery, the concrete supplier shall furnish to the Contracting Officer delivery tickets containing the following information: name and number of batch plant, serial number of ticket, date and truck number, name of Contractor, job name and location, class of concrete, volume of concrete in cubic yards, for central mixed concrete the time when first mixing was completed at the central mix plant, for transit mixed concrete and truck mixed concrete the time when the cement was introduced to the aggregates, name of any admixtures, and space to indicate time when discharge started and finished. The ticket shall be issued to the truck operator at the plant for each load.

3.2.6 Consolidation

Immediately after placement, each layer of concrete, including flowing concrete, shall be consolidated by internal vibrating equipment. Vibrators shall not be used to transport concrete within the forms. Hand spading may be required, if necessary, with internal vibrating along formed surfaces permanently exposed to view. Form or surface vibrators shall not be used unless specifically approved. The vibrator shall be inserted vertically at uniform spacing over the entire area of placement. The distance between insertions shall be approximately 1-1/2 times the radius of action of the vibrator. The vibrator shall penetrate rapidly to the bottom of the layer and at least 6 inches into the preceding unhardened layer if such exists. It shall be held stationary until the concrete is consolidated and then withdrawn slowly.

3.2.7 Placing Concrete in Congested Areas

Special care shall be used to ensure complete filling of the forms,

elimination of all voids, and complete consolidation of the concrete when placing concrete in areas congested with reinforcing bars, embedded items, and other tight spacing. An appropriate concrete mixture shall be used, and the nominal maximum size of aggregate (NMSA) shall meet the specified criteria when evaluated for the congested area. Vibrators with heads of a size appropriate for the clearances available shall be used, and the consolidation operation shall be closely supervised to ensure complete and thorough consolidation at all points. Where necessary, splices of reinforcing bars shall be alternated to reduce congestion. Where two mats of closely spaced reinforcing are required, the bars in each mat shall be placed in matching alignment to reduce congestion.

3.2.8 Placing Tremie Concrete

Deposit concrete in water by a tremie method. The methods and equipment used will be subject to approval. Concrete buckets will not be permitted for placement of concrete except to deliver concrete to the tremie. The tremie must be watertight and sufficiently large to permit a free flow of concrete. Keep the discharge end of the pump line or tremie shaft continuously submerged in the concrete with at least 10 ft of concrete head according to paragraph 3.13.3 of Section 31 63 29 DRILLED CONCRETE PIER AND SHAFT. Effect the underwater seal in a manner that will not produce undue turbulence in the water. Keep the tremie shaft full of concrete to a point well above the water surface. Provide a sufficient number tremie pipes so that the maximum horizontal flow is limited to 15 feet. Proceed with placement without interruption until the concrete has been brought to the required height. Do not move the tremie horizontally during a placement.

3.2.9 Placing Bridge Deck Concrete

Concrete for bridge decks shall be placed in one continuous operation for each span unless otherwise indicated.

3.2.10 Placing Load on Bridge Substructure

No concrete shall be placed in the superstructure until the column or drilled shaft forms have been stripped sufficiently to determine the character of the concrete. The load of the superstructure shall not be allowed to come upon the substructure until the substructure has been in place at least 14 days.

3.3 PLACING NONSHRINK GROUT

Nonshrink grout shall conform to the requirements of subparagraph "Nonshrink Grout" above. Water content shall be the minimum that will provide a flowable mixture and fill the space to be grouted without segregation, bleeding, or reduction of strength.

3.3.1 Mixing and Placing of Nonshrink Grout

Mixing and placing shall be in conformance with the material manufacturer's instructions and as specified. Ingredients shall be thoroughly dry-mixed before adding water. After adding water, the batch shall be mixed for 3 minutes. Batches shall be of size to allow continuous placement of freshly mixed grout. Grout not used within 30 minutes after mixing shall be discarded. Forms shall be of wood or other equally suitable material for retaining the grout and shall be removed after the grout has set. If grade "A" grout as specified in

ASTM C1107/C1107M is used, all surfaces shall be formed to provide restraint. The placed grout shall be worked to eliminate voids; however, overworking and breakdown of the initial set shall be avoided. Grout shall not be retempered or subjected to vibration from any source. Where clearances are unusually small, placement shall be under pressure with a grout pump. Temperature of the grout, and of surfaces receiving the grout, shall be maintained at 65 to 85 degrees F until after setting.

3.3.2 Treatment of Exposed Surfaces

After the grout has set, those types containing metallic aggregate shall have the exposed surfaces cut back 1 inch and immediately covered with a parge coat of mortar proportioned by mass of one part portland cement, two parts sand, and sufficient water to make the mixture placeable. The parge coat shall have a smooth, dense finish. The exposed surface of other types of nonshrink grout shall have a smooth, dense finish.

3.4 DECK PLACEMENT

A smooth, durable riding surface of uniform texture, true to the required grade and cross-section, shall be obtained on the bridge decks. Particular emphasis should be placed on proper vibration of the concrete to avoid honeycomb and voids, especially at construction joints, expansion joints, and valleys and ends of form sheets.

Prior to any deck concreting, a "pre-pour" conference shall be held with the Contractor and representatives of the Contracting Officer in attendance. At this time, the Contractor's plan and procedures for deck construction shall be presented. The plan shall be submitted for approval by the Contracting Officer and approved prior to the meeting as the [Deck Concrete Placement Procedure Plan](#). The plan at a minimum shall include pouring sequences, placement procedures, equipment used, contracting personnel involved, formwork design, screed support design and drawings, finishing equipment used, finishing procedures including scarification and grooving and approved concrete mixes.

3.5 TESTS AND INSPECTIONS

Submit test results and inspection reports, as required by applicable specification sections below. Test and inspect in accordance the following:

3.5.1 General

Perform the inspections and tests described below, and, based upon the results of these inspections and tests, take the action required and submit reports as required. The laboratory performing the tests shall be onsite and shall conform with the requirements given in [ASTM C1077](#). The individuals who sample and test concrete or the constituents of concrete as required in this specification shall have demonstrated a knowledge and ability to perform the necessary test procedures equivalent to the ACI minimum guidelines for certification of Concrete Field Testing Technicians, Grade I. The Government will inspect the laboratory, equipment, and test procedures prior to start of concreting operations and at least once per year thereafter for conformance with [ASTM C1077](#).

3.5.2 Testing and Inspection Requirements

3.5.2.1 Fine Aggregate

3.5.2.1.1 Grading

At least once during each shift when the concrete plant is operating, perform one sieve analysis and fineness modulus determination in accordance with [ASTM C136/C136M](#) and [COE CRD-C 104](#) for the fine aggregate or for each size range of fine aggregate if it is batched in more than one size or classification. The location at which samples are taken may be selected by the Contractor as the most advantageous for control. However, the Contractor is responsible for delivering fine aggregate to the mixer within specification limits.

3.5.2.1.2 Fineness Modulus Control Chart

Results for fineness modulus shall be grouped in sets of three consecutive tests, and the average and range of each group shall be plotted on a control chart. The upper and lower control limits for average shall be drawn 0.10 units above and below the target fineness modulus, and the upper control limit for range shall be 0.20 units above the target fineness modulus.

3.5.2.1.3 Corrective Action for Fine Aggregate Grading

When the amount passing on any sieve is outside the specification limits, immediately resample and retest the fine aggregate. If there is another failure on any sieve, immediately report to the Contracting Officer. Stop concrete batching and take immediate steps to correct the grading.

3.5.2.1.4 Moisture Content Testing

When in the opinion of the Contracting Officer the electric moisture meter is not operating satisfactorily, perform at least four tests for moisture content in accordance with [ASTM C566](#) during each 8-hour period of mixing plant operation. Randomly select the times for the tests within the 8-hour period. Make an additional test whenever the slump is shown to be out of the specified range or excessive variation in workability is reported by the placing foreman. When the electric moisture meter is operating satisfactorily, make at least two direct measurements of moisture content per week to check the calibration of the meter. Use the results of tests for moisture content to adjust the added water in the control of the batch plant.

3.5.2.1.5 Moisture Content Corrective Action

Whenever the moisture content of the fine aggregate changes by 0.5 percent or more, adjust the scale settings for the fine-aggregate batcher and water batcher (directly or by means of a moisture compensation device) if necessary to maintain the specified slump.

3.5.2.2 Coarse Aggregate

3.5.2.2.1 Grading

At least once during each shift in which the concrete plant is operating, conduct a sieve analysis in accordance with [ASTM C136/C136M](#) for each size of coarse aggregate. The location at which samples are taken may be

selected by the Contractor as the most advantageous for production control. A test record of samples of aggregate taken at the same locations must show the results of the current test as well as the average results of the five most recent tests including the current test. The Contractor may adopt limits for control which are coarser than the specification limits for samples taken at locations other than as delivered to the mixer to allow for degradation during handling.

3.5.2.2.2 Corrective Action for Grading

When the amount passing any sieve is outside the specification limits, immediately resample and retest the coarse aggregate. If the second sample fails on any sieve, report that fact to the Contracting Officer. Where two consecutive averages of five tests are outside specification limits, consider the operation out of the specified range and report to the Contracting Officer. Stop concrete batching and take immediate steps to correct the grading.

3.5.2.2.3 Coarse Aggregate Moisture Content

Make a test for moisture content of each size group of coarse aggregate at least twice per week. When two consecutive readings for smallest size coarse aggregate differ by more than 1.0 percent, increase frequency of testing to that specified above for fine aggregate, until the difference falls below 1.0 percent.

3.5.2.2.4 Coarse Aggregate Moisture Corrective Action

Whenever the moisture content of any size of coarse aggregate changes by 0.5 percent or more, adjust the scale setting for the coarse aggregate batcher and the water batcher, if necessary, to maintain the specified slump.

3.5.2.3 Quality of Aggregates

Submit aggregate quality tests at least 30 days prior to start of concrete placement.

3.5.2.3.1 Frequency of Quality Tests

Thirty days prior to the start of concrete placement, perform all tests for aggregate quality listed below. In addition, after the start of concrete placement, perform tests for aggregate quality in accordance with the frequency schedule shown below. Take samples tested after the start of concrete placement immediately prior to entering the concrete mixer.

FREQUENCY			
PROPERTY	FINE AGGREGATE	COARSE AGGREGATE	TEST
Specific Gravity	Every 3 months	Every 3 months	ASTM C127 ASTM C128
Absorption	Every 3 months	Every 3 months	ASTM C127 ASTM C128

FREQUENCY			
PROPERTY	FINE AGGREGATE	COARSE AGGREGATE	TEST
Durability Factor (using Procedure A)	Every 12 months	Every 12 months	COE CRD-C 114 ASTM C666/C666M
Clay Lumps and Friable Particles	Every 3 months	Every 3 months	ASTM C142/C142M
Material Finer than No. 200 Sieve	Every 3 months	Every 3 months	ASTM C117
Impurities	Every 3 months	Not applicable	ASTM C40/C40M ASTM C87/C87M
L.A. Abrasion	Not applicable	Every 6 months	ASTM C131/C131M ASTM C535
Chert, less than 2.40 specific gravity	Every 6 months	Every 6 months	ASTM C123/C123M
Coal and Lignite, less than 2.00 specific gravity	Every 6 months	Every 6 months	ASTM C123/C123M
Petrographic Examination	Every 6 months	Every 6 months	ASTM C295/C295M

3.5.2.3.2 Corrective Action for Aggregate Quality

If the result of a quality test fails to meet the requirements for quality immediately prior to start of concrete placement, change production procedures or materials and perform additional tests until the material meets the quality requirements prior to proceeding with either mixture proportioning studies or starting concrete placement. After concrete placement commences, whenever the result of a test for quality fails the requirements, immediately rerun the test. If the second test fails the quality requirement, report the fact to the Contracting Officer and take immediate steps to rectify the situation.

3.5.2.4 Scales

3.5.2.4.1 Weighing Accuracy

Check the accuracy of the scales by test weights prior to start of concrete operations and at least once every 3 months for conformance with the applicable requirements of paragraph 2.2.1 BATCHING EQUIPMENT. Also make such tests whenever there are variations in properties of the fresh concrete that could result from batching errors.

3.5.2.4.2 Batching and Recording Accuracy

Once a week check the accuracy of each batching and recording device during a weighing operation by noting and recording the required weight, recorded weight, and the actual weight batched. Confirm that the calibration devices described in paragraph 2.2 EQUIPMENT, for checking the accuracy of dispensed admixtures, are operating properly.

3.5.2.4.3 Scales Corrective Action

When either the weighing accuracy or batching accuracy does not comply with specification requirements, do not operate the plant until necessary adjustments or repairs have been made. Correct discrepancies in recording accuracies immediately.

3.5.2.5 Batch-Plant Control

Continuously control the measurement of all constituent materials including cementitious materials, each size of aggregate, water, and admixtures. Adjust the aggregate weights and amount of added water as necessary to compensate for free moisture in the aggregates. Adjust the amount of air-entraining agent to control air content within specified limits. Prepare a report indicating type and source of cement used, type and source of pozzolan or slag used, amount and source of admixtures used, aggregate source, the required aggregate and water weights per cubic yard, amount of water as free moisture in each size of aggregate, and the batch aggregate and water weights per cubic yard for each class of concrete batched during plant operation.

3.5.2.6 Concrete Mixture

3.5.2.6.1 Air Content Testing

Conduct air content tests at the same frequency as specified for slump tests. In addition, make at least two tests for air content on randomly selected batches of each separate concrete mixture produced during each 8-hour period of concrete production. Make additional tests when excessive variation in workability is reported by the placing foreman or Government quality assurance representative. Conduct tests in accordance with ASTM C231/C231M and ASTM C173/C173M for lightweight concrete only. Plot test results on control charts which at all times are readily available to the Government. Keep copies of the current control charts in the field, in close proximity to the Contractor's quality control representatives, and plot results as tests are made. When a single test result reaches either the upper or lower action limit immediately conduct a second test. Average the results of the two tests and use this average as the air content of the batch to plot on both the control chart for air content and the control chart for range, and for determining the need for any remedial action. Plot the result of each test, or average as noted in the previous sentence, on a separate chart for each mixture on which an "average line" is set at the midpoint of the specified air content range from paragraph AIR CONTENT. Set an upper warning limit and a lower warning limit line 1.0 percentage point above and below the average line. Set an upper action limit and a lower action limit line 1.5 percentage points above and below the average line, respectively. Plot the range between each two consecutive tests on a control chart for range where an upper warning limit is set at 2.0 percentage points and up upper action limit is set at 3.0 percentage points. Samples for air content may be taken at the mixer, however, the Contractor is responsible for delivering

the concrete to the placement site at the stipulated air content. If the materials or transportation methods cause air content loss between the mixer and the placement, take correlation samples at the placement site as required by the Contracting Officer and control the air content at the mixer.

3.5.2.6.2 Air Content Corrective Action

Whenever points on the control chart for percent air reach either warning limit, immediately adjust the amount of air-entraining admixture batched. As soon as is practical after each adjustment, conduct another test to verify the result of the adjustment. Whenever a point on the control chart range reaches the warning limit, recalibrate the admixture dispenser to ensure that it is operating accurately and with good reproducibility. Whenever a point on either control chart reaches an action limit line, the air content is considered out of control and immediately halt the concreting operation until the air content is under control. Make additional air content tests when concreting is restarted.

3.5.2.6.3 Slump Testing

Perform slump tests at the commencement of concrete placement, when test cylinders are made, and for each batch or every 20 cubic yards of concrete placed in accordance with ASTM C143/C143M. Also, make additional tests when excessive variation in workability is reported by the placing foreman or Government's quality assurance representative. Plot test results on control charts, which are readily available to the Government at all times. Keep copies of the current control charts in the field in close proximity to the Contractor's quality control representatives and plot results as tests are made. When a single slump test reaches or goes beyond either the upper or lower action limit, immediately make a second test on the same batch of concrete. Average the results of the two tests and use this average as the slump of the batch to plot on both the control chart for slump and the chart for range, and for determining the need for any remedial action. Set an upper warning limit at 1/2 inch below the maximum allowable slump on separate control charts for each type of mixture as specified in paragraph 1.2.4 SLUMP, and set an upper action limit line and lower action limit line at the maximum and minimum allowable slumps, respectively, as specified in the same paragraph. Plot the range between each consecutive slump test for each type of mixture on a single control chart for range on which an upper action limit is set at 2 inches. Take samples for slump at the mixer, however, the Contractor is responsible for delivering the concrete to the placement site at the stipulated slump. If the materials or transportation methods cause slump loss between mixer and the placement, take correlation samples at the placement site as required by the Contracting Officer and control the slump at the mixer.

3.5.2.6.4 Slump Corrective Action

Whenever points on the control chart for slump reach the upper warning limit, immediately adjust the batch weights of water and fine aggregate. Make adjustments so that the total water content does not exceed the amount allowed by the maximum W/C specified, based upon aggregates which are in a saturated surface-dry condition. When a single slump reaches the upper or lower action limit, no further concrete may be delivered to the placing site until proper adjustments have been made. Immediately after each adjustment, make another test to verify the correctness of the adjustment. Whenever two consecutive slump tests, made during a period

when there was no adjustment of batch weights, produce a point on the control chart for range at or above the upper action limit, immediately halt the concreting operation and take appropriate steps to bring the slump under control. Also, make additional slump tests as needed to comply with contract requirements.

3.5.2.6.5 Temperature

Measure the temperature of the concrete when compressive strength specimens are fabricated and when slump tests are conducted. Measure in accordance with [ASTM C1064/C1064M](#). Report the temperature along with the compressive strength data.

3.5.2.6.6 Compressive Strength Specimens

Obtain samples and test concrete for quality control during placement. Sample fresh concrete for testing in accordance with [ASTM C172/C172M](#). Make six test cylinders that are [6 inches by 12 inches](#). Test concrete for compressive strength at 7 and 28 days for each concrete design mix placed each day, not less than once a day, nor less than once for each [30 cu yd.](#) of concrete. Additional sets of test cylinders shall be made when the mixture proportions are changed or when low strengths have been detected. A [random sampling plan](#) shall be developed by the Contractor and approved by the Contracting Officer prior to the start of construction. The plan shall ensure that sampling is done in a completely random and unbiased manner. Test two cylinders at 7 days; two cylinders at 28 days; and hold two cylinders in reserve. Prepare concrete test specimens in accordance with [ASTM C31/C31M](#). Perform compressive strength testing in accordance with [ASTM C39/C39M](#). All [compressive strength tests](#) shall be reported immediately to the Contracting Officer. Quality control charts shall be kept for individual strength tests, moving average for strength, and moving average for range for each mixture. The charts shall be similar to those found in [ACI 214R](#).

3.5.2.6.7 Protection from Mechanical Injury

During curing period, protect concrete from damaging mechanical disturbances, particularly load stresses, heavy shock, and excessive vibration and from damage caused by rain or running water.

3.5.2.6.8 Strength of Concrete Structure

The strength of the concrete structure will be considered to be deficient if any of the following conditions are identified:

- a. Failure to meet compressive strength tests as evaluated.
- b. Reinforcement not conforming to requirements specified.
- c. Concrete which differs from required dimensions or location in such a manner as to reduce strength.
- d. Concrete curing and protection of concrete against extremes of temperature during curing, not conforming to requirements specified.
- e. Concrete subjected to damaging mechanical disturbances, particularly load stresses, heavy shock, and excessive vibration.
- f. Poor workmanship likely to result in deficient strength.

Where the strength of the concrete structure is considered deficient submit a mitigation or remediation plan for review and approval by the Contracting Officer.

3.5.2.6.9 Testing Concrete Structure for Strength

When there is evidence that strength of concrete structure in place does not meet specification requirements or there are non-conforming materials, make cores drilled from hardened concrete for compressive strength determination in accordance with [ASTM C42/C42M](#), and as follows:

- a. Take at least three representative cores from each member or area of concrete-in-place that is considered potentially deficient. Location of cores will be determined by the Contracting Officer.
- b. Test cores after moisture conditioning in accordance with [ASTM C42/C42M](#) if concrete they represent is more than superficially wet under service.
- c. Air dry cores, ([60 to 80 degrees F](#) with relative humidity less than 60 percent) for 7 days before test and test dry if concrete they represent is dry under service conditions.
- d. Strength of cores from each member or area are considered satisfactory if their average is equal to or greater than 85 percent of the 28-day design compressive strength of the class of concrete.
- e. Core specimens will be taken and tested by the Contractor in the presence of the Contracting Officer. If the results of core-boring tests indicate that the concrete as placed does not conform to the drawings and specification, the cost of such tests and restoration required must be borne by the Contractor.

Fill core holes solid with patching mortar and finished to match adjacent concrete surfaces.

Correct concrete work that is found inadequate by core tests in a manner approved by the Contracting Officer.

3.5.2.6.10 [Permeability](#)

Measure the permeability (Level 2) according to Table 934-2 in compliance with Special Provision Specification 934 - Structural Concrete of PRHTA and the frequency of testing as stated in [PRHTA SP 934-2.09](#).

3.5.2.6.11 Anti-Washout Testing

Comply with [COE CRD-C 61](#). Determine cumulative mass loss. Perform test once for each [350 cubic yards](#) of underwater concrete.

3.5.2.6.12 Aggregate for Travel Way

For concrete intended for use on the travel way for vehicular traffic, samples shall be taken of the fresh mix, the mortar shall be washed out and the remaining aggregates shall be delivered to the Government's Resident Office in San Juan, Puerto Rico to be tested for compliance with paragraph 2.1.2 MATERIALS - Aggregates as to manufactured sands and polishing value of course aggregate.

3.5.2.7 Inspection Before Placing

Inspect foundation or construction joints, forms, and embedded items for quality in sufficient time prior to each concrete placement to certify to the Contracting Officer that they are ready to receive concrete. Report the results of each inspection in writing.

3.5.2.8 Placing

3.5.2.8.1 Placing Inspection

The placing foreman shall supervise all placing operations, determine that the correct quality of concrete or grout is placed in each location as specified and is responsible for measuring and recording concrete temperatures and ambient temperature hourly during placing operations, weather conditions, time of placement, yardage placed, and method of placement.

3.5.2.8.2 Placing Corrective Action

The placing foreman shall not permit batching and placing to begin until he or she has verified that an adequate number of vibrators in working order and with competent operators are available. Discontinue placing if any concrete is inadequately consolidated. If any batch of concrete fails to meet the temperature requirements, take immediate steps to improve temperature controls.

3.5.2.9 Vibrators

3.5.2.9.1 Vibrator Testing and Use

Determine the frequency and amplitude of each vibrator in accordance with **COE CRD-C 521** prior to initial use and at least once a month when concrete is being placed. Make additional tests when a vibrator does not appear to be adequately consolidating the concrete. Determine the frequency at the same time the vibrator is operating in concrete with the tachometer held against the upper end of the vibrator head while almost submerged and just before the vibrator is withdrawn from the concrete. Determine the amplitude with the head vibrating in air. Take two measurements, one near the tip and another near the upper end of the vibrator head and average these results. Report the make, model, type, and size of the vibrator and frequency and amplitude results in writing.

3.5.2.9.2 Vibrator Corrective Action

Immediately remove any vibrator not meeting the requirements of paragraph 3.4.2.9 VIBRATORS from service and repair or replace.

3.5.2.10 **Mixer Uniformity**

Submit the results of the initial mixer uniformity tests, at least 30 days prior to the initiation of placing.

3.5.2.10.1 Stationary Mixers

Prior to the start of concrete placing and once every 6 months when concrete is being placed, or once for every **75,000 cubic yards** of concrete placed, whichever results in the longest time interval, determine

uniformity of concrete mixing in accordance with ASTM C94/C94M.

3.5.2.10.2 Truck Mixers

Prior to the start of concrete placing and at least once every 6 months when concrete is being placed, determine uniformity of concrete in accordance with ASTM C94/C94M. Randomly select the truck mixers for testing. When satisfactory performance is found in one truck mixer, the performance of mixers of the same design and condition of the blades may be regarded as satisfactory.

3.5.2.11 Mixer Uniformity Corrective Action

When a mixer fails to meet mixer uniformity requirements, either remove the mixer from service on the work, increase the mixing time, change the batching sequence, reduce the batch size, or adjust the mixer until compliance is achieved.

3.5.3 Reports

Report all results of tests or inspections conducted informally as they are completed and in writing daily. Prepare a weekly report for the updating of control charts covering the entire period from the start of the construction season through the current week. During periods of cold-weather protection, make reports of pertinent temperatures daily. These requirements do not relieve the Contractor of the obligation to report certain failures immediately as required in preceding paragraphs. Confirm such reports of failures and the action taken in writing in the routine reports. The Contracting Officer has the right to examine all test and inspection records.

3.6 REPAIR, REHABILITATION AND REMOVAL

Before the Owner accepts the structure and final payment is made the Contractor shall inspect the structure for cracks, damage, and substandard concrete placements that may adversely affect the service life of the structure. A report documenting these defects shall be prepared that includes recommendations for repair, removal and/or remediation which, and these plans shall be submitted to the COR for evaluation and approval.

3.6.1 Crack Repair

Prior to final acceptance, all cracks in excess of 0.02 inches wide shall be documented and repaired at the Contractor's expense. The proposed method and materials to repair the cracks shall be submitted to the Contracting Officer for approval. The proposal shall address the amount of movement expected in the crack due to temperature changes and loading.

3.6.2 Repair of Weak Surfaces

Weak surfaces are defined as mortar-rich, rain-damaged, uncured, or containing exposed voids or deleterious materials. Concrete surfaces with weak surfaces less than 1/4 inch thick shall be diamond ground to remove the weak surface. Surfaces containing weak surfaces greater than 1/4 inch thick shall be removed and replaced or mitigated in a manner acceptable to the Contracting Officer. Repairs will be made at the Contractor's expense.

3.6.3 Failure of Quality Control Test Results

Proposed mitigation efforts by the Contractor to restore the original design intent shall be submitted to the COR for evaluation and approval prior to proceeding.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 03 - CONCRETE

SECTION 03 35 00

CONCRETE FINISHING

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY ASSURANCE
 - 1.3.1 Sample Wall Panels

PART 2 PRODUCTS

- 2.1 LATEX BONDING COMPOUND
- 2.2 EPOXY RESIN

PART 3 EXECUTION

- 3.1 FINISHING FORMED SURFACES
 - 3.1.1 Class B Finish
 - 3.1.2 Class D Finish
 - 3.1.3 Finishes for Bridge Structure
- 3.2 REPAIRS
 - 3.2.1 Class B Finishes
 - 3.2.2 Class D Finish
 - 3.2.3 Material and Procedure for Repairs
 - 3.2.4 Repairs for Bridge Structure Finishes
- 3.3 FINISHING UNFORMED SURFACES
 - 3.3.1 General
 - 3.3.2 Float Finish
 - 3.3.3 Broom Finish

-- End of Section Table of Contents --

SECTION 03 35 00

CONCRETE FINISHING

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C1059/C1059M (2013) Standard Specification for Latex Agents for Bonding Fresh to Hardened Concrete

ASTM C881/C881M (2020a) Standard Specification for Epoxy-Resin-Base Bonding Systems for Concrete

COMMONWEALTH OF PUERTO RICO, DEPARTMENT OF TRANSPORTATION AND PUBLIC WORKS (PRHTA)

PRHTA Specifications (2005) Standard Specifications for Road and Bridge Construction

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where PRHTA Specifications Section, and/or Special Provision(s) is cited that contains references to other PRHTA sections, they shall also form a part of this specification to the extent referenced. Where reference is made to "Highway Authority" or "Authority's Representative" in

PRHTA Specifications shall mean "Government" or "Contracting Officer" as defined in the Contract Documents. Where PRHTA Specifications refer to the "Engineer", "Engineer of Tests", or "Division of Tests" it shall be understood to mean the "Contracting Officer" as stated in the Contract Documents. In case of conflict between the PRHTA specifications and the Specifications in this Section that follows, the Specifications in this Section shall govern.

Paragraphs titled "Method of Measurement" and "Basis of Payment" of referenced PRHTA Specifications shall not apply.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Designation R0 stands for Resident Office and designation D0 stands for District Office. Submit the following in accordance with Section **01 33 00** SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Finishing Plan; G, D0

SD-04 Samples

Sample Wall Panels
Latex Bonding Compound
Epoxy Resin

1.3 QUALITY ASSURANCE

1.3.1 Sample Wall Panels

Construct one sample panel at least 4 feet by 5 feet and 6 inches thick to demonstrate Class B formed finish. Each panel must include a full length and full width joint line and have at least two voids each at least 12 inches by 12 inches by 3 inches deep either impressed in the concrete as placed or chipped in the hardened concrete. After the concrete is 7 days old, patch the voids to demonstrate the effectiveness and the appearance of the Contractor's repair procedures.

PART 2 PRODUCTS

2.1 LATEX BONDING COMPOUND

Latex bonding compound agents for bonding fresh to hardened concrete shall conform to ASTM C1059/C1059M. Submit samples as required.

2.2 EPOXY RESIN

Epoxy resin for use in repairs shall conform to ASTM C881/C881M, Type III, Grade I or II. Submit samples as required.

PART 3 EXECUTION

3.1 FINISHING FORMED SURFACES

Forms, form materials, and form construction are specified in Section 03 11 13 STRUCTURAL CAST-IN-PLACE CONCRETE FORMING. Finishing of formed surfaces shall be as specified herein. Unless another type of architectural or special finish is specified, surfaces shall be left with the texture imparted by the forms except that defective surfaces shall be repaired at no cost to the Government. Maintain uniform color of the concrete by use of only one mixture without changes in materials or proportions for any structure or portion of structure that is exposed to view or on which a special finish is required. The form panels used to produce the finish shall be orderly in arrangement, with joints between panels planned in approved relation to openings, building corners, and other architectural features. Forms shall not be reused if there is any evidence of surface wear or defects that would impair the quality of the surface.

3.1.1 Class B Finish

Class B finish is required on surfaces exposed to view. Remove fins, ravelings, and loose material. All surface defects over 1/2 inch in diameter or more than 1/2 inch deep shall be repaired. Except as otherwise indicated or as specified in Section 03 11 13 STRUCTURAL CAST-IN-PLACE

CONCRETE FORMING, holes left by removal of form ties shall be reamed and filled. Defects more than 1/2 inch in diameter shall be cut back to sound concrete, but in all cases at least 1 inch deep. Prepare a sample panel for approval (as specified in PART 1) before commencing repair, showing that the surface texture and color match will be attained. Metal tools shall not be used to finish repairs in Class B surfaces.

3.1.2 Class D Finish

Class D finish is required on surfaces not exposed to view. Fins, ravelings, and loose material shall be removed, and, except as otherwise indicated or as specified in Section 03 11 13 STRUCTURAL CAST-IN-PLACE CONCRETE FORMING, holes left by removal of form ties shall be reamed and filled. Honeycomb and other defects more than 1/2 inch deep or more than 2 inches in diameter shall be repaired. Defects more than 2 inches in diameter shall be cut back to sound concrete, but in all cases at least 1 inch deep.

3.1.3 Finishes for Bridge Structure

Finishes for all components of bridge shall be in accordance with PRHTA Specifications Section 601-3.17 - Finishing Concrete Surfaces of the PRHTA Specifications.

A copy of the above-referenced Sections of the PRHTA Specifications is appended to the end of and made a part of this Specification.

Submit a general finishing plan indicating how all surfaces will be finished and what procedures will be used. All surfaces shall be true to elevation as shown on the drawings. The finished surfaces shall be protected from stains or abrasions. The concrete shall be thoroughly consolidated before finishing operations commence or before leaving it for future concrete or backfill placement. Finish must match adjacent finishes.

3.2 REPAIRS

After removal of forms, all ridges, lips, and bulges on surfaces permanently exposed shall be removed. All repairs shall be completed within 48 hours after form removal.

3.2.1 Class B Finishes

Surfaces listed in Section 03 11 13 STRUCTURAL CAST-IN-PLACE CONCRETE FORMING and as shown to have class B finish shall have surface defects repaired as follows: defective areas, voids, and honeycombs smaller than 16 square inches in area and less than 1/2 inch deep and bug holes exceeding 1/2 inch in diameter shall be chipped and filled with dry-packed mortar. Holes left by removal of tie rods shall be reamed and filled with dry-packed mortar as specified in paragraph MATERIAL AND PROCEDURE FOR REPAIRS below. Defective and unsound concrete areas larger than described shall be defined by 1/2 inch deep dovetailed saw cuts in a rectangular pattern with lines parallel to the formwork, the defective concrete removed by chipping, and the void repaired with replacement concrete. The prepared area shall be brush-coated with an epoxy resin meeting the requirements of paragraph EPOXY RESIN in PART 2, a latex bonding agent meeting the requirements of paragraph LATEX BONDING COMPOUND in PART 2, or a neat cement grout after dampening the area with water. The void shall be filled with replacement concrete in accordance with paragraph MATERIAL AND PROCEDURE FOR REPAIRS below.

3.2.2 Class D Finish

Surfaces listed in Section 03 11 13 STRUCTURAL CAST-IN-PLACE CONCRETE FORMING and as shown to have class D finish shall have surface defects repaired as follows: defective areas, voids, and honeycombs greater than 48 square inches in area or more than 2 inches deep shall be defined by 1/2 inch deep dovetailed saw cuts in a rectangular pattern, the defective concrete removed by chipping and the void repaired with replacement concrete. The prepared area shall be brush-coated with an epoxy resin meeting the requirements of paragraph EPOXY RESIN in PART 2, a latex bonding agent meeting the requirements of paragraph LATEX BONDING COMPOUND in PART 2, or a neat cement grout after dampening the area with water. The void shall be filled with replacement concrete as specified below.

3.2.3 Material and Procedure for Repairs

The cement used in the dry-packed mortar or replacement concrete shall be a blend of the cement used for production of project concrete and white portland cement properly proportioned so that the final color of the mortar or concrete will match adjacent concrete. Trial batches shall be used to determine the proportions required to match colors. Dry-packed mortar shall consist of one part cement to two and one-half parts fine aggregate. The fine aggregate shall be that used for production of project concrete. The mortar shall be remixed over a period of at least 30 minutes without addition of water until it obtains the stiffest consistency that will permit placing. Mortar shall be thoroughly compacted into the prepared void by tamping, rodding, ramming, etc. and struck off to match adjacent concrete. Replacement concrete shall be produced using project materials and shall be proportioned by the Contractor. It shall be thoroughly compacted into the prepared void by internal vibration, tamping, rodding, ramming, etc. and shall be struck off and finished to match adjacent concrete. Forms shall be used to confine the concrete. If an expanding agent is used in the repair concrete, the repair shall be thoroughly confined on all sides including the top surface. Metal tools shall not be used to finish permanently exposed surfaces. The repaired areas shall be cured for 7 days. The temperature of the in situ concrete, adjacent air, and replacement mortar or concrete shall be above 40 degrees F and below 90 degrees F during placement, finishing, and curing. Other methods and materials for repair may be used only when approved in writing by the Contracting Officer. Repairs of the so called "plaster-type" will not be permitted.

3.2.4 Repairs for Bridge Structure Finishes

Repairs to finishes for all components of bridge shall be in accordance with PRHTA Specifications Section 601-3.17 - Finishing Concrete Surfaces of the PRHTA Specifications.

A copy of the above-referenced Sections of the PRHTA Specifications is appended to the end of and made a part of this Specification.

3.3 FINISHING UNFORMED SURFACES

The finish of all unformed surfaces shall meet the requirements of paragraph Tolerances in 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS, when tested as specified herein.

3.3.1 General

The ambient temperature of spaces adjacent to unformed surfaces being finished and of the base on which concrete will be placed shall be not less than 40 degrees F. In hot weather all requirements of Section 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS paragraphs Hot Weather Placing and Drying Shrinkage shall be met. Unformed surfaces that are not to be covered by additional concrete or backfill shall have a float finish, with additional finishing as specified below, and shall be true to the elevation shown on the drawings. Surfaces to receive additional concrete or backfill shall be brought to the elevation shown on the drawings, properly consolidated, and left true and regular. Unless otherwise shown on the drawings, exterior surfaces shall be sloped for drainage. Joints shall be carefully made with a jointing or edging tool. The finished surfaces shall be protected from stains or abrasions. Grate tampers or "jitterbugs" shall not be used for any surfaces. The dusting of surfaces with dry cement or other materials or the addition of any water during finishing shall not be permitted. If bleedwater is present prior to finishing, the excess water shall be carefully dragged off or removed by absorption with porous materials such as burlap. During finishing operations, extreme care shall be taken to prevent over finishing or working water into the surface; this can cause "crazing" (surface shrinkage cracks which appear after hardening) of the surface. Any slabs with surfaces which exhibit significant crazing shall be removed and replaced. During finishing operations, surfaces shall be checked with a 10-foot straightedge, applied in both directions at regular intervals while the concrete is still plastic, to detect high or low areas.

3.3.2 Float Finish

Surfaces shall be screeded and darried or bullfloated to bring the surface to the required finish level with no coarse aggregate visible. No water, cement, or mortar shall be added to the surface during the finishing operation. The concrete, while still green but sufficiently hardened to bear a man's weight without deep imprint, shall be floated to a true and even plane. Floating may be performed by use of suitable hand floats or power-driven equipment. Hand floats shall be made of magnesium or aluminum.

3.3.3 Broom Finish

A broom finish shall be applied to the tops of cap beams. The concrete surface shall be finished with a float finish. The floated surface shall be broomed with a fiber-bristle brush in a direction transverse to the channel wall.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 03 - CONCRETE

SECTION 03 39 00

CONCRETE CURING

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 DELIVERY, STORAGE, AND HANDLING

PART 2 PRODUCTS

- 2.1 Concrete Curing Materials
 - 2.1.1 Impervious-Sheet
 - 2.1.2 Membrane-Forming Curing Compound
 - 2.1.3 Burlap and Cotton Mat
- 2.2 WATER

PART 3 EXECUTION

- 3.1 CURING AND PROTECTION
 - 3.1.1 Moist Curing
 - 3.1.2 Membrane Forming Curing Compounds
 - 3.1.2.1 Application Restrictions
 - 3.1.2.2 Pigmented Curing Compound
 - 3.1.2.3 Nonpigmented Curing Compound
 - 3.1.2.4 Application
 - 3.1.3 Impervious Sheeting (Evaporative Retardant)
 - 3.1.4 Ponding or Immersion
 - 3.1.5 Curing of Bridge Elements
- 3.2 TESTING AND INSPECTION FOR CQC (Contractor's Quality Control)
 - 3.2.1 Moist Curing Inspections
 - 3.2.2 Moist Curing Corrective Action
 - 3.2.3 Membrane Curing Inspection
 - 3.2.4 Membrane Curing Corrective Action
 - 3.2.5 Sheet Curing Inspection
 - 3.2.6 Sheet Curing Corrective Action

-- End of Section Table of Contents --

SECTION 03 39 00

CONCRETE CURING

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO M 182 (2005; R 2017) Standard Specification for
Burlap Cloth Made from Jute or Kenaf and
Cotton Mats

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 301 (2016) Specifications for Structural
Concrete

ACI 308.1 (2011) Specification for Curing Concrete

ASTM INTERNATIONAL (ASTM)

ASTM C1602/C1602M (2018) Standard Specification for Mixing
Water Used in Production of Hydraulic
Cement Concrete

ASTM C171 (2020) Standard Specification for Sheet
Materials for Curing Concrete

ASTM C309 (2011) Standard Specification for Liquid
Membrane-Forming Compounds for Curing
Concrete

COMMONWEALTH OF PUERTO RICO, DEPARTMENT OF TRANSPORTATION AND
PUBLIC WORKS (PRHTA)

PRHTA Specifications (2005) Standard Specifications for Road
and Bridge Construction

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where PRHTA Specifications Section, and/or Special Provision(s) is cited that contains references to other PRHTA sections, they shall also form a part of this specification to the extent referenced. Where reference is made to "Highway Authority" or "Authority's Representative" in

PRHTA Specifications shall mean "Government" or "Contracting Officer" as defined in the Contract Documents. Where PRHTA Specifications refer to the "Engineer", "Engineer of Tests", or "Division of Tests" it shall be

understood to mean the "Contracting Officer" as stated in the Contract Documents. In case of conflict between the PRHTA specifications and the Specifications in this Section that follows, the Specifications in this Section shall govern.

Paragraphs titled "Method of Measurement" and "Basis of Payment" of referenced PRHTA Specifications shall not apply.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Designation RO stands for Resident Office and designation DO stands for District Office. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Concrete Curing Plan; G, DO

SD-03 Product Data

Concrete Curing Materials; G, DO

SD-06 Test Reports

Testing and Inspection for CQC (Contractor's Quality Control; G, RO

SD-08 Manufacturer's Instructions

Curing Compound; G, DO

1.3 DELIVERY, STORAGE, AND HANDLING

Store materials in such a manner as to avoid contamination and deterioration. Materials must be capable of being accurately identified after bundles or containers are opened.

PART 2 PRODUCTS

2.1 Concrete Curing Materials

Provide concrete curing material in accordance with ACI 301 Section 5 and ACI 308.1 Section 2. Submit product data for concrete curing compounds. Submit manufactures instructions for placement of curing compound.

2.1.1 Impervious-Sheet

Impervious-sheet materials shall conform to ASTM C171, type optional, except that polyethylene sheet shall not be used.

2.1.2 Membrane-Forming Curing Compound

Membrane-Forming curing compound shall conform to ASTM C309, Type 1-D or 2, except that only a styrene acrylate or chlorinated rubber compound meeting Class B requirements shall be used for surfaces that are to be painted or are to receive bituminous roofing, or waterproofing, or floors that are to receive adhesive applications of resilient flooring. The

curing compound selected shall be compatible with any subsequent paint, roofing, waterproofing, or flooring specified. Nonpigmented compound shall contain a fugitive dye and shall have the reflective requirements in [ASTM C309](#) waived.

2.1.3 Burlap and Cotton Mat

Burlap and cotton mat used for curing shall conform to [AASHTO M 182](#).

2.2 WATER

Provide water for curing that is fresh, clean, potable, and free of injurious amounts of oil, acid, salt, or alkali, except that non-potable water may be used if it meets the requirements of [ASTM C1602/C1602M](#).

PART 3 EXECUTION

3.1 CURING AND PROTECTION

Prior to construction, submit a [concrete curing plan](#). Submit the proposed materials, methods, and duration for curing concrete in accordance with [ACI 308.1](#) and this specification.

Concrete shall be cured by an approved method for the period of time given below:

Type IS or Type IP	7 days
Type II Portland cement	10 days

For cast-in-place concrete other than drilled shafts, no vibratory compaction equipment, drilling or pile-driving equipment shall be operated within 100 feet horizontally of concrete less than 7 days old. Immediately after placement, protect concrete from premature drying, extremes in temperatures, rapid temperature change and mechanical injury for the duration of the curing period. Concrete shall be protected from the damaging effects of rain for 12 hours and from flowing water for 14 days. No fire or excessive heat including welding shall be permitted near or in direct contact with concrete or concrete embedments at any time. Maintain air and forms in contact with concrete at a temperature above [50 degrees F](#) for the first 3 days and at a temperature above [32 degrees F](#) for the remainder of the specified curing period. Exhaust fumes from combustion heating units shall be vented to the outside of the enclosure, and heaters and ducts shall be placed and directed so as not to cause areas of overheating and drying of concrete surfaces or to create fire hazards. Materials and equipment needed for adequate curing and protection shall be available and at the site prior to placing concrete. Except as otherwise permitted by paragraph Membrane Forming Curing Compounds in PART 2, moist curing shall be provided for any areas to receive floor hardener, any paint or other applied coating, or to which other concrete is to be bonded. Except for plastic coated burlap, impervious sheeting alone shall not be used for curing.

3.1.1 Moist Curing

Maintain concrete to be moist-cured, continuously wet for the entire

curing period, commencing immediately after finishing. If water or curing materials stain or discolor concrete surfaces which are to be permanently exposed, the concrete surfaces shall be cleaned as approved. When wooden forms are left in place during curing, they shall be kept wet at all times. If steel forms are used in hot weather, non-supporting vertical forms shall be carefully broken loose from the concrete, soon after the concrete hardens, and curing water continuously applied into the void so as to continuously saturate the entire concrete surface. If the forms are removed before the end of the curing period, curing shall be carried out as on unformed surfaces, using suitable materials. Horizontal surfaces shall be cured by ponding, by continuous sprinkling, by continuously saturated burlap or cotton mats, or by continuously saturated plastic-coated burlap. Burlap and mats shall be clean and free from any contamination and shall be completely saturated before being placed on the concrete. Provide an approved work system to ensure that moist curing is continuous 24 hours per day. Horizontal construction joints shall be kept wet for 12 hours immediately prior to the placing of the following lift.

3.1.2 Membrane Forming Curing Compounds

3.1.2.1 Application Restrictions

Concrete may be cured with an approved membrane-forming curing compound in lieu of moist curing except that membrane curing will not be permitted on any surface to which a grout-cleaned finish is to be applied or other concrete is to be bonded, on any surface containing protruding steel reinforcement, or on an abrasive aggregate finish. However, a styrene acrylate or chlorinated rubber compound meeting [ASTM C309](#), Class B requirements, may be used for surfaces which are to be painted or are to receive bituminous roofing or waterproofing, or floors that are to receive adhesive applications of resilient flooring. The curing compound selected shall be compatible with any subsequent paint, roofing, waterproofing or flooring specified. Membrane curing compound shall not be used on surfaces that are maintained at curing temperatures with free steam.

3.1.2.2 Pigmented Curing Compound

A pigmented curing compound meeting the requirements of the above paragraph may be used on surfaces that will not be exposed to view when the project is completed.

3.1.2.3 Nonpigmented Curing Compound

A nonpigmented curing compound containing a fugitive dye may be used on surfaces that will be exposed to view when the project is completed. Concrete cured with nonpigmented curing compound must be shaded from the sun for the first 3 days when the ambient temperature is [90 degrees F](#) or higher.

3.1.2.4 Application

Apply the curing compound to formed surfaces immediately after the forms are removed and prior to any patching or other surface treatment except the cleaning of loose sand, mortar, and debris from the surface. The surfaces shall be thoroughly moistened with water. The curing compound shall be applied to unformed surfaces as soon as free water has disappeared and bleeding has stopped. The curing compound shall be applied in a two-coat continuous operation by approved motorized

power-spraying equipment operating at a minimum pressure of 75 psi, at a uniform coverage of not more than 400 square feet per gallon for each coat, and the second coat shall be applied perpendicular to the first coat. Concrete surfaces that have been subjected to rainfall within 3 hours after curing compound has been applied shall be resprayed by the method and at the coverage specified. All concrete surfaces on which the curing compound has been applied shall be adequately protected for the duration of the entire curing period from pedestrian and vehicular traffic and from any other cause that will disrupt the continuity of the curing membrane.

3.1.3 Impervious Sheeting (Evaporative Retardant)

Use impervious-sheet curing only on horizontal or nearly horizontal surfaces. Surfaces shall be thoroughly wetted and be completely covered with the sheeting. Sheeting shall be at least 18 inches wider than the concrete surface to be covered. Covering shall be laid with light-colored side up. Covering shall be lapped not less than 12 inches and securely weighted down or shall be lapped not less than 4 inches and taped to form a continuous cover with completely closed joints. The sheet shall be weighted to prevent displacement so that it remains in contact with the concrete during the specified length of curing. Coverings shall be folded down over exposed edges of slabs and secured by approved means. Sheets shall be immediately repaired or replaced if tears or holes appear during the curing period. In all cases measures shall be implemented to prevent wind to flow in between the concrete surface and the sheets.

3.1.4 Ponding or Immersion

Concrete shall be continually immersed throughout the curing period. Water shall not be more than 20 degrees F less than the temperature of the concrete.

3.1.5 Curing of Bridge Elements

Curing of all components of bridge shall be in accordance with PRHTA Specifications Section 601-3.18 - Curing of Concrete.

A copy of the above-referenced Sections of the PRHTA Specifications is appended to the end of and made a part of this Specification.

3.2 TESTING AND INSPECTION FOR CQC (Contractor's Quality Control)

Perform the inspection and tests described below and, based upon the results of these inspections and tests, take the action required. Submit certified copies of laboratory test reports, including curing compound proposed for use on this project.

3.2.1 Moist Curing Inspections

At least once each shift, and not less than twice per day on both work and non-work days, inspect all areas subject to moist curing. Note and record the surface moisture condition.

3.2.2 Moist Curing Corrective Action

When a daily inspection report lists an area of inadequate curing, take immediate corrective action, and extend the required curing period for

those areas by 1 day.

3.2.3 Membrane Curing Inspection

Apply no curing compound until the Contractor has verified that the compound is properly mixed and ready for spraying. At the end of each operation, estimate the quantity of compound used by measurement of the container and the area of concrete surface covered, compute the rate of coverage in **square feet/gallon**, and note whether or not coverage is uniform.

3.2.4 Membrane Curing Corrective Action

When the coverage rate of the curing compound is less than that specified or when the coverage is not uniform, spray the entire surface again.

3.2.5 Sheet Curing Inspection

At least once each shift and once per day on non-work days, inspect all areas being cured using impervious sheets. Note and record the condition of the covering and the tightness of the laps and tapes.

3.2.6 Sheet Curing Corrective Action

When a daily inspection report lists any tears, holes, or laps or joints that are not completely closed, promptly repair the tears and holes or replace the sheets, close the joints, and extend the required curing period for those areas by 1 day.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 03 - CONCRETE

SECTION 03 41 00

PRECAST, PRESTRESSED CONCRETE BRIDGE GIRDERS

PART 1 GENERAL

- 1.1 SCOPE
- 1.2 REFERENCES
- 1.3 SUBMITTALS
- 1.4 DELIVERY, STORAGE, AND HANDLING
 - 1.4.1 Delivery and Storage
 - 1.4.2 Handling
- 1.5 QUALITY CONTROL
 - 1.5.1 General
 - 1.5.2 Reporting
- 1.6 MANUFACTURER QUALIFICATIONS

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Admixtures
 - 2.1.2 Aggregates
 - 2.1.3 Cement
 - 2.1.4 Pozzolan
 - 2.1.5 Prestressing Steel
 - 2.1.6 Reinforcing Steel
 - 2.1.7 Ties
 - 2.1.8 Water
- 2.2 FABRICATION
 - 2.2.1 Precast, Prestressed Florida I-Beam 36 Concrete Bridge Girders
 - 2.2.2 Forms
 - 2.2.3 Reinforcement and Embedments
 - 2.2.4 Concrete Mixture
 - 2.2.5 Concrete Work
 - 2.2.6 Pretensioning
 - 2.2.7 Detensioning
 - 2.2.8 Curing of Girders
 - 2.2.9 Girder Ends
 - 2.2.10 Top Surface Treatment
 - 2.2.11 Marking
 - 2.2.12 Finish
- 2.3 TESTS, INSPECTIONS, AND VERIFICATIONS
 - 2.3.1 Concrete Cylinders
 - 2.3.2 Certificates of Compliance

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Fabrication
 - 3.1.2 Transportation and Storage
 - 3.1.3 Erection
 - 3.1.4 Inspection

3.1.5 Defects and Repair

-- End of Section Table of Contents --

SECTION 03 41 00

PRECAST, PRESTRESSED CONCRETE BRIDGE GIRDERS

PART 1 GENERAL

1.1 SCOPE

The work covered by this section consists of furnishing all plant, equipment, labor, and materials, except materials specified to be furnished by the Government, and performing all operations in connection with the manufacture, and installation of Precast, Prestressed Florida I-Beam 36 Concrete Bridge Girders.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI SP-66 (2004) ACI Detailing Manual

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO LRFD (8th Edition; 2017) Bridge Design Specifications

ASTM INTERNATIONAL (ASTM)

ASTM A416/A416M (2019) Standard Specification for Low-Relaxation, Seven-Wire Steel Strand for Prestressed Concrete

ASTM A615/A615M (2020) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement

ASTM A706/A706M (2016) Standard Specification for Low-Alloy Steel Deformed and Plain Bars for Concrete Reinforcement

ASTM A1064/A1064M (2018a) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete

ASTM C39/C39M (2020) Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens

ASTM C597 (2009) Pulse Velocity Through Concrete

ASTM C666/C666M (2015) Resistance of Concrete to Rapid

Freezing and Thawing

ASTM C1047

(2019) Standard Specification for
Accessories for Gypsum Wallboard and
Gypsum Veneer Base

DEPARTMENT OF TRANSPORTATION AND PUBLIC WORKS, PUERTO RICO HIGHWAY
TRANSPORTATION AUTHORITY (PRHTA)

PRHTA Specifications

(2005) Standard Specifications for Road
and Bridge Construction

U.S. ARMY CORPS OF ENGINEERS (USACE)

COE CRD-C 400

(1963) Requirements for Water for Use in
Mixing or Curing Concrete

PRECAST/PRESTRESSED CONCRETE INSTITUTE (PCI)

PCI MNL-116

(1999) Manual for Quality Control for
Plants and Production of Structural
Precast Concrete Products, 4th Edition

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Designation R0 stands for Resident Office and designation D0 stands for District Office. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Precast, Prestressed Florida I-Beam 36 Concrete Bridge Girders; G, R0, D0

Submit detailed drawings for approval at least 30 days prior to commencement of work. Drawings shall show complete details including formwork, inserts, reinforcing steel, size and type of tensioning elements and anchorages, sequence of stressing, method of curing, quality of concrete, camber, erection details and methods, and complete stressing calculations, including steel and concrete stresses as well as stresses under the anchorages. Drawings shall indicate pick-up and support points.

Girder Placement and Tolerances; G, R0, D0

Girder placement plans shall be submitted for approval at least 30 days prior to delivery of girders to the jobsite. The plan shall include the sequence of erection, temporary shoring and bracing of girders, crane information and positioning confirming it can perform the lift detailed in the plan, lifting location on girders, rigging plan, rigging materials, calculations showing girders will not be overstressed during erection. This information must be reviewed, stamped and sealed by a professional engineer licensed in Puerto Rico.

Tolerances for the girders shall be in accordance with

PRHTA Specifications, Section 630 and this section.

SD-03 Product Data

Mixture Proportions; G, R0, D0

Mixture proportions shall be in accordance with PRHTA Specifications, Special Provision Specification 934-2. The proposed mixture proportions shall be submitted for approval thirty (30) days prior to casting girders.

Disposition Records: G, R0

A system of identification which shows the disposition of specific lots of approved tested materials in the work shall be established and submitted to the Contracting Officer before completing the contract.

Equipment Descriptions and Certifications; G, R0

Descriptions of tensioning jacks, gages, dynamometers, load cells or other devices for measuring stressing load, certified calibration records for each set of jacking equipment, and testing curves for stress measurement gages which show that the gages have been calibrated for the jacks for which they are used shall be submitted for approval prior to start of the tensioning operations. Calibration jacking equipment shall not be older than 6 months.

Tensioning and Detensioning Records; G, R0

Complete records of tensioning and detensioning shall be submitted to the Contracting Officer by the Contractor, prior to use of girders on the project.

Delivery, Storage, and Handling; G, R0

Delivery, storage, and handling plans for girders shall be submitted for approval at least 30 days prior to delivery of girders to the jobsite.

Curing of Girders; G, R0

Methods and details for curing girders shall be in accordance with PRHTA Specifications, Section 630 and shall be and shall be submitted for approval 30 days prior to casting girders.

SD-06 Test Reports

Test Reports; G, R0

Certified test reports of required material tests conforming to the applicable provisions of related work sections shall be submitted by the Contractor for approval by the Contracting Officer prior to use of the materials in the work. Reports shall be furnished for each shipment and shall be identified with specific lots.

Concrete Test Cylinders; G, R0

Complete records of cylinder tests as describe or referenced in **PRHTA Specifications**, Special Provision Specification 934-2.09 shall be submitted to the Contracting Officer for approval prior to use of girders on the project.

SD-07 Certificates

Certificates of Compliance

Certificates of compliance for admixtures, aggregates, cement, pozzolan, reinforcing steel, and prestressing steel shall be in accordance with the respective Section of this specification and **PRHTA Specifications**, Special Provision Specification 934-3. Certificates of compliance shall be submitted prior to commencing fabrication of girders. Certificates for admixtures, aggregates, cement, and pozzolan shall be submitted along with concrete mix proportions dated within 6 months of the submittal. Aggregate source and gradation information shall be submitted for aggregates.

Records; G, R0

Manufacturer Qualifications; G, R0

1.4 DELIVERY, STORAGE, AND HANDLING

Girders shall be stored, handled, and transported in accordance with **PCI MNL-116** and **PRHTA Specifications**, Special Provision Specification 934-3 and as described in this section. Methods used for handling and storage of girders shall be such that the girders are not subjected to excessive bending stress determined by the engineer, cracking, spalling, or other damage. Girders which are damaged during delivery, storage, or handling to the extent they are rendered unsuitable for the work, by the engineer, will be rejected and shall be removed from the work site at no cost to the Government.

1.4.1 Delivery and Storage

Girders shall be held at the plant until the specified ultimate compressive strength is obtained or 14 days, whichever is greater. Storage areas for girders shall be stabilized and suitable foundations provided so differential settlement or twisting of the girder does not occur. Store girders on only two points of support that are the same as when the girder is in its final position. Stacked girders shall be separated and supported by uniform load transferring material placed across the full width of each bearing point and in vertical planes between the girders.

1.4.2 Handling

Girders shall be lifted by means of a suitable bridle or slings attached to the girder at embedded or attached lifting devices. Unless special lifting devices are attached for pickup, pickup points shall be plainly marked on all girders after removal of the forms. Alternate pickup methods or locations shall be subject to approval prior to commencement of girder fabrication. Dragging of girders across the ground will not be permitted. The Contractor shall inspect each girder for distortion and structural damage such as cracking and spalling before transporting them

from the storage site to the installation area. The Contractor shall again check the girder for out of tolerance distortion and damage immediately prior to placement. Girders having excessive distortion shall not be used.

1.5 QUALITY CONTROL

1.5.1 General

The Contractor shall establish and maintain quality control for girder manufacturing and installation operations, **placement**, **tolerances** and assure compliance with contract specifications and maintain quality control **records** for all construction operations including, but not limited to, the following:

- (1) Testing and gradation of aggregates, permeability, concrete workability, and compressive strength of concrete as required, including batched proportions.
- (2) Setting and bracing of forms and checkout just prior to concrete placement, including accurate placement of reinforcing steel, prestressing strands, and other embedded items.
- (3) Casting, handling and storage of precast, prestressed girder: records of prestressing tension strands.
- (4) Curing method and duration.
- (5) Inspection, evaluation, **disposition**, and reporting of any girder defects.
- (6) Installation of all girders.

1.5.2 Reporting

The original and two copies of these records and reports, as well as corrective action taken, shall be furnished to the Government daily. Format of the report shall be as prescribed in the Section **01 45 04**, "CONTRACTOR QUALITY CONTROL."

1.6 MANUFACTURER QUALIFICATIONS

The precast concrete manufacturing plant shall be certified by the Prestressed Concrete Institute, Plant Certification Program, or the manufacturer shall establish a quality control program based on PCI MNL-116, prior to the start of production. Provide documentation of the qualifications in accordance with this section.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Admixtures

Chemical admixtures and air-entraining admixture shall conform to **PRHTA Specifications**, Special Provision Specifications 934-2.08. Calcium chloride or admixtures containing more than 0.05 percent chloride ions shall not be used.

2.1.2 Aggregates

Aggregates shall conform to the grading requirements of [PRHTA Specifications](#), Special Provision Specifications 934-2. Fine aggregates from different sources of supply shall not be mixed or stored in the same stockpile or used alternately in the same concrete mix or the same structure without approval. The fineness modulus of fine aggregate shall not be less than 2.40 or greater than 3.00. If the selected aggregates fail the soundness test, the aggregate source may be used if the concrete specimens made with the aggregates have a durability factor of not less than 80, based on 300 cycles of freezing and thawing, when tested in accordance with [ASTM C666/C666M](#).

2.1.3 Cement

Portland cement shall conform to [PRHTA Specifications](#), Special Provision Specifications 934-2.

2.1.4 Pozzolan

If used, pozzolans shall conform to [PRHTA Specifications](#), Special Provision Specifications 934-2.

2.1.5 Prestressing Steel

Prestressing steel for the Florida I-Beam Girders shall be the seven wire, [1/2 inch](#) diameter, low-relaxation strand, that conforms to AASHTO M203([ASTM A416/A416M](#)) Grade 270 with a minimum ultimate strength of 45.090 lbs on a nominal cross sectional area of 0.167 square inches. The strand location shall conform with the applicable details on the contract drawings. Steel shall be free from grease, oil, wax, paint, soil, dirt, loose rust, kinks, bends, and other defects.

2.1.6 Reinforcing Steel

Non-prestressing reinforcing steel shall conform to [ASTM A615/A615M](#) or [ASTM A706/A706M](#), Grade 60, deformed.

2.1.7 Ties

Steel for ties shall conform to [ASTM A1064/A1064M](#).

2.1.8 Water

Water for mixing concrete shall be fresh, clean, potable, and free from injurious amounts of oils, acids, alkalies, salts, organic materials, or other substances that may be deleterious to concrete or steel. Non-potable water may be used if it meets the requirements of [COE CRD-C 400](#). The time of set for concrete made with non-potable water may vary from one hour earlier to one and one-half hours later than a control sample made with distilled water.

2.2 FABRICATION

2.2.1 Precast, Prestressed Florida I-Beam 36 Concrete Bridge Girders

Prestressed concrete bridge girders shall be of the type indicated. Girders shall be cast as monolithic units of homogeneous concrete and pretensioned with prestressing steel. Embedded or attached lifting

devices for pick-up shall be as shown on the shop drawings. Pick-up points shall be marked on the girders at the indicated location after removal of the forms. Detail drawings per the Submittals described in this section shall be submitted for approval. The Contractor shall notify the Contracting Officer 1 week prior to the date casting of girders is to begin.

2.2.2 Forms

Forms shall be in accordance with **PRHTA Specifications**, Section 630 except as follows. Forms shall be of steel, braced and stiffened against deformation, accurately constructed, watertight, and supported on unyielding concrete casting beds. For Florida I-Beam 36 girders, use forms that do not have more than two horizontal joints. Form surfaces shall meet the following tolerances:

1. Horizontal alignment (horizontal deviation of side forms either side of a vertical plane within the length of a product) = 1/8 inch
2. Vertical alignment (vertical deviation of the bed or pallet from a horizontal plane within the length of a product) = 1/8 inch
3. Vertical joints (offset between adjacent form sections) = 1/8 inch
4. Horizontal joints (offset between adjacent form sections) = 1/16 inch

Forms shall permit movement of the girder without damage during release of the prestressing force. Voids shall be formed.

2.2.3 Reinforcement and Embedments

Reinforcing steel, prestressing steel, and embedded items shall be accurately positioned in the forms and secured to prevent movement during concrete placement. Steel shall have a minimum concrete cover of as indicated on the plans. Reinforcing steel details shall conform to **ACI SP-66**. Welding of reinforcing steel is not permitted.

2.2.4 Concrete Mixture

The concrete mix shall conform to **PRHTA Specifications**, Special Provision Specifications 934-2 except as specified herein. Corrosion inhibitors are required for the concrete mix for the precast, prestressed concrete girders.

2.2.5 Concrete Work

Concrete shall not be deposited in the forms until the placement of the reinforcement, strands, embedded items, and anchorages has been inspected and approved by the Contracting Officer. Conveying equipment shall be cleaned thoroughly before each run and the concrete conveyed from the mixer to the forms as rapidly as practicable using methods that will not cause segregation or loss of ingredients. Concrete shall be deposited as nearly as practicable in its final position in the forms. At any point in conveying, the free vertical drop of the concrete shall not exceed **3 feet**. Chuting will be permitted if the concrete is deposited into a hopper before being placed in the forms. Concrete that has segregated in conveying shall be removed. Each girder shall be produced of dense concrete with smooth surfaces. Each girder shall be a continuous pour until it is completed. Vibrator heads shall be smaller than the minimum

distance between steel pretensioning. Side forms shall not be removed until concrete has attained 3500 psi compressive strength. Dimensional tolerances shall conform to PCI MNL-116, except as shown in paragraph 3.1.1 FABRICATION of this section. The ends and corners of girders shall be chamfered as indicated.

2.2.6 Pretensioning

Pretensioning of girders shall be in accordance with PRHTA Specifications, Section 630 and this section. Anchorages for tensioning the prestressing steel shall be an approved type. Equipment to pretension the steel shall be certified. The tension to which the steel is to be pretensioned shall be measured by the elongation of the steel and also by the jack pressure reading on a gauge or by the use of an accurately calibrated dynamometer. The gauge or dynamometer shall have been calibrated by a calibration laboratory approved by the Contracting Officer within 12 months of commencing work and every 6 months thereafter during the term of the contract. Means shall be provided for measuring the elongation of the steel to the nearest 1/32 inch. The applied load determined from elongation measurements shall be computed using load-elongation curves for the steel used. When the difference between the results of measurement and gauge reading is more than 5 percent, the cause of the discrepancy shall be corrected. The tensioning steel shall be given a uniform prestress prior to being brought to design prestress. The same initial prestress shall be induced in each unit when several units of prestressing steel in a girder are stretched simultaneously. The prestress force shall be checked by accurately measuring tendon elongation between 1,000-pounds and the required maximum applied prestress force. Elongation requirements shall be obtained from load-elongation curves for the steel used and the applied prestress force shall be computed from the measured elongation. If the difference between the computed force and the measured jack force exceeds 5 percent, the cause of the discrepancy shall be ascertained and corrected. Strands shall not be spliced within the casting length of the girder. The prestressing strands shall be cut prior to reducing the temperature of the curing steam. Sheathing used to debond prestress strands shall be constructed of polyethylene having sufficient rigidity to prevent bonding of the pre-stressing strand and concrete. The sheathing shall be split type sheathing having a minimum thickness of 0.025 inch and shall be of sufficient width to maintain a 0.75 inch, ±1/16 inch overlap after being placed on the strand. The joints between segments of sheathing shall be taped to prevent leakage of concrete into the sheathing.

2.2.7 Detensioning

Releasing of prestressing force in pretensioned girders shall be performed in a manner that minimizes the eccentricity of the prestressing strands. The required concrete strength at which the prestressing force may be transferred to the concrete in a product shall be a minimum of 6,000 psi, . Verify the release strength by compressive strength cylinder tests or other approved means, no later than 24 hours after casting and every 24 hours thereafter until release strength is developed.

In lieu of every 24 hour testing, estimate the strength development of concrete using the maturity method in accordance with ASTM C1047, the pulse velocity method in accordance with ASTM C597, or any other nondestructive test method acceptable to the Contracting Officer, until the time of the detensioning.

Before detensioning, verify the concrete release strength by testing the

compressive strength test cylinders. Make a minimum of two compressive strength release test cylinders daily for each individual mix or for each LOT, or fraction thereof, of a given concrete mix design where the daily consumption exceeds this volume or when non-continuous batching or dissimilar curing is used. The release strength test, representing the LOT, is the average compressive strength of two test cylinders, cured under conditions similar to the product or match-cured test specimens, which are match cured until the time of release.

When side forms are loosened upon setting of concrete or removed before the 72 hour curing period is complete, the top flange dormant strands may be released after the concrete reaches a compressive strength of 2,000 psi.

2.2.8 Curing of Girders

All girders shall be cured in accordance with the applicable requirements of PRHTA Specifications, Section 630 and this section. Cure prestressed concrete shall be for a minimum duration of 72 hours. If forms are loosened upon setting of concrete and/or removed before the 72 hour curing period is complete, expand the curing to cover the newly exposed surfaces by either coating with curing compound or extending the continuous moist cure area. Maintain concrete surface moisture at all times until curing is begun. If a water sheen is not present, apply supplemental moisture by misting or prevent water sheen loss on flat work by use of an evaporation retarder.

After the finishing operations have been completed and as soon as the concrete has hardened sufficiently to permit the application of curing material without marring the exposed surface, cover the exposed surfaces of all prestressed concrete products by one of the following procedures or curing methods.

Continuous Moisture Method: Place burlap on the surface and keep it continuously saturated for the curing period by means of soil soakers, leaking pipes, or automatic sprinklers. Do not apply moisture manually. If side forms are removed during the curing period, extend the burlap to completely shield the sides of the product. Water flow may be metered to cycle repetitively for five minutes on and five minutes off during the 72 hour curing period. When it is not practical to apply moisture or curing compound inside the voided piles, cover their ends with wet burlap to prevent moisture loss.

Curing Blankets: Curing blankets may be used for curing the top surfaces of products. Do not use curing blankets which are torn or punctured. Securely fasten edges to provide as tight a seal as practical. Allow curing blankets to remain in place for the curing period. Should the system fail to maintain a moist condition on the concrete surface, discontinue the use of curing blankets and take immediate corrective action to prevent further loss of concrete moisture.

Curing shall be continued until the concrete has attained a minimum compressive strength of 8,000 psi for the bridge girders as tested in accordance with the requirements of ASTM C39/C39M. All cylinders shall be made and tested by Contractor and witnessed by the Contracting Officer. Concrete test cylinders shall be cured at the same location, under identical conditions, and by the identical method used to cure the girders from which the samples were taken. The girders shall not be shipped from the plant site until the full compressive strength as indicated on the Contract Plans has been attained.

2.2.9 Girder Ends

The ends of all girders shall be finished in accordance with paragraph 2.2.12 Finish of this specification..

2.2.10 Top Surface Treatment

The top surface of all girders shall be clean, free of laitance and intentionally roughened in the transverse direction to a full amplitude of approximately $\frac{1}{4}$ inch.

2.2.11 Marking

All precast units shall be plainly marked and identified for ready correlation with corresponding test specimens. Pickup points shall be plainly marked on all units. Units shall be marked to show the number, size and location of reinforcement bars and to identify the top face and the position on the placement drawings.

2.2.12 Finish

All surfaces of the girders other than the top surface shall be finished in accordance with this section. After placing and consolidating the concrete, strike-off all exposed surfaces to the lines and grades indicated in the Contract Plans in a manner that will leave a surface of uniform texture free of undesirable surface irregularities, cavities, and other defects. Cut back metal ties supporting reinforcement, conduit, and other appurtenances a minimum of $\frac{1}{2}$ inch from finished surface. Ensure that finished work in addition to that specified above is compatible and complementary to the class of surface finish required.

Remove all laitance, loose material, form oil and curing compound from exposed surfaces that do not require forming and from exposed surfaces requiring forming, after form removal. Remove fins and irregular projections flush with the surface. Clean, saturate with water, and fill all holes, tie cavities, honeycomb, chips and spalls. Prior to filling, prepare the surface to ensure that patching mortar will bond to the existing concrete. Exercise care during the roughening process to prevent excessive defacement and damage to the surface of the existing concrete. Use patching mortar blended from the mix ingredients of the existing concrete. Ensure the patching mortar closely matches the color and surface of the existing concrete when fully cured. Remove mortar leakage and stains to produce beams with a uniform appearance.

In the event unsatisfactory surfaces are obtained, repair these surfaces by methods approved by the Contracting Officer or the affected concrete will be rejected. Repair any surface or remove rejected concrete at no expense to the Government.

2.3 TESTS, INSPECTIONS, AND VERIFICATIONS

Testing during manufacture shall be performed by an approved commercial testing laboratory or by an approved laboratory maintained by the manufacturer of the material. Minimum requirements for testing during manufacture shall be as required in **PCI MNL-116** except as modified herein.

2.3.1 Concrete Cylinders

A minimum of four standard 6-inch by 12-inch concrete test cylinders per casting bed or per 30 cubic yards, whichever is less, shall be made to indicate transfer and 28-day strengths. The test cylinders may be 4-inch by 8-inch in lieu of the standard 6-inch by 12-inch cylinders when the required 28-day strength of the concrete is 8000 psi or above.

2.3.2 Certificates of Compliance

The Contractor shall certify that admixtures, aggregates, cement, and pozzolan used conform with the requirements of the specifications. Manufacturer's literature indicating conformance may be submitted for admixtures.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Fabrication

Fabrication of Precast, Prestressed Florida I-Beam 36 Concrete Bridge Girders shall follow the applicable provisions of PCI MNL-116, AASHTO LRFD Bridge Design Specifications, and PRHTA Specifications except as specified herein. Shop drawings must be approved by the Contracting Officer before any fabrication is begun. Unless otherwise shown on the contract drawings, the following dimensional tolerances of girders shall govern:

Length of girders	plus or minus 1/16 inch per 10 feet but not greater than 1 inch
Camber-Differential Between Adjacent Girders After Installation (except between stages of construction)	3/4 inch between two adjacent girders when erected
Exposed Beam Ends-Deviation from square or designated skew	Horizontal: plus or minus 1/8 inch per 12 inch width, plus or minus 1/2 inch maximum Vertical: plus or minus 1/8 inch per foot of girder depth
Location of Harp Points for Harped Strands from Design Location	Plus or minus 12 inches

The lengths of girders required are indicated on the drawings.

3.1.2 Transportation and Storage

Precast, Prestressed Florida I-Beam 36 Concrete Bridge Girders shall be transported in an upright position. Points of support and directions of reactions with respect to the girder shall be the same during transportation and storage as when the girder is in its final position. Care shall be taken during handling and storage of precast units to prevent damage. Units damaged by improper storage or handling shall be replaced by the Contractor at no additional cost to the Government.

Girders may be handled immediately after detensioning. Prestressed girders shall be held at the plant until the concrete has attained the specified 28-day compressive strength or the compressive strength attained after 14 days, whichever is greater. Adequate padding and bracing shall be applied during transport.

3.1.3 Erection

Points of support used in handling and erection of girders shall be as close as practical to the final points of bearing and shall be located within 3 feet of the ends of the girder. All erection details including shoring and bracing shall be clearly shown on the shop drawings. Adequately brace beams as necessary to maintain stability during erection and prior to releasing beam from cranes. Extreme care shall be taken during storage, hoisting and handling to prevent cracking and damage. Girders damaged during handling, storage or erection shall be replaced by the Contractor at no additional cost to the Government.

3.1.4 Inspection

The Contractor's facilities shall be open for inspection by the Contracting Officer at any time. The Contractor shall notify the Contracting Officer one week in advance of the first concrete placement. The Contractor shall maintain full and accurate records of all materials incorporated into prestressed concrete. Manufacturer's test reports shall be submitted on all prestressing steels, reinforcing steels, cement, admixtures, and curing materials. All concrete and cement testing shall be as provided herein and Section 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS. Complete tensioning records shall be maintained and furnished to the Contracting Officer prior to erection, as specified in paragraph 5.4.5 of PCI MNL-116.

3.1.5 Defects and Repair

Precast, Prestressed Florida I-Beam 36 Concrete Bridge Girders shall be evaluated for deficiencies and damage prior to and after installation. Surface deficiencies such as spalls, chips, bug holes, surface porosities, honeycombs, and shallow surface cracks must be recorded and reviewed by the Contracting Officer prior to acceptance of the beams. When the total surface area of all deficiencies within a single product exceeds 1.0% of the product's area that will be exposed to the environment, the deficiencies are considered major regardless of the type. Maintain the record of deficiencies and their repair methods. Ensure the record includes information about product description, unit serial number, date cast, defect description including dimensions, repair method and materials and defect discovery date. All records for deficiencies and repairs shall be maintained by the Contractor and transmitted to the Government and/or Contracting Officer. Contractor prescribed repair methods and major repairs must be approved by the Contracting Officer prior to use and acceptance and will be performed at the Contractor's expense.

Rejection of Precast, Prestressed Florida I-Beam 36 Concrete Bridge Girders shall follow the applicable provisions of PRHTA Specifications, section 630-2.07.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 50 13

MISCELLANEOUS METAL FABRICATIONS

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 WELDER QUALIFICATIONS
- 1.4 DELIVERY, STORAGE, AND PROTECTION
- 1.5 MISCELLANEOUS REQUIREMENTS
 - 1.5.1 Fabrication Drawings
 - 1.5.2 Installation Drawings

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Structural Carbon Steel
 - 2.1.2 Structural Tubing
 - 2.1.3 Steel Pipe
 - 2.1.4 Fittings for Steel Pipe
 - 2.1.5 Anchor Bolts
 - 2.1.5.1 Bolts, Nuts, Studs and Rivets
 - 2.1.5.2 Washers
 - 2.1.5.3 Welded Headed Shear Studs
 - 2.1.6 Eye Bolts, Tie Rods, and Stainless Steel Headed Stud
 - 2.1.7 Joint Ring
 - 2.1.8 Access Hatches
 - 2.1.8.1 Door Leafs
 - 2.1.8.2 Frames
 - 2.1.8.3 Hardware
- 2.2 FABRICATION FINISHES
 - 2.2.1 Galvanizing
 - 2.2.1.1 Surface Preparation
 - 2.2.2 Galvanize
 - 2.2.3 Repair of Zinc-Coated Surfaces
 - 2.2.4 Shop Cleaning and Painting
 - 2.2.4.1 Surface Preparation
 - 2.2.4.2 Pretreatment, Priming and Painting
 - 2.2.5 Nonferrous Metal Surfaces
 - 2.2.6 Aluminum Surfaces
 - 2.2.6.1 Surface Condition
 - 2.2.6.2 Aluminum Finishes
- 2.3 CORNER GUARDS

PART 3 EXECUTION

- 3.1 GENERAL INSTALLATION REQUIREMENTS
- 3.2 WORKMANSHIP
- 3.3 ANCHORAGE, FASTENINGS, AND CONNECTIONS

- 3.4 BUILT-IN WORK
- 3.5 WELDING
 - 3.5.1 Welding Process
 - 3.5.2 Welding Technique
 - 3.5.2.1 Filler Metal
 - 3.5.2.2 Preheat and Interpass Temperature
 - 3.5.2.3 Stress-Relief Heat Treatment
 - 3.5.3 Workmanship
 - 3.5.3.1 Preparation of Base Metal
 - 3.5.3.2 Temporary Welds
 - 3.5.3.3 Tack Welds
- 3.6 DISSIMILAR METALS
- 3.7 PREPARATION
 - 3.7.1 Material Coatings and Surfaces
 - 3.7.2 Environmental Conditions
- 3.8 ACCESS HATCH

-- End of Section Table of Contents --

SECTION 05 50 13

MISCELLANEOUS METAL FABRICATIONS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ALUMINUM ASSOCIATION (AA)

AA DAF45 (2003; Reaffirmed 2009) Designation System for Aluminum Finishes

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 303 (2016) Code of Standard Practice for Steel Buildings and Bridges

AISC 360 (2016) Specification for Structural Steel Buildings

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020) Structural Welding Code - Steel

AWS D1.6/D1.6M (2017) Structural Welding Code—Stainless Steel

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.2 (2015) Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

ASME B18.21.1 (2009; R 2016) Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)

ASME B18.21.2M (1999; R 2014) Lock Washers (Metric Series)

ASME B18.22M (1981; R 2017) Metric Plain Washers

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A47/A47M (1999; R 2018; E 2018) Standard Specification for Ferritic Malleable Iron Castings

ASTM A53/A53M (2020) Standard Specification for Pipe,

	Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A108	(2013) Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished
ASTM A123/A123M	(2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A153/A153M	(2016a) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A240/A240M	(2020) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A276/A276M	(2017) Standard Specification for Stainless Steel Bars and Shapes
ASTM A307	(2014; E 2017) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A500/A500M	(2020) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
ASTM A653/A653M	(2020) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A780/A780M	(2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM A924/A924M	(2020) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process
ASTM B221	(2014) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B928/B928M	(2015) Standard Specification for High Magnesium Aluminum-Alloy Products for Marine Service and Similar Environments
ASTM D1187/D1187M	(1997; E 2011; R 2011) Asphalt-Base Emulsions for Use as Protective Coatings for Metal
ASTM F1554	(2018) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength

MASTER PAINTERS INSTITUTE (MPI)

MPI 79 (2016) Primer, Alkyd, Anti-Corrosive for Metal

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 2 (2018) Hand Tool Cleaning

SSPC SP 3 (2018) Power Tool Cleaning

SSPC SP 6/NACE No.3 (2007) Commercial Blast Cleaning

SSPC SP 11 (2012) Power Tool Cleaning to Bare Metal

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health (SOH) Requirements

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Designation R0 stands for Resident Office and designation D0 stands for District Office. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Corner Guards

Filler Metal

Access Hatches

SD-06 Test Reports

Bolt Testing Reports

SD-07 Certificates

Welder Qualifications

1.3 WELDER QUALIFICATIONS

Prior to welding, submit certification for each welder stating the type of welding and positions qualified for, the code and procedure qualified under, date qualified, and the firm and individual certifying the qualification tests. If the qualification date of the welder or welding operator is more than 6 months old, accompany the welding operator's qualification certificate with a current certificate by the welder attesting to the fact that he has been engaged in welding since the date of certification, with no break in welding service greater than 6 months.

Qualify welders in accordance with AWS D1.1/D1.1M. Use procedures, materials, and equipment of the type required for the work.

1.4 DELIVERY, STORAGE, AND PROTECTION

Protect from corrosion, deformation, and other types of damage. Store items in an enclosed area free from contact with soil and weather. Remove and replace damaged items with new items.

1.5 MISCELLANEOUS REQUIREMENTS

1.5.1 Fabrication Drawings

Submit fabrication drawings showing layout(s), connections to structural system, and anchoring details as specified in [AISC 303](#).

1.5.2 Installation Drawings

Submit templates, erection, and installation drawings indicating thickness, type, grade, class of metal, and dimensions. Show construction details, reinforcement, anchorage, and installation in relation to the channel wall construction.

PART 2 PRODUCTS

2.1 MATERIALS

Provide exposed fastenings of compatible materials (avoid contact of dissimilar metals). Coordinate color and finish with the material to which fastenings are applied. Materials called out on the plans govern over what is shown below in the specifications.

2.1.1 Structural Carbon Steel

Provide in accordance with [ASTM A36/A36M](#).

2.1.2 Structural Tubing

Provide in accordance with [ASTM A500/A500M](#).

2.1.3 Steel Pipe

Provide in accordance with [ASTM A53/A53M](#), Type E or S, Grade B.

2.1.4 Fittings for Steel Pipe

Provide standard malleable iron fittings in accordance with [ASTM A47/A47M](#).

2.1.5 Anchor Bolts

Provide in accordance with [ASTM F1554](#). Where exposed, provide anchor bolts of the same material, color, and finish as the metal to which they are applied.

2.1.5.1 Bolts, Nuts, Studs and Rivets

Provide in accordance with [ASME B18.2.2](#) or [ASTM A307](#).

2.1.5.2 Washers

Provide plain washers in accordance with [ASME B18.22M](#), [ASME B18.21.1](#).

Provide beveled washers for American Standard beams and channels, square or rectangular, tapered in thickness, and smooth. Provide lock washers in accordance with [ASME B18.21.2M](#), [ASME B18.21.1](#).

2.1.5.3 Welded Headed Shear Studs

Provide in accordance with [ASTM A108](#).

2.1.6 Eye Bolts, Tie Rods, and Stainless Steel Headed Stud

Provide in accordance with [ASTM A276/A276M](#) Type 316. Nuts and washers in contact with eye bolts and tie rods shall be of same material and grade.

2.1.7 Joint Ring

Provide in accordance with [ASTM A240/A240M](#) Type 316.

2.1.8 Access Hatches

Provide manufacturer's product data for each size of access hatch. Components of the access hatch shall be the requirements listed within this specification and within the Contract Drawings.

2.1.8.1 Door Leafs

Provide door leafs in accordance with [ASTM B928/B928M](#) Grade 5086-H34 mill finish aluminum plates with diamond pattern. Reinforce with aluminum flat bars to attain a 300 psf live load.

2.1.8.2 Frames

Provide frame members in accordance with [ASTM B221](#) Grade 6005-T5 mill finish will built-in EPDM ushion/gasket and bituminous coating per manufacturer's requirements.

2.1.8.3 Hardware

Provide hinges, hold-open arms, slam locks, and any additional attaching hardware of Type 316 stainless steel.

2.2 FABRICATION FINISHES

2.2.1 Galvanizing

Hot-dip galvanize items specified to be zinc-coated, after fabrication where practicable. Provide galvanizing in accordance with [ASTM A123/A123M](#), [ASTM A153/A153M](#), [ASTM A653/A653M](#) or [ASTM A924/A924M](#), Z275 G90.

2.2.1.1 Surface Preparation

Cleaning steel to prepare for the hot-dip galvanized coating consists of three steps:

- a. Degreasing: First the steel is immersed in a degreasing bath such as an alkaline caustic solution to remove organic contaminants such as dirt, oil, and grease from the surface of the steel. After degreasing the steel is rinsed with water.
- b. Pickling: Next the steel is pickled in a dilute solution of either

hydrochloric or sulfuric acid, which removes oxides and mill scale. Once all oxidation has been removed from the steel, it is again rinsed with water and sent to the final step of the surface preparation.

- c. Fluxing: Finally, the steel is dipped in the flux. The flux is to clean the steel of all oxidation developed since the pickling of the steel and to create a protective coating to prevent any oxidation before entering the galvanizing kettle. One type of flux is contained in a separate tank, is slightly acidic, and contains a combination of zinc chloride and ammonium chloride. Another type of flux, top flux, floats on top of the liquid zinc in the galvanizing kettle, but serves the same purpose.

After degreasing, pickling, and fluxing, the surface of the steel shall be at a near white metal condition, clean and completely free of any oxides or other contaminants that might inhibit the reaction of the iron and molten zinc in the galvanizing kettle.

Mechanical means may be required to remove the following after chemical cleaning in accordance with SSPC SP7 or **SSPC SP 11**:

- Weld slag and other welding flux residues
- Weld splatter and anti-splatter
- Burrs (could include excessively rough edges from flame cutting)
- Very heavy or extremely adherent mill scale
- Mill coatings such as varnishes or lacquers present on some types of pipe
- Epoxies, types of vinyl, and asphalt
- Sand and other impurities on castings
- Oil-based paints and markers
- Crayon markers
- Very heavy or thick deposits of wax or grease

If certain areas cannot be reached with blasting or power tools due to space constraints, clean area with hand tools in accordance with **SSPC SP 2**.

2.2.2 Galvanize

Anchor bolts, grating fasteners, washers, and parts or devices necessary for proper installation, unless indicated otherwise.

2.2.3 Repair of Zinc-Coated Surfaces

Repair damaged surfaces with galvanizing repair method and paint in accordance with **ASTM A780/A780M** or by application of stick or thick paste material specifically designed for repair of galvanizing, as approved by Contracting Officer. Clean areas to be repaired and remove slag from welds. Heat, with a torch, surfaces to which stick or paste material will be applied. Heat to a temperature sufficient to melt the metals in the

stick or paste. Spread molten material uniformly over surfaces to be coated and wipe off excess material. Repairs will be at the Contractor's expense.

2.2.4 Shop Cleaning and Painting

2.2.4.1 Surface Preparation

Blast clean surfaces in accordance with **SSPC SP 6/NACE No.3**. Surfaces that will be exposed in spaces above ceiling or in attic spaces, crawl spaces, furred spaces, and chases may be cleaned in accordance with **SSPC SP 3** in lieu of being blast cleaned. Wash cleaned surfaces which become contaminated with rust, dirt, oil, grease, or other contaminants with solvents until thoroughly clean. Steel to be embedded in concrete must be free of dirt and grease prior to embed. Do not paint or galvanize bearing surfaces, including contact surfaces within slip critical joints. Shop coat these surfaces with rust prevention.

2.2.4.2 Pretreatment, Priming and Painting

Apply pre-treatment, primer, and paint in accordance with manufacturer's printed instructions. On surfaces concealed in the finished construction or not accessible for finish painting, apply an additional prime coat to a minimum dry film thickness of **1.0 mil**. Tint additional prime coat with a small amount of tinting pigment.

2.2.5 Nonferrous Metal Surfaces

Protect by plating, anodic, or organic coatings.

2.2.6 Aluminum Surfaces

2.2.6.1 Surface Condition

Before finishes are applied, remove roll marks, scratches, rolled-in scratches, kinks, stains, pits, orange peel, die marks, structural streaks, and other defects which will affect uniform appearance of finished surfaces.

2.2.6.2 Aluminum Finishes

Unexposed sheet, plate and extrusions may have mill finish as fabricated. Sandblast castings' finish, medium, **AA DAF45**. Unless otherwise specified, provide all other aluminum items with a standard mill finish. Provide a coating thickness not less than that specified for protective and decorative type finishes for items used in interior locations or architectural Class I type finish for items used in exterior locations. Provide in accordance with **AA DAF45**. Provide a polished satin finish on items to be anodized.

2.3 CORNER GUARDS

For jambs and sills of openings and edges of platforms provide steel shapes and plates anchored in masonry or concrete with welded steel straps or end-weld stud anchors.

PART 3 EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

Install items at locations indicated in accordance with applicable manufacturer's instructions. Verify all field dimensions prior to fabrication. Include materials and parts necessary to complete each assembly, whether indicated or not. Miss-alignment and miss-sizing of holes for fasteners is cause for rejection. Conceal fastenings where practicable. Joints exposed to weather must be watertight.

3.2 WORKMANSHIP

Provide miscellaneous metalwork that is true and accurate in shape, size, and profile. Make angles and lines continuous and straight. Make curves consistent, smooth and unfaceted. Provide continuous welding along the entire area of contact, and incorporate tack welding into the final weld in accordance with [AWS D1.1/D1.1M](#), except where tack welds are used solely for installation aid. Do not tack weld exposed connections. Unless otherwise indicated and approved, provide a smooth finish on exposed surfaces. Provide countersunk rivets where exposed. Provide coped and mitered corner joints aligned flush and without gaps.

3.3 ANCHORAGE, FASTENINGS, AND CONNECTIONS

Provide anchorage as necessary, whether indicated or not, for fastening miscellaneous metal items securely in place. Include slotted inserts, expansion shields, powder-driven fasteners, toggle bolts (when approved for concrete), through bolts for masonry, headed shear studs, machine and carriage bolts for steel, through bolts, lag bolts, and screws for wood. Do not use wood plugs. Provide non-ferrous attachments for non-ferrous metal. Provide exposed fastenings of compatible materials (avoid contact of dissimilar metals), that generally match in color and finish the surfaces to which they are applied. Conceal fastenings where practicable. Provide all fasteners flush with the surfaces they fasten, unless indicated otherwise. Test a minimum of 2 bolt, nut, and washer assemblies from each certified mill batch in a tension measuring device at the job site prior to the beginning of bolting start-up. Demonstrate that the bolts and nuts, when used together, can develop tension not less than the provisions specified in [AISC 360](#), depending on bolt size and grade. The bolt tension must be developed by tightening the nut. A representative of the manufacturer or supplier must be present to ensure that the fasteners are properly used, and to demonstrate that the fastener assemblies supplied satisfy the specified requirements. Submit [bolt testing reports](#).

3.4 BUILT-IN WORK

Where necessary and not otherwise indicated, form built-in metal work for anchorage with concrete or masonry. Provide built-in metal work in ample time for securing in place as the work progresses.

3.5 WELDING

Perform welding, welding inspection, and corrective welding in accordance with [AWS D1.1/D1.1M](#). For stainless steel items, perform welding, welding inspection, and corrective welding in accordance with [AWS D1.6/D1.6M](#). Use continuous welds on all exposed connections. Grind visible welds smooth in the finished installation. Provide welded headed shear studs in

accordance with AWS D1.1/D1.1M, Clause 7, except as otherwise specified. Provide in accordance with the safety requirements of EM 385-1-1.

3.5.1 Welding Process

Perform welding of structural steel by an electric arc welding process using a method which excludes the atmosphere from the molten metal and conforms to the applicable provisions of AWS D1.1/D1.1M. Minimize residual stresses, distortion and shrinkage from welding.

3.5.2 Welding Technique

3.5.2.1 Filler Metal

Provide the electrode, electrode-flux combination and grade of filler metal conforming to the appropriate AWS specification for the base metal and welding process being used or be as shown where a specific choice of AWS specification allowables is required. Submit filler metal product data. Include the AWS designation of the electrodes to be used in the schedule of welding procedures. Use only low hydrogen electrodes for manual shielded metal-arc welding regardless of the thickness of the steel. Use a controlled temperature storage oven at the job site as prescribed by AWS D1.1/D1.1M, Fabrication Clause to maintain low moisture of low hydrogen electrodes.

3.5.2.2 Preheat and Interpass Temperature

Perform preheating as required by AWS D1.1/D1.1M, Fabrication Clause or as otherwise specified except that the temperature of the base metal must be at least 70 degrees F. Slowly and uniformly preheat the joint area by approved means to the prescribed temperature, held at that temperature until the welding is completed and then permitted to cool slowly in still air.

3.5.2.3 Stress-Relief Heat Treatment

Where stress relief heat treatment is specified or shown, perform in accordance with the requirements of AWS D1.1/D1.1M, Fabrication Clause.

3.5.3 Workmanship

Perform welding workmanship in accordance with AWS D1.1/D1.1M, Fabrication Clause and other applicable requirements of these specifications.

3.5.3.1 Preparation of Base Metal

Prior to welding inspect surfaces to be welded to ensure compliance with AWS D1.1/D1.1M, Fabrication Clause.

3.5.3.2 Temporary Welds

Make temporary welds, required for fabrication and erection, under the controlled conditions prescribed for permanent work. Make temporary welds using low-hydrogen welding electrodes and by welders qualified for permanent work as specified in these specifications. Conduct preheating for temporary welds as required by AWS D1.1/D1.1M for permanent welds except that the minimum temperature must be 120 degrees F in any case. In making temporary welds, do not strike arcs in other than weld locations. Temporary welds shall be incorporated into the final weld..

3.5.3.3 Tack Welds

Tack welds that are to be incorporated into the permanent work are to exhibit the same quality requirements as the permanent welds; clean and thoroughly fuse them with permanent welds. Perform preheating as specified above for temporary welds. Provide cascaded ends on multiple-pass tack welds. Remove defective tack welds before permanent welding. Tack welders shall be qualified in accordance with AWS D1.1/D1.1M and the requirements of paragraph 1.3 WELDER QUALIFICATIONS.

3.6 DISSIMILAR METALS

Where dissimilar metals are in contact, protect surfaces with a coating in accordance with MPI 79 to prevent galvanic or corrosive action. Where aluminum is in contact with concrete, plaster, mortar, masonry, wood, or absorptive materials subject to wetting, protect in accordance with ASTM D1187/D1187M, asphalt-base emulsion. Clean surfaces with metal shavings from installation at the end of each work day.

3.7 PREPARATION

3.7.1 Material Coatings and Surfaces

Remove rust preventive coating just prior to field erection, using a remover approved by the metal manufacturer. Surfaces, when assembled, must be free of rust, grease, dirt and other foreign matter.

3.7.2 Environmental Conditions

Do not clean or paint surfaces when damp or exposed to foggy or rainy weather, when metallic surface temperature is less than minus 5 degrees F above the dew point of the surrounding air, or when surface temperature is below 45 degrees F or over 95 degrees F, unless approved by the Contracting Officer. Metal surfaces to be painted must be dry for a minimum of 48 hours prior to the application of primer or paint.

3.8 ACCESS HATCH

Construction and accessories as follows:

- a. Provide insulated cover and curb with mounting flanges for securing to roof deck. Provide curbs with integral metal cap flashing of the same gage and metal as the curb, fully welded and ground at corners for weather tightness.
- b. Provide hatches completely assembled, with pintle hinges, compression spring operators enclosed in telescopic tubes, positive snap latches with turn handles on inside and outside, and neoprene draft seals. Provide fasteners for padlocking from the inside. Provide covers with automatic hold-open arms complete with grip handle to permit one hand release. Cover action must be smooth through its entire range of motion with an operating pressure of approximately 30 pounds.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 51 00

METAL STAIRS

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
- 2.2 FABRICATION
 - 2.2.1 General Fabrication
 - 2.2.2 Floor Grating Treads and Platforms
 - 2.2.3 Protective Coating
- 2.3 COMPONENTS
 - 2.3.1 Steel Stair Components
 - 2.3.1.1 Design Loads
 - 2.3.1.2 Materials
 - 2.3.2 Concrete Anchors
 - 2.3.3 Fasteners
- 2.4 MATERIALS
 - 2.4.1 Structural Steel Plates and Shapes

PART 3 EXECUTION

- 3.1 PREPARATION
- 3.2 INSTALLATION
 - 3.2.1 Field Preparation
 - 3.2.2 Field Welding

-- End of Section Table of Contents --

SECTION 05 51 00

METAL STAIRS

PART 1 GENERAL

Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS applies to work specified in this section.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 360 (2016) Specification for Structural Steel Buildings

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.21.1 (2009; R 2016) Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A6/A6M (2017a) Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling

ASTM A27/A27M (2020) Standard Specification for Steel Castings, Carbon, for General Application

ASTM A29/A29M (2020) Standard Specification for General Requirements for Steel Bars, Carbon and Alloy, Hot-Wrought

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A47/A47M (1999; R 2018; E 2018) Standard Specification for Ferritic Malleable Iron Castings

ASTM A123/A123M (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A153/A153M (2016a) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel

Hardware

ASTM A283/A283M	(2013) Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates
ASTM A307	(2014; E 2017) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A653/A653M	(2020) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM MBG 531	(2017) Metal Bar Grating Manual
---------------	---------------------------------

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Iron and Steel Hardware; G, R0
Steel Shapes and Plates; G, R0
Metal Stair System; G, R0

SD-03 Product Data

Structural Steel Plates and Shapes; G, R0
Concrete Anchors; G, R0
Protective Coating; G, R0
Steel Stair Components; G, R0

SD-07 Certificates

SD-08 Manufacturer's Instructions

Structural Steel Plates and Shapes; G, R0
Protective Coating; G, R0

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Submit complete and detailed fabrication drawings for all [iron and steel hardware](#), and for all [steel shapes and plates](#) used in accordance with the design specifications referenced in this section.

2.2 FABRICATION

Preassemble items in the shop to the greatest extent possible.
Disassemble units only to the extent necessary for shipping and handling.
Clearly mark units for reassembly and coordinated installation.

For the fabrication of work exposed to view, use only materials that are smooth and free of surface blemishes, including pitting, seam marks, roller marks, rolled trade names, and roughness. Remove blemishes by grinding, or by welding and grinding, before cleaning and treating surfaces and applying surface finishes, including zinc coatings.

2.2.1 General Fabrication

Prepare and submit [metal stair system](#) shop drawings with detailed plans and elevations at scales not less than [1 inch to 1 foot](#) and with details of sections and connections at scales not less than [3 inches to 1 foot](#). Also detail the placement drawings, diagrams, and templates for installation of anchorages, including concrete inserts, anchor bolts, and miscellaneous metal items having integral anchorage devices.

Use materials of size and thicknesses indicated or, if not indicated, of the size and thickness necessary to produce a finished product that is strong enough and durable enough for its intended use. Work the materials to the dimensions indicated on approved detail drawings, using proven methods of fabrication and support. Use the type of materials indicated or specified for the various components of work.

Form exposed work true to line and level, with accurate angles and surfaces and with straight sharp edges. Ease exposed edges to a radius of approximately [1/32 inch](#), and bend metal corners to the smallest radius possible without causing grain separation or otherwise impairing the work.

Continuously weld corners and seams in accordance with the recommendations of [AWS D1.1/D1.1M](#). Grind exposed welds smooth and flush to match and blend with adjoining surfaces.

Form exposed connections with hairline joints that are flush and smooth, using concealed fasteners wherever possible. Use exposed fasteners of the type indicated or, if not indicated, use Phillips flat-head (countersunk) screws or bolts.

Provide and coordinate anchorage of the type indicated for the supporting structure. Fabricate anchoring devices, and space them as indicated and as necessary to provide adequate support for the intended use of the work.

2.2.2 Floor Grating Treads and Platforms

Provide floor grating treads and platforms conforming to [ASTM A6/A6M](#), [ASTM A29/A29M](#) and [NAAMM MBG 531](#), "Metal Bar Grating Manual." Provide the pattern, spacing, and bar sizes as indicated:

- a. Galvanized finish, conforming to **ASTM A123/A123M**.

Fabricate grating treads with steel plate nosings on one edge and with steel angle or steel plate carriers at each end for string connections. Secure treads to strings with bolts.

Match the nosings of grating platforms with the nosing of grating treads at landings. Provide toeplates where the open-sided edges of floor grating meet platform framing members.

2.2.3 Protective Coating

Hot-dip galvanize steelwork as indicated in accordance with **ASTM A123/A123M**. Touch up abraded surfaces and cut ends of galvanized members with zinc-dust, zinc-oxide primer, or an approved galvanizing repair compound.

2.3 COMPONENTS

2.3.1 Steel Stair Components

Provide steel stairs complete with stringers, grating treads, landings, columns, handrails, and necessary bolts and other fastenings. Hot-dip-galvanize steel stairs and accessories.

2.3.1.1 Design Loads

Design stairs to sustain a live load of not less than 100 **pounds per square foot**, or a concentrated load of 300 pounds applied where it is most critical. Except for a commercial product, design and fabricate steel stairs to conform to **AISC 360**.

2.3.1.2 Materials

Provide steel stairs of welded construction except that bolts may be used where welding is not practicable. Do not use screw or screw-type connections.

- a. Structural Steel: **ASTM A36/A36M**.
- b. Gratings for Treads and Landings: **NAAMM MBG 531** or Plank grating; **ASTM A653/A653M**, **G-90** for steel.
- c. Support steel grating on angle cleats welded to stringers or treads with integral cleats, welded or bolted to the stringer. Close exposed ends. For exterior stairs, form all exposed joints to exclude water.
- e. Before fabrication, obtain necessary field measurements and verify drawing dimensions.
- f. Clean metal surfaces free of mill scale, flake rust, and rust pitting before shop finishing. Weld permanent connections. Finish welds flush and smooth on surfaces that will be exposed after installation.

2.3.2 Concrete Anchors

Threaded-type concrete inserts consisting of galvanized ferrous castings, internally threaded to receive **3/4-inch** diameter machine bolts; either

malleable iron conforming to [ASTM A47/A47M](#) or cast steel conforming to [ASTM A27/A27M](#), and hot-dip-galvanized in accordance with [ASTM A153/A153M](#).

2.3.3 Fasteners

Select galvanized zinc-coated fasteners conforming to [ASTM A153/A153M](#) for exterior applications or where the fasteners are built into exterior walls or floor systems. Select the fasteners for the type, grade, and class required for the installation of steel stair items:

- a. Standard/regular hexagon-head bolts and nuts, conforming to [ASTM A307, Grade A](#).
- b. Plain, round, general-assembly-grade, carbon steel washers, conforming to [ASME B18.21.1](#).

2.4 MATERIALS

2.4.1 Structural Steel Plates and Shapes

Structural size shapes and plates, conforming to [ASTM A36/A36M](#), unless otherwise noted, except bent or cold-formed plates.

Steel plates - bent or cold-formed, conforming to [ASTM A283/A283M](#), Grade C.

PART 3 EXECUTION

3.1 PREPARATION

Clean surfaces thoroughly before installation. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions. Examine materials upon arrival at site. Notify the carrier and manufacturer of any damage.

Protect installed products until completion of project. Touch up, repair or replace, damaged products before substantial completion

3.2 INSTALLATION

Install in accordance with the manufacturer's instructions and approved submittals. Install in proper relationship with adjacent construction.

Install items at locations indicated, according to the manufacturer's instructions. Verify all measurements and take all field measurements necessary before fabrication. Ensure that exposed fastenings are compatible with generally match the color and finish of, and harmonize with the material to which they are applied. Include materials and parts necessary to complete each item, even though such work is not definitely shown or specified. Poor matching of holes for fasteners is cause for rejection. Conceal fastenings where practicable. Select thickness of metal and details of assembly and supports that adequately strengthen and stiffen the construction. Form joints exposed to the weather to exclude water.

3.2.1 Field Preparation

Remove rust-preventive coating just before field erection, using a remover approved by the coating manufacturer. Provide surfaces, when assembled,

free of rust, grease, dirt and other foreign matter.

3.2.2 Field Welding

Comply with AWS D1.1/D1.1M in executing manual shielded-metal arc welding, (for appearance and quality of new welds) and in correcting existing welding.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 52 00

METAL HAND RAILINGS

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 ADMINISTRATIVE REQUIREMENTS
 - 1.2.1 Preinstallation Meetings
- 1.3 SUBMITTALS
- 1.4 QUALITY CONTROL
 - 1.4.1 Handrails
 - 1.4.2 Welding Procedures
 - 1.4.3 Welder Qualification

PART 2 PRODUCTS

- 2.1 FABRICATION
 - 2.1.1 Steel Handrails
 - 2.1.2 Protective Coating
- 2.2 COMPONENTS
 - 2.2.1 Structural Steel Plates, Shapes And Bars
 - 2.2.2 Steel Pipe
 - 2.2.3 Fasteners
 - 2.2.4 Steel Railings And Handrails
 - 2.2.4.1 Steel Handrails

PART 3 EXECUTION

- 3.1 PREPARATION
- 3.2 INSTALLATION
 - 3.2.1 Steel Handrail
- 3.3 FIELD QUALITY CONTROL
 - 3.3.1 Field Welding

-- End of Section Table of Contents --

SECTION 05 52 00

METAL HAND RAILINGS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020) Structural Welding Code - Steel

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1 (2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)

ASME B18.6.1 (2016) Wood Screws (Inch Series)

ASME B18.6.3 (2013; R 2017) Machine Screws, Tapping Screws, and Machine Drive Screws (Inch Series)

ASME B18.21.1 (2009; R 2016) Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A53/A53M (2020) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A123/A123M (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A153/A153M (2016a) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A283/A283M (2013) Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates

ASTM A307 (2014; E 2017) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength

ASTM A500/A500M

(2020) Standard Specification for
Cold-Formed Welded and Seamless Carbon
Steel Structural Tubing in Rounds and
Shapes

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 521

(2001; R 2012) Pipe Railing Systems Manual

1.2 ADMINISTRATIVE REQUIREMENTS

1.2.1 Preinstallation Meetings

No later than 60 days prior to installation, submit fabrication drawings to the Contracting Officer for the following items:

- a. Iron and steel hardware
- b. Steel shapes, plates, bars and strips
- c. Steel railings and handrails
- d. Anchorage and fastening systems

Submit manufacturer's catalog data, including two copies of manufacturers' specifications, load tables, dimension diagrams, and anchor details for the following items:

- a. Structural-steel plates, shapes, and bars
- b. Structural-steel tubing
- c. Hot-rolled carbon steel bars
- d. Cold-drawn steel tubing
- e. Concrete inserts
- f. Protective coating
- g. Steel railings and handrails
- h. Anchorage and fastening systems

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Designation R0 stands for Resident Office and designation D0 stands for District Office. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fabrication Drawings; G, R0

Iron and Steel Hardware; G, R0

Steel Shapes, Plates, Bars and Strips; G, R0

SD-03 Product Data

Structural-Steel Plates, Shapes, and Bars; G, R0

Structural-Steel Tubing; G, R0

Cold-Drawn Steel Tubing; G, R0

Concrete Inserts; G, R0

Anchorage Devices; G, R0

Protective Coating; G, R0

Steel Railings and Handrails; G, R0

Anchorage and Fastening Systems; G, R0

SD-07 Certificates

Welding Procedures; G, D0

Welder Qualification; G, D0

SD-08 Manufacturer's Instructions

Installation Instructions

1.4 QUALITY CONTROL

1.4.1 Handrails

Provide handrails as type specified and shown herein. Furnish and install complete with all fittings, brackets, fasteners, sleeves, anchors, and other appurtenances as shown and as required for proper installation. Design handrails to resist a minimum concentrated load of 200 lbf in any direction at any point of the top of the rail or 50 lb/ft applied horizontally to top of the rail, whichever is more severe.

1.4.2 Welding Procedures

Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS applies to work specified in this section.

Submit results of welding procedures testing in accordance with AWS D1.1/D1.1M made in the presence of the Contracting Officer and by an approved testing laboratory at the Contractor's expense.

1.4.3 Welder Qualification

Submit certified welder qualification by tests in accordance with AWS D1.1/D1.1M, or under an equivalent approved qualification test. In addition, perform tests on test pieces in positions and with clearances equivalent to those actually encountered. If a test weld fails to meet requirements, conduct an immediate retest of two test welds and ensure that each test weld passes. Failure in the immediate retest will require that the welder be retested after further practice or training and make a

complete set of test welds.

PART 2 PRODUCTS

2.1 FABRICATION

Preassemble items in the shop to the greatest extent possible. Disassemble units only to the extent necessary for shipping and handling. Clearly mark units for reassembly and coordinated installation.

For the fabrication of work exposed to view, use only materials that are smooth and free of surface blemishes, including pitting, seam marks, roller marks, rolled trade names, and roughness. Remove blemishes by grinding, or by welding and grinding, before cleaning, treating, and applying surface finishes, including zinc coatings.

Provide railing and handrail detail plans and elevations at not less than 1 inch to 1 foot. Provide details of sections and connections at not less than 3 inches to 1 foot. Also detail setting drawings, diagrams, templates for installation of anchorages, including concrete inserts, anchor bolts, and miscellaneous metal items having integral anchors.

Use materials of size and thicknesses indicated or, if not indicated, of the size and thickness necessary to produce adequate strength and durability in the finished product for its intended use. Work the materials to the dimensions indicated on approved detail drawings, using proven details of fabrication and support. Use the type of materials indicated or specified for the various components of work.

Form exposed work true to line and level, with accurate angles and surfaces and straight sharp edges. Ensure that all exposed edges are eased to a radius of approximately 1/32 inch. Bend metal corners to the smallest radius possible without causing grain separation or otherwise impairing the work.

Weld corners and seams continuously and in accordance with the recommendations of AWS D1.1/D1.1M. Grind exposed welds smooth and flush to match and blend with adjoining surfaces.

Form the exposed connections with hairline joints that are flush and smooth, using concealed fasteners wherever possible. Use exposed fasteners of the type indicated or, if not indicated, use countersunk Phillips flathead screws or bolts.

Provide anchorage of the type indicated and coordinated with the supporting structure. Fabricate anchoring devices and space as indicated and as required to provide adequate support for the intended use of the work.

Use hot-rolled steel bars for work fabricated from bar stock unless work is indicated or specified to be fabricated from cold-finished or cold-rolled stock.

2.1.1 Steel Handrails

Fabricate joint posts, rail, and corners by one of the following methods:

- a. Flush-type rail fittings of commercial standard, welded and ground smooth, with railing splice locks secured with 3/8 inch

hexagonal-recessed-head setscrews.

- b. Mitered and welded joints made by fitting post to top rail and intermediate rail to post, mitering corners, groove-welding joints, and grinding smooth. Butt railing splices and reinforce them by a tight-fitting interior sleeve not less than 6 inches long.
- c. Railings may be bent at corners in lieu of jointing, provided that bends are made in suitable jigs and the pipe is not crushed.

2.1.2 Protective Coating

Provide hot-dipped galvanized steelwork as indicated in accordance with [ASTM A123/A123M](#). Touch up abraded surfaces and cut ends of galvanized members with zinc-dust, zinc-oxide primer, or an approved galvanizing repair compound.

2.2 COMPONENTS

2.2.1 Structural Steel Plates, Shapes And Bars

Provide structural-size shapes and plates, except plates to be bent or cold-formed, conforming to [ASTM A36/A36M](#), unless otherwise noted.

Provide steel plates, to be bent or cold-formed, conforming to [ASTM A283/A283M](#), Grade C.

Provide steel bars and bar-size shapes conforming to [ASTM A36/A36M](#), unless otherwise noted.

2.2.2 Steel Pipe

Provide pipe conforming to [ASTM A53/A53M](#), type as selected, Grade B; primed finish, unless galvanizing is required; standard weight Schedule 40.

2.2.3 Fasteners

Provide galvanized zinc-coated fasteners in accordance with [ASTM A153/A153M](#) used for exterior applications or where built into exterior walls or floor systems. Select fasteners for the type, grade, and class required for the installation of railings and handrails.

Provide standard hexagon-head bolts, conforming to [ASTM A307](#), Grade A.

Provide square-head lag bolts conforming to [ASME B18.2.1](#).

Provide cadmium-plated steel machine screws conforming to [ASME B18.6.3](#).

Provide flat-head carbon steel wood screws conforming to [ASME B18.6.1](#).

Provide plain round, general-assembly-grade, carbon steel washers conforming to [ASME B18.21.1](#).

Provide helical spring, carbon steel lockwashers conforming to [ASME B18.2.1](#).

2.2.4 Steel Railings And Handrails

Per [NAAMM AMP 521](#), provide the same size rail and post. Provide pipe collars of the same material and finish as the handrail and posts.

Provide series 300 stainless-steel pipe collars.

2.2.4.1 Steel Handrails

Provide steel handrails, including inserts in concrete, steel pipe conforming to [ASTM A53/A53M](#) or structural tubing conforming to [ASTM A500/A500M](#), Grade A or B of equivalent strength. Provide steel railings of [1 1/2 inch](#) nominal size, hot-dip galvanized.

Provide galvanized railings, including pipe, fittings, brackets, fasteners, and other ferrous metal components.

PART 3 EXECUTION

3.1 PREPARATION

Adjust handrails before securing in place in order to ensure proper matching at butting joints and correct alignment throughout their length. Space posts not more than [6 feet](#) on center. Plumb posts in each direction. Secure posts and rail ends to building construction as follows:

- a. Anchor posts to steel with oval steel flanges, angle type or floor type as required by conditions, welded to posts and bolted to the steel supporting members.
- b. Anchor rail ends into concrete and masonry with round steel flanges welded to rail ends and anchored into the wall construction with lead expansion shields and bolts.

Install toe boards and brackets where indicated. Make splices, where required, at expansion joints. Install removable sections as indicated.

3.2 INSTALLATION

Submit manufacturer's [installation instructions](#) for the following products to be used in the fabrication of hand rail work:

- a. Structural-steel plates, shapes, and bars
- b. Structural-steel tubing
- c. Cold-finished steel bars
- d. Hot-rolled carbon steel bars
- e. Cold-drawn steel tubing
- f. Protective coating
- g. [Anchorage devices](#)
- h. Steel railings and handrails
- i. Anchorage and fastening systems

Provide complete, detailed fabrication and installation drawings for all [iron and steel hardware](#), and for all [steel shapes, plates, bars, and strips](#) used in accordance with the design specifications cited in this section.

3.2.1 Steel Handrail

Install handrail by means of base plates bolted to stringers or structural-steel frame work. Secure rail ends by steel pipe flanges

3.3 FIELD QUALITY CONTROL

3.3.1 Field Welding

Ensure that procedures of welding, appearance and quality of welds made, and methods used in correcting welding work comply with [AWS D1.1/D1.1M](#).

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 06 - WOOD, PLASTICS, AND COMPOSITES

SECTION 06 73 01

FIBERGLASS REINFORCED PLASTIC (FRP) GRATING

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY CONTROL
- 1.4 DELIVERY, HANDLING, AND STORAGE

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
 - 2.1.1 Design Requirements
 - 2.1.2 Performance Requirements
 - 2.1.2.1 Structural Performance of Gratings
- 2.2 FABRICATION
 - 2.2.1 Molded FRP Grating
 - 2.2.2 Fasteners

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Anchorage, Fastenings, and Connections
- 3.2 CLOSEOUT ACTIVITIES
 - 3.2.1 Manufacturer's Warranty

-- End of Section Table of Contents --

SECTION 06 73 01

FIBERGLASS REINFORCED PLASTIC (FRP) GRATING

PART 1 GENERAL

This Section includes, but is not limited to, new fiberglass reinforced plastic (FRP) grating for elevated platforms and walkways.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

ASTM INTERNATIONAL (ASTM)

ASTM D635 (2018) Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position

ASTM D638 (2014) Standard Test Method for Tensile Properties of Plastics

ASTM D696 (2016) Standard Test Method for Coefficient of Linear Thermal Expansion of Plastics Between -30 degrees C and 30 degrees C With a Vitreous Silica Dilatometer

ASTM D790 (2017) Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials

ASTM D953 (2019) Standard Test Method for Pin-Bearing Strength of Plastics

ASTM D2344/D2344M (2016) Standard Test Method for Short-Beam Strength of Polymer Matrix Composite Materials and Their Laminates

ASTM D2863 (2019) Standard Test Method for Measuring the Minimum Oxygen Concentration to Support Candle-Like Combustion of Plastics (Oxygen Index)

ASTM E84 (2020) Standard Test Method for Surface Burning Characteristics of Building Materials

ASTM E662	(2019) Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials
ASTM G154	(2016) Standard Practice for Operating Fluorescent Light Apparatus for UV Exposure of Nonmetallic Materials
ASTM G155	(2013) Standard Practice for Operating Xenon Arc Light Apparatus for Exposure of Non-Metallic Materials

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2018) International Building Code
---------	------------------------------------

U.S. DEPARTMENT OF DEFENSE (DOD)

SAE CMH-17-36	(2012) Composite Materials Handbook, Vol. 3, Polymer Matrix Composites Material Usage, Design, and Analysis
---------------	---

UNDERWRITERS LABORATORIES (UL)

UL 94	(2013; Reprint Jun 2020) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances
-------	---

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Installation Drawings, Templates, and Directions; G, D0

SD-03 Product Data

FRP Grating; G, D0

Clips and Anchorage; G, D0

SD-06 Test Reports

Bearing Strength Testing

Flexural Properties

Ultraviolet Testing

Shear Strength

Tensile Properties

Toxicity Testing

Coefficient of Lineal Thermal Expansion

Flame Spread Testing

SD-07 Certificates

Manufacturer's Sample Warranty

Certification of Anchorage System compliance with [ASCE 7-16](#)

Proof Of Certification From A Minimum Of Two Quality Assurance Programs For Its Facilities Or Products (UL, Dnv, Abs, Uscg, Aarr)

SD-08 Manufacturer's Instructions

Shipping, Handling, And Erection Procedures

Care and Maintenance Instructions

SD-09 Manufacturer's Field Reports

Manufacturer's Certification of Installation

SD-11 Closeout Submittals

Manufacturer's Warranty

1.3 QUALITY CONTROL

Provide items by manufacturers having a minimum of 10 years of experience in the design and manufacture of similar products and systems. In addition, if requested, provide a record of at least five separate, similar, successful installations in the last 5 years. Submit manufacturer's catalog data, to include two copies of the manufacturer's specifications, load tables, dimension diagrams, and anchor details for the following items:

- a. [FRP Grating](#)
- b. [Clips and Anchorage](#)

Provide 3-year manufacturer's limited warranty on all FRP products against defects in materials and workmanship. Submit the [Manufacturer's Sample Warranty](#) before work starts.

Submit [installation drawings, templates, and directions](#) for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are embedded in concrete or masonry.

Submit [Certification of Anchorage System compliance with ASCE 7-16](#). Deliver such items to the Contracting Officer and the project site before installation starts.

Ensure that the manufacturer is certified to ISO 9001-2008. Submit the [Proof of Certification from a minimum of two quality assurance programs](#)

for its facilities or products (UL, DNV, ABS, USCG, AARR).

1.4 DELIVERY, HANDLING, AND STORAGE

Submit the manufacturer's recommendations for shipping, handling, and erection procedures, and care and maintenance instructions. Deliver manufactured materials in original, unbroken pallets, packages, containers, or bundles bearing the label of the manufacturer. Ensure that all adhesives, resins, and their catalysts and hardeners are crated or boxed separately, and noted as such.

Handle all materials to prevent abrasion, cracking, chipping, twisting, other deformations, and other types of damage. Store adhesives, resins and their catalysts in dry indoor facilities between 70 and 85 degrees F until they are required.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide gratings composed of continuous roving fiberglass reinforcement and resin in qualities, quantities, properties, arrangements, and dimensions as necessary to meet the design requirements and dimensions as specified. Provide grating with 1-1/2" x 1-1/2" grid spacing.

Provide resin of isophthalic polyester with chemical formulations as necessary to provide the corrosion resistance, strength, and other physical properties conforming to the specified requirements.

Ensure that all surfaces of FRP items and fabrications are nonslip grit, resin-rich, and free of voids, dry spots, cracks, and unreinforced areas. Completely cover all glass fibers with resin to protect against their exposure to ultraviolet light, wear, or weathering.

2.1.1 Design Requirements

Submit documentation for the following product tests before work starts:

Test	Standard	Structural Performance Requirements	Minimum Flexural Strength	Minimum Flexural Modulus
Bearing Strength Testing	ASTM D953			
Flexural Properties	ASTM D790		30,000 psi	(1.8 x 10 to power of 6) psi
Ultraviolet Testing	ASTM G155 ASTM G154			
Shear Strength	ASTM D2344/D2344M	4,500 psi		
Tensile Properties	ASTM D638	30,000 psi		

Test	Standard	Structural Performance Requirements	Minimum Flexural Strength	Minimum Flexural Modulus
Toxicity Testing	SAE CMH-17-36			
Coefficient of Lineal Thermal Expansion	ASTM D696	8.0 x 10 power minus 6 in/in/degree F		
Flame Spread Testing	ASTM D2863 ASTM E662 UL 94			

2.1.2 Performance Requirements

2.1.2.1 Structural Performance of Gratings

Provide gratings capable of withstanding the effects of gravity loads in accordance with ASCE 7-16, ICC IBC, and a uniform load of 100 lb/square foot.

Provide grating products with a flame spread rating of 25 or less per ASTM E84 Tunnel Test. Test gratings for burn time of less than 30 seconds and an extent of burn rate of less than or equal to 10 millimeters per ASTM D635.

2.2 FABRICATION

2.2.1 Molded FRP Grating

Ensure that all field-fabricated and shop-fabricated grating cuts are coated with vinyl ester resin to provide maximum corrosion resistance in accordance with the manufacturer's instructions.

Provide grating made as one-piece molded construction with tops and bottoms of bearing bars and cross bars in the same plane with a rectangular mesh pattern providing unidirectional strength and reinforced with continuous roving of an equal number of layers in each direction. Ensure that the top layer of reinforcement is no more than 1/8 inch below the top surface of the grating to provide maximum stiffness and prevent resin chipping of unreinforced surfaces having percentage of glass (by weight) not exceeding 35 percent.

Ensure that no dry glass fibers are visible on any surface of bearing bars or cross bars after molding, and that all bars are smooth and uniform, with no evidence of fiber orientation irregularities, interlaminar voids, porosity, resin-rich areas or resin-starved areas.

Fillet grating bar intersections to a minimum radius of 1/16 inch to eliminate local stress concentrations and the possibility of resin cracking at these locations.

Provide fire-retardant grating with a tested flame spread rating of 25 or less when tested in accordance with ASTM E84.

2.2.2 Fasteners

Provide Type 316 stainless-steel fasteners, clips, and anchorage for exterior use. Select fasteners for type, grade, and class required.

PART 3 EXECUTION

3.1 INSTALLATION

Install items at locations indicated, according to the manufacturer's instructions. Submit 3 copies of [manufacturer's Certification of Installation](#) to the Contracting Officer. Verify all measurements and take all field measurements necessary before fabrication. Include all materials and parts necessary to complete each item, even though such work is not definitely shown or specified. Perform cutting, drilling, and fitting required for installing gratings. Set units accurately in location, alignment, and elevation; measured from established lines; and levels and free of rack. Comply with recommendations of cited bar grating standards, including installation clearances and standard anchoring details.

- a. Attach removable units to supporting members with the type and size of clips and fasteners as indicated or, if not indicated, as recommended by the grating manufacturer for the type of installation conditions shown.
- b. Attach nonremovable units to supporting members by welding where both materials are same; otherwise, fasten by bolting as indicated above.

3.1.1 Anchorage, Fastenings, and Connections

Provide anchorage where necessary for fastening miscellaneous FRP items securely in place.

3.2 CLOSEOUT ACTIVITIES

3.2.1 [Manufacturer's Warranty](#)

Submit original and 2 copies of manufacturer's signed Warranty.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 06 - WOOD, PLASTICS, AND COMPOSITES

SECTION 06 82 14

FIBERGLASS REINFORCED PLASTIC (FRP) PIPE AND TUBE RAILINGS

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY CONTROL
 - 1.3.1 Qualifications of Manufacturer
 - 1.3.2 Qualifications of Engineer of Record
- 1.4 DELIVERY, HANDLING, AND STORAGE

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
 - 2.1.1 Installation Drawings
 - 2.1.2 Product Data
 - 2.1.3 Design Requirements
 - 2.1.4 Performance Requirements
 - 2.1.4.1 Structural Performance of Pipe and Tube Railings
- 2.2 FABRICATION
- 2.3 MATERIALS
 - 2.3.1 Fasteners
 - 2.3.2 Anchors
 - 2.3.3 Grout And Anchoring Cement
 - 2.3.4 Component Connections
 - 2.3.4.1 Lag Screws and Bolts
 - 2.3.4.2 Toggle Bolts
 - 2.3.4.3 Bolts, Nuts, Studs, and Rivets
 - 2.3.4.4 Powder-Driven Fasteners
 - 2.3.4.5 Screws
 - 2.3.4.6 Washers

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Anchorage, Fastenings, and Connections
 - 3.1.2 Workmanship
- 3.2 CLOSEOUT ACTIVITIES
 - 3.2.1 Warranty
 - 3.2.2 Manufacturer's Instructions

-- End of Section Table of Contents --

SECTION 06 82 14

FIBERGLASS REINFORCED PLASTIC (FRP) PIPE AND TUBE RAILINGS

PART 1 GENERAL

This Section includes new fiberglass reinforced plastic (FRP) pipe and tube railing/guards, mounting systems and accessories.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1 (2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)

ASME B18.2.2 (2015) Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

ASME B18.6.2 (1998; R 2010) Slotted Head Cap Screws, Square Head Set Screws, and Slotted Headless Set Screws: Inch Series

ASME B18.6.3 (2013; R 2017) Machine Screws, Tapping Screws, and Machine Drive Screws (Inch Series)

ASME B18.21.1 (2009; R 2016) Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)

ASME B18.21.2M (1999; R 2014) Lock Washers (Metric Series)

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.3 (2013) Safety Requirements for Powder-Actuated Fastening Systems American National Standard for Construction and Demolition Operations

ASTM INTERNATIONAL (ASTM)

ASTM A307 (2014; E 2017) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength

ASTM C1107/C1107M	(2020) Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
ASTM D430	(2006; R 2012) Standard Test Methods for Rubber Deterioration - Dynamic Fatigue
ASTM D638	(2014) Standard Test Method for Tensile Properties of Plastics
ASTM D696	(2016) Standard Test Method for Coefficient of Linear Thermal Expansion of Plastics Between -30 degrees C and 30 degrees C With a Vitreous Silica Dilatometer
ASTM D790	(2017) Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
ASTM D1148	(2013; R 2018) Standard Test Method for Rubber Deterioration—Discoloration from Ultraviolet (UV) or UV/Visible Radiation and Heat Exposure of Light-Colored Surfaces
ASTM D2344/D2344M	(2016) Standard Test Method for Short-Beam Strength of Polymer Matrix Composite Materials and Their Laminates
ASTM E84	(2020) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E488/E488M	(2015) Standard Test Methods for Strength of Anchors in Concrete and Masonry Elements

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2018) International Building Code
---------	------------------------------------

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 101	(2021) Life Safety Code
----------	-------------------------

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.23	(Nov 2016) Ladders
----------------	--------------------

29 CFR 1926	Safety and Health Regulations for Construction
-------------	--

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL

PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications of Manufacturer; G, D0

Qualifications of Engineer of Record; G, D0

SD-02 Shop Drawings

Installation Drawings; G, D0

SD-03 Product Data

FRP Pipe and Tube; G, D0

Railings/Guards; G, D0

Anchorage Materials; G, D0

Adhesives; G, D0

Resins; G, D0

Hardeners; G, D0

Manufacturer's Sample Warranty

SD-06 Test Reports

Ultraviolet Test Reports

Thermal Expansion Test Reports

Flame Spread Test Reports

SD-07 Certificates

Proof of Certification from a minimum of two quality assurance programs for its facilities or products (UL, DNV, ABS, USCG, AARR)

SD-08 Manufacturer's Instructions

Manufacturer's Instructions

SD-09 Manufacturer's Field Reports

Manufacturer's Certification of Installation

SD-11 Closeout Submittals

Manufacturer's Warranty

1.3 QUALITY CONTROL

1.3.1 Qualifications of Manufacturer

Submit Qualifications Of Manufacturer documentation certifying that the Fiberglass Reinforced Plastic (FRP) manufacturer has a minimum of 10 years

of experience in manufacturing FRP products.

Provide items within this section from manufacturers having a minimum of 10 years of experience in the design and manufacture of similar products and systems.

Submit the [Proof of Certification from a minimum of two quality assurance programs for its facilities or products \(UL, DNV, ABS, USCG, AARR\)](#).

Provide the [manufacturer's sample warranty](#) for all FRP products against defects in material and workmanship for a minimum of 5 years. Provide evidence of manufacturer's ISO 9001:2000 standard certification.

1.3.2 Qualifications of Engineer of Record

Submit [Qualifications of Engineer of Record](#) documentation that the Engineer of Record is currently licensed within the jurisdiction of the project.

- a. The design in accordance with International Building Code and [ASCE 7-16](#)
- b. [Installation drawings](#)

1.4 DELIVERY, HANDLING, AND STORAGE

Deliver all manufactured materials in original, unbroken pallets, packages, containers, or bundles bearing the label of the manufacturers, clearly marked and identified relative to the complete system. Provide all adhesives, resins and their catalysts and hardeners in clearly marked or noted crates or boxes. Store all manufactured materials in dry indoor facilities with a constant temperature range between [70 and 85 degrees F](#) until they are required.

Handle all materials to prevent abrasion, cracking, chipping, twisting, or other deformations, and other types of damage.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

2.1.1 Installation Drawings

Submit templates and erection and installation drawings indicating thickness, type, and dimensions. Show construction details, reinforcement, anchorage, and installation with relation to the building construction.

Include plans, elevations, sections, details, and attachments to other work. Indicate for installed products to comply with design loads. Include structural analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

2.1.2 Product Data

Submit the manufacturer's catalog data including two copies of manufacturer's specifications, load tables, dimension diagrams, and anchor details for the following items:

- a. [FRP Pipe and Tube](#)

- b. Railings/Guards
- c. Anchorage Materials
- d. Adhesives
- e. Resins
- f. Hardeners

2.1.3 Design Requirements

Ensure that all posts and rails are FRP structural shapes manufactured by the pultrusion process. Compose structural shapes of fiberglass reinforcement and resin in qualities, quantities, properties, arrangements, and dimensions as necessary to meet the design requirements in accordance with [ASCE 7-16](#), [29 CFR 1910.23](#), [NFPA 101](#), and dimensions specified.

Ensure that fiberglass reinforcements are a combination of continuous roving, continuous strand mat, and surfacing veil in sufficient quantities as needed by the application required, the physical properties required, or both.

Provide resins, with appropriate hardeners, of isophthalic polyester, with the chemical formulation necessary for corrosion resistance, strength, and other physical properties.

Ensure that all finished surfaces of FRP items, including FRP pipe and tube, railings/guards, anchorage materials, and fabrications, are smooth, resin-rich, and free of voids, dry spots, cracks, and unreinforced areas. Provide complete coverage of all glass fibers with resin to protect against their exposure to wear or weathering.

Protect all pultruded structural shapes from ultraviolet (UV) attack with:

- a. Integral UV inhibitors within the resin
- b. Synthetic surfacing veil to help produce a resin-rich surface
- c. UV-resistant coating for outdoor exposures

Provide FRP products that have a flame spread rating of 25 or less as specified in [ASTM E84](#), Tunnel Test. Submit 3 copies of [Flame Spread Test Reports](#) to the Contracting Officer.

Ensure that rails, posts, and kick plates are integrally pigmented yellow. Submit 3 copies of [Ultraviolet Test Reports](#) for FRP material, similar to the requirements of [ASTM D1148](#) for rubber deterioration, and [ASTM D430](#), to the Contracting Officer. Submit testing data relating to [Thermal Expansion Test Reports](#).

Provide structural shapes in the guardrail system to meet minimum longitudinal mechanical properties as follows:

Tensile Strength	ASTM D638	30,000 psi
Tensile Modulus	ASTM D638	2,500,000 psi
Flexural Strength	ASTM D790	30,000 psi
Flexural Modulus	ASTM D790	1,800,000 psi
Flexural Modulus-Full Section		2,800,000 psi
Short Beam Shear	ASTM D2344/D2344M	4,500 psi
Shear Modulus-Transverse		450,000 psi
Coefficient of Thermal Expansion	ASTM D696	.000008 in/in/F
Flame Spread	ASTM E84	25 or less

2.1.4 Performance Requirements

2.1.4.1 Structural Performance of Pipe and Tube Railings

Provide a pipe and tube railing system capable of withstanding the effects of gravity loads in accordance with **ASCE 7-16** and International Building Code, **ICC IBC** with the following loads and stresses within limits and under the conditions indicated:

a. Handrails:

- (1) Uniform load of **50 lbf/foot** applied in any direction.
- (2) Concentrated load of **200 lbf** applied in any direction.
- (3) Uniform and concentrated loads need not be assumed to act concurrently.

b. Top Rails of Guards:

- (1) Uniform load of **50 lbf/foot** applied in any direction.
- (2) Concentrated load of **200 lbf** applied in any direction.
- (3) Uniform and concentrated loads need not be assumed to act concurrently.

c. Infill of Guards:

- (1) Concentrated load of **50 lbf** applied horizontally on an area of **1 square foot**.
- (2) Uniform load of **25 lbf/square foot** applied horizontally.
- (3) Infill load and other loads need not be assumed to act concurrently.

2.2 FABRICATION

Perform fabrication of the handrail post/rail connection such that the rails are unbroken and continuous through the post without the use of packs or splices. Install the bottom rail through the post at a prepared hole made to fit the outside dimensions of the rail, and the top rail fit into a machined, u-shaped pocket formed into the top of the post such that the rail is located at the center of the post. Radius all exposed corners to eliminate sharp edges. Join the rails to the post through a combination of bonding and riveting. Ensure that no sharp, protruding edges remain after assembly of the handrail. Space the posts no more than [72 inches](#) apart. Attach post bases according to the construction contract drawings. Reinforce post bases to a height of [8 1/2 inches](#). Coat all field-fabricated and shop-fabricated cuts with a vinyl ester resin to provide maximum corrosion resistance.

2.3 MATERIALS

2.3.1 Fasteners

Provide Type 316 stainless-steel concealed fasteners, unless unavoidable or standard for railings indicated.

2.3.2 Anchors

Provide cast-in-place epoxy mechanical anchors, fabricated from corrosion-resistant materials with capability to sustain, without failure, a load equal to six times the design load imposed when installed in unit masonry and equal to four times the design load imposed when installed in concrete, as determined by testing as specified in [ASTM E488/E488M](#).

2.3.3 Grout And Anchoring Cement

Provide factory-packaged, nonshrink, nonmetallic grout complying with [ASTM C1107/C1107M](#); or water-resistant, nonshrink anchoring cement; recommended by the manufacturer for exterior use. Ensure that all other adhesives conform to the manufacturer's recommendations and instructions.

2.3.4 Component Connections

2.3.4.1 Lag Screws and Bolts

Provide lag screws and bolts conforming to [ASME B18.2.1](#), of the type and grade best suited for the purpose.

2.3.4.2 Toggle Bolts

Provide toggle bolts conforming to [ASME B18.2.1](#).

2.3.4.3 Bolts, Nuts, Studs, and Rivets

Provide bolts, nuts, studs, and rivets conforming to [ASME B18.2.2](#) and [ASTM A307](#).

2.3.4.4 Powder-Driven Fasteners

Follow safety provisions of [ASSP A10.3](#).

2.3.4.5 Screws

Provide screws conforming to [ASME B18.2.1](#), [ASME B18.6.2](#), and [ASME B18.6.3](#).

2.3.4.6 Washers

Provide plain washers conforming to [ASME B18.21.1](#). Provide beveled washers for American Standard beams and channels, square or rectangular, tapered in thickness, and smooth. Provide lock washers conforming to [ASME B18.21.2M](#) and [ASME B18.21.1](#).

PART 3 EXECUTION

3.1 INSTALLATION

Install items in accordance with [29 CFR 1910.23](#) and [29 CFR 1926](#) at locations indicated, according to the manufacturer's instructions. Verify all measurements and take all field measurements necessary before fabrication. Include all materials and parts necessary to complete each item, even though such work is not definitely shown or specified. Perform cutting, drilling, and fitting required for installing railings. Set railings accurately in location, alignment, and elevation. Submit 3 signed copies of [Manufacturer's Certification of Installation](#).

- a. Set posts plumb within a tolerance of [1/16 inch](#) in [3 feet](#).
- b. Align rails so variations from level for horizontal members and variations from parallel with rake of steps and ramps for sloping members do not exceed [1/4 inch](#) in [12 feet](#).

3.1.1 Anchorage, Fastenings, and Connections

Provide anchorage where necessary for fastening miscellaneous FRP items securely in place. Include for anchorage not otherwise specified or indicated slotted inserts, expansion shields, and powder-driven fasteners, when approved for concrete; toggle bolts and through-bolts for masonry; machine and carriage bolts for steel; through-bolts and screws. Conceal fastenings where practicable.

3.1.2 Workmanship

Ensure that FRP work is well formed to shape and size, with sharp lines and angles and true curves. Ensure that drilling and punching produces clean true lines and surfaces. Ensure that exposed surfaces of work-in-place to have smooth finishes. Mill joints where tight fits are required. Ensure that corner joints are coped or mitered, well formed, and in true alignment. Accurately set work to established lines and elevations and securely fasten in place. Ensure that the installation is in accordance with the manufacturer's installation instructions and the approved drawings, cuts, and details.

3.2 CLOSEOUT ACTIVITIES

3.2.1 Warranty

Submit 3 signed copies of the [Manufacturer's Warranty](#) 30 calendar days before final inspection.

3.2.2 Manufacturer's Instructions

Submit the [manufacturer's instructions](#) for shipping and handling, and procedures for erection, care, and maintenance upon completion of installation.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 97 02.03

PAINTING: COAL TAR EPOXY

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SAFETY, HEALTH, AND ENVIRONMENTAL REQUIREMENTS
 - 1.2.1 Safety
 - 1.2.1.1 Abrasive Blasting
 - 1.2.1.2 Workers Other Than Blasters
 - 1.2.1.3 Cleaning Before and After Abrasive Blasting
 - 1.2.1.4 Paint Mixing
 - 1.2.1.5 Paint Spraying
 - 1.2.1.6 Explosion Proof Equipment
 - 1.2.1.7 Further Precautions
 - 1.2.1.8 Ignition Sources
 - 1.2.2 Health
 - 1.2.2.1 Air Monitoring
 - 1.2.2.2 Medical Status
 - 1.2.2.3 Change in Medical Status
 - 1.2.3 Environmental Protection
 - 1.2.3.1 Waste Classification, Handling, and Disposal
 - 1.2.3.2 Water Quality
 - 1.2.3.3 Soil Quality
- 1.3 SUBMITTALS
- 1.4 QUALIFICATIONS
 - 1.4.1 Certified Environmental, Health, and Safety (EHS) Professionals
 - 1.4.2 Certified Lead Laboratory
 - 1.4.3 Qualified Painting Contractor
 - 1.4.4 Qualified Hazardous Paint Removal Contractor
 - 1.4.5 Coating Thickness Gage Qualification
- 1.5 DELIVERY, STORAGE, AND HANDLING
- 1.6 AMBIENT CONDITIONS

PART 2 PRODUCTS

- 2.1 SPECIAL PAINT FORMULAS
- 2.2 PAINT FORMULATIONS
 - 2.2.1 Formula C-200a, Coal Tar-Epoxy (Black) Paint

PART 3 EXECUTION

- 3.1 CLEANING AND PREPARATION OF SURFACES TO BE PAINTED
 - 3.1.1 General Requirements
 - 3.1.2 Ferrous Surfaces Subject to Atmospheric Exposures
- 3.2 PAINT APPLICATION
 - 3.2.1 General
 - 3.2.2 Mixing and Thinning
 - 3.2.3 Time Between Surface Preparation and Painting

- 3.2.4 Method of Paint Application
- 3.2.5 Coverage and Film Thickness
 - 3.2.5.1 Measurement on Ferrous Metal
- 3.2.6 Progress of Painting Work
- 3.2.7 Contacting Surfaces
- 3.2.8 Drying Time Prior to Immersion
- 3.2.9 Protection of Painted Surfaces
- 3.2.10 Coal Tar-Epoxy (Black) Paint (Formula C-200a)
 - 3.2.10.1 Mixing
 - 3.2.10.2 Application
 - 3.2.10.3 Subsequent Coats
 - 3.2.10.4 Ambient Temperature
 - 3.2.10.5 Safety
- 3.3 PAINT SYSTEMS APPLICATION
 - 3.3.1 Fabricated and Assembled Items
 - 3.3.2 Surface Preparation
 - 3.3.3 System No. 6-A-Z
 - 3.3.4 Protection of Nonpainted Items and Cleanup
- 3.4 INSPECTION
- 3.5 PAINTING SCHEDULES

-- End of Section Table of Contents --

SECTION 09 97 02.03

PAINTING: COAL TAR EPOXY

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D7091 (2013) Standard Practice for
Nondestructive Measurement of Dry Film
Thickness of Nonmagnetic Coatings Applied
to Ferrous Metals and Nonmagnetic,
Nondestructive Coatings Applied to
Non-Ferrous Metals

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z87.1 (2020) Occupational and Educational
Personal Eye and Face Protection Devices

ANSI/ISEA Z358.1 (2014) American National Standard for
Emergency Eyewash and Shower Equipment

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 4 2023; ERTA 5 2023; ERTA 6
2023) National Electrical Code

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

NIOSH 2003-154 (2003; 4th Ed; Supple 3) NIOSH Manual of
Analytical Methods

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC 7/NACE No.4 (2007) Brush-Off Blast Cleaning

SSPC Paint 16 (2006; R 2015; E 2015) Coal Tar
Epoxy-Polyamide Black (or Dark Red) Paint

SSPC QP 1 (2019) Standard Procedure for Evaluating
the Qualifications of Industrial/Marine
Painting Contractors (Field Application to
Complex Industrial Steel Structures and
Other Metal Components)

SSPC QP 2 (2019) Standard Procedure for Evaluating
the Qualifications of Industrial/Marine
Painting Contractors (Removal of Hazardous

	Coatings from Structures)
SSPC SP 1	(2015) Solvent Cleaning
SSPC SP 3	(2018) Power Tool Cleaning
U.S. ARMY CORPS OF ENGINEERS (USACE)	
EM 385-1-1	(2024) Safety and Occupational Health (SOH) Requirements
U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)	
29 CFR 1910	Occupational Safety and Health Standards
29 CFR 1910.20	Access to Employee Exposure and Medical Records
29 CFR 1910.94	Ventilation
29 CFR 1910.134	Respiratory Protection
29 CFR 1910.146	Permit-required Confined Spaces
29 CFR 1926.62	Lead
40 CFR 117	Determination of Reportable Quantities for Hazardous Substances
40 CFR 122	EPA Administered Permit Programs: The National Pollutant Discharge Elimination System
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 262.22	Number of Copies
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 302	Designation, Reportable Quantities, and Notification
40 CFR 355	Emergency Planning and Notification
49 CFR 171	General Information, Regulations, and Definitions

1.2 SAFETY, HEALTH, AND ENVIRONMENTAL REQUIREMENTS

Perform work in accordance with all applicable health, safety, and environmental requirements as well as EM 385-1-1. Submit matters of interpretation of these requirements to the Contracting Officer for resolution before starting work. If no clarifications are sought, then the submittal is not necessary. Where the regulations conflict, the most

stringent requirements shall apply. This paragraph supplements the health, safety, and environmental requirements of [EM 385-1-1](#).

1.2.1 Safety

Submit a [Safety Plan](#) in accordance with the requirements of Section 01 of [EM 385-1-1](#), including, but not limited to, each of the topic areas listed in Appendix A therein and the specified requirements. Develop each topic in a concise manner to include management and operational aspects.

1.2.1.1 Abrasive Blasting

For abrasive blasting comply with the requirements in Section 06.H of [EM 385-1-1](#). In addition to the requirements in Section 20 of [EM 385-1-1](#), use hoses and hose connections of a type to prevent shock from static electricity. Hose lengths shall be joined together by approved couplings of a material and type designed to prevent erosion and weakening of the couplings. The couplings and nozzle attachments shall fit on the outside of the hose and designed to prevent accidental disengagement.

1.2.1.2 Workers Other Than Blasters

Protect workers, other than blasting operators working in close proximity to abrasive blasting operations, by utilizing MSHA/NIOSH-approved half-face or full-face air purifying respirators equipped with high-efficiency particulate air (HEPA) filters, eye protection meeting or exceeding [ANSI/ISEA Z87.1](#) and hearing protectors (ear plugs and/or ear muffs) providing a noise reduction rating of at least 20 dBA or as needed to provide adequate protection. Personal protective equipment shall be provided where required by [29 CFR 1910.146](#) and in accordance with [29 CFR 1910](#), Subpart I.

1.2.1.3 Cleaning Before and After Abrasive Blasting

Cleaning with compressed air shall be in accordance with Section 20.B.5 of [EM 385-1-1](#) and personnel shall be protected as specified in [29 CFR 1910.134](#). When cleaning with solvents, provide ventilation where required by [29 CFR 1910.146](#) or where the concentration of solvent vapors exceeds 10 percent of the Lower Explosive Limit (LEL). Ventilation shall be in accordance with [29 CFR 1910.94](#), paragraph (c)(5).

1.2.1.4 Paint Mixing

Local exhaust ventilation shall be provided in the area where coatings are mixed. This ventilation system shall be capable of providing at least 100 linear fpm of capture velocity in the mixing zone. Exposure of skin and eyes shall be avoided by wearing appropriate chemically resistant gloves, apron, safety goggles, and face shields meeting or exceeding the requirements of [ANSI/ISEA Z87.1](#). A combination unit, comprised of an eyewash and deluge shower, within close proximity to the mixing operation shall be provided in accordance with [ANSI/ISEA Z358.1](#), paragraph (9). Individuals who have a history of, or develop a sensitivity to epoxy or polyurethane resin systems, shall not conduct work tasks or otherwise be exposed to such chemicals

1.2.1.5 Paint Spraying

Submit a comprehensive written [Respiratory Protection Plan](#) in accordance with [29 CFR 1910.134](#), [29 CFR 1926.62](#), and Section 05.G of [EM 385-1-1](#).

During all spray painting operations, spray painters shall use approved SCBA or SAR (air line) respirators, unless valid air sampling has demonstrated contaminant levels to be consistently within concentrations that are compatible with air-purifying respirator Assigned Protection Factor (APF). Persons with facial hair that interferes with the sealing surface of the facepiece to face seal or interferes with respirator valve function shall not be allowed to perform work requiring respiratory protection. Air-purifying chemical cartridge/canister half- or full-facepiece respirators that have a particulate prefilter and are suitable for the specific type(s) of gas/vapor and particulate contaminant(s) may be used for nonconfined space painting, mixing, and cleaning (using solvents). These respirators may be used provided the measured or anticipated concentration of the contaminant(s) in the breathing zone of the exposed worker does not exceed the APF for the respirator and the gas/vapor has good warning properties or the respirator assembly is equipped with a NIOSH-approved end of service life indicator for the gas(es)/vapor anticipated or encountered. Where paint contains toxic elements that may become airborne during painting in nonconfined spaces, air-purifying half- and full-facepiece respirators or powered air-purifying respirators equipped with appropriate gas vapor cartridges, in combination with a high-efficiency filter, or an appropriate canister incorporating a high-efficiency filter, shall be used.

1.2.1.6 Explosion Proof Equipment

Electrical wiring, lights, and other equipment located in the paint spraying area shall be of the explosion proof type designed for operation in Class I, Division 1, Group D, hazardous locations as required by the **NFPA 70**. Electrical wiring, motors, and other equipment, outside of but within **20 feet** of any spraying area, shall not spark and shall conform to the provisions for Class I, Division 2, Group D, hazardous locations. Electric motors used to drive exhaust fans shall not be placed inside spraying areas or ducts. Fan blades and portable air ducts shall be constructed of nonferrous materials. Motors and associated control equipment shall be properly maintained and grounded. The metallic parts of air-moving devices, spray guns, connecting tubing, and duct work shall be electrically bonded and the bonded assembly shall be grounded.

1.2.1.7 Further Precautions

- a. Workers must wear nonsparking safety shoes.
- b. Place and ground solvent drums, taken into the spraying area, on nonferrous surfaces. Maintain metallic bonding between containers and drums when materials are being transferred.
- c. Inspect insulation on all power and lighting cables to ensure that the insulation is in excellent working condition and is free of all cracks and worn spots. Cables shall be further inspected to ensure that no connections are within **50 feet** of the operation, that lines are not overloaded, and that they are suspended with sufficient slack to prevent undue stress or chafing.

1.2.1.8 Ignition Sources

Ignition sources, to include lighted cigarettes, cigars, pipes, matches, or cigarette lighters shall be prohibited in area of solvent cleaning, paint storage, paint mixing, or paint application.

1.2.2 Health

Prepare and submit a [Medical Surveillance Plan](#) and a statement from the examining physician indicating the name of each employee evaluated and any limitations which will preclude the employee from performing the work required. The statement shall include the date of the medical evaluation, the physician's name, signature, and telephone number.

1.2.2.1 Air Monitoring

Prepare and submit an [Air Monitoring Test Plan](#). Perform air sampling and testing as needed to assure that workers are not exposed to contaminants above the permissible exposure limit. In addition, provide the Contracting Officer with a copy of the [Air Monitoring Test Report](#) from the laboratory within five working days of the sampling date, including records of air monitoring plans and tests performed. Submit reports as soon as information is available. Also provide results from direct-reading instrumentation on the same day the samples are collected. Prepare and submit an [Airborne Sampling Plan](#) detailing the [NIOSH 2003-154](#), Factory Mutual, or Underwriters Laboratories approved equipment, equipment calibration procedures, sampling methods, sampling to be performed, and analytical procedures to be used based on the type of work to be performed and anticipated toxic contaminants to be generated. Include the name of the accredited laboratory, listed by the American Industrial Hygiene Association (AIHA), to be used to conduct the analysis of any collected air samples.

1.2.2.2 Medical Status

Prior to the start of work, and annually thereafter, submit a [Medical Status Report](#) including records of medical tests. Medically evaluate all Contractor employees working with or around paint systems, thinners, blast media, those required to wear respiratory protective equipment, and those who will be exposed to high noise levels for the particular type of exposure they may encounter. Maintain medical records as required by [29 CFR 1910.20](#). The evaluation shall include:

- a. Audiometric testing and evaluation of employees who will work in a noise environment with a time weighted average greater than or equal to 90 dBA.
- b. Vision screening (employees who use full-facepiece respirators shall not wear contact lenses).
- c. Medical evaluation shall include, but shall not be limited to, the following:
 - (1) Medical history including, but not limited to, alcohol use, with emphasis on liver, kidney, and pulmonary systems, and sensitivity to chemicals to be used on the job.
 - (2) General physical examination with emphasis on liver, kidney, and pulmonary system.
 - (3) Determination of the employee's physical and psychological ability to wear respiratory protective equipment and to perform job-related tasks.
 - (4) Determination of baseline values of biological indices for later

comparison to changes associated with exposure to paint systems and thinners or blast media, which include: liver function tests to include SGOT, SGPT, GGPT, alkaline phosphates, bilirubin, complete urinalysis, EKG (employees over age 40), blood urea nitrogen (bun), serum creatinine, pulmonary function test, FVC, and FEV, chest x-ray (if medically indicated), blood lead and ZPP (for individuals where it is known there will be an exposure to materials containing lead), other criteria that may be deemed necessary by the Contractor's physician, and Physician's statements for individual employees that medical status would permit specific task performance.

- (5) For lead-based paint removal, the medical requirements of 29 CFR 1926.62 shall also be included. Prepare and submit a Worker Protection Plan in accordance with the requirements of 29 CFR 1926.62, addressing all necessary aspects of worker protection and including activities emitting lead, means to achieve compliance, alternative technologies considered, air monitoring program, implementation schedule, work practice program, administrative controls, multi-Contractor site arrangements, and jobsite inspections.

1.2.2.3 Change in Medical Status

Any employee whose medical status has changed negatively due to work related chemical and/or physical agent exposure while working with or around paint systems and thinners, blast media, or other chemicals shall be evaluated by a physician, and obtain a physicians statement as described in paragraph MEDICAL STATUS prior to allowing the employee to return to those work tasks. Submit a Change in Medical Status Report detailing any negative changes in employee medical status and the results of the physicians reevaluation statement.

1.2.3 Environmental Protection

In addition to the requirements of Section 01 57 20 ENVIRONMENTAL PROTECTION, prepare an Environmental Protection Plan incorporating the submittals for Water Quality Plan, Containment Plan, Waste Disposal Plan, Soil Quality Plan, TSP Monitoring Plan, PM-10 Monitoring Plan, and Visible Emissions Monitoring Plan. The submitted plan shall also address all aspects of establishing and demarcating regulated areas, ventilation/containment system performance verification, and reporting of accidental releases. Comply with the following environmental protection criteria.

1.2.3.1 Waste Classification, Handling, and Disposal

Prepare and submit a Waste Disposal Plan in accordance with the requirements of 40 CFR 261 and 40 CFR 262 including classification and handling. The Contractor is responsible for assuring the proper disposal of all hazardous and nonhazardous waste generated during the project. Waste generated from abrasive blasting, lead-containing paints with recyclable steel or iron abrasives shall be disposed of as a hazardous waste or shall be stabilized with proprietary pre-blast additives regardless of the results of 40 CFR 261 App II, Mtd 1311. Where stabilization is preferred, employ a proprietary blast additive, that has been blended with the blast media prior to use. Place hazardous waste in properly labeled, closed containers shielded adequately to prevent dispersion of the waste by wind or water. Any evidence of improper

storage shall be cause for immediate shutdown of the project until corrective action is taken. Store nonhazardous waste in closed containers separate from hazardous waste storage areas. All hazardous waste shall be transported by a licensed transporter in accordance with 40 CFR 263 and 49 CFR 171, Subchapter C. Transport all nonhazardous waste in accordance with local regulations regarding waste transportation. In addition to the number of copies required by 40 CFR 262.22, supply one copy of each Waste Manifest to the Contracting Officer prior to transportation.

1.2.3.2 Water Quality

Prepare a Water Quality Plan for all job sites where lead-containing or other hazardous paint will be removed, including provisions for halting work if spills or emissions are observed entering into bodies of water or found in areas where storm water runoff could carry the debris into bodies of water or storm sewers. The plan shall also address cleanup and reporting procedures. Conduct operations in such a manner that lead-containing and other hazardous paint debris do not contaminate the water and so that NPDES permits in accordance with EPA regulation 40 CFR 122 are not required for the project. In the event that there are any releases of lead paint debris into the waterways, with reportable quantities of hazardous substances designated pursuant to Section 311 of the Clean Water Act, they shall be reported to the EPA in accordance with 40 CFR 117 and 40 CFR 355. Releases or spills that carry into waterways or storm sewers shall be thoroughly documented. The documentation shall include the time and location of the release, amount of material released, actions taken to clean up the debris, amount of debris recovered, and corrective action taken to avoid a reoccurrence. Releases shall also be reported to the Coast Guard and other commonwealth and local authorities as appropriate. If the release is equivalent to 10 pounds or more of lead-containing material in a 24-hour period, it is considered to be a reportable quantity under CERCLA. Comply with 40 CFR 302.

1.2.3.3 Soil Quality

Prepare a Soil Quality Plan for all job sites where lead-containing or other hazardous paint will be removed. The plan shall include provisions for halting the work should soil contamination occur, correcting the deficiencies responsible for the contamination, and provide procedures for removing and replacing contaminated soil. Establish and implement practices and procedures for preventing contamination of the soil from the removal of lead-containing or other hazardous paints. Soil shall be considered to have been contaminated by the Contractor's operation if an increase in the total lead content of 100 PPM or greater over background levels occurs. For purposes of computing the increase compute the mean background levels and the mean post-removal levels. The 100 PPM criteria is met if the difference between the means is less than 100 PPM plus the 95 percent confidence limit. Soil sampling and testing shall be conducted prior to the beginning of the project and after the project is completed. Interim testing may also be performed in the event the Contractor or Contracting Officer wants to confirm that the containment system and work practices continue to provide satisfactory protection of the soil. The following minimum test locations shall be selected for soil analysis. Two locations shall be selected beneath or immediately adjacent to the structure being prepared, and additional samples shall be taken within 100 feet in each direction of the project (i.e., N, S, E, W) in which soil is present. The number of soil sample locations shall be sufficient to adequately characterize the soil contaminant levels within and around the project area. Five composite samples shall be collected at each

location. Each of the five samples shall be comprised of five individual plugs of soil combined in a single bag. The composite samples at each location shall be collected using the following procedure:

- a. Place a 1-square foot template at each location.
- b. Remove a sample of soil 3/4 inch in diameter and 1/2 inch in depth at the center of the template and at each of the four corners. Place the five soil plugs into a single bag. This represents one of the three samples to be removed at a given location.
- c. Move the template 1 inch in any direction and repeat the process to collect the second sample. Place all plugs in a separate bag. Move the template 1 inch farther to collect the third sample.
- d. Identify each sample bag with the date, specific location of the sample, name and signature of the sampling technician, and complete chain of custody records.
- e. It is critical that the specific location of each sample be thoroughly measured and documented as the final project testing (and any interim testing) shall be sampled in the precise locations.

Three samples collected at each location shall be analyzed. One of the remaining two samples shall be maintained by the Contractor for the duration of the project and the other by the Contracting Officer in the event reanalysis is required. Lead-containing samples shall be analyzed in accordance with EPA testing guidance as published in 40 CFR 261, App III, by a laboratory listed by the American Industrial Hygiene Association (AIHA) as being proficient in conducting the test. Note that if it is determined that contamination of the soil has occurred as a result of the paint removal operations, TCLP testing will be employed to determine if the soil shall be handled and disposed of as a hazardous waste. The initial sampling of the soil for total lead content does not establish whether the soil would be considered hazardous by TCLP testing. As a result, at the Contractor's option, additional prework soil samples may be removed (minimum of 105 grams is required for a single test at each site) to conduct TCLP testing to establish whether the soil would be classified as hazardous prior to project startup. In the event that there is a release of lead paint debris onto the soil and if the release is 10 pounds or more of lead-containing material in a 24-hour period, it is considered to be a reportable quantity under CERCLA. Comply with 40 CFR 302. Thoroughly document the occurrence of any spills of lead debris into the soil. The documentation shall include the time and location of the release, amount of material released, actions taken to clean up the debris, amount of debris reclaimed, and corrective action taken to avoid a reoccurrence. Provide the documentation to the Contracting Officer and include the Soil Quality Test Report with results of the prework and post work soil quality tests.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Designation R0 stands for Resident Office and designation D0 stands for District Office. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Safety, Health, and Environmental Requirements; G, DO
Safety Plan; G, DO
Respiratory Protection Plan; G, DO
Airborne Sampling Plan; G, DO
Medical Surveillance Plan; G, DO
Worker Protection Plan; G, DO
Environmental Protection Plan; G, DO
Waste Manifest
Waste Disposal Plan; G, DO
Water Quality Plan; G, DO
Soil Quality Plan; G, DO

SD-03 Product Data

Manufacturer's Product Data Sheet; G, DO

SD-04 Samples

Product Samples; G, DO
Special Paint Formulas; G, DO

SD-06 Test Reports

Soil Quality Test Report
Inspection Reports
Medical Status Report
Change in Medical Status Report
Air Monitoring Test Plan; G, DO
Air Monitoring Test Report

SD-07 Certificates

Certified EHS Professional
Certified Lead Laboratory
SSPC QP 1 Certificate
SSPC QP 2 Certificate
Qualified Hazardous Paint Removal Contractor; G, DO
Coating Thickness Gage Qualification

1.4 QUALIFICATIONS

Qualifications and experience shall comply with the following.

1.4.1 Certified Environmental, Health, and Safety (EHS) Professionals

Provide a certificate for each [Certified EHS Professional](#); submit qualifications and experience of qualified and competent persons employed to provide preconstruction and onsite environmental, safety, and health services. Obtain acceptance of this submission prior to the submission of other required environmental, safety, and health submittal items. Utilize a qualified and competent person as defined in Section 01 of [EM 385-1-1](#) to develop the required safety and health submittal and to provide onsite safety and health services during the contract period. The person shall be a Certified Industrial Hygienist (CIH), an Industrial Hygienist (IH), or a Certified Safety Professional (CSP) with a minimum of 3 years of demonstrated experience in similar related work. The CIH, IH, or CSP may utilize other qualified and competent persons, as defined in [EM 385-1-1](#),

to conduct on-site safety and health activities as long as these persons have a minimum of 2 years of demonstrated experience in similar related work and are under the direct supervision of the CIH, IH, or CSP. For lead containing jobsites, the competent and qualified person shall have successfully completed an EPA or commonwealth accredited lead-based paint abatement Supervisor course specific to the work to be performed and shall possess current and valid commonwealth and/or local government certification, as required.

1.4.2 Certified Lead Laboratory

Provide documentation which includes the name, address, and telephone number of the laboratories to be providing services. In addition, the documentation shall indicate that each laboratory is an EPA National Lead Laboratory Accreditation Program (NLLAP) accredited laboratory and that each is rated proficient in the NIOSH/EPA Environmental Lead Proficiency Analytical Testing Program (ELPAT) and will document the date of current accreditation. Certification shall include accreditation for heavy metal analysis, list of experience relevant to analysis of lead in air, and a Quality Assurance and Quality Control Program. Submit a certificate for the Certified Lead Laboratory.

1.4.3 Qualified Painting Contractor

Provide a certified SSPC QP 1 Painting Contractor. Submit a copy of the SSPC QP 1 Certificate.

1.4.4 Qualified Hazardous Paint Removal Contractor

Provide a certified SSPC QP 2 Painting Contractor. Provide a copy of the SSPC QP 2 Certificate.

1.4.5 Coating Thickness Gage Qualification

Submit Coating Thickness Gage Qualification documentation of manufacturer's certification for all coating thickness gages. Magnetic flux thickness gages as described in ASTM D7091 shall be used to make all coating thickness measurements on ferrous metal substrates. . Gages shall have an accuracy of +/- 3 percent or better. Gages to be used on the job shall be certified by the manufacturer as meeting these requirements.

1.5 DELIVERY, STORAGE, AND HANDLING

Process and package paints to ensure that within a period of one year from date of manufacture, they will not gel, liver, or thicken deleteriously, or form gas in the closed container. Paints, unless otherwise specified or permitted, shall be packaged in standard containers not larger than 5 gal, with removable friction or lug-type covers. . Each container of paint or separately packaged component thereof shall be labeled to indicate the purchaser's order number, date of manufacture, manufacturer's batch number, quantity, color, component identification and designated name, and formula or specification number of the paint together with special labeling instructions, when specified. Paint shall be delivered to the job in unbroken containers. Paints that can be harmed by exposure to cold weather shall be stored in ventilated, heated shelters. All paints shall be stored under cover from the elements and in locations free from sparks and flames.

1.6 AMBIENT CONDITIONS

Paint shall be applied only to surfaces that are above the dew point temperature and that are completely free of moisture as determined by sight and touch. Paint shall not be applied to surfaces upon which there is detectable frost or ice. Except as otherwise specified, the temperature of the surfaces to be painted and of air in contact therewith shall be not less than 45 degrees F during paint application nor shall paint be applied if the surfaces can be expected to drop to 32 degrees F or lower before the film has dried to a reasonably firm condition. During periods of inclement weather, painting may be continued by enclosing the surfaces and applying artificial heat, provided the minimum temperatures and surface dryness requirements prescribed previously are maintained. Paint shall not be applied to surfaces heated by direct sunlight or other sources to temperatures that will cause detrimental blistering, pinholing, or porosity of the film.

PART 2 PRODUCTS

Product Samples of each batch of thinner, solvent, and paint shall be submitted to the Government for testing. Submit Manufacturer's Product Data Sheet for each type of paint used; for products that are specified to be applied in accordance with the manufacturer's recommendations, submit the paint manufacturer's product data sheet or other written instructions for those products. Submit samples of all special paint formula, Military, Master Painter Institute, and SSPC paints and samples of solvents or thinners used to reduce the viscosity of the paint. Allow at least 30 days for sampling and testing of samples of paints and thinners. Sampling may be at the jobsite or source of supply. Notify the Contracting Officer when the paint and thinner are available for sampling. Sampling of each batch will be witnessed by the Contracting Officer unless otherwise specified or directed. Submit a 1-quart sample of paint and thinner for each batch proposed for use. The sample shall be labeled to indicate formula or specification number and nomenclature, batch number, batch quantity, color, date made, and applicable project contract number. Testing will be performed by the Contractor at Contractor's expense. Costs for retesting rejected material will be at the Contractor's expense.

2.1 SPECIAL PAINT FORMULAS

Special paints shall have the composition as indicated in the formulas listed herein. Where so specified, certain components of a paint formulation shall be packaged in separate containers for mixing on the job. If not specified or otherwise prescribed, the color shall be that naturally obtained from the required pigmentation.

2.2 PAINT FORMULATIONS

Special paint formulas shall comply with the following:

2.2.1 Formula C-200a, Coal Tar-Epoxy (Black) Paint

The paint shall conform to SSPC Paint 16 manufactured with Type 1 pitch. In addition to standard labeling, container labels shall include the term, Corps of Engineers Formula C-200a.

PART 3 EXECUTION

3.1 CLEANING AND PREPARATION OF SURFACES TO BE PAINTED

3.1.1 General Requirements

Clean surfaces to be painted before applying paint or surface treatments. Remove deposits of grease or oil in accordance with **SSPC SP 1**, prior to mechanical cleaning. Solvent cleaning shall be accomplished with mineral spirits or other low toxicity solvents having a flash point above **100 degrees F**. Use clean cloths and clean fluids to avoid leaving a thin film of greasy residue on the surfaces being cleaned. Protect items not to be prepared or coated from damage by the surface preparation methods. Machinery shall be protected against entry of blast abrasive and dust into working parts. Cleaning and painting shall be so programmed that dust or other contaminants from the cleaning process do not fall on wet, newly painted surfaces, and surfaces not intended to be painted shall be suitably protected from the effects of cleaning and painting operations. Welding of, or in the vicinity of, previously painted surfaces shall be conducted in a manner to prevent weld spatter from striking the paint and to otherwise reduce coating damage to a minimum; paint damaged by welding operations shall be restored to original condition. Surfaces to be painted that will be inaccessible after construction, erection, or installation operations are completed shall be painted before they become inaccessible.

3.1.2 Ferrous Surfaces Subject to Atmospheric Exposures

Ferrous surfaces that are to be continuously in exterior or interior atmospheric exposure shall be cleaned by means of dry blasting to a commercial grade. Cleaning and priming shall be done in the shop unless otherwise directed or permitted. Power tool cleaning shall conform to the requirements of **SSPC SP 3**. Brush-off blast cleaning shall conform to the requirements of **SSPC 7/NACE No.4**. Welds and adjoining surfaces within a few inches (centimeters) thereof shall be cleaned of weld flux, spatter, and other harmful deposits by blasting, power impact tools, power wire brush, or such combination of these and other methods as may be necessary for complete removal of each type of deposit. The combination of cleaning methods need not include blasting when preparation of the overall surfaces is carried out by the power tool method. However, brush scrubbing and rinsing with clean water, after mechanical cleaning is completed, will be required unless the latter is carried out with thoroughness to remove all soluble alkaline deposits. Wetting of the surfaces during water-washing operations shall be limited to the weld area required to be treated, and such areas shall be dry before painting. Welds and adjacent surfaces cleaned thoroughly by blasting alone will be considered adequately prepared provided that weld spatter not dislodged by the blast stream shall be removed with impact or grinding tools. All surfaces shall be primed as soon as practicable after cleaning and in all cases prior to contamination or deterioration of the prepared surfaces. To the greatest degree possible, steel surfaces shall be cleaned (and primed) prior to lengthy outdoor storage.

3.2 PAINT APPLICATION

3.2.1 General

The finished coating shall be free from holidays, pinholes, bubbles, runs, drops, ridges, waves, laps, excessive or unsightly brush marks, and

variations in color, texture, and gloss. Application of initial or subsequent coatings shall not commence until the Contracting Officer has verified that atmospheric conditions and the surfaces to be coated are satisfactory. Each paint coat shall be applied in a manner that will produce an even, continuous film of uniform thickness. Edges, corners, crevices, seams, joints, welds, rivets, corrosion pits, and other surface irregularities shall receive special attention to ensure that they receive an adequate thickness of paint. Spray equipment shall be equipped with traps and separators and where appropriate, mechanical agitators, pressure gauges, pressure regulators, and screens or filters. Air caps, nozzles, and needles shall be as recommended by the spray equipment manufacturer for the material being applied. Airless-type spray equipment may be used only on broad, flat, or otherwise simply configured surfaces, except that it may be employed for general painting if the spray gun is equipped with dual or adjustable tips of proper types and orifice sizes. Airless-type equipment shall not be used for the application of vinyl paints.

3.2.2 Mixing and Thinning

Paints shall be thoroughly mixed, strained where necessary, and kept at a uniform composition and consistency during application. Dry-powder pigments specified to be added at the time of use shall, with the aid of powered stirrers, be incorporated into the vehicle or base paint in a manner that will produce a smooth, homogeneous mixture free of lumps and dry particles. Where necessary to suit conditions of the surface temperature, weather, and method of application, the paint may be thinned immediately prior to use. Thinning shall generally be limited to the addition of not more than 1 pint per gal of the proper thinner; this general limitation shall not apply when more specific thinning instructions are provided. Paint that has been stored at low temperature, shall be brought up to at least 70 degrees F before being mixed and thinned, and its temperature in the spray tank or other working container shall not fall below 60 degrees F during the application. Paint that has deteriorated in any manner to a degree that it cannot be restored to essentially its original condition by customary field-mixing methods shall not be used and shall be removed from the project site. Paint and thinner that is more than 1 year old shall be resampled and resubmitted for testing to determine its suitability for application. Moisture cure urethane paint shall be resampled and resubmitted for testing to determine its suitability for application whenever the paint is more than six months old as indicated by the date of manufacture on the container.

3.2.3 Time Between Surface Preparation and Painting

Surfaces that have been cleaned and/or otherwise prepared for painting shall be primed as soon as practicable after such preparation has been completed but, in any event, prior to any deterioration of the prepared surface.

3.2.4 Method of Paint Application

Unless otherwise specified, paint shall be applied by brush, roller, or spray to ferrous and nonferrous metal surfaces. Special attention shall be directed toward ensuring adequate coverage of edges, corners, crevices, pits, rivets, bolts, welds, and similar surface irregularities. Other methods of application to metal surfaces shall be subject to the specific approval of the Contracting Officer. Paint on plaster, concrete, or other nonmetallic surfaces shall be applied by brush, roller, and/or spray.

3.2.5 Coverage and Film Thickness

Film thickness or spreading rates shall be as specified hereinafter. Where no spreading rate is specified, the paint shall be applied at a rate consistent with the manufacturer's written instructions. In any event, the combined coats of a specified paint system shall completely hide base surface and the finish coats shall completely hide undercoats of dissimilar color.

3.2.5.1 Measurement on Ferrous Metal

Where dry film thickness requirements are specified for coatings on ferrous surfaces, measurements shall be made with a gage qualified in accordance with paragraph Coating Thickness Gage Qualification. They shall be calibrated and used in accordance with [ASTM D7091](#). Prior to each use the Base Metal Reading (BMR) shall be established for the gage as specified in the test method. Accuracy of the gage shall be verified using plastic shims as specified by the test method both prior to and following each set of measurements. Frequency of measurements shall be as recommended for field measurements by [ASTM D7091](#), except that measurements shall be performed on all areas of the structure being coated. Thickness measurements shall be reported as the mean for each spot determination.

3.2.6 Progress of Painting Work

Where field painting on any type of surface has commenced, the complete painting operation, including priming and finishing coats, on that portion of the work shall be completed as soon as practicable, without prolonged delays. Sufficient time shall elapse between successive coats to permit them to dry properly for recoating, and this period shall be modified as necessary to suit adverse weather conditions. Paint shall be considered dry for recoating when it feels firm, does not deform or feel sticky under moderate pressure of the finger, and the application of another coat of paint does not cause film irregularities such as lifting or loss of adhesion of the undercoat. All coats of all painted surfaces shall be unscarred and completely integral at the time of application of succeeding coats. At the time of application of each successive coat, undercoats shall be cleaned of dust, grease, overspray, or foreign matter by means of airblast, solvent cleaning, or other suitable means. Cement and mortar deposits on painted steel surfaces, not satisfactorily removed by ordinary cleaning methods, shall be brush-off blast cleaned and completely repainted as required. Undercoats of high gloss shall, if necessary for establishment of good adhesion, be scuff sanded, solvent wiped, or otherwise treated prior to application of a succeeding coat. Field coats on metal shall be applied after erection except as otherwise specified and except for surfaces to be painted that will become inaccessible after erection.

3.2.7 Contacting Surfaces

When riveted or ordinary bolted contact is to exist between surfaces of ferrous or other metal parts of substantially similar chemical composition, such surfaces will not be required to be painted, but any resulting crevices shall subsequently be filled or sealed with paint. Contacting metal surfaces formed by high-strength bolts in friction-type connections shall not be painted. Where a nonmetal surface is to be in riveted or bolted contact with a metal surface, the contacting surfaces of the metal shall be cleaned and given three coats of the specified primer. Unless otherwise specified, corrosion-resisting metal surfaces, including

cladding therewith, shall not be painted.

3.2.8 Drying Time Prior to Immersion

Minimum drying periods after final coat prior to immersion shall be: epoxy and moisture cure urethane systems at least 5 days, vinyl-type paint systems at least 3 days, and cold-applied coal tar systems at least 7 days. Minimum drying periods shall be increased twofold if the drying temperature is below 65 degrees F and/or if the immersion exposure involves considerable abrasion.

3.2.9 Protection of Painted Surfaces

Where shelter and/or heat are provided for painted surfaces during inclement weather, such protective measures shall be maintained until the paint film has dried and discontinuance of the measures is authorized. Items that have been painted shall not be handled, worked on, or otherwise disturbed until the paint coat is fully dry and hard. All metalwork coated in the shop or field prior to final erection shall be stored out of contact with the ground in a manner and location that will minimize the formation of water-holding pockets; soiling, contamination, and deterioration of the paint film, and damaged areas of paint on such metalwork shall be cleaned and touched up without delay. The first field coat of paint shall be applied within a reasonable period of time after the shop coat and in any event before weathering of the shop coat becomes extensive.

3.2.10 Coal Tar-Epoxy (Black) Paint (Formula C-200a)

3.2.10.1 Mixing

Component B shall be added to previously stirred Component A and thoroughly mixed together with a heavy-duty mechanical stirrer just prior to use. The use of not more than 1 pint of xylene thinner per 1 gal of paint will be permitted to improve application properties and extend pot life. The pot life of the mixed paint, extended by permissible thinning, may vary from 2 hours in very warm weather to 5 or more hours in cool weather. Pot life in warm weather may be extended by precooling the components prior to mixing; cooling the mixed material; and/or by slow, continuous stirring during the application period. The mixed material shall be applied before unreasonable increases in viscosity take place.

3.2.10.2 Application

Spray guns shall be of the conventional type equipped with a fluid tip of approximately 0.09 inch in diameter and external atomization, seven-hole air cap. Material shall be supplied to the spray gun from a bottom withdrawal pot or by means of a fluid pump; hose shall be 1/2 inch in diameter. Atomization air pressure shall not be less than 80 psi. High-pressure airless spray equipment may be used only on broad, simply configured surfaces. Brush application shall be with a stiff-bristled tool heavily laden with material and wielded in a manner to spread the coating smoothly and quickly without excessive brushing. The coverage rate of the material is approximately 110 square feet/gal per coat to obtain 20 mils (dry thickness) in a two-coat system. The paint shall flow together and provide a coherent, pinhole-free film. The direction of the spray passes (or finish strokes if brushed) of the second coat shall be at right angles to those of the first where practicable.

3.2.10.3 Subsequent Coats

Except at the high temperatures discussed later in this paragraph, the drying time between coal tar-epoxy coats shall not be more than 72 hours, and application of a subsequent coat as soon as the undercoat is reasonably firm is strongly encouraged. Where the temperature for substrate or coating surfaces during application or curing exceeds or can be expected to exceed 125 degrees F as the result of direct exposure to sunlight, the surfaces shall be shaded by overhead cover or the interval between coats shall be reduced as may be found necessary to avoid poor intercoat adhesion. Here, poor intercoat adhesion is defined as the inability of two or more dried coats of coal tar-epoxy paint to resist delamination when tested aggressively with a sharp knife. Under the most extreme conditions involving high ambient temperatures and sun-exposed surfaces, the drying time between coats shall not exceed 10 hours, and the reduction of this interval to a few hours or less is strongly encouraged. Where the curing time of a coal tar-epoxy undercoat exceeds 72 hours of curing at normal temperatures, 10 hours at extreme conditions, or where the undercoat develops a heavy blush, it shall be given one of the following treatments before the subsequent coat is applied:

- a. Etch the coating surface lightly by brush-off blasting, using fine sand, low air pressure, and a nozzle-to-surface distance of approximately 3 feet.
- b. Remove the blush and/or soften the surface of the coating by wiping it with cloths dampened with 1-methyl-2-pyrrolidone. The solvents may be applied to the surface by fog spraying followed by wiping, but any puddles of solvent shall be mopped up immediately after they form. Apply the subsequent coat in not less than 15 minutes or more than 3 hours after the solvent treatment.

3.2.10.4 Ambient Temperature

Coal tar-epoxy paint shall not be applied when the receiving surface or the ambient air is below 50 degrees F nor unless it can be reasonably anticipated that the average ambient temperature will be 50 degrees F or higher for the 5-day period subsequent to the application of any coat.

3.2.10.5 Safety

In addition to the safety provisions in paragraph SAFETY, HEALTH, AND ENVIRONMENT, other workmen as well as painters shall avoid inhaling atomized particles of coal tar-epoxy paint and contact of the paint with the skin.

3.3 PAINT SYSTEMS APPLICATION

The required paint systems and the surfaces to which they shall be applied are shown in this paragraph, and/or in the drawings. Supplementary information follows.

3.3.1 Fabricated and Assembled Items

Items that have been fabricated and/or assembled into essentially their final form and that are customarily cleaned and painted in accordance with the manufacturer's standard practice will be exempted from equivalent surface preparation and painting requirements described herein, provided that:

- a. Surfaces primed (only) in accordance with such standard practices are compatible with specified field-applied finish coats.
- b. Surfaces that have been primed and finish painted in accordance with the manufacturer's standard practice are of acceptable color and are capable of being satisfactorily touched up in the field.
- c. Items expressly designated herein to be cleaned and painted in a specified manner are not coated in accordance with the manufacturer's standard practice if different from that specified herein.

3.3.2 Surface Preparation

The method of surface preparation and pretreatment shown in the tabulation of paint systems is for identification purposes only. Cleaning and pretreatment of surfaces prior to painting shall be accomplished in accordance with detailed requirements previously described.

3.3.3 System No. 6-A-Z

Apply epoxy zinc-rich primer 19C in accordance with the manufacturer's directions in two single, half-lapped spray coats to an average dry film thickness of a minimum of **3.0 mils**. The thickness at any spot must not be less than **2.5 mils** or greater **6 mils** for the primer. After a minimum drying period of 6 hours and no more than 96 hours, apply at least two coats of coal tar epoxy paint to provide a minimum thickness at any spot of **16 mils** for the completed system. Any spot having an excess of coal tar paint, here defined as more than **20 mils** in a single coat or **35 mils** in multiple coats must be repaired by sanding, grinding or abrasive blasting the excess material from the surface and reapplying the coatings to the above specified requirements. If the epoxy zinc-rich paint has been applied in the shop or otherwise has been permitted to cure for longer than 96 hours, it must be abraded and recoated with an additional thin tack coat of the zinc-rich paint, which in turn must be overcoated within 96 hours with the first coat of coal tar-epoxy paint. The specified film thicknesses must be attained in any event, and any additional coats needed to attain specified thickness must be applied at no additional cost to the Government.

3.3.4 Protection of Nonpainted Items and Cleanup

Walls, equipment, fixtures and all other items in the vicinity of the surfaces being painted shall be maintained free from damage by paint or painting activities. Paint spillage and painting activity damage shall be promptly repaired at no cost to the Government.

3.4 INSPECTION

Inspect and document all work phases and operations on a daily basis. Submit daily **Inspection Reports**. As a minimum the daily report shall contain the following:

- a. Inspections performed, including the area of the structure involved and the results of the inspection.
- b. Surface preparation operations performed, including the area of the structure involved, the mode of preparation, the kinds of solvent, abrasive, or power tools employed, and whether contract requirements

were met.

- c. Thinning operations performed, including thinners used, batch numbers, and thinner/paint volume ratios.
- d. Application operations performed, including the area of the structure involved, mode of application employed, ambient temperature, substrate temperature, dew point, relative humidity, type of paint with batch numbers, elapsed time between surface preparation and application, elapsed time for recoat, condition of underlying coat, number of coats applied, and if specified, measured dry film thickness or spreading rate of each new coating.

3.5 PAINTING SCHEDULES

SYSTEM NO. 6			
Items or surfaces to be coated:		Permanent Steel Sheet Piling	
SURFACE PREPARATION	1st COAT	2nd COAT	3rd COAT
White metal blast cleaning	Coal tar epoxy C-200a (black)	Coal tar epoxy C-200a (black)	Coal tar epoxy C-200a (black) (if needed to attain required thickness)

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 00 00

BASIC ELECTRICAL MATERIALS AND METHODS

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED REQUIREMENTS
- 1.3 DEFINITIONS
- 1.4 ELECTRICAL CHARACTERISTICS
- 1.5 ADDITIONAL SUBMITTALS INFORMATION
 - 1.5.1 Shop Drawings (SD-02)
 - 1.5.2 Product Data (SD-03)
- 1.6 QUALITY ASSURANCE
 - 1.6.1 Regulatory Requirements
 - 1.6.2 Standard Products
 - 1.6.2.1 Material and Equipment Manufacturing Date
- 1.7 WARRANTY
- 1.8 POSTED OPERATING INSTRUCTIONS
- 1.9 MANUFACTURER'S NAMEPLATE
- 1.10 FIELD FABRICATED NAMEPLATES
- 1.11 ELECTRICAL REQUIREMENTS
- 1.12 INSTRUCTION TO GOVERNMENT PERSONNEL

PART 2 PRODUCTS

- 2.1 FACTORY APPLIED FINISH

PART 3 EXECUTION

- 3.1 FIELD APPLIED PAINTING
- 3.2 FIELD FABRICATED NAMEPLATE MOUNTING

-- End of Section Table of Contents --

SECTION 26 00 00

BASIC ELECTRICAL MATERIALS AND METHODS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D709 (2017) Standard Specification for
Laminated Thermosetting Materials

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 100 (2000; Archived) The Authoritative
Dictionary of IEEE Standards Terms

IEEE C2 (2023) National Electrical Safety Code

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment
(1000 Volts Maximum)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 4 2023; ERTA 5 2023; ERTA 6
2023) National Electrical Code

1.2 RELATED REQUIREMENTS

This section applies to certain sections of Division 02, EXISTING CONDITIONS. This section applies to all sections of Division 26 and 33, ELECTRICAL and UTILITIES, of this project specification unless specified otherwise in the individual sections. This section has been incorporated into, and thus, does not apply to, and is not referenced in the following sections.

Section **26 56 00** EXTERIOR LIGHTING

1.3 DEFINITIONS

- a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, shall be as defined in **IEEE 100**.
- b. The technical sections referred to herein are those specification sections that describe products, installation procedures, and equipment operations and that refer to this section for detailed description of submittal types.

- c. The technical paragraphs referred to herein are those paragraphs in PART 2 - PRODUCTS and PART 3 - EXECUTION of the technical sections that describe products, systems, installation procedures, equipment, and test methods.

1.4 ELECTRICAL CHARACTERISTICS

Electrical characteristics for this project shall be 38 kV and 4.16 kV primary, three phase, three or four wire where indicated, 60 Hz, and 120/240 volts secondary, single phase, three wire. Final connections to the power distribution system shall be made by the Contractor as directed by the Contracting Officer and/or the respective Governing Utility, LUMA Energy, unless noted otherwise.

1.5 ADDITIONAL SUBMITTALS INFORMATION

Submittals required in other sections that refer to this section must conform to the following additional requirements as applicable.

1.5.1 Shop Drawings (SD-02)

Include wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a coordinated installation. Wiring diagrams shall identify circuit terminals and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Drawings shall indicate adequate clearance for operation, maintenance, and replacement of operating equipment devices.

1.5.2 Product Data (SD-03)

Submittal shall include performance and characteristic curves.

1.6 QUALITY ASSURANCE

1.6.1 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship shall be in accordance with the mandatory and advisory provisions of NFPA 70 unless more stringent requirements are specified or indicated.

1.6.2 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products shall have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period shall include applications of equipment and materials under similar circumstances and of similar size. The product shall have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items

shall be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in the technical section.

1.6.2.1 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site shall not be used, unless specified otherwise.

1.7 WARRANTY

The equipment items shall be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.8 POSTED OPERATING INSTRUCTIONS

Provide for each system and principal item of equipment as specified in the technical sections for use by operation and maintenance personnel. The operating instructions shall include the following:

- a. Wiring diagrams, control diagrams, and control sequence for each principal system and item of equipment.
- b. Start up, proper adjustment, operating, lubrication, and shutdown procedures.
- c. Safety precautions.
- d. The procedure in the event of equipment failure.
- e. Other items of instruction as recommended by the manufacturer of each system or item of equipment.

Print or engrave operating instructions and frame under glass or in approved laminated plastic. Post instructions where directed. For operating instructions exposed to the weather, provide weather-resistant materials or weatherproof enclosures. Operating instructions shall not fade when exposed to sunlight and shall be secured to prevent easy removal or peeling.

1.9 MANUFACTURER'S NAMEPLATE

Each item of equipment shall have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

1.10 FIELD FABRICATED NAMEPLATES

ASTM D709. Provide laminated plastic nameplates for each equipment enclosure, relay, switch, and device; as specified in the technical sections or as indicated on the drawings. Each nameplate inscription shall identify the function and, when applicable, the position. Nameplates shall be melamine plastic, **0.125 inch** thick, white with black center core. Surface shall be matte finish. Corners shall be square. Accurately align lettering and engrave into the core. Minimum size of nameplates shall be **one by 2.5 inches**. Lettering shall be a minimum of

0.25 inch high normal block style.

1.11 ELECTRICAL REQUIREMENTS

Electrical installations shall conform to IEEE C2, NFPA 70, and requirements specified herein.

1.12 INSTRUCTION TO GOVERNMENT PERSONNEL

Where specified in the technical sections, furnish the services of competent instructors to give full instruction to designated Government personnel in the adjustment, operation, and maintenance of the specified systems and equipment, including pertinent safety requirements as required. Instructors shall be thoroughly familiar with all parts of the installation and shall be trained in operating theory as well as practical operation and maintenance work. Instruction shall be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished shall be as specified in the individual section. When more than 4 man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with equipment or system. When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instructions to acquaint the operating personnel with the changes or modifications.

PART 2 PRODUCTS

2.1 FACTORY APPLIED FINISH

Electrical equipment shall have factory-applied painting systems which shall, as a minimum, meet the requirements of NEMA 250 corrosion-resistance test and the additional requirements specified in the technical sections.

PART 3 EXECUTION

3.1 FIELD APPLIED PAINTING

Paint electrical equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Corrosion resistance of field applied painting shall equal manufacturer's factory applied coating.

3.2 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 08 00

APPARATUS INSPECTION AND TESTING

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED REQUIREMENTS
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Qualifications
 - 1.4.2 Acceptance Tests and Inspections Reports
 - 1.4.3 Acceptance Test and Inspections Procedure

PART 2 PRODUCTS

PART 3 EXECUTION

- 3.1 ACCEPTANCE TESTS AND INSPECTIONS
- 3.2 SYSTEM ACCEPTANCE
- 3.3 PLACING EQUIPMENT IN SERVICE

-- End of Section Table of Contents --

SECTION 26 08 00

APPARATUS INSPECTION AND TESTING

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS (2021) Standard for Acceptance Testing
Specifications for Electrical Power
Equipment and Systems

1.2 RELATED REQUIREMENTS

Section 26 00 00 BASIC MATERIALS AND METHODS applies to this section with additions and modifications specified herein.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Acceptance Tests and Inspections; G, D0

SD-07 Certificates

Qualifications of Organization, and Lead Engineering Technician; G, D0

Acceptance Test and Inspections Procedure; G, D0

1.4 QUALITY ASSURANCE

1.4.1 Qualifications

Contractor shall engage the services of a qualified testing organization to provide inspection, testing, calibration, and adjustment of the electrical distribution system and generation equipment listed in paragraph entitled "Acceptance Tests and Inspections" herein. Organization shall be independent of the supplier, manufacturer, and installer of the equipment. The organization shall be a first tier subcontractor. No work required by this section of the specification shall be performed by a second tier subcontractor.

- a. Submit name and qualifications of organization. Organization shall have been regularly engaged in the testing of electrical materials, devices, installations, and systems for a minimum of 5 years. The organization shall have a calibration program, and test instruments used shall be calibrated in accordance with **NETA ATS**.
- b. Submit name and qualifications of the lead engineering technician performing the required testing services. Include a list of three comparable jobs performed by the technician with specific names and telephone numbers for reference. Testing, inspection, calibration, and adjustments shall be performed by an engineering technician, certified by NETA (Level III) or the National Institute for Certification in Engineering Technologies (NICET) with a minimum of 5 years' experience inspecting, testing, and calibrating electrical distribution and generation equipment, systems, and devices.

1.4.2 Acceptance Tests and Inspections Reports

Submit certified copies of inspection reports and test reports. Reports shall include certification of compliance with specified requirements, identify deficiencies, and recommend corrective action when appropriate. Type and neatly bind test reports to form a part of the final record. Submit test reports documenting the results of each test not more than 10 days after test is completed.

1.4.3 Acceptance Test and Inspections Procedure

Submit test procedure reports for each item of equipment to be field tested at least 45 days prior to planned testing date. Do not perform testing until after test procedure has been approved.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 ACCEPTANCE TESTS AND INSPECTIONS

Testing organization shall perform acceptance tests and inspections. Test methods, procedures, and test values shall be performed and evaluated in accordance with **NETA ATS**, the manufacturer's recommendations, and paragraph entitled "Field Quality Control" of each applicable specification section. Tests identified as optional in **NETA ATS** are not required unless otherwise specified. Equipment shall be placed in service only after completion of required tests and evaluation of the test results have been completed. Contractor shall supply to the testing organization complete sets of shop drawings, settings of adjustable devices, and other information necessary for an accurate test and inspection of the system prior to the performance of any final testing. Contracting Officer shall be notified at least 14 days in advance of when tests will be conducted by the testing organization. Perform acceptance tests and inspections on applicable equipment and systems specified in the following sections:

3.2 SYSTEM ACCEPTANCE

Final acceptance of the system is contingent upon satisfactory completion

of acceptance tests and inspections.

3.3 PLACING EQUIPMENT IN SERVICE

A representative of the approved testing organization shall be present when equipment tested by the organization is initially energized and placed in service.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 56 00

EXTERIOR LIGHTING

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED REQUIREMENTS
- 1.3 DEFINITIONS
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Drawing Requirements
 - 1.5.1.1 Luminaire Drawings
 - 1.5.2 Luminaire Design Data
 - 1.5.3 ANSI/IES LM-79 Test Report
 - 1.5.4 ANSI/IES LM-80 Test Report
 - 1.5.5 ANSI/IES TM-21 Test Report
 - 1.5.6 Photometric Plan
 - 1.5.7 Test Laboratories
 - 1.5.8 Regulatory Requirements
 - 1.5.9 Standard Products
 - 1.5.9.1 Material and Equipment Manufacturing Date
- 1.6 WARRANTY
 - 1.6.1 Luminaire Warranty
- 1.7 OPERATION AND MAINTENANCE MANUALS
 - 1.7.1 Lighting System
 - 1.7.2 Exterior Lighting Control System

PART 2 PRODUCTS

- 2.1 PRODUCT COORDINATION
- 2.2 LUMINAIRES
 - 2.2.1 Luminaires
- 2.3 LIGHT SOURCES
 - 2.3.1 LED Light Sources
- 2.4 LED DRIVERS
- 2.5 LIGHTING CONTROLS
 - 2.5.1 System
 - 2.5.2 Devices
 - 2.5.2.1 Photosensors
- 2.6 BRACKETS AND SUPPORTS
- 2.7 EQUIPMENT IDENTIFICATION
 - 2.7.1 Manufacturer's Nameplate
 - 2.7.2 Labels
- 2.8 FACTORY APPLIED FINISH

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Luminaires

- 3.1.2 LED Drivers
- 3.1.3 Field-Applied Painting
- 3.1.4 Grounding
- 3.2 FIELD QUALITY CONTROL
 - 3.2.1 Tests
 - 3.2.1.1 Lighting Control Verification Test
- 3.3 CLOSEOUT ACTIVITIES
 - 3.3.1 Training
 - 3.3.1.1 Maintenance Staff Training
 - 3.3.1.2 End-User Training

-- End of Section Table of Contents --

SECTION 26 56 00

EXTERIOR LIGHTING

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM B117 (2019) Standard Practice for Operating
Salt Spray (Fog) Apparatus

EUROPEAN UNION (EU)

Directive 2011/65/EU (2011) Restriction of the Use of Certain
Hazardous Substances in Electrical and
Electronic Equipment

ILLUMINATING ENGINEERING SOCIETY (IES)

ANSI/IES LM-79 (2019) Approved Method: Electrical and
Photometric Measurements of Solid State
Lighting Products

ANSI/IES LM-80 (2020) Approved Method: Measuring Luminous
Flux and Color Maintenance of LED
Packages, Arrays and Modules

ANSI/IES LS-1 (2020) Lighting Science: Nomenclature and
Definitions for Illuminating Engineering

ANSI/IES RP-8 (2018; Addenda 1 2020; Errata 1-2 2021)
Recommended Practice for Design and
Maintenance of Roadway and Parking
Facility Lighting

ANSI/IES TM-15 (2020) Technical Memorandum: Luminaire
Classification System for Outdoor
Luminaires

ANSI/IES TM-21 (2019) Technical Memorandum: Projecting
Long-Term Lumen, Photon, and Radiant Flux
Maintenance of LED Light Sources

IES Lighting Library IES Lighting Library

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 100 (2000; Archived) The Authoritative
Dictionary of IEEE Standards Terms

IEEE C62.41.2 (2002) Recommended Practice on
Characterization of Surges in Low-Voltage
(1000 V and Less) AC Power Circuits

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C136.3 (2020) Roadway and Area Lighting Equipment
- Luminaire Attachments

ANSI C136.13 (2020) Roadway and Area Lighting Equipment
- Metal Brackets for Wood Poles

ANSI C136.21 (2014) American National Standard for
Roadway and Area Lighting Equipment -
Vertical Tenons Used with Post-Top-Mounted
Luminaires

NEMA 250 (2020) Enclosures for Electrical Equipment
(1000 Volts Maximum)

NEMA ANSLG C78.377 (2017) Electric Lamps- Specifications for
the Chromaticity of Solid State Lighting
Products

NEMA C82.77-10 (2020) Harmonic Emission Limits - Related
Power Quality Requirements

NEMA C136.10 (2017) American National Standard for
Roadway and Area Lighting
Equipment-Locking-Type Photocontrol
Devices and Mating Receptacles--Physical
and Electrical Interchangeability and
Testing

NEMA C136.31 (2018) Roadway and Area Lighting Equipment
- Luminaire Vibration

NEMA IEC 60529 (2004) Degrees of Protection Provided by
Enclosures (IP Code)

NEMA SSL 1 (2016) Electronic Drivers for LED Devices,
Arrays, or Systems

NEMA SSL 3 (2011) High-Power White LED Binning for
General Illumination

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 4 2023; ERTA 5 2023; ERTA 6
2023) National Electrical Code

UNDERWRITERS LABORATORIES (UL)

UL 773 (2016; Reprint Jul 2020) UL Standard for
Safety Plug-In, Locking Type Photocontrols
for Use with Area Lighting

UL 773A (2016; Reprint Jun 2020) UL Standard for
Safety Nonindustrial Photoelectric

Switches for Lighting Control

- UL 1310 (2018) UL Standard for Safety Class 2 Power Units
- UL 1598 (2021; Reprint Jun 2021) Luminaires
- UL 8750 (2015; Reprint Sep 2021) UL Standard for Safety Light Emitting Diode (LED) Equipment for Use in Lighting Products

1.2 RELATED REQUIREMENTS

Materials not considered to be luminaires, luminaire accessories, or lighting equipment are specified in Section 33 71 01 OVERHEAD DISTRIBUTION.

1.3 DEFINITIONS

- a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications and on the drawings must be as defined in IEEE 100 and ANSI/IES LS-1.
- b. For LED luminaire light sources, "Useful Life" is the operating hours before reaching 70 percent of the initial rated lumen output (L70) with no catastrophic failures under normal operating conditions. This is also known as 70 percent "Rated Lumen Maintenance Life" as defined in ANSI/IES LM-80.
- c. For LED luminaires, "Luminaire Efficacy" (LE) is the appropriate measure of energy efficiency, measured in lumens/watt. This is gathered from LM-79 data for the luminaire, in which absolute photometry is used to measure the lumen output of the luminaire as one entity, not the source separately and then the source and housing together.
- d. Total Harmonic Distortion (THD) is the Root Mean Square (RMS) of all the harmonic components divided by the total fundamental current.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Luminaire Drawings; G, D0

SD-03 Product Data

Luminaires; G, D0

Light Sources; G, D0

LED Drivers; G, D0

Luminaire Warranty; G, D0

Photosensors; G, D0

Brackets; G, D0

SD-05 Design Data

Luminaire Design Data; G, D0

Photometric Plan; G, D0

SD-06 Test Reports

ANSI/IES LM-79 Test Report; G, D0

ANSI/IES LM-80 Test Report; G, D0

ANSI/IES TM-21 Test Report; G, D0

SD-10 Operation and Maintenance Data

Lighting System, Data Package 5; G, D0

Exterior Lighting Control System, Data Package 5; G, D0

Maintenance Staff Training Plan; G, D0

End-User Training Plan; G, D0

1.5 QUALITY ASSURANCE

Data, drawings, and reports must employ the terminology, classifications and methods prescribed by the IES Lighting Library as applicable, for the lighting system specified.

1.5.1 Drawing Requirements

1.5.1.1 Luminaire Drawings

Include dimensions, effective projected area (EPA), weight, accessories, and installation and construction details. Photometric data, including CRI, CCT, TM-15-11 BUG rating, LED driver type, aiming diagram, zonal lumen data, and candlepower distribution data per LM-79 must accompany shop drawings.

1.5.2 Luminaire Design Data

- a. Provide distribution data according to IES classification type as defined in IES Lighting Library and ANSI/IES RP-8.
- b. B.U.G. rating for the installed position as defined by ANSI/IES TM-15 and shielding as defined by ANSI/IES RP-8.
- c. Provide safety certification and file number for the luminaire family. Include listing, labeling and identification in accordance with NFPA 70 (NEC). Applicable testing bodies are determined by the

US Occupational Safety Health Administration (OSHA) as Nationally Recognized Testing Laboratories (NRTL) and include: CSA (Canadian Standards Association), ETL (Edison Testing Laboratory), and UL (Underwriters Laboratories).

- d. Provide long term lumen maintenance projections for each LED luminaire in accordance with [ANSI/IES TM-21](#). Data used for projections must be obtained from testing in accordance with [ANSI/IES LM-80](#).
- e. Provide wind loading calculations for luminaires mounted on poles. Weight and effective projected area (EPA) of luminaires and mounting brackets must not exceed maximum rating of pole as installed in particular wind zone area.

1.5.3 [ANSI/IES LM-79 Test Report](#)

Submit test report on manufacturer's standard production model of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data as outlined under "14.0 Test Report" in [ANSI/IES LM-79](#).

1.5.4 [ANSI/IES LM-80 Test Report](#)

Submit report on manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data as outlined under "8.0 Test Report" in [ANSI/IES LM-80](#).

1.5.5 [ANSI/IES TM-21 Test Report](#)

Submit test report on manufacturer's standard production LED light source (package, array or module) of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data, as well as required interpolation information as outlined under "7.0 Report" in [ANSI/IES TM-21](#).

1.5.6 [Photometric Plan](#)

For roadways include computer-generated photometric analysis of the "designed to" values in accordance with [ANSI/IES RP-8](#) for the "end of useful life" of the luminaire installation using a light loss factor of 0.81. Provide photometric plans that meet criteria in the Basis of Design in the project plans. Include the following in the submittal:

- a. Horizontal illuminance measurements at finished grade, taken at a maximum grid size of [10 feet](#) by [10 feet](#).
- b. Vertical illuminance measurements at [5 feet](#) above finished grade at all sidewalks and crosswalks, taken at a maximum of [10 feet](#).
- c. Minimum and maximum [footcandle](#) levels.
- d. Average maintained [footcandle](#) level.
- e. Maximum to minimum ratio for horizontal illuminance only.
- f. Average maintained luminance in candela per square meter.

1.5.7 Test Laboratories

Test laboratories for the ANSI/IES LM-79 and ANSI/IES LM-80 test reports must be one of the following:

- a. National Voluntary Laboratory Accreditation Program (NVLAP) accredited for solid-state lighting testing as part of the Energy-Efficient Lighting Products laboratory accreditation program.
- b. One of the qualified labs listed on the Department of Energy - Energy Efficiency & Renewable Energy, Solid-State Lighting web site.
- c. One of the EPA-Recognized Laboratories listed at for LM-80 testing.

1.5.8 Regulatory Requirements

Equipment, materials, installation, and workmanship must be in accordance with the mandatory provisions of NFPA 70 unless more stringent requirements are specified or indicated. Provide luminaires and assembled components that are approved by and bear the label of UL for the applicable location and conditions unless otherwise specified.

1.5.9 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for six months prior to bid opening. The six-month period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the six-month period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.9.1 Material and Equipment Manufacturing Date

Do not use products manufactured more than six months prior to date of delivery to site, unless specified otherwise.

1.6 WARRANTY

Support all equipment items by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.6.1 Luminaire Warranty

Provide and transfer to the government the original LED luminaire manufacturers standard commercial warranty for each different luminaire manufacturer used in the project.

- a. Provide a written five year minimum replacement warranty for material, luminaire finish, and workmanship. Provide written warranty document that contains all warranty processing information needed, including customer service point of contact, whether or not a return

authorization number is required, return shipping information, and closest return location to the luminaire location.

(1) Finish warranty must include failure and substantial deterioration such as blistering, cracking, peeling, chalking, or fading.

(2) Material warranty must include:

(a) All LED drivers and integral control equipment.

(b) Replacement when more than 15 percent of LED sources in any lightbar or subassembly(s) are defective, non-starting, or operating below 70 percent of specified lumen output.

b. Warranty period must begin in accordance with the manufacturer's standard warranty starting date.

c. Provide replacements that are promptly shipped, without charge, to the using Government facility point of contact and that are identical to or an improvement upon the original equipment. All replacements must include testing of new components and installation.

1.7 OPERATION AND MAINTENANCE MANUALS

1.7.1 Lighting System

Provide operation and maintenance manuals for the lighting system in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA that provide basic data relating to the design, operation, and maintenance of the lighting system. Include the following:

a. Manufacturers' operating and maintenance manuals.

b. Luminaire shop drawings for modified and custom luminaires.

c. Luminaire Manufacturers' standard commercial warranty information as specified in paragraph LUMINAIRE WARRANTY.

1.7.2 Exterior Lighting Control System

Provide operation and maintenance manuals for the exterior lighting control system in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA that provide basic data relating to the design, operation, and maintenance of the exterior lighting control system. Include the following:

a. Product data for all devices, including installation and programming instructions.

b. Training materials, such as videos or in-depth manuals, that cover basic operation of the lighting control system and instructions on modifying the control system. Training materials must include calibration, adjustment, troubleshooting, maintenance, repair, and replacement.

PART 2 PRODUCTS

2.1 PRODUCT COORDINATION

2.2 LUMINAIRES

UL 1598, NEMA C82.77-10. Provide luminaires as indicated in the luminaire schedule and XL plates or details on project plans, complete with light source, wattage, and lumen output indicated. All luminaires of the same type must be provided by the same manufacturer. Luminaires must be specifically designed for use with the LED driver and light source provided.

2.2.1 Luminaires

UL 8750, ANSI/IES LM-79, ANSI/IES LM-80. For all luminaires, provide:

- a. Complete system with LED drivers and light sources.
- b. Housing constructed of non-corrosive materials. All new aluminum housings must be anodized or powder-coated. All new steel housings must be treated to be corrosion resistant.
- c. ANSI/IES TM-21, ANSI/IES LM-80. Minimum L70 lumen maintenance value of 50,000 hours unless otherwise indicated in the luminaire schedule. Luminaire drive current value must be identical to that provided by test data for luminaire in question.
- d. Minimum efficacy as specified in the luminaire schedule. Theoretical models of initial lamp lumens per watt are not acceptable. If efficacy values are not listed in the luminaire schedule, provide luminaires that meet the following minimum values:

Luminaire Style	Minimum Luminaire Efficacy
Area and Roadway (pole mounted, arm mounted)	60 LPW
- e. Product rated for operation within an ambient temperature range of minus 22 degrees F to 122 degrees F.
- f. UL listed for wet locations. Optical compartment for LED luminaires must be sealed and rated a minimum of IP65 per NEMA IEC 60529.
- g. IES Lighting Library. Light distribution and NEMA field angle classifications as indicated in luminaire schedule on project plans.
- h. Housing finish that is baked-on enamel, anodized, or baked-on powder coat paint. Finish must be capable of surviving ASTM B117 salt fog environment testing for 2500 hours minimum without blistering or peeling.
- i. LED driver and light source package, array, or module are accessible for service or replacement without removal or destruction of luminaire.
- j. ANSI/IES TM-15. Does not exceed the BUG ratings as listed in the luminaire schedule.. If BUG ratings are not listed in the luminaire schedule, provide luminaires that meet the following minimum values

for each application and mounting conditions:

Lighting Application	Mounting Conditions	BUG Rating
Area and Roadway	All	B3-U0-G3

- k. Fully assembled and electrically tested prior to shipment from factory.
- l. Finish color is as indicated in the luminaire schedule or detail on the project plans.
- m. Lenses constructed of clear tempered glass or UV-resistant acrylic. Provide polycarbonate vandal-resistant lenses.
- n. All factory electrical connections are made using crimp, locking, or latching style connectors. Twist-style wire nuts are not acceptable.
- o. **NEMA C136.31**. Comply with 3G vibration testing.
- p. Luminaire arm bolts constructed from 304 stainless steel or zinc-plated steel.
- q. Wiring compartment on pole-mounted, street and area luminaires is accessible without the use of hand tools to manipulate small screws, bolts, or hardware.
- r. Incorporate modular electrical connections, and construct luminaires to allow replacement of all or any part of the optics, heat sinks, LED drivers, surge suppressors and other electrical components using only a simple tool, such as a manual or cordless electric screwdriver.
- s. **ANSI C136.3**. For all roadway and area luminaires, provide products with an integral tilt adjustment of plus or minus 5 degrees to allow the unit to be leveled.

2.3 LIGHT SOURCES

NEMA ANSLG C78.377, NEMA SSL 3. Provide type, lumen rating, and wattage as indicated in luminaire schedule on project plans.

2.3.1 LED Light Sources

Provide LED light sources that meet the following requirements:

- a. **NEMA ANSLG C78.377**. Emit white light and have a nominal Correlated Color Temperature (CCT) of 4000 Kelvin.
- b. Minimum Color Rendering Index (CRI) of 70.
- c. **Directive 2011/65/EU**. Restriction of Hazardous Substances (RoHS) compliant.
- d. Light source color consistency by utilizing a binning tolerance within a 4-step McAdam ellipse.

2.4 LED DRIVERS

NEMA SSL 1, UL 1310. Provide LED Drivers that are electronic, UL Class 1

or Class 2, constant-current type and meet the following requirements:

- a. The combined LED driver and LED light source system is greater than or equal to the minimum luminaire efficacy values as listed in the luminaire schedule provided.
- b. Operate at a voltage of 120-277 volts at 50/60 hertz, with input voltage fluctuations of plus or minus 10 percent.
- c. Power Factor (PF) greater than or equal to 0.90 at full input power and across specified dimming range.
- d. Maximum Total Harmonic Distortion (THD) less than or equal to 20 percent at full input power and across specified dimming range.
- e. Operates for at least 50,000 hours at maximum case temperature and 90 percent non-condensing relative humidity.
- f. Meets the "Elevated" (10kV/10kA) requirements per IEEE C62.41.2 -2002. Manufacturer must indicate whether failure of the electrical immunity system can possibly result in disconnect of power to luminaire. Provide surge protection that is integral to the LED driver.
- g. Contains integral thermal protection that reduces the output power to protect the driver and light source from damage if the case temperature approaches or exceeds the driver's maximum operating temperature.
- h. Complies with the requirements of the Federal Communications Commission (FCC) rules and regulations, Title 47 CFR part 15, Non-Consumer (Class A) for EMI/RFI (conducted and radiated).
- i. Class A sound rating for all drivers mounted under a covered structure, such as a canopy, or where otherwise appropriate.
- j. Directive 2011/65/EU. Restriction of Hazardous Substances (RoHS) compliant.
- k. UL listed for wet locations typical of exterior installations.
- l. Non-dimmable.
- m. Rated to operate between ambient temperatures of minus 22 degrees F and 104 degrees F 122 degrees F.

2.5 LIGHTING CONTROLS

Provide a control system interface within each luminaire that is compatible with the energy management or control system used by the utility department in charge of the project area for control of site lighting.

2.5.1 System

Provide exterior lighting control system that operates the exterior lighting system as described in the exterior lighting control strategies in the project plans.

2.5.2 Devices

2.5.2.1 Photosensors

UL 773, UL 773A. Provide Photosensors that meet the following requirements:

- a. Hermetically sealed, silicon diode light sensor type, rated at 100 watts, 240 volts, 50/60 Hz with single-pole, single-throw contacts.
- b. Turns ON at 1 to 3 footcandles and turns OFF at 3 to 15 footcandles.
- c. Designed to fail to the ON position.
- d. Housing is constructed of polycarbonate die cast aluminum, or UV stabilized polypropylene, rated to operate within a temperature range of minus 40 to 158 degrees F.
- e. Time delay that prevents accidental switching from transient light sources.
- f. Directional lens in front of the cell to prevent fixed light sources from creating a turnoff condition.
- g. Designed for 20-year service to match life expectancy of long-life LED fixtures and exceed 15,000 operations at full load. Provide photosensors with zero-cross technology to withstand severe in-rush current and extend relay life.
- i. **NEMA C136.10.** Twist-lock receptacle type. Provide with solid brass prongs and voltage markings and color coding on exterior of housing.
- j. Provide photosensors with metal oxide varistor (MOV) type surge protection.

2.6 BRACKETS AND SUPPORTS

ANSI C136.3, ANSI C136.13, and ANSI C136.21. Provide pole brackets that are not less than 1 1/4 inch galvanized steel pipe secured to pole. Slip-fitter or pipe-threaded brackets may be used, but brackets must be coordinated to luminaires provided, and brackets for use with one type of luminaire must be identical. Brackets for pole-mounted street lights must correctly position luminaire no lower than mounting height indicated. Mount brackets not less than 24 feet above street. Provide special mountings or brackets as indicated and of metal which will not promote galvanic reaction with luminaire head.

2.7 EQUIPMENT IDENTIFICATION

2.7.1 Manufacturer's Nameplate

Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.7.2 Labels

UL 1598. Luminaires must be clearly marked for operation of specific

light sources and drivers according to proper light source type. Note the following luminaire characteristics in the format "Use Only ____":

- a. Correlated color temperature (CCT) and color rendering index (CRI) for all luminaires.
- b. Driver and dimming protocol.

Markings related to light source type must be clear and located to be readily visible to service personnel, but unseen from normal viewing angles when light sources are in place. LED drivers must have clear markings indicating dimming type and indicate proper terminals for the various outputs.

2.8 FACTORY APPLIED FINISH

NEMA 250. Provide all luminaires and lighting equipment with factory-applied painting system that as a minimum meets requirements of corrosion-resistance testing.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Luminaires

Install all luminaires in accordance with the luminaire manufacturer's written instructions. Install all luminaires at locations and heights as indicated on the project plans. Level all luminaires in accordance to manufacturer's written instructions.

3.1.2 LED Drivers

Provide LED drivers integral to luminaire as constructed by the manufacturer.

3.1.3 Field-Applied Painting

Paint lighting equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Corrosion resistance of field applied painting shall match that of manufacturer's factory applied painting.

3.1.4 Grounding

Ground noncurrent-carrying parts of equipment including metal poles, luminaires, mounting arms, brackets, and metallic enclosures as specified in Section **33 71 01** OVERHEAD DISTRIBUTION. Where copper grounding conductor is connected to a metal other than copper, provide specially treated or lined connectors suitable for this purpose.

3.2 FIELD QUALITY CONTROL

3.2.1 Tests

Upon completion of installation, verify that equipment is properly installed, connected, and adjusted. Perform initial operational test, consisting of the entire system energized for 72 consecutive hours without any failures of any kind occurring in the system. All circuits must test

clear of faults, grounds, and open circuits.

3.2.1.1 Lighting Control Verification Test

Verify lighting control system and devices operate according to approved sequence of operations. Verification tests are to be completed after commissioning.

3.3 CLOSEOUT ACTIVITIES

3.3.1 Training

Provide on-site training to the Owner's personnel in the operation and maintenance of lighting and lighting control system. Provide training that includes calibration, adjustment, troubleshooting, maintenance, repair, and replacement.

3.3.1.1 Maintenance Staff Training

Submit a [Maintenance Staff Training Plan](#) at least 30 calendar days prior to training session that describes training procedures for Owner's personnel in the operation and maintenance of lighting and lighting control system. Provide on-site training which demonstrate full system functionality, assigning schedules, calibration adjustments for light levels and sensor sensitivity, integration procedures for connecting to third-party devices, and manual override including information on appropriate use. Provide protocols for troubleshooting, maintenance, repair, and replacement, and literature on available system updates and process for implementing updates.

3.3.1.2 End-User Training

Submit a [End-User Training Plan](#) at least 30 calendar days prior to training session that describes training procedures for end-users on the lighting control system. Provide demonstration for each type of user interface. Provide users with the curfew schedule as currently commissioned, including conditional programming based on astronomic time clock functionality. Provide users with the correct contact information for maintenance personnel who will be available to address any lighting control issues.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 31 - EARTHWORK

SECTION 31 00 00

EARTHWORK

PART 1 GENERAL

- 1.1 SCOPE
- 1.2 REFERENCES
- 1.3 DEFINITIONS
 - 1.3.1 Satisfactory Materials
 - 1.3.2 Unsatisfactory Materials
 - 1.3.3 Cohesionless and Cohesive Materials
 - 1.3.4 Degree of Compaction
 - 1.3.5 Topsoil
 - 1.3.6 Hard/Unyielding Materials
 - 1.3.7 Unstable Material
 - 1.3.8 Expansive Soils
- 1.4 SYSTEM DESCRIPTION
 - 1.4.1 Classification of Excavation
 - 1.4.2 Blasting
 - 1.4.3 Water Control Plan
- 1.5 SUBMITTALS
- 1.6 IN-CHANNEL EXCAVATION RESTRICTIONS
 - 1.6.1 Order of Work
 - 1.6.2 Transportation of Material
- 1.7 APPROVED DISPOSAL SITE

PART 2 PRODUCTS

- 2.1 REQUIREMENTS FOR OFFSITE SOILS
- 2.2 GENERAL FILL MATERIAL
- 2.3 GRANULAR FILL MATERIALS
 - 2.3.1 Select Fill Material
 - 2.3.2 Structural Fill Material
- 2.4 UNDERWATER PLACEMENT GRANULAR MATERIAL
- 2.5 WALL DRAINAGE MATERIAL
- 2.6 BEDDING MATERIAL
- 2.7 UTILITY BEDDING AND BACKFILL
- 2.8 BURIED WARNING AND IDENTIFICATION TAPE
- 2.9 DETECTION WIRE FOR NON-METALLIC PIPING
- 2.10 MATERIAL FOR RIP-RAP
- 2.11 PIPE CASING
 - 2.11.1 Utility Casing Pipe
 - 2.11.2 Wood Supports
- 2.12 GEOTEXTILES

PART 3 EXECUTION

- 3.1 STRIPPING OF TOPSOIL
- 3.2 GENERAL EXCAVATION

- 3.2.1 Ditches, Gutters, and Channel Changes
- 3.2.2 Drainage Structures
- 3.2.3 Drainage
- 3.2.4 Dewatering
- 3.2.5 Trench Excavation Requirements
 - 3.2.5.1 Bottom Preparation
 - 3.2.5.2 Removal of Unyielding Material
 - 3.2.5.3 Removal of Unstable Material
 - 3.2.5.4 Excavation for Appurtenances
 - 3.2.5.5 Jacking, Boring, and Tunneling
- 3.2.6 Underground Utilities
- 3.2.7 Structural Excavation
- 3.2.8 Drilled Shaft Wall Monitoring
- 3.3 IN-CHANNEL EXCAVATION PLAN
 - 3.3.1 Required Depth
 - 3.3.2 Structural Offsets
- 3.4 SELECTION OF BORROW MATERIAL
- 3.5 OPENING AND DRAINAGE OF EXCAVATION AND BORROW PITS
- 3.6 SHORING AND SHEETING
 - 3.6.1 General Requirements
 - 3.6.2 Contractor's Geotechnical Engineer
- 3.7 GRADING AREAS
- 3.8 GROUND SURFACE PREPARATION
- 3.9 UTILIZATION OF EXCAVATED MATERIALS
- 3.10 BURIED DETECTION WIRE
 - 3.10.1 Buried Detection Wire
- 3.11 BACKFILLING AND COMPACTION
 - 3.11.1 Trench Backfill
 - 3.11.1.1 Replacement of Unyielding Material
 - 3.11.1.2 Replacement of Unstable Material
 - 3.11.1.3 Utility Bedding and Initial Backfill
 - 3.11.1.4 Final Backfill
 - 3.11.1.4.1 Roadways
 - 3.11.2 Backfill for Appurtenances
- 3.12 SPECIAL REQUIREMENTS
 - 3.12.1 Gas Distribution
 - 3.12.2 Water Lines
 - 3.12.3 Electrical Distribution System
- 3.13 ROADWAY EMBANKMENTS
 - 3.13.1 Roadway Earth Embankments
- 3.14 SUBGRADE PREPARATION
 - 3.14.1 Proof Rolling
 - 3.14.2 Subgrade Preparation
 - 3.14.2.1 Roadway Subgrade
 - 3.14.3 Compaction
- 3.15 FINISHING
 - 3.15.1 Subgrade and Embankments
 - 3.15.2 Grading Around Structures
- 3.16 PLACING TOPSOIL
- 3.17 TESTING
 - 3.17.1 Fill and Backfill Material Gradation
 - 3.17.2 In-Place Density Tests
 - 3.17.3 Moisture Contents
 - 3.17.4 Optimum Moisture and Laboratory Maximum Density
 - 3.17.5 Tolerance Tests for Subgrades
 - 3.17.6 Displacement of Sewers
- 3.18 DISPOSITION OF SURPLUS MATERIAL

-- End of Section Table of Contents --

SECTION 31 00 00

EARTHWORK

PART 1 GENERAL

1.1 SCOPE

The work covered by this section consists of furnishing all labor, materials, and equipment, and performing excavation and filling activities required within the project limits and to define requirements of the various materials. Excavations covered under this specification include general excavation, structural excavations, and land-based methods of in-channel excavation, as well as, requirements for material designated for disposal within the La Chuleta upland disposal site or an approved disposal site. [Refer to the Contract drawings for additional information related to the limits of material designated for offshore versus upland disposal. For requirements of barge-based operations and offshore disposal, see specification section 35 20 23 DREDGING.](#)

The approved disposal sites are available for excavated material disposal and are being utilized by multiple projects and shall be managed by the Government. The Contractor is responsible for hauling operations including placing material in locations instructed by the Government. The Government is responsible for, but not limited to, stormwater pollution prevention, containment, moisture control operations, dust control, and grading. Contractor shall coordinate with Government to ensure efficient hauling operations.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

[AASHTO M 145](#) (1991; R 2012) Standard Specification for Classification of Soils and Soil-Aggregate Mixtures for Highway Construction Purposes

[AASHTO T 180](#) (2017) Standard Method of Test for Moisture-Density Relations of Soils Using a 4.54-kg (10-lb) Rammer and a 457-mm (18-in.) Drop

[AASHTO T 224](#) (2010) Standard Method of Test for Correction for Coarse Particles in the Soil Compaction Test

AMERICAN WATER WORKS ASSOCIATION (AWWA)

[AWWA C600](#) (2017) Installation of Ductile-Iron Mains and Their Appurtenances

AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)

AWPA P5 (2015) Standard for Waterborne
Preservatives

ASTM INTERNATIONAL (ASTM)

ASTM A139/A139M (2016) Standard Specification for
Electric-Fusion (ARC)-Welded Steel Pipe
(NPS 4 and over)

ASTM A252 (2010) Standard Specification for Welded
and Seamless Steel Pipe Piles

ASTM C33/C33M (2023) Standard Specification for Concrete
Aggregates

ASTM C136/C136M (2019) Standard Test Method for Sieve
Analysis of Fine and Coarse Aggregates

ASTM D1140 (2017) Standard Test Methods for
Determining the Amount of Material Finer
than 75- μ m (No. 200) Sieve in Soils by
Washing

ASTM D1557 (2012; E 2015) Standard Test Methods for
Laboratory Compaction Characteristics of
Soil Using Modified Effort (56,000
ft-lbf/ft³) (2700 kN-m/m³)

ASTM D2216 (2019) Standard Test Methods for
Laboratory Determination of Water
(Moisture) Content of Soil and Rock by Mass

ASTM D2434 (1968; R 2006) Permeability of Granular
Soils (Constant Head)

ASTM D2487 (2017; E 2020) Standard Practice for
Classification of Soils for Engineering
Purposes (Unified Soil Classification
System)

ASTM D2974 (2020; E 2020) Moisture, Ash, and Organic
Matter of Peat and Other Organic Soils

ASTM D4253 (2016; E 2019) Standard Test Methods for
Maximum Index Density and Unit Weight of
Soils Using a Vibratory Table

ASTM D4318 (2017; E 2018) Standard Test Methods for
Liquid Limit, Plastic Limit, and
Plasticity Index of Soils

ASTM D4972 (2018) Standard Test Methods for pH of
Soils

ASTM D5268 (2019) Topsoil Used for Landscaping
Purposes

ASTM D6913/D6913M	(2017) Standard Test Methods for Particle-Size Distribution (Gradation) of Soils Using Sieve Analysis
ASTM D6938	(2017a) Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)
U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)	
EPA 600/4-79/020	(1983) Methods for Chemical Analysis of Water and Wastes
EPA SW-846.3-3	(1999, Third Edition, Update III-A) Test Methods for Evaluating Solid Waste: Physical/Chemical Methods
COMMONWEALTH OF PUERTO RICO, DEPARTMENT OF TRANSPORTATION AND PUBLIC WORKS (PRHTA)	
PRHTA Specifications	(2005) Standard Specifications for Road and Bridge Construction

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where PRHTA Specifications Section, and/or Special Provision(s) is cited that contains references to other PRHTA sections, they shall also form a part of this specification to the extent referenced. Where reference is made to "Highway Authority" or "Authority's Representative" in PRHTA Specifications shall mean "Government" or "Contracting Officer" as defined in the Contract Documents. Where PRHTA Specifications refer to the "Engineer", "Engineer of Tests", or "Division of Tests" it shall be understood to mean the "Contracting Officer" as stated in the Contract Documents. In case of conflict between the PRHTA specifications and the Specifications in this Section that follows, the Specifications in this Section shall govern.

Paragraphs titled "Method of Measurement" and "Basis of Payment" of referenced PRHTA Specifications shall not apply.

1.3 DEFINITIONS

1.3.1 Satisfactory Materials

Satisfactory materials are onsite or imported materials that are shown to meet the requirements for fill, backfill, and bedding materials provided in Part 2 of this Specification through applicable laboratory testing.

1.3.2 Unsatisfactory Materials

Unsatisfactory materials are those which do not comply with the requirements for fill, backfill, and bedding material provided in Part 2 of this specification and the following requirements. Unsatisfactory materials also include materials that consist of or contain man-made fills; trash; refuse; backfills from previous construction; roots, organic matter, stones larger than 3 inches in any dimension, or other deleterious material. Notify the Contracting Officer when encountering any contaminated materials.

1.3.3 Cohesionless and Cohesive Materials

Cohesionless materials include materials classified in [ASTM D2487](#) as GW, GP, SW, and SP. Cohesive materials include materials classified as GC, SC, ML, CL, MH, and CH. Materials classified as GM and SM will be identified as cohesionless only when the fines are non-plastic. Perform testing, required for classifying materials, in accordance with [ASTM D4318](#), [ASTM C136/C136M](#) and [ASTM D1140](#).

1.3.4 Degree of Compaction

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum density obtained by the test procedure presented in [ASTM D1557](#), [AASHTO T 180](#), or [ASTM D4253](#) abbreviated as a percent of laboratory maximum density. [ASTM D1557](#) and [AASHTO T 180](#) are used interchangeably in this specification. The maximum density test for soils used shall be selected based on the proportion of material retained on the [3/4-inch](#) sieve. [ASTM D1557](#) shall be used for maximum density determinations if soils have 30 percent or less by weight of their particles retained on the [3/4 inch](#) sieve. If soils have more than 30 percent by weight of their particles retained on the [3/4 inch](#) sieve, the maximum density shall be determined in accordance with [AASHTO T 180](#), Method D and the reported maximum density shall be corrected in accordance with [AASHTO T 224](#). To maintain the same percentage of coarse material, use the "remove and replace" procedure as described in NOTE 8 of Paragraph 7.2 in [AASHTO T 180](#). [ASTM D4253](#) shall be used for maximum density determinations of coarse aggregates as defined by [ASTM C33/C33M](#).

1.3.5 Topsoil

Material suitable for topsoil is defined as: Natural, friable soil representative of productive, well-drained soils in the area, free of subsoil, stumps, rocks larger than [one inch](#) diameter, brush, weeds, toxic substances, and other material detrimental to plant growth. The composition of topsoil shall be in general accordance with [ASTM D5268](#). Amend topsoil pH range to obtain a pH of 5.5 to 7, tested in accordance with [ASTM D4972](#). Topsoil shall have an organic matter content between 3 and 20 percent when tested by [ASTM D2974](#). Topsoil shall have chemical attributes and nutrient content that is amenable to emergent vegetation and surface cover vegetation as specified. Topsoil shall have a maximum particle size of 1 inch in any dimension and shall have at least 15 percent but not more than 80 percent by weight passing the No 200 sieve when tested by [ASTM D1140](#).

1.3.6 Hard/Unyielding Materials

Hard/Unyielding materials comprise weathered rock, dense consolidated deposits, or conglomerate materials with stones greater than 3 [inches](#) in any dimension or as defined by the pipe manufacturer, whichever is smaller. These materials usually require the use of heavy excavation equipment, ripper teeth, or jack hammers for removal.

1.3.7 Unstable Material

Unstable materials are too wet to properly support the utility pipe, conduit, or appurtenant structure.

1.3.8 Expansive Soils

Expansive soils are defined as soils that have a plasticity index equal to or greater than 30 when tested in accordance with [ASTM D4318](#).

1.4 SYSTEM DESCRIPTION

Boring logs provided in Section [00 31 32](#) GEOTECHNICAL DATA REPORT. Variations may exist in the subsurface between boring locations.

1.4.1 Classification of Excavation

Finish the specified excavation on an unclassified basis.

1.4.2 Blasting

Blasting will not be permitted.

1.4.3 [Water Control Plan](#)

Submit procedures for accomplishing dewatering work and managing surface water. The water control plan shall be prepared by a professional engineer licensed in the Commonwealth of Puerto Rico.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section [01 33 00](#) SUBMITTAL PROCEDURES:

[SD-01 Preconstruction Submittals](#)

[Shoring and Sheet piling; G, D0](#)

[Water Control Plan](#)

Water control plan shall address dewatering and surface water management. Dewatering work plan shall include, but not be limited to: Identifying dewatering areas and dewatering activities, equipment list, narrative discussing means and methods for controlling surface and groundwater along with detailed calculations for anticipated water quantity and pump capacity, anticipated permits, measuring and monitoring plan for groundwater levels surrounding excavations, and anticipated treatment requirements. The plan should also provide calculations that demonstrate that the temporary excavation bottoms and sides are stable against groundwater pressures.

[In-Channel Excavation Plan; G, D0](#)

[Geotechnical Engineer Qualifications; G, D0](#)

[Borrow Site; G, D0](#)

[Disposal Site; G, D0](#)

[SD-03 Product Data](#)

Utilization of Excavated Materials; G, D0
Opening of any Excavation or Borrow Pit

SD-06 Test Reports

Testing

Borrow Site Testing

Within 24 hours of conclusion of physical tests, submit electronic copies of test results.

SD-07 Certificates

Testing

Laboratory Qualifications; G, D0

1.6 IN-CHANNEL EXCAVATION RESTRICTIONS

1.6.1 Order of Work

Final depth excavation of the channel shall be completed from downstream to upstream (North to South) within the project limits, except for material removed for establishment of the required Bypass Channel as shown on the drawings. If necessary, for coordination purposes with other contracts or project schedule requirements, exceptions may be approved by the Contracting Officer.

1.6.2 Transportation of Material

Water and excavated material shall not be permitted to overflow or spill onto haul routes during transport to the disposal site. Contractor shall supply watertight hauling equipment and truck washdown racks at the project site. Cleanup of spilled material on the haul route shall be done by the Contractor at no cost to the Government. Any spills shall be promptly cleaned up and comply with all Federal, Commonwealth, and local environmental regulations.

1.7 APPROVED DISPOSAL SITE

The La Chuleta disposal site is approved for disposal of excavated soil material only and shall be managed by the Government. The Government may designate additional disposal sites within a 2 mile radius from the Rio Puerto Nuevo Contract 3 Project Site which shall be considered as Approved Disposal Sites. Construction waste, demolition debris, and clear and grubbing debris shall not be disposed of within La Chuleta. The Contractor is responsible for hauling operations including placing material in locations instructed by the Government. The Government is responsible for, but not limited to, stormwater pollution prevention, containment, moisture control operations, dust control, and grading. Contractor shall coordinate with Government to ensure efficient hauling operations.

The Contractor may utilize other disposal sites for excavated soil material and the Contractor shall be responsible for ensuring compliance with all Federal, Commonwealth, and local regulations. All operations including, but not limited to, hauling, placing, grading, moisture control, stormwater pollution prevention, containment, and dust control

shall be the responsibility of the Contractor. Provide location, route, and provisions for meeting Federal, Commonwealth, and local regulations prior to the start of excavation operations.

PART 2 PRODUCTS

2.1 REQUIREMENTS FOR OFFSITE SOILS

Soil materials imported from offsite shall be free of environmental contaminants and must meet federal, state, and/or local environmental requirements for imported fill material. Tests shall be performed to document compliance with applicable environmental regulations before they are brought to the site.

At a minimum, offsite soils for use as fill shall be tested for Total Petroleum Hydrocarbons (TPH); Benzene, Toluene, Ethyl Benzene, and Xylene (BTEX). Fill shall contain a maximum of 100 parts per million (ppm) of total petroleum hydrocarbons (TPH) and a maximum of 10 ppm of the sum of Benzene, Toluene, Ethyl Benzene, and Xylene (BTEX), or shall meet applicable federal, state, and/or local requirements, whichever are more stringent. Determine TPH concentrations by using EPA 600/4-79/020 Method 418.1. Determine BTEX concentrations by using EPA SW-846.3-3 Method 5030/8020. Provide Borrow Site Testing for TPH and, BTEX from a composite sample of material from the borrow site, with at least one test from each borrow site. Do not bring material onsite until test results have been approved by the Contracting Officer.

Contractor shall provide documentation of physical properties of soils from the borrow source including ASTM D6913/D6913M and ASTM D1140 grain size, ASTM D4318 Atterberg limits, ASTM D2487 USCS classification, ASTM D2216 moisture content, ASTM D2974 organic content, and ASTM D1557 Proctor results.

Imported offsite soils shall meet the requirements in paragraph SATISFACTORY MATERIALS.

2.2 GENERAL FILL MATERIAL

General Fill may be used as fill or backfill in areas to landscaped and shall not be used for structural backfill or support. General fill must be thoroughly mixed and free from debris, clods, seeds of nuisance or exotic species, or stones of a size larger than that specified in PARAGRAPH "Unsatisfactory Materials". In areas where general fill is placed at the ground surface where grass, sod, or landscaping material will be placed, general fill material shall not have stones larger than 1.5 inches in any dimension within the upper 1 foot of the general fill. General fill shall have an organic content of less than 5% by weight.

2.3 GRANULAR FILL MATERIALS

2.3.1 Select Fill Material

Select granular material consists of materials classified as GW, GP, SW, SP, SP-SM, and SP-SC by ASTM D2487 where indicated. The organic content shall be 2 percent or less when tested in accordance with ASTM D2974. Not more than 10 percent by weight may be finer than No. 200 sieve when tested in accordance with ASTM D1140. Provide a minimum coefficient of permeability of 0.002 feet per minute when tested in accordance with ASTM D2434.

2.3.2 Structural Fill Material

ASTM D2487, classification GW, GP, GM, GC, GP-GM, GP-GC, SW, SP, SM, SC, SW-SM, SW-SC, or SP-SC with a maximum ASTM D4318 liquid limit of 35, a maximum ASTM D4318 plasticity index of 10, a maximum of 10 percent by weight passing ASTM D1140 No. 200 sieve, and a maximum of 2 percent when tested in accordance with ASTM D2974 organic content.

2.4 UNDERWATER PLACEMENT GRANULAR MATERIAL

Underwater placement granular material consists of materials classified as GW, GP, SW, and SP by ASTM D2487 where indicated. Material shall have a maximum particle size of 3 inches and consist of clean granular material that is free of debris and other deleterious material. Material shall have a maximum organic content of 2 percent when tested in accordance with ASTM D2974. Not more than 5 percent by weight may be finer than No. 200 sieve when tested in accordance with ASTM D1140.

2.5 WALL DRAINAGE MATERIAL

Wall drainage material consists of granular material meeting the requirements of PRHTA Specifications Section 209 - Sheathing, Part 209-2.01(a).

2.6 BEDDING MATERIAL

Bedding material for applications other than utility bedding shall consist of a granular mixture of well-graded sand and crushed gravel or aggregate, whose average size shall not exceed 0.5 inches and its gradation spans from 0.75 inches to Sieve No. 4. Compose material of tough, durable particles. Not more than 10 percent by weight may be finer than No. 200 sieve when tested in accordance with ASTM D1140. Fines passing the No. 200 standard sieve shall have a plasticity index less than six. (AASHTO A-1-a based on AASHTO M 145)

2.7 UTILITY BEDDING AND BACKFILL

Provide bedding for buried piping in accordance with AWWA C600, Type 4, except as specified herein or as indicated on the drawings. Utility bedding shall consist of crushed stone material in accordance with ASTM C33/C33M Gradation No. 67. Provide materials as follows:

- a. Bedding and Haunching: Bedding and haunching shall meet material requirements for bedding material.
- b. Initial Backfill: Initial backfill shall meet material requirements for bedding material.
- c. Final Backfill: Final backfill shall meet material requirements for structural fill.

2.8 BURIED WARNING AND IDENTIFICATION TAPE

Buried warning and identification tape requirements are provided in Section 33 11 00 WATER UTILITY DISTRIBUTION PIPING.

2.9 DETECTION WIRE FOR NON-METALLIC PIPING

Insulate a single strand, solid copper detection wire with a minimum of 12 AWG.

2.10 MATERIAL FOR RIP-RAP

Rip-rap requirements are provided in Section 35 31 19 STONE, CHANNEL, SHORELINE/COASTAL PROTECTION FOR STRUCTURES.

2.11 PIPE CASING

2.11.1 Utility Casing Pipe

ASTM A139/A139M, Grade B, or ASTM A252, Grade 2, smooth wall pipe. Match casing size to the outside diameter and wall thickness as indicated on the drawings. Protective coating is not required on casing pipe.

2.11.2 Wood Supports

Locally available, rough, structural grade. Provide wood with nonleaching water-borne pressure preservative (ACA or CCA) and treatment conforming to AWPA P5. Secure wood supports to carrier pipe with stainless steel or zinc-coated steel bands.

2.12 GEOTEXTILES

Geotextile requirements are provided in Section 31 05 19 GEOTEXTILES.

PART 3 EXECUTION

3.1 STRIPPING OF TOPSOIL

Where indicated or directed, strip topsoil to a depth of the root zone or 4 inches, whichever is greater. Spread topsoil on areas already graded and prepared for topsoil, or transport and deposit in stockpiles convenient to areas that are to receive application of the topsoil later, or at locations indicated or specified. Keep topsoil separate from other excavated materials, brush, litter, objectionable weeds, roots, stones larger than 2 inches in diameter, and other materials that would interfere with planting and maintenance operations. Remove from the site any surplus of topsoil from excavations and gradings.

3.2 GENERAL EXCAVATION

Perform excavation of every type of material encountered within the limits of the project to the lines, grades, and elevations indicated and as specified. Perform the grading in accordance with the typical sections shown and the tolerances specified in paragraph FINISHING. Transport satisfactory excavated materials that meet requirements for structural fill and place in fill or embankment within the limits of the work. Excavate unsatisfactory materials encountered within the limits of the work below grade and replace with structural fill. Include such excavated material and the structural fill ordered as replacement in excavation. Dispose surplus and unsatisfactory excavated material in accordance with paragraph UTILIZATION OF EXCAVATED MATERIALS and local and Commonwealth regulations. During construction, perform EARTHWORK in a manner and sequence that will provide proper drainage at all times. Excavate material required for fill or embankment in excess of that produced by

excavation within the grading limits from the borrow areas indicated or from other approved areas selected by the Contractor as specified.

All excavations shall comply with Occupational Safety and Health Administration excavation, slopes and benching, and trenching and shoring regulations.

3.2.1 Ditches, Gutters, and Channel Changes

Finish excavation of ditches, gutters, and channel changes by cutting accurately to the cross sections, grades, and elevations shown on the drawings. Do not excavate ditches and gutters below grades shown. Backfill the excessive open ditch or gutter excavation with satisfactory, thoroughly compacted, material or with suitable stone or cobble to grades shown at the Contractor's expense. Dispose excavated material as shown or as approved, except in no case allow material to be deposited within 4 feet from edge of a ditch. Maintain excavations free from detrimental quantities of leaves, brush, sticks, trash, and other debris until final acceptance of the work.

3.2.2 Drainage Structures

Make excavations to the lines, grades, and elevations shown. Provide trenches and foundation pits of sufficient size to permit the placement and removal of forms for the full length and width of structure footings and foundations as shown. Clean rock or other hard foundation material of loose debris and cut to a firm, level, stepped, or serrated surface. Remove loose disintegrated rock and thin strata. Do not disturb the bottom of the excavation when concrete or masonry is to be placed in an excavated area. Any disturbed areas shall be repaired with bedding material compacted to a minimum 95 percent density as determined by AASHTO T 180 or ASTM D1557 as indicated in Section 1.2.4 of this specification at a moisture content within the range of -3 to +3 percent of optimum moisture content. Do not excavate to the final grade level until just before the concrete or masonry is to be placed. Repair of disturbed areas is incidental to installation of drainage structures and shall be performed by the Contractor at the Contractor's expense.

3.2.3 Drainage

Provide for the collection and disposal of surface and subsurface water encountered during construction. Completely drain construction site during periods of construction to keep soil materials sufficiently dry. Construct storm drainage features at the earliest stages of site development, and throughout construction grade the construction area to provide positive surface water runoff away from the construction activity, or provide temporary ditches, swales, and other drainage features and equipment as required to maintain dry soils. When unsuitable working platforms for equipment operation and unsuitable soil support for subsequent construction features develop, remove unsuitable material and replace with compacted structural fill material. It is the responsibility of the Contractor to assess the soil and groundwater conditions presented by the plans and specifications and to employ necessary measures to permit construction to proceed.

3.2.4 Dewatering

Dewatering for dry excavation is prohibited within the channel banks. The language herein applies to structural excavations and utility trenches

outside of the channel limits.

Control groundwater flowing toward or into excavations to prevent caving of excavation slopes and walls, boils, uplift and heave in the excavation and to eliminate interference with orderly progress of construction. Do not permit French drains, sumps, ditches, or trenches within 3 feet of the foundation of any structure, except with specific written approval, and after specific contractual provisions for restoration of the foundation area have been made. Take control measures by the time the excavation reaches the water level in order to maintain the integrity of the in-situ material. While the excavation is open, maintain the water level continuously, at least 2 feet below the working level. Operate dewatering system continuously until construction work below existing water levels is complete. Submit performance records weekly. Measure and record performance of dewatering system at same time each day by use of observation wells or piezometers installed in conjunction with the dewatering system. Relieve hydrostatic head in pervious zones below subgrade elevation in layered soils to prevent uplift.

Dewatering systems shall be designed by a registered Geotechnical Engineer in the Commonwealth of Puerto Rico with at least 10 years of experience related to dewatering systems. The dewatering system may include sumps, pumps, or well points capable of meeting the project dewatering requirements.

3.2.5 Trench Excavation Requirements

Excavate the trench as recommended by the manufacturer of the pipe to be installed. Slope trench walls below the top of the pipe, or make vertical, and of such width as recommended in the manufacturer's printed installation manual. Provide vertical trench walls in the pipe zone where no manufacturer's printed installation manual is available. Shore trench walls more than 4 feet high, cut back to a stable slope, or provide with equivalent means of protection for employees who may be exposed to moving ground or cave in. Shore vertical trench walls more than 4 feet high, or excavate trench walls to the slopes as recommended by OSHA or the angle of repose, whichever results in a flatter slope. Give special attention to slopes which may be adversely affected by weather or moisture content. For pipes with outside diameter (O.D.) of 15 inches or smaller, the trench width shall be 24 inches greater than the pipe O.D. For pipes with outside diameter (O.D.) of 16 inches or larger, the trench width shall be 36 inches greater than the pipe O.D. Where recommended trench widths are exceeded, Contractor shall provide a redesign of the pipe segment that is certified by an engineer licensed in Puerto Rico that includes stronger pipe or special installation procedures to achieve the design ground surface loading with the same safety factor as the original design. The Contractor is responsible for the cost of redesign, stronger pipe, or special installation procedures without any additional cost to the Government.

3.2.5.1 Bottom Preparation

Grade the bottoms of trenches accurately to provide uniform bearing and support for the bottom quadrant of each section of the pipe. Excavate bell holes to the necessary size at each joint or coupling to eliminate point bearing. Remove stones of 3 inch or greater in any dimension, or as recommended by the pipe manufacturer, whichever is smaller, to avoid point bearing.

3.2.5.2 Removal of Unyielding Material

Where unyielding material is encountered in the bottom of the trench, remove such material 6 inches below the required grade and replace with select granular material as provided in paragraph BACKFILLING AND COMPACTION

3.2.5.3 Removal of Unstable Material

Where unstable material is encountered in the bottom of the trench, remove such material to the depth directed and replace it to the proper grade with select granular material as provided in paragraph BACKFILLING AND COMPACTION.

3.2.5.4 Excavation for Appurtenances

Provide excavation for manholes, catch-basins, inlets, or similar structures sufficient to leave at least 12 inches clear or adequate space to allow for compaction equipment and testing between the outer structure surfaces and the face of the excavation or support members. Clean loose debris and cut to a firm surface either level, stepped, or serrated, as shown. Remove loose disintegrated rock and thin strata. Remove and replace unstable material as required to comply with contract requirements. When concrete or masonry is to be placed in an excavated area, take special care not to disturb the bottom of the excavation. Do not excavate to the final grade level until just before the concrete or masonry is to be placed.

3.2.5.5 Jacking, Boring, and Tunneling

Unless otherwise indicated, provide excavation by open cut. Contractor may propose installing pipes by jacking, boring, or tunneling if work can be conducted safely and pipe and backfill can be properly installed and compacted, subject to the approval of the Contracting Officer.

3.2.6 Underground Utilities

The Contractor is responsible for movement of construction machinery and equipment over pipes and utilities during construction. Perform work adjacent to non-Government utilities as indicated in accordance with procedures outlined by utility company. Excavation made with power-driven equipment is not permitted within 2 feet of known utility or subsurface construction. For work immediately adjacent to or for excavations exposing a utility or other buried obstruction, excavate by hand. Start hand excavation on each side of the indicated obstruction and continue until the obstruction is uncovered or until clearance for the new grade is assured. Support uncovered lines or other existing work affected by the contract excavation until approval for backfill is granted by the Contracting Officer. Report damage to utility lines or subsurface construction immediately to the Contracting Officer.

3.2.7 Structural Excavation

Ensure that foundation subgrades have been inspected and approved by the Contracting Officer prior to concrete placement.

3.2.8 Drilled Shaft Wall Monitoring

Excavation and material compaction shall be performed in a manner to avoid

damage to the drilled shafts walls. Drilled shafts walls shall be monitored in accordance with Section 31 63 29 DRILLED CONCRETE PIERS AND SHAFTS. Prevent damage to drilled shafts caused by any excavation and backfill near drilled shaft walls. Repair any damage to drilled shafts at the Contractor's expense as specified in Section 31 63 29 DRILLED CONCRETE PIERS AND SHAFTS.

3.3 IN-CHANNEL EXCAVATION PLAN

Provide a description of equipment and methods utilized in the removal of the material within the channel banks. The plan shall include provisions for maintaining stable slopes, grade control methods, hauling equipment, method for transporting wet materials, truck washing, and methods for excavating near structures as describe in the paragraph STRUCTURAL OFFSETS below. Refer to specification section 35 20 23 DREDGING for requirements of hydraulic or barge-based mechanical dredging if method is elected for use by the Contractor.

3.3.1 Required Depth

The designated areas to be excavated, to final channel invert as shown on the Contract Drawings, will be estimated and paid for in accordance with the provisions contained in the subparagraphs "Measurement" and "Payment" of Section 01 22 00 MEASUREMENT AND PAYMENT.

3.3.2 Structural Offsets

Structural elements shall be protected from damage during adjacent material removal. Damage to structural elements during material removal shall be repaired or replaced at the discretion of the Contracting Officer at no additional cost to the Government. Provide within the in-channel excavation plan provisions to excavate within 5 feet of structural elements that includes a description of equipment and methods to be used. Use equipment and methods for excavation and removal of material will minimize the potential for damage to structures. Damaged caused to existing structures and drilled shaft concrete shall be repaired at no expense to the Government. Turbidity and monitoring in Section 01 57 25 TURBIDITY AND DISPOSAL MONITORING requirements shall be met regardless of method utilized.

3.4 SELECTION OF BORROW MATERIAL

Select borrow material to meet the requirements and conditions of the particular fill or embankment for which it is to be used. Obtain borrow material from the required excavation areas within the limits of the project site, selected by the Contractor, or from approved off-site sources identified by the Contractor. Unless otherwise provided in the contract, the Contractor is responsible for obtaining the right to procure material, pay royalties and other charges involved, and bear the expense of developing the sources, including rights-of-way for hauling from the owners. Borrow material from approved sources on Government-controlled land may be obtained without payment of royalties. Unless specifically provided, do not obtain borrow within the limits of the project site without prior written approval. Contractor shall be responsible for necessary clearing, grubbing, and satisfactory drainage of borrow pits and the disposal of debris related to the borrow excavation operations.

3.5 OPENING AND DRAINAGE OF EXCAVATION AND BORROW PITS

Notify the Contracting Officer 7 days in advance of the opening of any excavation or borrow pit or borrow areas to permit elevations and measurements of the undisturbed ground surface to be taken. Provide documentation of material types and physical properties within the source, including grain size by ASTM D6913/D6913M, Atterberg limits by ASTM D4318, USCS classification by ASTM D2487, moisture content by ASTM D2216, organic content by ASTM D2974, and Proctor results by ASTM D1557. Provide documentation that material is free from contaminants and other deleterious matter. Except as otherwise permitted, excavate borrow pits and other excavation areas by providing adequate drainage at all times. Transport overburden and other spoil material to designated spoil areas or otherwise dispose of as approved. Provide neatly trimmed and drained borrow pits after the excavation is completed. Ensure that excavation of any area, operation of borrow pits, or dumping of spoil material results in minimum detrimental effects on natural environmental conditions.

3.6 SHORING AND SHEETING

3.6.1 General Requirements

Submit a Shoring and Sheet piling plan for approval prior to starting work in accordance with PARAGRAPH 1.4 of this specification. Submit drawings and calculations, certified by an experienced professional engineer licensed in Puerto Rico, describing the methods for shoring and sheet piling of excavations.

- a. Prevent undermining of pavements, foundations and slabs.
- b. Prevent slippage or movement in banks or slopes adjacent to the excavation.
- c. Allow for the abandonment of shoring and sheet piling materials in place in critical areas as the work is completed. In these areas, backfill the excavation to within 3 feet of the finished grade and remove the remaining exposed portion of the shoring before completing the backfill.

Finish shoring, including sheet piling, and install as necessary to protect workmen, banks, adjacent paving, structures, and utilities. Remove shoring, bracing, and sheet piling as excavations are backfilled in a manner to prevent deformation or caving.

3.6.2 Contractor's Geotechnical Engineer

Hire a Professional Geotechnical Engineer licensed in the Commonwealth of Puerto Rico to provide inspection of excavations and soil/groundwater conditions throughout construction. The Geotechnical Engineer shall have at least 10 years of professional experience performing similar work and is responsible for performing pre-construction and periodic site visits throughout construction to assess site conditions. The Geotechnical Engineer is responsible for updating the excavation, sheet piling, and dewatering plans as construction progresses to reflect changing conditions and submit an updated plan if necessary. Submit a monthly written report, informing the Contractor and Contracting Officer of the status of the plan and an accounting of the Contractor's adherence to the plan addressing any present or potential problems. The Contractor shall make the Geotechnical Engineer available to meet with the Contracting Officer upon the Contracting Officer's request throughout the contract duration.

3.7 GRADING AREAS

Where indicated, divide work into grading areas within which satisfactory excavated material will be placed in embankments, fills, and required backfills. Do not haul satisfactory material excavated in one grading area to another grading area except when so directed in writing. Place and grade stockpiles of satisfactory materials as specified. Keep stockpiles in a neat and well drained condition giving due consideration to drainage at all times. Clear, grub, and seal by rubber-tired equipment, the ground surface at stockpile locations. Protect stockpiles of satisfactory materials from contamination which may destroy the quality and fitness of the stockpiled material. If the Contractor fails to protect the stockpiles, and any material becomes unsatisfactory, remove and replace such material with satisfactory material from approved sources at no additional cost to the Government.

3.8 GROUND SURFACE PREPARATION

The ground surface shall be prepared in accordance with the subgrade preparation requirements in paragraph SUBGRADE PREPARATION.

3.9 UTILIZATION OF EXCAVATED MATERIALS

Dispose surplus and unsatisfactory excavated material offsite in accordance with all local and Commonwealth regulations. Dispose debris removed from areas to be dredged or excavated off-site at an approved landfill. Dry material obtained above the mean higher high water mark (approximately El. +1.0) shall be designated for disposal at the La Chuleta site or an approved disposal site. Excavated material obtained below the mean higher high water mark will have high water content and shall be disposed at the designated Ocean Dredged Material Disposal Site (ODMDS). See specification section 35 20 23 DREDGING for additional requirements related to disposal at the ODMDS.

Use satisfactory material removed from excavations that meets material specifications in the construction of fills, embankments, subgrades, bedding, backfill, and for similar purposes. Submit procedure for disposal of unused satisfactory material. Do not waste any satisfactory excavated material without specific written authorization. Do not dispose excavated material to obstruct the flow of any stream, endanger a partly finished structure, impair the efficiency or appearance of any structure, or be detrimental to the completed work in any way.

3.10 BURIED DETECTION WIRE

3.10.1 Buried Detection Wire

Bury detection wire directly above non-metallic piping at a distance not to exceed 12 inches above the top of pipe. Extend the wire continuously and unbroken, from manhole to manhole. Terminate the ends of the wire inside the manholes at each end of the pipe, with a minimum of 3 feet of wire, coiled, remaining accessible in each manhole. Furnish insulated wire over its entire length. Install wires at manholes between the top of the corbel and the frame and extend up through the chimney seal between the frame and the chimney seal. For force mains, terminate the wire in the valve pit at the pump station end of the pipe.

3.11 BACKFILLING AND COMPACTION

Except where otherwise directed in this Section, place backfill adjacent to any and all types of structures in successive horizontal layers of loose material not more than 8 inches in depth. Restrict maximum lift thickness to 6 inches in narrow placement areas or adjacent to structures where manually operated compaction equipment is used such as tamper, jumping jacks, walk-behind rollers, and other similar equipment. Compact to at least 95 percent laboratory maximum density per AASHTO T 180 or ASTM D1557 as indicated in paragraph DEGREE OF COMPACTION of this specification. Prevent wedging action or eccentric loading upon or against the structure. Backfill material must be within the range of -3 to +3 percent of optimum moisture content at the time of compaction.

Heavy equipment for spreading and compacting fill material shall not be operated closer than 5 feet from drilled shaft walls, 15 feet from sheet pile walls, or a distance equal to the retained soil height for other features. Compaction close to these structures shall be performed using manually-operated small compaction equipment such as tampers, jumping jacks, or walk-behind rollers, or similar equipment.

No embankment backfill will be allowed above jet grouted locations until after 28 days of curing time or test results indicate the jet grout has achieved a compressive strength of 100 psi at 28 days.

Prepare ground surface on which backfill is to be placed and compact backfill materials in conformance with the applicable portions of paragraphs GROUND SURFACE PREPARATION. Finish compaction by sheepfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment as required to achieve the specified compaction.

In areas that require fill below the water level within adjacent channels, underwater placement granular material shall be placed in 18-inch thick lifts and spread and compacted using the equipment placing the fill such as the arm of a backhoe. Underwater fill shall continue until reaching 2 feet above the water level to obtain a bridge lift. Any additional fill to be placed above the bridge lift will be placed and compacted to at least 95 percent laboratory maximum density per AASHTO T 180 or ASTM D1557 as indicated in Section 1.2.4 of this specification at a moisture content within the range of -3 to +3 percent of optimum moisture content..

3.11.1 Trench Backfill

Backfill trenches to the grades provided in specification and on drawings. Backfill the trench to 2 feet above the top of pipe prior to performing the required pressure tests. Leave the joints and couplings uncovered during the pressure test. Do not backfill the trench until all specified tests are performed.

3.11.1.1 Replacement of Unyielding Material

Replace unyielding material removed from the bottom of the trench with select granular material or initial backfill material.

3.11.1.2 Replacement of Unstable Material

Replace unstable material removed from the bottom of the trench or excavation with select granular material placed in layers not exceeding 8

inches loose thickness and compacted to 95 percent of AASHTO T 180 or ASTM D1557 maximum density as indicated in Section 1.2.4 of this specification at a moisture content within the range of -3 to +3 percent of optimum moisture content.

3.11.1.3 Utility Bedding and Initial Backfill

Provide bedding of the type and thickness shown. Place initial backfill material in maximum 6-inch loose lifts and compact lifts with approved light-weight manually operated compaction equipment, such as tampers and jumping jacks to a height of at least one foot above the utility pipe or conduit. Bring up the backfill evenly on both sides of the pipe for the full length of the pipe. Take care to ensure thorough compaction of the fill under the haunches of the pipe. Compact utility bedding and initial backfill to 90 percent of ASTM D4253 maximum density as indicated in paragraph DEGREE OF COMPACTION of this specification.

3.11.1.4 Final Backfill

Fill the remainder of the trench, except for special materials for roadways with final backfill material. Place final backfill material in maximum 8 inch loose lifts and compact. Compact to at least 95 percent laboratory maximum density AASHTO T 180 or ASTM D1557 as indicated in paragraph DEGREE OF COMPACTION of this specification. Prevent wedging action or eccentric loading upon or against structures. Backfill material must be within the range of -3 to +3 percent of optimum moisture content at the time of compaction.

3.11.1.4.1 Roadways

Place final backfill up to the required elevation as specified. Do not permit water flooding or jetting methods of compaction.

3.11.2 Backfill for Appurtenances

After the manhole, catch basin, inlet, or similar structure has been constructed and the concrete has been allowed to cure for 14 days, place backfill in such a manner that the structure is not be damaged by falling earth. Deposit the backfill material, compact it as specified for final backfill, and bring up the backfill evenly on all sides of the structure to prevent eccentric loading and excessive stress.

3.12 SPECIAL REQUIREMENTS

Special requirements for both excavation and backfill relating to the specific utilities are as follows:

3.12.1 Gas Distribution

Excavate trenches to a depth that will provide a minimum 18 inches of cover in rock excavation and a minimum 24 inch of cover in other excavation.

3.12.2 Water Lines

Excavate trenches to a depth that provides a minimum cover of 2 feet from the existing ground surface, or from the indicated finished grade, whichever is lower, to the top of the pipe.

3.12.3 Electrical Distribution System

Provide a minimum cover of 24 inches from the finished grade to direct burial cable and conduit or duct line, unless otherwise indicated.

3.13 ROADWAY EMBANKMENTS

3.13.1 Roadway Earth Embankments

Construct roadway earth embankments from structural fill material free of organic material and rocks with any dimension greater than 3 inches. Place the material in successive horizontal layers of loose material not more than 8 inches in depth. Spread each layer uniformly on a soil surface that has been moistened or aerated as necessary, and scarified to a depth of not less than 2 inches or otherwise broken up so that the fill will bond with the surface on which it is placed. After spreading, plow, disk, or otherwise break up each layer; moisten or aerate as necessary; thoroughly mix; and compact to at least 95 percent laboratory maximum density from AASHTO T 180 or ASTM D1557 as indicated in Section 1.2.4 of this specification. Backfill and fill material must be within the range of -3 to +3 percent of optimum moisture content at the time of compaction.

Compaction requirements for the upper portion of earth embankments forming subgrade for pavements are identical with those requirements specified in paragraph SUBGRADE PREPARATION. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment as required to achieve the specified compaction.

No embankment backfill will be allowed above jet grouted locations until after 28 days of curing time or test results indicate the jet grout has achieved a compressive strength of 100 psi at 28 days.

3.14 SUBGRADE PREPARATION

3.14.1 Proof Rolling

Proof rolling shall be done on an exposed subgrade free of surface water (wet conditions resulting from rainfall and runoff) which would promote degradation of an otherwise acceptable subgrade. After stripping, proof roll the existing subgrade with six passes of a dump truck loaded with 212 cubic feet of soil or 15-ton, pneumatic-tired roller. Operate the roller or truck in a systematic manner to ensure the number of passes over all areas, and at speeds between 2 1/2 to 3 1/2 miles per hour. Notify the Contracting Officer a minimum of 3 days prior to proof rolling. Proof rolling shall be performed in the presence of the Contracting Officer. Rutting or pumping of material shall be undercut and replaced with satisfactory fill material compacted as specified.

3.14.2 Subgrade Preparation

Shape subgrade to line, grade, and cross section, and compact as specified. Include plowing, disking, and any moistening or aerating required to obtain specified compaction for this operation. Remove soft or otherwise unsatisfactory material and replace with satisfactory fill material or other approved material as directed.

Excavate rock encountered in the cut section to a depth of 6 inches below finished grade for the subgrade. Bring up low areas resulting from

removal of unsatisfactory material or excavation of unyielding material to required grade with structural fill material, and shape the entire subgrade to line, grade, and cross section.

Scarify and recompact the subgrade of areas to receive embankments, structures, or roadways to a depth of 6-inches to at least 95 percent of laboratory maximum density per AASHTO T 180 or ASTM D1557 as indicated in paragraph DEGREE OF COMPACTION of this specification at a moisture content within the range of -3 to +3 percent of optimum moisture content. Roughen the subgrade surface prior to placing fill. Plow, step, bench, or break up sloped surfaces steeper than 1 vertical to 4 horizontal so that the fill material will bond with the existing subgrade material.

3.14.2.1 Roadway Subgrade

After rolling, the surface of the subgrade for roadways shall not show deviations greater than 1/2 inch when tested with a 12-foot straightedge applied both parallel and at right angles to the centerline of the area. Do not vary the elevation of the finish subgrade more than 0.05 foot from the established grade and cross section.

3.14.3 Compaction

Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment. Compact each layer of the embankment to at least 95 percent of laboratory maximum density per AASHTO T 180 or ASTM D1557 as indicated in Section 1.2.4 of this specification at a moisture content within the range of -3 to +3 percent of optimum moisture content.

3.15 FINISHING

Finish the surface of excavations, embankments, and subgrades to a smooth and compact surface in accordance with the lines, grades, and cross sections or elevations shown. Provide the degree of finish for graded areas within 0.1 foot of the grades and elevations indicated for that the degree of finish for subgrades specified in paragraph SUBGRADE PREPARATION. Finish gutters and ditches in a manner that will result in effective drainage. Repair graded, topsoiled, or backfilled areas prior to acceptance of the work, and re-establish grades to the required elevations and slopes.

3.15.1 Subgrade and Embankments

During construction, keep embankments and excavations shaped and drained. Maintain ditches and drains along subgrade to drain effectively at all times. Do not disturb the finished subgrade by traffic or other operation. Protect and maintain the finished subgrade in a satisfactory condition until ballast, subbase, base, or pavement is placed.

3.15.2 Grading Around Structures

Construct areas within 5 feet outside of each building and structure line true-to-grade, shape to drain, and maintain free of trash and debris until final inspection has been completed and the work has been accepted.

3.16 PLACING TOPSOIL

On areas to receive topsoil, prepare the compacted subgrade soil to a 2

inches depth for bonding of topsoil with subsoil. Spread topsoil evenly to a thickness of 4 inches and grade to the elevations and slopes shown. Do not spread topsoil when frozen or excessively wet or dry. Obtain material required for topsoil in excess of that produced by excavation within the grading limits from offsite areas from an approved source at the Contractor's expense.

3.17 TESTING

Perform testing by a USACE validated commercial testing laboratory or the Contractor's testing facility validated by USACE. Submit **laboratory qualifications** of the USACE validated commercial testing laboratory or the Contractor's validated testing facilities. If the Contractor elects to establish testing facilities, do not permit work requiring testing until the Contractor's facilities have been inspected, Corps validated and approved by the Contracting Officer.

- a. Determine field in-place density in accordance with **ASTM D6938**.
- b. Check the calibration curves furnished with the moisture gauges along with density calibration checks as described in **ASTM D6938**; perform verification checks in accordance with **ASTM D6938** Annex A1 and A2 at the beginning of a job and on each different type of material encountered. If test results indicate that the instrument does not meet the requirements of **ASTM D6938** and has been used to determine acceptance of in-place material, re-test areas since last calibration or verification and replace and recompact to meet specification requirements.
- c. Perform tests on recompacted areas to determine conformance with specification requirements. A licensed professional geotechnical engineer shall certify inspections and test results. These certifications shall state that the tests and observations were performed by or under the direct supervision of the engineer and that the results are representative of the materials or conditions being certified by the tests. The following number of tests, if performed at the appropriate time, will be the minimum acceptable for each type operation.

3.17.1 Fill and Backfill Material Gradation

One test per 10,000 cubic yards stockpiled or in-place source material. Determine gradation of fill and backfill material in accordance with **ASTM C136/C136M**.

3.17.2 In-Place Density Tests

- a. One test per 5,000 square feet, or fraction thereof, of each lift of fill or backfill areas compacted by other than hand-operated machines.
- b. One test per 1,000 square feet, or fraction thereof, of each lift of fill or backfill areas compacted by hand-operated machines.
- c. One test per 50 linear feet, or fraction thereof, of each lift of embankment or backfill for trench backfill.

3.17.3 Moisture Contents

In the stockpile, excavation, or borrow areas, perform a minimum of two moisture content tests per day per type of material or source of material being placed during stable weather conditions. During unstable weather, perform tests as dictated by local conditions and approved by the Contracting Officer. Moisture content shall be tested in accordance with [ASTM D2216](#).

3.17.4 Optimum Moisture and Laboratory Maximum Density

Perform tests in accordance with [AASHTO T 180](#) or [ASTM D1557](#) as indicated in Section 1.2.4 of this specification for each type of material or source of material [including borrow material](#) to determine the optimum moisture and laboratory maximum density values. One representative test per 25,000 cubic [yards](#) of each type fill and backfill, or when any change in material occurs which may affect the optimum moisture content or laboratory maximum density.

3.17.5 Tolerance Tests for Subgrades

Perform continuous checks on the degree of finish specified in paragraph SUBGRADE PREPARATION during construction of the subgrades.

3.17.6 Displacement of Sewers

After other required tests have been performed and the trench backfill compacted to the finished grade surface, inspect the pipe to determine whether significant displacement has occurred. Conduct this inspection in the presence of the Contracting Officer. Inspect pipe sizes larger than [36 inches](#) and document with photographs or video. Inspect pipes smaller than [36 inches](#) in diameter by shining a light or laser between manholes or manhole locations or by the use of television cameras passed through the pipe. If the interior of the pipe shows poor alignment or any other defects that would cause improper functioning or poor performance of the system, replace or repair the defects at no additional cost to the Government.

3.18 DISPOSITION OF SURPLUS MATERIAL

Remove surplus material or other soil material not required or suitable for filling or backfilling, and brush, refuse, stumps, roots, and timber from Government property to an approved location at the Contractor's expense.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 31 - EARTHWORK

SECTION 31 05 19

GEOTEXTILE

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 REFERENCES
- 1.3 SUBMITTALS
- 1.4 DELIVERY, STORAGE, AND HANDLING
 - 1.4.1 Delivery
 - 1.4.2 Storage
 - 1.4.3 Handling
 - 1.4.4 Warranty

PART 2 PRODUCTS

- 2.1 RAW MATERIALS
 - 2.1.1 Geotextile
 - 2.1.2 Thread
- 2.2 MANUFACTURING QUALITY CONTROL SAMPLING AND TESTING

PART 3 EXECUTION

- 3.1 QUALITY ASSURANCE SAMPLES AND TESTS
 - 3.1.1 Quality Assurance Samples
 - 3.1.2 Quality Assurance Tests
- 3.2 INSTALLATION
 - 3.2.1 Subgrade Preparation
 - 3.2.2 Placement
- 3.3 SEAMS
 - 3.3.1 Overlap Seams
 - 3.3.2 Sewn Seams
- 3.4 PROTECTION
- 3.5 REPAIRS
- 3.6 PENETRATIONS
- 3.7 COVERING

-- End of Section Table of Contents --

SECTION 31 05 19

GEOTEXTILE

PART 1 GENERAL

1.1 SUMMARY

Non-woven geotextile as filter material under riprap, under gabion baskets, under aggregate road surfacing, behind wall drainage material, and at stormwater pipe/headwall joints.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D1557	(2012; E 2015) Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft ³) (2700 kN-m/m ³)
ASTM D3786/D3786M	(2018) Standard Test Method for Bursting Strength of Textile Fabrics-Diaphragm Bursting Strength Tester Method
ASTM D4354	(2012) Sampling of Geosynthetics for Testing
ASTM D4355/D4355M	(2014) Deterioration of Geotextiles from Exposure to Light, Moisture and Heat in a Xenon-Arc Type Apparatus
ASTM D4491/D4491M	(2017) Standard Test Methods for Water Permeability of Geotextiles by Permittivity
ASTM D4533	(2011) Trapezoid Tearing Strength of Geotextiles
ASTM D4632/D4632M	(2015a) Grab Breaking Load and Elongation of Geotextiles
ASTM D4751	(2020) Standard Test Method for Determining Apparent Opening Size of a Geotextile
ASTM D4759	(2011; R 2018) Standard Practice for Determining the Specification Conformance of Geosynthetics
ASTM D4833/D4833M	(2007; R 2020) Standard Test Method for Index Puncture Resistance of Geomembranes

and Related Products

ASTM D4873/D4873M

(2017) Standard Guide for Identification, Storage, and Handling of Geosynthetic Rolls and Samples

ASTM D5261

(2010; R 2018) Standard Test Method for Measuring Mass Per Unit Area of Geotextiles

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Layout; G, D0, AE

Geotextile layout plan with proposed size, number, position, and sequencing of geotextile rolls and direction of all field seams.

Seams; G, D0, AE

Contractor shall submit details of the proposed sewing method for field seams of each type of geotextile (type of stitch, number of stitches per inch, number of stitching rows, distance from the edge of the geotextile to the stitches, and weight and type of thread) and a minimum 3 feet long sample of the sewn seam for each type of geotextile for approval by the Contracting Officer.

SD-03 Product Data

Thread; G, D0, AE

Submit proposed thread type for sewn seams along with data sheets showing the physical properties of the thread.

Manufacturing Quality Control Sampling and Testing; G, D0, AE

SD-04 Samples

Quality Assurance Samples

SD-06 Test Reports

Quality Assurance Tests; G, D0, AE

SD-07 Certificates

Geotextile; G, D0, AE

Manufacturer's documentation that raw materials and roll materials comply with required geotextile physical properties. For needle punched geotextile, the Manufacturer shall certify that the geotextile has been continuously inspected using permanent on-line full-width metal detectors and does not contain any needles which could damage other geosynthetic layers.

1.4 DELIVERY, STORAGE, AND HANDLING

Deliver, store, and handle geotextile in accordance with [ASTM D4873/D4873M](#).

1.4.1 Delivery

Notify the Contracting Officer a minimum of 24 hours prior to delivery and unloading of geotextile rolls, which shall be packaged in opaque, waterproof, protective plastic wrapping. The plastic wrapping shall not be removed until deployment. If quality assurance samples are collected, immediately rewrap rolls with the plastic wrapping. Geotextile or plastic wrapping damaged during storage or handling shall be repaired or replaced. Label each roll with the manufacturer's name, geotextile type, roll number, roll dimensions (length, width, gross weight), and date manufactured.

1.4.2 Storage

Protect rolls of geotextile from construction equipment, chemicals, sparks and flames, temperatures in excess of [160 degrees F](#), or any other environmental condition that may damage the physical properties of the geotextile. To protect geotextile from becoming saturated, either elevate rolls off the ground or place them on a sacrificial sheet of plastic in an area where water will not accumulate.

1.4.3 Handling

Handle and unload geotextile rolls with load carrying straps, a forklift with a stinger bar, or an axial bar assembly. Rolls shall not be dragged along the ground, lifted by one end, or dropped to the ground.

1.4.4 Warranty

The Manufacturer shall warrant the EQUIPMENT, MATERIALS and PRODUCTS specified in this section against defective materials and workmanship with the Manufacturer's standard warranty, but for no less than two years from the date of Substantial Completion. If the Manufacturer's standard warranty is less than the stipulated period, the Manufacturer shall provide a special Manufacturer's extended warranty for the stipulated period, or a Maintenance Bond in the form attached herein, to extend the Manufacturer's warranty period for the stipulated period.

The Contractor shall warranty the work against defects for one year from the date of Substantial Completion.

PART 2 PRODUCTS

2.1 RAW MATERIALS

Submit manufacturer's certificate of compliance stating that the geotextile meets the requirements of this section in accordance with the schedule in PARAGRAPH 1.3 of this specification. For needle punched geotextiles, the manufacturer shall also certify that the geotextile has been continuously inspected using permanent on-line full-width metal detectors and does not contain any needles inadvertently embedded in the geotextile during the manufacturing process. The certificate of compliance shall be attested to by a person having legal authority to bind the geotextile manufacturer.

2.1.1.1 Geotextile

Provide geotextile that is a nonwoven pervious sheet of polymeric material consisting of long-chain synthetic polymers composed of at least 95 percent by weight polyolefins, polyesters, or polyamides. The use of woven slit film geotextiles (i.e., geotextiles made from yarns of a flat, tape-like character) will not be allowed. Add stabilizers and/or inhibitors to the base polymer, as needed, to make the filaments resistant to deterioration by ultraviolet light, oxidation, heat exposure, acid and alkaline conditions, oils, microorganisms and insects. Regrind material, which consists of edge trimmings and other scraps that have never reached the consumer, may be used to produce the geotextile. Post-consumer recycled material shall not be used. Class 1 geotextile for use beneath aggregate road surfacing, behind wall drainage material, and at stormwater pipe/headwall joints shall be formed into a network such that the filaments or yarns retain dimensional stability relative to each other, including the edges. Geotextiles shall meet the requirements specified in Table 1. Class A geotextiles for use beneath riprap and under gabion baskets shall meet the requirements specified in Table 2. Where applicable, Table 1 and Table 2 property values represent minimum average roll values (MARV) in the weakest principal direction. Values for apparent opening size (AOS) represent maximum average roll values. Slit film or slit tape fabrics are not permitted.

During shipment and storage, the rolls of fabric shall be protected from exposure to the sun, heat, dirt, and other detrimental conditions. Except for fabrics designed for slope stabilization and soil erosion control work, the geotextiles shall not be exposed to ultraviolet radiation for more than seven days.

The following definitions apply to geotextiles:

- a. Machine Direction - the long (or warp) direction of the geotextile. The cross-machine (or fill) direction is perpendicular to the machine direction.
- b. Mean Roll Values - The mean roll value of any specific geotextile property is the average of the test results from any roll within a lot.
- c. Nonwoven Geotextile - A textile produced by bonding or interlocking of fibers, or both, accomplished by mechanical, heat, or chemical means.
- d. Seam Allowance - The minimum distance from the edge of a geotextile to the stitch line nearest to that edge.
- e. Seam Type - A designation relating to the essential characteristics of geotextile positioning and rows of stitching in a specified sewn seam.
- f. Selvage - The finished edge of a geotextile parallel to the machine direction.
- g. Stitch type - A designation relating to the essential characteristics of the interlacing of sewing thread(s) in a specified seam.

h. Woven Geotextile - A textile comprising two or more sets of filaments or yarns interlaced in such a way that they result in a uniform pattern.

TABLE 1 REQUIREMENTS FOR CLASS 1 GEOTEXTILE FOR USE BENEATH AGGREGATE ROAD SURFACING, BEHIND WALL DRAINAGE MATERIAL, AND AT STORMWATER PIPE/HEADWALL JOINTS (1)				
PROPERTY	UNITS	ACCEPTABLE VALUES (2)	N/A	TEST METHOD
WEIGHT	OZ/SY	8		ASTM D5261
GRAB STRENGTH	LBS	200	N/A	ASTM D4632/D4632M
ELONGATION AT BREAK	%	>50	N/A	ASTM D4632/D4632M
SEAM STRENGTH (3)	LBS	180	N/A	ASTM D4632/D4632M
PUNCTURE STRENGTH CBR PUNCTURE STRENGTH (4)	LBS	80 430	N/A	ASTM D4833/D4833M ASTM D6241
BURST STRENGTH	PSI	250	N/A	ASTM D3786/D3786M
TRAPEZOID TEAR	LBS	80	N/A	ASTM D4533
MAXIMUM APPARENT OPENING SIZE (AOS) - US STD SIEVE	-	#70		ASTM D4751
MINIMUM PERMITTIVITY	1/SEC	0.1		ASTM D4491/D4491M
ULTRAVIOLET DEGRADATION	PERCENT	70		ASTM D4355/D4355M

(1) Minimum - Use value in weaker principal direction. All numerical values represent minimum average roll value (i.e., test results from any sampled roll in a lot shall meet or exceed the minimum values in the Table). Stated values are for non-critical, non-severe applications. Lots sampled according to ASTM D4354.

(2) Class A Drainage applications for fabrics are where installation stresses are more severe, i.e., very coarse sharp angular aggregate is used, a heavy degree of compaction (90% ASTM D1557 or greater) is specified or depth of trench is greater than 10 feet.

(3) Values apply to both field and manufactured seams.

(4) CBR puncture strength may be used to show compliance with this specification if puncture strength is unavailable from the manufacturer.

TABLE 2 REQUIREMENTS FOR CLASS A GEOTEXTILE FOR USE BENEATH RIPRAP, BENEATH ACCESS ROADS, BENEATH ACB REVETMENTS, WITHIN MARINE MATTRESSES, AND UNDER GABION BASKETS (1)			
PROPERTY	UNITS	ACCEPTABLE VALUES (2)	TEST METHODS
WEIGHT	OZ/SY	12	ASTM D5261
GRAB STRENGTH	LBS	300	ASTM D4632/D4632M
ELONGATION AT BREAK	%	>50	ASTM D4632/D4632M
SEAM STRENGTH (3)	LBS	180	ASTM D4632/D4632M
PUNCTURE STRENGTH CBR PUNCTURE STRENGTH (4)	LBS	180 800	ASTM D4833/D4833M ASTM D6241
BURST STRENGTH	PSI	320	ASTM D3786/D3786M
TRAPEZOID TEAR	LBS	115	ASTM D4533
MAXIMUM APPARENT OPENING SIZE (AOS) - US STD SIEVE	-	#70	ASTM D4751
MINIMUM PERMITTIVITY	1/SEC	0.1	ASTM D4491/D4491M
ULTRAVIOLET DEGRADATION	PERCENT	70	ASTM D4355/D4355M

(1) Minimum - Use value in weaker principal direction. All numerical values represent minimum average roll value (i.e., test results from any sampled roll in a lot shall meet or exceed the minimum values in the table). Stated values are for non-critical, non-severe conditions. Lot sampled according to ASTM D4354.

(2) Stone placement height shall be less than 3 feet and stone weights shall not exceed 250 pounds for the specified geotextile material.

(3) Values apply to both field and manufactured seams.

(4) CBR puncture strength may be used to show compliance with this specification if puncture strength is unavailable from the manufacturer.

2.1.2 Thread

Submit proposed thread type for sewn seams along with data sheets showing the physical properties of the thread in accordance with Section 1.3 of this specification. Construct sewn seams with high-strength polymeric thread, consisting of polyolefins, polyester, or polyamides, and shall be as resistant to deterioration and ultraviolet light as the geotextile being sewn. Nylon threads will not be allowed. Thread color shall contrast with the geotextile.

2.2 MANUFACTURING QUALITY CONTROL SAMPLING AND TESTING

The Manufacturer is responsible for establishing and maintaining a quality control program to assure compliance with the requirements of the specification. Submit manufacturer's quality control manual minimum of 30 days prior to scheduled use. Documentation describing the quality control program shall be made available upon request. Perform manufacturing quality control sampling and testing in accordance with the manufacturer's approved quality control manual. Results shall include or bracket the rolls delivered for use in the work. Submit all manufacturer's quality control conformance test results. As a minimum, geotextiles shall be randomly sampled by the manufacturer for testing in accordance with [ASTM D4354](#), Procedure A. Acceptance of geotextile shall be in accordance with [ASTM D4759](#). Tests not meeting the specified requirements will result in the rejection of applicable rolls.

Sampling, testing, certification, and acceptance shall conform to the following requirements:

- a. Acceptance of geotextile material shall be based on [ASTM D4759](#).
- b. Tests shall be performed by the manufacturer in accordance with ASTM methods to determine geotextile properties specified herein for the intended application(s).
- c. The Contractor shall furnish the geotextile manufacturer's certified test results attesting that the geotextile and all factory seams meet the requirements stated in these specifications for the intended application. The certification shall state that all the rolls being furnished of the named product conform to the Authority's requirements.
 1. The certification shall give the name and address of the manufacturer and the testing agency and the date of tests; and shall set forth the means of identification which will permit field determination of the product delivered to the project as being the product covered by the certification.
 2. The certification shall be in duplicate and shall be sent with the shipment of the covered product to the Engineer.

PART 3 EXECUTION

3.1 QUALITY ASSURANCE SAMPLES AND TESTS

3.1.1 Quality Assurance Samples

Provide assistance to the Contracting Officer in the collection of quality

assurance samples for quality assurance testing; the testing schedule will be in accordance with Section 1.3 of this specification. Collect samples upon delivery to the site in accordance with [ASTM D4354](#), Procedure B. Lot size for quality assurance sampling shall be the shipment quantity of the product or a truckload of the product, whichever is smaller. The unit size shall be considered one roll of geotextile. Quality assurance sampling shall be performed at a frequency of at least one per [100,000 square feet](#) of geotextile. Identify samples with a waterproof marker by manufacturer's name, product identification, lot number, roll number, and machine direction. The date and a unique sample number shall also be noted on the sample. Discard the outer layer of the geotextile roll prior to sampling a roll. Samples shall then be collected by cutting the full-width of the geotextile sheet a minimum of [3 feet](#) long in the machine direction. Rolls which are sampled shall be immediately resealed in their protective covering.

3.1.2 Quality Assurance Tests

Provide quality assurance samples to an Independent Laboratory. Samples will be tested to verify that geotextile meets the requirements specified in Table 1 and Table 2. Test method [ASTM D4355/D4355M](#) shall not be performed on the collected samples. Geotextile product acceptance shall be based on [ASTM D4759](#). Tests not meeting the specified requirements will result in the rejection of applicable rolls. Testing is incidental to geotextile installation.

3.2 INSTALLATION

Install geotextile in accordance with manufacturer's written recommendations and the [layout](#) plan. Hand place geotextile when above water. No equipment will be permitted to traffic in direct contact with the geotextile. Lay geotextile smooth, free of tensile stresses, folds, and wrinkles. Use adequate ballast (e.g., sandbags) to prevent uplift by wind. Do not leave geotextile uncovered for more than 7 days.

3.2.1 Subgrade Preparation

The surface underlying the geotextile shall be smooth and free of ruts or protrusions which could damage the geotextile. Subgrade materials and compaction requirements shall be in accordance with Section [31 00 00](#) Earthwork.

3.2.2 Placement

Notify the Contracting Officer a minimum of 24 hours prior to installation of geotextile. Geotextile rolls which are damaged or contain imperfections shall be repaired or replaced at no cost to the Government. The geotextile shall be laid flat and smooth so that it is in direct contact with the subgrade. The geotextile shall also be free of tensile stresses, folds, and wrinkles. On slopes steeper than 1 vertical on 10 horizontal, lay the geotextile with the machine direction of the fabric parallel to the slope direction. Overlaps shall be placed so that upstream sheets are placed over downstream or downslope sheets.

For geotextile placement under gabion baskets filled with riprap, geotextile shall be connected to the bottom of the gabion baskets prior to riprap placement. Connections between gabion baskets and geotextile shall follow manufacturer's recommendations.

3.3 SEAMS

3.3.1 Overlap Seams

Continuously overlap geotextile panels a minimum of **12 inches** at all longitudinal and transverse joints. Where seams must be oriented across the slope, lap the upper panel over the lower panel. If approved, sewn seams may be used instead of overlapped seams.

Install geotextile vertically from the top of slopes to the bottom with the upstream geotextile panel overlapping the downstream panel by at least 12 inches (minimum). For geotextile placed beneath turf reinforcement mat, overlap as shown on the Drawings. On horizontal seams the upper panel will overlap the lower panel by 12 inches (minimum).

3.3.2 Sewn Seams

Factory and field seams shall be continuously sewn on all slopes steeper than 1 vertical on 4 horizontal. The stitch type used shall be a 401 locking chain stitch or as recommended by the manufacturer. For field and factory seams which are sewn, provide at least a 6-foot sample of sewn seam before the geotextile is installed. For seams that are field sewn, the seams shall be sewn using the same equipment and procedures as will be used for the production seams. If seams are sewn in both the machine and cross machine direction, provide samples of seams from both directions. Seam strength shall meet the minimum requirements specified in Table 1 and Table 2. The thread at the end of each seam run shall be tied off to prevent unraveling. Skipped stitches or discontinuities shall be sewn with an extra line of stitching with a minimum of **18 inches** of overlap.

3.4 PROTECTION

Protect the geotextile during installation from clogging, tears, and other damage. Damaged geotextile shall be repaired or replaced.

3.5 REPAIRS

Repair torn or damaged geotextile. Clogged areas of geotextile shall be removed and replaced. Perform repairs by placing a patch of the same type of geotextile over the damaged area. The patch shall extend a minimum of **12 inches** beyond the edge of the damaged area. Patches shall be continuously fastened using approved methods. The machine direction of the patch shall be aligned with the machine direction of the geotextile being repaired. Remove and replace geotextile rolls which cannot be repaired. Repairs shall be performed at no additional cost to the Government.

3.6 PENETRATIONS

Construct engineered penetrations of the geotextile by methods recommended by the geotextile manufacturer.

3.7 COVERING

Do not cover geotextile prior to inspection and approval by the Contracting Officer. Place cover soil in a manner that prevents soil from entering the geotextile overlap zone, prevents tensile stress from being mobilized in the geotextile, and prevents wrinkles from folding over onto

themselves. On side slopes, soil backfill shall be placed from the bottom of the slope upward. Cover soil shall not be dropped onto the geotextile from a height greater than 3 feet. No equipment shall be operated directly on top of the geotextile without approval of the Contracting Officer. Use equipment with ground pressures less than 7 psi to place the first lift over the geotextile. A minimum of 12 inches of soil shall be maintained between full-scale construction equipment and the geotextile. Cover soil material type, compaction, and testing requirements are described in Section 31 00 00 EARTHWORK. Equipment placing cover soil shall not stop abruptly, make sharp turns, spin their wheels, or travel at speeds exceeding 5 mph.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 31 - EARTHWORK

SECTION 31 11 00

CLEARING AND GRUBBING

PART 1 GENERAL

1.1 SUBMITTALS

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Tree Wound Paint

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Protection

3.1.1.1 Roads and Walks

3.1.1.2 Monuments and Markers

3.1.1.3 Trees, Shrubs, and Existing Facilities

3.1.1.4 Utility and Stormwater Lines

3.2 CLEARING

3.2.1 Tree Removal

3.2.2 Pruning

3.2.3 Grubbing

3.3 DISPOSAL OF MATERIALS

3.3.1 Saleable Timber

3.4 Nonsaleable Materials

-- End of Section Table of Contents --

SECTION 31 11 00
CLEARING AND GRUBBING

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Tree Wound Paint

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Tree Wound Paint

Use bituminous based paint from standard manufacture specially formulated for tree wounds.

Deliver materials to the site, and handle in a manner which will maintain the materials in their original manufactured or fabricated condition until ready for use.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Protection

3.1.1.1 Roads and Walks

Keep roads and walks free of dirt and debris at all times.

3.1.1.2 Monuments and Markers

No land monuments, property markers, or official datum points shall be damaged or removed until the Contracting Officer has witnessed or otherwise referenced their location and approved their removal.

3.1.1.3 Trees, Shrubs, and Existing Facilities

Provide protection in accordance with Section 01 57 20 ENVIRONMENTAL PROTECTION. Protect trees and vegetation to be left standing from damage incident to clearing, grubbing, and construction operations by the erection of barriers or by such other means as the circumstances require.

3.1.1.4 Utility and Stormwater Lines

Protect existing utility and stormwater lines that are indicated to remain from damage. Notify the Contracting Officer immediately of damage to or an encounter with an unknown existing utility or stormwater line. The Contractor is responsible for the repair of damage to existing utility or stormwater lines that are indicated or made known to the Contractor prior to start of clearing and grubbing operations. When utility or stormwater lines which are to be removed are encountered within the area of operations, notify the Contracting Officer in ample time to minimize interruption of the service. Refer to Section 01 30 00 ADMINISTRATIVE PROCEDURES.

3.2 CLEARING

Clearing consists of the felling, trimming, and cutting of trees into sections and the satisfactory disposal of the trees and other vegetation designated for removal, including downed timber, snags, brush, and rubbish occurring within the areas to be cleared. Clearing also includes the removal and disposal of structures that obtrude, encroach upon, or otherwise obstruct the work. Cut off flush with or below the original ground surface trees, stumps, roots, brush, and other vegetation in areas to be cleared, except such trees and vegetation as may be indicated or directed to be left standing. Trim dead branches 1-1/2 inches or more in diameter on trees designated to be left standing within the cleared areas and trim all branches to the heights indicated or specified. Neatly cut close to the bole of the tree or main branches, limbs and branches to be trimmed. Paint, with an approved tree-wound paint, cuts more than 1-1/2 inches in diameter. Apply herbicide in accordance with the manufacturer's label to the top surface of stumps designated not to be removed.

3.2.1 Tree Removal

Where indicated or directed, trees and stumps that are designated as trees shall be removed from areas outside those areas designated for clearing and grubbing. This work includes the felling of such trees and the removal of their stumps and roots as specified in paragraph GRUBBING. Dispose of trees as specified in paragraph DISPOSAL OF MATERIALS.

3.2.2 Pruning

Trim trees designated to be left standing within the cleared areas of dead branches 1-1/2 inches or more in diameter; and trim branches to heights and in a manner as indicated. Neatly cut limbs and branches to be trimmed close to the bole of the tree or main branches. Paint cuts more than 1-1/4 inches in diameter with an approved tree wound paint.

3.2.3 Grubbing

Grubbing consists of the removal and disposal of stumps, roots larger than 3 inches in diameter, and matted roots from the designated grubbing areas. Remove material to be grubbed, together with logs and other organic or metallic debris not suitable for foundation purposes, to a depth of not less than 18 inches below the original surface level of the ground in areas indicated to be grubbed and in areas indicated as construction areas under this contract, such as areas for buildings, and areas to be paved. Fill depressions made by grubbing with suitable material and compact to make the surface conform with the original adjacent surface of the ground.

3.3 DISPOSAL OF MATERIALS

Dispose of excess materials in accordance with the approved solid waste management permit and include those materials in the solid waste management report.

All wood or wood like materials, except for salable timber, remaining from clearing, pruning or grubbing such as limbs, tree tops, roots, stumps, logs, rotten wood, and other similar materials shall become the property of the Contractor and disposed of as specified. All non-saleable timber and wood or wood like materials remaining from timber harvesting such as limbs, tree tops, roots, stumps, logs, rotten wood, and other similar materials shall become the property of the Contractor and disposed as specified.

3.3.1 Saleable Timber

All timber on the project site from which sawn logs, posts, ties, pulpwood, or fuelwood can be produced, shall become the property of the Contractor and in the interest of conservation it is required that the Contractor make a reasonable effort to dispose of material for these purposes. The Contractor may cut timber into convenient lengths at the site, but approval of the Contracting Officer must be secured prior to the operation of saw mills within the site.

3.4 Nonsaleable Materials

Logs, stumps, roots, brush, rotten wood, and other refuse from the clearing and grubbing operations, except for salable timber, shall be disposed of outside the limits of Government-controlled land at the Contractor's responsibility. In no case shall material be thrown into or left in the channels or on site.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 31 - EARTHWORK

SECTION 31 23 19

UNWATERING

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SCOPE
- 1.3 DEFINITIONS
- 1.4 UNWATERING REQUIREMENTS
- 1.5 PIEZOMETERS
- 1.6 RELIEF WELLS
- 1.7 SUBMITTALS
- 1.8 GENERAL CRITERIA
 - 1.8.1 Unwatering Design Package
- 1.9 QUALITY CONTROL
 - 1.9.1 General
 - 1.9.2 Reporting
- 1.10 INITIAL TESTING

PART 2 PRODUCTS

PART 3 EXECUTION

- 3.1 WATER LEVEL PRIOR TO OPERATION
- 3.2 OPERATION
- 3.3 MAINTENANCE AND SERVICING
- 3.4 REMOVAL

-- End of Section Table of Contents --

SECTION 31 23 19

UNWATERING

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 1110-2-2100 (2005) Stability Analysis of Concrete Structures

1.2 SCOPE

The work provided for herein consists of furnishing all plant, equipment, labor and materials; performing all operations required for designing, furnishing, installing, and operating a system to unwater the area within the sanitary siphon temporary retaining structure (TRS) or coffer cell; maintaining these areas free from standing water during boring and jacking and other construction operations; emergency flooding; rewatering the area under controlled conditions at the termination of the unwatering and removing the system. This work shall be performed concurrently with Section 31 52 10 TEMPORARY RETAINING STRUCTURES.

1.3 DEFINITIONS

- (1) Unwatering system defines the machinery, equipment, and appurtenances necessary for and related to the accomplishment of unwatering, and the collection and disposal of all water within the TRS area.
- (2) Flooding of the TRS is defined as the controlled process of filling the coffer cell with water to a specified elevation and at a specified rate.
- (3) Unwatering is defined as the process of removing all water within the TRS by means of sumps and pumps.
- (4) Rewatering is defined as the controlled process of placing water in the completed structure and/or TRS to its naturally occurring elevation, at a specified rate, when the construction is completed, and the unwatering system is no longer required.
- (5) Uplift (Flotation) Factor of Safety as defined in Section 3-8 of EM 1110-2-2100

1.4 UNWATERING REQUIREMENTS

The unwatering system shall be of a type and capacity to accomplish all requirements specified herein.

(1) The unwatering system design shall be based on the geotechnical properties in Section 00 31 32 Geotechnical Data Report. The borings in the vicinity of the TRS are over 100 feet from the planned locations and may not be representative of conditions encountered during construction. The Contractor may elect to perform additional borings at the TRS locations. The system shall be designed so that it will: 1) remove any rainwater which may enter the cofferdam (rain intensity up to 2 inches per hour); and, 2) remove any water which enters the coffer cell through the sheet pile interlocks, remove under seepage, and any seepage through the entrance/exit rings and steel casing during boring and jacking operations in order to keep the coffer cell free of any standing water. Unwatering is required during placement of bottom tremie seal and after flooding from high water events. Such unwatering system may be a system of sump pumps, relief wells, well point system, or other approved devices. Drawings and notes show a TRS concept that is provided for bidding purposes only. The Contractor shall submit a [work plan](#) for review and approval by the Contracting Officer.

(2) The unwatering system shall be designed, installed, and operated to unwater the TRS down to the unwatering depths needed for proposed construction with Puerto Nuevo River stages up to El. 11.5 feet. The unwatering system must also include standby pumping and power supply such that a continuously operable system is available during any power outages, pump failures, etc.

(3) The coffer cell slab shall be maintained free of standing water as specified in the contract documents and approved work plan so that construction operations can be performed without interruption due to wet conditions.

(4) After the system is installed, relief pumps shall work to keep pressures under the tremie slab to a minimum. At no time during construction shall the Factor of Safety against flotation of the tremie slab be less than 1.3.

(5) No uncontrolled upward or vertical flow of water into the excavated area will be permitted at any time. The Contracting Officer shall be notified immediately if any upward or vertical flow of water is observed from foundation soils.

(6) The unwatering system may consider installing horizontal drains or seepage barriers using jet-grouting and/or other similar methods as needed to control and reduce infiltration and underseepage.

(7) The rate of unwatering or rewatering the excavation shall meet the construction sequencing requirements and provide for a stable TRS. In the event of a hurricane or other unusual/extreme high river flood condition, the coffer cell may be flooded, to provide a stable excavation and TRS, in accordance with the Contractor's approved unwatering, coffer cell and excavation work plan. Flooding shall commence at the direction of the Contracting Officer.

(8) The TRS shall be emergency flooded in anticipation of significant and impending storms. Emergency flooding shall be accomplished within 6 hours of direction to flood. A minimum 24-hour advance notice will be given for emergency flooding by the Contracting Officer. If flooding is directed by the Contracting Officer, the Contractor shall be compensated for damages in accordance with the Contract Documents.

(9) The tremie slab shall have a crushed rock gravel subbase of an effective thickness such that underseepage can flow freely to standpipes and piezometers and also to provide a stable foundation for the tremie slab. An aggregate bedding consisting of a minimum 18-inch thick gravel material in accordance with the contract documents, shall be provided under the tremie seal and bottom of standpipes or relief wells. An additional purpose of the aggregate bedding is to: 1) provide a stable foundation for the tremie seal; and, 2) prevent any mud wave when placing the tremie seal. The design of the aggregate layer shall consider the consistency of the associated insitu soil and shall include any geotextile fabric and sand layer as necessary.

The Contractor is responsible for the proper design, operation, and performance of the dewatering or unwatering system. Any modification to the dewatering system needed during construction or operation shall be at no cost to the Government.

1.5 PIEZOMETERS

a. A piezometer system shall be installed by the Contractor to monitor the piezometric conditions within and below the aggregate layer which is to be installed underneath the tremie slab. A minimum of three (3) piezometers shall be installed; one (1) piezometer near the center of the TRS and two (2) piezometers (at opposite corners of the TRS).

b. The Contractor shall provide all required personnel and equipment to install piezometers in accordance with normal geotechnical procedures to monitor groundwater levels and Contractor's approved work plan.

c. Piezometer depths should be adjusted in the field as needed to assure that the piezometer is seated in the layer required such that the piezometric head is being measured in the desired stratum.

d. The Contractor shall furnish all materials required to install the piezometers. The materials shall include, but are not limited to, a well screen or optional piezometer type, riser pipe, and filter material. The well screen shall be a Johnson type or equivalent, 1.5-inch I.D. and 5 feet long. The riser pipe shall be 1.5-inch I.D., schedule 80 PVC pipe and filter material shall be cleaned filter sand, protected by filter fabric. The piezometers shall not be sanded full depth; only the well screen shall be sanded. The annulus surrounding the piezometer riser and above the monitored soil zone shall be seated with bentonite pellets.

e. The Contractor shall make a minimum of two readings per piezometer, per working day, a minimum of 8 hours apart, and not to exceed two (2) calendar days between readings at any time. These piezometer readings, along with corresponding channel stage readings, shall be recorded and reported to the Contracting Officer within 4 hours after they are obtained.

f. The piezometers shall be installed within a 6-inch steel pipe casing through the tremie slab and shall at all times be protected from being damaged by construction operations

1.6 RELIEF WELLS

a. A relief well system shall be installed by the Contractor to relieve groundwater underseepage pressure within and below the aggregate layer which is to be installed underneath the tremie slab. A minimum of four (4) wells shall be installed, one at each corner of the TRS).

b. The Contractor shall provide all required personnel and equipment to install relief wells in accordance with normal geotechnical procedures for well installation and Contractor's approved work plan.

c. Well depths should be adjusted in the field as needed to assure that the well is seated in the gravel subbase layer as required to facilitate flow.

d. The Contractor shall furnish all materials required to install the relief wells. The materials shall include, but are not limited to, a well screen, riser pipe, and filter material. Filter material shall be cleaned filter sand. The wells shall not be sanded full depth; only the well screen shall be sanded. The annulus surrounding the well riser shall be seated with bentonite pellets.

e. The wells shall be installed within steel pipe casing of sufficient diameter to allow submersible relief pumps to be located at the level of the tremie slab and shall at all times be protected from being damaged by construction operations.

f. The Contractor shall provide sufficient pumping capacity to maintain uplift pressures under the tremie slab to a minimum and the uplift (flotation) factor of safety factor greater than 1.3.

g. The Contractor shall provide 100 percent pumping redundancy on site should pumps fail to operate properly.

1.7 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Unwatering Design Package; G, D0

Work Plan; G, D0

The contractor shall develop and submit a work plan that shall at minimum include the following:

a. Construction sequence.

b. Description of means and methods of construction, and unwatering and rewatering operations.

c. Details of collection, maintenance, and disposal of water

- d. Details of piezometer installation and monitoring
- e. Details of relief well installation and monitoring
- f. Description of equipment to be used to perform the work.
- g. Proposed staging and laydown area.
- h. Safety controls for the equipment/material use and containment of any water and spoils.
- i. Removal of equipment after completion of work.
- j. Boring logs and laboratory test reports should the Contractor elect to perform additional borings in vicinity of the temporary retaining structures.

SD-06 Test Reports

Fabrication And Workmanship Records

Installation, Operation, And Removal Records

Inspection

Piezometer Installation Records

Relief Well Installation Records

1.8 GENERAL CRITERIA

All permanent work under this Contract, except as otherwise specified, shall be carried on in areas free of water. The Contractor shall design, furnish, install, operate, and maintain such facilities necessary to accomplish the following:

- (1) Collect and dispose of all water in the TRS regardless of source.
- (2) Install and monitor construction piezometers inside the temporary retaining structure.
- (3) Emergency flood the cofferdam as specified within this section of the Specifications.

1.8.1 Unwatering Design Package

The Contractor shall submit an original of the complete unwatering design package with details of the proposed facilities. These details must be presented in the form of shop drawings including details of the type of system, planned layout and sizes of headers, including all lengths requiring burial, well points, collectors, ditches, piezometers, sumps and pumps; number, type, location, elevation, and proposed methods of testing of piezometers; provisions for disposal of water; Maintenance of Flow plan; and plan of operation, including flooding and rewatering plans. This submittal shall be submitted no later than 120 days prior to installation of the system.

The Contracting Officer's initial review of the Contractor's proposed facilities will be for the purpose of determining: (1) the acceptability of the general design concept and layout of the system; (2) the gross capacity of the system at the design stage; and, (3) the acceptability of the flooding and rewatering plans.

If the Contracting Officer determines, based on the above-mentioned review, that the system appears adequate to accomplish the required results, the system will be approved for installation. If the Contracting Officer's review determines that the Contractor's proposed facilities are either inadequate or inappropriate to accomplish the required results, the Contractor will be so notified in writing, and the basis for rejection will be included. Approval of the plan for installation, either as submitted or revised as a result of the review, should not be interpreted as the Contracting Officer accepting responsibility for the performance of the system, and shall not relieve the Contractor of full responsibility for the proper design, installation, maintenance, operation, and actual performance of both the individual system components and the entire system.

After approval of installation, the Contractor shall install the entire system and shall make no alteration to the planned system without the prior written approval of the Contracting Officer. If, during the progress of the work, the installed system proves inadequate to meet the requirements specified, including piezometers, the Contractor shall, at its expense, furnish, install, and operate such additional facilities and/or make such changes, either in features of the system or the plan of operation, as may be necessary to perform the required unwatering without additional cost to the Government.

1.9 QUALITY CONTROL

1.9.1 General

The Contractor shall establish and maintain quality control for all unwatering operations to assure compliance with Contract requirements and maintain records of his quality control for all construction operations, including but not limited to the following:

- (1) Designing.
- (2) Fabrication And Workmanship Records.
- (3) Installation, Operation, And Removal Records.
- (4) Piezometer Installation Records.
- (5) Relief Well Installation Records

1.9.2 Reporting

An original and two (2) copies of these records and tests, as well as the corrective action taken, shall be furnished to the Contracting Officer daily. Format of the report shall be as prescribed in Specification Section 01 45 04, CONTRACTOR QUALITY CONTROL. Reports of operation and inspection shall include the following data: piezometric elevation, channel stages, time of operation of each pump, effluent discharge, problems encountered, proposed actions, and any other pertinent data.

1.10 INITIAL TESTING

Upon installation of the system, the Contractor shall test and evaluate the completed system to demonstrate that the system is, in fact, capable of performing the intended unwatering operation as outlined herein. This testing shall include complete falling-head tests to be conducted on each piezometer, if appropriate. The Contractor shall give the Contracting Officer 24-hour advance notice of its intention to perform testing. The documentation of results of the test shall be provided to the Contracting Officer within 48 hours of completion.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 WATER LEVEL PRIOR TO OPERATION

During construction of the TRS and Tremie Seal, water level inside and outside the TRS shall be maintained at equal levels until the tremie has attained the appropriate strength (minimum compressive strength of 2,500 psi at 7 days) so as to act as a compression strut.

3.2 OPERATION

The Contractor shall perform unwatering and maintain the work areas free of standing water as necessary for the work under this Contract. Once an area is unwatered, it shall be maintained in a unwatered condition until all work in that area is completed, unless flooding is directed by the Contracting Officer. In the event that flooding is deemed necessary by the Contracting Officer, the protected area shall be flooded in accordance with the sequence of flooding proposed by the Contractor and approved by the Contracting Officer in the unwatering plan. However, the Contractor shall not flood the TRS without the approval by the Contracting Officer. If flooding is directed by the Contracting Officer, the Contractor will be compensated for damages in accordance with the applicable requirements of the Contract Documents. If flooding occurs because of the Contractor's fault, negligence, or convenience, all costs resulting from such flooding shall be borne by the Contractor. Commencement of unwatering subsequent to flooding will be subject to prior approval of the Contracting Officer.

3.3 MAINTENANCE AND SERVICING

The Contractor shall be responsible for the maintenance, servicing, and repairs of the entire unwatering system and appurtenances during the life of the Contract, including replacement of any equipment performing unsatisfactorily.

3.4 REMOVAL

The unwatering facilities required to maintain a dry condition within the protected area shall be maintained until completion of the work within the TRS, and then shall be completely removed. However, no unwatering facilities of any kind shall be removed without prior approval of the Contracting Officer. All equipment and appurtenances employed in the unwatering system and all materials shall remain the property of the Contractor and shall be removed from the site of the work. All holes

created by removal of unwatering facilities shall be plugged in accordance with San Juan and Puerto Rico water well closure criteria. Any approval of the implementation and/or removal plans by the Contracting Officer does not shift the responsibility for the removal of the system from the Contractor to the Government. Nor does it relieve the Contractor of his responsibility to provide a removal plan, which comports with industry standards and prudent construction practices.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 31 - EARTHWORK

SECTION 31 32 23

JET GROUTING

PART 1 GENERAL

- 1.1 SCOPE
- 1.2 RELATED SECTIONS
- 1.3 REFERENCES
- 1.4 DEFINITIONS
- 1.5 SUBMITTALS
- 1.6 QUALITY CONTROL
 - 1.6.1 General
 - 1.6.1.1 Experience Profile
 - 1.6.2 Drilling Deviation Measurements
 - 1.6.3 Jet Grout Samples
 - 1.6.4 Reporting
 - 1.6.5 Sample Log
 - 1.6.6 Metering Equipment
- 1.7 DELIVERY, STORAGE, AND HANDLING
- 1.8 PROJECT/SITE CONDITIONS
- 1.9 CONTRACTOR'S RESPONSIBILITY
- 1.10 QUALITY ASSURANCE
- 1.11 Grouting Plans

PART 2 PRODUCTS

- 2.1 GENERAL
- 2.2 GROUTING MATERIAL
 - 2.2.1 Water
 - 2.2.2 Cement
 - 2.2.3 Pozzolans
 - 2.2.4 Admixtures
 - 2.2.5 Bentonite

PART 3 EXECUTION

- 3.1 JET GROUTING
 - 3.1.1 General
 - 3.1.2 Demonstration of Jet Grout Seal and Soil Improvement
 - 3.1.3 Jet Grouting Containment Plan
 - 3.1.4 Equipment, Materials and Procedures
 - 3.1.5 Jet Grout Seals and Columns
 - 3.1.6 Cleanup
- 3.2 JET GROUTING PERFORMANCE TESTING
 - 3.2.1 Materials Testing
 - 3.2.2 Field and Laboratory Testing
 - 3.2.3 Performance Testing - Seals
 - 3.2.4 Core Drilling, Uniformity, and Core Samples
 - 3.2.5 Borehole Camera Surveys
- 3.3 RECORDS
 - 3.3.1 Final Jet Grouting Reports

- 3.3.2 Record Drawings of Jet Grout Installation
- 3.3.3 360-degree Video Recording
- 3.4 COMMUNICATIONS
- 3.5 EMBANKMENT BACKFILL
- 3.6 DRIVING SHEETPILE RESTRICTION
- 3.7 REMEDIAL MEASURES
- 3.8 QUALITY CONTROL

-- End of Section Table of Contents --

SECTION 31 32 23

JET GROUTING

PART 1 GENERAL

1.1 SCOPE

The work covered by this Section consists of furnishing all plant, materials, and performing all operations in connection with furnishing and placing jet grout seals for seepage reduction and grout columns for ground improvement as shown on the drawings and specified herein. The grout seals are to be installed to the size and depths indicated on the drawings. When approved by the Contracting Officer, jet grouting may be used for mitigation of gaps between adjacent drilled shafts according to Specification 31 63 29. For ground improvement, jet grout columns are to be installed in areas under planned stormwater pipes as shown in the drawings. The intent of the jet grouting operation is to inject the soil foundation with a jet grouted mass to 1)prevent loss of soil through potential gaps between adjacent drilled shafts and 2)improve soils to limit settlement below stormwater pipes. All jet grouting will be performed in accordance with these specifications and applicable drawings. The Contractor will demonstrate its jet grouting means and methods at the demonstration wall section. Jet grouting means and methods shall be as identified in the Contractor's preconstruction submittals.

The jet grout demonstration section shall be conducted along the demonstration shaft wall as shown on the plans. The jet grout demonstration section shall demonstrate that a jet grout seal with a cross-sectional area of at least two square feet is formed in the soil between and in intimate contact with tangent shafts. Methods of measurement of the cross-sectional area shall be determined and documented. The seals formed during the demonstration section shall conform to all requirements in this specification.

The jet grout quality control program for this project shall include real time data collection of the drilling and jet grouting parameters defined in this specification. Every jet grout seal and column shall have drilling deviation measurements taken for the purpose of assessing the as-built geometry and diameter of the jet grout column. Validation of the jet grout placement shall be determined by core holes as specified herein and performed and paid for by the Contractor.

1.2 RELATED SECTIONS

00 31 32 GEOTECHNICAL DATA REPORT
31 63 29 DRILLED CONCRETE PIERS AND SHAFTS

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C618 (2019) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete

ASTM D2166/D2166M (2016) Standard Test Method for Unconfined Compressive Strength of Cohesive Soil

1.4 DEFINITIONS

1. Jet Grouting: The process of creating stabilized jet grout columns in place. Jet grouting, as specified herein may consist of a single, double, or triple jet system, in which the jet rod injects cement grout at high velocity through horizontal radial nozzles(s) to directly erode and mix with in-situ soil. The jet rod is rotated and incrementally lifted at a constant speed to achieve thorough mixing and a consistent continuous column. This jet grouting process will be used to form a homogeneous mixing of the soil and grout to create full column.
2. Jet Grout Seals: Vertical columns of at least 2 square feet in areas installed with intimate contact to the adjacent tangent drilled shafts preventing sediment migration from occurring between gaps in tangent drilled shafts.
3. Jet Grout Columns: Volume of required ground improvement by jet grouting for mitigating settlement below required stormwater outfall piping and structures.
4. Monitor: A rod appliance attached to the end of a drilling string and designed to deliver air and fluid of the Jet Grouting process.
5. **Drilling Deviation Measurements:** A drilling deviation measurement is a method of surveying and documenting the plumbness of the drill hole and provides a way of determining the location of the actual bottom of drill hole with respect to the theoretical or intended location. Drilling deviation measurements shall be performed for every jet grout drill hole and quality control coring using an instrument capable of biaxial deviation measurement to determine the magnitude and azimuth of all down-the-hole deviations. The drilling deviation measurements shall be obtained over the entire length of the drill hole. The instrumentation used in drilling deviation measurement shall have the capability of measuring down a depth of at least 75 feet. Use an instrument capable of controlling or measuring the azimuth and recording the magnitude of the deviation of the instrument as it is raised inside of the jetting rods. The azimuth and magnitude of the drilling deviation measurement shall be displayed on an LCD screen that will enable the Government inspector to manually record the raw drilling deviation data. The Contractor shall use the drilling deviation data to determine the as-built geometry of the jet grout seal and column to further verify that the jet grouting meets specification requirements.
6. Jet Grout Waste. Jet Grout waste is material generated during the jet grouting process including jet grouting cuttings; soil, water, and cement, wash water, unused grout; and any other liquids and debris generated in the jet grouting process. These waste products, after being allowed to solidify, will be considered ordinary construction

debris for the purposes of disposal requirements. Disposal of jet grout waste shall be the responsibility of the Contractor.

7. Uniformity. The amount of uniformly mixed grout material measured by core recovery and percent-treated as described in paragraph CORE DRILLING, UNIFORMITY, AND CORE SAMPLES.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Designation RO stands for Resident Office and designation DO stands for District Office. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Jet Grout Experience; G, DO

The Contractor shall have at least five years jet grouting experience over the last ten years; and have completed at least five (5) jet grouting projects with at least two (2) projects having objectives and jet grouting system similar to those of this project and in the same type of soil.

Grouting Plans; G, DO

At least 30 calendar days prior to jet grouted seal and column installation, contractor shall submit jet grout installation plans for the demonstration section and for production work. Details shall include a description of the jetting procedures, drilling methods, method of supporting bore holes, drilling deviation measuring and data reduction techniques, jetting parameters, all nozzle sizes, pump pressures, flow rates, rod rotation rates, lift speed, seal/column size estimates, grout formulations, volumes of waste generation, and installation schedule. A proposed method for creating a safety corridor to convey the high pressure line(s) from the drill location to the batch plant site shall be included in the plan.

Plans shall outline the spacing, location, depth and quantity of grout to achieve the specified criteria of this specification. The plan must describe the sequence of operations; jet grout seal and column installation procedures; jet grout column size and layout scheme; grout batch plant and high-pressure pumping certification program; safety procedures; control and disposal of jet grout waste; and survey procedures including drilling deviation measurements. Work procedures, sequence and control criteria shall be included.

The grouting plan submittal shall include the means and methods to be used to repair any deficient, non-conformance grout seals or columns. Details of the Quality Control (QC) testing and inspection procedures and parameters shall also be provided, including experience of QC personnel. As part of this submittal, the Contractor shall provide the procedures that will be used in obtaining, transporting, storing, and testing jet grout samples including equipment and procedures to be used in obtaining cored samples.

The plan must include a narrative describing the Contractor's understanding of the site subsurface conditions. The Contractor shall reference the available subsurface data provided in Section 00 31 32, GEOTECHNICAL DATA REPORT.

Grout Mix Proportions; G, D0

At least 30 calendar days prior to placement of grout, the Contractor shall submit the mix design for the project indicating sources and types of grout materials, with volumetric proportions, and field test data from previous projects indicating compressive strength achieved.

The Contractor shall submit proportions and properties of the proposed design mix including proportions of cement, bentonite, additives, and water; and viscosity, density, and unconfined compressive strength to the Contracting Officer. Submit either a new project specific laboratory design mix program or the results of previous laboratory test programs (if the materials are similar to those proposed for construction).

Jet Grouting Containment Plan; G, D0

At least 30 calendar days prior to placement of grout, the Contractor shall submit a containment plan. The Jet Grouting Containment Plan shall consist of the Contractor's methods for wash water and tailings containment. The Contractor shall keep the site clean during jet grouting operations.

Drilling Deviation Measurement Plan; G, D0

At least 30 calendar days prior to jet grouted seal and column installation, the Contractor shall submit a detailed plan describing the use of in-rod drilling deviation measurement instruments and recording system. The Contractor shall provide sketches of the drilling deviation inclinometer intended to fit inside the jetting rods for down-the-hole measurements. In particular, the Contractor shall provide details concerning the operation and data reduction of the raw hole survey data. Contractor shall submit a detailed sample calculation and illustrative examples from previous work showing how the raw deviation data is used to calculate the as-built location of the drill hole in the field.

As part of this submittal, the Contractor shall also describe the data form for transferring all drilling deviation digitized data to the Government Inspector immediately after the data are collected and analyzed at the drill. The digitized data provided shall also contain an as-built location of the installed drill hole. The exact format of the digitized data shall be coordinated with the Contracting Officer.

Sample Log; G, D0

Submit sample log to be used to record seal installation and ground improvement. The log shall contain the following information in American units:

- a. Project name.
- b. Machine number.
- c. Type of mixing tool.
- d. Date and time of grouting.
- e. Verification boring number and reference drawing number.

- f. Column diameter or seal area.
- g. Elevation of top of grout.
- h. Column or seal length.
- i. Grout rates and takes for each grout seal or column.
- j. Rate of withdrawal of mixing tool in feet/revolution, reported at each foot depth.
- k. A description of obstructions or other interruptions of binder injections during installation.

SD-02 Shop Drawings

Jet Grouting; G, D0

The Contractor shall submit shop drawings and a jet grout narrative at least 30 calendar days prior to anticipated beginning of jet grout operations for approval by the Contracting Officer. Submit shop drawings including sequence time schedule of the jet grouting operations for seals between drilled shafts along shaft wall and ground improvement columns beneath stormwater pipes. Grout hole locations shall be dimensionally referenced in the contract drawings. Drawing locations of the jet grouting shall be shown on a drawing not smaller than 1 inch = 10 feet scale to show the details of the layout required to satisfy the geometry required. The drawings shall include the following:

- a. Plant, equipment, and material descriptions.
- b. Arrangement of grout mixing and injection equipment, location of boreholes, and other necessary details
- c. Methods for drilling and supporting boreholes
- d. Sequence of jet grout installation
- e. Grout mix design, sources of mix materials, and material property data sheets demonstrating compliance with the specified mix material described herein
- f. Layout and jet grouting parameters, including sampling and testing to determine quality and properties of grout materials.

SD-03 Product Data

Drilling and Grouting Equipment; G, D0

Details and data on the drilling and grouting equipment, including details, catalog-cut sheets and all ancillary equipment.

Grout Plant; G, D0

A detailed plan showing equipment and grout plant layout proposed for mixing and placing grout.

Borehole Camera; G, D0

The borehole camera is for making 360 degree log of the borehole when required. The camera shall be able to operate in either a dry or water-filled borehole to a maximum depth of 75 feet. The camera shall meet the following minimum requirements:

TYPE: Digital Optical Borehole Televiewer
FRAME RATE: 20Hz
EXPOSURE: 6
LIGHTING: 10
RESOLUTION: 900 pixels
LOGGING SPEED: 2.9 feet/minute

Submit manufacturer's data specification sheets and operating

manuals for the borehole camera to be used for the borehole surveys. Indicate in the submittal the capabilities of the encoder to provide date and time stamps on the images.

SD-06 Test Reports

Field and Laboratory Testing; G, D0

The original and two copies of these records, as well as the records of any corrective action taken, shall be furnished to the Government weekly. Contractor shall maintain electronic and printed records of data and quality control testing. Format of report shall be as prescribed in Section 01 45 04 CONTRACTOR QUALITY CONTROL. The laboratory performing the testing shall be certified by USACE for ASTM D2166/D2166M.

Borehole Camera

Copies of camera surveys shall be uploaded to the Contractor provided SFTP within 24 hours of performing the survey.

Metering Equipment

Provide calibration test results for metering equipment. Calibrations shall be current, not older than one year, and follow manufacturer's recommend procedures. These shall include:

- a. Factory Calibrations on Instruments Used for Control of Jet Grouting
- b. Factory Calibration on Drilling Deviation and Borehole Surveying Instruments

Instrument For Drilling Deviation Measurements; G, D0

Factory Calibrations On Instruments Used For Control Of Jet Grouting; G, D0

Provide certificates of calibration for all instrumentation used for jet grouting. Calibration shall not be older than 6 months.

Factory Calibration On Drilling Deviation And Borehole Surveying Instruments; G, D0

Provide descriptions of instruments and certificates of calibration for all instrumentation used for drilling deviation and borehole surveying. Calibration shall not be older than 6 months.

SD-11 Closeout Submittals

Final Demonstration Test Report; G, D0

Final Jet Grouting Reports; G, D0

Record Drawings Of Jet Grout Installation; G, D0

1.6 QUALITY CONTROL

1.6.1 General

The Contractor shall establish and maintain quality control for the grouting operation to assure compliance with contract requirements and maintain quality control records for all construction operations including but not limited to the following:

- a. Batching and Mixing Equipment: Verify type, size and suitability for construction of the prescribed work.
- b. Grout: Verify grout mix is appropriate for intended use and meets requirements.
- c. Spoil: Ensure continuous spoil return during all jet grouting operations.
- d. Grout Injection: Grout injection, rotation and extraction rates shall be sufficient to produce grout seals, ground improvement columns, or mitigation of gaps between adjacent drilled shafts meeting requirements specified herein.
- e. Geometric Tolerances. The maximum horizontal deviation of the as-installed center of any grout seal or column at the ground surface shall have a maximum value of 1-inch from the layout center coordinate, shown on the Contractor's submittal. The vertical alignment of the drill hole shall have a maximum value of not more than one (1) percent from vertical.
- f. The minimum cross-section area of the jet grout seals between tangent shafts is 2 square feet as shown in the plans.
- g. The maximum diameter of the ground improvement columns is 36 inches and the maximum center-to-center spacing of the columns is two times the column diameter. The Contractor may propose larger diameter columns by providing the Government with calculations that show criteria will be met.
- h. Top of grout seals and columns, including mitigation jet grout seals, shall be no lower than the top elevation shown in the plans.
- i. Bottom of grout seals and columns, including mitigation jet grout seals, shall be no higher than the bottom elevation shown in the plans.
- j. Damaged Grout: Any jet grout seal or column damaged as the result of mechanical failure of equipment, inadequacy of grout supplies, or improper drilling or injection procedures shall be backfilled with cement grout and replaced by another grout hole, drilled and injected by the Contractor at no additional cost to the Government.
- k. Core Samples: A minimum 10 percent of the grout seals and ground improvement columns shall be cored. Selected samples shall be tested for permeability or compressive strength as described in paragraph FIELD TESTING. Uniformity is acceptable if percent treatment is at least 80 percent for every 5-ft core run as defined in paragraph CORE DRILLING, UNIFORMITY, AND CORE SAMPLES.
- l. Control Testing: The individuals who monitor, sample and test the grout mix and grout treatment zone shall have a demonstrated knowledge and ability to perform the necessary testing specified herein. The laboratory performing the sampling and testing shall be certified by USACE for ASTM D2166/D2166M.

1.6.1.1 Experience Profile

The Contractor shall submit a detailed experience profile demonstrating experience and competence to perform jet grouting in accordance with these specifications. Submit evidence and performance records from at least five previous projects that all equipment used for drilling injection holes; drilling deviation measurements; lowering, raising and rotating the jet rods; mixing grout; supplying pressurized fluids to the nozzle(s); and jetting monitors and rods used to construct the jet grout columns are capable of executing the required jet grouting work in accordance with these specifications. The jet grouting company shall have at least five years of prior experience performing similar projects over the past 10 years. Submittal shall include experience of individuals who monitor, sample, and test the grout mix.

1.6.2 Drilling Deviation Measurements

Drilling deviation measurements shall be performed for every jet grout drill hole, coring, and rotary exploratory hole using an instrument capable of compass and biaxial deviation measurements to determine the magnitude and azimuth of all down-the-hole deviations. Contractor shall demonstrate that it can be used for every hole required for the project. The Contractor shall provide evidence that [instrument for drilling deviation measurements](#) is capable of both measuring and recording the compass direction and biaxial deviations of the drill rods to determine the resultant magnitude and azimuth of all hole deviations.

1.6.3 Jet Grout Samples

Jet grout samples shall consist of cored samples as described herein. The coring technique selected by the Contractor shall be capable of obtaining samples to jet grouting depths as shown on the plans and planned depths of grout columns for ground improvement. Cored samples of jet grout shall be 4-inch diameter samples of hardened material obtained in a maximum of 5-foot long runs. Split inner tubes will be used for core removal. The core hole sample locations and time of coring on the cured grout will be determined by the Contracting Officer and shall include at a minimum 10 percent of the grout seals and columns. The contractor shall also provide one core hole per jet grout column and seal at the finished demonstration section.

1.6.4 Reporting

All results of tests, inspections, daily logs and corrective actions shall be reported informally as they are completed and formally in writing (original, 2 copies and electronic copies) to the Government daily. Accurate daily records shall be made of all jet grouting locations, depths of treatment, start and stop times, all jetting parameters, and amount of grout injected and amount of spoil returned for each location. Electronic and printed records of data and quality control testing shall be maintained by the Contractor. Format of the daily reports shall be as specified in Section 01 45 04 CONTRACTOR QUALITY CONTROL. The Contracting Officer has the right to examine all test and inspection records.

1.6.5 Sample Log

Submit sample log to be used to record jet grout seal and column installation.

1.6.6 Metering Equipment

Provide calibration test results for metering equipment. Calibrations shall be current, not older than one year, and follow manufacturer's recommend procedures. These shall include:

- a. Factory Calibrations On Instruments Used For Control Of Jet Grouting
- b. Factory Calibration On Drilling Deviation And Borehole Surveying Instruments

1.7 DELIVERY, STORAGE, AND HANDLING

Enough cement shall be stored at or near the site of the work to ensure that grouting operations will not be delayed by shortage of cement. In the event the cement is found to contain lumps or foreign matter of a nature and in amounts which may be deleterious to the grouting operations, screening through a standard 100 mesh screen may be required. No payment will be made for such screening.

1.8 PROJECT/SITE CONDITIONS

The proposed jet grout seal location, depth, features and characteristics as depicted on the plans and these specifications are based on currently available information. Grouting mixes, pressures, injection rate and the sequence in which the holes are drilled and grouted shall be approved by the Government prior to installation of production seals, columns, or mitigation of gaps between adjacent drilled shafts.

1.9 CONTRACTOR'S RESPONSIBILITY

Contractor shall perform a demonstration section to verify the capabilities of the jet grout means and methods. The demonstration section shall be performed along the demonstration shaft wall as required in SECTION 31 63 29 DRILLED CONCRETE PIERS AND SHAFTS.

The objective of the jet grout seal is to prevent loss of soil through gaps between tangent drilled shafts. The effectiveness of the jet grout is dependent on the intimacy of contact between the jet grout and the adjacent tangent drilled shafts. The contractor shall be prepared to adjust the jet grout operation as required to achieve the desired contact between the jet grout seal and the adjacent tangent drilled shafts. This objective also applies to mitigation of gaps between adjacent drilled shafts. All grouting work to mitigate gaps due to a non-conformance shall be at no additional cost to the Government.

The objective of the jet grout columns is to improve properties of the soils to minimize differential settlement along stormwater pipes. The contractor shall be prepared to adjust the jet grout operation to achieve stabilization of the soils within the required footprint. The footprint of ground improvement shall extend as shown on the Contract Drawings.

The Contractor is responsible for quality control of all grouting operations to assure compliance with contract requirements.

The jet grouting Contractor shall become acquainted and satisfied as to the character, quality, and quantity of surface and subsurface materials by inspecting the site and by evaluating information from available

exploratory work and soil borings provided in section 00 31 32. Furthermore, the jet grouting Contractor shall locate existing utilities through excavation, probing, or other suitable methods prior to jet grouting. Existing utilities are shown on the drawings to the extent they are known.

1.10 QUALITY ASSURANCE

The Government will collect and perform quality assurance testing. The Government testing will in no way relieve the Contractor of the responsibility of performing tests necessary to meet the construction requirements. All testing procedures and results conducted by the Contractor shall be available for inspection by the Contracting Officer at any time.

1.11 Grouting Plans

Submit grouting plans meeting requirements within the paragraph SUBMITTALS herein.

PART 2 PRODUCTS

2.1 GENERAL

The Contractor shall obtain and maintain at the job site spare parts and backup equipment to maintain jet grouting equipment in satisfactory operating condition to prevent loss of time due to mechanical breakdown or equipment failure.

Jet grout drilling equipment shall be of the type and capacity suitable for drilling in materials at the site and of the required hole sizes and depths. Jet grout drilling equipment shall be suitable for achieving depths required, then raising and rotating the jet rods to the depths and at the rates required for jet grout seal or column construction. The jetting rods shall be of sufficient inside diameter to allow access of the drilling deviation instrument for in-rod measurements. The borehole shall be stable and have sufficient annular space between the jet grout drill rods and the side walls of the drill hole to be able to maintain a constant flush of cuttings to the surface. Drilling technique shall be able to achieve the required horizontal and vertical tolerances. The equipment shall be capable of drilling and jet grouting to the depths shown on the plans.

If difficult drilling is encountered, the Contractor may be authorized to pre-drill to fully fracture the soil volume before injecting the grout. Such pre-drilling will only be performed at the authorization of the Contracting Officer. Contractor shall fully investigate overhead clearance to address solutions in the work plan.

The Contractor is alerted to the fact that specialized jet grouting equipment may be required for working below overhead obstructions such as trees and utilities and power lines.

The jet grout drill shall have an integrated data acquisition system, which during drilling will continuously monitor or calculate the following:

- a. Clock Time
- b. Depth Below Ground
- c. Drilling Speed (feet per second)

- d. Thrust on Drill Rods (lbf)
- e. Torque (lbf-ft)
- f. Drill Rod RPM's (revolutions per minute)
- g. Drilling Fluid Pressure (psi)
- h. Specific Energy (lbf-ft/in³)
- i. Delivery of Grout per Incremental Foot of Seal or Column

2.2 GROUTING MATERIAL

Provide grout composed of water and cement, pozzolans, blast furnace slag, admixtures, and bentonite. The [grout mix proportions](#) shall be designed by the Contractor. The grout shall be mixed in a grout plant that combines dry cement and water in predetermined proportions. Grout mixes shall be proportioned to provide the required strength described in the paragraph FIELD TESTING. The various materials to be furnished by the Contractor shall conform to the specifications.

2.2.1 Water

The water used in the grout shall be furnished by the Contractor. It shall be fresh potable water, clean and free from of sewage, oil, acid, alkali, salts, or organic matter.

2.2.2 Cement

Cement used in grout shall conform to the requirements of paragraph 2.1.1 of SECTION [03 31 01](#) CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS in addition to Type I. The use of bulk cement will be permitted provided the Contractor employs methods of handling, transporting, and storage that are satisfactory to the Contracting Officer, otherwise only cement furnished in cloth or paper bags will be accepted to use in the work. Storage of cement shall be in accordance with paragraph DELIVERY, STORAGE, AND HANDLING.

The maximum temperature of the water and grout delivered at the drill for injection shall be less than 90 Deg F but greater than 40 Deg F. Contractor shall take pro-active measures to keep the fresh grout within these temperature ranges.

Contractor shall make available a sampling port at the jet grout drill rig that can be used to obtain a one-gallon sample of the water and fresh grout for testing. This sample shall be obtained when directed by the Contracting Officer's field representative. Jet sampling at the batch plant is not acceptable.

The fresh grout shall not be stored at the grout plant or other temporary holding tank for longer than one hour prior to injection. Fresh grout beyond this time limit shall be discarded in a manner acceptable to the Contracting Officer. Fresh cement grout shall be constantly agitated to keep the solids well distributed in the water.

2.2.3 Pozzolans

If used, pozzolans shall be fly ash or other raw or calcined natural pozzolans conforming to [ASTM C618](#) and paragraph 2.1.1 of SECTION [03 31 01](#) CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS. It shall be transported, handled, and stored to avoid damage, waste, or absorption of moisture. For any pozzolans used in the grout, provide a certification from the supplier to the Contracting Officer. The manufacturer's

certification shall be updated as appropriate, but not less than once per month.

2.2.4 Admixtures

The Contractor is responsible for determining the need for admixtures. If required, admixtures shall be added to the grout immediately before or during its mixing and will consist of accelerators, retarders, water reducers, and fluidifiers. The Contractor is responsible for determining the need for admixtures. All admixtures shall have MSDS sheets on file and available for review by the Contracting Officer.

2.2.5 Bentonite

The Contractor is responsible for determining the amount of bentonite required to produce a slurry viscous enough to thoroughly fill the voids. Bentonite shall be pre-hydrated for a minimum of 8-hours before mixing with the cement slurry. If the Contractor proposes to eliminate pre-hydration of the bentonite, the Contractor shall demonstrate through a laboratory test program that the design criteria for strength will be met.

PART 3 EXECUTION

3.1 JET GROUTING

3.1.1 General

Submit a detailed plan showing [drilling and grouting equipment](#) and [grout plant](#) layout proposed for mixing and placing grout. Jet Grouting shall be performed in accordance with the approved submittal.

3.1.2 Demonstration of Jet Grout Seal and Soil Improvement

Contractor shall perform demonstrations of grout wall-seal and ground improvement installations to verify the Contractor's jet grout means and methods and the required 28-day minimum strength gaining as follows. All grout seal samples tested shall have a minimum 28-day unconfined compressive strength of 100 psi accord to [ASTM D2166/D2166M](#). All ground improvement grout column samples tested shall have a minimum 28-day unconfined compressive strength of 150 psi according to [ASTM D2166/D2166M](#).

Demonstration of wall seal grouting shall be performed along the demonstration shaft wall. Approximate locations for the demonstration shaft walls are identified in the Contract Drawings and shall be adjusted and/or approved by the Contracting Officer prior to the mobilization of equipment.

The demonstration of ground improvement grouting shall be performed in a minimum of 10-ft by 15-ft area with equal (spacing/diameter/depth) grouting columns using means and methods to be used during production phase. Approximate locations of the ground improvement demonstration areas are identified in the contract drawings and shall be adjusted and/or approved by the Contracting Officer prior to the mobilization of equipment. A supplementary demonstration of the ground improvement grouting shall be performed in a new test area if the soil-grout mix of the prior demonstration columns fail to achieve the required minimum 28-day strength of 150 psi. The size and location for supplementary demonstration work shall be confirmed by the Contracting Officer.

The means and methods to install the demonstration section jet grout seals must be reproducible during the installation of the production phase. Therefore, all submittal documents describing the proposed demonstration seal installation procedures and parameters shall simulate the conditions for work at the site. The Contractor shall provide a plan discussing the expected results from the demonstration section. The production submittals shall be based on the result of the demonstration section. Contractor shall not proceed with jet grouting operations until Grouting Plan has been approved.

All the above details shall be submitted as part of the [Final Demonstration Test Report](#) for the demonstration section means and methods. The submittal shall also respond to the following items for the locations specified.

3.1.3 Jet Grouting Containment Plan

The Jet Grouting Containment Plan shall consist of the Contractor's methods for wash water and tailings containment. The Contractor shall keep the site clean during jet grouting operations.

3.1.4 Equipment, Materials and Procedures

All equipment used for drilling boreholes; lowering, raising and rotating jet monitors, mixing grout, supplying pressurized grout and air to jet monitors, and jet monitors used to perform the mixing shall have proven performance records for use in jet grouting work. Maintain jet grouting equipment in satisfactory operating condition at all times during the jet grouting work.

3.1.5 Jet Grout Seals and Columns

Install jet grout seals and columns, such that continuous spoil return up the borehole annulus is achieved during all work. Corrective action for mitigating gaps between adjacent drilled shafts are in SECTION 31 63 29 DRILLED CONCRETE PIERS AND SHAFTS and are at no additional cost to the Government. The amount of spoil returned shall be reported to the Contracting Officer daily. The jet grout seals shall be installed in intimate contact with the two adjacent previously installed tangent drilled shafts so that there are no voids or gaps in the finished wall area. The jet grout columns shall be installed as shown in the plans. If jet-grouting is interrupted during the execution of a seal or column, the re-start of the jetting shall be undertaken at least 6 inches below the stopping point.

3.1.6 Cleanup

At completion of daily jet grouting operations, thoroughly clean site and properly dispose of all spoil debris, water, and spilled material. Spoil stockpiling overnight is permitted prior to transfer to a waste disposal site.

The Contractor shall be responsible for all jet grout spoil material waste management including loading/unloading, transport, disposal, disposal fees, and any addition of materials or chemical such as cement, chemical absorbents, or acid (for lowering the pH) as required to facilitate disposal in a timely manner and in accordance with applicable regulations.

3.2 JET GROUTING PERFORMANCE TESTING

3.2.1 Materials Testing

- a. Bentonite: Certificate of Compliance with the specification shall be obtained from the manufacturer for each shipment of bentonite delivered to the site.
- b. Cement: Certificate of Compliance with the specification shall be obtained from the manufacturer for each shipment of cement delivered to the site. Cement shall be Portland Type I or Type II.
- c. Fresh Slurry: A complete series of tests shall be conducted from the mixer or tank containing fresh grout slurry ready for introduction in the borehole at least twice per shift.

3.2.2 Field and Laboratory Testing

The Contractor shall use the QC sampling specified below:

Unconfined Compressive Strength of Jet Grout: Unconfined compression tests shall be performed on core samples of the jet-grouted material. Test specimens from core samples shall be selected at a minimum of one test specimen from the top, middle, and bottom of each seal cored. All grout seal samples tested shall have a minimum 28-day unconfined compressive strength of 100 psi according to [ASTM D2166/D2166M](#). All ground improvement grout column samples tested shall have a minimum 28-day unconfined compressive strength of 150 psi according to [ASTM D2166/D2166M](#).

The following requirements apply to the samples:

- a. The grout core samples shall be wrapped in plastic film and properly labeled to indicate depth and location.
- b. The samples shall be stored in a cool, dry location according to ASTM requirements.
(preferably in a climate-controlled trailer on site the day that they drilled).
- c. The Contractor will make the samples accessible to the Contracting Officer's representative for viewing.
- d. The Government reserves the right to select a small number of samples for compressive strength testing. The unconfined compressive strength for grout seals shall equal or exceed 100 psi at 28 days of curing. The unconfined compressive strength for grout ground improvement columns shall equal or exceed 150 psi at 28 days of curing.
- e. Installed grout material not meeting the required strength or uniformity requirements must be drilled out and replaced. Contractor is required to submit a plan to replace the deficient jet grout seal or column without damaging the existing tangent drilled shafts or utilities.

3.2.3 Performance Testing - Seals

After drilling for the jet grout seals is completed to final depth, the

Contractor shall measure the drilling deviations for each hole. For grout seals, the vertical alignment of the jet grout drill hole shall be compared to the as-built vertical alignment of the two adjacent tangent drilled shafts. If the comparison of vertical alignments indicates that there will not be intimate contact of the jet grout seal for its full depth with the two adjacent tangent drilled shafts, the Contractor shall take measures as required to correct any misalignment. On completion of the seal, the top of the jet grout material shall be shown to be in intimate contact with the adjacent tangent drilled shafts.

For jet grout seals shown to be in intimate contact with the two adjacent tangent drilled shafts, the performance requirements for jet grout seals shall be strength and uniformity as demonstrated by cores from completed seals.

3.2.4 Core Drilling, Uniformity, and Core Samples

During the coring operation, obtain information about the characteristics of the jet grout material that may or may not be apparent from the core recovered from the hole such as but not limited to observing and reporting the drilling action to describe a complete picture of the consistency of the jet grout. A minimum 10 percent of the grout seals and ground improvement shall be cored. No drilling or coring shall be performed in the jet grout (seals and soil improvements) being tested until the material has cured for a minimum of 28 days or to the set-time determined from the approved grout mix design.

When coring, the Contractor's quality control representative shall note the amount of water return compared to the amount being injected through the drill rods to detect potential defects in the jet grout seals or columns. Note and record the drill action and rate of penetration while the drill rod is rotating. Changes in drilling rate can be related to changes in wall composition and provide complimentary data in areas of poor core recovery. Basic information to be included on each core boring log shall include size and type of core bit and barrel used; bit changes; depth; length and time for each run; and amount of core recovered and rock quality designation value (RQD).

All cored samples shall be minimum continuous 4-inch diameter core. Coring of the jet grout seals or columns shall be performed by the Contractor or sub-Contractor performing the jet grout work or drilling contractor, experienced in coring cemented soil, having equipment and procedures that minimize drilling deviations, and capable of obtaining undisturbed core samples. Core recovery for the jet grout columns is critical to the verification of jet grout seal/column continuity. Coring equipment and procedures require Government approval. All core holes shall be backfilled with cement grout.

Full-depth core samples retrieved by the contractor shall be used to evaluate uniformity. Core recovery (expressed as a percentage) is equal to the total length of recovered intact grout core divided by the total core run length. Length of recovered core includes lengths of treated and untreated soil. The minimum core recovery shall be 80 percent. If recovery is lower than required, the Contracting Officer may require additional coring at Contractor's expense.

Percent treatment is calculated as the total length of recovered core minus the sum of the lengths of unmixed soil regions or lumps that extend across the entire diameter of the core divided by the total core run

length expressed as a percentage.

Uniformity is acceptable if percent-treatment is at least 80 percent for every 5-ft core run and unmixed inclusions meet the following criteria:

- a. Full core width unmixed inclusions shall be no more than 6" long.
- b. Unmixed inclusions with a maximum width of half the core diameter shall be no more than 12" long.

If the contractor uses core runs shorter than 5 ft (e.g., 3 ft), then the recovery and percent treatment can be calculated by adding equal amounts of core run length on either side of the short core run length to make up a total 5-ft run length for calculation purposes.

If the minimum percent treatment cannot be confirmed by coring, downhole camera/video can be used to confirm percent treated and uniformity as described in paragraph BOREHOLE CAMERA SURVEYS.

All recovered core not used for testing shall be wrapped in plastic film, preserved, and stored in a laboratory moisture room having constant temperature, saturated environment meeting the requirements of ASTM C 511 until tested or until otherwise directed by the Contracting Officer. All core holes shall be measured for deviations after all samples have been extracted from the coring operation. Location and scheduling of coring shall be as approved by the Contracting Officer.

Test specimens from core samples shall be selected at a minimum of one test specimen from the top, middle, and bottom of each seal cored.

3.2.5 Borehole Camera Surveys

When the minimum 80 percent treatment cannot be confirmed by coring, downhole camera/video may be used at the Government's discretion to visually assess compliance for uniformity. The Contractor shall camera survey core holes where core recovery cannot be achieved. Downhole camera information does not negate the requirement for column replacement at the Contractor's expense if the Government is not satisfied with the uniformity.

- a. Borehole camera surveys shall be taken at the completion of each individual core hole. The core hole shall be flushed with fresh water until clean from drill cuttings prior to any camera surveys.
- b. The Contractor shall upload, on a weekly basis, copies of the visual inspection for each borehole camera survey, to the Contractor provided SFTP and in a format approved by the Contracting Officer.

3.3 RECORDS

3.3.1 Final Jet Grouting Reports

The Contractor shall keep accurate records of all grouting operations and quality control data; such as a log of the grout holes, results of washing and pressure testing operations, time of each change of grouting operation, pressure, rate of pumping, amount of cement for each change in water/cement ratio, and other data as deemed by the Contracting Officer to be necessary. Grout delivery per incremental foot of seal/column shall be reported. Submit these records prior to project closeout. Submit final reports for the following:

- Drilled Shaft Seals
- Drilled Shaft Gap Mitigation (if required)
- Ground Improvement at Stormwater Pipes

3.3.2 Record Drawings of Jet Grout Installation

Submit record drawings showing the location and depths of all jet grout seals, columns, and seals to mitigate gaps between adjacent drilled shafts installed.

3.3.3 360-degree Video Recording

Make a 360 degree video recording of all boreholes performed in jet grout seals and columns, projected as a "flat core", and showing orientation and depth. Format of final video logs shall be high resolution quality.

3.4 COMMUNICATIONS

When, for its own convenience, the Contractor has the individual elements of the plant so located that communication by normal voice between these elements is not feasible or satisfactory, the Contracting Officer may require the Contractor to install a satisfactory alternative means of communications, such as a telephone, radio, or other suitable communications device.

3.5 EMBANKMENT BACKFILL

Backfilling over the Jet-grout treated soil areas shall be performed in phased increments based on the strength-gaining rate of the mix such that a given percent of the design backfill weight is applied only after an equal percent of soil-grout mix strength gaining. The 100 percent level of backfill shall not be placed until the curing time of the soil-grout mix has been shown to achieve a minimum of 150-psi strength. The Contractor shall develop and submit for the Government's review an anticipated strength-gain versus time curve and proposed backfill rate of loading based on demonstration grouting and testing results.

3.6 DRIVING SHEETPILE RESTRICTION

No sheet pile shall be driven within **30 feet** of the jet grouted locations until after 28 days of curing time or test results indicate the jet grout has achieved a compressive strength of **100 psi**.

3.7 REMEDIAL MEASURES

Seals or columns, and seals to mitigate gaps between adjacent drilled shafts not satisfying the acceptance criteria shall be replaced by the Contractor at no extra cost to the Government by the Contractor, using methods approved by the Contracting Officer's representative. If a test fails, the Contractor may test the same seal or column in another quadrant of the grout material at the contractor's expense. If a grout does not meet strength and uniformity, a replacement seal or column shall be installed at the Contractor's expense without damaging the existing tangent drilled shafts or utilities.

3.8 QUALITY CONTROL

The Contractor shall be responsible for project quality control records.

Observations, measurements, and tests described in these specifications shall be performed for quality control. All quality control records, routine testing procedures, observations, and measurements shall be available for inspection by the Contracting Officer's Representative at any time. The laboratory performing the sampling and testing shall be certified by USACE. No work requiring testing will be permitted until laboratory test facilities have been approved and or inspected by the Contracting Officer.

All required quality control testing and reports shall be furnished by the Contractor. The Contractor shall furnish records of all observations, measurements, and tests performed, identified by location and time of testing. These records shall be furnished no later than 24 hours after the tests, measurements, and/or observations were made. All test results used for Quality Control shall be maintained in an electronic data base system. Specification values shall be shown with the test results and shall be updated weekly and provided. Upon completion of the job, an electronic copy shall be submitted.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 31 - EARTHWORK

SECTION 31 36 00

GABION MATTRESS

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 REFERENCES
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Samples
 - 1.4.2 Test Report or Documents
 - 1.4.3 Double Twisted Wire Mesh Gabions
- 1.5 DELIVERY, STORAGE, AND HANDLING

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Double twisted wire mesh Gabions
 - 2.1.1.1 Metallic Coating
 - 2.1.1.2 PVC for Coating
 - 2.1.1.2.1 Specific Gravity
 - 2.1.1.2.2 Tensile Strength
 - 2.1.1.2.3 Modulus of Elasticity
 - 2.1.1.2.4 Hardness
 - 2.1.1.2.5 Brittleness Temperature
 - 2.1.1.2.6 Resistance to Abrasion
 - 2.1.1.2.7 Salt Spray Exposure and Ultra Violet Light Exposure
 - 2.1.1.2.8 Evaluation of Coating After Salt Spray and Ultraviolet Exposure Test
 - 2.1.1.3 Wire Tensile Strength
 - 2.1.1.4 Wire Elongation
 - 2.1.1.5 Mesh Strength and Panel to Panel Joint Strength
 - 2.1.1.6 Ring Fasteners
 - 2.1.1.6.1 Salt Spray Test
 - 2.1.1.6.2 Pull-Apart Resistance Test
 - 2.1.2 Testing
 - 2.1.3 Stone Fill
 - 2.1.3.1 General

PART 3 EXECUTION

- 3.1 FOUNDATION PREPARATION
- 3.2 ASSEMBLY
 - 3.2.1 Double twisted wire mesh Gabions
- 3.3 LACING OPERATIONS
 - 3.3.1 Double Twisted Wire Mesh Gabions
 - 3.3.1.1 Steel Wire Ring Fasteners
- 3.4 INSTALLATION AND FILLING
 - 3.4.1 Placement Tolerances

- 3.4.2 Double Twisted Wire Mesh Gabions
- 3.5 CLOSING

-- End of Section Table of Contents --

SECTION 31 36 00

GABION MATTRESS

PART 1 GENERAL

1.1 SUMMARY

The work under this specification includes furnishing, assembling, filling and tying open wire mesh rectangular compartmented gabions for underwater placement, as specified, and in accordance with the lines, grades, and dimensions shown or otherwise established in the field. Work includes Gabion placement in the wet, such as the erosion protection of the middle pier at planned bridges.

- a. Gabions are wire mesh containers of variable sizes, uniformly partitioned into internal cells, interconnected with other similar units, and filled with stone at the project site to form flexible, permeable, monolithic structures. Gabions shall be manufactured with all components mechanically connected at the production facility with the exception of the mattress lid, which is produced separately from the base. The supply to the jobsite of unassembled individual wire mesh components (panels) forming gabions will not be permitted.
- b. Definitions of terms specific to this specification and to all materials furnished on the jobsite, with the exception of the rock to fill the baskets and the filter material, shall refer and be in compliance with [ASTM A975](#) for double twisted wire mesh Gabions

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A313/A313M	(2017) Standard Specification for Stainless Steel Spring Wire
ASTM A370	(2020) Standard Test Methods and Definitions for Mechanical Testing of Steel Products
ASTM A641/A641M	(2019) Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire
ASTM A764	(2007; R 2017) Standard Specification for Metallic Coated Carbon Steel Wire, Coated at Size and Drawn to Size for Mechanical Springs
ASTM A975	(2011; R 2016) Standard Specification for Double-Twisted Hexagonal Mesh Gabions and Revet Mattresses (Metallic-Coated Steel

	Wire or Metallic-Coated Steel Wire With Poly(Vinyl Chloride) (PVC) Coating
ASTM B117	(2019) Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM D412	(2016) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension
ASTM D746	(2014) Standard Test Method for Brittleness Temperature of Plastics and Elastomers by Impact
ASTM D792	(2013) Density and Specific Gravity (Relative Density) of Plastics by Displacement
ASTM D1499	(2013) Filtered Open-Flame Carbon-Arc Type Exposures of Plastics
ASTM D2240	(2015; E 2017) Standard Test Method for Rubber Property - Durometer Hardness
ASTM G152	(2013) Operating Open Flame Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Installation Plan; G, D0

SD-03 Product Data

Product Data; G, R0

SD-04 Samples

Gabions
Wire Mesh

SD-06 Test Reports

Test Reports; G, R0

SD-07 Certificates

Stone Fill; G, R0
Certificate Of Compliance; G, R0

SD-09 Manufacturer's Field Reports

Mill Certifications; G, R0

1.4 QUALITY ASSURANCE

1.4.1 Samples

Furnish samples of materials used to fabricate the [gabions](#) to the Contracting Officer 60 days prior to start of installation. Samples will be tested in accordance with specification [ASTM A975](#). The Government reserves the right to test additional samples to verify the submitted test records at the Government's expense. When the first test results indicate that the fasteners do not meet the specified requirements, the additional test will be at the Contractor's expense. The fasteners will be rejected after two tests failing to meet the requirements.

1.4.2 Test Report or Documents

Copies of all test results shall be furnished to the Technical Representative of this specification, USACE District, Jacksonville, 701 San Marco Blvd., Jacksonville, FL 32207-8175.

1.4.3 Double Twisted Wire Mesh Gabions

Manufacturer shall demonstrate at least 10 years of continuous experience in manufacturing gabions.

The following tests shall be carried out on minimum three randomly selected field samples of lacing wire

- a. Wire thickness
- b. Tensile strength
- c. PVC coating thickness
- d. Ring fastener individual pull apart strength

Submit the following in addition to above listed documents to the Contracting Officer at least 60 days prior to start of installation for approval:

- a. Manufacturer's [product data](#), technical specifications, and product installation instructions. submit [wire mesh](#)
- b. Samples at least 12 inch by 12 inch to include edge and selvedge wires
- c. Written manufacturer's [certificate of compliance](#). Manufacturer's Certificate of Compliance shall be signed by person authorized to bind the manufacturer's certifications and must have Manufacturer's name and product manufacturing location.
- d. [Test reports](#) from a third party test laboratory to verify the product compliance with ASTM A975.
- e. [Mill certifications](#) of the wire used in manufacturing the product dated within 12 months of the submittal.
- f. Manufacturer's Quality Control Manual along with certificate signed by person authorized to bind the manufacturer's certifications.

1.5 DELIVERY, STORAGE, AND HANDLING

Gabions shall be delivered with all components mechanically connected at the production facility. The front, base, back and lid of the gabions shall be woven into a single unit. The ends and diaphragm(s) shall be factory connected to the base. All perimeter edges of the mesh forming the basket and top, or lid, shall be selvedge with wire having a larger diameter. The gabion is divided into cells by means of diaphragms positioned at approximately 3 ft (1 m) centers. The diaphragms shall be secured in position to the base so that no additional lacing is necessary at the jobsite.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Double twisted wire mesh Gabions

Double twisted wire mesh gabions shall be manufactured with a non-raveling mesh made by twisting continuous pairs of wires through three half turns (commonly called double twisted) to form a hexagonal-shaped opening with a nominal opening size measuring 2-1/2 inch X 3-1/4 inch (nominal). Gabion sizes, wire diameters, mesh opening sizes, and tolerances shall comply with the requirements of [ASTM A975](#) (Tables 1, 3, 4, 5, 6, and Sections 9). Dimensions of the gabion baskets shall be as follows.

a. Width: 3 feet

b. Height: The thickness of individual gabion baskets shall be 1 foot

c. Length: The lengths of empty gabion baskets shall be 6 feet. However, the gabion baskets shall be subdivided into 2 cells.

2.1.1.1 Metallic Coating

The coating weights shall conform to the requirements of [ASTM A641/A641M](#), Class 3, soft temper coating. The adhesion of the zinc coating to the wire shall be such that, when the wire is wrapped six turns around a mandrel having four times the diameter of the wire, it does not flake or crack when rubbing it with the bare fingers, in accordance with [ASTM A641/A641M](#).

2.1.1.2 PVC for Coating

The PVC coating shall show no cracks or breaks after the wires are twisted in the fabrication of the mesh. The initial properties of PVC coating material shall have a demonstrated ability to conform to the following requirements:

2.1.1.2.1 Specific Gravity

In the range from 1.30 to 1.35 dN/dm³, when tested in accordance with test method [ASTM D792](#)

2.1.1.2.2 Tensile Strength

Not less than 2985 psi when tested in accordance with test method ASTM D412

2.1.1.2.3 Modulus of Elasticity

Not less than 2700 psi when tested in accordance with test method ASTM D412

2.1.1.2.4 Hardness

Shore "D" between 50 and 60, when tested in accordance with test method ASTM D2240

2.1.1.2.5 Brittleness Temperature

Not higher than 15 degrees F, or lower temperature when specified by the purchaser, when tested in accordance with test method ASTM D746.

2.1.1.2.6 Resistance to Abrasion

The percentage of the weight loss shall be less than 12 percent according to ASTM D1242.

2.1.1.2.7 Salt Spray Exposure and Ultra Violet Light Exposure

The PVC shall show no effect after 3,000 h of salt spray exposure in accordance with ASTM B117. The PVC shall show no effect of exposure to ultra violet light with test exposure of 3,000 h, using apparatus Spectral Irradiance of Open Flame Carbon Arc with Daylight Filters and 145 degrees F, when tested in accordance with practice ASTM D1499 and ASTM G152

2.1.1.2.8 Evaluation of Coating After Salt Spray and Ultraviolet Exposure Test

After the salt spray test and exposure to ultraviolet light, the PVC coating shall not show cracks nor noticeable change of color, or blisters or splits. In addition, the specific gravity, hardness, tensile strength, modulus of elasticity, resistance to abrasion, and brittleness temperature shall not change more than 6 percent, 10 percent, 25 percent, 25 percent, 10 percent, and +64 degrees F respectively, from their initial values.

2.1.1.3 Wire Tensile Strength

The tensile strength of the wire used for the double twisted mesh and lacing wire, when tested in accordance with Test Methods and definitions ASTM A370, shall have a maximum tensile strength of 75,000 psi (515 MPa), in accordance with the requirements of ASTM A641/A641M for soft temper wire.

2.1.1.4 Wire Elongation

Elongation test must be carried out on a sample at least 12 in. (30 cm) long. Elongation shall not be less than 12%, in accordance with ASTM A370.

2.1.1.5 Mesh Strength and Panel to Panel Joint Strength

The minimum strength requirements of the mesh, selvedge wire to mesh connection, panel to panel connection, and punch test, when tested in accordance with ASTM A975 Section 13.1, shall be as shown in Table 1. The

strength values reported in lb/ft are referred to the unitary width of the specimen. The panel to panel test shall demonstrate the ability of the fastening system to achieve the required strength, and indicate the number of wire revolutions for the lacing wire or the ring spacing for ring fasteners used. The same number of wire revolutions or ring spacing shall be used in the field installation. Pleating the based panel to obtain internal panels is prohibited.

TABLE 1 Minimum Strength Requirements of Mesh and Connections	
Test Description	Gabions, PVC coated
Mesh Tensile Strength	3425 lb/ft
Connection to selvages	1200 lb/ft
Panel to panel (using lacing wire or ring fasteners)	1200 lb/ft
Punch Test	5300 lb

2.1.1.6 Ring Fasteners

The tensile strength of the zinc-coated steel wire used for fasteners shall be in accordance with the requirements of [ASTM A764](#), Type A, B, or C, Table 2 or Table 3. Stainless Steel overlapping fasteners may be used in lieu of, or in conjunction with lacing wire for basket assembly and installation. The tensile strength of stainless steel wire used for fasteners shall be in accordance with the requirements of [ASTM A313/A313M](#), Type 302, Table 2. Any fastener system shall give the number of fasteners required to comply with TABLE 1, in accordance with ASTM A975 (Section 13.1.2) for woven wire gabions. Ring fasteners shall not be installed more than 4 inches apart. Each fastener type shall be closed and the free ends of the fastener shall overlap a minimum of 1 inch. The Manufacturer or supplier shall state the number of fasteners required for all vertical and horizontal connections for single and multiple basket joining. Approved ring fasteners including fasteners made of stainless steel shall be subject to the salt spray test and pull-apart resistance test and shall be documented by actual testing of panel to panel connections within the last year by validated laboratories.

2.1.1.6.1 Salt Spray Test

A set of two identical rectangular gabion panels, each with a width about 10-1/2 mesh openings along a selvedge wire, shall be joined by properly installed wire fasteners along the two selvedge wires so that each fastener confines two selvedge and two mesh wires. If the fasteners are also to be used to joint two individual empty gabion baskets, two additional selvedge wires which are each mechanically wrapped with mesh wires shall be included so that each fastener confines four selvedge and four mesh wires. The set of the jointed panels shall be subject to salt spray test, [ASTM B117](#), for a period of not less than 48 hours. At the end of the test, the fasteners, the selvedge, or mesh wires confined by the fasteners shall show no rusty spots on any part of the surface excluding

the cut ends. A properly installed fastener shall meet the following requirements:

- a. Each interlocking fastener shall be in a locked and closed position.
- b. Each ring fastener shall be closed, and the free ends of the fastener shall overlap a minimum of 1 inch.

2.1.1.6.2 Pull-Apart Resistance Test

A new set of the jointed panels, which are prepared by the same method as specified in the salt spray test but without being subject to the 48-hour salt spray test, shall be mounted on a loading machine with grips or clamps such that the panels are uniformly secured along the full width. The grips or clamps shall be designed to transmit only tension forces. The load will then be applied at a uniform rate of 50 lbs/sec until failure occurs. The failure is defined as when the maximum load is reached and a drop of strength is observed with subsequent loading or the opening between any two closest selvedge wires, applicable to a fastener confining either two or four selvedge wires, becomes greater than 2 inches at any place along the panel width. The strength of the jointed panels at failure shall have a minimum as indicated in TABLE 1.

2.1.2 Testing

Test records made within one year by certified laboratories and Government agencies will be used to determine the acceptability of the fastening system. Samples of wire fasteners and samples of material for fabricating the gabions with their certified test records shall be submitted at least 60 days in advance to the Contracting Officer for approval. The Government reserves the right to test additional samples to verify the submitted test records at the Government's expense. When the first test results indicate that the fasteners do not meet the specified requirements, Government reserves the right to request additional testing which shall be at the Contractor expense. The fasteners will be rejected after two tests failing to meet the requirements.

2.1.3 Stone Fill

Submit a certificate or affidavit signed by a legally authorized official of the supplier of the stone fill that it meets the quality required and gradation limits specified.

2.1.3.1 General

Unless otherwise specified, the rock used to fill gabions shall conform to the requirements for Bedding Material under section 35 31 19 STONE AND CHANNEL PROTECTION FOR STRUCTURES. For gabions, the ability to function properly depends upon their stability, which is partly depending upon the rocks filling them. Rock sizes should be chosen to prevent them from falling through the mesh of the gabions. The rock has also to withstand natural weathering processes during the life of the project that would cause it to breakdown to sizes smaller than the wire mesh opening dimensions. Rock to fill gabions shall be durable and of suitable quality to ensure permanence in the structure and climate in which it is to be used.

PART 3 EXECUTION

3.1 FOUNDATION PREPARATION

Gabions shall be placed underwater and divers or other suitable methods shall be used to ensure that the bed is free of logs, large rocks, construction materials, or other blocky materials that would create voids beneath the system. Immediately prior to placing the gabions, the prepared subgrade must be inspected.

3.2 ASSEMBLY

3.2.1 Double twisted wire mesh Gabions

The gabions shall be assembled as directed by the Manufacturer.

3.3 LACING OPERATIONS

3.3.1 Double Twisted Wire Mesh Gabions

Either lacing wire or ring fasteners are permitted to lace double twisted wire mesh gabions as required and directed by the Manufacturer.

3.3.1.1 Steel Wire Ring Fasteners

When steel wire ring fasteners are used, the rings shall be installed at the top and bottom connections of the end and center diaphragms. The ring spacing shall be based on the minimum pull apart strength as specified in TABLE 1. In any case, the maximum ring spacing along the edges shall not exceed 6 inches. The use of either a mechanical or a pneumatic fastening tool for steel wire ring fasteners is required. Ring fasteners shall be galvanized, stainless steel or Zn-5 percent aluminum-mischmetal alloy coated.

3.4 INSTALLATION AND FILLING

Prior to construction, Contractor shall coordinate with gabion manufacturer to develop and submit an [installation plan](#) to the Contracting Officer specifying Contractor's means and methods of gabion unit placement and verification of proper placement. Contractor must receive approval of the plan from the Contracting Officer prior to installation of gabion units. Empty gabion units shall be assembled per Manufacturer recommendations and placed on the approved surface to the lines and grades as shown or as approved, with the sides, ends, and diaphragms erected in such a manner to ensure the correct position of all creases and that the tops of all sides are level. Geotextile as shown in the contract drawings and as specified in SECTION 31 05 19 GEOTEXTILE shall be attached to the bottom of gabion units using ring fasteners prior to placement for any gabion units in contact with the channel bottom. The second vertical layer of gabion units resting on the first vertical layer does not require geotextile fabric to be attached. All gabion units shall be properly staggered horizontally and vertically as shown in the construction drawings. Finished gabion structures shall have no gaps along the perimeter of the contact surfaces between adjoining units. All adjoining empty gabion units shall be connected along the perimeter of their contact surfaces in order to obtain a monolithic structure. All lacing wire terminals shall be securely fastened. All joining shall be made through selvedge-to-selvedge or selvedge-to-edge wire connection; mesh-to-mesh or selvedge-to-mesh wire connection is prohibited except in the case where

baskets are offset or stacked and selvedge-to-mesh or mesh-to-mesh wire connection would be necessary. As a minimum, a fastener shall be installed at each mesh opening at the location where mesh wire meets selvedge or edge wire.

- a. The initial line of basket units shall be placed on the prepared foundation and adjoining empty baskets set to line and grade, and common sides with adjacent units thoroughly laced or fastened. They shall be placed in a manner to remove any kinks from the mesh and to a uniform alignment. The basket units then shall be filled to provide anchorage against deformation and displacement during the filling operation. The stone shall be placed in the units as specified by the Manufacturer.
- b. Undue deformation and bulging of the mesh shall be corrected prior to further stone filling. Care shall be taken, when placing the stone by hand or machine, to assure that the PVC coating on gabions will not be damaged. All visible faces shall be filled with some hand placement to ensure a neat and compact appearance and that the void ratio is kept to a minimum.
- c. Uniformly overfill gabions by about 1 to 2 inches to compensate for future rock settlements. Gabions can be filled by any kind of earth-filling equipment, such as a backhoe, gradall, crane, etc. The maximum height from which the stones may be dropped into the baskets shall be 3 to 4 feet. If PVC coated materials are used, no work shall take place unless the ambient temperature is above 20 degrees F.

3.4.1 Placement Tolerances

The Contractor shall document the correct placement of gabion underwater to within the tolerances specified by using one of the following methods: sounding surveys using a sounding pole or sounding basket on a lead line, divers, or sonar bottom profiles.

Gabions shall be placed to the lines and grades shown on the contract drawings within (+\-) 2 inches vertically and (+\-) 12 inches horizontally.

3.4.2 Double Twisted Wire Mesh Gabions

After the foundation has been prepared and inspected, the pre-assembled gabions shall be placed in their proper location to form the structure. Gabions shall be connected together and aligned before filling the baskets with rock. All connections (panel-to-panel) and basket-to-basket shall be already carried out as described in paragraph ASSEMBLY. Stone fill shall have a gradation of 4 to 8 inches, as described in paragraph Gradation, and shall be placed in 1 foot lifts. Cells shall be filled to a depth not exceeding 1 foot at a time. The fill layer should never be more than 1 foot higher than any adjoining cell. Stiffeners or internal cross ties shall be installed in all front and side of the gabions at 1/3 and 2/3 of the height for 3 feet or higher gabions, as the cell is being filled. Stiffeners shall be installed in the center of the cells. In 1.5 foot high units, stiffeners or internal crossties are not required. Internal cross ties, or alternatively the preformed stiffeners, shall be looped around three twisted wire mesh openings at each basket face and the wire terminals shall be securely twisted to prevent their loosening. The number of voids shall be minimized by using a well-graded stone in order to achieve a dense, compact stone fill. All corners shall be securely

connected to the neighboring baskets of the same layer before filling the units. When more than one layer of gabions is required, in order for the individual units to become incorporated into one continuous structure, the next layer of gabions shall be connected to the layer underneath after this layer has been securely closed. Gabions shall be uniformly overfilled by about 1 to 2 inches to compensate for future rock settlements.

3.5 CLOSING

Lids shall be tightly secured along all edges, ends and diaphragms in the same manner as described for assembling. Adjacent lids may be securely attached simultaneously. The panel edges shall be pulled to be connected using the appropriate closing tools where necessary. Single point leverage tools, such as crowbars, may damage the wire mesh and shall not be used. All end wires shall then be turned in.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 31 - EARTHWORK

SECTION 31 41 16

METAL SHEET PILING

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 TEMPORARY SHEET PILE DESIGN
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Material Certificates
- 1.5 QUALITY CONTROL
 - 1.5.1 Work Plan
 - 1.5.2 Shop Drawings
 - 1.5.3 Monitoring Plan
 - 1.5.4 Sheet Pile Contractor Experience
 - 1.5.5 Surveyor Experience
 - 1.5.6 Supervisor Experience
 - 1.5.7 Rig Operator Experience
- 1.6 DELIVERY, STORAGE, AND HANDLING
- 1.7 PROJECT CONDITIONS

PART 2 PRODUCTS

- 2.1 STEEL SHEET PILING
 - 2.1.1 Interlocks
 - 2.1.2 General Requirements
- 2.2 APPURTENANT METAL MATERIALS
- 2.3 TESTS, INSPECTIONS, AND VERIFICATIONS
 - 2.3.1 Materials Tests

PART 3 EXECUTION

- 3.1 EARTHWORK
- 3.2 PROTECTION OF EXISTING STRUCTURES AND MONITORING
 - 3.2.1 Movement Monitoring and Protection of Existing Structures
 - 3.2.2 Vibration Monitoring
 - 3.2.3 Movement Monitoring of Sheet Pile Wall and Excavation
- 3.3 SHEET PILE INSTALLATION
 - 3.3.1 Pile Driving Equipment
 - 3.3.1.1 Driving Hammers
 - 3.3.1.2 Jetting Equipment
 - 3.3.2 Placing and Driving
 - 3.3.2.1 Placing
 - 3.3.2.2 Driving
 - 3.3.3 Refusal Criteria
 - 3.3.4 Cutting-Off and Splicing
 - 3.3.5 Inspection of Driven Piling
 - 3.3.6 Pulling and Redriving
- 3.4 EXCAVATION AND BACKFILL

3.5 REMOVAL

3.5.1 Pulling

3.5.2 Sorting, Cleaning, Inventorying and Storing for Reuse

3.6 INSTALLATION RECORDS

-- End of Section Table of Contents --

SECTION 31 41 16

METAL SHEET PILING

PART 1 GENERAL

This section applies to all metal sheet piling work except the Siphon coffer cell temporary retaining structure to be located at approximately Station 100+00. The Siphon temporary retaining structure works are addressed in Section 31 52 10 TEMPORARY RETAINING STRUCTURES.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A6/A6M (2017a) Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling

ASTM A572/A572M (2021; E 2021) Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 1110-2-2504 (1994) Design of Sheet Pile Walls

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Designation R0 stands for Resident Office and designation D0 stands for District Office. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Work Plan; G, D0

Monitoring Plan; G, D0

SD-02 Shop Drawings

Steel Sheet Piling; G, D0

Sheet Pile Splicing; G, D0

SD-03 Product Data

Driving
Pile Driving Equipment; G, D0
Pulling and Redriving; G, D0

SD-05 Design Data

Design Calculations; G, D0
The contractor shall submit for approval, the design calculation and design data if any additional or redesign of sheet pile wall is performed by the contractor.

SD-06 Test Reports

Materials Tests

SD-07 Certificates

Sheet Pile Contractor Experience; G, D0

Sheet Pile Design Experience; G, D0

Surveyor Experience; G, D0

Supervisor Experience; G, D0

Rig Operator Experience; G, D0

SD-11 Closeout Submittals

Pile Driving Record

1.3 TEMPORARY SHEET PILE DESIGN

The temporary sheet piles as shown on the drawings, are part of a recommended discretionary construction sequence to the contractor. The sheet piles were designed in compliance with EM 1110-2-2504 and all the applicable references. The contractor has the option of using other hot-rolled sheet pile sections that meet the minimum requirements as shown on the drawings. The contractor shall be responsible for modifications to the design of the temporary sheet piles if any variation occurs on the alignment and/or design details of the sheet piles presented on the drawings. Modifications to the temporary sheet piles shall be designed by geotechnical and/or structural engineers, licensed in the Commonwealth of Puerto Rico, with a minimum of 10 years experience with sheet pile design and construction. Any variation on the temporary sheet pile design and alignment proposed by the Contractor shall be provided to the Government for review and shall be at the Contractor's expense.

Variations to the temporary sheet piles shall be designed in compliance with EM 1110-2-2504 and all the applicable references. It shall be designed for several load conditions as specified in the EM 1110-2-2504, as minimum; construction, flood drawdown load, and earthquake load conditions that were analyzed for the current design of the sheet piles shall be considered. For the construction case, the construction surcharge loadings shall be based on the contractor's equipment, but it should not exceed the surcharge load used in the current design as specified on the

plans. The construction load case shall be designed with and without applicable live loads for both short-term (Q-Case) and long-term (S-Case) soil strength conditions with groundwater surface elevation on the land side at minimum of 5.0 feet MSL and the channel water level at 0.0 feet MSL. For the flood drawdown event, the contractor shall design the temporary sheet pile for a water surface elevation on the land side at the working platform elevation with the channel water level lowered to 0.0 Feet MSL. For the earthquake event, the contractor shall design the temporary sheet pile for pseudo-static seismic coefficient of 0.13g with the water levels used for the construction case. The elevations of the temporary sheet piles are presented on the drawings. Any deviation of the sheet pile alignment to the one presented on the drawings shall be submitted to the Contracting Officer as a variation for evaluation and approval.

1.4 QUALITY ASSURANCE

1.4.1 Material Certificates

For each shipment, submit certificates identified with specific lots prior to installing piling. Include in the identification data piling type, dimensions, chemical composition, mechanical properties, section properties, heat number, and mill identification mark.

1.5 QUALITY CONTROL

1.5.1 Work Plan

The contractor shall develop and submit a work plan that shall at minimum include the following:

- a. Construction sequence.
- b. Description of means and methods of construction.
- c. Description of procedures to be used for underwater construction tasks including removal of underwater sheet piles.
- d. Description of equipment to be used to perform the sheet pile installation work. Results of pile drivability studies used for selecting pile driving method and equipment. Also include, in accordance with Section SHEET PILE INSTALLATION, verification of refusal criteria based on the manufacturer recommendations and drivability studies of the selected method/equipment.
- e. Proposed staging and laydown area.
- f. Safety controls for the equipment/material use and containment of drill spoils.

1.5.2 Shop Drawings

- a. **Steel Sheet Piling:** Submit detail drawings for sheet piling, including fabricated sections, showing complete piling dimensions and driving sequence. Include in the drawings the details of top protection, special reinforcing tips, tip protection, lagging, splices, fabricated additions to plain piles, cut-off method, corrosion protection, and dimensions of templates and other temporary guide structures for installing piling. Provide details

of the method for handling piling to prevent permanent deflection, distortion or damage to piling interlocks.

- b. **Sheet Pile Splicing:** Submit detail drawings of pile splices. Clearly indicate welds using standard AWS notations and symbols, and clearly show or indicate size, length, and type of each weld, as applicable. Welding of splices shall conform to the requirements of Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS.
- c. **Redesigned Steel Sheet Piling:** Submit complete details of Steel Sheet Piling if any redesigned by the Contractor as per Sections TEMPORARY SHEET PILE DESIGN. The shop drawings shall bear the stamp and signature of the Registered Professional Engineer, licensed in the Commonwealth of Puerto Rico. These drawings shall clearly show:
 - 1. All pertinent dimensions and locations of the redesigned sheetings with reference to the project centerline (Wall-line, Baseline, etc.).
 - 2. Material grade, weight, length and designation of steel sheet pile section(s) used.
 - 3. Bracing if considered and all connection details as applicable

1.5.3 Monitoring Plan

In conjunction with specification Section 02 22 13 VIBRATION AND NOISE MONITORING REQUIREMENTS, a monitoring plan shall be submitted and approved, at least 30 days prior to the start of any land disturbing work. Provide a written monitoring plan with but not limited to the following:

- a. Schedule and sequencing of monitoring activities and reporting as required for Section titled PROTECTION OF EXISTING STRUCTURES AND MONITORING.
- b. Drawings indicating reference points of existing structures or utilities.
- c. Description of pre-construction survey methods, documenting the details of the structures identified for the survey.
- d. Description of monitoring scheme proposed for the sheet pile installation and excavations.
- e. Quantity and location of all instrumentation and survey monitoring points.
- f. Maximum allowable movements as specified and procedures for detection of vertical or horizontal movement of sheet pile wall, bottom of excavations for heave, and in existing structures as/if required.
- g. Description of how monitoring monuments and instrumentation will be protected from damage and vandalism, and the details of repair or replacement procedure if any monuments and instrumentation are damaged.

h. Examples of daily field data recording sheets.

1.5.4 Sheet Pile Contractor Experience

Sheet Pile Design Experience: The contractor has the option to design the temporary sheet pile walls if/where needed and submit **design calculations** as detailed in Section 1.3. The temporary sheet pile shall be designed by geotechnical and/or structural engineers, licensed in the Commonwealth of Puerto Rico, with a minimum of 10 years' experience with sheet pile design and construction. The qualifications and experience of sheet pile designers shall be submitted to the government for review.

1.5.5 Surveyor Experience

The contractor shall employ qualified Surveyor, licensed in the Commonwealth of Puerto Rico, with a minimum of 5 years' experience with construction surveying and monitoring, to monitor and record monitoring data. The qualifications and experience of the surveyor shall be submitted to the government for review.

1.5.6 Supervisor Experience

On-site supervisors for each sheet piling crew shall have a minimum of five (5) years' experience in same capacity in construction of similar in water projects. The work experience shall be direct supervisory responsibility for the on-site sheet pile construction operation. The Government reserves the right to suspend the construction work if unapproved personnel are substituted for one or more of the approved supervisors. The Contractor shall be fully liable for the additional costs resulting from the suspension of work and no adjustments in the contract time resulting from the suspension of the work will be allowed.

1.5.7 Rig Operator Experience

Rig operators shall have a minimum of five (5) years prior experience as a rig operator for driving sheeting in similar site conditions.

1.6 DELIVERY, STORAGE, AND HANDLING

Materials delivered to the site for permanent sheet piles shall be new and undamaged and shall be accompanied by certified test reports. Provide the manufacturer's logo and mill identification mark on the sheet piling as required by the referenced specifications. Store and handle sheet piling in the manner recommended by the manufacturer to prevent permanent deflection, distortion or damage to the interlocks; as a minimum, support on level blocks or racks spaced not more than **10 feet** apart and not more than **2 feet** from the ends. Storage of sheet piling should also facilitate required inspection activities and prevent damage to coatings and corrosion protection prior to installation. Handle sheet piling over **80 feet** in length using a minimum of two pickup points.

Temporary sheet piling shall be undamaged (new or used), inspected and approved for use, accompanied by certified test reports. The utilization of previously used sheet piles shall be approved on a case-by-case basis, and shall require Government inspection. The Contractor shall be liable for any unforeseen complications or delays caused by reuse of sheet piles.

1.7 PROJECT CONDITIONS

- a. Existing Utilities: Do not interrupt utilities unless allowed by the Contracting Officer.
- b. The contractor shall obtain measurements at the work sites to verify layout information and dimensions for sheet piling in relation to reference points indicated in the contract documents.

PART 2 PRODUCTS

2.1 STEEL SHEET PILING

Submit detail drawings for sheet piling, including fabricated sections, showing complete piling dimensions and details, driving sequence and location of installed piling.

- a. Metal sheet piling shall be hot-rolled steel sections conforming to **ASTM A572/A572M**, Grade 50
- b. Metal sheet piling shall be of the type and dimensions indicated on the contract documents and be of a design such that when in place they will be continuously interlocked throughout their entire length.
- c. Sheet piling shall be provided with standard pulling holes located approximately 4 inches below the top of the pile unless otherwise shown or directed.
- d. For protection of all permanent sheet pile walls, including the channel transition and wing walls, that are identified on the contract drawing C0001, coat both sides of the sheet piles up to El. -5 or 15-ft below the finish grade, whichever is deeper, in accordance with **Section 09 97 02 PAINTING: COAL TAR EPOXY**. The coating work shall be performed, as indicated on the contract drawing C0001, for the sheet piles of all permanent walls, channel transition walls, and the wing walls. The temporary sheet piling, that are to be pulled out upon completion of construction, shall remain uncoated.

2.1.1 Interlocks

The interlocks of sheet piling shall be free-sliding, provide a swing angle suitable for the intended installation but not less than 5 degrees when interlocked, and maintain continuous interlocking when installed.

2.1.2 General Requirements

Sheet piling shall be full-length sections of the dimensions shown. Provide fabricated sections conforming to the requirement and the piling manufacturer's recommendations for fabricated sections. Provide sheet piling with standard pulling holes. Metalwork fabrication for sheet piling shall be as specified in **Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS**.

2.2 APPURTENANT METAL MATERIALS

Provide metal plates, shapes, bolts, nuts, rivets and other appurtenant fabrication and installation materials conforming to manufacturer's standards and to the requirements specified in the respective sheet piling standards and in **Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS**.

2.3 TESTS, INSPECTIONS, AND VERIFICATIONS

Requirements for material tests, workmanship and other measures for quality assurance shall be as specified and in Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS.

2.3.1 Materials Tests

Submit certified materials tests reports showing that sheet piling and appurtenant metal materials meet the specified requirements, for each shipment and identified with specific lots prior to installing materials. Material test reports shall meet the requirements of ASTM A6/A6M. Perform materials tests conforming to the following requirements. Sheet piling and appurtenant materials shall be tested and certified by the manufacturer to meet the specified chemical, mechanical and section property requirements prior to delivery to the site. Testing of sheet piling for mechanical properties shall be performed after the completion of all rolling and forming operations. Testing of sheet piling shall meet the requirements of ASTM A6/A6M.

PART 3 EXECUTION

3.1 EARTHWORK

Perform in accordance with Section 31 00 00 EARTHWORK. Backfill as indicated.

3.2 PROTECTION OF EXISTING STRUCTURES AND MONITORING

3.2.1 Movement Monitoring and Protection of Existing Structures

a. The Contractor shall take all reasonable precautions to prevent damage to existing structures, ground loss or ground heave. The Contractor shall monitor structures for movement in a manner approved by the Contracting Officer, recording elevations to 0.001 foot. Monitor any existing structures within a distance of 100-ft of the sheet pile wall alignment.

b. The Contractor is responsible for evaluating the need for, design of, and providing any necessary features to protect adjacent structures. When sheeting and shoring are used, employ a Qualified Engineer, licensed in the Commonwealth of Puerto Rico, with a minimum of 10 years' experience with sheet pile design and construction, to design the sheeting and shoring, and to sign and seal the plans and specification requirements. Send these designs to the Contracting Officer for his record before beginning construction.

c. In conjunction with specification Section 02 22 13 VIBRATION AND NOISE MONITORING REQUIREMENTS, The contractor shall establish and identify monitoring points on existing structures, located within 100 feet of the sheet pile wall alignment, and on other structures or locations as determined by the Government.

d. The contractor shall survey the structures and the reference points and their pre-construction condition and elevation details prior to commencing any construction work and shall perform continuous daily monitoring during the sheet pile driving and

excavation period.

e. Adequately document the condition of existing structures and all existing cracks with descriptions and pictures. In conjunction with specification Section 02 22 13 VIBRATION AND NOISE MONITORING REQUIREMENTS, prepare two reports documenting the condition of the structures; one report before beginning foundation construction operations and a second report after completing construction operations. The Government will take ownership of both reports.

f. Qualified Surveyor shall be employed to survey all structures and the monitoring points.

g. All data shall be recorded by the Surveyor, and furnished to the contracting officer. Date-stamped photographs and records documenting the conditions at monitoring of the identified monitoring points shall be submitted along with the monitoring reports unless otherwise approved by the contracting officer.

h. Notify the Contracting Officer of any movements of existing structures detected in excess of 1/16 inch and immediately take any remedial measures required to prevent damage to the existing structures.

3.2.2 Vibration Monitoring

a. Provisions for monitoring nearby structures for vibrations shall conform to Section 02 22 13 VIBRATION AND NOISE MONITORING REQUIREMENTS.

b. If measured vibrations exceed the limits, the contractor shall immediately stop work and notify the Government and the contracting officer and take necessary actions to reduce the vibrations to acceptable limits. Work shall not proceed until the contractor has the approval of the Government.

3.2.3 Movement Monitoring of Sheet Pile Wall and Excavation

a. The contractor shall establish reference points to monitor vertical and horizontal movement of the sheet pile walls and any settlement and heave respectively outside and inside the excavations. Employ qualified Surveyor to monitor and record the data.

b. Fixed reference points not subjected to movement due to the work activities shall be established. The reference points shall be maintained and protected during the project duration. The reference points may be used to reestablish the monitoring points if needed.

c. The contractor shall monitor any heave (vertical deformations) of the excavation bottom using magnet extensometers. No heave monitoring is required for below water excavations.

d. The contractor shall immediately stop work and notify the Government and the contracting officer of any measured sheet pile deflection exceeding 0.33 feet or upon detecting settlement or heave of 0.25 feet. Work shall not proceed until the contractor has the approval of the Government.

3.3 SHEET PILE INSTALLATION

3.3.1 Pile Driving Equipment

Submit complete descriptions of sheet piling driving equipment including hammers, extractors, protection caps and other installation appurtenances, prior to commencement of work. Descriptive information includes manufacturer's name, model numbers, capacity, rated energy, hammer details, cushion material, helmet, and templates. Provide pile driving equipment conforming to the following requirements.

3.3.1.1 Driving Hammers

Hammers shall be steam, air, or diesel drop, single-acting, double-acting, differential-acting, hydraulic (press-in), or vibratory type. The driving energy of the hammers shall be sized as recommended by the manufacturer for the piling weights and subsurface materials to be encountered. Repair damage to piling caused by use of a pile hammer with excess delivered force or energy.

3.3.1.2 Jetting Equipment

Jetting shall not be used unless otherwise authorized by the Contracting Officer.

3.3.2 Placing and Driving

3.3.2.1 Placing

- a. Pilings shall be carefully located as indicated on the contract drawings. Pilings properly placed and driven shall be interlocked throughout their length with adjacent pilings to form a continuous diaphragm throughout the length or run of piling wall.
- b. Pilings shall be placed plumb with out-of-plumbness not exceeding 1/8 inch per foot of length and true to line. Place the pile so the face will not be more than 6 inches from vertical alignment plumb at any point. Top of pile at elevation of cut-off shall be within 1/2 inch horizontally and 2 inches vertically of the location indicated. Manipulation of piles to force them into position will not be permitted. Check all piles for heave. Redrive all heaved piles to the required tip elevation.
- c. Provide temporary wales, templates, or guide structures to ensure that the pilings are placed and driven to the correct alignment. Use a system of structural framing sufficiently rigid to resist lateral and driving forces and to adequately support the sheet piling until design tip elevation is achieved. Use two templates, at least, when placing each piling. Templates shall not move when supporting sheet piling. Fit templates with wood blocking to bear against the web of each alternate sheet pile and hold the sheet pile at the design location alignment. Provide outer template straps or other restraints as necessary to prevent the sheets from warping or wandering from the alignment. Mark template for the location of the leading edge of each alternate sheet pile. If in view, also mark the second level to assure that the piles are vertical and in position. If two guide marks cannot be seen, other means shall be used to keep the sheet pile vertical along its leading edge.

3.3.2.2 Driving

All sheet pilings shall be driven to the required minimum tip elevations indicated on the contract drawings unless otherwise approved by the Contracting Officer. Submit records of the completed sheet piling driving operations, including a system of identification which shows the disposition of approved piling in the work, driving equipment performance data, piling penetration rate data, piling dimensions and top and bottom elevations of installed piling. Prior to driving pilings in water, paint a horizontal line on both sides of each piling at a fixed distance from the bottom so that it will be visible above the water line after installation. This line shall indicate the profile of the bottom elevation of installed pilings and potential problem areas can be identified by abrupt changes in its elevation. Drive pilings with the proper size hammer and by approved methods so as not to subject the pilings to damage and to ensure proper interlocking throughout their lengths.

- a. Maintain driving hammers in proper alignment during driving operations by use of leads or guides attached to the hammer. Caution shall be taken in the sustained use of vibratory hammers when a hard driving condition is encountered to avoid interlock-melt or damages. Discontinue the use of vibratory hammers when penetration rate due to vibratory loading is one foot or less per minute, and then use impact hammers to reach the desired tip elevation.
- b. Employ a protecting cap in driving when using impact hammers to prevent damage to the tops of pilings. Use cast steel shoe to prevent damage to the tip of the sheet piling. Remove and replace pilings damaged during driving or driven out of interlock at the Contractor's expense.
- c. Drive pilings without the aid of a water jet unless otherwise authorized by the Contracting Officer.
- d. Take adequate precautions to ensure that pilings are driven plumb. Where possible, drive Z-pile with the ball end leading. If an open socket is leading, a bolt or similar object placed in the bottom of the interlock will minimize packing material into it and ease driving for the next sheet. If at any time the forward or leading edge of the piling wall is found to be out-of-plumb in the plane of the wall the piling being driven shall be driven to the required depth and tapered pilings shall be provided and driven to interlock with the out-of-plumb leading edge or other approved corrective measures shall be taken to insure the plumbness of succeeding pilings. The maximum permissible taper for any tapered piling shall be **1/8 inch per foot** of length.
- e. Pilings in each run or continuous length of piling wall shall be driven alternately in increments of depth to the required depth or elevation. No piling shall be driven to a lower elevation than those behind it in the same run except when the pilings behind it cannot be driven deeper. Incrementally sequence driving of individual piles such that the tip of any sheet pile shall not be more than **4 feet** below that of any adjacent sheet pile. When the penetration resistance exceeds five blows per **inch**, the tip of any sheet pile shall not be more than **2 feet** below any adjacent sheet pile. If the piling next to the one being driven tends to follow below final

elevation it may be pinned to the next adjacent piling.

- f. As indicated on Contract documents, the design tip elevations of selected sheet pile sections extend into hard materials including weathered rock encountered within residual soils. As such, difficult driving conditions to achieve the required tip elevation are anticipated. If obstructions restrict driving a piling to the specified penetration, the obstructions shall be removed or penetrated with a chisel beam or alternative methods approved by the Contracting Officer. A tolerance of 3 inches above the indicated top elevation will be permitted.
- g. Pilings shall not be driven within 100 feet of concrete less than 7 days old.
- h. No sheet pile shall be driven within 30 feet of the jet grouted locations until after 28 days of curing time or test results indicate the jet grout has achieved a compressive strength of 100 psi.
- i. Pre-augering or spudding of piles shall not be used unless otherwise authorized by the Contracting Officer.

3.3.3 Refusal Criteria

- a. Refusal criteria for sheet piling, depending on the driving method and equipment to be used, should be verified with drivability studies and manufacturer recommendations, and should be submitted for approval as per paragraph WORK PLAN.
- b. The followings shall be considered when selecting the driving method, equipment size, and determining the associated driving refusal:
 - 1. Able to drive the sheet pilings to the required minimum tip elevations, with no damage to the pilings and maintaining proper pile interlocking throughout their lengths.
 - 2. Verification with wave equation analysis that the selected driving system will produce no overstressing of the pile.
 - 3. The following conditions do not constitute practical driving refusal:
 - i. The approved hammer system is not operating correctly or
 - ii. The hammer is not operated to the maximum rated operating energy as recommended by the hammer manufacturer.
- c. Sheet pile penetration of 1 inch per 20 blows shall be considered as the refusal limit when driving the sheet piles with impact hammers if used for the project. The refusal criteria for the selected hammer should be verified based on the hammer manufacturer recommendations and drivability studies.

3.3.4 Cutting-Off and Splicing

Pilings driven to the required tip elevation or a refusal point where additional penetration cannot be attained and are extending above the required top elevation in excess of the specified tolerance shall be cut off to the required elevation upon obtaining authorization from the

Contracting Officer. Pilings driven below the required top elevation and pilings damaged by driving and cut off to permit further driving shall be extended as required to reach the top elevation by splicing at no additional cost to the Government. If directed, pilings shall be spliced as required to drive them to depths greater than shown and extend them up to the required top elevation.

- a. Pilings adjoining spliced pilings shall be full length unless otherwise approved. Splicing of pilings shall be as indicated. Ends of pilings to be spliced shall be squared before splicing to eliminate dips or camber. Pilings shall be spliced together with concentric alignment of the interlocks so that there are no discontinuities, dips or camber at the abutting interlocks. Spliced pilings shall be free sliding and able to obtain the maximum swing with contiguous pilings. Welding of splices shall conform to the requirements of Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS. Shop and field welding, qualification of welding procedures, welders, and welding operators shall be in accordance with AWS D1.1/D1.1M.
- b. The tops of pilings excessively battered during driving shall be trimmed when directed, at no cost to the Government. Piling cut-offs shall become the property of the Contractor and shall be removed from the site.
- c. Cut holes in pilings for bolts, drains or utilities in a neat and workmanlike manner, as shown or as approved. Use a straight edge in cuts made by burning to avoid abrupt nicks. Bolt holes in steel piling shall be drilled or may be burned and reamed by approved methods which will not damage the surrounding metal. Holes other than bolt holes shall be reasonably smooth. Do not use explosives for cutting.

3.3.5 Inspection of Driven Piling

Perform continuous inspection during pile driving. Inspect all piles for compliance with tolerance requirements. Bring any unusual problems which may occur to the attention of the Contracting Officer. Inspect the interlocked joints of driven pilings extending above ground. Pilings found to be out of interlock shall be removed and replaced at the Contractor's expense.

3.3.6 Pulling and Redriving

Submit the proposed method of pulling sheet piling, prior to pulling any piling. Pull selected pilings after driving to determine the condition of the underground portions of pilings. Any piling so pulled and found to be damaged, to the extent that its usefulness in the structure is impaired, shall be removed and replaced at the Contractor's expense. Pilings pulled and found to be in satisfactory condition shall be redriven when directed at no additional cost to the Government.

3.4 EXCAVATION AND BACKFILL

- a. Upon completion of the installation and approval of the sheet piles, excavation and backfill as indicated on contract documents shall be performed in accordance with Section 31 00 00 EARTHWORK.

3.5 REMOVAL

The removal of sheet pilings shall consist of pulling, sorting, cleaning the interlocks, inventorying, and storing previously installed sheet pilings as shown and specified. Use of pre auguring, water jetting or other methods to reduce bonding between the sheet pile and soils may be necessary to remove the temporary sheet piles. Proposed methods other than pulling shall be submitted to the Contracting Officer for approval.

3.5.1 Pulling

All temporary bypass and transverse sheet piles shall be pulled out no earlier than 28 days after completion of adjacent drilled shaft concrete placement and authorization by the Contracting Officer. Temporary sheet piles with no adjacent drilled shafts shall be pulled out as soon as the associated construction works are completed and as authorized by the Contracting Officer. Removal time and the method of pulling piling shall be approved by the Contracting Officer. Provide pulling holes in pilings, as required. Extractors shall be of suitable type and size. Care shall be exercised during pulling of pilings to avoid damaging piling interlocks and adjacent construction. If the Contracting Officer determines that adjacent permanent construction has been damaged during pulling, the Contractor will be required to repair this construction at no cost to the Government. Pull pilings one sheet at a time. Pilings fused together shall be separated prior to pulling, unless the Contractor demonstrates, to the satisfaction of the Contracting Officer, that the pilings cannot be separated. The Contractor will not be able to reuse pilings damaged beyond structural use.

3.5.2 Sorting, Cleaning, Inventorying and Storing for Reuse

Pulled pilings shall be sorted, cleaned, inventoried and stored by type into groups as:

- a. Piling usable without reconditioning.
- b. Piling requiring reconditioning.
- c. Piling damaged beyond structural use.

3.6 INSTALLATION RECORDS

Maintain a [pile driving record](#) for each sheet pile driven. Indicate on the installation record: installation dates and times, type and size of hammer, rate of operation, total driving time, dimensions of driving helmet and cap used, blows required per [foot](#) for each [foot](#) of penetration, final driving resistance in blows for final [6 inches](#), pile locations, tip elevations, ground elevations, cut-off elevations, and any reheading or cutting of piles. Record any unusual pile driving problems during driving. Submit complete records to the Contracting Officer within 7 days of pile installation.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 31 - EARTHWORK

SECTION 31 52 10

TEMPORARY RETAINING STRUCTURES

PART 1 GENERAL

- 1.1 SCOPE
- 1.2 REFERENCES
- 1.3 SUBMITTALS
- 1.4 TEMPORARY RETAINING STRUCTURE DESIGN CALCULATIONS
 - 1.4.1 Design Procedures
 - 1.4.2 Elevations
 - 1.4.3 Sheet Pile Wall Design
 - 1.4.3.1 Load Conditions
 - 1.4.3.2 Retaining Wall Members
 - 1.4.4 Underwater Concrete (Tremie)
 - 1.4.5 Designs and Modifications
- 1.5 QUALITY CONTROL
 - 1.5.1 General
 - 1.5.2 Reporting
- 1.6 DELIVERY, STORAGE AND HANDLING OF MATERIALS
- 1.7 CONTRACTOR EXPERIENCE
 - 1.7.1 Temporary Retaining Structure (TRS) Contractor Experience
 - 1.7.2 Concrete Tremie Slab Contractor Experience
- 1.8 TRS WORK PLAN

PART 2 PRODUCTS

- 2.1 SHEET PILING
- 2.2 Tremie Concrete
 - 2.2.1 Granular Subbase

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Placing
 - 3.1.2 Driving
 - 3.1.3 Refusal Criteria
 - 3.1.4 Emergency Locking System on Pile Driving Head
 - 3.1.5 Inspection of Driven Piling
 - 3.1.6 Void Backfill
- 3.2 TREMIE CONCRETE PLACEMENT
- 3.3 REMOVAL OF MATERIAL
 - 3.3.1 Removal Criteria
 - 3.3.2 Safety

-- End of Section Table of Contents --

SECTION 31 52 10

TEMPORARY RETAINING STRUCTURES

PART 1 GENERAL

1.1 SCOPE

This work shall consist of designing, furnishing, installing, maintaining and subsequently removing all temporary retaining structures required for the installation of the two sanitary sewer siphon coffer cells. The Contractor shall be solely responsible for the design, layout, construction, maintenance and subsequent removal and disposal of all elements of the temporary retaining structures. This section applies only to temporary retaining structures (TRS) related to the siphon at approximate Station 100+00. All other temporary retaining structures are addressed in SECTION 31 41 16 METAL SHEET PILING.

The coffer cell elements shown in the Contract Drawings are provided as a concept for bidding purposes, however, certain elements and dimensions are stated as minimum requirements.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325 (2017) Steel Construction Manual

ASTM INTERNATIONAL (ASTM)

ASTM A572/A572M (2021; E 2021) Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health (SOH) Requirements

EM 1110-2-1902 (2003) Slope Stability

EM 1110-2-2502 (1989) Retaining and Flood Walls

EM 1110-2-2504 (1994) Design of Sheet Pile Walls

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

TRS Work Plan; G, D0

Tremie Concrete Placement Plan; G, D0

SD-02 Shop Drawings

Shop Drawings; G, D0

A detailed layout of temporary retaining structures on standard size (22 inches x 34 inches) sheets. These shop drawings shall bear the stamp and signature of the Registered Professional Engineer. These drawings shall clearly show:

(a) All pertinent dimensions and locations of these structures with reference to the project centerline (Wall-line, Baseline, etc.).

(b) Material grade, weight, length and designation of steel sheet pile section(s) used.

(c) Waler beams and bracing details at all levels and structural stiffeners at all entry and exit ring openings.

(d) Detailed drawings of pile splices. Clearly indicate welds using standard AWS notations and symbols, and clearly show or indicate size, length, and type of each weld, as applicable.

(e) Excavation sequence and procedure.

(f) Provisions made for unwatering, indicating stage of excavation vs. necessary drawdown, water loading conditions, soil and equipment loads, and provisions for sealing sheet pile interlocks.

(g) Any other items incidental or significant to this work.

(h) Equipment Description. Complete hammer, extractors and other installation appurtenances.

SD-05 Design Data

Temporary Retaining Structure Design Calculations; G, D0

Tremie Mix Design; G, D0

SD-07 Certificates

Concrete Tremie Slab Contractor Experience; G, D0

Temporary Retaining Structure (TRS) Contractor Experience; G, D0

1.4 TEMPORARY RETAINING STRUCTURE DESIGN CALCULATIONS

1.4.1 Design Procedures

The Contractor shall follow design procedures using the wedge-type method

of developing soil pressure for estimating the external forces, set forth in Chapter 4 of the U.S. Army Corps of Engineers Manual EM 1110-2-2504 "Design of Sheet Pile Walls". The design performed by the Contractor must evaluate the overall stability and sizing of the sheet piling and other structural elements for the temporary retaining structures. The Contractor shall submit for approval [shop drawings](#) of the proposed design. The Contractor shall use and rely upon the design shear strength profile(s) and unit weight data presented in the plans and/or in section 00 31 32 GEOTECHNICAL DATA REPORT. The temporary retaining structure (TRS) shall be based on the geotechnical properties in Section 00 31 32 Geotechnical Data Report. The Contractor may elect to perform additional borings at the TRS locations. The structure shall meet all the requirements of [EM 385-1-1](#) for fall protection and ingress and egress.

1.4.2 Elevations

The resulting temporary retaining structure shall have sufficient height to retain soil and groundwater up to El. +8 as described below in Section "LOAD CONDITIONS", and have a minimum sheet pile tip elevation -44.0. In addition, the Contractor shall design and implement additional means and methods to protect the siphon coffer cell from potential flood waters up to El. +11.5.

1.4.3 Sheet Pile Wall Design

The temporary sheet piles and the waler/bracing support system of the coffer cell shall be designed in compliance with [EM 1110-2-2504](#) and all the applicable references. The design of the sheet pile wall shall be developed using a method of analysis indicated in paragraph "Design Procedures," with the safety factor applied to the soil strengths on both sides of the wall, such as a free earth or fixed earth method. The soil properties used shall be those presented in the drawings and/or in section 00 31 32 GEOTECHNICAL DATA REPORT.

1.4.3.1 Load Conditions

Design analyses shall be performed for load conditions as specified in the [EM 1110-2-2504](#), as minimum; the following loading cases representing construction, drawdown/dewatering, and earthquake conditions shall be considered in the design analyses. Analyses shall be performed for rotational stability and global stability of all loading cases, and for anticipated seepage quantities associated with the unwatering loading cases.

1. Construction Loading with Short-term (Q) Strength - prior to installing Level 1 waler/bracing support:
 - a. Excavation and unwatering down to 2-feet below the Level 1 waler/bracing level
 - b. Groundwater Level in the Retaining Side: El. +4
 - c. Q soil strength (short-term)
 - d. Surcharge: shall be based on the anticipated loadings of construction materials and equipment.
2. Construction Loading with Short-term (Q) Strength - prior to installing Level 2 waler/bracing support:
 - a. Excavation and unwatering down to 2-feet below the Level 2 waler/bracing level
 - b. Groundwater Level in the Retaining Side: El. +4

- c. Q soil strength (short-term)
 - d. Surcharge: shall be based on the anticipated loadings of construction materials and equipment.
3. Construction Loading with Short-term (Q) Strength - prior to installing Level 3 waler/bracing support:
- a. Excavation and unwatering down to 2-feet below the Level 3 waler/bracing level
 - b. Groundwater Level in the Retaining Side: El. +4
 - c. Q soil strength (short-term)
 - d. Surcharge: shall be based on the anticipated loadings of construction materials and equipment.
4. Construction Loading with Q Strength and Maximum Excavation w/ Bottom at EL. -34:
- a. Excavation down to EL. -34, the bottom of proposed gravel base layer.
 - b. Consider unwatering depth per contractor's unwatering plan for this excavation case.
 - c. Groundwater Level in the Retaining Side: El. +4
 - d. Q soil strength (short-term)
 - e. Surcharge: shall be based on the anticipated loadings of construction materials and equipment.
5. Construction Loading with Long-term (S) Strength - Completion of Excavation w/ Bottom at EL. -26:
- a. Excavation and unwatering down to EL. -26, the top of the tremie concrete slab.
 - b. Groundwater Level in the Retaining Side: El. +4
 - c. S soil strength (long-term)
 - d. Surcharge: shall be based on the anticipated loadings of construction materials and equipment.
6. Construction Loading with Flood at Top of Wall (TOW) and S Strength - Completion of Excavation with Bottom at EL. -26.0
- a. Excavation and unwatering down to -26.0, the top of the tremie concrete slab.
 - b. Water Level in the Retaining Side at TOW
 - c. S soil strength (long-term)
 - d. Surcharge: shall be based on the anticipated loadings of construction materials and equipment.

Note: Contractor shall design and implement additional means and methods to protect the siphon coffer cell from potential flood water up to El. +11.5. Alternatively, the Contractor can design the coffer cell extending to the top of wall to minimum El. +11.5 and analyze with flood at El. +11.5.

7. Drawdown Loading with R Strength:
- a. Excavation and unwatering down to the deepest depth that would be subject to dewatering/unwatering loading.
 - b. Groundwater Level in the Retaining Side: El. +4
 - c. R soil strength
 - d. Surcharge: shall be based on the anticipated loadings of construction materials and equipment.
8. Reduced (50% MDE) Earthquake Loading with 80% R Strength (No Liquefaction:
- a. Excavation and unwatering down to EL. -26, the top of the tremie

- concrete slab.
- b. Groundwater Level in the Retaining Side: El. +4
- c. 80%R soil strength; no soil liquefaction on the retaining side of the wall
- d. Surcharge: shall be based on the anticipated loadings of construction materials and equipment.
- e. Consider dynamic active earth pressure in addition to the static pressures for pseudo-static seismic coefficient of 0.13g, which represents 50% MDE earthquake loading.

If the designer elects to account for wall friction and adhesion in the determination of soil pressures, the following criteria shall be used:

- (1) a maximum value of 50% of the "allowable" soil friction angle in granular materials may be used for wall friction,
- (2) a maximum value of 50% of the "allowable" soil cohesion may be used for adhesion and
- (3) no wall friction or adhesion may be applied to the soils above the dredge line.

In accordance with EM 1110-2-2504, Factor of Safety (FOS) as summarized below shall be considered on the passive earth pressures to estimate required minimum sheet pile embedment, total length, and tip elevation.

Analysis Case	FOS
Construction Loading with Q Strength	2.00
Construction Loading with S Strength	1.50
Construction Loading with Flood at TOW and S Strength	1.25
Drawdown Loading with R Strength	2.00
Earthquake Loading (Extreme) with 80% R Strength	1.10

In accordance with EM 1110-2-1902 and EM 1110-2-2502, the FOS values as summarized below shall be considered for global stability analyses.

Analysis Case	FOS
Construction Loading with Q Strength	1.30
Construction Loading with S Strength	1.30
Drawdown Loading with R Strength	1.10
Earthquake Loading (Extreme) with 80% R Strength	1.10

1.4.3.2 Retaining Wall Members

The structural design of the temporary retaining structure excluding the

sheet pile shall be designed using **AISC 325**. The earth pressures for the structural design of these wall members shall be based upon a wedge-type method applying a safety factor chosen by the designer which shall be no lower than 1.0.

1.4.4 Underwater Concrete (Tremie)

The underwater concrete (also referred to as Tremie Slab or Tremie Seal), must be designed to perform 3 functions: 1) It must provide an adequate structural system to prevent a blow-out in the coffer cell floor in the unwatered condition for the duration of the work. 2) it must act as a lateral compression strut to provide internal lateral bracing to the sheet piles. 3) it must allow relief of groundwater uplift pressure and facilitate uplift pressure monitoring via piezometers as specified under Section **31 23 19** Unwatering. Drawings and notes show a TRS concept that is provided for bidding purposes only. The tremie design shall discount the upper and lower 6 inches of concrete and shall have a minimum of 6.5 ft of effective thickness. A minimum uplift factor of safety of 1.3 is required for the tremie slab to resist uplift.

1.4.5 Designs and Modifications

All designs and any subsequent modifications to the design presented above shall be performed, certified and stamped by a Professional Engineer licensed in the Commonwealth of Puerto Rico and submitted to the Contracting Officer for review and approval. The Registered Professional Engineer shall be present at the Contractor Quality Control preparatory and initial inspections. The Contractor shall, as a part of the Quality Control, furnish a signed statement by the design Professional Engineer stating that the installation is in conformance with the Contractor's approved design.

1.5 QUALITY CONTROL

1.5.1 General

The Contractor shall establish and maintain quality control for all operations to assure compliance with contract specifications and maintain records of its quality control for all construction operations, including but not limited to the following:

- (1) Designing.
- (2) Materials (type, strength, etc.).
- (3) Fabrication, installation and workmanship.
- (4) Full and proper engagement of interlock (inspection and strength).
- (5) Placing (location, alignment, etc.).
- (6) Driving (pile hammer and rate of operation).
- (7) Cutting.
- (8) Welding.
- (9) Final sheet pile position; depth of penetration; tip and top elevations.

(10) Stockpiling and Storage.

(11) Removal and disposal of damaged piles.

1.5.2 Reporting

The original and two (2) copies of these records and tests, as well as the corrective action taken, shall be furnished the Government daily. Format of the report shall be as prescribed in Section 01 45 04 CONTRACTOR QUALITY CONTROL.

1.6 DELIVERY, STORAGE AND HANDLING OF MATERIALS

Materials delivered to the site shall be undamaged and shall be accompanied by certified test reports. Sheet piling shall be stored and handled in the manner recommended by the manufacturer to prevent permanent deflection, distortion or damage to the interlocks. Storage of sheet piling should also facilitate inspection.

1.7 CONTRACTOR EXPERIENCE

1.7.1 Temporary Retaining Structure (TRS) Contractor Experience

Submit documentation showing that the Contractor or subcontractor has completed a minimum of three similar installations of TRS. Each representative TRS project shall be of similar or larger size and depth and completed within the past 10 years.

1.7.2 Concrete Tremie Slab Contractor Experience

Submit documentation showing that the Contractor or subcontractor has successfully completed a minimum of three similar successful installations of tremie slabs. Each representative tremie placement project shall be of similar size or larger and depth and completed within the past 10 years.

1.8 TRS WORK PLAN

The contractor shall develop and submit a work plan that shall at minimum include the following:

- a. Construction sequence.
- b. Description of means and methods of construction, and unwatering and rewatering operations per Section 31 23 19 UNWATERING.
- c. Details of collection, maintenance, and disposal of water removed from coffer cells.
- d. Description of equipment to be used to perform the work.
- e. Proposed staging and laydown area.
- f. Safety controls for the equipment/material use and containment of any water and spoils.
- g. Boring logs and laboratory test reports should the Contractor elect to perform additional borings in vicinity of the temporary

retaining structures.

h. Provisions for cutting and removing bypass coffer cell sheets underwater.

PART 2 PRODUCTS

2.1 SHEET PILING

The sheet pile for the siphon coffer cell may be uncoated and shall be hot rolled sheet pile conforming to [ASTM A572/A572M](#), Grade 50 as shown on the drawings or as required by design. All other sheet piles are addressed in Section [31 41 16](#) METAL SHEET PILING.

2.2 Tremie Concrete

The tremie concrete shall be highly flowable, non-segregating, self-consolidating mixture, and placed in a continuous manner in its entirety. Submit [tremie mix design](#). Concrete mix, strength testing, and chemical admixtures additives shall conform to the requirements of Section [03 31 01](#) CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS except as follows

- a. Maximum coarse aggregate size shall be 3/4 of an inch.
- b. Concrete compressive strength shall be a minimum of 2,500 psi at 7 days.
- c. The maximum water-cementitious ratio shall be 0.42. The slump of the concrete as delivered to the point of placement shall be 8 to 10 inches. Slump flow shall be limited between 16 to 23 inches.
- d. Washout shall be tested in accordance with CRD C61-89A and be limited to 5% or less by mass. Flowing water is not anticipated, therefore anti-wash additives are not required. Bleeding of concrete placed underwater shall be tested in accordance with ASTM C 232, Method A and restricted to 2%.
- e. Concrete temperature at the point of placement shall not be more than 70 degrees. Cooling of the mixing water and aggregates with ice or nitrogen may be required.
- f. Permeability testing is not required.

2.2.1 Granular Subbase

The tremie slab shall have a crushed rock gravel subbase of an effective thickness such that groundwater can flow freely to standpipes and piezometers and provide a stable foundation for the tremie slab.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Placing

Suitable temporary wales, templates, guide structures, or other approved methods shall be provided to insure that the sheet piles are placed and driven to the correct alignment as shown on the Contractor's shop drawings. Piles shall be placed with each pile interlocked with adjoining piles for its entire length, so as to form a continuous diaphragm throughout the length of each run of piling wall. Interlocks shall be properly engaged. The Contractor's personnel shall not sit or place themselves on top of the sheet piling during the handling, installation,

and removal of the piling.

3.1.2 Driving

All sheet piles shall be driven to the depths shown on the Contractor's shop drawings. Pilings shall be driven by approved methods so as not to subject the pilings to damage and to insure proper interlocking throughout their lengths. Pilings damaged during driving or driven out of interlock shall be removed and replaced. All piles shall be driven without the aid of a water jet, unless otherwise authorized. Unless specifically indicated otherwise, each run of piling wall shall be driven to grade progressively from the start and pilings in each run shall be driven alternately in increments of depth to the required depth or elevation. On each day of sheet pile driving, the Contractor shall stab only the number of piles that can be driven to grade by the end of the day, and all piling stabbed shall be driven to grade by the end of each working day except that the last two piles may remain tapered up to receive the next days piles. If the piling next to the one being driven tends to follow below final grade, it may be pinned to the next adjacent piling. The Contractor is advised that buried stumps or similar debris may be encountered periodically on the sheet pile wall alignment and appropriate consideration should be given to hard driving conditions should they occur. Piles shall not be driven within 100-feet of concrete less than 7 days old nor within 30- feet of concrete less than 28 days old.

3.1.3 Refusal Criteria

- a. Refusal criteria for sheet piling, depending on the driving method and equipment to be used, shall be verified with drivability studies and manufacturer recommendations.
- b. The followings shall be considered when selecting the driving method, equipment size, and determining the associated driving refusal:
 1. Able to drive the sheet pilings to the required minimum tip elevations, with no damage to the pilings and maintaining proper pile interlocking throughout their lengths.
 2. Verification with wave equation analysis that the selected driving system will produce no overstressing of the pile.
 3. The following conditions do not constitute practical driving refusal:
 - i. The approved hammer system is not operating correctly or
 - ii. The hammer is not operated to the maximum rated operating energy as recommended by the hammer manufacturer.
- c. Sheet pile penetration of 1 inch per 20 blows shall be considered as the refusal limit when driving the sheet piles with impact hammers if used for the project. However, the refusal criteria for the selected hammer should be verified based on the hammer manufacturer recommendations and drivability studies.

3.1.4 Emergency Locking System on Pile Driving Head

All pile driving equipment shall be equipped so as to prevent piles from

falling when a single or multiple power failure occurs after the pile driving head is attached to the pile. The jaws of vibratory hammers shall be equipped with devices such that upon loss of hydraulic pressure, the jaws will not release the pile.

3.1.5 Inspection of Driven Piling

The Contractor shall inspect the interlocked joints of driven pilings extending above ground. Pilings found to be damaged or driven out of interlock shall be removed and replaced.

3.1.6 Void Backfill

Where voids adjacent to the steel sheet piling are induced by pile driving operations, the Contractor shall pump out all seepage and rain water and backfill with a tremie-placed slurry. The slurry shall consist of one part cement, two parts bentonite, and six parts sand mixed with enough water to produce a slurry viscous enough to thoroughly fill the voids.

3.2 TREMIE CONCRETE PLACEMENT

Provide a [tremie concrete placement plan](#) to the Contracting Officer for review and approval at least 60 days prior to the start of placement of any tremie concrete. The plan shall describe the methods and equipment proposed for the underwater concrete work. Placement of the tremie concrete shall meet the following requirements:

- a. Underwater placement of concrete shall have a maximum tolerance of 6 inches for the top elevation of tremie slab. Provide sounding methods and working platforms to monitor and achieve the top placement tolerance.
- b. Throughout tremie placement, the tip of the tremie pipe shall always remain inside the fresh concrete with a minimum 24-inch cover. At no time shall concrete be allowed to fall through water. Do not move the tremie horizontally more than 15 feet during a placing operation and provide a sufficient number of tremies so that the maximum horizontal flow is limited to 15 feet. The use of end valves shall be used to relocate the tremie pipe to maintain the tremie seal.
- c. After curing of the tremie slab and unwatering the coffer cell, levelling concrete shall be placed to meet the requirements of the auger boring jacking rails and equipment.

3.3 REMOVAL OF MATERIAL

3.3.1 Removal Criteria

Removal of the temporary retaining structures requires approval of the Contracting Officer. The temporary retaining structures shall not be removed until suitable backfill, between the finished structure and the steel sheet pile wall of the temporary retaining structures is satisfactorily placed and compacted to an elevation approximately [one \(1\) foot](#) below the finished surface. Piles shall not be pulled within 100-feet of concrete less than 7 days old nor within 30- feet of concrete less than 28 days old. Nor, shall the temporary retaining structures be removed until the completed permanent structure and/or excavations are rewatered. Where voids are induced by removal operations, the Contractor shall pump out all seepage and rainwater and backfill to within [3 feet](#) of

the ground surface with a tremie-placed slurry. The slurry shall consist of one part cement, two parts bentonite and six parts sand mixed with enough water to produce a slurry viscous enough to thoroughly fill the voids but have no less than 12 pounds of solids per gallon. The upper 3 feet shall be filled with suitable backfill. Suitable backfill and compaction requirements shall be as defined in Section 31 00 00 EARTHWORK. All Contractor-furnished temporary retaining structures shall be removed from the site of work upon completion of work.

3.3.2 Safety

The removal of the temporary retaining structures shall be accomplished in a manner not injurious to the properties adjacent to and in the proximity of the project excavations.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 31 - EARTHWORK

SECTION 31 63 29

DRILLED CONCRETE PIERS AND SHAFTS

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DESCRIPTION
- 1.3 SUBMITTALS
- 1.4 APPROVALS
- 1.5 SUBSURFACE DATA
- 1.6 QUALITY CONTROL
 - 1.6.1 Shaft Construction Tolerances
 - 1.6.1.1 Horizontal
 - 1.6.1.2 Vertical
 - 1.6.2 Frequent Checks on Plumb
 - 1.6.3 Concreting Tolerances
 - 1.6.4 Access to Shafts
 - 1.6.5 Satisfactory Field Performance
 - 1.6.6 Failure to Meet Required Tolerance
 - 1.6.7 Drilled Shaft Contractor Experience
 - 1.6.7.1 Site Superintendent
 - 1.6.7.2 Supervisor Experience
 - 1.6.7.3 Drill Operator Experience
 - 1.6.7.4 Cross Hole Sonic Logging (CSL) Testing Experience
 - 1.6.7.5 Ultrasonic or Sonar Shaft Excavation Measuring Device Testing System Experience
 - 1.6.7.6 Shaft Inspection Device Experience
 - 1.6.8 Welding Qualifications
 - 1.6.9 Quality Control Personnel Qualifications
 - 1.6.10 Survey
 - 1.6.11 Geotechnical Engineer Qualifications
 - 1.6.12 Structural Engineer Qualifications
- 1.7 NONDESTRUCTIVE TESTING OF SHAFTS

PART 2 PRODUCTS

- 2.1 CASING
- 2.2 REINFORCING STEEL
- 2.3 CONCRETE
- 2.4 SLURRY
- 2.5 CROSS HOLE SONIC LOGGING (CSL) TESTING TUBES
 - 2.5.1 Typical Cross Hole Sonic Logging (CSL) Test Equipment
- 2.6 EQUIPMENT
 - 2.6.1 Cranes
 - 2.6.2 Equipment for Casing Installation
 - 2.6.3 Excavating Equipment
 - 2.6.4 Correlation and Confirmation Boring Drilling

PART 3 EXECUTION

- 3.1 CORRELATION AND CONFIRMATION BORING DRILLING
- 3.2 SHAFT INSTALLATION
 - 3.2.1 Wet Construction Method
 - 3.2.2 Protection of Existing Structures
 - 3.2.3 Demonstration Wall
 - 3.2.4 Demonstration Shafts for Wall
 - 3.2.5 Casing Advancement
 - 3.2.6 Shaft Alignment and Depth
 - 3.2.6.1 Casing Alignment
 - 3.2.6.2 Temporary Casing
 - 3.2.6.3 Temporary Casing Removal for Channel Walls and Utility Poles
 - 3.2.6.4 Casing for Bridge Middle Pier Drilled Shaft
- 3.3 EXCAVATION
 - 3.3.1 Excavation Depth
 - 3.3.2 Excavation Stability
 - 3.3.3 Drilled Shaft Excavation Logging
- 3.4 CLEANUP AND FINAL EXCAVATION CHECKS
 - 3.4.1 De-sanding
 - 3.4.2 Bottom Sediment
- 3.5 SLURRY
 - 3.5.1 Slurry Conformance Requirements
 - 3.5.2 Slurry Mixing and Placement
 - 3.5.2.1 Freshly Mixed Slurry
 - 3.5.2.2 Slurry Supplied to the Drilled Shaft Excavation
 - 3.5.2.3 Slurry Test Reports
 - 3.5.3 Slurry Level
 - 3.5.4 Failure of Wet Construction Method
 - 3.5.5 Containment of Slurries
- 3.6 DISPOSAL
- 3.7 PROTECTION
- 3.8 DRILLED SHAFT GROUND IMPROVEMENTS
 - 3.8.1 Jet Grouted Seals
- 3.9 INDIVIDUAL SHAFT INSTALLATION RECORDS
- 3.10 DRILLED SHAFT COMPLETION
 - 3.10.1 Inspection
 - 3.10.2 Depth
 - 3.10.3 Shaft Cleanliness Requirements
 - 3.10.4 Time of Excavation
- 3.11 ASSEMBLY AND PLACEMENT OF REINFORCING STEEL
 - 3.11.1 Reinforcing Cage
 - 3.11.2 Reinforcement Positioning
 - 3.11.2.1 Vertical
 - 3.11.2.2 Horizontal (Along Circumference)
 - 3.11.2.3 Concrete Spacers
 - 3.11.2.4 Requirements for Utility Pole Drilled Shafts
 - 3.11.3 Placement of Reinforcing Bars
 - 3.11.4 Reinforcing Splice
- 3.12 CSL ACCESS TUBES
- 3.13 CSL LOGGING PROCEDURES
 - 3.13.1 CSL Testing Results
 - 3.13.2 Evaluation of CSL Test Results
 - 3.13.3 Core Drilling of Drilled Shaft Concrete
- 3.14 PLACING CONCRETE IN SHAFT
 - 3.14.1 Fluid in Excavation at Time of Concrete Placement
 - 3.14.2 Immediate Placement
 - 3.14.3 Concrete Placement
 - 3.14.4 Placement Time

- 3.14.5 Removal of Sediment at Construction Joints
- 3.14.6 Removal of Excess Concrete
- 3.14.7 Beginning Work on Adjacent Shafts
- 3.15 GROUTING THE CSL TESTING TUBES
- 3.16 INSPECTION OF EXPOSED CONCRETE SURFACE
- 3.17 FINAL APPROVAL AND ACCEPTANCE
- 3.18 SAMPLE FORMS

ATTACHMENTS:

Sample shaft installation record form

slurry data

concreting data forms

-- End of Section Table of Contents --

SECTION 31 63 29

DRILLED CONCRETE PIERS AND SHAFTS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO LRFDCONS (3rd Edition; 2010, 2011, 2012, 2014, 2015, and 2016 Int) Bridge Construction Specifications

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318 (2019; R 2022) Building Code Requirements for Structural Concrete (ACI 318-19) and Commentary (ACI 318R-19)

ACI 336.1 (2001) Specification for the Construction of Drilled Piers

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M (2020) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM D1586 (2018) Penetration Test and Split-Barrel Sampling of Soils

ASTM D1586/D1586M (2018) Standard Test Method for Standard Penetration Test (SPT) and Split-Barrel Sampling of Soils

ASTM D2166/D2166M (2016) Standard Test Method for Unconfined Compressive Strength of Cohesive Soil

ASTM D2488 (2017; E 2018) Standard Practice for Description and Identification of Soils (Visual-Manual Procedure)

ASTM D6760 (2016) Standard Test Method for Integrity Testing of Concrete Deep Foundations by Ultrasonic Crosshole Testing

AMERICAN PETROLEUM INSTITUTE (API)

API RP 13B-1

(2009; R 2016) Recommended Practice for
Field Testing Water-Based Drilling Fluids

1.2 DESCRIPTION

This item of work shall consist of furnishing all materials, labor, tools, equipment, services and incidentals required to construct drilled shafts in accordance with the plans, approved shop drawings, and contract specifications, including Crosshole Sonic Logging (CSL) testing. Concrete testing shall conform to the requirements in Section 03 31 01

CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS. The drilled shaft work shall follow ACI 336.1 and AASHTO LRFD CONS, except as modified in this specification. Drilled shaft walls are designed as tangent shafts with intimate contact between adjacent shafts. All channel wall shafts shall require a temporary casing to the depth of stable materials or to a minimum tip elevation of -50 feet, whichever is deeper. Note that all elevations are based on Puerto Rico Vertical Datum of 2002.

This item of work shall also include furnishing all materials, labor, tools, equipment, services, and incidentals necessary to construct a demonstration wall and cap as specified herein. On satisfactory completion of the Demonstration Wall the location is to be restored as shown on the plans.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Designation R0 stands for Resident Office and designation D0 stands for District Office. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Confirmation Boring Drilling Plan At Middle Pier; G, D0

The Contractor shall submit shop drawings and a drilling plan narrative for the utility bridge middle pier confirmation boring at least 30 calendar days prior to anticipated beginning of the drilling. The plan shall include a single boring within 30 feet of the middle pier and terminate at EL. -95 feet.

Correlation Boring Drilling Plan; G, D0

The Contractor shall submit shop drawings and a drilling plan narrative for correlation boring at least 30 calendar days prior to anticipated beginning of the drilling. The plan shall include single borings within 30 feet of the Demonstration Wall unless existing borings to weathered rock are present within the same distance.

Demonstration Wall Installation Plan; G, D0

The Contractor shall submit shop drawings and a demonstration shaft wall and cap installation narrative at least 60 calendar days prior to anticipated beginning of the drilled shaft

operation. This narrative shall provide similar information as the Shaft Installation Plan.

Survey; G, D0

SD-02 Shop Drawings

Shaft Installation Plan; G, D0

The Contractor shall submit shop drawings and a shaft installation narrative at least sixty (60) calendar days prior to anticipated beginning of the drilled shaft operation for approval by the Contracting Officer. This narrative shall provide at least the following information:

a. An overall construction operation sequence and the sequence of drilled shaft construction including operation and sequence of drilled shaft construction at the bridge middle pier. Submit information that demonstrates that crane loading will not exceed the maximum surcharge load shown on the plans and that the crane used for constructing the drilled shafts at the bridge middle pier is sized and equipped to safely reach the required shaft locations without overloading the **ground or structures**. Refer to contract drawing for sequencing notes for bridge middle pier.

b. The Contractor's understanding of the site subsurface conditions. The Contractor shall reference the available subsurface data provided in Section **00 31 32 GEOTECHNICAL DATA REPORT (GDR)**.

c. Lists and descriptions, including capacities, of proposed equipment, including but not limited to, cranes, drills, augers, bailing buckets, final cleaning equipment and drilling units. Explain why the equipment was selected and describe the equipment's suitability for the anticipated site and subsurface conditions. The equipment shall be capable of delivering the vibratory, impact, or rotational/torque capacity to install the temporary and permanent casings. The narrative shall include a project history of the drilling equipment demonstrating the successful use of the equipment proposed by the Contractor to construct shafts of equal or greater size in similar soil with artesian water or rock conditions. The equipment shall be capable of performing the installation of the casing, excavation, and shaft construction as specified and as shown on the contract drawings.

d. Details of shaft excavation methods, including proposed drilling methods, methods for cleanout of the shafts and a disposal plan for excavated material in accordance with all applicable local and federal environmental regulations. This shall include a review of the method's suitability for the anticipated site and subsurface conditions. Detailed procedures for temporary casing installation shall be provided. The Contractor shall have adequate equipment or other tools/techniques to penetrate obstructions (e.g. concrete, rock, debris, etc.) not only for the temporary casing installation but for drilling excavation that allows to go through subsurface conditions in order to reach the required tip elevation.

- e. Description of the material to be used to temporarily backfill a drilled shaft excavation hole during a stoppage of the excavation operation, as well as the method used to place and remove the material.
- f. Details of the method(s) to be used to ensure shaft stability (i.e., prevention of caving, bottom heave, etc.) during excavation and concrete placement as well as during periods of inactivity before completion of the shaft. This shall include a review of the method's suitability for the anticipated site and subsurface conditions. Detailed procedures for temporary casing installation shall be provided.
- g. Details of the equipment and method(s) to be used to check plumbness, alignment, and dimensions of the shaft during excavation and to inspect the bottom of shaft excavation.
- h. Detailed procedures for containing and disposing of the material removed during concreting and from shaft excavation shall be provided.
- i. Reinforcing steel shop drawings per Section 03 20 00 CONCRETE REINFORCING and details of reinforcement placement (including centering and lifting methods) and the methods used to assure the reinforcing cage remains in final position during and following concrete placement.
- j. Details of concrete placement, including proposed operational procedures for pumping and/or tremie methods. Describe the method to be used to form an emergency horizontal construction joint during concrete placement. Describe procedures to ensure that concrete contaminated with slurry is removed during pumping and that concrete at the top of the shaft is free of slurry or other contamination.
- k. Details of operational procedures for maintaining a dry work platform, including dewatering. Describe methods and equipment (including pump capacity) used to maintain a dry work platform and work area during the most severe rainfall events that may be reasonably expected during the duration of the work. Construction delays and the work required to clean up and repair any damage to the work area caused by rainfall events shall be at no additional cost to the Government. Refer to section 01 50 02 and 31 00 00 for surface water control requirements.
- l. Details of temporary and permanent casing, including finish and material, diameter, and thickness.
- m. Details and shop drawings of contractor's proposed template or guide for initial positioning of casing prior to installation.

Casing Installation Plan - General; G, D0

This submittal shall include a description of the method of positioning the casings for all shafts, including the shaft to be installed at the pier location at the middle of the bridge. The submittal shall describe positioning casings at the specified locations within the specified tolerances; method of monitoring the casing alignment as it is advanced to ensure compliance with

specified tolerances; a description of the equipment and methods to be employed in advancing the casing to the specified tip elevation; number, size and type of casing advancement equipment; special procedures to be employed to install the casing to the specified tip elevation; special procedures to be employed if the casing drifts out of plumb during advancement; methods and equipment to be used for drilling; description of casing cleanup equipment; description of reinforcement cage fabrication support frames; reinforcement cage installation methods; and description of concrete supplier, transportation method, and concrete placement method.

This submittal shall describe specific materials, equipment, and procedures for installing the casing for the bridge middle pier shafts and columns. If permanent casing is planned, this submittal shall include description of removing the casing above El. 0 feet after concrete has hardened.

This submittal shall include evidence that the equipment used for casing installation has satisfactorily completed installation of similarly sized drilled shafts to similar depths in similar ground conditions on at least three previous projects.

Excavation; G, D0

This submittal shall include descriptions of the proposed method and type, size and number of each piece of excavation equipment proposed for removing materials within the casings; and the proposed method of minimizing heave of the soil. The submittal shall also include a description of the methods and equipment to be used for monitoring the excavation depth and heaving.

Drilled Shaft Concreting Plan; G, D0

At least 60 days prior to any concrete being placed, the Contractor shall submit a drilled shaft concreting narrative for approval by the Contracting Officer. The Contracting Officer will evaluate the shaft concreting plan for conformance with the contract Drawings and Specifications. This narrative shall provide at least the following information:

- a. Details of holding the reinforcement steel in place during concrete placement and the method and timing to assure the reinforcement cage stays at its final position as indicated on the plans.
- b. Details of concrete placement, including proposed operational procedures for pumping and/or tremie methods, plans for batching the concrete, and plans for transporting the concrete to the shafts. Concrete shall be placed in the wet without dewatering the casings.
- c. The method to be used to form an emergency horizontal construction joint during concrete placement.
- d. A detailed slurry mix design, and a discussion of its suitability for the anticipated subsurface conditions.

Sheeting and Shoring; G, D0

Submit designs for Sheeting and Shoring to the Contracting Officer at least thirty (30) days prior to construction.

SD-05 Design Data

Correlation Boring Report

Prior to commencing drilling operations for a demonstration wall, the Contractor shall submit a copy of the Correlation Boring Report to the Contracting Officer for approval. The Correlation Boring Report shall describe the results from the Correlation Boring and provide guidance for monitoring the subsurface conditions encountered during drilled shaft construction.

Confirmation Boring Report At Middle Pier

Prior to commencing drilling operations for the middle pier shaft of the utility bridge, the Contractor shall submit a copy of the Boring Report to the Contracting Officer for approval. The Confirmation Boring Report shall describe the results from the middle pier Confirmation Boring.

Demonstration Wall Reports; G, D0

Shaft Installation Record Form; G, D0

The Contractor shall submit a copy of the proposed shaft installation record form to the Contracting Officer for approval thirty (30) calendar days prior to commencement of shaft construction. A sample of the form is appended at the end of this section. This form may be modified by the Contractor with the approval of the Contracting Officer.

Slurry Data; G, D0

The information provided shall include suppliers' test reports on the physical and chemical properties of the slurry and any proposed slurry chemical additives including Material Safety Data Sheets. The Contractor shall submit a copy of the proposed slurry data form to the Contracting Officer for approval thirty (30) calendar days prior to commencement of shaft construction. A sample of the form is appended at the end of this section. This form may be modified by the Contractor with the approval of the Contracting Officer.

Slurry Processing; G, D0

The Contractor shall submit a description of the proposed slurry processing to the Contracting Officer for approval thirty (30) calendar days prior to commencement of shaft construction. This submittal shall include details of the slurry plant and mixing of slurry; equipment and procedures for removing spoils and slurry from the casing excavation with no spillage into the river; slurry testing equipment and procedures; equipment and all additives to be used for the agitating, treating, and reconditioning the slurry; procedures for disposing of the spoils and slurry removed from the casing complying with all Federal, Commonwealth, and local environmental regulations.

Cleanup of Casing; G, D0

Submit a description of the proposed cleanup of casing to the Contracting Officer for approval thirty (30) calendar days prior to commencement of shaft construction. This submittal shall include details of cleanup methods to be used following excavation to prepare the casing for concrete placement; equipment and methods for treating slurry and for cleaning the bottom; and determining if any protrusions are present in the casing. This submittal shall also include details of the measuring device for casing length and the proposed grid system on which the final measurements will be taken.

Concreting Data; G, D0

The Contractor shall submit a copy of the proposed form for recording concrete mix, truck tickets, placement techniques, concrete test results, concreting volume, and concreting curve (e.g., Vol./cubic yard vs. depth). The form shall be submitted to the Contracting Officer for approval thirty (30) calendar days prior to commencement of shaft construction. A sample of the form is appended at the end of this section. This form may be modified by the Contractor with the approval of the Contracting Officer.

SD-06 Test Reports

Slurry Test Reports; G, D0

This daily submittal shall include a slurry composition, viscosity, density, sand content, pH and water hardness.

Slump Loss Test Report; G, D0

The slump loss test shall be performed for a minimum period of the placement time using the concrete and temperatures appropriate for site conditions.

CSL Test Reports; G, D0

Condition of Existing Structures; G, D0

The Contractor shall prepare 2 reports regarding existing conditions. One report thirty (30) calendar days prior to beginning foundation construction operations and the second report after completing foundation construction operations. Observations shall include settlement, ground heave, and vibration damage. Vibrations shall be monitored according to Section 02 22 13 VIBRATION AND NOISE MONITORING REQUIREMENTS.

Individual Shaft Installation Records; G, D0

This submittal shall include vertical alignment (ultrasonic or sonar measuring device results) and bottom inspection test reports. The shaft installation record submittal shall include logs of all activities including excavation, logging of cuttings, casing installation, cleanup work, slurry placement / elevations, estimated groundwater elevations, steel cage placement, concrete placement, etc. The record shall also include time, duration, and dates of these activities and documentation of any problems

encountered.

SD-07 Certificates

Drilled Shaft Contractor Experience; G, D0

Site Superintendent; G, D0

Supervisor Experience; G, D0

Drill Operator Experience; G, D0

Cross Hole Sonic Logging (CSL) Testing Experience; G, D0

Ultrasonic or Sonar Shaft Excavation Measuring Device Testing
System Experience; G, D0

Shaft Inspection Device Experience; G, D0

Soil Testing Laboratory; G, D0

The Contractor shall submit, for approval, a qualified soil testing laboratory to perform control tests on the slurry mixture and concrete. Testing shall be by a Corps-validated commercial testing laboratory or the Contractor's testing facility that has been Corps validated.

Quality Control Personnel Qualifications; G, D0

Geotechnical Engineer Qualifications; G, D0

Structural Engineer Qualifications; G, D0

Welding Qualifications; G, D0

SD-08 Manufacturer's Instructions

Mechanical Butt Splicing Device; G, D0

1.4 APPROVALS

Work shall not begin until all the required submittals have been approved in writing by the Contracting Officer. All procedures and approvals given by the Contracting Officer will be subject to trial in the field and shall not relieve the Contractor of the responsibility to satisfactorily completing the work. The Contracting Officer will approve or reject the Contractor's qualifications and field personnel within thirty (30) calendar days after receipt of the submission. Work shall not be started on any shaft until the Contractor's qualifications are approved by the Contracting Officer. The Contracting Officer may suspend the shaft construction and CSL testing if the Contractor substitutes personnel that have not been previously approved by the Contracting Officer. The Contractor shall be fully liable for the additional costs resulting from the suspension of work and no adjustments in contract time resulting from the suspension of work will be allowed.

1.5 SUBSURFACE DATA

Soil boring logs and laboratory test data are shown in Section 00 31 32

GEOTECHNICAL DATA REPORT. Representative samples of materials taken from the subsurface investigations may be examined by contacting the Contracting Officer. These data represent the best subsurface information available; however, variations may exist in the subsurface between boring locations.

1.6 QUALITY CONTROL

1.6.1 Shaft Construction Tolerances

1.6.1.1 Horizontal

The shaft shall be installed so that the shaft centerline at the design top of the shaft is within 1 inch in any direction of the specified location. A template, or approved similar method, shall be used for horizontal control.

1.6.1.2 Vertical

All drilled shafts shall be installed with a vertical alignment that is within 0.5 percent of vertical plumb to the bottom of casing. Below the bottom of casing, drilled shaft shall be installed with a vertical alignment that is within 1 percent of vertical plumb to the bottom of drilled shaft.

For drilled shafts without required jet grout seals, as shown on the Contract Drawings, install such that no gaps greater than 1.5 inches occur within the drainage layer.

1.6.2 Frequent Checks on Plumb

During advancement of the casing, drilling and excavation of the shaft, the Contractor's Quality Control personnel shall make frequent checks on and document the plumbness, alignment, and dimensions of the shaft. These checks shall be performed and documented at intervals less than 10 feet and checks may be required at closer spacings at the direction of the Contracting Officer. After drilling is completed, the shaft excavation shall be surveyed using an Ultrasonic or Sonar Shaft Excavation measuring device (SONICaliper or approved equivalent). Survey measurements shall be taken at 5 foot intervals from top of casing to the bottom of the excavation.

1.6.3 Concreting Tolerances

After all concrete is placed, the top of the reinforcing steel cage shall be no more than 2 inches above and no more than 1 inch below plan position. Concrete cover shall be as shown on the contract drawings. The top elevation of the shaft shall be no more than 0.5 inches above the plan top of shaft elevation.

1.6.4 Access to Shafts

The Contractor shall provide the Contracting Officer's representative with means and/or equipment access to all shafts at any time. All equipment, labor, and materials necessary to provide this access shall be considered incidental to shaft installation. This includes access to the shaft/column for the bridge middle pier.

1.6.5 Satisfactory Field Performance

The objectives of the installation of the demonstration wall are to verify that the Contractor's installation plan, including equipment, materials, labor, tools, services, and incidentals, is capable of constructing drilled shafts that meet all specified requirements. The Contractor will be required to make necessary adjustments to the Contractor's installation plan, including equipment, materials, labor, tools, services, and incidentals, if the Demonstration Wall fails to meet the requirements specified herein.

The Contractor's selected installation methods, temporary casing length, and slurry density are subject to satisfactory field performance. The casing length and slurry density shall be modified as required to obtain satisfactory field results that prevent deterioration of the drilled shaft holes. The use of casings and slurry during drilled shaft construction procedures is considered incidental to the drilled shaft construction.

1.6.6 Failure to Meet Required Tolerance

The objective of the drilled shaft wall is to, among other things, provide intimate contact between adjacent shafts mainly for the portion of the wall above the proposed dredge line to reduce the potential for seepage or loss of soil through gaps between adjacent shafts. If a drilled shaft casing is installed such that its verticality and spacing does not meet the specified tolerance, or if during inspection of the drilled shaft installation procedures and review of the verticality records per DRILLED SHAFT COMPLETION - INSPECTION paragraph it becomes apparent that there is a potential for a gap, the Contractor shall propose remedial measures to fully close all known or suspected gaps at no cost to the Government. Gaps exceeding 1.5 inches adjacent to the drainage layer shall be a non-conformance gap and require corrective action to seal the gap between adjacent shafts. A non-conformance gap requires the Contractor to propose remedial measures to fully close the gap at no cost to the Government.

Such remedial measures shall include 1) removal and reinstallation of the casing so that it is within the specified tolerance; or 2) post-installation of a jet grouted seal immediately adjacent to the shaft to seal the gap; or 3) other measures approved by the Contracting Officer.

All remedial measures proposed by the Contractor shall be subject to review and approval by the Government prior to implementation. The Contractor shall, at the Contractor's expense, promptly undertake all work required to reinstall the casing, or to install jet grouted seals at no additional expense to the Government. Jet grouted seals installed as remedial measures shall follow the requirements set forth in Section 31 32 23 JET GROUTING.

1.6.7 Drilled Shaft Contractor Experience

The Contractor or subcontractor shall have successfully completed a minimum of three (3) similar in-water, large diameter (equal or greater than 5 feet diameter), minimum 80-foot depth shaft foundation projects within the past ten years and that the company has the key personnel and equipment in which these personnel successfully filled the same role and successfully met the same requirement. Each representative drilled shaft project shall be at least 25 shafts of equal to or similar size and depth.

1.6.7.1 Site Superintendent

On-site superintendent that supervises all shaft drilling operations shall have a minimum of ten (10) years of experience in same capacity in construction of similar in-water, large diameter shaft (equal or greater than 5-feet diameter), minimum 80-foot depth foundation projects. The work experience shall be direct supervisory responsibility for the on-site drilled shaft construction operation. The Government reserves the right to suspend the construction work if unapproved personnel are substituted for site superintendent. The Contractor shall be fully liable for the additional costs resulting from the suspension of work and no adjustments in the contract time resulting from the suspension of the work will be allowed.

1.6.7.2 Supervisor Experience

On-site supervisors for each drilled shaft crew shall have a minimum of five (5) years of experience in same capacity in construction of similar in-water, large diameter (equal or greater than 5-feet diameter), minimum 80-foot depth shaft foundation projects. The work experience shall be direct supervisory responsibility for the on-site drilled shaft construction operation. The Government reserves the right to suspend the construction work if unapproved personnel are substituted for one or more of the approved supervisors. The Contractor shall be fully liable for the additional costs resulting from the suspension of work and no adjustments in the contract time resulting from the suspension of the work will be allowed.

1.6.7.3 Drill Operator Experience

Drill operators shall have a minimum of five (5) years prior experience as a drill operator in similar materials for in-water, large diameter shafts.

1.6.7.4 Cross Hole Sonic Logging (CSL) Testing Experience

- a. The testing organization shall have performed CSL tests in accordance with **ACI 336.1** on a minimum of three large diameter (4 feet in diameter or greater) drilled shaft projects in the last five (5) years. Test depths of projects should be at least 70 feet.
- b. Personnel conducting the tests for the testing organization shall have a minimum of three (3) years of experience in CSL testing.
- c. Evidence that the testing organization meets these qualifications shall be provided to the Contracting Officer in writing at least thirty (30) calendar days prior to beginning shaft construction.

1.6.7.5 Ultrasonic or Sonar Shaft Excavation Measuring Device Testing System Experience

- a. The testing organization shall have performed ultrasonic or sonar tests on a minimum of three large diameter (4 feet diameter or greater) drilled shaft projects in the last five (5) years. Test depths of projects should be at least 70 feet
- b. Personnel conducting the tests for the testing organization shall have a minimum of one (1) year of experience with the

ultrasonic or sonar testing system.

c. Evidence that the testing organization meets these qualifications shall be provided to the Contracting Officer in writing at least thirty (30) calendar days prior to beginning shaft construction.

1.6.7.6 Shaft Inspection Device Experience

The personnel shall have performed bottom sediment inspection on a minimum of three large diameter (4 feet diameter or greater) drilled shaft projects in the last five (5) years. Test depths should be at least 70 feet. Personnel conducting the inspection for the contractor shall have a minimum of one (1) year of experience with the mini-SID or similar method. Evidence that the testing organization meets these qualifications shall be provided to the Contracting Officer in writing at least thirty (30) calendar days prior to beginning shaft construction.

1.6.8 Welding Qualifications

Provide and maintain qualified procedures and personnel according to **AWS D1.1/D1.1M**. Submit Welding Certificates to the Contracting Officer a minimum of 30 days before beginning welding work.

1.6.9 Quality Control Personnel Qualifications

On-site quality control personnel related to drilled shaft construction shall have a minimum of five (5) years of experience in same capacity in construction of similar in- water, large diameter shaft foundation projects with similar complexity. Refer to Section **01 45 04 CONTRACTOR QUALITY CONTROL** for additional requirements for CQC personnel.

1.6.10 Survey

Submit a description of the proposed survey to the Contracting Officer for approval thirty (30) calendar days prior to commencement of shaft construction. This submittal shall include a description of the drilled shaft drilling template, guide, and survey equipment and procedures to be employed for positioning the casing, ensuring that it is within the specified tolerances, and monitoring the horizontal position and verticality of the casing during installation. Shaft excavations shall be surveyed using a ultrasonic or sonar testing system (SONICaliper or approved equivalent). Provide a sample form for gap width analysis. This submittal shall also include the accuracy of all devices used and proof that the accuracy is sufficient to monitor the casing in both the horizontal and vertical planes, and to detect deviations smaller than the specified tolerances. Submit certificate of calibration showing that the equipment has been calibrated current within twelve months prior to use.

1.6.11 Geotechnical Engineer Qualifications

The Geotechnical Engineer shall be a registered Professional Engineer in the Commonwealth of Puerto Rico with at least ten (10) years of experience and at least five (5) years of experience in drilled shaft installation.

1.6.12 Structural Engineer Qualifications

The Structural Engineer shall be a registered Professional Engineer in the Commonwealth of Puerto Rico with at least ten (10) years of experience and

at least five (5) years of experience in bridge construction and drilled shaft installation.

1.7 NONDESTRUCTIVE TESTING OF SHAFTS

All drilled shafts for the bridge abutments, all shafts for the Demonstration Wall, and middle pier, and twenty-five (25) percent of channel wall completed drilled shafts shall be tested with a nondestructive testing (NDT) method called Crosshole Sonic Logging (CSL) after at least three (3) days of curing time but before thirty (30) calendar days of curing time. If the quality control/assurance personnel or COR observe problems during shaft construction and the 25% of the tests were already performed, it is the Contractors responsibility to perform additional NDT tests to verify the integrity of the shaft at no additional cost to the Government. The purpose of CSL testing is to establish the integrity of the concrete structure and locate any void in the concrete. CSL testing shall be performed in accordance with [ASTM D6760](#). Upon completion of CSL testing, approval of test results, and after being approved by the Contracting Officer all water shall be removed from the access tubes. The tubes shall then be filled with a non-shrink grout. The CSL tubes for the demonstration test shafts do not need to be backfilled with grout following CSL testing as specified for production shafts. The CSL tests shall be performed by an approved independent testing organization hired by the Contractor. This independent testing organization shall meet the requirements indicated in paragraph SUBMITTALS above. The standard for acceptance of the shafts is sound, solid concrete without voids. If voids are detected, then the exact locations of the voids shall be further investigated and evaluated for repair. Repair measures shall be subject to the approval of the Contracting Officer. All required repairs and shaft replacements shall be at no additional cost to the Government.

PART 2 PRODUCTS

2.1 CASING

All casing used in the construction of the shafts shall be of steel with ample strength to resist handling transportation and installation stresses and the external pressure of the foundation material. The casing shall be watertight and shall be clean prior to placement in the excavation. The inside diameter of the casing shall be the specified diameter of the shaft. For a 5-foot nominal shaft diameter, the inside diameter of the casing shall be 60 inches. For a 6-foot nominal shaft diameter, the inside diameter of the casing shall be 72 inches. [If permanent casing is selected for the bridge middle pier, new steel shall be used for the casing.](#)

2.2 REINFORCING STEEL

Reinforcing steel used in the construction of shafts shall conform to Section [03 20 00](#) CONCRETE REINFORCING.

2.3 CONCRETE

Concrete used in construction of the shafts shall conform to Section [03 31 01](#) CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS.

2.4 SLURRY

Polymer or mineral slurry shall be used in conformance with the manufacturer's recommendations.

2.5 CROSS HOLE SONIC LOGGING (CSL) TESTING TUBES

Four (4) tubes shall be securely fastened and installed in every 5-foot diameter shaft and five (5) tubes shall be securely fastened and installed in every 6-foot diameter shaft at the inside perimeter of the reinforcing cage to permit access for CSL. The tubes shall be 2 inch inside diameter schedule 40 steel pipe conforming to [ASTM A53/A53M](#) Grade A or B. The pipes shall have a round, regular internal diameter free from defects or obstructions, including any at pipe joints. All pipe joints shall be threaded. If couplings reused, the couplings shall be fitted to provide a smooth interior surface to permit the free, unobstructed passage of a 1.3-inch diameter source and receiver probes. The tubes shall be watertight and free from corrosion with clean internal and external faces to ensure passage of the probes and a good bond between the concrete and the tubes. All access tubes shall be close-ended at the bottom and fitted with a removable, threaded cap on the top. After placing the concrete, exercise care when removing the caps from the access tubes so as not to apply excess torque, hammering, or other stresses which could break the bond between the tubes and the concrete. Label each access tube with a unique identifier at the top of the tube. Casing may be extended upward when the working platform is at a higher elevation than the top of the drilled shaft. The CSL tubes shall be protected with sleeves to prevent damage during demolition of the added concrete. CSL tubes installed in the drilled shaft at the bridge middle pier shall extend to the top of the column. At other locations, the tubes shall extend from the bottom of the reinforcement cage to at least two (2) feet above the top of the shaft.

2.5.1 Typical Cross Hole Sonic Logging (CSL) Test Equipment

Typical CSL test equipment includes but is not limited to the following components:

- a. A microprocessor based CSL system for display of individual CSL records, analog-digital conversion and recording of CSL data, analysis of receiver responses, and printing of CSL logs.
- b. Ultrasonic source and receiver probes for 2-inch I.D. pipe, as appropriate.
- c. An ultrasonic voltage pulsar to excite the source with a synchronized triggering system to start the recording system.
- d. A depth measurement device to determine and record depths.
- e. Appropriate filter/amplification and cable systems for CSL testing.

2.6 EQUIPMENT

2.6.1 Cranes

Cranes and drill rigs certified with suitable capacity and boom length shall be provided to perform the required work. Net crane surcharge shall not exceed the maximum surcharge load shown on the plans. If the

Contractor elects to use a crane that exceeds the maximum surcharge shown, submit documentation and design calculations prepared by a qualified professional engineer demonstrating that the larger surcharge loads may be accommodated by both the temporary and permanent construction that will be affected by surcharge loads. Such documentation and design calculations are subject to the approval by the Contracting Officer. Crane matting or load platforms, if used, shall be designed by a qualified professional engineer registered in the Commonwealth of Puerto Rico based on the maximum surcharge load and design calculations shall be submitted for approval by the Contracting Officer.

Cranes used for constructing the drilled shaft at the middle pier of the bridge shall be appropriately sized and equipped with attachments to safely reach the required shaft locations and shall not overload the working platform used to support the crane.

2.6.2 Equipment for Casing Installation

Equipment for casing installation shall be provided by the Contractor. The type and size of the equipment shall be capable of advancing the casing through the subsurface materials described in the boring logs to the specified depth without damage to the casing. The equipment shall be capable of advancing the casing through weathered rock and dense material as described in the boring logs. The Contractor shall demonstrate to the satisfaction of the Government with similar size and weight of casing; soil materials; and depths, that the equipment is capable of performing the work including advancing through obstructions including concrete excess, bulging from adjacent shafts, and metal components of piles and sheet piles.

2.6.3 Excavating Equipment

Suitable excavating equipment shall be provided to perform the required work. This equipment may consist of but is not limited to augers, chisels, bailing buckets, clamshells, and air lifts. The equipment shall be able to excavate materials described in the boring logs. The excavation equipment used shall be designed to prevent adverse differential hydrostatic pressures from developing in the casing as the equipment is lifted from the bottom. The excavation equipment shall be capable of constructing shafts to depths up to 20 feet below the deepest shaft shown in the plans. The equipment shall be capable of advancing through obstructions including concrete excess, bulging from adjacent shafts, and metal components of piles and sheet piles

2.6.4 Correlation and Confirmation Boring Drilling

Drilling equipment shall be capable of performing standard penetration tests in accordance with ASTM D1586/D1586M and rock coring to depths of at least 130 feet.

PART 3 EXECUTION

3.1 CORRELATION AND CONFIRMATION BORING DRILLING

The purpose of correlation boring drilling is to evaluate and document subsurface conditions to be encountered during construction by establishing a correlation between the drilled shaft rig drilling resistance and drill cuttings with the correlation boring log data.

The purpose of confirmation boring drilling is to evaluate and document subsurface conditions to ensure they are as assumed for design. The Contractor shall perform one confirmation boring at the middle pier of the utility bridge.

The Confirmation Boring Report shall be prepared by or under the direction of the Contractor's Geotechnical Engineer and submitted to the Government. The Correlation Boring Report shall present the detailed log of the confirmation boring for evaluation by the Government.

- a. Standard Penetration Tests (SPTs) shall be performed in accordance with ASTM D1586 every 5 feet from ground surface to an elevation 10 feet above the proposed shaft tip elevation. Continuous SPTs shall be performed from 10 feet above the proposed tip elevation to a depth 10 feet below the proposed tip elevation. If SPT refusal is encountered (defined as $N > 50$ blows/6 inches), NX-size coring of material shall be performed in lieu of SPTs.
- b. Contractor's Geotechnical Engineer shall visually log the correlation boring in accordance with ASTM D2488 as its drilled and record soil or rock type, relative density or consistency, soil wetness, stratum breaks, SPT N-values, recovery, elevation of decomposed rock ($N > 60$ bpf), and elevation of SPT refusal (50 blows/6 inches). Soils shall be classified according to the Unified Soil Classification System (USCS). For rock core samples the rock type, recovery, Rock Quality Designation (RQD) values, hardness, weathering and fracture characteristics shall be documented. Groundwater elevations shall be logged during drilling and before abandoning borehole. Drilling, sampling, and presentation of results should be in accordance with USACE South Atlantic Division DM-1110-1-1.
- c. Provide a Boring Location plan showing proposed location of boring and drilling instructions containing specific requirements for drilling and backfilling of boreholes.
- d. Right-of-entry and drilling permits shall be obtained by the Contractor where required. No drilling shall be performed prior to obtaining utility clearance.
- e. The method of locating boring should be planned and coordinated. Accurate determination of boring locations and elevations including any required surveys shall be the Contractor's responsibility. Datum and units of location coordinates and elevation shall be indicated on the boring logs and boring location map. All changes to proposed boring locations at any stage shall be documented.
- f. Correlation boring shall be abandoned by backfilling with grout. All grout materials shall be pumped into the borehole from the bottom to the top by pressure grouting with the positive displacement method (tremie method).
- g. The Correlation Boring Report shall be prepared by or under the direction of the Contractor's Geotechnical Engineer and submitted to the Government. The Correlation Boring Report shall present the detailed log of the correlation boring along with an assessment of the drilled shaft drilling resistance and cuttings compare to the correlation boring log data. The report shall include the Contractor's proposed methods and criteria for correlating the

subsurface conditions to the drilled shaft drilling resistance and cuttings. The report narrative shall also compare correlation boring log data to previous boring logs provided in Section 00 31 32, GEOTECHNICAL DATA REPORT.

3.2 SHAFT INSTALLATION

3.2.1 Wet Construction Method

The wet construction method consists of drilling the shaft excavation below the water table, keeping the shaft filled with slurry, de-sanding and cleaning the slurry, and final cleaning of the excavation by means of a bailing bucket, air lift, submersible pump, or other approved devices. Place the shaft concrete with a tremie or concrete pump extending to the shaft bottom, which displaces the water or slurry during concreting of the shaft excavation. Except where permanent casing is required, provide temporary casings to El. -50 feet or deeper, or as required and approved by the Contracting Officer, to aid shaft alignment and position and to prevent caving of the shaft sidewalls. Extend casing from the design top of wall elevation up to the working platform level where required. The Contractor shall use a template or approved similar means to facilitate alignment and position of the shafts. If a lean concrete template is used, all template concrete shall be dislodged and removed after shaft completion.

3.2.2 Protection of Existing Structures

Vibrations shall be monitored according to Section 02 22 13 VIBRATION AND NOISE MONITORING REQUIREMENTS. The Contractor shall take all reasonable precautions to prevent damage to existing structures, ground loss, or ground heave. The Contractor shall monitor structures for settlement in a manner approved by the Contracting Officer, recording elevations to 0.01 foot. Monitor any existing structures within 100-ft of the wall centerline.

Take elevations of existing structures being monitored:

- (1) Before beginning construction.
- (2) Daily during the driving of any casings.
- (3) During drilled shaft excavation.

Notify the Contracting Officer of movements detected greater than 0.01 foot and immediately take any remedial measures required to prevent damage to the existing structures.

Adequately document the [condition of existing structures](#) and all existing cracks with descriptions and pictures. Prepare two reports documenting the condition of the structures. Submit one report before beginning foundation construction operations, and submit a second report after completing foundation construction operations. The Government will take ownership of both reports.

Employ qualified Quality Control Personnel as specified in paragraph QUALITY CONTROL of this specification to monitor and record performance of existing structures during shaft excavation.

Upon detecting settlement or heave of 0.01 foot or damage to the structure, immediately stop the source of settlements, backfill any open drilled shaft excavations, and contact the Contracting Officer for instructions.

The Contractor is responsible for evaluating the need for, design of, and providing any necessary features to protect adjacent structures from the work or from deformations caused by excavations. When sheeting and shoring are used, employ a licensed Engineer experienced in the design of excavation support systems to design the sheeting and shoring, and to sign and seal the plans and specification requirements. Metal sheet piling design and installation shall be in accordance with Specification Section 31 41 16 METAL SHEET PILING.

Submit sheeting and shoring designs in accordance with Specification Section 31 41 16 METAL SHEET PILING to the Contracting Officer for his record before beginning construction.

The Contractor is responsible for evaluating the need for, design of, and providing all reasonable precautionary features to prevent damage, including, but not limited to, selecting construction methods and procedures that will prevent caving of the shaft excavation. The Contractor is responsible for monitoring and controlling the vibrations from construction activities, including driving of casing.

3.2.3 Demonstration Wall

The Contractor shall construct a Demonstration Wall consisting of three (3) drilled shafts, as shown on the drawings. The demonstration shafts top and tip elevations shall be as shown on the drawings. The Contractor shall implement the same installation and construction techniques, equipment, template, concrete mix, materials, CSL tubes, key personnel and subcontractors, and timing as planned for the production drilled shafts.

After the construction of the Demonstration Wall, the Contractor shall excavate soil in the upper 7.0 feet around wall and remove any template for Government inspection and evaluation. The final approval of the demonstration wall shall be based on the installation being within the prescribed timeframe, construction tolerances and the quality of the demonstration shafts, including but not limited to, drilling records and CSL results. Shaft excavation survey results, CSL results, casing installation records, vertical plumbness records, heave records, and concrete placement records shall be submitted to the Contracting Officer for evaluation and approval. The Demonstration Wall and demonstration drilled shaft shall comply with the contract specifications.

Each demonstration drilled shaft shall meet the requirements specified in subparagraph DEMONSTRATION SHAFT below. Installation of the production shafts will not be permitted until the Demonstration Wall and demonstration shafts have been approved by the Contracting Officer. If the Contractor fails to demonstrate to the Contracting Officer's satisfaction the adequacy of his personnel, methods, or equipment, and alterations are required, the Contractor shall make appropriate modifications and provide an additional Demonstration Wall at no expense to the Government. If a change in personnel (including subcontractors), methods or equipment is required, the Contractor shall provide an additional Demonstration Wall at no expense to the Government. An additional correlation boring shall be performed for subsequent demonstration walls to correlate drill rig performance with subsurface conditions. A correlation boring is not required if the first shaft location is within 20 feet of an existing boring.

Install a complete drainage system, including two groundwater weeps,

drainage material, and geotextile filter material, and jet grouted seal, as shown on the plans for the Demonstration Wall. All grout seal samples tested shall have a minimum 28-day unconfined compressive strength of 100 psi according to [ASTM D2166/D2166M](#). Construct a demonstration cap on top of the demonstration shafts as shown on the plans. The cap should demonstrate the installation of typical cap reinforcement, one expansion joint, and embedment for two guardrail posts.

Within 14 days of completing the Demonstration Wall, the Contractor shall submit a copy of the [demonstration wall report](#) to the Contracting Officer for approval. The demonstration wall report shall describe the results of the demonstration wall and include recommendations for correlating shaft drilling resistance and drill cuttings with encountered subsurface conditions.

After the Demonstration Wall and cap have been accepted by the Contracting Officer, restore the Demonstration Wall and cap as shown on the plans. Clean up the site of the Demonstration Wall and cap as required.

3.2.4 Demonstration Shafts for Wall

The Contracting Officer shall use the construction of the shafts along the Demonstration Wall (Demonstration Shafts) to determine if the methods and equipment used by the Contractor are sufficient to produce a shaft meeting the requirements of the Contract Documents. Demonstration Shafts shall be cased with temporary steel casing to a minimum elevation of -50.0 feet. Casing shall be installed using the same methods and equipment as the production drilled shafts. Jetting shall not be allowed. Demonstration Shafts shall be constructed under a similar differential head condition as the production drilled shafts.

During Demonstration Shaft excavations, the Contracting Officer will evaluate the ability to control dimensions and alignment of excavations within tolerances; to control the size of the excavation under caving conditions by the use of slurry or by other means; to properly clean the completed shaft excavation; to determine the elevation of ground water; to place reinforcing steel and concrete meeting the requirements of these specifications within the prescribed time frame; and to execute any other necessary construction operations. Contractor shall follow requirements of DRILLED SHAFT COMPLETION paragraphs during installation of demonstration shafts.

Revise the methods and equipment as necessary at any time during the construction of the Demonstration Shaft when unable to satisfactorily carry out any of the necessary operations described above or when unable to control the dimensions and alignment of the shaft excavation within tolerances. All deviations from the approved [Shaft Installation Plan](#) shall be subject to approval by the Contracting Officer.

Drill the Demonstration Shafts at the location shown on the plans or as approved by the Contracting Officer. Ensure that the diameter and depth of the Demonstration Shafts are the same diameters and maximum depths as the production drilled shafts. Construction methods shall result in the top portion of the concrete shaft being free of contamination. If the Contractor fails to demonstrate to the Contracting Officer's satisfaction the adequacy of his personnel, methods, or equipment, and alterations are required, make appropriate modifications, and provide additional demonstration shafts or walls at no expense to the Government. Make no changes in personnel, methods, or equipment after initial approval without

the consent of the Contracting Officer. If a change in personnel, methods or equipment is required or if additional crews are utilized, the Contractor shall provide an additional demonstration shaft or wall at no expense to the Government.

During drilled shaft drilling process for the Demonstration Shafts, the Contractor's Geotechnical Engineer shall log the drill bit cuttings and monitor drilling resistance and compare observations with the boring log material descriptions, N-values, recovery, and RQD. Contractor's Geotechnical Engineer shall identify correlations between drill cuttings, drilling characteristics and resistance/difficulty to the boring log data with depth as noted for each drill rig and crew used for drilled shaft construction. If casing refusal occurs, drilling resistance and the corresponding strata shall be fully documented. Contractor's Geotechnical Engineer shall prepare and submit to the Contractor Officer, a Demonstration Shaft Report documenting the above results with recommendations for correlating shaft drilling resistance and drill cuttings with encountered subsurface conditions.

3.2.5 Casing Advancement

The casing shall be positioned as specified, held in place, and advanced open ended with no internal plates into the underlying materials. Unless refusal is met or the casing is deflected, the casing shall be advanced its full length before commencing excavation. If casing refusal is encountered or the casing is deflected outside of specified tolerance, the Contractor shall propose alternative methods to restore casing advancement or to adjust the casing to within specified tolerance. The alternative methods shall be subject to the approval of the Contracting Officer and shall be implemented by the Contractor at no cost to the Government.

3.2.6 Shaft Alignment and Depth

3.2.6.1 Casing Alignment

During the advancement of the casing, the Contractor shall make frequent checks on the plumbness and alignment of the casing. Any deviation exceeding the allowable tolerances shall be corrected with approved procedures. The Contractor shall keep a record of the casing alignment that shall include all alignment measurements taken and the corresponding depth of the casing tip.

3.2.6.2 Temporary Casing

All drilled shafts, except those shafts requiring permanent casing as shown, shall be temporarily cased from the top of the shaft to a minimum elevation of -50 feet or as required and approved by the Contracting Officer. The Contractor shall determine the casing wall thickness, the splice type, and the splice locations to carry loads that will be experienced from transportation, handling, installation, extraction, and construction. All welding regarding the casing splicing shall conform to the requirements of Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS. The casing depth shall be monitored during the advancement. Cuttings brought to the surface during advancement shall be visually logged by the Contractor's Geotechnical Engineer in accordance with ASTM D2488.

Advance the excavation by drilling in a slurry unless otherwise approved by the Contracting Officer. The Contracting Officer may require overreaming to the outside diameter of the casing. Over-reaming, if

required, shall be at no additional cost to the Government. Take whatever steps are required to prevent caving during shaft excavation including installation of deeper casings as required.

If a casing is removed and replaced with a longer casing through caving soils, stabilize the excavation with slurry or backfill the excavation. The Contractor may use soil previously excavated or soil from the site if backfilling the excavation. The Contractor may use other approved methods that will control the size of the excavation and protect the integrity of the foundation soils to excavate through caving layers.

Casing is to be measured and marked at 1-foot increments for the portion of the casing anticipated to be above the ground as installation proceeds. Monitoring intervals shall be sufficient to determine casing advancement rates. The excavation method(s), depths, volumes, and measured heaves of the excavation bottom shall be reported as well as any modification of the slurry.

3.2.6.3 Temporary Casing Removal for Channel Walls and Utility Poles

Casings shall be removed during or after concrete placement, but before concrete starts to harden.

3.2.6.4 Casing for Bridge Middle Pier Drilled Shaft

Temporary or Permanent is required to construct the drilled shaft and column for the bridge middle pier. **Permanent casing shall be comprised of new steel.** Casing tip shall extend to El. -44 feet. The Contractor shall determine the casing wall thickness, the splice type, and the splice locations to carry loads that will be experienced from transportation, handling, installation, extraction, and construction. Top of casing shall be adequately supported and braced to construct columns within construction tolerances. Bolted casing molds are permitted above ground level.

The casing depth shall be monitored during the advancement. Cuttings brought to the surface during advancement shall be visually logged by the Contractor's Geotechnical Engineer in accordance with **ASTM D2488**.

Advance the excavation by drilling in a slurry unless otherwise approved by the Contracting Officer. The Contracting Officer may require overreaming to the outside diameter of the casing. Over-reaming, if required, shall be at no additional cost to the Government. Take whatever steps are required to prevent caving during shaft excavation including installation of deeper casings as required. The extension of casing below El. -44 feet due to soil instability shall be at no cost to the Government.

If a casing is removed and replaced with a longer casing through caving soils, stabilize the excavation with slurry or backfill the excavation. The Contractor may use soil previously excavated or soil from the site if backfilling the excavation. The Contractor may use other approved methods that will control the size of the excavation, protect the integrity of the foundation soils to excavate through caving layers, and prevent damage or displacement of shafts previously placed.

Casing is to be measured and marked at 1-foot increments for the portion of the casing anticipated to be above the ground as installation proceeds. Monitoring intervals shall be sufficient to determine casing

advancement rates. The excavation method(s), depths, volumes, and measured heaves of the excavation bottom shall also be reported as well as any modification of the slurry.

The steel casing shall be removed above El. 0 feet after the column concrete has reached at least 85 percent of its design strength. The method of removing the casing at the cut-off elevation shall be subject to the approval of the Contracting Officer. Torch cutting and carbon arc gouging are not permitted. Casing below El. 0 feet shall remain in place.

Horizontal and vertical construction tolerances of the middle pier shafts shall be in accordance with paragraph SHAFT CONSTRUCTION TOLERANCES.

3.3 EXCAVATION

All soil and rock materials shall be excavated to the specified tip elevation. Use equipment or methods to excavate near the casing tip so as not to cause blow-ins, quick soil conditions, scouring, caving or other disturbance to the surrounding in situ materials. Cuttings brought to the surface during excavation shall be visually logged by the Contractor's Geotechnical Engineer in accordance with [ASTM D2488](#).

3.3.1 Excavation Depth

The excavation depth shall be monitored during excavation. The frequency and precision of the depth monitoring shall be sufficient to determine if heaving is occurring at the bottom of the excavation. The bottom of the shaft excavation shall be normal to the axis of the shaft within a tolerance of 3/8 inch per foot of shaft diameter. In conjunction with measuring the excavation depth, the verticality of the shaft at the measured depth shall be determined.

3.3.2 Excavation Stability

Temporary casing shall be extended below the minimum tip elevation at Contractor's expense if signs of shaft instability are observed.

Excavation shall be performed using equipment and methods that will minimize the potential for loss of in situ strength of materials beyond the casing such as by heaving and flowing of materials into excavated area due to negative differential hydrostatic pressures generated by the lifting of tight fitting excavating tools. Stability at the shaft tip shall be monitored and evaluated based on the amount of heave that occurs at each excavation stage. Immediately before lifting an excavation tool off the bottom of the excavation, the bottom of the excavation elevation shall be determined to an accuracy of 3 inches. The bottom of the excavation shall be remeasured to the specified level of precision after the excavation tool has been completely removed from the shaft.

Monitor, evaluate, and correct any deficiencies to the existing conditions of soil level inside the casing, slurry level, slurry properties, excavation techniques, and the outside soil and water level. Deficiencies in specified requirements shall be corrected to eliminate the negative differential pressures contributing to heave before excavation resumes. If heave reoccurs, the causes cannot be resolved, and there is no sign of settlement to the existing structure, the Contracting Officer may allow excavation to continue. Prepare a complete record of heave for each excavation sequence for the full depth of the shaft.

Existing structures shall be monitored during excavation in accordance with these specifications.

3.3.3 Drilled Shaft Excavation Logging

Cuttings brought to the surface during drilled shaft excavation advancement shall be visually logged with depth by the Contractor's Geotechnical Engineer. Contractor's Geotechnical Engineer shall also note any abrupt changes in drilling conditions (e.g. loss of slurry, sudden drop or increase in excavated depth indicating void or collapse, etc.).

Tangent drilled shaft wall design was based in part on the top of decomposed to weathered rock (materials with consistent minimum N-value of 60 blows per foot) at the approximate elevations shown in the table below. The top of decomposed to weathered rock should be encountered at or above the elevation shown in the table for the corresponding range of stations.

Begin Station (Approximate)	End Station (Approximate)	Approximate Decomposed/Weathered Rock Elevation (ft)
97+50	99+25	-80
99+25	109+00	-48
100+00 (Siphon)		-80
109+00	109+25	-44
109+25	111+80	-48
111+80	117+00	-40
117+00	118+50	-36
123+50	131+00	-49
131+00	138+50	-45
138+50	147+40	-70

If decomposed to weathered rock is encountered at elevations more than five (5) feet deeper than the elevations noted in the table, the Contractor's Geotechnical Engineer shall notify the Contracting Officer. This notice shall be given during the drilling process as soon as it is learned that the encountered top of decomposed to weathered rock is more than five (5) feet deeper than the stated elevation values. The Contractor shall continue to drill, report the actual depth to weathered rock, finish drilling the shaft to the design bottom elevation, and await direction from the Contracting Officer prior to completing the drilled shaft construction. If soft rock or soil is encountered at the design bottom elevation the Contractor's Geotechnical Engineer shall notify the Contracting Officer.

3.4 CLEANUP AND FINAL EXCAVATION CHECKS

Cleanup shall immediately follow the completion of excavation, including desanding and reconditioning the slurry media within the casing; removing any materials adhering to the sides of the casing; removing accumulated sediment from the bottom of the excavation; and performing the required tests and measurements to assure that the cleanup work has been performed as specified.

3.4.1 De-sanding

De-sanding shall consist of removing excessive sand from suspension to prevent further sediment buildup on the bottom.

3.4.2 Bottom Sediment

For the bridge middle piers, no more than 50% of the shaft bottom shall have 1 inch or less of sediments with no sediments exceeding 1.5 inches in thickness prior to placing concrete. Shafts for walls and bridge abutments shall have no more than an average of 2 inches of loose soil or sediments at the bottom of the shaft excavation and no more than 3 inches at each measurement. The excavated shaft shall be inspected and approved by the Contracting Officer prior to proceeding with steel cage installation. Approval shall be based on sounding the bottom of the excavated shaft with an approved device on a pre-established grid pattern, spaced not more than 2 feet by 2 feet, as well as at the edge. Soundings shall be compared to readings previously taken immediately following the end of the excavation work. The sounding device shall consist of a Shaft Inspection Device (mini-SID) or alternate device approved by the Contracting Officer. The design of the sounding device, the number of soundings, and grid spacing shall be approved by the Contracting Officer.

3.5 SLURRY

3.5.1 Slurry Conformance Requirements

Slurry shall be sampled and tested before use and during shaft excavation.

3.5.2 Slurry Mixing and Placement

Use slurry having a grain size such that it will remain in suspension and having sufficient viscosity and gel characteristics to transport excavated material to a suitable screening system. Use a percentage and specific gravity of the material to make the suspension sufficient to maintain the stability of the excavation and to allow proper placement of concrete.

Ensure that the material used to make the slurry is not detrimental to concrete or surrounding ground strata. During construction, maintain the level of the slurry at a height sufficient to prevent caving of the hole. In the event of a sudden loss of slurry such that the slurry level cannot practically be maintained by adding slurry to the hole, document depth of shaft at time of slurry loss, and delay the construction of that foundation until an alternate construction procedure has been approved. Such delays and corrective procedures shall be at no cost to the Government.

Thoroughly premix the slurry with clean, fresh water prior to introduction into the shaft excavation. Ensure that the percentage of slurry used to make the suspension is such as to maintain the stability of the shaft

excavation. The Contracting Officer will require adequate water and/or slurry tanks when necessary to perform the work in accordance with these specifications. Excavated pits for slurry shall not be used without the written permission of the Contracting Officer.

Provide adequate de-sanding equipment as specified. Take steps necessary to prevent the slurry from "setting up" in the shaft, including but not limited to agitation, circulation, and/or adjusting the composition and properties of the slurry. Dispose of all waste slurry in a manner meeting all federal and local requirements pertaining to pollution.

Provide a qualified professional soil testing laboratory approved by the Contracting Officer to perform control tests using suitable apparatus on the slurry mixture to determine the following:

3.5.2.1 Freshly Mixed Slurry

Measure the density of the freshly mixed slurry regularly as a check on the quality of the suspension being formed using a measuring device calibrated to read within plus or minus 0.5 pounds/cubic foot.

3.5.2.2 Slurry Supplied to the Drilled Shaft Excavation

Perform the following tests on the slurry supplied to the shaft excavation and ensure that the results are within the ranges stated in the table below:

MINERAL SLURRY

<u>Item to be Measured</u>	<u>Range of Results at 68 degrees F</u>	<u>Test Method</u>
Density (pcf)	64-73 pounds/cubic foot or as required for proper performance	Mud Weight (Density), API RP 13B-1 Section 1
Viscosity (s/qt)	28 to 40 seconds	Marsh Funnel and Cup API RP 13B-1 Section 2.2
pH	8 to 11	Glass Electrode, pH Meter or pH Paper
Sand Content	4 percent or less	Sand, API RP 13B-1 Section 5

POLYMER SLURRY

<u>Item to be Measured</u>	<u>Range of Results at 68 degrees F</u>	<u>Test Method</u>
Density (pcf)	as required for proper performance	Mud Weight (Density), API RP 13B-1 Section 1
Viscosity (s/qt)	32 to 135 seconds	Marsh Funnel and Cup API RP 13B-1 Section 2.2
pH	8 to 11.5	Glass Electrode, pH Meter or pH Paper
Sand Content	1.0 percent or less	Sand, API RP 13B-1 Section 5

The Contractor may adjust the limits in the above table when field conditions warrant if approved by the Contracting Officer. The Contracting Officer must approve all changes in writing before the Contractor can apply them.

Perform tests to determine density, viscosity, and pH value to establish a consistent working pattern accounting for the mixing process and blending of freshly mixed slurry and previously used slurry. Perform a minimum of four sets of tests to determine density, viscosity, and pH value during the first 8 hours mineral slurry is in use.

When the results show consistent behavior, discontinue the tests for pH value, and perform tests to determine density and viscosity during each four hours slurry is in use. If the consistent working pattern changes, reintroduce the additional tests for pH value for the time required to establish consistency of the test values within the required range.

3.5.2.3 Slurry Test Reports

Furnish reports of all slurry tests required above to the Contracting Officer on completion of each drilled shaft. Reports shall be signed and sealed by a Quality Control Personnel representing the soil testing laboratory.

3.5.3 Slurry Level

During construction, maintain the level of slurry in the shaft excavation at a level not less than 4 feet above the highest expected groundwater level adjacent to the shaft or as determined by excavation stability calculations performed by the Contractor's Geotechnical engineer.

3.5.4 Failure of Wet Construction Method

At any time the wet construction method of stabilizing excavation fails to meet criteria and fails to produce the desired final result, discontinue

this method of construction, and propose modifications in procedure or alternate means of construction for approval by the Contracting Officer. Any modification to the method of construction will be at the Contractor's expense and satisfaction of the Contracting Officer.

3.5.5 Containment of Slurries

Prevent all slurry and excavated materials from entering the river. Approved protective devices and measures shall be used to contain all potential spillage including drippings from the excavation tools. Slurry and excavated materials shall be contained in water-tight containers. The containers shall be emptied and the materials transported to the disposal site at regular intervals to prevent overflowing. No slurry or excavated materials shall be allowed to spill on the site. Slurry and waste materials shall be transported to the disposal site in water-tight containers and prevented from spilling. Any spills shall be cleaned up immediately.

3.6 DISPOSAL

The disposal of the excavated materials and waste slurry shall be at offsite disposal areas. Comply with all local environmental regulations in the disposal of the materials. Materials shall be disposed of in accordance with the approved disposal plan, or as approved by the Contracting Officer.

3.7 PROTECTION

If the excavation or other work is stopped before completion, the casing shall remain supported and held within the contract tolerances until the concrete placement is completed. Following the construction, the CSL tubing shall be protected from damage from construction activities or vandalism.

3.8 DRILLED SHAFT GROUND IMPROVEMENTS

Ground improvement measures may be necessary around shafts where the adjacent in situ soil has been disturbed by caving or heaving so that the strength of the soil can no longer be counted on, either for an individual layer or the entire sequence. The need for ground improvements shall be determined by the Contracting Officer for the heaving records and other records generated during the advancement of each casing and during excavation. Ground improvement methods shall be as approved by the Contracting Officer. All costs for ground improvement shall be borne by the Contractor.

3.8.1 Jet Grouted Seals

Jet grouted seals shall be installed between shafts at the bridge abutments and at designated locations as shown on the drawings. In addition, when approved by the Government, jet grouted seals may be employed to mitigate gaps between adjacent tangent drilled shafts or for shafts that otherwise do not meet the specified construction tolerances. Jet grout seals used to correct gaps between tangent drilled shafts or to correct for shafts that do not meet specified construction tolerances shall be installed at the Contractor's expense.

Jet grouted seals are a mitigation option for drilled shafts that do not meet the horizontal and/or vertical plumbness tolerances above El. -20

feet. When approved by the Contracting Officer, jet grouted seals shall be installed behind the tangent point of the drilled shaft and the adjacent drilled shaft on each side. The jet grouted seal shall be installed from the bottom of the cap beam down to El. -20 and shall have intimate contact with the drilled shaft on each side of the jet grouted seal. Headwalls above stormwater outfalls require the seal to be from the concrete shelf down to El. -20. Jet grouted seals shall be installed before wall drainage material is placed. Jet grouted seals shall be installed in accordance with Section 31 32 23 JET GROUTING. All grout seal samples tested shall have a minimum 28-day unconfined compressive strength of 100 psi according to ASTM D2166/D2166M. All jet grouting installed due to drilled shafts not meeting construction tolerances shall be at no cost to the Government.

3.9 INDIVIDUAL SHAFT INSTALLATION RECORDS

Throughout the shaft installation operation, comprehensive advancement, excavation and associated data shall be recorded. The records shall be neat, orderly and complete. All records for an individual shaft shall be submitted to the Contracting Officer after the shaft excavation is complete. The recorded data for each shaft shall be on a form approved by the Contracting Officer. At a minimum, the recorded data shall include:

- a. Shaft identification.
- b. Penetration of casing (depth and rate).
- c. Detailed description of materials removed from each shaft including depth, and size and shape of cobbles and boulders.
- d. Interruptions in advancement, including initial "setup" time.
- e. Vertical plumbness of shaft including electronic data readout. Provide a calculation or diagram showing the gap width along the length of the drilled shaft when installation occurs adjacent to a previously constructed drilled shaft.
- f. Problems occurring during shaft construction such as but not limited to evidence of caving, loss of slurry, bottom heave, etc.
- g. River conditions during shaft installation.
- h. Concreting data forms.
- i. Final results of bottom inspection.

3.10 DRILLED SHAFT COMPLETION

3.10.1 Inspection

Prior to concrete placement, the Contractor shall document the depth, diameter, center offset/encroachment, and plumbness of each shaft after excavation with an ultrasonic or sonar measuring device (e.g. SoniCaliper, SHAPE, or similar) in the presence of the Contracting Officer. The Contractor shall inspect the shaft bottom using an electronic Shaft Inspection Device (SID) in the presence of the Contracting Officer. Each excavated shaft shall be inspected by the Contractor and approved by the Contracting Officer prior to proceeding with [concreting](#).

3.10.2 Depth

The depth of the shaft during drilling shall be referenced to appropriate marks on the Kelly bar or other suitable methods. Final shaft depths shall be documented with an ultrasonic/sonar measuring device or other approved method after final cleaning.

3.10.3 Shaft Cleanliness Requirements

Immediately prior to placing concrete, perform cleaning operations so that the bottom of each shaft will meet the following criteria. For the bridge middle piers, no more than 50% of the shaft bottom shall have 1 inch or less of sediments with no sediments exceeding 1.5 inches in thickness prior to placing concrete. Shafts for walls and bridge abutments shall have no more than an average of 2 inches of loose soil or sediments at the bottom of the shaft excavation and a maximum sediment depth of 3 inches for any measurement. Clean and remove any contaminated concrete at the top of each shaft tremie concrete pour before cast-in-place concrete cap formwork or reinforcement is in place.

3.10.4 Time of Excavation

If shaft excavation work lasts more than 36 elapsed hours (measured from the beginning of excavation below the casing), the Contractor may be required to over-ream the sidewalls to the depth of softening or to remove excessive slurry cake buildup as indicated by samples taken by the sidewall sampler or other test methods employed by the Contracting Officer. Ensure that over-reaming performed is no less than the shaft diameter plus 1/2 inch and is no more than the shaft diameter plus 3 inches. Provide any overreaming required at no expense to the Government when exceeding the 36-hour limit unless the time limit is exceeded solely to accomplish excavating deeper than the elevation shown in the plans as ordered by the Contracting Officer.

When using slurry, adjust excavation operations so that the maximum time that slurry is in contact with the uncased portion of the shaft (from time of drilling to concreting) does not exceed 12 hours. If exceeding the 12-hour time limit, overream the uncased portion of the shaft at no additional expense to the Government prior to performing other operations in the shaft.

3.11 ASSEMBLY AND PLACEMENT OF REINFORCING STEEL

3.11.1 Reinforcing Cage

The reinforcing cage shall be 100 percent tied and shall be rigidly braced to retain its configuration during handling and construction. Individual or loose bars will not be permitted. The cage of reinforcing steel consisting of longitudinal bars, ties, CSL tubes, and cage stiffener bars shall be completely assembled and placed as a unit immediately after the shaft excavation is inspected and accepted and immediately prior to concrete placement. All intersections of drilled shaft reinforcing steel shall be tied with cross ties or "figure 8" ties. Double strand ties or ties with larger tie wire shall be used when necessary. Final approval of the cage construction and placement is subject to satisfactory performance in the field.

3.11.2 Reinforcement Positioning

The reinforcement shall be carefully positioned and securely fashioned to provide the minimum clearances shown on the contract drawings and to ensure that no displacement of the reinforcing steel bars occurs during placement of the concrete. Spacer wheels or noncorrosive spacing devices shall be used to ensure concentric spacing for the entire length of the cage. Block type spacers will not be allowed. The spacer shall be positioned as follows for drilled shaft channel walls and middle pier shafts:

3.11.2.1 Vertical

- Within the length of the steel casing: 6-foot maximum spacing
- Below the steel casing: 8-foot maximum spacing except that for the bottom 10-feet, the maximum spacing shall be 2 feet

3.11.2.2 Horizontal (Along Circumference)

- For 5-foot diameter shafts: 5 spacers equally spaced at every level
- For 6-foot diameter shafts: 6 spacers equally spaced at every level

3.11.2.3 Concrete Spacers

Concrete spacers shall be provided at the bottom of the drilled shaft reinforcing cage to ensure that the specified distance between the bottom of the cage and the bottom of the shaft is maintained. The spacers shall be constructed of approved material equal in quality and durability to the concrete specified for the shaft. The Contractor shall submit details of the proposed concrete spacers along with the shop drawings.

The elevation of the top of the steel cage shall be checked before and after the concrete is placed. If the cage is not maintained within the specified tolerances, corrections shall be made by the Contractor. No additional shafts shall be constructed until the Contractor has modified the cage support in a manner satisfactory to the Contracting Officer.

3.11.2.4 Requirements for Utility Pole Drilled Shafts

Place, tie and/or clip cage symmetrically about the axis of the shaft. Use centering devices securely attached to the cage to clear the shaft walls and maintain the cage in place throughout the concrete placement operations.

Cooperate with other trades in setting of anchor bolts, inserts, and other embedded items. Where conflicts occur between reinforcing and embedded items, notify the Contracting Officer in order to reconcile conflicts before concrete placement. Position and support anchors and embedded items with appropriate accessories.

Use templates to set anchor bolts, leveling plates and other accessories required for structure erection. Provide blocking and/or holding devices to maintain required anchoring positions during final concrete placement.

3.11.3 Placement of Reinforcing Bars

Reinforcing bar placement shall be as shown in the drawings.

3.11.4 Reinforcing Splice

Reinforcement splices shall conform to **ACI 318** and shall be made only as required or indicated. Splicing shall be by lapping or by mechanical means except that lap splices shall not be used for bars larger than No. 11 unless otherwise indicated. Lapped bars shall be placed in contact and securely tied or spaced transversely apart to permit the embedment of the entire surface of each bar in concrete. Lapped bars shall be spaced as specified in Section **03 20 00 CONCRETE REINFORCING**. **Mechanical butt** splices shall be in accordance with the recommendation of the manufacturer of the mechanical splicing device. Mechanical coupler splices shall be as specified in Section **03 20 00 CONCRETE REINFORCING**.

3.12 CSL ACCESS TUBES

- a. Four (4) access tubes for CSL testing shall be installed in each 5-foot diameter drilled shaft and five (5) access tubes shall be installed in each 6-foot diameter shaft. The tubes shall have a round, regular inside diameter, free of defects or obstruction, as specified.
- b. The tubes shall be securely attached to the interior of the reinforcement cage. The tubes shall be installed in each drilled shaft equally spaced. The tubes shall be installed in straight alignment and as near to parallel to the vertical axis of the reinforcement cage as possible. The tubes shall extend from the bottom of the reinforcement cage to at least two (2) feet above the top of the shaft except at the bridge middle pier. At the bridge middle pier, the tubes shall extend to the top of column. Joints required to achieve full length tubes shall be watertight. The CSL tubes installed in **the** drilled shaft for the middle pier of the bridge shall extend from the bottom of the reinforcement cage to the top of the column. Prevent damaging the tubes during reinforcement cage installation operations in the drilled shaft excavation. The tubes shall be filled with potable water as soon as possible after concrete placement (but no later than one (1) day) and the tube tops shall be capped.
- c. The tubes shall be placed around the shaft, inside the spiral or hoop reinforcement, at a uniform spacing, measured along the circle passing through the centers of the tubes.

3.13 CSL LOGGING PROCEDURES

- a. Before the placement of concrete, a minimum of one tube per shaft shall be plumbed and the tube length recorded, including a notation of the stickup of the tubes above the shaft tops. Information on the shaft bottom and top elevations and/or length, along with construction dates shall be provided to the Contracting Officer and the approved testing organization before the CSL tests.
- b. CSL tests shall be conducted between the pairs of tubes encompassing the perimeter and the major diagonals. Additional logs shall be conducted at no additional cost in the event anomalies are detected. For shafts with five tubes, test all pairs of adjacent tubes around the perimeter, as approved by the Contracting Officer.
- c. The CSL tests shall be carried out with the source and receiver

probes in the same horizontal plane unless test results indicate potential defects in which case the questionable zone may be further evaluated with angled tests (source and receiver vertically offset in the tubes). CSL measurements shall be made at depth intervals of 4 inches or less, and shall be done from the bottom of the tubes to the top of each shaft. The probes shall be pulled simultaneously, starting from the bottoms of the tubes, over a depth measuring device.

- d. Remove all slack from the cables prior to pulling to provide for accurate depth measurements. Any defects indicated by longer pulse arrival times and significantly lower amplitude/energy signals shall be reported to the Contracting Officer and further tests shall be conducted as required to evaluate the extent of such defects. Additional NDT methods that may be used to evaluate possible defects include Single Hole Sonic Logging, Gamma-Gamma Nuclear Density Logging, and/or Surface Sonic Echo and Impulse Response tests. Additional NDT method shall be implemented as approved by the Contracting Officer and at the Contractor's expense.

3.13.1 CSL Testing Results

The CSL results shall be presented to the Contracting Officer in a [CSL test report](#) within 7 working days after completion of the test. The test results shall include CSL logs with analyses of:

- Initial pulse arrival time versus depth
- Pulse energy/amplitude versus depth

A CSL log shall be presented for each tube pair tested. All observed defects shall be indicated on the logs and discussed in the test report as appropriate.

3.13.2 Evaluation of CSL Test Results

The Contracting Officer will evaluate the CSL test results and determine whether or not the drilled shaft construction is acceptable. The rating of CSL test will generally follow the criteria listed below:

RATING	NDT RESULTS INDICATIVE OF DRILLED SHAFT CONCRETE CONDITION
Good (G)	No signal distortion and decrease in signal velocity of 10% or less are indicative of good quality concrete.
Questionable (Q)	Minor signal distortion and a lower signal amplitude with a decrease in signal velocity between 10 percent and 20 percent. Results indicative of minor contamination or intrusion and/or questionable quality concrete.

RATING	NDT RESULTS INDICATIVE OF DRILLED SHAFT CONCRETE CONDITION
Poor/Defect (P/D)	Severe signal distortion and much lower signal amplitude with a decrease in signal velocity of 20 percent or more. Results indicative of water slurry contamination or soil intrusion and/or poor quality concrete.
No Signal (NS)	No signal was received. Highly probable that a soil intrusion or other severe defect has absorbed the signal
Water (W)	A measured signal velocity of nominally $V = 4,790$ to $4,990$ fps. This is indicative of a water intrusion or of a water filled gravel intrusion with few or no fines present.

Ratings other than "Good (G)" will be considered unacceptable. If the Contracting Officer determines that the drilled shaft is unacceptable based on the CSL tests, the shaft shall be cored as specified to enable further evaluation of the shaft. Coring shall be performed by the Contractor without additional compensation. The results of the evaluation shall be utilized to develop the repair program or to determine a need for replacement of the drilled shaft. The replacement or repair of the drilled shaft shall be made to the satisfaction of the Contracting Officer at no additional cost to the Government.

If the concrete quality was rated other than "Good (G)" based on CSL test results, the Contracting Officer may request additional CSL testing on other shafts to ensure good quality of the concrete placed. These additional tests shall be at no additional cost to the Government.

3.13.3 Core Drilling of Drilled Shaft Concrete

a. Drilled shafts that are determined to be unacceptable by the CSL tests may be cored to determine the quality of the concrete. A 3-inch core hole should be drilled in any questionable quality shaft (as determined from CSL test results) to explore the shaft condition. A coring method that provides complete core recovery and minimizes abrasion and erosion of the core (i.e., double- or triple-tube core barrels) shall be used. At each irregularity identified by CSL testing, obtain a 3-inch diameter core sample at each defective area for the full depth of the irregularities and for three feet above and below the irregularity. No compensation for time and related cost associated with the coring, repair, or replacement of the shaft will be granted by the Government. All inspection pipes and cored holes shall be grouted in accordance with paragraph GROUTING THE CSL TESTING TUBES.

b. The core drill shall be capable of obtaining a high percentage of core recovery for visual inspection and testing. The core drill shall be in good condition and capable of delivering a smooth flow of power to the bit, both in rotation and down thrust. The pump shall be positive displacement and shall be in good condition. The pump shall be capable of delivering a minimum of fifteen gallons of water per minute at two hundred psi. It shall be equipped with a relief valve set to release at a maximum of 200 psi. It shall be equipped with a pressure gauge with range from zero through 300 psi.

3.14 PLACING CONCRETE IN SHAFT

3.14.1 Fluid in Excavation at Time of Concrete Placement

Prior to placing concrete in any shaft excavation, ensure that heavily contaminated suspensions, which could impair the free flow of concrete from the tremie pipe, have not accumulated in the bottom of the shaft. Take samples of the slurry in the shaft from the base of the shaft and at intervals not exceeding 10 feet up the shaft using an approved sampling tool. Ensure that the density of the slurry in the shaft excavation, prior to concreting is less than 75 pounds per cubic foot. Modify the slurry in the shaft excavation prior to placing the concrete to bring the slurry within the specification requirements.

When using slurry, tests shall be performed by an approved soil testing laboratory engaged by the Contractor or by an experienced person furnished by the Contractor and approved by the Contracting Officer. Provide comparison tests when requested by the Contracting Officer as listed in paragraph SLURRY TEST REPORTS.

3.14.2 Immediate Placement

Concrete placement shall commence after completion of excavation, inspection, reinforcing placement, and CSL access tube installation or as approved by the Contracting Officer. Concrete placement shall be in one continuous operation to the top of the shaft. There shall be no work stoppage during this operation until all concrete in the shaft has been placed. Any unforeseen stoppage of work may require a horizontal construction joint during shaft construction. For this reason, an emergency construction joint method shall be submitted to the Contracting Officer for approval prior to starting shaft construction.

3.14.3 Concrete Placement

- a. The maximum temperature of the concrete when it is delivered to the shaft shall be 90 degrees F.
- b. Place concrete immediately after completing all excavation, cleaning the shaft excavation, inspection, and placing reinforcing steel.
- c. Concrete for drilled shaft shall be deposited under water or slurry by means of the tremie method. Concrete placement shall commence within a period of 8 hours after the shaft has been cleaned out.
- d. Placing concrete in each shaft shall be a continuous operation

lasting not more than the time specified in PLACEMENT TIME paragraph between placing the first load of concrete and completion of placing the final load of concrete, unless otherwise approved in writing by the Contracting Officer. The concrete shall be placed with concrete pumps and a gravity fed delivery tube system of adequate number and size to complete the placing of concrete in the time specified. The delivery tube system shall consist of a tremie tube or tubes with diameters not less than 5 inches. Concrete pumps may be used to charge suitably sized funnels or hoppers to facilitate the transfer of concrete to the gravity fed tremie. No more than one tube shall be supplied from a single funnel or hopper.

e. Concrete may be delivered to the tremie hopper by pump or other means that will provide a continuous flow of concrete. Direct pumping of the concrete into the tremie delivery tubes (without a hopper) may be used if the placement requirements are met. The concrete shall be maintained at a higher head than the slurry and the slurry head shall be higher than the ground water head.

f. The delivery tube system shall consist of watertight tubes with gasketed joints and with sufficient rigidity to keep the ends always in the mass of concrete being placed. Tremie tubes shall be marked to allow the distance from the surface of the water to the mouth of the tremie to be determined. If only one delivery tube is utilized to place the concrete, the tube shall be placed near the center of the drilled hole. If multiple tubes are used, they shall be uniformly spaced in the hole. Internal bracing for the steel reinforcing cage shall accommodate the delivery tube system.

g. A fully operational standby concrete delivery system, adequate to complete the work in the time specified, shall be provided at the site during concrete placement.

h. Spillage of concrete into the shaft during concrete placing operations is not allowed. Delivery tubes shall be capped at the end of the tube with a water tight cap, or plugged above the water level with a good quality, tight fitting, non-collapsing moving plug that will expel the water from the tube as it is charged with concrete. The cap or plug shall be designed to be released as the tube is charged. The tremie tube shall extend to the bottom of the hole before charging the tube with concrete. After charging the delivery tube system with concrete, the flow of concrete through a tube shall be induced by slightly raising the discharge end. During concrete placement, the tip of the delivery tube shall be maintained to prevent reentry of the water into the tube. Until at least 10 feet of concrete has been placed, the tip of the delivery tube shall be within 12 inches of the bottom of the drilled hole, and then the minimum 10-foot embedment of the tip shall be maintained during concrete placement. The Contractor shall continuously monitor the top of concrete while placing concrete. Rapid raising or lowering of the delivery tube is not permitted. If the seal is lost or the delivery tube becomes plugged and must be removed, the tube shall be withdrawn and cleaned and the tip of the tube capped to prevent entrance of the water. Charge the tube with concrete, and restart the operation by pushing the capped tube a minimum of 5 feet (or to the bottom

of the drilled shaft, which ever is less) into the concrete and then reinitiating the flow of concrete. Moving plugs shall not be used for restarting concrete placement. Before withdrawing the casing, ensure that the fresh concrete is at such a level that the fluid trapped behind the casing is displaced upward. As the casing is withdrawn, maintain the level of concrete within the casing so that fluid trapped behind the casing is displaced upward out of the shaft excavation without mixing with or displacing the shaft concrete.

i. The concrete delivery tubes shall remain fixed horizontally while concrete is flowing.

j. Concrete shall be free of contamination from slurry or residual foundation material.

k. A log of the placing of the concrete shall be maintained by the Contractor on a form approved by the Contracting Officer when concrete is deposited under water. The log shall show the shaft location, tip elevation, dates of excavation and concrete placement, total quantity of concrete deposited, length and tip elevation of any steel casing, and details of any hole stabilization method and materials used. The log shall include an 8 1/2" x 11" sized graph of the concrete placed versus depth of hole filled. The graph shall be plotted continuously throughout placing of concrete. The depth of concrete placed shall be plotted vertically with the shaft tip oriented at the bottom and the quantity of concrete shall be plotted horizontally. Readings shall be made at least at each 5 foot of depth, and the time of the reading shall be indicated. The graph shall be labeled with the location, tip elevation, cutoff elevation, and the dates of excavation and concrete placement. The log shall be delivered to the Government within one working day of completion of placing concrete in the shaft.

3.14.4 Placement Time

a. The concrete shall maintain a slump within the range specified in paragraph SLUMP of SECTION 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS throughout concrete placement time. The admixtures in the concrete mix shall be adjusted as approved by the Contracting Officer for the conditions encountered on the job so that the concrete remains in a workable plastic state throughout the placement. Satisfactory slump loss tests that demonstrate that the concrete will maintain a 7 inch or greater slump for the time equal to the estimated transport plus placement shall be provided before drilled shaft construction begins.

b. The placing time shall not be more than 4 hours for 5-foot diameter shafts and not more than 6 hours for 6-foot diameter shafts. The Contractor may increase the placement time by up to 1 hour provided the concrete mix maintains a slump of 7 inches or greater over the longer placement time as demonstrated by slump loss tests. The slump loss tests shall be conducted using concrete and ambient temperatures appropriate for site conditions. Slump loss test results shall be submitted to the Contracting Officer.

3.14.5 Removal of Sediment at Construction Joints

Before placing any new concrete against concrete deposited in water, the Contractor shall remove all scum, laitance, loose gravel, contaminated concrete, and sediment from the upper surface of the concrete deposited in water, and chip off any high spots on the upper surface of the existing concrete that would prevent the shaft reinforcing from being placed in the position required by the contract drawings.

3.14.6 Removal of Excess Concrete

After attaining the initial set, clean and remove any excess laitance and weak concrete above the design top elevation. The top surface of the finished drilled shaft shall have sound and clean concrete as specified herein.

3.14.7 Beginning Work on Adjacent Shafts

Do not begin work on installing casing on a drilled shaft until the concrete in the adjacent shaft(s) has attained at least 70 percent of its specified compressive strength.

3.15 GROUTING THE CSL TESTING TUBES

All CSL tubes for production drilled shafts shall be grouted after all testing is completed with a Good (G) rating. If the CSL testing results in a rating other than Good (G), or if the shaft is otherwise unacceptable, grouting shall not occur until approved by the Contracting Officer. The Contractor shall pump the grout from one end at a pressure of not more than 250 psig. The Contractor shall leave all plugs and valves in place and closed for at least 24 hours after grouting. Grouting equipment shall:

- a. Include a pressure gauge with an upper end readout of between 275 and 325 psig.
- b. Screen the grout before it enters the pump with an easily reached screen that has clear openings of no more than 0.125 inches.
- c. Be gravity fed from an attached, overhead hopper kept partly full during pumping.

The grout shall be nonshrink grout and meet the requirements of Section 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS.

3.16 INSPECTION OF EXPOSED CONCRETE SURFACE

Remove material and clean the drilled shaft wall face to be able to inspect the integrity of concrete surface above the low mean water level. Inspect and repair concrete in accordance with the requirements within specification 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS.

3.17 FINAL APPROVAL AND ACCEPTANCE

Final approval for the first production drilled shaft constructed will be given after the CSL tests, casing installation records, shaft excavation survey/vertical plumbness records, heave records, jet grout seal installation or other mitigation records (if necessary), compressive strength cylinders test results, and concrete placement records and the

results have been submitted to the Contracting Officer for evaluation and approval. Construction of the first shaft will not be permitted until the correlation boring and Demonstration Wall have been completed and approved. The Contracting Officer will provide a response to the CSL test results within (30) thirty calendar days after receiving the CSL test results and analysis.

The final acceptance of each subsequent drilled shaft shall be determined by the Contracting Officer and will be based on the CSL test results, casing installation records, vertical plumbness records, heave records, compressive strength cylinders test results, concrete placement records, and satisfactory installation of jet grout seals as required. Final acceptance will be provided to the Contractor within (30) thirty calendar days after receiving the test results and analysis by the testing organization.

For any shaft determined to be unacceptable, but approved for repair at the Contractor's expense, the Contractor shall submit a plan for remedial action to the Contracting Officer for approval. Any remedial correction procedures or designs shall be approved by the Contracting Officer prior to repair operations. After remedial correction procedures have been completed, they shall be documented and submitted to the Contracting Officer for approval.

3.18 SAMPLE FORMS

Sample [shaft installation record form](#), [slurry data](#), and [concreting data](#) forms are appended after the end of this section.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 01 16

COLD MILLING ASPHALT PAVING

PART 1 GENERAL

1.1 QUALITY ASSURANCE

1.1.1 Grade

1.1.2 Surface Smoothness

1.1.3 Traffic Control

1.2 EQUIPMENT, TOOLS, AND MACHINES

1.2.1 Cold-Milling Machine

1.2.2 Cleaning Equipment

1.2.3 Straightedge

1.3 ENVIRONMENTAL REQUIREMENTS

PART 2 PRODUCTS

PART 3 EXECUTION

3.1 MILLING OPERATION

3.2 GRADE AND SURFACE-SMOOTHNESS TESTING

3.2.1 Grade-Conformance Tests

3.2.2 Surface-Smoothness Tests

3.3 REMOVAL OF MILLED MATERIAL

-- End of Section Table of Contents --

SECTION 32 01 16

COLD MILLING ASPHALT PAVING

PART 1 GENERAL

1.1 QUALITY ASSURANCE

1.1.1 Grade

Mill pavement such that the finished surface conforms to the lines, grades, and cross sections indicated. The maximum allowable deviation of the finished milled pavement surfaces from the established plan grade line and elevation will be $\frac{1}{4}$ inch. The deviations from the plan grade line and elevation will not be permitted in areas of pavements where closer conformance with planned grade and elevation is required for the proper functioning of appurtenant structures involved.

1.1.2 Surface Smoothness

The maximum allowable deviation of the finished surfaces from the testing edge in the transverse or longitudinal direction will be $\frac{1}{4}$ inch.

1.1.3 Traffic Control

Provide all necessary traffic controls during milling operations.

1.2 EQUIPMENT, TOOLS, AND MACHINES

Maintain in a satisfactory working condition equipment, tools, and machines used in the performance of the work.

1.2.1 Cold-Milling Machine

Provide a cold-milling machine which is self-propelled, capable of milling the pavement to a specified depth and smoothness and of establishing grade control; with means of controlling transverse slope and dust produced during the pavement milling operation. Machine will have capability of adding water in front of equipment to minimize dust during milling operation. The machine will have the ability to remove the millings or cuttings from the pavement and load them into a truck. The milling machine will not damage any part of the pavement structure that is not to be removed.

1.2.2 Cleaning Equipment

Provide cleaning equipment suitable for removing and cleaning loose material from the pavement surface.

1.2.3 Straightedge

Furnish and maintain at the site, in good condition, one $\frac{1}{2}$ -foot straightedge or other suitable device for each milling machine, for testing the finished surface. Make straightedge available for Government use. Use straightedges constructed of aluminum or other lightweight metal,

with blades of box or box-girder cross section with flat bottom reinforced to insure rigidity and accuracy. Use straightedges with handles to facilitate movement on the pavement.

1.3 ENVIRONMENTAL REQUIREMENTS

Do not perform milling when there is accumulation of snow or ice on the pavement surface.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 MILLING OPERATION

A minimum of seven days notice is required, prior to start work, for the Contracting Officer to coordinate the milling operation with other activities at the site. Make sufficient passes so that the designated area is milled to the grades and cross sections indicated. Mill the pavement in depth increments that will not damage the pavement below the designated finished grade. If scabbing occurs, the surface will not meet smoothness requirements. Take steps to modify the process as needed to prevent scabbing from occurring. Repair or replace items damaged during milling such as manholes, valve boxes, utility lines, pavement that is torn, cracked, gouged, broken, or undercut. Remove the milled material from the pavement and load into trucks.

3.2 GRADE AND SURFACE-SMOOTHNESS TESTING

3.2.1 Grade-Conformance Tests

Test the finished milled surface of the pavement for conformance with the plan-grade requirements and for acceptance by the Contracting Officer by running lines of levels at intervals of 25 feet longitudinally and 20 feet transversely to determine the elevation of the completed pavement. Correct variations from the designated grade line and elevation in excess of the plan-grade requirements. Skin patching for correcting low areas will not be permitted. Remove and replace the deficient low area. Remove sufficient material to allow at least 1 inch of asphalt concrete to be placed.

3.2.2 Surface-Smoothness Tests

After completion of the final milling, the finished milled surface will be tested by the Government with a straightedge. Other approved devices may be used, provided that when satisfactorily and properly operated, such devices reveal all surface irregularities exceeding the tolerances specified. Correct surface irregularities that depart from the testing edge by more than 1/4 inch. Skin patching for correcting low areas will not be permitted. Remove and replace the deficient low area. Remove sufficient material to allow at least 1 inch of asphalt concrete to be placed.

3.3 REMOVAL OF MILLED MATERIAL

Material that is removed will become the property of the Contractor and removed from the site.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 11 23

AGGREGATE BASE COURSES

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Aggregate Base Course
 - 1.2.2 Graded-Crushed Aggregate Base Course
 - 1.2.3 Subbase Course
 - 1.2.4 Compaction
- 1.3 SUBMITTALS
- 1.4 EQUIPMENT, TOOLS, AND MACHINES
- 1.5 QUALITY ASSURANCE

PART 2 PRODUCTS

- 2.1 AGGREGATES
 - 2.1.1 Coarse Aggregate
 - 2.1.1.1 Aggregate Base Course
 - 2.1.1.2 Graded-Crushed Aggregate Base Course
 - 2.1.2 Fine Aggregate
 - 2.1.2.1 Aggregate Base Course
 - 2.1.2.2 Graded-Crushed Aggregate Base Course
 - 2.1.3 Gradation Requirements
- 2.2 LIQUID LIMIT AND PLASTICITY INDEX
- 2.3 TESTS, INSPECTIONS, AND VERIFICATIONS

PART 3 EXECUTION

- 3.1 GENERAL REQUIREMENTS
- 3.2 OPERATION OF AGGREGATE SOURCES
- 3.3 STOCKPILING MATERIAL
- 3.4 PREPARATION OF UNDERLYING COURSE OR SUBGRADE
- 3.5 GRADE CONTROL
- 3.6 MIXING AND PLACING MATERIALS
- 3.7 LAYER THICKNESS
- 3.8 COMPACTION
- 3.9 FINISHING
- 3.10 SMOOTHNESS TEST
- 3.11 FIELD QUALITY CONTROL
 - 3.11.1 In-Place Tests
 - 3.11.2 Approval of Material
- 3.12 TRAFFIC
- 3.13 MAINTENANCE
- 3.14 DISPOSAL OF UNSATISFACTORY MATERIALS

-- End of Section Table of Contents --

SECTION 32 11 23

AGGREGATE BASE COURSES

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO M 145 (1991; R 2012) Standard Specification for
Classification of Soils and Soil-Aggregate
Mixtures for Highway Construction Purposes

ASTM INTERNATIONAL (ASTM)

ASTM C88 (2018) Standard Test Method for Soundness
of Aggregates by Use of Sodium Sulfate or
Magnesium Sulfate

ASTM C131/C131M (2020) Standard Test Method for Resistance
to Degradation of Small-Size Coarse
Aggregate by Abrasion and Impact in the
Los Angeles Machine

ASTM D2487 (2017; E 2020) Standard Practice for
Classification of Soils for Engineering
Purposes (Unified Soil Classification
System)

ASTM D5821 (2013; R 2017) Standard Test Method for
Determining the Percentage of Fractured
Particles in Coarse Aggregate

COMMONWEALTH OF PUERTO RICO, DEPARTMENT OF TRANSPORTATION AND
PUBLIC WORKS (PRHTA)

PRHTA Specifications (2005) Standard Specifications for Road
and Bridge Construction

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where PRHTA Specifications Section, and/or Special Provision(s) is cited that contains references to other PRHTA sections, they shall also form a part of this specification to the extent referenced. Where reference is made to "Highway Authority" or "Authority's Representative" in PRHTA Specifications shall mean "Government" or "Contracting Officer" as defined in the Contract Documents. Where PRHTA Specifications refer to the "Engineer", "Engineer of Tests", or "Division of Tests" it shall be understood to mean the "Contracting Officer" as stated in the Contract Documents. In case of

conflict between the PRHTA specifications and the Specifications in this Section that follows, the Specifications in this Section shall govern.

Paragraphs title "Method of Measurement" and "Basis of Payment" of referenced PRHTA Specifications shall not apply.

1.2 DEFINITIONS

For the purposes of this specification, the following definitions apply.

1.2.1 Aggregate Base Course

Aggregate base course (ABC) is well graded, durable aggregate uniformly moistened and mechanically stabilized by compaction. The aggregate shall be in accordance with [PRHTA Specifications](#), section 304.

1.2.2 Graded-Crushed Aggregate Base Course

Graded-crushed aggregate (GCA) base course is well graded, crushed, durable aggregate uniformly moistened and mechanically stabilized by compaction. The aggregate shall be in accordance with [PRHTA Specifications](#), section 304.

1.2.3 Subbase Course

Material for subbase courses shall be a well graded granular soil-aggregate mixture classifiable as an A-1 or A-2-4 material as per [AASHTO M 145](#), or a soil or soil-aggregate mixture having less than 6% passing the No. 200 sieve. The subbase course shall be in accordance with [PRHTA Specifications](#), Section 301.

1.2.4 Compaction

Aggregate Base Course material compaction requirements shall be in accordance with the PRHTA Specification, Section 304. Subbase material compaction requirements shall be in accordance with the PRHTA Specification, Section 301.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section [01 33 00 SUBMITTAL PROCEDURES](#). Submittals shall be made according to requirements of this section and in accordance with the requirements of the [PRHTA Specifications](#).

[SD-03 Product Data](#)

[Plant, Equipment, and Tools; G, R0](#)

[SD-06 Test Reports](#)

[In-Place Tests; G, R0](#)

1.4 EQUIPMENT, TOOLS, AND MACHINES

All [plant, equipment, and tools](#) used in the performance of the work will

be subject to approval by the Contracting Officer before the work is started. Maintain all plant, equipment, and tools in satisfactory working condition at all times. Submit a list of proposed equipment, including descriptive data. Use equipment capable of minimizing segregation, producing the required compaction, meeting grade controls, thickness control, and smoothness requirements as set forth herein.

1.5 QUALITY ASSURANCE

Sampling and testing are the responsibility of the Contractor. Perform sampling and testing using a laboratory approved in accordance with Section QUALITY CONTROL. Work requiring testing will not be permitted until the testing laboratory has been inspected and approved. Test the materials to establish compliance with the specified requirements and perform testing at the specified frequency. The Contracting Officer may specify the time and location of the tests. Furnish copies of test results to the Contracting Officer within 24 hours of completion of the tests. All Sampling and Testing shall be in accordance with [PRHTA Specifications](#), Section 301 and 304.

PART 2 PRODUCTS

2.1 AGGREGATES

Provide ABC and GCA consisting of clean, sound, durable particles of crushed stone, crushed slag, crushed gravel, crushed recycled concrete, angular sand, or other approved material as indicated in the [PRHTA Specifications](#), Section 304 and 703. Provide ABC that is free of lumps of clay, organic matter, and other objectionable materials or coatings. Provide GCA that is free of silt and clay as defined by [ASTM D2487](#), organic matter, and other objectionable materials or coatings. The portion retained on the [No. 4](#) sieve is known as coarse aggregate; that portion passing the [No. 4](#) sieve is known as fine aggregate. When the coarse and fine aggregate is supplied from more than one source, provide aggregate from each source that meets the specified requirements.

2.1.1 Coarse Aggregate

Provide coarse aggregates with angular particles of uniform density. Separately stockpile coarse aggregate supplied from more than one source.

- a. Crushed Gravel: Provide crushed gravel that has been manufactured by crushing gravels and that meets all the requirements specified below.
- b. Crushed Stone: Provide crushed stone consisting of freshly mined quarry rock, meeting all the requirements specified below.

2.1.1.1 Aggregate Base Course

The percentage of loss of ABC coarse aggregate must not exceed 50 percent when tested in accordance with [ASTM C131/C131M](#). Provide aggregate that contains no more than 30 percent flat and elongated particles. A flat particle is one having a ratio of width to thickness greater than 3; an elongated particle is one having a ratio of length to width greater than 3. In the portion retained on each sieve specified, the crushed aggregates must contain at least 50 percent by weight of crushed pieces having two or more freshly fractured faces determined in accordance with [ASTM D5821](#). When two fractures are contiguous, the angle between planes

of the fractures must be at least 30 degrees in order to count as two fractured faces. Manufacture crushed gravel from gravel particles 50 percent of which, by weight, are retained on the maximum size sieve listed in TABLE 1.

2.1.1.2 Graded-Crushed Aggregate Base Course

The percentage of loss of GCA coarse aggregate must not exceed 50 percent loss when tested in accordance with [ASTM C131/C131M](#). Provide GCA coarse aggregate that does not exhibit a loss greater than 18 percent weighted average, at five cycles, when tested for soundness in magnesium sulfate, or 12 percent weighted average, at five cycles, when tested in sodium sulfate in accordance with [ASTM C88](#). Provide aggregate that contains no more than 20 percent flat and elongated particles for the fraction retained on the [1/2 inch](#) sieve nor 20 percent for the fraction passing the [1/2 inch](#) sieve. A flat particle is one having a ratio of width to thickness greater than 3; an elongated particle is one having a ratio of length to width greater than 3. In the portion retained on each sieve specified, the crushed aggregate must contain at least 90 percent by weight of crushed pieces having two or more freshly fractured faces determined in accordance with [ASTM D5821](#). When two fractures are contiguous, the angle between planes of the fractures must be at least 30 degrees in order to count as two fractured faces. Manufacture crushed gravel from gravel particles 90 percent of which by weight are retained on the maximum size sieve listed in TABLE 1.

2.1.2 Fine Aggregate

Provide fine aggregates consisting of angular particles of uniform density.

2.1.2.1 Aggregate Base Course

Provide ABC fine aggregate that consists of screenings, angular sand, crushed recycled concrete fines, or other finely divided mineral matter processed or naturally combined with the coarse aggregate.

2.1.2.2 Graded-Crushed Aggregate Base Course

Provide GCA fine aggregate consisting of angular particles produced by crushing stone, slag, or gravel that meets the requirements for wear and soundness specified for GCA coarse aggregate. Produce fine aggregate by crushing only particles larger than [No. 4](#) sieve in size. Provide fine aggregate that contains at least 90 percent by weight of particles having two or more freshly fractured faces in the portion passing the [No. 4](#) sieve and retained on the [No. 10](#) sieve, and in the portion passing the [No. 10](#) sieve and retained on the [No. 40](#) sieve.

2.1.3 Gradation Requirements

Apply the specified gradation requirements to the completed base course. Provide aggregates that are continuously well graded within the limits specified in TABLE 703-5 of the [PRHTA Specifications](#)

2.2 LIQUID LIMIT AND PLASTICITY INDEX

Apply liquid limit and plasticity index requirements to the completed course and to any component that is blended to meet the required

gradation. The portion of any component or of the completed course passing the No. 40 sieve must be either nonplastic or have a liquid limit not greater than 25 and a plasticity index not greater than 5.

2.3 TESTS, INSPECTIONS, AND VERIFICATIONS

Tests, Inspections, and Verifications shall be in accordance with PRHTA Specifications, section 301 and 703.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

When the ABC or GCA is constructed in more than one layer, clean the previously constructed layer of loose and foreign matter by sweeping with power sweepers or power brooms, except that hand brooms may be used in areas where power cleaning is not practicable. Provide adequate drainage during the entire period of construction to prevent water from collecting or standing on the working area.

3.2 OPERATION OF AGGREGATE SOURCES

Condition aggregate sources on private lands in accordance with local laws or authorities. Clearing, stripping, and excavating are the responsibility of the Contractor. Condition aggregate sources on Government property to readily drain and leave in a satisfactory condition upon completion of the work.

3.3 STOCKPILING MATERIAL

Clear and level storage sites prior to stockpiling of material. Stockpile all materials, including approved material available from excavation and grading, in the manner and at the locations designated. Stockpile aggregates on the cleared and leveled areas designated by the Contracting Officer to prevent segregation. Stockpile materials obtained from different sources separately.

3.4 PREPARATION OF UNDERLYING COURSE OR SUBGRADE

Clean the underlying course or subgrade of all foreign substances prior to constructing the base course(s). Do not construct base course(s) on underlying course or subgrade that is frozen. Construct the surface of the underlying course or subgrade to meet specified compaction and surface tolerances. Correct ruts or soft yielding spots in the underlying courses, areas having inadequate compaction, and deviations of the surface from the specified requirements set forth herein by loosening and removing soft or unsatisfactory material and adding approved material, reshaping to line and grade, and recompacting to specified density requirements. For cohesionless underlying courses or subgrades containing sands or gravels, as defined in ASTM D2487, stabilize the surface prior to placement of the base course(s). Stabilize by mixing ABC or GCA into the underlying course and compacting by approved methods. Consider the stabilized material as part of the underlying course and meet all requirements of the underlying course. Do not allow traffic or other operations to disturb the finished underlying course and maintain in a satisfactory condition until the base course is placed.

3.5 GRADE CONTROL

Provide a finished and completed base course conforming to the lines, grades, and cross sections shown. Place line and grade stakes as necessary for control.

3.6 MIXING AND PLACING MATERIALS

Mix the coarse and fine aggregates in a stationary plant, or in a traveling plant or bucket loader on an approved paved working area. Make adjustments in mixing procedures or in equipment to obtain true grades, to minimize segregation or degradation, to obtain the required water content, and to insure a satisfactory base course meeting all requirements of this specification. Place the mixed material on the prepared subgrade or subbase in layers of uniform thickness with an approved spreader. Place the layers so that when compacted they will be true to the grades or levels required with the least possible surface disturbance. Where the base course is placed in more than one layer, clean the previously constructed layers of loose and foreign matter by sweeping with power sweepers, power brooms, or hand brooms. Make adjustments in placing procedures or equipment to obtain true grades, to minimize segregation and degradation, to adjust the water content, and to insure an acceptable base course.

3.7 LAYER THICKNESS

Layer thickness shall comply with [PRHTA Specifications](#), Section 301, 304 and 703.

3.8 COMPACTION

Compact each layer of the base course as indicated in the [PRHTA Specifications](#) Section 301, 304 and 703

3.9 FINISHING

Finish the surface of the top layer of base course after final compaction by cutting any overbuild to grade and rolling with a steel-wheeled roller. Do not add thin layers of material to the top layer of base course to meet grade. If the elevation of the top layer of base course is [1/2 inch](#) or more below grade, scarify the top layer to a depth of at least [3 inches](#) and blend new material in and compact to bring to grade. Make adjustments to rolling and finishing procedures to minimize segregation and degradation, obtain grades, maintain moisture content, and insure an acceptable base course. Should the surface become rough, corrugated, uneven in texture, or traffic marked prior to completion, scarify the unsatisfactory portion and rework and recompact it or replace as approved by the Contracting Officer.

3.10 SMOOTHNESS TEST

Construct the top layer so that the surface shows no deviations in excess of [3/8 inch](#) when tested with a [12 foot](#) straightedge. Take measurements in successive positions parallel to the centerline of the area to be paved. Also take measurements perpendicular to the centerline at [50 foot](#) intervals. Correct deviations exceeding this amount by removing material and replacing with new material, or by reworking existing material and compacting it to meet these specifications.

3.11 FIELD QUALITY CONTROL

3.11.1 In-Place Tests

Perform each of the following tests on samples taken from the placed and compacted ABC and GCA. Take samples and test at the rates indicated. Perform sampling and testing of recycled concrete aggregate at twice the specified frequency until the material uniformity is established.

- a. Perform density tests on every lift of material placed and at a frequency of one set of tests for every 250 square yards, or portion thereof, of completed area.
- b. Perform sieve analysis including 0.02 mm size material on every lift of material placed and at a frequency of one sieve analysis for every 500 square yards, or portion thereof, of material placed.
- c. Perform liquid limit and plasticity index tests at the same frequency as the sieve analysis.
- d. Measure the thickness of the base course at intervals providing at least one measurement for each 500 square yards of base course or part thereof. Measure the thickness using test holes, at least 3 inch in diameter through the base course.

3.11.2 Approval of Material

Final approval of the materials will be based on tests for gradation, liquid limit, and plasticity index performed on samples taken from the completed and fully compacted course(s).

3.12 TRAFFIC

Completed portions of the base course may be opened to limited traffic, provided there is no marring or distorting of the surface by the traffic. Do not allow heavy equipment on the completed base course except when necessary for construction. When it is necessary for heavy equipment to travel on the completed base course, protect the area against marring or damage to the completed work.

3.13 MAINTENANCE

Maintain the base course in a satisfactory condition until the full pavement section is completed and accepted. Immediately repair any defects and repeat repairs as often as necessary to keep the area intact at no cost to the Government. Retest any base course that was not paved over prior to the onset of winter to verify that it still complies with the requirements of this specification. Rework or replace any area of base course that is damaged as necessary to comply with this specification.

3.14 DISPOSAL OF UNSATISFACTORY MATERIALS

Dispose of any unsuitable materials that have been removed . No additional payments will be made for materials that have to be replaced.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 12 16

HOT-MIX ASPHALT (HMA) FOR ROADS

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 ENVIRONMENTAL REQUIREMENTS

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
 - 2.1.1 Asphalt Mixing Plant
 - 2.1.1.1 Truck Scales
 - 2.1.1.2 Testing Facilities
 - 2.1.1.3 Inspection of Plant
 - 2.1.1.4 Storage bins
 - 2.1.2 Hauling Equipment
 - 2.1.3 Asphalt Pavers
 - 2.1.3.1 Receiving Hopper
 - 2.1.3.2 Automatic Grade Controls
 - 2.1.4 Rollers
- 2.2 AGGREGATES
 - 2.2.1 Coarse Aggregate
 - 2.2.2 Fine Aggregate
 - 2.2.3 Mineral Filler
 - 2.2.4 Aggregate Gradation
- 2.3 ASPHALT CEMENT BINDER
- 2.4 MIX DESIGN

PART 3 EXECUTION

- 3.1 CONSTRUCTION REQUIREMENTS
- 3.2 SAMPLING AND TESTING

-- End of Section Table of Contents --

SECTION 32 12 16

HOT-MIX ASPHALT (HMA) FOR ROADS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

- | | |
|--------------|---|
| AASHTO M 156 | (2013; R 2017) Standard Specification for Requirements for Mixing Plants for Hot-Mixed, Hot-Laid Bituminous Paving Mixtures |
| AASHTO M 320 | (2017) Standard Specification for Performance-Graded Asphalt Binder |
| AASHTO T 304 | (2011; R 2015) Standard Method of Test for Uncompacted Void Content of Fine Aggregate |

ASTM INTERNATIONAL (ASTM)

- | | |
|-----------------|--|
| ASTM C29/C29M | (2017a) Standard Test Method for Bulk Density ("Unit Weight") and Voids in Aggregate |
| ASTM C88 | (2018) Standard Test Method for Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate |
| ASTM C131/C131M | (2020) Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine |
| ASTM C142/C142M | (2017) Standard Test Method for Clay Lumps and Friable Particles in Aggregates |
| ASTM D140/D140M | (2016) Standard Practice for Sampling Asphalt Materials |
| ASTM D242/D242M | (2009; R 2014) Mineral Filler for Bituminous Paving Mixtures |
| ASTM D2419 | (2014) Sand Equivalent Value of Soils and Fine Aggregate |
| ASTM D4791 | (2019) Flat Particles, Elongated Particles, or Flat and Elongated Particles |

in Coarse Aggregate

U.S. ARMY CORPS OF ENGINEERS (USACE)

COE CRD-C 171

(1995) Standard Test Method for
Determining Percentage of Crushed
Particles in Aggregate

COMMONWEALTH OF PUERTO RICO, DEPARTMENT OF TRANSPORTATION AND
PUBLIC WORKS (PRHTA)

PRHTA Specifications

(2005) Standard Specifications for Road
and Bridge Construction

PRHTA SP 959

(2017) Specifications Special Provisions
959

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where PRHTA Specifications Section, and/or Special Provision(s) is cited that contains references to other PRHTA sections, they shall also form a part of this specification to the extent referenced. Where reference is made to "Highway Authority" or "Authority's Representative" in PRHTA Specifications shall mean "Government" or "Contracting Officer" as defined in the Contract Documents. Where PRHTA Specifications refer to the "Engineer", "Engineer of Tests", or "Division of Tests" it shall be understood to mean the "Contracting Officer" as stated in the Contract Documents. In case of conflict between the PRHTA specifications and the Specifications in this Section that follows, the Specifications in this Section shall govern.

Paragraphs title "Method of Measurement" and "Basis of Payment" of referenced PRHTA Specifications shall not apply.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES. Submittals shall be made according to requirements of this section and in accordance with the requirements of the PRHTA Specifications.

SD-03 Product Data

Mix Design; G, R0

SD-04 Samples

Asphalt Cement Binder

Aggregates

SD-06 Test Reports

Aggregates; G, R0

SD-07 Certificates

Asphalt Cement Binder; G, R0

1.3 ENVIRONMENTAL REQUIREMENTS

Do not place the hot-mix asphalt upon a wet surface or when the surface temperature of the underlying course is less than specified in Table 3. The temperature requirements may be waived by the Contracting Officer, if requested; however, meet all other requirements, including compaction.

Table 3. Surface Temperature Limitations of Underlying Course	
Mat Thickness, inches	Degrees F
3 or greater	40
Less than 3	45

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Perform the work consisting of pavement courses composed of mineral aggregate and asphalt material heated and mixed in a central mixing plant and placed on a prepared course. HMA designed and constructed in accordance with this section shall conform to PRHTA SP 959, the lines, grades, thicknesses, and typical cross sections indicated. Construct each course to the depth, section, or elevation required by the drawings and roll, finish, and approve it before the placement of the next course.

2.1.1 Asphalt Mixing Plant

Plants used for the preparation of hot-mix asphalt shall conform to the requirements of AASHTO M 156 with the following changes:

2.1.1.1 Truck Scales

Weigh the asphalt mixture on approved, certified scales at the Contractor's expense. Ensure the truck scales have been inspected and certified within the last year by an approved calibration laboratory.

2.1.1.2 Testing Facilities

Provide laboratory facilities at the plant for the use of the Government's acceptance testing and the Contractor's quality control testing.

2.1.1.3 Inspection of Plant

Provide the Contracting Officer with access at all times, to all areas of the plant for checking adequacy of equipment; inspecting operation of the plant; verifying weights, proportions, and material properties; checking the temperatures maintained in the preparation of the mixtures and for taking samples. Provide assistance as requested, for the Government to procure any desired samples.

2.1.1.4 Storage bins

Use of storage bins for temporary storage of hot-mix asphalt will be permitted as follows:

- a. The asphalt mixture may be stored in non-insulated storage bins for a period of time not exceeding 3 hours.
- b. The asphalt mixture may be stored in insulated storage bins for a period of time not exceeding 8 hours. The mix drawn from bins shall meet the same requirements as mix loaded directly into trucks.

2.1.2 Hauling Equipment

Provide trucks for hauling hot-mix asphalt having tight, clean, and smooth metal beds. To prevent the mixture from adhering to them, the truck beds shall be lightly coated with a minimum amount of paraffin oil, lime solution, or other approved material. Petroleum based products shall not be used as a release agent. Each truck shall have a suitable cover to protect the mixture from adverse weather. When necessary to ensure that the mixture will be delivered to the site at the specified temperature, truck beds shall be insulated or heated and covers (tarps) shall be securely fastened.

2.1.3 Asphalt Pavers

Provide asphalt pavers which are self-propelled, with an activated screed, heated as necessary, and capable of spreading and finishing courses of hot-mix asphalt which will meet the specified thickness, smoothness, and grade. The paver shall have sufficient power to propel itself and the hauling equipment without adversely affecting the finished surface.

2.1.3.1 Receiving Hopper

Provide paver with a receiving hopper of sufficient capacity to permit a uniform spreading operation and equipped with a distribution system to place the mixture uniformly in front of the screed without segregation. The screed shall effectively produce a finished surface of the required evenness and texture without tearing, shoving, or gouging the mixture.

2.1.3.2 Automatic Grade Controls

Equip the paver with a control system capable of automatically maintaining the specified screed elevation. The control system shall be automatically actuated from either a reference line and/or through a system of mechanical sensors or sensor-directed mechanisms or devices which will maintain the paver screed at a predetermined transverse slope and at the proper elevation to obtain the required surface. The transverse slope controller shall be capable of maintaining the screed at the desired slope within plus or minus 0.1 percent. A transverse slope controller shall not be used to control grade. Provide controls capable of working in conjunction with any of the following attachments:

- a. Ski-type device of not less than 30 feet in length.
- b. Taut stringline set to grade.
- c. Short ski or shoe for joint matching.
- d. Laser control.

2.1.4 Rollers

Rollers shall be in good condition and shall be operated at slow speeds to avoid displacement of the asphalt mixture. The number, type, and weight of rollers shall be sufficient to compact the mixture to the required density while it is still in a workable condition. Do not use equipment which causes excessive crushing of the aggregate.

2.2 AGGREGATES

Provide aggregates consisting of crushed stone, crushed gravel, crushed slag, screenings, natural sand and mineral filler, as required. Submit sufficient materials to produce 200 lb of blended mixture for mix design verification. The portion of material retained on the No. 4 sieve is coarse aggregate. The portion of material passing the No. 4 sieve and retained on the No. 200 sieve is fine aggregate. The portion passing the No. 200 sieve is defined as mineral filler. Submit all aggregate test results and samples to the Contracting Officer at least 30 days prior to start of construction.

2.2.1 Coarse Aggregate

Provide coarse aggregate consisting of sound, tough, durable particles, free from films of material that would prevent thorough coating and bonding with the asphalt material and free from organic matter and other deleterious substances. All individual coarse aggregate sources shall meet the following requirements:

- a. The percentage of loss shall not be greater than 40 percent after 500 revolutions when tested in accordance with ASTM C131/C131M.
- b. The percentage of loss shall not be greater than 18 percent after five cycles when tested in accordance with ASTM C88 using magnesium sulfate or 12 percent when using sodium sulfate.
- c. At least 75 percent by weight of coarse aggregate shall have at least two or more fractured faces when tested in accordance with COE CRD-C 171. Fractured faces shall be produced by crushing.
- d. The particle shape shall be essentially cubical and the aggregate shall not contain more than 20 percent, by weight, of flat and elongated particles (3:1 ratio of maximum to minimum) when tested in accordance with ASTM D4791.
- e. Slag shall be air-cooled, blast furnace slag, with a compacted weight of not less than 75 lb/cu ft when tested in accordance with ASTM C29/C29M.
- f. Clay lumps and friable particles shall not exceed 0.3 percent, by weight, when tested in accordance with ASTM C142/C142M.

2.2.2 Fine Aggregate

Fine aggregate shall consist of clean, sound, tough, durable particles free from coatings of clay, silt, or any objectionable material and containing no clay balls.

- a. All individual fine aggregate sources shall have a sand equivalent value not less than 45 when tested in accordance with ASTM D2419.

- b. The fine aggregate portion of the blended aggregate shall have an uncompacted void content not less than 45.0 percent when tested in accordance with [AASHTO T 304](#) Method A.
- c. The quantity of natural sand (noncrushed material) added to the aggregate blend shall not exceed 25 percent by weight of total aggregate.
- d. Clay lumps and friable particles shall not exceed 0.3 percent, by weight, when tested in accordance with [ASTM C142/C142M](#)

2.2.3 Mineral Filler

Mineral filler shall be nonplastic material meeting the requirements of [ASTM D242/D242M](#).

2.2.4 Aggregate Gradation

The aggregate gradation shall conform to gradations specified in [PRHTA Specifications](#), Section 703, Tables 703-1 through 703-5.

2.3 ASPHALT CEMENT BINDER

Asphalt cement binder shall conform to [AASHTO M 320](#) Performance Grade (PG) 70-22. Asphalt Binder shall be consistent with the [PRHTA Specifications](#) Sections, 401, 402, 407 and 408 respectively. Test data indicating grade certification shall be provided by the supplier at the time of delivery of each load to the mix plant. Submit copies of these certifications to the Contracting Officer. The supplier is defined as the last source of any modification to the binder. Obtain samples for this verification testing in accordance with [ASTM D140/D140M](#). Submit samples of the asphalt cement specified for approval not less than 14 days before start of the test section. Submit copies of certified test data, amount, type and description of any modifiers blended into the asphalt cement binder.

2.4 MIX DESIGN

- a. Develop the mix design. The asphalt mix shall be composed of a mixture of well-graded aggregate, mineral filler if required, and asphalt material. The aggregate fractions shall be sized, handled in separate size groups, and combined in such proportions that the resulting mixture meets the grading requirements of the job mix formula (JMF). Submit proposed JMF; do not produce hot-mix asphalt for payment until a JMF has been approved. The hot-mix asphalt shall be designed in accordance with [PRHTA Specifications](#) and [PRHTA SP 959](#).

PART 3 EXECUTION

3.1 CONSTRUCTION REQUIREMENTS

Comply with the [PRHTA Specifications](#), Section 401 and [PRHTA SP 959](#).

3.2 SAMPLING AND TESTING

Comply with the [PRHTA Specifications](#), Section 401 and [PRHTA SP 959](#).

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 15 00

AGGREGATE SURFACING

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEGREE OF COMPACTION
- 1.3 SUBMITTALS
- 1.4 EQUIPMENT, TOOLS, AND MACHINES
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Sampling
 - 1.5.2 Testing
 - 1.5.2.1 Sieve Analysis
 - 1.5.2.2 Liquid Limit and Plasticity Index
 - 1.5.2.3 Moisture-Density Determinations
 - 1.5.2.4 Field Density Tests
 - 1.5.2.5 Wear Test
- 1.6 ENVIRONMENTAL REQUIREMENTS

PART 2 PRODUCTS

- 2.1 AGGREGATES
 - 2.1.1 Coarse Aggregates
 - 2.1.2 Fine Aggregates
 - 2.1.3 Gradation Requirements
- 2.2 LIQUID LIMIT AND PLASTICITY INDEX
- 2.3 TESTS, INSPECTIONS, AND VERIFICATIONS
 - 2.3.1 Initial Tests
 - 2.3.2 Approval of Material

PART 3 EXECUTION

- 3.1 OPERATION OF AGGREGATE SOURCES
- 3.2 STOCKPILING MATERIAL
- 3.3 PREPARATION OF UNDERLYING SUBGRADE
- 3.4 GRADE CONTROL
- 3.5 MIXING AND PLACING MATERIALS
- 3.6 LAYER THICKNESS
- 3.7 COMPACTION
- 3.8 PROOF ROLLING
- 3.9 EDGES OF AGGREGATE SURFACE COURSE
- 3.10 SMOOTHNESS TEST
- 3.11 FIELD QUALITY CONTROL
 - 3.11.1 In-Place Tests
 - 3.11.2 Approval of Material
- 3.12 MAINTENANCE

-- End of Section Table of Contents --

SECTION 32 15 00

AGGREGATE SURFACING

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO T 180 (2017) Standard Method of Test for
Moisture-Density Relations of Soils Using
a 4.54-kg (10-lb) Rammer and a 457-mm
(18-in.) Drop

AASHTO T 224 (2010) Standard Method of Test for
Correction for Coarse Particles in the
Soil Compaction Test

ASTM INTERNATIONAL (ASTM)

ASTM C117 (2017) Standard Test Method for Materials
Finer than 75-um (No. 200) Sieve in
Mineral Aggregates by Washing

ASTM C131/C131M (2020) Standard Test Method for Resistance
to Degradation of Small-Size Coarse
Aggregate by Abrasion and Impact in the
Los Angeles Machine

ASTM C136/C136M (2019) Standard Test Method for Sieve
Analysis of Fine and Coarse Aggregates

ASTM D75/D75M (2019) Standard Practice for Sampling
Aggregates

ASTM D1556/D1556M (2015; E 2016) Standard Test Method for
Density and Unit Weight of Soil in Place
by Sand-Cone Method

ASTM D1557 (2012; E 2015) Standard Test Methods for
Laboratory Compaction Characteristics of
Soil Using Modified Effort (56,000
ft-lbf/ft³) (2700 kN-m/m³)

ASTM D2167 (2015) Density and Unit Weight of Soil in
Place by the Rubber Balloon Method

ASTM D4318 (2017; E 2018) Standard Test Methods for
Liquid Limit, Plastic Limit, and

Plasticity Index of Soils

ASTM D6938

(2017a) Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)

ASTM E11

(2020) Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves

1.2 DEGREE OF COMPACTION

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum laboratory dry density obtained by the test procedure presented in [ASTM D1557](#) abbreviated as a percent of laboratory maximum dry density. Since [ASTM D1557](#) applies only to soils that have 30 percent or less by weight of their particles retained on the [3/4 inch](#) sieve, the degree of compaction for material having more than 30 percent by weight of their particles retained on the [3/4 inch](#) sieve will be expressed as a percentage of the laboratory maximum dry density in accordance with [AASHTO T 180](#) Method D and corrected with [AASHTO T 224](#).

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section [01 33 00](#) SUBMITTAL PROCEDURES:

[SD-03 Product Data](#)

[Plant, Equipment, and Tools; G, R0](#)

[SD-06 Test Reports](#)

[Initial Tests; G, R0](#)
[In-Place Tests; G, R0](#)

1.4 EQUIPMENT, TOOLS, AND MACHINES

All [plant, equipment, and tools](#) used in the performance of the work will be subject to approval by the Contracting Officer before the work is started. Maintain all plant, equipment, and tools in satisfactory working condition at all times. Submit a list of proposed equipment, including descriptive data. Provide adequate equipment having the capability of minimizing segregation, producing the required compaction, meeting grade controls, thickness control, and smoothness requirements as set forth herein.

1.5 QUALITY ASSURANCE

Sampling and testing are the responsibility of the Contractor. Perform sampling and testing using a laboratory approved in accordance with Section [01 45 00](#) CONTRACTOR QUALITY CONTROL. Work requiring testing will not be permitted until the testing laboratory has been inspected and approved. Test the materials to establish compliance with the specified requirements and perform testing at the specified frequency. The Contracting Officer may specify the time and location of the tests. Furnish copies of test results to the Contracting Officer within 24 hours

of completion of the tests.

1.5.1 Sampling

Take samples for laboratory testing in conformance with [ASTM D75/D75M](#). When deemed necessary, the sampling will be observed by the Contracting Officer.

1.5.2 Testing

1.5.2.1 Sieve Analysis

Perform sieve analysis in conformance with [ASTM C117](#) and [ASTM C136/C136M](#) using sieves conforming to [ASTM E11](#).

1.5.2.2 Liquid Limit and Plasticity Index

Determine liquid limit and plasticity index in accordance with [ASTM D4318](#).

1.5.2.3 Moisture-Density Determinations

Determine the laboratory maximum dry density and optimum moisture content in accordance with the previous paragraph DEGREE OF COMPACTION.

1.5.2.4 Field Density Tests

Measure field density in accordance with [ASTM D1556/D1556M](#), [ASTM D2167](#) or [ASTM D6938](#). For the method presented in [ASTM D1556/D1556M](#) use the base plate as shown in the drawing. For the method presented in [ASTM D6938](#) check the calibration curves and adjust them, if necessary, using only the sand cone method as described in paragraph Calibration, of the ASTM publication. Tests performed in accordance with [ASTM D6938](#) result in a wet unit weight of soil and [ASTM D6938](#) will be used to determine the moisture content of the soil. Also check the calibration curves furnished with the moisture gauges along with density calibration checks as described in [ASTM D6938](#). Make the calibration checks of both the density and moisture gauges using the prepared containers of material method, as described in paragraph Calibration of [ASTM D6938](#), on each different type of material being tested at the beginning of a job and at intervals as specified. Submit calibration curves and related test results prior to using the device or equipment being calibrated.

1.5.2.5 Wear Test

Perform wear tests on aggregate surface course material in conformance with [ASTM C131/C131M](#).

1.6 ENVIRONMENTAL REQUIREMENTS

Perform construction when the atmospheric temperature is above [35 degrees F](#). It is the responsibility of the Contractor to protect, by approved method or methods, all areas of surfacing that have not been accepted by the Contracting Officer. Bring surfaces damaged by freeze, rainfall, or other weather conditions to a satisfactory condition.

PART 2 PRODUCTS

2.1 AGGREGATES

Provide aggregates consisting of clean, sound, durable particles of natural gravel, crushed gravel, crushed stone, sand, slag, soil, or other approved materials processed and blended or naturally combined. Provide aggregates free from lumps and balls of clay, organic matter, objectionable coatings, and other foreign materials. The Contractor is responsible for obtaining materials that meet the specification and can be used to meet the grade and smoothness requirements specified herein after all compaction and proof rolling operations have been completed.

2.1.1 Coarse Aggregates

The material retained on the No. 4 sieve is known as coarse aggregate. Use only coarse aggregates that are reasonably uniform in density and quality. Use only coarse aggregate having a percentage of wear not exceeding 50 percent after 500 revolutions as determined by ASTM C131/C131M. The amount of flat and/or elongated particles must not exceed 20 percent. A flat particle is one having a ratio of width to thickness greater than three; an elongated particle is one having a ratio of length to width greater than three. When the coarse aggregate is supplied from more than one source, aggregate from each source must meet the requirements set forth herein.

2.1.2 Fine Aggregates

The material passing the No. 4 sieve is known as fine aggregate. Fine aggregate consists of screenings, sand, soil, or other finely divided mineral matter that is processed or naturally combined with the coarse aggregate.

2.1.3 Gradation Requirements

Gradation requirements specified in TABLE I apply to the completed aggregate surface. It is the responsibility of the Contractor to obtain materials that will meet the gradation requirements after mixing, placing, compacting, and other operations. TABLE I shows permissible gradings for granular material used in aggregate surface roads and airfields. Use sieves conforming to ASTM E11.

TABLE I. GRADATION FOR AGGREGATE SURFACE COURSES Percentage by Weight Passing Square-Mesh Sieve				
Sieve Designation	No. 1	No. 2	No. 3	No. 4
1 inch	100	100	100	100
3/8 inch	50-85	60-100	--	--
No. 4	35-65	50-85	55-100	70-100
No. 10	25-50	40-70	40-100	55-100

TABLE I. GRADATION FOR AGGREGATE SURFACE COURSES Percentage by Weight Passing Square-Mesh Sieve				
Sieve Designation	No. 1	No. 2	No. 3	No. 4
No. 40	15-30	24-45	20-50	30-70
No. 200	8-15	8-15	8-15	8-15

2.2 LIQUID LIMIT AND PLASTICITY INDEX

The portion of the completed aggregate surface course passing the No. 40 sieve must have a maximum liquid limit of 35 and a plasticity index of 4 to 9.

2.3 TESTS, INSPECTIONS, AND VERIFICATIONS

2.3.1 Initial Tests

Perform one of each of the following tests, on the proposed material prior to commencing construction, to demonstrate that the proposed material meets all specified requirements when furnished. Complete this testing for each source if materials from more than one source are proposed.

- Sieve Analysis.
- Liquid limit and plasticity index.
- Moisture-density relationship.
- Wear.

Submit certified copies of test results for approval not less than 30 days before material is required for the work.

2.3.2 Approval of Material

Tentative approval of material will be based on initial test results.

PART 3 EXECUTION

3.1 OPERATION OF AGGREGATE SOURCES

Perform clearing, stripping, and excavating. Operate the aggregate sources to produce the quantity and quality of materials meeting these specification requirements in the specified time limit. Upon completion of the work, leave aggregate sources on Government property in a satisfactory condition so that they readily drain. Finalize aggregate sources on private lands in agreement with local laws or authorities.

3.2 STOCKPILING MATERIAL

Prior to stockpiling the material, clear and level the storage sites. Stockpile all materials, including approved material available from excavation and grading, in the manner and at the locations designated. Stockpile aggregates in such a manner that will prevent segregation. Stockpile aggregates and binders obtained from different sources

separately.

3.3 PREPARATION OF UNDERLYING SUBGRADE

Clean the subgrade and shoulders of all foreign substances. Do not construct the surface course on subgrade that is frozen material. Correct ruts or soft yielding spots in the subgrade, areas having inadequate compaction and deviations of the surface from the requirements set forth herein by loosening and removing soft or unsatisfactory material and by adding approved material, reshaping to line and grade and recompacting to density requirements specified in Section 31 00 00 EARTHWORK . Do not allow traffic or other operations to disturb the completed subgrade and maintain in a satisfactory condition until the surface course is placed.

3.4 GRADE CONTROL

During construction, maintain the lines and grades including crown and cross slope indicated for the aggregate surface course by means of line and grade stakes placed by the Contractor in accordance with the SPECIAL CONTRACT REQUIREMENTS.

3.5 MIXING AND PLACING MATERIALS

Mix and place the materials to obtain uniformity of the material and a uniform optimum water content for compaction. Make adjustments in mixing, placing procedures, or in equipment to obtain the true grades, to minimize segregation and degradation, to obtain the desired water content, and to ensure a satisfactory surface course.

3.6 LAYER THICKNESS

Place the aggregate material on the subgrade in layers of uniform thickness. Compact the completed aggregate surface course to the thickness indicated. No individual layer may be thicker than 6 inches nor be thinner than 3 inches in compacted thickness. Compact the aggregate surface course to a total thickness that is within 1/2 inch of the thickness indicated. Where the measured thickness is more than 1/2 inch deficient, correct such areas by scarifying, adding new material of proper gradation, reblading, and recompacting. Where the measured thickness is more than 1/2 inch thicker than indicated, the course will be considered as conforming to the specified thickness requirements. The average job thickness will be the average of all thickness measurements taken for the job and must be within 1/4 inch of the thickness indicated. Measure the total thickness of the aggregate surface course at intervals of one measurement for each 500 square yards of surface course. Measure total thickness using 3 inch diameter test holes penetrating the aggregate surface course.

3.7 COMPACTION

Degree of compaction is a percentage of the maximum density obtained by the test procedure presented in ASTM D1557 abbreviated herein as percent laboratory maximum density. Compact each layer of the aggregate surface course with approved compaction equipment, as required in the following paragraphs. Maintain the water content during the compaction procedure at optimum. Compaction at a moisture content other than optimum moisture must be approved by the Contracting Officer. Compact the mixture with mechanical tampers in locations not accessible to rollers. Continue compaction until each layer through the full depth is compacted to at

least 97 percent of laboratory maximum density. Remove any materials that are found to be unsatisfactory and replace them with satisfactory material or rework them to produce a satisfactory material.

3.8 PROOF ROLLING

In addition to the compaction specified above, proof roll the designated areas by application of 30 coverages of a heavy rubber-tired roller having four tires abreast with each tire loaded to 30,000 pounds and tires inflated to 150 psi. In the areas designated, proof roll the top lift of layer on which surface course is laid and to each layer of the surface course. Maintain the water content of the lift of the layer on which the surface course is placed and each layer of the aggregate surface course at optimum or at the percentage directed from the start of compaction to the completion of a proof rolling. Remove and replace materials in the aggregate surface course or underlying materials indicated to be unacceptable by the proof rolling with acceptable materials.

3.9 EDGES OF AGGREGATE SURFACE COURSE

Place approved aggregate material along the edges of the aggregate surface course in such quantity as to compact to the thickness of the course being constructed. Simultaneously roll and compact at least 1 foot of shoulder width with the rolling and compacting of each layer of the surface course when the course is being constructed in two or more layers.

3.10 SMOOTHNESS TEST

Construct each layer so that the surface shows no deviations in excess of 3/8 inch when tested with a 10 foot straightedge applied both parallel with and at right angles to the centerline of the area to be paved. Correct deviations exceeding this amount by removing material, replacing with new material, or reworking existing material and compacting.

3.11 FIELD QUALITY CONTROL

3.11.1 In-Place Tests

Perform each of the following tests on samples taken from the placed and compacted aggregate surface course. Take samples and test at the rates indicated.

- a. Perform density tests on every lift of material placed and at a frequency of one set of tests for every 250 square yards, or portion thereof, of completed area.
- b. Perform sieve analysis on every lift of material placed and at a frequency of one sieve analysis for every 500 square yards, or portion thereof, of material placed.
- c. Perform liquid limit and plasticity index tests at the same frequency as the sieve analysis.
- d. Measure the thickness of the aggregate surface course at intervals providing at least one measurement for each 500 square yards of surface course or part thereof. Measure the thickness using test holes, at least 3 inch in diameter through the aggregate surface course.

3.11.2 Approval of Material

Final approval of the materials will be based on tests for gradation, liquid limit, and plasticity index performed on samples taken from the completed and full compacted aggregate surface course.

3.12 MAINTENANCE

Maintain the aggregate surface course in a condition that will meet all specification requirements until accepted.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 16 19

CONCRETE CURBS, GUTTERS AND SIDEWALKS

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 EQUIPMENT, TOOLS, AND MACHINES
 - 1.3.1 General Requirements
 - 1.3.2 Slip Form Equipment
- 1.4 ENVIRONMENTAL REQUIREMENTS
 - 1.4.1 Placing During Cold Weather
 - 1.4.2 Placing During Warm Weather

PART 2 PRODUCTS

- 2.1 CONCRETE
 - 2.1.1 Air Content
 - 2.1.2 Slump
 - 2.1.3 Reinforcement Steel
- 2.2 CONCRETE CURING MATERIALS
 - 2.2.1 Impervious Sheet Materials
 - 2.2.2 Burlap
 - 2.2.3 White Pigmented Membrane-Forming Curing Compound
- 2.3 CONCRETE PROTECTION MATERIALS
- 2.4 JOINT FILLER STRIPS
 - 2.4.1 Contraction Joint Filler for Curb and Gutter
 - 2.4.2 Expansion Joint Filler, Premolded
- 2.5 JOINT SEALANTS
- 2.6 FORM WORK
 - 2.6.1 Wood Forms
 - 2.6.2 Steel Forms
 - 2.6.3 Sidewalk Forms
 - 2.6.4 Curb and Gutter Forms
 - 2.6.5 Biodegradable Form Release Agent
- 2.7 Detectable Warning System

PART 3 EXECUTION

- 3.1 SUBGRADE PREPARATION
 - 3.1.1 Sidewalk Subgrade
 - 3.1.2 Curb and Gutter Subgrade
 - 3.1.3 Maintenance of Subgrade
- 3.2 FORM SETTING
 - 3.2.1 Sidewalks
 - 3.2.2 Curbs and Gutters
- 3.3 SIDEWALK CONCRETE PLACEMENT AND FINISHING
 - 3.3.1 Formed Sidewalks
 - 3.3.2 Concrete Finishing

- 3.3.3 Edge and Joint Finishing
- 3.3.4 Surface and Thickness Tolerances
- 3.4 CURB AND GUTTER CONCRETE PLACEMENT AND FINISHING
 - 3.4.1 Formed Curb and Gutter
 - 3.4.2 Curb and Gutter Finishing
 - 3.4.3 Concrete Finishing
 - 3.4.4 Joint Finishing
 - 3.4.5 Surface and Thickness Tolerances
- 3.5 SIDEWALK JOINTS
 - 3.5.1 Sidewalk Contraction Joints
 - 3.5.2 Sidewalk Expansion Joints
 - 3.5.3 Reinforcement Steel Placement
- 3.6 CURB AND GUTTER JOINTS
 - 3.6.1 Contraction Joints
 - 3.6.2 Expansion Joints
- 3.7 CURING AND PROTECTION
 - 3.7.1 General Requirements
 - 3.7.1.1 Mat Method
 - 3.7.1.2 Impervious Sheeting Method
 - 3.7.1.3 Membrane Curing Method
 - 3.7.2 Backfilling
 - 3.7.3 Protection
 - 3.7.4 Protective Coating
 - 3.7.4.1 Application
 - 3.7.4.2 Precautions
- 3.8 FIELD QUALITY CONTROL
 - 3.8.1 General Requirements
 - 3.8.2 Concrete Testing
 - 3.8.2.1 Strength Testing
 - 3.8.2.2 Air Content
 - 3.8.2.3 Slump Test
 - 3.8.3 Thickness Evaluation
 - 3.8.4 Surface Evaluation
- 3.9 SURFACE DEFICIENCIES AND CORRECTIONS
 - 3.9.1 Thickness Deficiency
 - 3.9.2 High Areas
 - 3.9.3 Appearance
- 3.10 DETECTABLE WARNING SYSTEM

-- End of Section Table of Contents --

SECTION 32 16 19

CONCRETE CURBS, GUTTERS AND SIDEWALKS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO M 182 (2005; R 2017) Standard Specification for
Burlap Cloth Made from Jute or Kenaf and
Cotton Mats

ASTM INTERNATIONAL (ASTM)

ASTM A615/A615M (2020) Standard Specification for Deformed
and Plain Carbon-Steel Bars for Concrete
Reinforcement

ASTM A1064/A1064M (2018a) Standard Specification for
Carbon-Steel Wire and Welded Wire
Reinforcement, Plain and Deformed, for
Concrete

ASTM C31/C31M (2019a) Standard Practice for Making and
Curing Concrete Test Specimens in the Field

ASTM C94/C94M (2020) Standard Specification for
Ready-Mixed Concrete

ASTM C143/C143M (2020) Standard Test Method for Slump of
Hydraulic-Cement Concrete

ASTM C171 (2020) Standard Specification for Sheet
Materials for Curing Concrete

ASTM C172/C172M (2017) Standard Practice for Sampling
Freshly Mixed Concrete

ASTM C173/C173M (2016) Standard Test Method for Air
Content of Freshly Mixed Concrete by the
Volumetric Method

ASTM C231/C231M (2017a) Standard Test Method for Air
Content of Freshly Mixed Concrete by the
Pressure Method

ASTM C309 (2011) Standard Specification for Liquid
Membrane-Forming Compounds for Curing

Concrete

ASTM C920

(2018) Standard Specification for
Elastomeric Joint Sealants

ASTM D1751

(2004; E 2013; R 2013) Standard
Specification for Preformed Expansion
Joint Filler for Concrete Paving and
Structural Construction (Nonextruding and
Resilient Bituminous Types)

ASTM D1752

(2018) Standard Specification for
Preformed Sponge Rubber, Cork and Recycled
PVC Expansion Joint Fillers for Concrete
Paving and Structural Construction

ASTM D5893/D5893M

(2016) Standard Specification for Cold
Applied, Single Component, Chemically
Curing Silicone Joint Sealant for Portland
Cement Concrete Pavements

COMMONWEALTH OF PUERTO RICO, DEPARTMENT OF TRANSPORTATION AND
PUBLIC WORKS (PRHTA)

PRHTA Specifications

(2005) Standard Specifications for Road
and Bridge Construction

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where PRHTA Specifications Section, and/or Special Provision(s) is cited that contains references to other PRHTA sections, they shall also form a part of this specification to the extent referenced. Where reference is made to "Highway Authority" or "Authority's Representative" in PRHTA Specifications shall mean "Government" or "Contracting Officer" as defined in the Contract Documents. Where PRHTA Specifications refer to the "Engineer", "Engineer of Tests", or "Division of Tests" it shall be understood to mean the "Contracting Officer" as stated in the Contract Documents. In case of conflict between the PRHTA specifications and the Specifications in this Section that follows, the Specifications in this Section shall govern.

Paragraphs title "Method of Measurement" and "Basis of Payment" of referenced PRHTA Specifications shall not apply.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES. Submittals shall be made according to requirements of this section and in accordance with the requirements of the PRHTA Specifications.

SD-03 Product Data

Concrete

Biodegradable Form Release Agent

SD-06 Test Reports

Field Quality Control

1.3 EQUIPMENT, TOOLS, AND MACHINES

1.3.1 General Requirements

Plant, equipment, machines, and tools used in the work will be subject to approval and must be maintained in a satisfactory working condition at all times. Use equipment capable of producing the required product, meeting grade controls, thickness control and smoothness requirements as specified. Discontinue using equipment that produces unsatisfactory results. Allow the Contracting Officer access at all times to the plant and equipment to ensure proper operation and compliance with specifications.

1.3.2 Slip Form Equipment

Slip form paver or curb forming machines, will be approved based on trial use on the job and must be self-propelled, automatically controlled, crawler mounted, and capable of spreading, consolidating, and shaping the concrete to the desired cross section in one pass.

1.4 ENVIRONMENTAL REQUIREMENTS

1.4.1 Placing During Cold Weather

Do not place concrete when the air temperature reaches 40 degrees F and is falling, or is already below that point. Placement may begin when the air temperature reaches 35 degrees F and is rising, or is already above 40 degrees F. Make provisions to protect the concrete from freezing during the specified curing period. If necessary to place concrete when the temperature of the air, aggregates, or water is below 35 degrees F, placement and protection must be approved in writing. Approval will be contingent upon full conformance with the following provisions. Prepare and protect the underlying material so that it is entirely free of frost when the concrete is deposited. Heat mixing water and aggregates as necessary to result in the temperature of the in-place concrete being between 50 and 85 degrees F. Methods and equipment for heating must be approved. Use only aggregates that are free of ice, snow, and frozen lumps before entering the mixer. Provide covering or other means as needed to maintain the concrete at a temperature of at least 50 degrees F for not less than 72 hours after placing, and at a temperature above freezing for the remainder of the curing period.

1.4.2 Placing During Warm Weather

The temperature of the concrete as placed must not exceed 85 degrees F except where an approved retarder is used. Cool the mixing water and aggregates as necessary to maintain a satisfactory placing temperature. The placing temperature must not exceed 95 degrees F at any time.

PART 2 PRODUCTS

2.1 CONCRETE

Provide concrete conforming to the applicable requirements of ASTM C94/C94M and complies with PRHTA Specifications except as otherwise specified.

Concrete must have a minimum compressive strength of **3500 psi** at 28 days. Size of aggregate must not exceed **1-1/2 inches**. Submit copies of certified delivery tickets for all concrete used in the construction.

2.1.1 Air Content

Use concrete mixtures that have an air content by volume of concrete of 5 to 7 percent, based on measurements made immediately after discharge from the mixer.

2.1.2 Slump

Use concrete with a slump of **3 inches** plus or minus **1 inch** for hand placed concrete or **1 inch** plus or minus **1/2 inch** for slipformed concrete as determined in accordance with **ASTM C143/C143M**.

2.1.3 Reinforcement Steel

Use reinforcement bars conforming to **ASTM A615/A615M**. Use wire mesh reinforcement conforming to **ASTM A1064/A1064M**.

2.2 CONCRETE CURING MATERIALS

2.2.1 Impervious Sheet Materials

Use impervious sheet materials conforming to **ASTM C171**, type optional, except that polyethylene film, if used, must be white opaque.

2.2.2 Burlap

Use burlap conforming to **AASHTO M 182**.

2.2.3 White Pigmented Membrane-Forming Curing Compound

Use white pigmented membrane-forming curing compound conforming to **ASTM C309**, Type 2.

2.3 CONCRETE PROTECTION MATERIALS

Use concrete protection materials consisting of a linseed oil mixture of equal parts, by volume, of linseed oil and either mineral spirits, naphtha, or turpentine. At the option of the Contractor, commercially prepared linseed oil mixtures, formulated specifically for application to concrete to provide protection against the action of deicing chemicals may be used, except that emulsified mixtures are not acceptable.

2.4 JOINT FILLER STRIPS

2.4.1 Contraction Joint Filler for Curb and Gutter

Use hard-pressed fiberboard contraction joint filler for curb and gutter.

2.4.2 Expansion Joint Filler, Premolded

Unless otherwise indicated, use **1/2 inch** thick remolded expansion joint filler conforming to **ASTM D1751** or **ASTM D1752**.

2.5 JOINT SEALANTS

Use cold-applied joint sealant conforming to [ASTM C920](#) or [ASTM D5893/D5893M](#).

2.6 FORM WORK

Design and construct form work to ensure that the finished concrete will conform accurately to the indicated dimensions, lines, and elevations, and within the tolerances specified. Use wood or steel forms that are straight and of sufficient strength to resist springing during depositing and consolidating concrete.

2.6.1 Wood Forms

Use forms that are surfaced plank, [2 inches](#) nominal thickness, straight and free from warp, twist, loose knots, splits or other defects. Use forms with a nominal length of [10 feet](#). Radius bends may be formed with [3/4 inch](#) boards, laminated to the required thickness.

2.6.2 Steel Forms

Use channel-formed sections with a flat top surface and welded braces at each end and at not less than two intermediate points. Use forms with interlocking and self-aligning ends. Provide flexible forms for radius forming, corner forms, form spreaders, and fillers as needed. Use forms with a nominal length of [10 feet](#) and that have a minimum of 3 welded stake pockets per form. Use stake pins consisting of solid steel rods with chamfered heads and pointed tips designed for use with steel forms.

2.6.3 Sidewalk Forms

Use sidewalk forms that are of a height equal to the full depth of the finished sidewalk.

2.6.4 Curb and Gutter Forms

Use curb and gutter outside forms that have a height equal to the full depth of the curb or gutter. Use rigid forms for curb returns, except that benders or thin plank forms may be used for curb or curb returns with a radius of [10 feet](#) or more, where grade changes occur in the return, or where the central angle is such that a rigid form with a central angle of 90 degrees cannot be used. Back forms for curb returns may be made of [1-1/2 inch](#) benders, for the full height of the curb, cleated together. In lieu of inside forms for curbs, a curb "mule" may be used for forming and finishing this surface, provided the results are approved.

2.6.5 Biodegradable Form Release Agent

Use form release agent that is colorless and biodegradable and that is composed of at least 87 percent biobased material. Provide product that does not bond with, stain, or adversely affect concrete surfaces and does not impair subsequent treatments of concrete surfaces. Provide form release agent that does not contain diesel fuel, petroleum-based lubricating oils, waxes, or kerosene.

2.7 Detectable Warning System

Detectable Warning Systems shown on the Contract plans are to meet requirements of MUTCD Section 3B.18, 8D.04 and the "Americans with

Disabilities Act Accessibility Guidelines for Buildings and Facilities (ADAAG)" (see Section 1A.11) contains specifications for design and placement of detectable warning surfaces.

PART 3 EXECUTION

3.1 SUBGRADE PREPARATION

Construct subgrade to the specified grade and cross section prior to concrete placement.

3.1.1 Sidewalk Subgrade

Place and compact the subgrade in accordance with Section 31 00 00 EARTHWORK . Test the subgrade for grade and cross slope with a template extending the full width of the sidewalk and supported between side forms.

3.1.2 Curb and Gutter Subgrade

Place and compact the subgrade in accordance with Section 32 11 23 AGGREGATE BASE COURSES. Test the subgrade for grade and cross section by means of a template extending the full width of the curb and gutter. Use subgrade materials equal in bearing quality to the subgrade under the adjacent pavement.

3.1.3 Maintenance of Subgrade

Maintain subgrade in a smooth, compacted condition in conformity with the required section and established grade until the concrete is placed. The subgrade must be in a moist condition when concrete is placed. Prepare and protect subgrade so that it is free from frost when the concrete is deposited.

3.2 FORM SETTING

Set forms to the indicated alignment, grade and dimensions. Hold forms rigidly in place by a minimum of 3 stakes per form placed at intervals not to exceed 4 feet. Use additional stakes and braces at corners, deep sections, and radius bends, as required. Use clamps, spreaders, and braces where required to ensure rigidity in the forms. Remove forms in a manner that will not injure the concrete. Do not use bars or heavy tools against the concrete when removing the forms. Promptly and satisfactorily repair concrete found to be defective after form removal. Clean forms and coat with form oil or biodegradable form release agent each time before concrete is placed. Wood forms may, instead, be thoroughly wetted with water before concrete is placed, except that with probable freezing temperatures, oiling is mandatory.

3.2.1 Sidewalks

Set forms for sidewalks with the upper edge true to line and grade with an allowable tolerance of 1/8 inch in any 10 foot long section. After forms are set, grade and alignment must be checked with a 10 foot straightedge. Sidewalks must have a transverse slope Unless otherwise indicated, construct sidewalks that are located adjacent to curbs with the low side adjacent to the curb. Do not remove side forms less than 12 hours after finishing has been completed.

3.2.2 Curbs and Gutters

Remove forms used along the front of the curb not less than 2 hours nor more than 6 hours after the concrete has been placed. Do not remove forms used along the back of curb until the face and top of the curb have been finished, as specified for concrete finishing. Do not remove gutter forms while the concrete is sufficiently plastic to slump in any direction.

3.3 SIDEWALK CONCRETE PLACEMENT AND FINISHING

3.3.1 Formed Sidewalks

Place concrete in the forms in one layer. When consolidated and finished, the sidewalks must be of the thickness indicated. Consolidate concrete by tamping and spading or with an approved vibrator. Following the placement of concrete, it shall be struck-off to conform to the cross sections shown on the plans and to an elevation such that when the concrete is properly consolidated and finished, the surface of the concrete will be at the elevation shown in the plans.

3.3.2 Concrete Finishing

When most of the water sheen has disappeared, and just before the concrete hardens, finish the surface with a wood or magnesium float or darby to a smooth and uniformly fine granular or sandy texture free of waves, irregularities, or tool marks. Produce a scored surface by brooming with a fiber-bristle brush in a direction transverse to that of the traffic, followed by edging.

3.3.3 Edge and Joint Finishing

Finish all slab edges, including those at formed joints, with an edger having a radius of $\frac{1}{8}$ inch. Edge transverse joints before brooming. Eliminate the flat surface left by the surface face of the edger with brooming. Clean and solidly fill corners and edges which have crumbled and areas which lack sufficient mortar for proper finishing with a properly proportioned mortar mixture and then finish.

3.3.4 Surface and Thickness Tolerances

Finished surfaces must not vary more than $\frac{5}{16}$ inch from the testing edge of a 10-foot straightedge. Permissible deficiency in section thickness will be up to $\frac{1}{4}$ inch.

3.4 CURB AND GUTTER CONCRETE PLACEMENT AND FINISHING

3.4.1 Formed Curb and Gutter

Place concrete to the required section in a single lift. Consolidate concrete using approved mechanical vibrators. Curve shaped gutters must be finished with a standard curb "mule".

3.4.2 Curb and Gutter Finishing

Approved slipformed curb and gutter machines may be used in lieu of hand placement.

3.4.3 Concrete Finishing

Float and finish exposed surfaces with a smooth wood float until true to grade and section and uniform in texture. Brush floated surfaces with a fine-hair brush using longitudinal strokes. Round the edges of the gutter and top of the curb with an edging tool to a radius of $\frac{1}{2}$ inch. Immediately after removing the front curb form, rub the face of the curb with a wood or concrete rubbing block and water until blemishes, form marks, and tool marks have been removed. Brush the front curb surface, while still wet, in the same manner as the gutter and curb top. Finish the top surface of gutter to grade with a wood float.

3.4.4 Joint Finishing

Finish curb edges at formed joints as indicated in [PRHTA Specifications](#) Section 608 and 609.

3.4.5 Surface and Thickness Tolerances

Finished surfaces must not vary more than $\frac{1}{4}$ inch from the testing edge of a 10-foot straightedge. Permissible deficiency in section thickness will be up to $\frac{1}{4}$ inch.

3.5 SIDEWALK JOINTS

Construct sidewalk joints to divide the surface into rectangular areas. Space transverse contraction joints at a distance equal to the sidewalk width or 5 feet on centers, whichever is less, and continuous across the slab. Construct longitudinal contraction joints along the centerline of all sidewalks 10 feet or more in width. Construct transverse expansion joints at sidewalk returns and opposite expansion joints in adjoining curbs. Where the sidewalk is not in contact with the curb, install transverse expansion joints as indicated in [PRHTA specifications](#) Section 608 and 609. Form expansion joints around structures and features which project through or into the sidewalk pavement, using joint filler of the type, thickness, and width indicated. Expansion joints are not required between sidewalks and curb that abut the sidewalk longitudinally.

3.5.1 Sidewalk Contraction Joints

Form contraction joints in the fresh concrete by cutting a groove in the top portion of the slab to a depth of at least one-fourth of the sidewalk slab thickness. Unless otherwise approved or indicated, either use a jointer to cut the groove or saw a groove in the hardened concrete with a power-driven saw. Construct sawed joints by sawing a groove in the concrete with a $\frac{1}{8}$ inch blade. Provide an ample supply of saw blades on the jobsite before concrete placement is started. Provide at least one standby sawing unit in good working order at the jobsite at all times during the sawing operations.

3.5.2 Sidewalk Expansion Joints

Form expansion joints using $\frac{1}{2}$ inch joint filler strips. Joint filler in expansion joints surrounding structures and features within the sidewalk may consist of preformed filler material conforming to [ASTM D1752](#) or building paper. Hold joint filler in place with steel pins or other devices to prevent warping of the filler during floating and finishing. Immediately after finishing operations are completed, round joint edges using an edging tool having a radius of $\frac{1}{8}$ inch. Remove any concrete

over the joint filler. At the end of the curing period, clean the top of expansion joints and fill with cold-applied joint sealant. Use joint sealant that is gray or stone in color. Thoroughly clean the joint opening before the sealing material is placed. Do not spill sealing material on exposed surfaces of the concrete. Apply joint sealing material only when the concrete at the joint is surface dry and atmospheric and concrete temperatures are above 50 degrees F. Immediately remove any excess material on exposed surfaces of the concrete and clean the concrete surfaces.

3.5.3 Reinforcement Steel Placement

Accurately and securely fasten reinforcement steel in place with suitable supports and ties before the concrete is placed.

3.6 CURB AND GUTTER JOINTS

Construct curb and gutter joints at right angles to the line of curb and gutter.

3.6.1 Contraction Joints

Construct contraction joints directly opposite contraction joints in abutting portland cement concrete pavements and spaced so that monolithic sections between curb returns will not be less than 5 feet nor greater than 10 feet in length.

- a. Construct contraction joints (except for slip forming) by means of 1/8 inch thick separators and of a section conforming to the cross section of the curb and gutter. Remove separators as soon as practicable after concrete has set sufficiently to preserve the width and shape of the joint and prior to finishing.
- b. When slip forming is used, cut the contraction joints in the top portion of the gutter/curb hardened concrete in a continuous cut across the curb and gutter, using a power-driven saw. Cut the contraction joint to a depth of at least one-fourth of the gutter/curb depth using a 1/8 inch saw blade.

3.6.2 Expansion Joints

Form expansion joints by means of preformed expansion joint filler material cut and shaped to the cross section of curb and gutter. Construct expansion joints in curb and gutter directly opposite expansion joints of abutting portland cement concrete pavement using the same type and thickness of joints as joints in the pavement. Where curb and gutter do not abut portland cement concrete pavement, provide expansion joints at least 1/2 inch in width at intervals not less than 30 feet nor greater than 120 feet. Seal expansion joints immediately following curing of the concrete or as soon thereafter as weather conditions permit. Seal expansion joints and the top 1 inch depth of curb and gutter contraction-joints with joint sealant. Thoroughly clean the joint opening before the sealing material is placed. Do not spill sealing material on exposed surfaces of the concrete. Concrete at the joint must be surface dry and atmospheric and concrete temperatures must be above 50 degrees F at the time of application of joint sealing material. Immediately remove excess material on exposed surfaces of the concrete and clean concrete surfaces.

3.7 CURING AND PROTECTION

3.7.1 General Requirements

Protect concrete against loss of moisture and rapid temperature changes for at least 7 days from the beginning of the curing operation. Protect unhardened concrete from rain and flowing water. All equipment needed for adequate curing and protection of the concrete must be on hand and ready for use before actual concrete placement begins. Protect concrete as necessary to prevent cracking due to temperature changes during the curing period.

3.7.1.1 Mat Method

Cover the entire exposed surface with two or more layers of burlap. Overlap mats at least 6 inches. Thoroughly wet the mat with water prior to placing on concrete surface and keep the mat continuously in a saturated condition and in intimate contact with concrete for not less than 7 days.

3.7.1.2 Impervious Sheeting Method

Wet the entire exposed surface with a fine spray of water and then cover with impervious sheeting material. Lay sheets directly on the concrete surface with the light-colored side up and overlapped 12 inches when a continuous sheet is not used. Use sheeting that is not less than 18-inches wider than the concrete surface to be cured. Secure sheeting using heavy wood planks or a bank of moist earth placed along edges and laps in the sheets. Satisfactorily repair or replace sheets that are torn or otherwise damaged during curing. Sheeting must remain on the concrete surface to be cured for not less than 7 days.

3.7.1.3 Membrane Curing Method

Apply a uniform coating of white-pigmented membrane-curing compound to the entire exposed surface of the concrete as soon after finishing as the free water has disappeared from the finished surface. Coat formed surfaces immediately after the forms are removed and in no case longer than 1 hour after the removal of forms. Do not allow concrete surface to dry before application of the membrane. If drying has occurred, moisten the surface of the concrete with a fine spray of water and apply the curing compound as soon as the free water disappears. Apply curing compound in two coats by hand-operated pressure sprayers at a coverage of approximately 200 square feet/gallon for the total of both coats. Apply the second coat in a direction approximately at right angles to the direction of application of the first coat. The compound must form a uniform, continuous, coherent film that will not check, crack, or peel and must be free from pinholes or other imperfections. If pinholes, abrasion, or other discontinuities exist, apply an additional coat to the affected areas within 30 minutes. Respray concrete surfaces that are subjected to heavy rainfall within 3 hours after the curing compound has been applied by the method and at the coverage specified above. Respray areas where the curing compound is damaged by subsequent construction operations within the curing period. Take precautions necessary to ensure that the concrete is properly cured at sawed joints, and that no curing compound enters the joints. Tightly seal the top of the joint opening and the joint groove at exposed edges before the concrete in the region of the joint is resprayed with curing compound. Use a method used for sealing the joint groove that prevents loss of moisture from the joint during the entire specified curing

period. Provide approved standby facilities for curing concrete pavement at a location accessible to the jobsite for use in the event of mechanical failure of the spraying equipment or other conditions that might prevent correct application of the membrane-curing compound at the proper time. Adequately protect concrete surfaces to which membrane-curing compounds have been applied during the entire curing period from pedestrian and vehicular traffic, except as required for joint-sawing operations and surface tests, and from other possible damage to the continuity of the membrane.

3.7.2 Backfilling

After curing, remove debris and backfill, grade, and compact the area adjoining the concrete to conform to the surrounding area in accordance with lines and grades shown on the Contract plans.

3.7.3 Protection

Protect completed concrete from damage until accepted. Repair damaged concrete and clean concrete discolored during construction at no cost to the Government. Remove and reconstruct concrete that is damaged for the entire length between regularly scheduled joints. Refinishing the damaged portion will not be acceptable. Dispose of removed material as required to comply with contract requirements.

3.7.4 Protective Coating

Apply a protective coating of linseed oil mixture to the exposed-to-view concrete surface after the curing period, if concrete will be exposed to de-icing chemicals within 6 weeks after placement. Moist cure concrete to receive a protective coating.

3.7.4.1 Application

Complete curing and backfilling operation prior to applying two coats of protective coating. Concrete surface must be dry and clean before each application. Spray apply at a rate of not more than 50 square yards/gallon for first application and not more than 70 square yards/gallon for second application, except that the number of applications and coverage for each application for commercially prepared mixture must be in accordance with the manufacturer's instructions. Protect coated surfaces from vehicular and pedestrian traffic until dry.

3.7.4.2 Precautions

Do not heat protective coating by direct application of flame or electrical heaters and protect the coating from exposure to open flame, sparks, and fire adjacent to open containers or applicators. Do not apply material at ambient or material temperatures lower than 50 degrees F.

3.8 FIELD QUALITY CONTROL

Submit copies of all test reports within 24 hours of completion of the test.

3.8.1 General Requirements

Perform the inspection and tests described and meet the specified requirements for inspection details and frequency of testing. Based upon

the results of these inspections and tests, take the action and submit reports as required below, and additional tests to ensure that the requirements of these specifications are met.

3.8.2 Concrete Testing

3.8.2.1 Strength Testing

Take concrete samples in accordance with [ASTM C172/C172M](#) not less than once a day nor less than once for every [250 cubic yards](#) of concrete placed. Mold cylinders in accordance with [ASTM C31/C31M](#) for strength testing by an approved laboratory. Each strength test result must be the average of 2 test cylinders from the same concrete sample tested at 28 days, unless otherwise specified or approved. Concrete specified on the basis of compressive strength will be considered satisfactory if the averages of all sets of three consecutive strength test results equal or exceed the specified strength, and no individual strength test result falls below the specified strength by more than [500 psi](#).

3.8.2.2 Air Content

Determine air content in accordance with [ASTM C173/C173M](#) or [ASTM C231/C231M](#). Use [ASTM C231/C231M](#) with concretes and mortars made with relatively dense natural aggregates. Make two tests for air content on randomly selected batches of each class of concrete placed during each shift. Make additional tests when excessive variation in concrete workability is reported by the placing foreman or the Government inspector. Notify the placing foreman if results are out of tolerance. The placing foreman must take appropriate action to have the air content corrected at the plant. Additional tests for air content will be performed on each truckload of material until such time as the air content is within the tolerance specified.

3.8.2.3 Slump Test

Perform two slump tests on randomly selected batches of each class of concrete for every [250 cubic yards](#), or fraction thereof, of concrete placed during each shift. Perform additional tests when excessive variation in the workability of the concrete is noted or when excessive crumbling or slumping is noted along the edges of slip-formed concrete.

3.8.3 Thickness Evaluation

Determine the anticipated thickness of the concrete prior to placement by passing a template through the formed section or by measuring the depth of opening of the extrusion template of the curb forming machine. If a slip form paver is used for sidewalk placement, construct the subgrade true to grade prior to concrete placement. The thickness will be determined by measuring each edge of the completed slab.

3.8.4 Surface Evaluation

Provide finished surfaces for each category of the completed work that are uniform in color and free of blemishes and form or tool marks.

3.9 SURFACE DEFICIENCIES AND CORRECTIONS

3.9.1 Thickness Deficiency

When measurements indicate that the completed concrete section is deficient in thickness by more than $\frac{1}{4}$ inch the deficient section will be removed, between regularly scheduled joints, and replaced.

3.9.2 High Areas

In areas not meeting surface smoothness and plan grade requirements, reduce high areas either by rubbing the freshly finished concrete with carborundum brick and water when the concrete is less than 36 hours old or by grinding the hardened concrete with an approved surface grinding machine after the concrete is 36 hours old or more. The area corrected by grinding the surface of the hardened concrete must not exceed 5 percent of the area of any integral slab, and the depth of grinding must not exceed $\frac{1}{4}$ inch. Remove and replace pavement areas requiring grade or surface smoothness corrections in excess of the limits specified.

3.9.3 Appearance

Exposed surfaces of the finished work will be inspected by the Contracting Officer and deficiencies in appearance will be identified. Remove and replace areas which exhibit excessive cracking, discoloration, form marks, or tool marks or which are otherwise inconsistent with the overall appearances of the work.

3.10 DETECTABLE WARNING SYSTEM

Install Detectable Warning Systems required by the Contract plans in accordance with MUTCD Section 3B.18, 8D.04 and ADAAG).

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 17 23

PAVEMENT MARKINGS

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY ASSURANCE
 - 1.3.1 Regulatory Requirements
 - 1.3.2 Qualifications
- 1.4 DELIVERY AND STORAGE
- 1.5 PROJECT/SITE CONDITIONS
 - 1.5.1 Environmental Requirements
 - 1.5.1.1 Weather Limitations for Application
 - 1.5.1.2 Weather Limitations for Removal of Pavement Markings on Roads and Automotive Parking Areas
 - 1.5.2 Traffic Controls
 - 1.5.3 Lighting

PART 2 PRODUCTS

- 2.1 EQUIPMENT
 - 2.1.1 Surface Preparation and Paint Removal
 - 2.1.1.1 Surface Preparation Equipment for Roads and Automotive Parking Areas
 - 2.1.1.1.1 Sandblasting Equipment
 - 2.1.1.1.2 Waterblasting Equipment
 - 2.1.1.1.3 Grinding or Scarifying Equipment
 - 2.1.1.1.4 Chemical Removal Equipment
 - 2.1.2 Application Equipment
 - 2.1.2.1 Paint Application Equipment
 - 2.1.2.1.1 Hand-Operated, Push-Type Machines
 - 2.1.2.1.2 Self-Propelled or Mobile-Drawn Spraying Machines
 - 2.1.2.1.2.1 Road Marking
 - 2.1.2.1.2.2 Hand Application
 - 2.1.2.2 Thermoplastic Application Equipment
 - 2.1.2.2.1 Thermoplastic Material
 - 2.1.2.2.2 Application Equipment
 - 2.1.2.2.3 Mobile Application Equipment
 - 2.1.2.2.4 Portable Application Equipment
 - 2.1.2.3 Reflective Media Dispenser
 - 2.1.2.4 Preformed Tape Application Equipment
- 2.2 MATERIALS

PART 3 EXECUTION

- 3.1 EXAMINATION
 - 3.1.1 Testing for Moisture
 - 3.1.2 Surface Preparation Demonstration

- 3.1.3 Test Stripe Demonstration
- 3.1.4 Application Rate Demonstration
- 3.1.5 Level of Performance Demonstration
- 3.2 EXTERIOR SURFACE PREPARATION
 - 3.2.1 Early Painting of Rigid Pavements
 - 3.2.2 Early Painting of Asphalt Pavements
- 3.3 APPLICATION
 - 3.3.1 Roads
 - 3.3.2 Thermoplastic Compound
 - 3.3.2.1 Primer
 - 3.3.2.2 Reflective Media
 - 3.3.3 Raised Pavement Markers
 - 3.3.4 Preformed Tape
 - 3.3.5 Cleanup and Waste Disposal
- 3.4 FIELD QUALITY CONTROL
 - 3.4.1 Sampling and Test Reports

-- End of Section Table of Contents --

SECTION 32 17 23

PAVEMENT MARKINGS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO M 247 (2013) Standard Specification for Glass
Beads Used in Pavement Markings

INTERNATIONAL CONCRETE REPAIR INSTITUTE (ICRI)

ICRI 03732 (1997) Selecting and Specifying Concrete
Surface Preparation for Sealers, Coatings,
and Polymer Overlays

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)

MUTCD (2015) Manual on Uniform Traffic Control
Devices

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FS TT-B-1325 (Rev D; Notice 1; Notice 2 2017) Beads
(Glass Spheres) Retro-Reflective (Metric)

COMMONWEALTH OF PUERTO RICO, DEPARTMENT OF TRANSPORTATION AND
PUBLIC WORKS (PRHTA)

PRHTA Specifications (2005) Standard Specifications for Road
and Bridge Construction

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where PRHTA Specifications Section, and/or Special Provision(s) is cited that contains references to other PRHTA sections, they shall also form a part of this specification to the extent referenced. Where reference is made to "Highway Authority" or "Authority's Representative" in PRHTA Specifications shall mean "Government" or "Contracting Officer" as defined in the Contract Documents. Where PRHTA Specifications refer to the "Engineer", "Engineer of Tests", or "Division of Tests" it shall be understood to mean the "Contracting Officer" as stated in the Contract Documents. In case of conflict between the PRHTA specifications and the Specifications in this Section that follows, the Specifications in this Section shall govern.

Paragraphs title "Method of Measurement" and "Basis of Payment" of referenced PRHTA Specifications shall not apply.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only or as otherwise designated. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES. Submittals shall be made according to requirements of this section and in accordance with the requirements of the **PRHTA Specifications**.

SD-03 Product Data

Surface Preparation Equipment List; G, R0

Application Equipment List; G, R0

Exterior Surface Preparation

Safety Data Sheets; G, D0

Materials; G, R0

Thermoplastic compound; G, R0

SD-06 Test Reports

Materials; G, R0

Thermoplastic Compound; G, R0

Test Reports

SD-07 Certificates

Qualifications; G, D0

Materials; G, D0

Volatile Organic Compound, (VOC); G, D0

Thermoplastic Compound

SD-08 Manufacturer's Instructions

Thermoplastic Compound; G, R0

1.3 QUALITY ASSURANCE

1.3.1 Regulatory Requirements

Submit certificate stating that the proposed pavement marking paint meets the **Volatile Organic Compound**, (VOC) regulations of the local Air Pollution Control District having jurisdiction over the geographical area in which the project is located. Submit **Safety Data Sheets** for each product.

1.3.2 Qualifications

Submit documentation certifying that pertinent personnel are qualified for equipment operation and handling of applicable chemicals. The documentation should include experience on five projects of similar size and scope with references for all personnel.

1.4 DELIVERY AND STORAGE

Deliver paint materials, thermoplastic compound materials, and reflective media in original sealed containers that plainly show the designated name, specification number, batch number, color, date of manufacture, manufacturer's directions, and name of manufacturer.

Provide storage facilities at the job site for maintaining materials at temperatures recommended by the manufacturer. Make available paint stored at the project site or segregated at the source for sampling not less than 30 days prior to date of required approval for use to allow sufficient time for testing. Notify the Contracting Officer when paint is available for sampling.

1.5 PROJECT/SITE CONDITIONS

1.5.1 Environmental Requirements

1.5.1.1 Weather Limitations for Application

Apply pavement markings to clean, dry surfaces, and unless otherwise approved, only when the air and pavement surface temperature is at least 5 degrees F above the dew point and the air and pavement temperatures are within the limits recommended by the pavement marking manufacturer. Allow pavement surfaces to dry after water has been used for cleaning or rainfall has occurred prior to striping or marking. Test the pavement surface for moisture before beginning work each day and after cleaning. Do not commence marking until the pavement is sufficiently dry and the pavement condition has been approved by the Contracting Officer. Employ the "plastic wrap method" to test the pavement for moisture as specified in paragraph TESTING FOR MOISTURE.

1.5.1.2 Weather Limitations for Removal of Pavement Markings on Roads and Automotive Parking Areas

Pavement surface must be free of snow, ice, or slush; with a surface temperature of at least 40 degrees F and rising at the beginning of operations, except those involving shot or sand blasting or grinding. Cease operation during thunderstorms, or during rainfall, except for waterblasting and removal of previously applied chemicals. Cease waterblasting where surface water accumulation alters the effectiveness of material removal.

1.5.2 Traffic Controls

Place warning signs conforming to MUTCD near the beginning of the worksite and well ahead of the worksite for alerting approaching traffic from both directions. Place small markers along newly painted lines or freshly placed raised markers to control traffic and prevent damage to newly painted surfaces or displacement of raised pavement markers. Mark painting equipment with large warning signs indicating slow-moving

painting equipment in operation.

When traffic must be rerouted or controlled to accomplish the work, provide necessary warning signs, flag persons, and related equipment for the safe passage of vehicles.

1.5.3 Lighting

When night operations are necessary, provide all necessary lighting and equipment. The Government reserves the right to accept or reject night work on the day following night activities by the Contractor.

PART 2 PRODUCTS

2.1 EQUIPMENT

2.1.1 Surface Preparation and Paint Removal

2.1.1.1 Surface Preparation Equipment for Roads and Automotive Parking Areas

Submit a [surface preparation equipment list](#) by serial number, type, model, and manufacturer. Include descriptive data indicating area of coverage per pass, pressure adjustment range, tank and flow capacities, and safety precautions required for the equipment operation. Mobile equipment must allow for removal of markings without damaging the pavement surface or joint sealant. Maintain machines, tools, and equipment used in the performance of the work in satisfactory operating condition.

2.1.1.1.1 Sandblasting Equipment

Use mobile sandblasting equipment capable of producing a pressurized stream of sand and air that effectively removes paint from the surface without creating voids in asphalt or tar pavements or removing joint sealants in Portland cement concrete pavements. Include with the equipment an air compressor, hoses, and nozzles of adequate size and capacity for removing paint. Equip the compressor with traps and coalescing filters that maintain the compressed air free of oil and water.

2.1.1.1.2 Waterblasting Equipment

Use mobile waterblasting equipment capable of producing a pressurized stream of water that effectively removes paint from the pavement surface without significantly damaging the pavement. Provide equipment, tools, and machinery which are safe and in good working order at all times.

2.1.1.1.3 Grinding or Scarifying Equipment

Use equipment capable of removing surface contaminants, paint build-up, or extraneous markings from the pavement surface without leaving any residue. Clean the surface by waterblasting to remove surface contaminants and ash after a weed torch is used to remove paint.

2.1.1.1.4 Chemical Removal Equipment

Use chemical equipment capable of applying and removing chemicals and paint from the pavement surface, leaving only non-toxic biodegradable residue without scarring or other damage to the pavement or joints and joint seals.

2.1.2 Application Equipment

Submit [application equipment list](#) appropriate for the material(s) to be used. Include manufacturer's descriptive data and certification for the planned use that indicates area of coverage per pass, pressure adjustment range, tank and flow capacities, and all safety precautions required for operating and maintaining the equipment. Provide and maintain machines, tools, and equipment used in the performance of the work in satisfactory operating condition or remove them from the work site. Provide mobile and maneuverable application equipment to the extent that straight lines can be followed and normal curves can be made in a true arc.

2.1.2.1 Paint Application Equipment

2.1.2.1.1 Hand-Operated, Push-Type Machines

Provide hand-operated push-type applicator machine of a type commonly used for application of water based paint or two-component, chemically curing paint, thermoplastic, or preformed tape, to pavement surfaces for small marking projects, such as legends and cross-walks, automotive parking areas, or surface painted signs. Provide applicator machine equipped with the necessary tanks and spraying nozzles capable of applying paint uniformly at coverage specified. Hand operated spray guns may be used in areas where push-type machines cannot be used.

2.1.2.1.2 Self-Propelled or Mobile-Drawn Spraying Machines

Provide self-propelled or mobile-drawn spraying machine with suitable arrangements of atomizing nozzles and controls to obtain the specified results. Provide machine having a speed during application capable of applying the stripe widths indicated at the paint coverage rate specified herein and of even uniform thickness with clear-cut edges.

2.1.2.1.2.1 Road Marking

Provide equipment used for marking roads capable of placing the prescribed number of lines at a single pass as solid lines, intermittent lines, or a combination of solid and intermittent lines using a maximum of three different colors of paint as specified.

2.1.2.1.2.2 Hand Application

Provide spray guns for hand application of paint in areas where the mobile paint applicator cannot be used.

2.1.2.2 Thermoplastic Application Equipment

2.1.2.2.1 Thermoplastic Material

Apply thermoplastic material with equipment that is capable of providing continuous uniformity in the dimensions and reflectorization of the marking.

2.1.2.2.2 Application Equipment

- a. Provide application equipment capable of continuous mixing and agitation of the material, with conveying parts which prevent accumulation and clogging between the main material reservoir and the

extrusion shoe or spray gun. All parts of the equipment which come into contact with the material must be easily accessible and exposed for cleaning and maintenance. All mixing and conveying parts up to and including the extrusion shoes and spray guns must maintain the material at the required temperature with heat-transfer oil or electrical-element-controlled heat.

- b. Provide application equipment constructed to ensure continuous uniformity in the dimensions of the stripe. Provide an applicator with a means for cleanly cutting off stripe ends squarely and providing a method of applying "skiplines." Provide equipment capable of applying varying widths of traffic markings.
- c. Provide mobile and maneuverable application equipment allowing straight lines to be followed and normal curves to be made in a true arc. Provide equipment used for the placement of thermoplastic pavement markings of two general types: mobile applicator and portable applicator.
- d. Equip the applicator with a pressurized or drop-on type bead dispenser capable of uniformly dispensing reflective glass spheres at controlled rates of flow. The bead dispenser must operate automatically to begin flow prior to the flow of binder to assure that the strip is fully reflectorized.

2.1.2.2.3 Mobile Application Equipment

Provide a truck-mounted, self-contained pavement marking machine that is capable of hot applying thermoplastic by either the extrusion or spray method.

- a. Equip the unit to apply the thermoplastic marking material at temperatures according to the manufacturer's instructions, at widths varying from 3 to 12 inches, with an automatic pressurized or drop-on bead dispensing system, capable of operating continuously, and of installing a minimum of 20,000 lineal feet of longitudinal markings in an 8-hour day.
- b. Equip the mobile unit with a melting kettle which holds a minimum of 6000 pounds of molten thermoplastic material; capable of heating the thermoplastic composition to temperatures as recommended by the manufacturer. Use a thermostatically controlled heat transfer liquid. Heating of the composition by direct flame is not allowed. Oil and material temperature gauges must be visible at both ends of the kettle.
- c. Equip mobile units for application of extruded markings with a minimum of two extrusion shoes; located one on each side of the truck, capable of marking simultaneous edge line and centerline stripes; each being a closed, oil-jacketed unit; holding the molten thermoplastic at a temperature as recommended by the manufacturer; and capable of extruding a line of 3 to 8 inches in width; and at a thickness of not less than 0.120 inch nor more than 0.190 inch, of generally uniform cross section.
- d. Equip mobile units for application of spray markings with a spray gun system capable of marking simultaneous edgeline and centerline stripes. Surround (jacket) the spray system with heating oil to maintain the molten thermoplastic at a temperature of 375 to 425

degrees F, capable of spraying a stripe of 3 to 12 inches in width, and in thicknesses varying from 0.060 inch to 0.098 inch, of generally uniform cross section.

- e. Equip the mobile unit with an electronic programmable line pattern control system, capable of applying skip or solid lines in any sequence, through any and all of the extrusion shoes, or the spray guns, and in programmable cycle lengths. In addition, equip the mobile unit with an automatic counting mechanism capable of recording the number of lineal feet of thermoplastic markings applied to the pavement surface with an accuracy of 0.5 percent.

2.1.2.2.4 Portable Application Equipment

Provide portable hand-operated equipment, specifically designed for placing special markings such as crosswalks, stop bars, legends, arrows, and short lengths of lane, edge and centerlines; and capable of applying thermoplastic pavement markings by the extrusion method. Equip the portable applicator with all the necessary components, including a materials storage reservoir, bead dispenser, extrusion shoe, and heating accessories, capable of holding the molten thermoplastic at the temperature recommended by the manufacturer, and of extruding a line of 3 to 12 inches in width, and in thickness of not less than 0.120 inch nor more than 0.190 inch and of generally uniform cross section.

2.1.2.3 Reflective Media Dispenser

Attach the dispenser for applying the reflective media to the paint dispenser and designed to operate automatically and simultaneously with the applicator through the same control mechanism. The bead applicator must be capable of adjustment and designed to provide uniform flow of reflective media over the full length and width of the stripe at the rate of coverage specified in paragraph APPLICATION.

2.1.2.4 Preformed Tape Application Equipment

Provide and use mechanical application equipment for the placement of preformed marking tape which is a mobile pavement marking machine specifically designed for use in applying pressure-sensitive pavement marking tape of varying widths. Equip the applicator with rollers, or other suitable compaction device to provide initial adhesion of the material with the pavement surface. Use additional tools and devices as needed to properly seat the applied material as recommended by the manufacturer.

2.2 MATERIALS

All materials shall be in accordance with the requirements of the PRHTA Specifications, section 618.

PART 3 EXECUTION

3.1 EXAMINATION

3.1.1 Testing for Moisture

Test the pavement surface for moisture before beginning pavement marking after each period of rainfall, fog, high humidity, or cleaning, or when the ambient temperature has fallen below the dew point. Do not commence

marking until the pavement is sufficiently dry and the pavement condition has been approved by the Contracting Officer or authorized representative.

Employ the "plastic wrap method" to test the pavement for moisture as follows: Cover the pavement with a 12 inch by 12 inch section of clear plastic wrap and seal the edges with tape. After 15 minutes, examine the plastic wrap for any visible moisture accumulation inside the plastic. Do not begin marking operations until the test can be performed with no visible moisture accumulation inside the plastic wrap. Re-test surfaces when work has been stopped due to rain.

3.1.2 Surface Preparation Demonstration

Prior to surface preparation, demonstrate the proposed procedures and equipment. Prepare areas large enough to determine cleanliness, adhesion of remaining coating and rate of cleaning. Perform a demonstration removal of pavement marking in an area designated by the Contracting Officer. Approved demonstration area establishes the standard for the remainder of the work.

3.1.3 Test Stripe Demonstration

Prior to paint application, demonstrate test stripe application within the work area using the proposed materials and equipment. Apply separate test stripes in each of the line widths and configurations required herein using the proposed equipment. Make the test stripes long enough to determine the proper speed and operating pressures for the vehicle(s) and machinery, but not less than 50 feet long.

3.1.4 Application Rate Demonstration

During the Test Stripe Demonstration, demonstrate compliance with the application rates specified herein. Document the equipment speed and operating pressures required to meet the specified rates in each configuration of the equipment and provide a copy of the documentation to the Contracting Officer prior to proceeding with the work.

3.1.5 Level of Performance Demonstration

The Contracting Officer will be present at the application demonstrations to observe the results obtained and to validate the operating parameters of the vehicle(s) and equipment. If accepted by the Contracting Officer, the test stripe is the measure of performance required for this project. Do not proceed with the work until the demonstration results are satisfactory to the Contracting Officer.

3.2 EXTERIOR SURFACE PREPARATION

Allow new pavement surfaces to cure for a period of not less than 30 days before application of marking materials. Thoroughly clean surfaces to be marked before application of the paint. Remove dust, dirt, and other granular surface deposits by sweeping, blowing with compressed air, rinsing with water, or a combination of these methods as required. Remove rubber deposits, existing paint markings, residual curing compounds, and other coatings adhering to the pavement by water blasting.

- a. For Portland Cement Concrete pavement, grinding, light shot blasting, or light scarification, to a resulting profile equal to ICRI 03732 CSP 2, CSP 3, and CSP 4, respectively, can be used in addition to water

blasting on most pavements, to either remove existing coatings, or for surface preparation.

- b. Scrub affected areas, where oil or grease is present on old pavements to be marked, with several applications of trisodium phosphate solution or other approved detergent or degreaser and rinse thoroughly after each application. After cleaning oil-soaked areas, seal with shellac or primer recommended by the manufacturer to prevent bleeding through the new paint. Do not commence painting in any area until pavement surfaces are dry and clean.

3.2.1 Early Painting of Rigid Pavements

Pretreat rigid pavements that require early painting with an aqueous solution containing 3 percent phosphoric acid and 2 percent zinc chloride. Apply the solution to the areas to be marked.

3.2.2 Early Painting of Asphalt Pavements

For asphalt pavement systems requiring painting application at less than 30 days, apply the paint and beads at half the normal application rate, followed by a second application at the normal rate after 30 days.

3.3 APPLICATION

Apply pavement markings to dry pavements only. Application of materials shall be in accordance with the requirements of the [PRHTA Specifications](#), section 618.

For reflectorized markings, apply paint and glass spheres at the following rates:

TABLE III

Bead Type	Paint Type	Paint Application Rate	Bead Application Rate
Type I (Gradation A)	Type I, II, III	121 plus or minus 6 Sq Ft/Gallon	8 plus or minus 1 lb/gallon
Type III	Type I, II, III	121 plus or minus 6 Sq Ft/Gallon	10 plus or minus 1 lb/gallon
Type IV (Gradation A)	Type III	76 plus or minus 12 Sq Ft/Gallon	8 plus or minus 1 lb/gallon
Type IV (Gradation B)	Type III	98 plus or minus 9 Sq Ft/Gallon	8 plus or minus 1 lb/gallon

3.3.1 Roads

Apply paint at a rate of 105 plus or minus 5 square feet per gallon. Apply [FS TT-B-1325 Type I \(Gradation A\)](#) beads at a rate of 7 plus or minus 0.5 pounds of glass spheres per gallon.

3.3.2 Thermoplastic Compound

Place thermoplastic pavement markings, free from dirt or tint, upon dry pavement. The temperature must be a minimum of 40 degrees F and rising at the time of installation. Apply all centerline, skipline, edgeline, and other longitudinal type markings with a mobile applicator. Place all special markings, crosswalks, stop bars, legends, arrows, and similar patterns with a portable applicator, using the extrusion method.

3.3.2.1 Primer

After surface preparation has been completed, prime the asphalt or concrete pavement surface with spray equipment. Allow primer materials to "set-up" prior to applying the thermoplastic composition. Allow the asphalt concrete primer to dry to a tack-free condition, usually occurring in less than 10 minutes. Allow the Portland cement concrete primer to dry in accordance with the thermoplastic manufacturer recommendations. To shorten the curing time of the epoxy resins, an infrared heating device may be used on the concrete primer. Apply asphalt concrete primer to all asphalt concrete pavements at a wet film thickness of 0.005 inch, plus or minus 0.001 inch (265 to 400 square feet per gallon). Apply Portland cement concrete primer to all concrete pavements (including concrete bridge decks) at a wet film thickness of between 0.04 to 0.05 inch 320 to 400 square feet per gallon.

After the primer has "set-up", apply the thermoplastic at temperatures no lower than 375 degrees F nor higher than 425 degrees F at the point of deposition. Apply all extruded thermoplastic markings at the specified width and at a thickness of not less than 0.125 inch nor more than 0.190 inch. Apply all sprayed thermoplastic markings at the specified width and the thickness designated in the contract plans. If the plans do not specify a thickness, apply centerline markings at a wet thickness of 0.090 inch, plus or minus 0.005 inch, and edgeline markings at a wet thickness of 0.060 inch, plus or minus 0.005 inch.

3.3.2.2 Reflective Media

Immediately after installation of the thermoplastic material, mechanically apply drop-on reflective glass spheres conforming to AASHTO M 247 Type 1 at the rate of one pound per 20 square feet such that the spheres are held by and imbedded in the surface of the molten material. Accomplish drop-on application of the glass spheres to ensure even distribution at the specified rate of coverage. If there is a malfunction of either thermoplastic applicator or reflective media dispenser, discontinue operations until deficiency is corrected.

3.3.3 Raised Pavement Markers

Align prefabricated markers carefully at the spacing indicated on the drawings and permanently fix in place by means of epoxy adhesives. To ensure good bond prior to applying adhesive, thoroughly clean all areas where markers are to be set by water blasting and use of compressed air.

3.3.4 Preformed Tape

The pavement surface and ambient air temperature must be a minimum of 60 degrees F and rising. Place the preformed markings in accordance with the manufacturer's written instructions.

3.3.5 Cleanup and Waste Disposal

Keep the worksite clean and free of debris and waste from the removal and application operations. Dispose of debris at approved sites.

3.4 FIELD QUALITY CONTROL

3.4.1 Sampling and Test Reports

Sampling and Testing of Pavement Marking shall be in accordance with the requirements of the PRHTA Specifications, Section 618.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 31 00

FENCES AND GATES

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY CONTROL
 - 1.3.1 Certificates of Compliance
- 1.4 DELIVERY, STORAGE, AND HANDLING
- 1.5 REINFORCED CONCRETE OR REINFORCED CONCRETE BLOCK PRIVACY WALL DESIGN

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
- 2.2 COMPONENTS
 - 2.2.1 Fabric
 - 2.2.1.1 Top and Bottom Selvages
 - 2.2.2 Line Posts
 - 2.2.3 End, Corner, and Pull Posts
 - 2.2.4 Top Rail
 - 2.2.5 Center Rails Between Line Posts
 - 2.2.6 Post-Brace Assembly
 - 2.2.7 Stretcher Bars
 - 2.2.8 Stretcher Bar Bands
 - 2.2.9 Post Tops
 - 2.2.10 Gate Posts
 - 2.2.11 Gates
 - 2.2.12 Gate Hardware and Accessories
 - 2.2.13 Miscellaneous Hardware
 - 2.2.14 Wire Ties
 - 2.2.15 Barbed Wire
 - 2.2.16 Barbed Wire Supporting Arms
 - 2.2.17 Concrete Cap Beam Mounting Hardware
 - 2.2.17.1 Anchor Bolts
 - 2.2.17.2 Anchor Nuts
 - 2.2.17.3 Anchor Washers
 - 2.2.17.4 Anchor Plate Washers
 - 2.2.17.5 Base Plates
- 2.3 MATERIALS
 - 2.3.1 Zinc Coating
 - 2.3.2 Tension Wire
 - 2.3.3 Concrete
 - 2.3.4 Grout
 - 2.3.5 Privacy Screen

PART 3 EXECUTION

- 3.1 PREPARATION
 - 3.1.1 Clearing and Grading
- 3.2 INSTALLATION
 - 3.2.1 Fence Installation
 - 3.2.1.1 Post Spacing
 - 3.2.1.2 Bottom Tension Wire
 - 3.2.2 Excavation
 - 3.2.3 Setting Posts
 - 3.2.3.1 Footings
 - 3.2.3.2 Bracing
 - 3.2.4 Concrete Strength
 - 3.2.5 Top Rails
 - 3.2.6 Center Rails
 - 3.2.7 Brace Assembly
 - 3.2.8 Tension Wire Installation
 - 3.2.9 Fabric Installation
 - 3.2.10 Stretcher Bar Installation
 - 3.2.11 Gate Installation
 - 3.2.12 Tie Wires
 - 3.2.13 Fasteners
 - 3.2.14 Zinc-Coating Repair
 - 3.2.15 Accessories Installation
 - 3.2.15.1 Post Caps
 - 3.2.15.2 Padlocks
 - 3.2.16 Grounding
 - 3.2.17 Barbed Wire and Supporting Arms
 - 3.2.17.1 Supporting Arms
 - 3.2.17.2 Barbed Wire
 - 3.2.18 Mounting Posts at Cap Beam
 - 3.2.19 Tolerances
 - 3.2.20 Privacy Screen
- 3.3 CLOSEOUT ACTIVITIES

-- End of Section Table of Contents --

SECTION 32 31 00

FENCES AND GATES

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318 (2019; R 2022) Building Code Requirements for Structural Concrete (ACI 318-19) and Commentary (ACI 318R-19)

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-22 (2022; Supp 1 2023; Supp 2 2023) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A90/A90M (2013; R 2018) Standard Test Method for Weight (Mass) of Coating on Iron and Steel Articles with Zinc or Zinc-Alloy Coatings

ASTM A116 (2011) Standard Specification for Metallic-Coated, Steel Woven Wire Fence Fabric

ASTM A121 (2013) Standard Specification for Metallic-Coated Carbon Steel Barbed Wire

ASTM A153/A153M (2016a) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A563M (2007; R 2013) Standard Specification for Carbon and Alloy Steel Nuts (Metric)

ASTM A702 (2013) Standard Specification for Steel Fence Posts and Assemblies, Hot Wrought

ASTM A780/A780M (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings

ASTM C94/C94M (2020) Standard Specification for Ready-Mixed Concrete

ASTM F567	(2014a) Standard Practice for Installation of Chain Link Fence
ASTM F626	(2014) Standard Specification for Fence Fittings
ASTM F844	(2007a; R 2013) Washers, Steel, Plain (Flat), Unhardened for General Use
ASTM F1043	(2018) Standard Specification for Strength and Protective Coatings on Steel Industrial Fence Framework
ASTM F1083	(2018) Standard Specification for Pipe, Steel, Hot-Dipped Zinc Coated (Galvanized) Welded, for Fence Structures
ASTM F1554	(2018) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength

INTERNATIONAL CODE COUNCIL (ICC)

ICC PBC	(2018) Puerto Rico Building Code
---------	----------------------------------

THE MASONRY SOCIETY (TMS)

TMS MSJC	(2016) Masonry Standard Joint Committee's (MSJC) Book - Building Code Requirements and Specification for Masonry Structures, Containing TMS 402/ACI 530/ASCE 5, TMS 602/ACI 530.1/ASCE 6, and Companion Commentaries
----------	--

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FS RR-F-191/3	(Rev E; Am 1) Fencing, Wire and Post, Metal (Chain-Link Fence Posts, Top Rails and Braces)
---------------	--

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fence Assembly; G, R0

Location of Gate, Corner, End, and Pull Posts; G, R0

Gate Assembly; G, R0

Gate Hardware and Accessories; G, R0

Erection/Installation Drawings; G, R0

SD-03 Product Data

Fence Assembly; G, R0

Gate Assembly; G, R0

Gate Hardware and Accessories; G, R0

Zinc Coating; G, R0

Fabric; G, R0

Stretcher Bars; G, R0

Concrete; G, R0

Barbed Wire; G, R0

Barbed Wire Supporting Arms; G, R0

Privacy Screen; G, R0

SD-04 Samples

Fabric; G, R0

Braces; G, R0

Line Posts; G, R0

Top Rail; G, R0

Tension Wire; G, R0

Stretcher Bars; G, R0

Gate Posts; G, R0

Gate Hardware and Accessories; G, R0

Wire Ties; G, R0

Privacy Screen; G, R0

SD-05 Design Data

Reinforced Concrete Or Reinforced Concrete Block Privacy Wall; G,
D0

SD-07 Certificates

Certificates of Compliance

SD-08 Manufacturer's Instructions

Fence Assembly

Gate Assembly

Hardware Assembly

Accessories

SD-11 Closeout Submittals

Recycled Material Content

1.3 QUALITY CONTROL

1.3.1 Certificates of Compliance

Submit [certificates of compliance](#) in accordance with the applicable reference standards and descriptions of this section for the following:

- a. Zinc coating
- b. Fabric
- c. Stretcher bars
- d. Gate hardware and accessories
- e. Concrete

1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials to site in an undamaged condition. Store materials off the ground to provide protection against oxidation caused by ground contact.

1.5 REINFORCED CONCRETE OR REINFORCED CONCRETE BLOCK PRIVACY WALL DESIGN

Contractor shall submit the design of the "Reinforced concrete or reinforced concrete block privacy wall in compliance with the latest editions of [ACI 318](#), the Puerto Rico Building Code ([ICC PBC](#)), and [TMS MSJC](#) (if applicable). Design for wind load per [ASCE 7-22](#) (ASCE Wind Design Geodatabase) and earthquake load ($k_h = 0.15$ for unusual load case and $k_h = 0.26$ for extreme load case). Provide design calculations, materials, and drawings of the proposed wall system.

The reinforced concrete or reinforced concrete block privacy wall must be designed by a structural engineer, licensed in the Commonwealth of Puerto Rico, with a minimum of 10 years' experience. The [reinforced concrete or reinforced concrete block privacy wall](#) must be submitted to the Contracting Officer for evaluation and approval.

Provide connection details of reinforced concrete or reinforced concrete block privacy wall to the drilled shaft cap beam and footing details for portions of wall located outside of drilled shaft cap beam limits. Provide overlapping expansion joints where the wall location departs from the drilled shaft cap beam which allow for 8-inches of potential seismic differential movement normal to the wall and 2-inches of differential settlement. Also provide expansion or contraction joints coincident with any expansion or contraction joints in the drilled shaft cap beam.

Provide a reinforced concrete or reinforced concrete block privacy wall

matching the appearance of the existing privacy wall in finish and in color. Include drainage blockouts at the base of the wall of at least 6x12 inches spaced a maximum of 15-feet on-center.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide fencing materials conforming to the requirements of [ASTM A116](#), [ASTM A702](#), [ASTM F626](#).

Submit reports listing chain-link fencing and accessories regarding weight in [ounces](#) for zinc coating.

Submit manufacturer's catalog data for complete [fence assembly](#), [gate assembly](#), [hardware assembly](#) and [accessories](#).

2.2 COMPONENTS

2.2.1 [Fabric](#)

Provide fabric consisting of [No. 9-gage](#) wires woven into a [2 inch](#) diamond mesh, with dimensions of fabric and wire conforming to [ASTM A116](#), with [2.0 ounces per square foot](#) zinc galvanizing.

Provide one-piece fabric widths for fence heights up to [12 feet](#).

2.2.1.1 Top and Bottom Selvages

Provide knuckled selvages at top and bottom for fabric with [2 inch](#) mesh and up to [60 inches](#) high, and if over [60 inches](#) high, provide twisted and barbed top selvege and knuckled bottom selvege.

Provide knuckled top and bottom selvages for [1-3/4 inch](#) and [1 inch](#) mesh fabric.

2.2.2 [Line Posts](#)

Minimum acceptable line posts are as follows:

Up to [6 feet](#) high:

Grade A: [1.900 inch](#) O.D. pipe weighing [2.72 pounds per linear foot](#).

Grade B: [2.375 inch](#) O.D. pipe weighing [3.12 pounds per linear foot](#).

Over [6 feet](#) high:

[2.375 inch](#) O.D. pipe weighing [3.65 pounds per linear foot](#).

2.2.3 End, Corner, and Pull Posts

Provide minimally acceptable end, corner, and pull posts as follows:

Up to [6 feet](#) high:

Grade A: [2.375 inch](#) O.D. pipe weighing [3.65 pounds per linear foot](#).

Grade B: [2.375 inch](#) O.D. pipe weighing [3.12 pounds per linear foot](#).

Over 6 feet high:

Grade A: 2.875 inch O.D. pipe weighing 5.79 pounds per linear foot.

Grade B: 2.875 inch O.D. pipe weighing 4.64 pounds per linear foot.

2.2.4 Top Rail

Provide top rails with a minimum of 1.660 inches O.D. , Grade B weighing 1.82 pounds per linear foot. Provide expansion couplings 6 inches long at each joint in top rails.

2.2.5 Center Rails Between Line Posts

For fencing over 6-feet high, provide 1.660 inches O.D. pipe center rails, Grade B weighing 1.82 pounds per linear foot.

2.2.6 Post-Brace Assembly

Provide horizontal braces at mid-height of fabric consisting of 1.660 inches O.D. pipe Grade A weighing 2.27 pounds per linear foot and 3/8 inch adjustable truss rods and turnbuckles. Provide at end and gate posts and at both sides of corner and pull posts

2.2.7 Stretcher Bars

Provide bars that have one-piece lengths equal to the full height of the fabric with a minimum cross section of 3/16 by 3/4 inch, in accordance with ASTM F626.

2.2.8 Stretcher Bar Bands

Provide bar bands for securing stretcher bars to posts that are steel, wrought iron, or malleable iron spaced not over 15 inches on center. Bands may also be used in conjunction with special fittings for securing rails to posts. Provide bands with projecting edges chamfered or eased.

2.2.9 Post Tops

Provide tops that are steel, wrought iron, or malleable iron designed as a weathertight closure cap. Provide one cap for each post. Provide caps with an opening to permit through passage of the top rail.

2.2.10 Gate Posts

Provide a gate post for supporting each gate leaf as follows:

Up to 6-feet wide:

2.875 inch O.D. pipe Grade A weighing 5.79 pounds per linear foot.

Over 6 feet wide and up to 13 feet wide:

2.875 inch O.D. pipe Grade A weighing 5.79 pounds per linear foot.

Over 13-feet and up to 18-feet wide:

Provide 6.625 inch O.D. pipe weighing 18.97 pounds per linear foot.

2.2.11 Gates

For gate leaves up to 6 feet high or 6 feet wide, provide perimeter gate frames of 1.66 inch O.D. pipe Grade A weighing 2.27 pounds per linear foot.

For gate leaves over 6 feet high or 6 feet wide, provide perimeter gate frames of 1.90 inch O.D. pipe Grade A weighing 2.72 pounds per linear foot.

Provide gate frame assembly that is welded or assembled with special malleable or pressed-steel fittings and rivets to provide rigid connections. Install fabric with stretcher bars at vertical edges; stretcher bars may also be used at top and bottom edges. Attach stretcher bars and fabric to gate frames on all sides at intervals not exceeding 15 inches. Attach hardware with rivets or by other means which provides equal security against breakage or removal.

Provide diagonal cross-bracing, consisting of 3/8 inch diameter adjustable-length truss rods on welded gate frames, where necessary to obtain frame rigidity without sag or twist. Provide nonwelded gate frames with diagonal bracing.

2.2.12 Gate Hardware and Accessories

Provide gate hardware and accessories that conforms to ASTM A116 and ASTM F626, and be as specified:

Provide pressed steel hinges to suit gate size, non-lift-off type, offset to permit 180-degree opening.

Provide latch that permits operation from either side of the gate, with a padlock eye provided as an integral part of the latch.

Provide stops and holders of malleable iron for vehicular gates. Provide stops that automatically engage the gate and hold it in the open position until manually released.

Provide double gates with a cane bolt and ground-set keeper, with latch or locking device and padlock eye designed as an integral part.

2.2.13 Miscellaneous Hardware

Provide miscellaneous hot-dip galvanized hardware as required.

2.2.14 Wire Ties

Provide 9-gage galvanized steel wire for tying fabric to line posts, spaced 12 inches on center. For tying fabric to rails and braces, space wire ties 24 inches on center. For tying fabric to tension wire, space 0.105-inch hog rings 24 inches on center.

Manufacturer's standard procedure will be accepted if of equal strength and durability.

Provide wire ties constructed of the same material as the fencing fabric.

2.2.15 Barbed Wire

ASTM A121 zinc-coated, Type Z, Class 3

2.2.16 Barbed Wire Supporting Arms

Provide barbed wire supporting arms of the 45 degree outward angle 3-strand arm type and of the design required for the post furnished. Secure arms by riveting.

2.2.17 Concrete Cap Beam Mounting Hardware

2.2.17.1 Anchor Bolts

ASTM F1554 Gr 36, Class 1A.

2.2.17.2 Anchor Nuts

ASTM A563M, Grade A, hex style.

2.2.17.3 Anchor Washers

ASTM F844.

2.2.17.4 Anchor Plate Washers

ASTM A36/A36M

2.2.17.5 Base Plates

ASTM A36/A36M.

2.3 MATERIALS

2.3.1 Zinc Coating

Provide hot-dip galvanized (after fabrication) ferrous-metal components and accessories, except as otherwise specified.

Provide zinc coating of weight not less than 1.94 ounces per square foot, as determined from the average result of two specimens, when tested in accordance with ASTM A90/A90M.

Provide zinc coating conforming to the requirements of the following:

- a. Pipe: FS RR-F-191/3 Class 1 Grade A in accordance with ASTM F1083 .
- b. Hardware and accessories: ASTM A153/A153M, Table 1
- c. Surface: ASTM F1043
- d. External: Type B-B surface zinc with organic coating, 0.97 ounce per square foot minimum thickness of acrylated polymer.
- e. Internal: Surface zinc coating of 0.97 ounce per square foot minimum.

Provide galvanizing repair material that is cold-applied zinc-rich coating conforming to ASTM A780/A780M.

2.3.2 Tension Wire

Provide galvanized, coiled spring wire, No. 7-gage. Provide zinc coating that weighs not less than 2.0 ounces per square foot.

2.3.3 Concrete

Provide concrete conforming to ASTM C94/C94M, and obtaining a minimum 28-day compressive strength of 3,000 psi.

2.3.4 Grout

Provide grout of proportions one part portland cement to three parts clean, well-graded sand and a minimum amount of water to produce a workable mix.

2.3.5 Privacy Screen

Provide a green, knitted high-density polyethylene screen with material weight of at least 9.0 ounces per square yard. Edges shall be hemmed at least 1.75 and reinforced with polypropylene webbing or vinyl. Corrosion resistance grommet for attaching the screen to the fence shall be spaced a maximum of 24 inches on center. Screen shall provide at minimum of 75 percent opacity. Submit manufacturer's product data and a sample which includes the same material, hems, and grommets used in the final product.

PART 3 EXECUTION

Submit manufacturer's erection/installation drawings and instructions that detail proper assembly and materials in the design for fence, gate, hardware and accessories.

Provide complete installation conforming to ASTM F567.

3.1 PREPARATION

Ensure final grading and established elevations are complete prior to commencing fence installation.

3.1.1 Clearing and Grading

Clear fence line of trees, brush, and other obstacles to install fencing for a distance of 3 feet inside; and 3 feet outside the fence. Establish a graded, compacted fence line prior to fencing installation.

3.2 INSTALLATION

3.2.1 Fence Installation

Install fence on prepared surfaces to line and grade indicated. Install fence in accordance with fence manufacturer's written installation instructions except as modified herein.

3.2.1.1 Post Spacing

Provide line posts spaced equidistantly apart, not exceeding 10 feet on center. Provide gate posts spaced as necessary for size of gate openings. Do not exceed 500 feet on straight runs between braced posts. Provide corner or pull posts, with bracing in both directions, for changes

in direction of 15 degrees or more, or for abrupt changes in grade.
Submit drawings showing location of gate, corner, end, and pull posts.

3.2.1.2 Bottom Tension Wire

Install bottom tension wires before installing chain-link fabric, and pull wires taut. Place bottom tension wires within 4 inches of respective fabric line.

3.2.2 Excavation

Provide excavations for post footings which are in virgin or compacted soil, of minimum sizes as indicated.

Space footings for line posts 10 feet on center maximum and at closer intervals when indicated, with bottoms of the holes approximately 3 inches below the bottoms of the posts. Set bottom of each post not less than 36 inches below finished grade when in firm, undisturbed soil. Set posts deeper, as required, in soft and problem soils and for heavy, lateral loads.

Uniformly spread soil from excavations adjacent to the fence line or on areas of Government property.

3.2.3 Setting Posts

Remove loose and foreign materials from holes and moisten the soil prior to placing concrete.

Provide tops of footings that are trowel finished and sloped or domed to shed water away from posts. Set hold-open devices, sleeves, and other accessories in concrete.

Keep exposed concrete moist for at least 7 calendar days after placement or cured with a membrane curing material, as approved.

Maintain vertical alignment of posts in concrete construction until concrete has set.

3.2.3.1 Footings

Provide concrete bases of dimensions indicated on the manufacturers installation drawings. Compact concrete to eliminate voids, and finish to a dome shape.

3.2.3.2 Bracing

Brace gate, corner, end, and pull posts to nearest post with a horizontal brace used as a compression member, placed at midheight of fabric with a diagonal tension rod.

3.2.4 Concrete Strength

Provide concrete that has attained at least 75 percent of its minimum 28-day compressive strength, but in no case sooner than 7 calendar days after placement, before rails, tension wire, or fabric are installed. Do not stretch fabric and wires or hang gates until the concrete has attained its full design strength.

Take samples and test concrete to determine strength as specified.

3.2.5 Top Rails

Provide top rails that run continuously through post caps or extension arms, bending to radius for curved runs. Provide expansion couplings as recommended by the fencing manufacturer.

3.2.6 Center Rails

For fencing over 6-feet high, provide single piece center rails between posts set flush with posts on the fabric side, using special offset fittings where necessary.

3.2.7 Brace Assembly

Provide bracing assemblies at end and gate posts and at both sides of corner and pull posts, with the horizontal brace located at midheight of the fabric.

Install brace assemblies so posts are plumb when the diagonal rod is under proper tension.

Provide two complete brace assemblies at corner and pull posts where required for stiffness and as indicated.

3.2.8 Tension Wire Installation

Install tension wire by securing the wire to the fabric with 10-gage ties or clips spaced 24 inches on center.

3.2.9 Fabric Installation

Provide fabric in single lengths between stretch bars with bottom barbs placed approximately 1-1/2 inches above the ground line. Pull fabric taut and tied to posts, rails, and tension wire with wire ties and bands.

Install fabric on the security side of fence, unless otherwise directed.

Ensure fabric remains under tension after the pulling force is released.

3.2.10 Stretcher Bar Installation

Thread stretcher bars through or clamped to fabric 4 inches on center and secured to posts with metal bands spaced 15 inches on center.

3.2.11 Gate Installation

Install gates plumb, level, and secure, with full opening without interference. Install ground set items in concrete for anchorage as recommended by the fence manufacturer. Adjust hardware for smooth operation and lubricated where necessary.

3.2.12 Tie Wires

Provide tie wires that are U-shaped to the pipe diameters to which attached. Twist ends of tie wires not less than two full turns and bent so as not to present a hazard.

3.2.13 Fasteners

Install nuts for tension bands and hardware on the side of the fence opposite the fabric side. Peen ends of bolts to prevent removal of nuts.

3.2.14 Zinc-Coating Repair

Clean and repair galvanized surfaces damaged by welding or abrasion, and cut ends of fabric, or other cut sections with specified galvanizing repair material applied in strict conformance with the manufacturer's printed instructions. Repairs shall be at the Contractor's expense.

3.2.15 Accessories Installation

3.2.15.1 Post Caps

Install post caps as recommended by the manufacturer.

3.2.15.2 Padlocks

Provide padlocks for gate openings and provide chains that are securely attached to gate or gate posts. Provide padlocks keyed alike, and provide two keys for each padlock.

3.2.16 Grounding

Ground fencing as indicated on drawings.

Ground all fences crossed by overhead power lines in excess of 600 volts, and all electrical equipment attached to the fence. Ground fences on each side of all gates, at each corner, at the closest approach to each building located within 50 feet of the fence, and where the fence alignment changes more than 15 degrees. Grounding locations can not exceed 650 feet. Bond each gate panel with a flexible bond strap to its gate post. Ground fences crossed by power lines of 600 volts or more at or near the point of crossing and at distances not exceeding 150 feet on each side of crossing. Provide ground conductor consisting of No. 6 AWG solid copper wire. Provide copper-clad steel rod grounding electrodes 3/4 inch by 10 foot long. Drive electrodes into the earth so that the top of the electrode is at least 6 inches below the grade. Where driving is impracticable, bury electrodes a minimum of 12 inches deep and radially from the fence, with top of the electrode not less than 2 feet or more than 8 feet from the fence. Clamp ground conductor to the fence and electrodes with bronze grounding clamps to create electrical continuity between fence posts, fence fabric, and ground rods. Total resistance of the fence to ground cannot exceed 25 ohms.

3.2.17 Barbed Wire and Supporting Arms

Barbed wire security only to be installed on top of fence in location indicated on the Project Drawings.

3.2.17.1 Supporting Arms

Install barbed wire supporting arms and barbed wire as indicated on the drawings and as recommended by the manufacturer. Anchor supporting arms with 3/8 inch diameter plain pin rivets or, at the Contractor's option, with studs driven by low-velocity explosive-actuated tools for steel, wrought iron, ductile iron, or malleable iron. Do not use

explosive-actuated tool to drive studs into gray iron or other material that can be fractured. Use a minimum of two studs per support arm. Pull barbed wire taut and attach to the arms with clips or other means that will prevent easy removal.

3.2.17.2 Barbed Wire

Install wire on arms at spacing indicated on plans. Pull wire taut to provide a smooth uniform appearance, free from sag. Fasten wire to supporting arms as indicated.

3.2.18 Mounting Posts at Cap Beam

Base plates shall be plumbed by placing a grout shim below the plate and leveling. Do not torque or apply load to the anchors until the full cure time has passed.

3.2.19 Tolerances

Provide posts that are straight and plumb within a vertical tolerance of **1/4 inch** after the fabric has been stretched. Provide fencing and gates that are true to line with no more than **1/2 inch** deviation from the established centerline between line posts. Repair defects as approved at no cost to the Government.

3.2.20 Privacy Screen

Install screen per the manufacturer's instructions.

3.3 CLOSEOUT ACTIVITIES

Remove waste fencing materials and other debris from the work site.

Submit manufacturer's data indicating percentage of **recycled material content** in protective fence materials, including chain link fence, fabric, and gates to verify affirmative procurement compliance.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 92 19

SEEDING

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Stand of Turf
- 1.3 RELATED REQUIREMENTS
- 1.4 SUBMITTALS
- 1.5 DELIVERY, STORAGE, AND HANDLING
 - 1.5.1 Delivery
 - 1.5.1.1 Seed Protection
 - 1.5.1.2 Fertilizer and Lime Delivery
 - 1.5.2 Storage
 - 1.5.2.1 Seed, Fertilizer and Lime Storage
 - 1.5.2.2 Topsoil
 - 1.5.2.3 Handling
- 1.6 TIME RESTRICTIONS AND PLANTING CONDITIONS
 - 1.6.1 Restrictions
- 1.7 TIME LIMITATIONS
 - 1.7.1 Seed

PART 2 PRODUCTS

- 2.1 SEED, FERTILIZER, AND TOPSOIL

PART 3 EXECUTION

- 3.1 PREPARATION
- 3.2 SEEDING
- 3.3 FERTILIZING AND WATERING
- 3.4 PROTECTION OF TURF AREAS
- 3.5 ESTABLISHMENT
- 3.6 INSPECTION AND ACCEPTANCE
- 3.7 RESTORATION

-- End of Section Table of Contents --

SECTION 32 92 19

SEEDING

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

COMMONWEALTH OF PUERTO RICO, DEPARTMENT OF TRANSPORTATION AND
PUBLIC WORKS (PRHTA)

PRHTA Specifications (2005) Standard Specifications for Road
and Bridge Construction

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where PRHTA Standard Specifications, and/or Special Provision(s) is cited that contains references to other PRHTA sections, they shall also form a part of this specification to the extent referenced. Where reference is made to "Highway Authority" or "Authority's Representative" in PRHTA Standard Specifications shall mean "Government" or "Contracting Officer" as defined in the Contract Documents. Where PRHTA Standard Specifications refer to the "Engineer", "Engineer of Tests", or "Division of Tests" it shall be understood to mean the "Contracting Officer" as stated in the Contract Documents.

1.2 DEFINITIONS

1.2.1 Stand of Turf

95 percent ground cover of the established species.

1.3 RELATED REQUIREMENTS

Section **31 00 00** EARTHWORK and Section **32 92 23** SODDING applies to this section for pesticide use and plant establishment requirements, with additions and modifications herein.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section **01 33 00** SUBMITTAL PROCEDURES:

SD-03 Product Data

Fertilizer

SD-06 Test Reports

Topsoil

SD-07 Certificates

Seed

1.5 DELIVERY, STORAGE, AND HANDLING

1.5.1 Delivery

1.5.1.1 Seed Protection

Protect from drying out and from contamination during delivery, on-site storage, and handling.

1.5.1.2 Fertilizer and Lime Delivery

Deliver to the site in original, unopened containers bearing manufacturer's chemical analysis, name, trade name, trademark, and indication of conformance to commonwealth and federal laws. Instead of containers, fertilizer and lime may be furnished in bulk with certificate indicating the above information.

1.5.2 Storage

1.5.2.1 Seed, Fertilizer and Lime Storage

Store in cool, dry locations away from contaminants.

1.5.2.2 Topsoil

Prior to stockpiling topsoil, treat growing vegetation with application of appropriate specified non-selective herbicide. Clear and grub existing vegetation three to four weeks prior to stockpiling topsoil.

1.5.2.3 Handling

Do not drop or dump materials from vehicles.

1.6 TIME RESTRICTIONS AND PLANTING CONDITIONS

1.6.1 Restrictions

Do not plant when the ground is muddy, or when air temperature exceeds 90 degrees Fahrenheit.

1.7 TIME LIMITATIONS

1.7.1 Seed

Apply seed within twenty four hours after seed bed preparation.

PART 2 PRODUCTS

2.1 SEED, FERTILIZER, AND TOPSOIL

All products shall comply with PRHTA Specifications Section 627 Turf Establishment.

Submit product data, samples, test reports, and/or certificates in

accordance with PRHTA Specifications Section 627 Turf Establishment.

PART 3 EXECUTION

3.1 PREPARATION

Seeding preparation shall comply with PRHTA Specifications Section 627 Turf Establishment.

3.2 SEEDING

Seeding shall conform to the requirements of PRHTA Specifications Section 627 Turf Establishment.

3.3 FERTILIZING AND WATERING

Fertilize and water seeded areas in accordance with PRHTA Specifications Section 627 Turf Establishment.

3.4 PROTECTION OF TURF AREAS

Immediately after turfing, protect area against traffic and other use.

3.5 ESTABLISHMENT

Turf will be considered established and completed when the areas to be sodded have produced the required grass species over a minimum of 95 percent of the entire area. Bare areas with diameters larger than 24 inches are to be considered unacceptable and shall be noted for repairs at the Contractor's expense.

3.6 INSPECTION AND ACCEPTANCE

Acceptance inspections of the entire turfed area shall be performed by the Contracting Officer by visual inspections. If inspection determines that turf establishment is not complete, the Contractor shall meet with the Contracting Owner at the job site to identify bare spots, eroded areas and rutting damage and to discuss the Contractor's plan of operation for completing new turf establishment. Prior to acceptance of turfed areas by the Owner, the Contractor shall restore any damaged areas resulting from the Contractor's operations or by natural forces at no additional cost to the Government.

3.7 RESTORATION

Restore to original condition existing turf areas which have been damaged during turf installation operations at the Contractor's expense. Keep clean at all times at least one paved pedestrian access route and one paved vehicular access route to each building. Clean other paving when work in adjacent areas is complete.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 92 23

SODDING

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Stand of Turf
- 1.3 RELATED REQUIREMENTS
- 1.4 SUBMITTALS
- 1.5 DELIVERY, STORAGE, AND HANDLING
 - 1.5.1 Delivery
 - 1.5.1.1 Sod Protection
 - 1.5.1.2 Fertilizer Gypsum Sulfur Iron and Lime Delivery
 - 1.5.2 Storage
 - 1.5.2.1 Sod Storage
 - 1.5.2.2 Topsoil
 - 1.5.2.3 Handling
- 1.6 TIME RESTRICTIONS AND PLANTING CONDITIONS
 - 1.6.1 Restrictions
- 1.7 TIME LIMITATIONS
 - 1.7.1 Sod

PART 2 PRODUCTS

- 2.1 SODS
 - 2.1.1 Classification
 - 2.1.2 Purity
 - 2.1.3 Planting Dates
 - 2.1.4 Composition
- 2.2 TOPSOIL
 - 2.2.1 On-Site Topsoil
 - 2.2.2 Off-Site Topsoil
 - 2.2.3 Composition
- 2.3 FERTILIZER
 - 2.3.1 Granular Fertilizer
- 2.4 WATER

PART 3 EXECUTION

- 3.1 PREPARATION
 - 3.1.1 Extent Of Work
 - 3.1.2 Soil Preparation
- 3.2 SODDING
 - 3.2.1 Finished Grade and Topsoil
 - 3.2.2 Sodding Slopes and Ditches
 - 3.2.3 Finishing
 - 3.2.4 Rolling
 - 3.2.5 Watering

- 3.2.6 Fertilizing
- 3.3 PROTECTION OF TURF AREAS
- 3.4 SOD ESTABLISHMENT PERIOD
 - 3.4.1 Commencement
 - 3.4.2 Satisfactory Stand of Grass Plans
 - 3.4.3 Maintenance During Establishment Period
 - 3.4.3.1 Mowing
 - 3.4.3.2 Post-Fertilization
 - 3.4.3.3 Repair
 - 3.4.3.4 Maintenance Record
- 3.5 INSPECTION AND ACCEPTANCE
- 3.6 RESTORATION

-- End of Section Table of Contents --

SECTION 32 92 23

SODDING

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

TURFGRASS PRODUCERS INTERNATIONAL (TPI)

TPI GSS (1995) Guideline Specifications to
Turfgrass Sodding

COMMONWEALTH OF PUERTO RICO, DEPARTMENT OF TRANSPORTATION AND
PUBLIC WORKS (PRHTA)

PRHTA Specifications (2005) Standard Specifications for Road
and Bridge Construction

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where PRHTA Specifications Section, and/or Special Provision(s) is cited that contains references to other PRHTA sections, they shall also form a part of this specification to the extent referenced. Where reference is made to "Highway Authority" or "Authority's Representative" in PRHTA Specifications shall mean "Government" or "Contracting Officer" as defined in the Contract Documents. Where PRHTA Specifications refer to the "Engineer", "Engineer of Tests", or "Division of Tests" it shall be understood to mean the "Contracting Officer" as stated in the Contract Documents. In case of conflict between the PRHTA specifications and the Specifications in this Section that follows, the Specifications in this Section shall govern.

Paragraphs title "Method of Measurement" and "Basis of Payment" of referenced PRHTA Specifications shall not apply.

1.2 DEFINITIONS

1.2.1 Stand of Turf

100 percent ground cover of the established species.

1.3 RELATED REQUIREMENTS

Section **31 00 00** EARTHWORK applies to this section for pesticide use and plant establishment requirements, with additions and modifications herein.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following

in accordance with Section 01 33 00 SUBMITTAL PROCEDURES. Submittals shall be made according to requirements of this section and in accordance with the requirements of the PRHTA Specifications.

SD-03 Product Data

Fertilizer

Include physical characteristics, and recommendations.

Topsoil composition (reports and recommendations).

SD-07 Certificates

Sod farm certification for sods. Indicate type of sod in accordance with TPI GSS.

1.5 DELIVERY, STORAGE, AND HANDLING

1.5.1 Delivery

1.5.1.1 Sod Protection

Protect from drying out and from contamination during delivery, on-site storage, and handling.

1.5.1.2 Fertilizer Gypsum Sulfur Iron and Lime Delivery

Deliver to the site in original, unopened containers bearing manufacturer's chemical analysis, name, trade name, trademark, and indication of conformance to commonwealth and federal laws. Instead of containers, fertilizer gypsum sulphur iron and lime may be furnished in bulk with certificate indicating the above information.

1.5.2 Storage

1.5.2.1 Sod Storage

Lightly sprinkle with water, cover with moist burlap, straw, or other approved covering; and protect from exposure to wind and direct sunlight until planted. Provide covering that will allow air to circulate so that internal heat will not develop. Do not store sod longer than 24 hours. Do not store directly on concrete or bituminous surfaces.

1.5.2.2 Topsoil

Prior to stockpiling topsoil, treat growing vegetation with application of appropriate specified non-selective herbicide. Clear and grub existing vegetation three to four weeks prior to stockpiling topsoil.

1.5.2.3 Handling

Do not drop or dump materials from vehicles.

1.6 TIME RESTRICTIONS AND PLANTING CONDITIONS

1.6.1 Restrictions

Do not plant when the ground is muddy, or when air temperature exceeds 90

degrees Fahrenheit.

1.7 TIME LIMITATIONS

1.7.1 Sod

Place sod a maximum of thirty six hours after initial harvesting, in accordance with **TPI GSS** as modified herein.

PART 2 PRODUCTS

2.1 SODS

2.1.1 Classification

Nursery grown, certified as classified in the **TPI GSS**. Machine cut sod at a uniform thickness of **3/4 inch** within a tolerance of **1/4 inch**, excluding top growth and thatch. Each individual sod piece shall be strong enough to support its own weight when lifted by the ends. Broken pads, irregularly shaped pieces, and torn or uneven ends will be rejected. Wood pegs and wire staples for anchorage shall be as recommended by sod supplier.

2.1.2 Purity

Sod species shall be genetically pure, free of weeds, pests, and disease.

2.1.3 Planting Dates

Prior to sodding, the contractor shall notify the Contracting officer and receive approval to place the sod in designed areas if weather or soil conditions are favorable for placement.

2.1.4 Composition

Sod will be as indicated in the **PRHTA Specifications** Section 713.

2.2 TOPSOIL

2.2.1 On-Site Topsoil

Surface soil stripped and stockpiled on site and modified as necessary to meet the requirements specified for topsoil in paragraph entitled "Composition." When available topsoil shall be existing surface soil stripped and stockpiled on-site in accordance with Section **31 00 00** EARTHWORK.

2.2.2 Off-Site Topsoil

Conform to requirements specified in paragraph entitled "Composition." Additional topsoil shall be furnished by the Contractor or obtained from topsoil borrow areas indicated by the Contracting Officer.

2.2.3 Composition

Topsoil composition will be as indicated in the **PRHTA Specifications** Section 625 and 628. Topsoil shall be free of sticks, stones, roots, and other debris and objectionable materials.

2.3 FERTILIZER

2.3.1 Granular Fertilizer

Fertilizer shall be as indicated in the PRHTA Specifications Section 628.

2.4 WATER

Source of water shall be approved by Contracting Officer and of suitable quality for irrigation containing no element toxic to plant life.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Extent Of Work

Provide soil preparation (including soil conditioners), fertilizing, and sodding of all newly graded finished earth surfaces, unless indicated otherwise, and at all areas inside or outside the limits of construction that are disturbed by the Contractor's operations. All work shall be according to requirements of this section and in accordance with the requirements of the PRHTA Specifications, Section 628.

3.1.2 Soil Preparation

Provide 4 inches of off-site topsoil or on-site topsoil to meet indicated finish grade. After areas have been brought to indicated finish grade, incorporate fertilizer pH adjusters soil conditioners into soil a minimum depth of 4 inches by disking, harrowing, tilling or other method approved by the Contracting Officer. Remove debris and stones larger than 3/4 inch in any dimension remaining on the surface after finish grading. Correct irregularities in finish surfaces to eliminate depressions. Protect finished topsoil areas from damage by vehicular or pedestrian traffic.

3.2 SODDING

3.2.1 Finished Grade and Topsoil

Prior to the commencement of the sodding operation, the Contractor shall verify that finished grades are as indicated on drawings; the placing of topsoil, smooth grading, and compaction requirements have been completed in accordance with Section 31 00 00 EARTHWORK.

The prepared surface shall be a maximum 1 inch below the adjoining grade of any surfaced area. New surfaces shall be blended to existing areas. The prepared surface shall be completed with a light raking to remove from the surface debris and stones over a minimum 5/8 inch in any dimension.

3.2.2 Sodding Slopes and Ditches

For slopes 2:1 and greater, lay sod with long edge perpendicular to the contour. For V-ditches and flat-bottomed ditches, lay sod with long edge perpendicular to flow of water. Anchor each piece of sod with wood pegs or wire staples maximum 2 feet on center. On slope areas, start sodding at bottom of the slope.

3.2.3 Finishing

After completing sodding, blend edges of sodded area smoothly into surrounding area. Air pockets shall be eliminated and a true and even surface shall be provided. Frayed edges shall be trimmed and holes and missing corners shall be patched with sod.

3.2.4 Rolling

Immediately after sodding, firm entire area except for slopes in excess of 3 to 1 with a roller not exceeding 90 pounds for each foot of roller width.

3.2.5 Watering

Start watering areas sodded as required by daily temperature and wind conditions. Apply water at a rate sufficient to ensure thorough wetting of soil to minimum depth of 6 inches. Run-off, puddling, and wilting shall be prevented. Unless otherwise directed, watering trucks shall not be driven over turf areas. Watering of other adjacent areas or plant material shall be prevented.

3.2.6 Fertilizing

Fertilize sodded areas in accordance with PRHTA Specifications Section 627 Turf Establishment.

3.3 PROTECTION OF TURF AREAS

Immediately after turfing, protect area against traffic and other use.

3.4 SOD ESTABLISHMENT PERIOD

3.4.1 Commencement

The sod establishment period to obtain a healthy stand of grass plants shall begin on the first day of work under this contract and shall end 3 months after the last day of sodding operation. Written calendar time period shall be furnished for the sod establishment period. When there is more than 1 sod establishment period, the boundaries of the sodded area covered for each period shall be described and provided by the Contractor. The sod establishment period shall be modified for inclement weather, shut down periods, or for separate completion dates of areas.

3.4.2 Satisfactory Stand of Grass Plans

Grass plants shall be evaluated for species and health. A satisfactory stand of grass plants from the sodding operation shall be living sod uniform in color and leaf texture. Bare spots shall be a maximum 2 inch square. Joints between sod pieces shall be tight and free from weeds and other undesirable growth.

3.4.3 Maintenance During Establishment Period

Maintenance of the sodded areas shall include eradicating weeds, insects and diseases; protecting embankments and ditches from surface erosion; maintaining erosion control materials and mulch; protecting installed areas from traffic; mowing; watering; and post-fertilization. Weeds shall not exceed 10 percent of established grass, as measured by surface area, at the conclusion of the establishment period.

3.4.3.1 Mowing

Mowing shall not begin until the grass has produced seed. Areas shall be mowed to a range of 4 to 8 inches in height. Frequency of mowing shall be as necessary to maintain this height range. Clippings shall be removed when the amount cut prevents sunlight from reaching the ground surface. The Contractor shall be responsible for determining if proper maintenance requires more frequent mowing.

3.4.3.2 Post-Fertilization

The application rate shall be 275 pounds per acre. A maximum 1/2 pound per 1000 square feet of actual available nitrogen shall be provided to the grass plants. The application shall be timed prior to the advent of winter dormancy and shall be made without burning the installed grass plants.

3.4.3.3 Repair

Unsatisfactory stand of grass plants shall be repaired or reinstalled, and eroded areas shall be repaired in accordance with paragraph PREPARATION.

3.4.3.4 Maintenance Record

A record of each site visit shall be furnished which describes the maintenance work performed; areas repaired or reinstalled; and diagnosis for unsatisfactory stand of grass plants.

3.5 INSPECTION AND ACCEPTANCE

Acceptance inspections of the entire turfed area shall be performed by the Contracting Officer by visual inspections. If inspection determines that turf establishment is not complete, the Contractor shall meet with the Contracting Owner at the job site to identify bare spots, eroded areas and rutting damage and to discuss the Contractor's plan of operation for completing new turf establishment. Prior to acceptance of turfed areas by the Owner, the Contractor shall restore any damaged areas resulting from the Contractor's operations or by natural forces at no additional cost to the Government.

3.6 RESTORATION

Restore to original condition existing turf areas which have been damaged during turf installation operations. Keep clean at all times at least one paved pedestrian access route and one paved vehicular access route to each building. Clean other paving when work in adjacent areas is complete.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 05 07

PILOT TUBE GUIDED BORING AND JACKING

PART 1 GENERAL

- 1.1 SCOPE
- 1.2 REFERENCES
- 1.3 SUBMITTALS
- 1.4 DESIGN AND PERFORMANCE REQUIREMENTS
 - 1.4.1 System Components
 - 1.4.2 Two Phase System
 - 1.4.3 Driving and Receiving Coffercells
 - 1.4.4 Grade and Alignment
 - 1.4.5 Tunneling Overcut
 - 1.4.6 Ground Movement
- 1.5 QUALITY ASSURANCE
 - 1.5.1 GUIDED BORING AND JACKING PLAN
 - 1.5.2 Pipe Casing
- 1.6 DELIVERY, STORAGE, AND HANDLING
- 1.7 WARRANTY

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Casing Pipe
- 2.2 ANNULAR SPACE GROUT

PART 3 EXECUTION

- 3.1 METHOD OF CONSTRUCTION
 - 3.1.1 Protection of Existing Features
 - 3.1.2 Excavation
 - 3.1.2.1 Ventilation
 - 3.1.3 Tolerances
 - 3.1.4 End Closures/Bulkheads and Grouting of Casing Pipe
- 3.2 GROUND MONITORING
- 3.3 FIELD TESTING AND INSPECTION
- 3.4 FINAL CLEANUP

-- End of Section Table of Contents --

SECTION 33 05 07

PILOT TUBE GUIDED BORING AND JACKING

PART 1 GENERAL

1.1 SCOPE

The work provided for herein consists of furnishing all plant, equipment, labor and materials; performing all operations required for designing, furnishing, installing, and operating a system to install the sanitary sewer siphon pipes below the channel limits and between the drilled shaft walls. Pilot tube guided boring shall be used where indicated on the plans to install steel casing pipe for ductile iron sanitary sewer pipes which require high accuracy in line and grade. Guided boring shall employ augers for excavation and soil removal and a jacking system for pushing the casing and ductile iron pipes.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A139/A139M (2016) Standard Specification for Electric-Fusion (ARC)-Welded Steel Pipe (NPS 4 and over)

ASTM C33/C33M (2018) Standard Specification for Concrete Aggregates

ASTM C150/C150M (2020) Standard Specification for Portland Cement

U.S. ARMY CORPS OF ENGINEERS (USACE)

GDR Section 00 31 32, Geotechnical Data Report
Rio Puerto Nuevo Supplemental Contract 3

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. **Submittals with a "PR" designation shall also be submitted to the PRASA for review and approval.** Submit the following in accordance with Section **01 33 00** SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Guided Boring and Jacking Plan; G, D0, PR

SD-06 Test Reports

Pilot Guide And Driven Casing Test; G D0, PR

SD-07 Certificates

Certified Of Compliance

SD-11 Closeout Submittals

Driving Records

1.4 DESIGN AND PERFORMANCE REQUIREMENTS

The system utilized for the pilot tube guide boring machine shall be a two phase system as described below and shall consist but not be limited to the major components listed below.

The Contractor shall examine the geotechnical report and borings provided in section 00 31 32 GEOTECHNICAL DATA REPORT and obtain additional information as required to assure that the system provided will be capable of operating successfully given groundwater conditions, soil type, potential for obstructions and all geotechnical parameters pertinent to this type of tunneling. Geotechnical information shall be relied upon by the contractor for determination of means and methods used by contractor for bidding purposes. The Contractor may elect to perform additional borings at the location of the planned pipes.

When using this method of construction, the Contractor shall be responsible for the orderly and timely progression of work. If obstructions with a maximum lateral dimension of less than one foot are encountered, the Contractor shall be capable of advancing past said obstructions. No additional compensation shall be provided for such work.

Design of coffercells, excavations and all other features required to complete pipe installation is provided as part of specification 31 52 10 TEMPORARY RETAINING STRUCTURES.

1.4.1 System Components

a. Line and Grade Control System - The control system shall include but not be limited to a theodolite, lighted target, camera, and monitor screen. The lighted target shall consist of LEDs arrayed to compose two circle and one line from the center. The lighted target shall be capable of battery operation of at lasting a minimum of 10 days when charged.

b. Steering Head and Auger - The guided boring system shall employ a steering head connected to pilot tubes. Steering head shall allow for horizontal and vertical adjustment to match required grades. After initial pilot tubes are installed, utilize a reamer and auger casing with augers inside to enlarge the borehole and to allow for the product pipe to be installed.

c. Jacking Frame - The jacking frame shall possess adequate strength to advance the pilot tube, the casing and the string of product pipe from the drive coffercell to the receiver coffercell. The jacking force shall be easily regulated down to the safe working load rating of the pipe. The frame shall develop a uniform distribution of jacking forces on the end of the pipe. The auger motor shall possess adequate torque to steer the pilot tube and adequate torque and speed to effectively auger the excavated material from the face of the bore to the drive coffercell.

d. Pilot Tube - The pilot tubes shall be constructed of steel in rigid but short sections to accommodate the small drive and receiver coffercells. The tubes shall rigidly connect to each other, the steering tip and the casing and have a clear inside diameter large enough to adequately view the lighted target. The tubes shall withstand the torque encountered in the steering process.

e. Casing - The casing shall be constructed of steel to a diameter as shown in the plans and have a leading connection compatible with the pilot tube. The leading face of the casing shall possess several large openings for the soil to enter as it advances along the proposed alignment. An internal auger chamber shall funnel the excavated material into the temporary full diameter casings of the internal auger casings of the Two-Phase Process.

Structural members shall connect the leading edge of the casing to the pilot tube connections.

f. Soil Transportation System - The soil transportation system shall consist of an auger train operating inside the full diameter temporary steel casings and an internal casing and auger train operating inside the product pipe. The internal casings of the Two-Phase Process shall be manufactured to minimize leakage of the excavated material into the product pipe.

g. Groundwater Control - The Contractor shall provide means and methods to improve the soils to counter balance groundwater discharge at the entrance and exit rings and at the front of the casing installation.

h. Soil Removal - A soil removal system shall be provided to safely remove the excavated material from the drive coffercell to the surface.

i. Hydraulic Power Unit - The hydraulic power unit can rest on the surface and be connected to the jacking frame by hoses. The unit shall meet all applicable noise standards.

j. Lubrication System - A lubrication system can be employed to minimize pipe friction to insure that pipe can be installed from the drive coffercell to the receiver coffercell within the safe working load rating of the pipe. The system may also be required to minimize the torque required to transport the excavated material to the drive coffercell.

k. Monitoring equipment shall be provided capable of continuously monitoring:

- The jacking pressure and advancement of the boring head.
- Deviation of the pilot tube microtunneling machine.
- Indication of the steering head.

1.4.2 Two Phase System

Phase 1 - A rigid steel pilot tube in approximately 3-foot lengths shall be installed through the ground from the drive coffercell to the receiver coffercell by earth displacement with the microtunnel-jacking frame. The alignment of the pilot tube shall be established with a theodolite mounted at the rear of the drive coffercell and accurately set to the desired line and grade. The theodolite shall view a lighted target in the lead or steering pilot tube. A camera shall be fitted to the theodolite and shall transmit the image of the crosshair and the target onto a monitor screen to be viewed in the drive coffercell by the operator. As the operator advances the pilot tube through the earth the center of the target will drift from the crosshair as a result of the biased or slanted leading tip of the pilot tube. The operator shall rotate the pilot tube as required to orient the slanted steering tip toward the crosshair and continue to advance the pilot tube until it reaches the receiver coffercell.

Phase 2 - A casing shall be rigidly connected to the final pilot tube and advanced into the earth behind the pilot tube. An auger shall be used inside the casing to remove the material being excavated. The auger shall be contained inside the limits of the casing as it progresses along the proposed alignment. The pilot tubes shall be recovered in the receiver coffercell as the casings are installed. The ductile iron pipe shall then be installed directly inside the casing pipe with the jacking frame. The augers shall be recovered in the receiver coffercell as the product pipe is installed. Careful sequencing of the installation of the wall thimble sleeves is required for installation of the casing ends through the sleeves of both concrete walls.

1.4.3 Driving and Receiving Coffercells

Coffercell size shall be minimized however shall be commensurate with safe working practices. Coffercells shall be located as shown on the plans and as approve by the Contracting Officer. The Contractor shall provide equipment as required to keep the coffercells free of flowing or standing water while in use. The groundwater control system shall not result in a loss of fines for the soil surrounding the coffercell. The Contractor shall assess during the design phase if the lowering of the groundwater in the vicinity of the coffercells will have unacceptable impacts and if true, shall preclude the lowering of the groundwater table. Installation of horizontal drains are an acceptable method to control groundwater seepage. Groundwater may be disposed of into the combined sewer or as approved by Contracting Officer. Design and construction technique for drive and receiving coffercells shall be the responsibility of the Contractor per specification 31 52 10 TEMPORARY RETAINING STRUCTURES and, shall be in compliance with all applicable requirements. The Government shall assume no responsibility for coffercell design and or construction. Coffercell design shall include all devices required for safe working conditions and shall be in compliance with OSHA and Commonwealth requirements. The Contractor shall make provisions to prevent groundwater, if present, from flowing along the pipe into the coffercell and prevent pipe lubricant from escaping into the coffercell.

1.4.4 Grade and Alignment

Acceptance criteria for the installed casing pipe shall be within 0.10 foot horizontally and vertically from the locations and grades shown on the plans at the entry and exit rings, such that it will have continuity through the thimble sleeve locations. Excessive steering should be avoided. Steering correction when required should not exceed 1 inch in 250 inches.

1.4.5 Tunneling Overcut

The overcut of the casing shall be limited to no more than 1 inch of the diameter of the product pipe to be installed.

1.4.6 Ground Movement

- a. The machine shall provide full support to the tunnel without the use of ground stabilization or other ground support techniques to allow the installation of the sewer line.
- b. Drive and receiving coffercells shall be designed in a manner to preclude settlement of the adjacent areas.
- c. Repair of ground losses and or damage to structures due to ground losses that are a result of the pilot tube microtunneling operation shall be the responsibility of the Contractor.
- d. Repair of voids created during the pilot tube microtunneling operation shall be the responsibility of the Contractor. A plan of repair shall be prepared by the Contractor for review and approval by the Government.

1.5 QUALITY ASSURANCE

1.5.1 GUIDED BORING AND JACKING PLAN

Submit for review complete working drawings showing details of the proposed method of construction and the sequence of operations to be performed during construction. Show the method of guided boring, including the guided boring system to be used, location of working coffercells, including method of excavation, shoring and bracing, and de-watering techniques that are proposed to be used. The following is not intended to limit, but to provide the minimum of, details which must be included:

- a. Quality control plan that includes at a minimum how supervision of installation shall be accomplished and procedures for assuring line and grade requirements are met. Include forms to be used to determine work progress.
- b. Manufacturer's literature describing in detail the tunneling system to be used. Include the method of controlling line and grade of the Pilot Tube Guided Boring Machine and details of the pipe installation. Detailed description of projects on which this system has been successful used, including the names, addresses and telephone numbers of owner's representative for these projects.
- c. Method of spoils removal including surface storage and disposal

- d. Method of controlling groundwater at the coffercells and at the front of casing excavation by the tunneling machine. Extents of soil improvement and/or dewatering wells shall be designed, signed, and sealed by a geotechnical engineer registered in the Commonwealth of Puerto Rico.
- e. Schedule of work.
- f. The manner in which the non-metallic spacers shall be installed and annular grout materials and method of placement, description of equipment used and grout pressures employed..
- g. Manufacturer's literature on type of steel pipe to be used. Pipe submittal should include Manufacturer's literature describing the casing pipe, indicating allowable safe jacking loads with the designated safety factor. Anticipated jacking loads include the manner in which the anticipated jacking loads were generated. Provide details of pipe compression rings to safely distribute jacking forces and instructions on storage, handling, transporting and installation requirements.
- h. Contractors Contingency Plan
- i. Miscellaneous materials to be used on the job.
- j. Plan for monitoring ground movement and action plan in case of potential settlement.

1.5.2 Pipe Casing

Quality assurance procedures shall be performed by the pipe manufacturer fully in accordance with the requirements of this specification. The certification shall include certified laboratory data confirming that said tests have been performed on a sample of the pipe to be provided under this contract, or pipe from that production run, and that satisfactory results were obtained.

The Contractor shall furnish a [certified of compliance](#) affidavit for all pipe and fittings furnished confirming that the materials supplied fully conform to the requirements specified herein.

1.6 DELIVERY, STORAGE, AND HANDLING

The Contractor shall exercise special care during the unloading, handling, and storage of all pipes to ensure that the steel or ductile iron pipes are not cracked, gouged, scored or other- wise damaged. Any pipe segment which has cracks in the pipe wall shall be removed from the site at the Contractor's cost. The pipe shall be stored in a way that does not damage the pipe.

1.7 WARRANTY

The Contractor shall provide to the Government a warranty to be in force and effect for a period of one (1) year from the date of final acceptance by the Government. The warranty shall require the Contractor to repair or replace the pipe should leakage, separation, collapse or other failure result from faulty materials or installation as determined by the Contracting Officer.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Casing Pipe

Casing pipes used for guided boring shall be in conformance with [ASTM A139/A139M](#), Grade B and must be capable of withstanding all forces imposed upon them during the construction phase as well as the final in-place loading conditions. All pipe must be able to withstand a compressive loading greater than the jacking load anticipated on this project.

Casing pipe of the thickness shown on the plans or larger shall have an allowable safe jacking load, with a minimum safety factor of 1.5.

The driving ends of the pipe and intermediate points must be protected against damage. The detailed method proposed to cushion and distribute the jacking force at the joint is subject to approval by the Contracting Officer. Any pipe showing signs of failure may be required to be jacked through to the reception coffercell and removed. Other methods of pipe repair may be used subject to approval of the Contracting Officer.

The pipe manufacturer shall be designated at the time of the bid. Any subsequent change of pipe manufacturer must be approved by the Contracting Officer in writing. A record of experience and product information shall be provided by the Contractor at the time of the bid.

The joints shall be welded one-side bevel of full penetration such that there is no resulting roughness of the inside of the pipe. The welding shall conform to the requirements of [AWS D1.1/D1.1M](#). Non-destructive testing (NDT) or ultrasonic testing (UT) shall be performed on 100-percent of welds. Submit welder qualifications per Specification Section [05 50 13 MISCELLANEOUS METAL FABRICATIONS](#). The casing pipe shall be sound and free from other imperfections that would impair their installation or use. The base metal distortion shall be minimized to facilitate compliance with specified tolerances.

The allowable jacking strength capacity of pipe shall be capable of withstanding the maximum jacking forces imposed by the operations. The allowable jacking capacity of the pipe shall be 1.5 times greater than the maximum jacking forces imposed by jacking operations as identified by theoretical calculations.

Submittals shall include shop drawings and calculation of jacking strength of the pipe. The drawings shall show dimensions of pipes including inside diameter and wall thickness, details of pipe joints and gaskets showing cushion packing ring and laying length of each pipe.

Spacers and centralizers for the product ductile iron pipe shall be non-metallic, virgin polypropylene, and installed as shown on the plans.

Deviations from a constant line and grade of the jacked pipe shall ensure induced point loading from jacking operations does not exceed the capacity of the pipe

2.2 ANNULAR SPACE GROUT

Provide cement grout for pressure grouting to fill the voids around the

casing and for filling the interior annular space between carrier pipe and the casing composed of Portland cement conforming to [ASTM C150/C150M](#), Type II, and sand meeting requirements of [ASTM C33/C33M](#) for fine aggregate, sufficiently fluid to inject through the casing and fill voids, with prompt setting to control grout flow. Utilize a grout with a minimum compressive strength of [100 psi](#) attained within 24 hours. Admixtures are to be free of chlorides, corrosive or other material detrimental to the materials the grout contacts.

PART 3 EXECUTION

3.1 METHOD OF CONSTRUCTION

Auger borings will be coordinated with the methods of construction for the coffercells, concrete wall, or other components of the construction such as to ensure the safety of the work, Contractor's employees, the public and adjacent property, whether public or private. All damage to property shall be restored to equal or better condition than prior to construction.

Blasting will not be permitted.

The tunnel machine operator shall be fully trained on other tunneling projects on the use of the machinery on this project or at all times be under the direct supervision of a fully trained and designated operator.

The machine shall be operated so as to prevent either surface heave or loss of ground during tunneling and shall be steerable to maintain line and grade within the tolerances specified.

The thrust reaction backstop shall be properly designed and constructed. The backstop shall be normal to the proposed pipe alignment. The thrust wall shall be designed to support the maximum obtainable jacking pressure developed by the main jacking system and shall include a safety factor. Failure of the thrust block system and any corrective action required shall be the responsibility of the Contractor.

The pipe shall be jacked in place without damaging the pipe joints or completed pipe section. Any pipe, which has been damaged during installation, shall be replaced by the Contractor at no additional cost to the Government. A written record of each drive is to be kept. This record shall include the jacking force and drive length of each individual drive. Submit [driving records](#) to the Contracting Officer.

The jacking system shall develop a uniform distribution of jacking forces on the end of the pipe. If less than the full jacking surface of the pipe is used, the maximum allowable jacking forces shall be decreased by the proportional amount. The jacking force shall be applied to the jacking surface of the pipe in a way that eliminates misalignment, distortion, and eccentricity.

The jacking force applied to the pipe shall at no time exceed that allowed by the pipe manufacturer with a Safety Factor of 1.5.

3.1.1 Protection of Existing Features

Before beginning construction at any location, the Contractor must adequately protect existing structures, utilities, trees, shrubs and other permanent objects. The repair for damage to permanent facilities due negligence for lack of adequate protection on the part of the Contractor

will be at no cost to the Government.

The Contractor shall conduct his operations in such a fashion that trucks and other equipment does not create a dirt nuisance in the streets. The Contractor shall immediately remove and dispose of any spillage or excess dirt on the roadway.

3.1.2 Excavation

Selection of an excavation technique, excavation support system, groundwater controls, and dewatering system is the contractor's responsibility. The Contractor is responsible to ensure all excavations are in compliance with all Federal, State and Local regulations.

All work of excavation, shoring and bracing and tunneling shall be so executed that settlement is minimized.

All excavated material from the tunnel and coffercell construction shall be disposed of away from the construction site. No stockpiling of material on the job-site will be permitted. Material shall be removed at regular intervals not exceeding 48 hours.

3.1.2.1 Ventilation

All excavations shall be adequately ventilated. Air monitoring of the coffercells or pits shall be conducted on a continuous basis. All coffercells and pits shall conform to applicable Trench Safety Standards and OSHA excavation, trenching, and shoring standards, which are contained in the Code of Federal Regulations 29 (CFR) 1926.650-1926.653.

Coffercells and pits shall be adequately ventilated. Air monitoring of the coffercells or pits shall be conducted on a continuous basis. Threshold limits of the gas concentrations monitored shall be:

Carbon Monoxide < 0.005%
Methane < 0.25%
Hydrogen Sulfide < 0.001%
Oxygen > 20.0%

3.1.3 Tolerances

The maximum deviation from line and grade shall be 0.10 foot. When the excavation is offline or grade, return to the plan line/grade at a rate of not more than 1 inch per 25 feet. If the pipe deviates sufficiently off plan line and/or grade to require a redesign of the sewer or appurtenances, the Contractor shall have the system redesigned at no cost to the Government.

Submit pilot guide and driven casing test results and alignment showing that the tolerances are within the limits specified herein to the Contracting Officer.

3.1.4 End Closures/Bulkheads and Grouting of Casing Pipe

- a. Closures: Seal ends of casing with brick and mortar.
- b. After installing, inspecting and acceptance of the carrier pipe and spacers within the casing pipe, pressure fill the annular space between the carrier pipe and the casing pipe, with cement grout

specified herein. Regulate pump pressures in accordance with those approved in the Guided Boring and Jacking Plan. Place grout in a sequence and manner that will preclude voids or pockets of entrapped air or water. Use a refusal pressure equal to 0.5 psf per foot of overburden.

3.2 GROUND MONITORING

The Contractor shall monitor all ground movements associated with the work and maintain these within permissible tolerances. A baseline survey shall be prepared by the Contractor prior to undertaking the tunneling work. This work shall be coordinated in conjunction with vibration monitoring as described in Section 02 22 13 VIBRATION AND NOISE MONITORING REQUIREMENTS.

Contractor shall provide generalized plans of action to be implemented in the event of any potential settlement. The plans of action shall be positive measures by the Contractor to control settlement (e.g. using alternative tunneling methods).

While available geotechnical data is supplied within the GDR, choice of cutting heads and resulting efficiency of the guided boring operation will be the Contractor's responsibility.

3.3 FIELD TESTING AND INSPECTION

The Contracting Officer will conduct field inspections and witness field tests specified in this section. The Contractor shall perform field tests and provide labor, equipment, and incidentals required for testing, and be able to produce evidence, when required, that each item of work has been constructed in accordance with the drawings and specifications.

3.4 FINAL CLEANUP

Prior to final acceptance and final inspection of the pipe by the Contracting Officer, the Contractor shall flush and clean all parts of the system by removing all accumulated construction debris, rocks, gravel, sand, silt, and other foreign material from the pipe.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 11 00

WATER UTILITY DISTRIBUTION PIPING

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Water Transmission Mains
 - 1.2.2 Water Mains
 - 1.2.3 Water Service Lines
 - 1.2.4 Additional Definitions
- 1.3 SUBMITTALS
- 1.4 QUALITY CONTROL
 - 1.4.1 Regulatory Requirements
 - 1.4.2 QUALITY ASSURANCE
- 1.5 DELIVERY, STORAGE, AND HANDLING
 - 1.5.1 Delivery and Storage
 - 1.5.2 Handling

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Pipe, Fittings, Joints And Couplings
 - 2.1.1.1 Ductile-Iron Piping
 - 2.1.1.1.1 Ductile Iron Pipe and Fittings
 - 2.1.1.1.1.1 Wall Penetrations
 - 2.1.1.1.2 Coating And Linings
 - 2.1.1.1.3 Identification
 - 2.1.1.1.4 Joints and Jointing Material
 - 2.1.2 Valves
 - 2.1.2.1 Butterfly Valves
 - 2.1.2.1.1 Materials
 - 2.1.2.1.2 Operators
 - 2.1.2.1.3 Painting
 - 2.1.2.1.4 Maintenance
 - 2.1.2.2 Air Release, Air/Vacuum, and Combination Air Valves
 - 2.1.2.2.1 Connections
 - 2.1.2.2.2 Design
 - 2.1.2.2.3 Materials
 - 2.1.2.2.4 Cross Contamination and Security Protection
 - 2.1.2.2.5 Manufacturer
 - 2.1.2.3 Gate Valves
 - 2.1.2.3.1 Maintenance
 - 2.1.3 Valve Appurtenances
 - 2.1.3.1 Surface Preparation And Shop Painting
 - 2.1.4 Disinfection
- 2.2 ACCESSORIES
 - 2.2.1 Pipe Restraint
 - 2.2.2 Tapping Sleeves

- 2.2.3 Sleeve-Type Mechanical Couplings
- 2.2.4 Detectable Marking Tape

PART 3 EXECUTION

- 3.1 PREPARATION
 - 3.1.1 Connections to Existing System
 - 3.1.2 Operation of Existing Valves
 - 3.1.3 Earthwork
- 3.2 INSTALLATION
 - 3.2.1 Piping
 - 3.2.1.1 General Requirements
 - 3.2.1.1.1 Termination of Water Lines
 - 3.2.1.1.2 Pipe Laying and Jointing
 - 3.2.1.1.3 Sewer Manholes
 - 3.2.1.1.4 Water Piping Crossing Sewer Piping
 - 3.2.1.1.5 Penetrations
 - 3.2.1.1.6 Flanged Pipe
 - 3.2.1.2 Ductile-Iron Piping
 - 3.2.1.2.1 Buried Piping
 - 3.2.1.2.2 Detectable Marking Tape
 - 3.2.1.2.3 Installation Above Grade
 - 3.2.1.2.4 Painting
 - 3.2.2 Valves
 - 3.2.2.1 Field Test and Adjustments
- 3.3 FIELD TESTING AND DISINFECTION
 - 3.3.1 General
 - 3.3.2 Flushing
 - 3.3.3 Testing of Pressurized Piping System
 - 3.3.4 Disinfection
- 3.4 MANUFACTURER'S SERVICES
- 3.5 SYSTEM STARTUP
- 3.6 CLEANUP

ATTACHMENTS:

Example Detectable Marking Tape

Maximum Deflection Diagram

-- End of Section Table of Contents --

SECTION 33 11 00

WATER UTILITY DISTRIBUTION PIPING

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA B300	(2018) Hypochlorites
AWWA B301	(2010) Liquid Chlorine
AWWA C104/A21.4	(2016) Cement-Mortar Lining for Ductile-Iron Pipe and Fittings for Water
AWWA C105/A21.5	(2018) Polyethylene Encasement for Ductile-Iron Pipe Systems
AWWA C110/A21.10	(2012) Ductile-Iron and Gray-Iron Fittings for Water
AWWA C111/A21.11	(2017) Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings
AWWA C115/A21.15	(2020) Flanged Ductile-Iron Pipe With Ductile-Iron or Gray-Iron Threaded Flanges
AWWA C150/A21.50	(2014) Thickness Design of Ductile-Iron Pipe
AWWA C151/A21.51	(2017) Ductile-Iron Pipe, Centrifugally Cast
AWWA C153/A21.53	(2019) Ductile-Iron Compact Fittings for Water Service
AWWA C219	(2017) Bolted Sleeve-Type Couplings for Plain-End Pipe
AWWA C504	(2015) Standard for Rubber-Seated Butterfly Valves
AWWA C512	(2015) Air-Release, Air/Vacuum, and Combination Air Valves for Water and Wastewater Service
AWWA C515	(2015) Reduced-Wall, Resilient-Seated Gate Valves for Water Supply Service
AWWA C550	(2017) Protective Interior Coatings for

Valves and Hydrants

AWWA C600	(2017) Installation of Ductile-Iron Mains and Their Appurtenances
AWWA C601	(1981) Standard for Disinfecting Water Mains
AWWA C606	(2015) Grooved and Shouldered Joints
AWWA C651	(2014) Standard for Disinfecting Water Mains
AWWA C800	(2014) Underground Service Line Valves and Fittings
AWWA M41	(2009; 3rd Ed) Ductile-Iron Pipe and Fittings

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.1	(2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250
------------	---

ASTM INTERNATIONAL (ASTM)

ASTM A126	(2004; R 2019) Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings
ASTM A276/A276M	(2017) Standard Specification for Stainless Steel Bars and Shapes
ASTM A307	(2014; E 2017) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A354	(2017; E 2017; E 2018) Standard Specification for Quenched and Tempered Alloy Steel Bolts, Studs, and Other Externally Threaded Fasteners
ASTM A536	(1984; R 2019; E 2019) Standard Specification for Ductile Iron Castings
ASTM A564/A564M	(2019) Standard Specification for Hot-Rolled and Cold-Finished Age-Hardening Stainless Steel Bars and Shapes
ASTM B62	(2017) Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM D2000	(2018) Standard Classification System for Rubber Products in Automotive Applications

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 9001	(2015) Quality Management Systems-Requirements
----------	--

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS
INDUSTRY (MSS)

MSS SP-25 (2018) Standard Marking System for Valves,
Fittings, Flanges and Unions

NSF INTERNATIONAL (NSF)

NSF 372 (2016) Drinking Water System Components -
Lead Content

NSF/ANSI 14 (2019) Plastics Piping System Components
and Related Materials

NSF/ANSI 61 (2020) Drinking Water System Components -
Health Effects

Puerto Rico Aqueduct and Sewer Authority (PRASA)

PRASA Specifications Puerto Rico Aqueduct and Sewer Authority
Standard Specifications

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where PRASA Specifications Section, and/or Special Provision(s) is cited that contains references to other PRASA sections, they shall also form a part of this specification to the extent referenced. Where reference is made to "authorized representative of P.R.A.S.A." in PRASA Specifications shall mean "Government" or "Contracting Officer" as defined in the Contract Documents. Where PRASA Specifications refer to the "Engineer", "Engineer of Tests", or "Division of Tests" it shall be understood to mean the "Contracting Officer" as stated in the Contract Documents. In case of conflict between the PRASA specifications and the Specifications in this Section that follows, the Specifications in this Section shall govern.

1.2 DEFINITIONS

1.2.1 Water Transmission Mains

Water transmission mains include water piping having diameters greater than 14 inch, specific materials, methods of joining and any appurtenances deemed necessary for a satisfactory system.

1.2.2 Water Mains

Water mains include water piping having diameters 4 through 14 inch, specific materials, methods of joining and any appurtenances deemed necessary for a satisfactory system.

1.2.3 Water Service Lines

Water service lines include water piping from a water main to a building service at a point approximately 5 feet from building or the point indicated on the drawings, specific materials, methods of joining and any appurtenances deemed necessary for a satisfactory system.

1.2.4 Additional Definitions

For additional definitions refer to the definitions in the applicable referenced standard.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. **Submittals with a "PR" designation shall also be submitted to the PRASA for review and approval.** Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Disinfection Method; G, DO, PR

48" Existing Watermain Work Plan; G, DO, PR

SD-02 Shop Drawings

Pipe, Fittings, Joints and Couplings; G, RO, PR

Butterfly Valve; G, RO, PR

Combination Air Valve; G, RO, PR

Extension Stems; G, RO, PR

SD-03 Product Data

Pipe, Fittings, Joints and Couplings; G, RO, PR

Valves; G, RO, PR

Valve Boxes; G, RO, PR

Pipe Restraint; G, RO, PR

Tapping Sleeves; G, RO, PR

Corporation Stops; G, RO, PR

Detectable Marking Tape; G, RO, PR

SD-06 Test Reports

Leakage Test

Hydrostatic Pressure Test

SD-07 Certificates

Pipe, Fittings, Joints and Couplings

Valves

SD-08 Manufacturer's Instructions

Ductile Iron Piping

1.4 QUALITY CONTROL

1.4.1 Regulatory Requirements

Comply with NSF/ANSI 14 or NSF/ANSI 61 and NSF 372 for materials for potable water systems; comply with lead content requirements for "lead-free" plumbing as defined by the U.S. Safe Drinking Water Act effective January 2014. Provide materials bearing the seal of the National Sanitation Foundation (NSF) for potable water service.

1.4.2 QUALITY ASSURANCE

The manufacturer shall furnish a certified statement that the pipes, valves and fittings have been inspected and pressure tested in accordance with AWWA Standards and the specifications to which the material is manufactured.

Ductile iron pipe and fittings shall be furnished by manufacturers who are fully experienced, reputable, and qualified in the manufacture of the materials to be furnished.

Inspection of the pipe will be made by the Contracting Officer after delivery. Pipe or fittings shall be subject to rejection at any time on account of failure to meet any of the Specification requirements, even though they may have been accepted as satisfactory at the place of manufacture.

Pipe or fittings rejected after delivery shall be marked for identification and shall immediately be removed from the job.

1.5 DELIVERY, STORAGE, AND HANDLING

1.5.1 Delivery and Storage

Inspect materials delivered to site for damage. Unload and store with minimum handling and in accordance with manufacturer's instructions. Store materials on site in enclosures or under protective covering. Store plastic piping, jointing materials and rubber gaskets under cover out of direct sunlight. Do not store materials directly on the ground. Keep inside of pipes, fittings, valves, and other accessories free of dirt and debris.

1.5.2 Handling

Handle pipe, fittings, valves, and other accessories in accordance with manufacturer's instructions and in a manner to ensure delivery to the trench in sound undamaged condition. Avoid injury to coatings and linings on pipe and fittings; make repairs if coatings or linings are damaged at no expense to the Government. Do not place other material, hooks, or pipe inside a pipe or fitting after the coating has been applied. Inspect the pipe for defects before installation. Carry, do not drag pipe to the trench. Use of pinch bars and tongs for aligning or turning pipe will be permitted only on the bare ends of the pipe. Clean the interior of pipe

and accessories of foreign matter before being lowered into the trench and keep them clean during laying operations by plugging. Replace defective material without additional expense to the Government. Store rubber gaskets, not immediately installed, under cover or out of direct sunlight.

Handle ductile iron pipe, fittings, and accessories in accordance with AWWA C600 and AWWA M41.

Pipes, fittings, special pieces and joint materials should be delivered along sides of the pipe trench, near the places where they are to be installed, as shown on the project plans.

Inspection of the pipe will be made by the Contracting Officer after delivery. Pipe or fittings shall be subject to rejection at any time on account of failure to meet any of the Specification requirements, even though they may have been accepted as satisfactory at the place of manufacture.

Pipe or fittings rejected after delivery shall be marked for identification and shall immediately be removed from the job.

Handle all valves and appurtenances very carefully. Valves which are cracked, dented or otherwise damaged or dropped will not be acceptable.

Store all valves and appurtenances in approved enclosed shelter and off the ground.

PART 2 PRODUCTS

2.1 MATERIALS

Provide all materials in accordance with AWWA C800 and as indicated herein. Provide valves and fittings with pressure ratings equivalent to the pressure ratings of the pipe.

2.1.1 Pipe, Fittings, Joints And Couplings

Submit manufacturer's standard drawings or catalog cuts, except submit both drawings and cuts for push-on and rubber-gasketed bell-and-spigot joints. Submit a certified statement from the manufacturer stating that the products have been inspected and pressure tested in accordance with the applicable AWWA standards and manufactured in accordance with applicable material specifications. Include information concerning gaskets with submittal for joints and couplings including shop drawings depicting the following information:

- a. Piping layouts in full detail
- b. Location of pipe supports and hangers
- c. Location and type of device to prevent joint separation, such as backup block
- d. Large scale details of all wall penetrations and special castings
- e. Schedules of all pipe, fittings, special castings, couplings, expansion joints, and other appurtenances.

2.1.1.1.1 Ductile-Iron Piping

2.1.1.1.1.1 Ductile Iron Pipe and Fittings

Unless otherwise noted in the Construction Drawings, Pipe shall be Class 350. Pipe, except flanged pipe, shall comply with AWWA C151/A21.51, Pressure Class 350. Flanged pipe shall comply with AWWA C115/A21.15. Fittings, AWWA C110/A21.10 or AWWA C153/A21.53; fittings with push-on joint ends are to meet the same requirements as fittings with mechanical-joint ends, except for the factory modified bell design. Provide fittings with pressure ratings equivalent to that of the pipe. Provide compatible pipe ends and fittings for the specified joints. Provide cement-mortar lining, AWWA C104/A21.4, standard thickness on pipe and fittings.

Ductile iron pipe shall follow all references stated below:

AWWA C104/A21.4: Cement-Mortar Lining for Ductile-Iron Pipe and Fittings for water, AWWA C150/A21.50: Polyethylene Encasement for Ductile-Iron Piping for Water and other Liquids, AWWA C110/A21.10: Ductile-Iron and Gray-Iron Fittings, 3-in. through 48-in., for Water and other Liquids, AWWA C111/A21.11: Rubber-Gasket Joints for Ductile-Iron and Gray-Iron Pressure Pipe and Fittings, AWWA C115/A21.15: Flanged Ductile-Iron Pipe with threaded Flanges, AWWA C150/A21.50: Thickness Design of Ductile-Iron Pipe, AWWA C151/A21.51: Ductile Iron Pipe, Centrifugally Cast in Metal Molds or Sand Lined Molds, for Water, AWWA C153/A21.53: Ductile-Iron Compact Fittings, 3-in. through 16-in., for Water and other Liquids, AWWA C600: Installation of Ductile-Iron Water Mains and their Appurtenances, AWWA C606: Grooved and Shouldered Joints, AWWA C651: Disinfecting Water Mains.

Provide fittings, valves, flanges and unions with the manufacturer's trademark affixed in accordance with MSS SP-25 so as to permanently identify the manufacturer.

- a. Fittings shall be ductile iron and shall conform to AWWA C110/A21.10.
- b. Compact fittings from 3-inch through 12-inch shall be also in accordance with AWWA C110/A21.10 and/or AWWA C111/A21.11 of latest revision. All necessary corten bolts gaskets and ductile iron glands must be included. All accessories shall be individually packaged.
- c. All fittings shall be bituminous coated outside and cement lined inside, in accordance with AWWA C104/A21.4 Cement-Mortar Lining for Ductile-Iron Pipe and Fittings for Water.

2.1.1.1.1.1.1 Wall Penetrations

Wall penetration shall be of the sizes and type as shown in the Construction Drawings.

Wall penetration shall be provided with a watertight mechanical elastomeric modular seal with steel wall sleeves with an integrated 1/2" water stop-ring.

2.1.1.1.1.2 Coating And Linings

Pipe and fittings shall have a cement mortar lining and asphaltic seal coat on the inside in accordance with AWWA C104/A21.4. All pipe and fittings, except where painting is specified, shall have an asphaltic coating on the outside in accordance with AWWA C151/A21.51.

Surface preparation and primers for exterior metal surfaces are specified in PRASA Specifications Section 099100, Painting.

2.1.1.1.3 Identification

The weight, class or nominal thickness, and casting, period shall be shown on each pipe. The manufacturer's mark, the year in which the pipe was produced and the letters "DI" of "ductile", shall be cast or stamped on the pipe.

Gaskets shall be marked with name of manufacturer, size, and proper insertion direction.

2.1.1.1.4 Joints and Jointing Material

Provide push-on joints for pipe and fittings unless otherwise indicated. Flanged joints locations are indicated on the Drawings.

- a. Push-On Joints: Shape of pipe ends and fitting ends, gaskets, and lubricant for joint assembly as recommended in AWWA C111/A21.11.
- b. Mechanical Joints: Dimensional and material requirements for pipe ends, glands, bolts and nuts, and gaskets as recommended in AWWA C111/A21.11.
- c. Flanged Joints: Bolts, nuts, and gaskets for flanged connections as recommended in Appendix A of AWWA C115/A21.15. Provide AWWA C115/A21.15 ductile iron flanges and conform to ASME B16.1, Class 125. Provide ASTM A536 epoxy coated steel set screw flanges. Gasket and lubricants for set screw flanges, in accordance with mechanical-joint gaskets specified in AWWA C111/A21.11.
- d. Sleeve-Type Mechanical Coupled Joints: As specified in the paragraph SLEEVE-TYPE MECHANICAL COUPLINGS.
- e. Mechanical joints and push-on joints shall conform to AWWA C111/A21.11. The design and shape of the gasket and the annular recess shall be such that the gasket is locked in place against displacement, as the joint is assembled. The Contractor shall furnish the necessary gaskets, lubricants and other accessories for jointing rubber gasket bell and spigot joints in quantities ten percent (10%) in excess of the theoretical requirement.
- f. Buried joints shall be push-on or mechanical joint type, as shown in the Construction Drawings. Joints above finished grade or in vaults shall be flanged unless shown otherwise in the Construction Drawings.
- g. Restrained joints for push-on pipes shall be by means of a restraint hardness and mechanical joints shall consist of a gland restraint system, or approved equal.

Restrained joints for push-on joints shall consist of a wedge action restraint ring on the spigot joined to a split ductile iron ring behind the bell. The restraint ring shall have individually actuated wedges that increase their resistance to pull-out as pressure or external forces increase. The restraint ring and its wedging components shall be made of minimum grade 65-45-12 ductile iron conforming to ASTM A536. The wedges shall be heat treated to a minimum

hardness of 370 BHN. Torque limiting twist off nuts shall be used to insure proper actuation of the restraining wedges. The split ring shall be made of a minimum grade of 65-45-12 ductile iron conforming to [ASTM A536](#). The connecting tie rods that join the two rings shall be made of low alloy steel that conforms to [AWWA C111/A21.11](#).

Mechanical joint fittings restraint system shall consist of a mechanical joint gland restrained by means of multiple gripping wedges mechanical system meeting standards [AWWA C110/A21.10](#). Mechanical joint fitting and appurtenances shall also conform to [AWWA C153/A21.53](#). The gland body, wedges and other components shall be cast from grade 65-45-12 ductile iron in accordance with [ASTM A536](#).

- h. Flanges shall have a 3.2-mm (1/8") thick flat inserted rubber gaskets. Flanges shall be flat faced with a serrated finish.
- i. Filler flanges and belimed filler flanges shall be furnished and install as required. Filler flanges and belimed filler flanges shall be furnished faced and drilled complete with extra length bolts.

2.1.2 Valves

Provide a protective interior coating in accordance with [AWWA C550](#). Conform to the requirements of each valve as shown on the construction drawings.

All valves shall have manufacturer's name and working pressure cast in raised letters on valve body.

All manual valve operators shall turn clockwise to close the valves. An arrow with the word "close" shall be cast on each operator to indicate direction of closing.

All flanged valves shall have ends conforming to [ASME B16.1](#), Class 125.

All bolts, nuts and washers required to connect buried and submerged flanged valves shall be of Type 304 stainless steel.

Exposed flanged valve connections shall be with bolts, nuts, and washers conforming to [ASTM A307](#), Grade B; or [ASTM A354](#).

All bolts and nuts shall have hexagon heads. Gasket material shall be suitable for the specified service.

[Extension stems](#) with nuts shall be provided. Operating nuts about 2 inches (5.08 cm) square shall be included with each extension stem and located in floor box, as required. Provide manufacturer's specifications and installation diagrams.

Provide manufacturer's literature, illustrations, specifications, and engineering data including dimensions, material, size and weight. Also include fabrication, assembly, installation diagrams, setting drawings, templates, and directions for the installation of anchor bolts and other anchorages.

Certificates of shop testing of valves as specified for all valves shall be shop tested at test pressures in accordance with the requirements of applicable AWWA and PRASA Standards. The manufacturer shall certify that

the required tests on the various materials and on the completed valves were performed and the valves were found conforming to the requirements of AWWA and PRASA Standards.

2.1.2.1 Butterfly Valves

Provide rubber-seated butterfly valves and wafer type valves that match the performance requirements of AWWA C504. Wafer type valves not meeting laying length requirements are acceptable if supplied and installed with a spacer, providing the specified laying length. Meet all tests required by AWWA C504. Flanged-end valves are required in a pit. Provide a union or sleeve-type coupling in the pit to permit removal. Direct-bury mechanical-end valves 3 through 10 inches in diameter. Provide a valve box, means for manual operation, and an adjacent pipe joint to facilitate valve removal. Provide valve operators that restrict closing to a rate requiring approximately 60 seconds, from fully open to fully closed.

- a. The construction of the valve and operating mechanisms shall be of such design that the disc can be held in open, partially open or shut position for manual operation or for automatic operation. When the disc is so held, there shall be no chatter or vibration of the vane or operating mechanism.
- b. This specification is designed to cover the design, manufacture, and testing of AWWA Class 150B (3"- 108") and AWWA Class 250B (3"-48") butterfly valves.
- c. The valves shall be designed, manufactured, and tested in accordance with American Water Works Association Standard AWWA C504.
- d. Valves shall be proof of design tested in accordance with AWWA C504 and C516, and certified to NSF/ANSI 61 Drinking Water System Components - Health Effects and certified to be Lead-Free in accordance with NSF/ANSI 61 - Annex G.
- e. Manufacturer shall have a quality management system that is certified to ISO 9001:2008 by an accredited, certifying body.
- f. Flanged end connections shall fully conform with ASME B16.1 for Class 125, Class 250 iron flanges, or AWWA C207 Class D. Both 125 and 250 flanges shall be flat faced.
- g. Water end connections shall be designed for installation between ASME B16.11 Class 125 iron flanges or between ISO 7005-2 PN10 or PN16 flanges.
- h. Valve shafts shall be of the through-type for sizes 3"-24". 30" and larger shall be of the stub type design. Shafts shall be locked to the disc by O-ring sealed taper pins retained with stainless steel nuts. Through-type shafts shall be supplied on 30" and larger valves when specified.
- i. Valve discs shall be of the solid type without external ribs or vanes to obstruct flow.
- j. Resilient seats shall be located on the valve disc and shall provide a 360° continuous, uninterrupted seating surface. Seats shall be

mechanically retained with a stainless-steel retaining ring and stainless-steel cap screws which shall pass through both the resilient seat and the retaining ring. The retaining ring shall be continuous or investment cast with overlapping sections, serrated grooves, and shoulders providing a Tri-Loc™ system. The resilient seat's mating surface shall be to a 360° continuous, uninterrupted stainless-steel body seat ring. Resilient seats shall be field adjustable and replaceable without removing the valve from the line and shall not require epoxy, syringes, needles, or pressure vessels to replace or adjust.

- k. Sleeve bearings shall be provided in the valve hubs and shall be nylatron or woven teflon, fiberglass backed. They shall be self-lubricating.
- l. Thrust bearings shall be provided and shall be adjustable on valves 30" and larger.
- m. Shaft seals shall be of the V-type and shall be replaceable without removal of the valve from the line or the shaft from the valve.

2.1.2.1.1 Materials

- a. Body: Class 150B valve bodies shall be **ASTM A126**, Class B gray iron or **ASTM A536** Grade 65-45-12 ductile iron. Class 250B valve bodies shall be **ASTM A536** Grade 65-45-12 ductile iron.
- b. Disc: Valve disc shall be **ASTM A536** Grade 65-45-12 ductile iron.
- c. Shafts: Shafts shall be **ASTM A276/A276M** type 304, or **ASTM A564/A564M**, Type 630 Stainless Steel.
- d. Seat: Resilient seat shall be Buna-N and mate to a Type 316 Stainless Steel body seat ring.
- e. Hardware: All seat retaining hardware shall be Type 316 stainless steel.

2.1.2.1.2 Operators

Manual actuators shall be of the traveling nut design with characterized closure per **AWWA C504** and equipped with externally adjustable closed position stops capable of withstanding 450 ft-lbs. Actuators shall be lubricated with EP-2 grease and fully enclosed in an iron housing sealed against the entry of water.

2.1.2.1.3 Painting

- a. Valve exteriors for above ground service shall be coated with a universal, alkyd primer.
- b. Valve exteriors for buried service shall be coated with an epoxy coating.
- c. Valve interiors shall be coated with an NSF/ANSI 61 epoxy coating approved for potable water. Fusion bonded epoxy shall be supplied on the exterior and interior when specified. Ferrous metal internals of valves for potable water service shall be painted in accordance with the applicable AWWA Standards.

- d. Surface Preparation and Shop Priming: All valves shall be painted in the shop. Surface preparation and shop painting are included herein. Surface preparation and primers for exterior metal surfaces are specified in [PRASA Specifications](#) Section 099100, Painting.
- e. All exterior ferrous metal surfaces of the valves shall be cleaned and shall receive one coat of a rust inhibiting priming paint.
- f. All prime coat paint materials shall be compatible with the finish coat materials to be furnished under [PRASA Specifications](#) Section 099100, Painting. The Contractor shall be responsible for the coordination of all paint materials to be supplied in the shop and field, and shall submit a complete list of all such paint materials to the Contracting Officer for approval prior to their application in either field or shop, with certification from the material suppliers that such prime and finish coat materials are compatible.
- g. Bearing surfaces and similar unpainted ferrous surfaces shall receive a heavy application of rust-resistant coating which shall be maintained during storage and until equipment begins operation.

2.1.2.1.4 Maintenance

The recommended interval for exercising is every six months or annually if the valve is regularly operated. The exercising is achieved by fully opening and closing the valve to verify smooth operation. If operation is difficult, it may be necessary to flush sediment from the valve by opening and closing the valve several times under flowing conditions or checking the lubricant in the gear actuator.

2.1.2.2 Air Release, Air/Vacuum, and [Combination Air Valves](#)

- a. Air release valves shall consist of a kinetic air and vacuum valve and an air release valve contained in a single body housing.
- b. Air release valves shall be of the kinetic float type and shall be installed at high points in piping systems as required, and as shown on the Contract Drawings and specified herein. Each valve shall maintain a closed position to prevent the loss of water by the positive seating of a stainless-steel float ball against the exhaust orifice. The valve shall automatically provide for the escape of air to the atmosphere without the loss of liquid when the special ball float moves away from the orifice seat.
- c. Each valve shall exhaust to a pipe which shall extend to drain. The exhaust pipe shall be of the same diameter as the exhaust orifice on the valve.
- d. Each air release valve shall be equipped with an isolating butterfly valve. Valves shall be furnished with flanged ends. Flanges shall be [ASME B16.1](#), Class 125.
- e. Valves shall be manufactured and tested in accordance with [AWWA C512](#).
- f. Valves used in potable water service shall be certified to [NSF/ANSI 61](#) Drinking Water System Components - Health Effects.
- g. Manufacturer shall have a quality management system that is certified

to ISO 9001 by an accredited, certifying body.

2.1.2.2.1 Connections

- a. Valves shall have ANSI Class 125 flanged inlets.
- b. The valve shall have two additional NPT connections for the addition of gauges, testing, and draining.

2.1.2.2.2 Design

The cover shall be bolted to the valve body and sealed with a flat gasket. Resilient seats shall be replaceable and provide drop tight shut off to the full valve pressure rating.

Floats shall be unconditionally guaranteed against failure including pressure surges. Mechanical linkage shall provide sufficient mechanical advantage so that the valve will open under full operating pressure. Simple lever designs shall consist of a single pivot arm and a resilient orifice button. Compound lever designs shall consist of two levers and an adjustable threaded resilient orifice button.

2.1.2.2.3 Materials

- a. The valve body and cover shall be constructed of ASTM A126 Class B cast iron for working pressures up to 300 psig. Higher pressure rated valves shall be constructed of ASTM A536 Grade 65-45-12 ductile iron.
- b. The orifice, float and linkage mechanism shall be constructed of Type 316 stainless steel. Non-metallic floats or linkage mechanisms are not acceptable. The orifice button shall be Viton for simple lever valves and Buna-N for compound lever designs.

2.1.2.2.4 Cross Contamination and Security Protection

All Air (Release, Vacuum, etc.) Valves installed in vaults or flood prone locations shall include an inflow preventer to prevent the introduction of contaminated water through the air valve outlet. The inflow preventer shall allow the admittance and exhausting of air while preventing contaminated water from entering during normal operating conditions. The inflow preventer shall be flow tested by an independent third party to certify performance. The third party shall be an approved testing lab of the American Society of Sanitary Engineers.

2.1.2.2.5 Manufacturer

- a. The manufacturer shall demonstrate a minimum of five (5) years' experience in the manufacture of air valves. When requested, the manufacturer shall provide test certificates, dimensional drawings, parts list drawings, and operation and maintenance manuals.
- b. The exterior of the valve shall be coated with a universal alkyd primer.

2.1.2.3 Gate Valves

- a. Valves between 2 and 12 inches in size shall be of a resilient wedge design.

- b. The valve shall have a cold water rated working pressure of 250 psig.
- c. All cast ferrous components shall be ductile iron and shall be manufactured in compliance with the lasted edition of [AWWA C515](#).
- d. The valve shall also be UL listed and FM approved in applicable configurations. The valve design shall be lightweight, easy to handle, and constructed with wall thickness as per Table 2 of [AWWA C515](#). Heavy wall and/or cast gray-iron bodies are not acceptable.
- e. The valve shall have a smooth and oversized water way and have the marking D.I. or "Ductile Iron" casted on the body.
- f. The valve wedge shall be constructed of ductile iron and provided with protective wedge guide covers in sizes 4 to 12 inches. All wedges 3 inches and smaller shall be constructed of bronze. All wedges shall be fully encapsulated with EPDM rubber. The valves shall be certified to NSF 51-G or NSF 372, as applicable.
- g. Valves stems shall be sealed by three O-rings. Two of the O-rings shall reside above the thrust collar. O-rings set in a cartridge shall not be allowed. The valve shall also be equipped with thrust washer above and below the stem thrust collar for reduction of operating torque.
- h. All exterior valve body bolting shall be type 304 stainless steel and shall be provided with hexagonal heads, with dimensions conforming to ANSI B 18.2.1. Metric size and/or socket head cap screws or bolts, are not allowed. The operating nut shall be 2 inches square and shall be constructed of ductile iron fitted to a square tapered stem to help ensure even valve operation input torque. All body gaskets shall be of the pressure energized O-ring style design.
- i. All internal and external ferrous surfaces of the valve body and bonnet shall have a fusion-bonded epoxy coating, complying with [AWWA C550](#). The coating shall be electro-statically applied prior to assembly.
- j. All valves shall be outside stem and yoke, handwheel operated, or chain operated.
- k. All gate valves shall open counter-clockwise, shall be furnished without bypasses and shall be suitable for vertical or horizontal mounting.
- l. Buried gate valves shall be standard AWWA nut operated, non-rising stem.
- m. All gate valves in water service not buried shall be drilled, tapped and plugged for drainage on both sides of this gate. Taps shall be maximum size recommended by valve manufacturer. Locations shall also be as recommended by this manufacturer.
- n. Valves installed higher than five feet (1.52 m) above the operating floor:
 - 1. Chainwheels, sprockets, gearing where required, and galvanized steel chain shall be provided for valves mounted more than five feet above the operating floor.

2. The chain shall extend to three feet above the operating floor.
 3. Gearing, if provided, shall be enclosed in a semi-steel or cast iron housing and be suitable for running in a lubricant, with seals provided on all shafts to prevent entry of dirt and water into the operator.
- o. Where shown on the Drawings the gate valves shall be furnished and installed with extension stems and floorstands. The stems shall be of extra strong steel pipe and the couplings shall be fabricated of high grade malleable iron. Floorstands and pedestals shall be of cast iron or steel, equipped with handwheels and not less than 32 inches (0.81 m) high. Floorstands for non-rising stem valves shall have valve position indicators.
 - p. Buried valves shall be provided with valve boxes.

2.1.2.3.1 Maintenance

The recommended interval for exercising is every six months or annually if the valve is regularly operated. The exercising is achieved by fully opening and closing the valve to verify smooth operation. If operation is difficult, it may be necessary to flush sediment from the valve by opening and closing the valve several times under flowing conditions or checking the lubricant in the gear actuator.

2.1.3 Valve Appurtenances

Valves and appurtenances shall follow the references [AWWA C504](#), Rubber-Seated Butterfly valves, Shaft - Or Trunnion-Mounted, 6 Inch Through 48 inch, For Water Pressures Up To 300 PSI, [AWWA C111/A21.11](#), Rubber Gasket Joints for Ductile Iron and Fittings, [ASTM A307](#), Carbon Steel Externally Threaded Standard Fasteners, [ASTM A536](#), Ductile Iron Castings, [ASTM B62](#), AGMA Standards, NEMA, National Electrical Manufacturers Association, and ISA, Instrument Society of America.

- a. Extension Stems, Stem Guides, Wrenches and Keys:
 1. Extension stem shall be at least as large as the valve stem it operates.
 2. Intermediate stem guide shall be installed for extensions more than 8 feet long.
 3. Stem brackets and guides shall be made of cast iron and have fully adjustable bronzed bushed guide block. Fasten brackets to walls with approved expansion bolts.
 4. Operating nuts about 2 inches (5.08 cm) square shall be included with each extension stem and located in floor box or grating recess, as required.
 5. Provide operating key or wrench of suitable length and size for each valve.
- b. Floor Boxes: Provide cast iron floor boxes for all valves which are to be operated from floor above valve. Boxes shall be equal in depth to

floor slab. Boxes shall have cast iron covers and be fitted with bronze bushing for stem alignment. Cast the word "WATER" on the lid. Factory coat the box with a heavy coat of bituminous paint.

- c. Identification: Identify each valve 4 inches and larger with a brass or stainless-steel nameplate stamped with the approved designation. Nameplate shall be permanently fastened to valve body at the factory. Stenciled designations are acceptable for buried valves.

2.1.3.1 Surface Preparation And Shop Painting

- a. Surface preparation and shop painting are required for all ferrous metals, equipment and accessories, and shall be as specified under [PRASA Specifications](#) Section 099100, Painting.
- b. Ferrous metal internals of valves for cold non-potable and potable water service shall be painted in accordance with the applicable AWWA Standards.

2.1.4 Disinfection

Chlorinating materials are to conform to: Chlorine, Liquid: [AWWA B301](#); Hypochlorite, Calcium and Sodium: [AWWA B300](#).

2.2 ACCESSORIES

2.2.1 Pipe Restraint

Restrained joints for push-on pipes shall be by means of a restraint hardness and mechanical joints shall consist of a gland restraint system, or approved equal.

Restrained joints for push-on joints shall consist of a wedge action restraint ring on the spigot joined to a split ductile iron ring behind the bell. The restraint ring shall have individually actuated wedges that increase their resistance to pull-out as pressure or external forces increase. The restraint ring and its wedging components shall be made of minimum grade 65-45-12 ductile iron conforming to [ASTM A536](#). The wedges shall be heat treated to a minimum hardness of 370 BHN. Torque limiting twist off nuts shall be used to insure proper actuation of the restraining wedges. The split ring shall be made of a minimum grade of 65-45-12 ductile iron conforming to [ASTM A536](#). The connecting tie rods that join the two rings shall be made of low alloy steel that conforms to [AWWA C111/A21.11](#).

Mechanical joint fittings restraint system shall consist of a mechanical joint gland restrained by means of multiple gripping wedges mechanical system meeting standards [AWWA C110/A21.10](#). Mechanical joint fitting and appurtenances shall also conform to [AWWA C153/A21.53](#). The gland body, wedges and other components shall be cast from grade 65-45-12 ductile iron in accordance with [ASTM A536](#).

2.2.2 Tapping Sleeves

Provide cast gray, ductile, malleable iron or stainless steel, split-sleeve type tapping sleeves of the sizes indicated for connection to existing main with flanged or grooved outlet, and with bolts, follower rings and gaskets on each end of the sleeve. Utilize similar metals for

bolts, nuts, and washers to minimize the possibility of galvanic corrosion. Provide dielectric gaskets where dissimilar metals adjoin. Provide a tapping sleeve assembly with a maximum working pressure of 350 psi. Provide bolts with square heads and hexagonal nuts. Longitudinal gaskets and mechanical joints with gaskets as recommended by the manufacturer of the sleeve. When using grooved mechanical tee, utilize an upper housing with full locating collar for rigid positioning which engages a machine-cut hole in pipe, encasing an elastomeric gasket which conforms to the pipe outside diameter around the hole and a lower housing with positioning lugs, secured together during assembly by nuts and bolts as specified, pre-torqued to 50 foot-pound.

2.2.3 Sleeve-Type Mechanical Couplings

Joint Restraint to prevent axial separation shall be incorporated into the design of the sleeve or coupling used to connect two plain pipe ends. The restraint mechanism shall consist of a plurality of individually actuated gripping surfaces to maximize restraint capability. Torque limiting twist off nuts shall be used to insure proper actuating of the restraint devices. Ductile Iron components shall be of a minimum of 65-45-12 ductile iron meeting the requirements of ASTM A536 of the latest revision and shall be tested in accordance with the stated standard. The restrained joining system shall meet the applicable requirements of AWWA C219, AWWA C111/A21.11, and ASTM D2000.

2.2.4 Detectable Marking Tape

This section includes furnishing and laying all required detectable marking tape, to be used above all water pipe and all sanitary force main, and/or cross country as shown in the Construction Drawings or as required to comply with contract requirements including all other materials, labor, tools and appliances necessary to complete the work.

Detectable marking tape shall consist of one 35 gauge layer of aluminum foil laminated between two layers of inert plastic film; top side to be polyester for strength and the underside to be a minimum of 2.5 polyethylene. The extrusion-laminated bond should be strong enough that the layers cannot be separated by hand. Tape shall have a minimum of 5 mils thick with a minimum tensile strength of 5,600 psi per 3 inches width strip. Tape shall be imprinted in both English and Spanish with a continuous warning message repeated every 16 to 36 inches. See attached drawing. Tape shall be of three (3) inches width. Tape shall be blue color with black imprint for Aqueduct Systems. Tape shall be green color with black imprint for Sanitary Sewer Systems.

Attachment: Example Detectable Marking Tape

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Connections to Existing System

The Contractor shall furnish all necessary materials and labor to make all connections to existing water mains, shown in the Construction Drawings or that may be ordered by the Contracting Officer, including excavation, backfilling, restoration of existing pavement to original conditions, materials, accessories, tools, equipment and sleeves. Perform all

connections to the existing water system in the presence of the Contracting Officer and a PRASA Representative.

The existing 48" watermain crossing the Rio Piedras at Sta. 109+50 may or may not be in service at the time of construction. Coordinate with PRASA to determine if pipe line has been put back in service. If pipe is in service, develop and submit a 48" existing watermain work plan and sequencing for the installation of the sheet piles and bridge shafts on the left bank; provisions for service outage during shaft or sheet installation; provisions for riprap erosion and flood protection of soils in the gap between sheet piles; provisions for removal of the watermain; and for the installation of the remaining gap sheets and shafts.

Connections to existing pipes shall be made using suitable joints and fittings for the conditions encountered. Each connection with an existing pipe shall be made at the time and under conditions which will least interfere with operation. Connections with buried existing flanges shall require removal of grout from the flanges. Remove concrete thrust block in when encountered for connections to existing pipes.

Contractor shall provide for proper dewatering and for disposal of all water removed from the dewatering lines and excavations without damage to adjacent property.

Existing valves will leak when in closed position. Contractor shall provide whatever means and equipment is necessary to control water leakage during construction.

Connection to existing water main shall be made at the proper grade elevations, with utmost care so as to avoid leaks and other imperfections, and to the satisfaction of the Contracting Officer.

All connections, once that the pipe, sleeves, valves and other appurtenance are laid, installed and supported, shall be tested by the Contractor. The Contractor shall furnish the necessary tools, equipment and labor to carry out the tests.

3.1.2 Operation of Existing Valves

Do not operate valves within or directly connected to the existing water system unless approved by the PRASA to do so and Contracting Officer.

3.1.3 Earthwork

Perform earthwork operations in accordance with Section 31 00 00 EARTHWORK.

3.2 INSTALLATION

Install all materials in accordance with the applicable reference standard, manufacturer's instructions and as indicated herein.

3.2.1 Piping

3.2.1.1 General Requirements

Install pipe, fittings, joints, and couplings in accordance with the applicable referenced standard, the manufacturer's instructions and as specified herein.

3.2.1.1.1 Termination of Water Lines

Terminate the work covered by this section at a point approximately 5 feet from the building, unless otherwise indicated.

Do not lay water lines in the same trench with gas lines, fuel lines, electric wiring, or any other utility. Where nonferrous metallic pipe (i.e., copper tubing) crosses any ferrous piping, provide a minimum vertical separation of 12 inches between pipes.

3.2.1.1.2 Pipe Laying and Jointing

Remove fins and burrs from pipe and fittings. Before placing in position, clean pipe, fittings, valves, and accessories, and maintain in a clean condition. Provide proper facilities for lowering sections of pipe into trenches. Under no circumstances is it permissible to drop or dump pipe, fittings, valves, or other water line material into trenches. Cut pipe cleanly, squarely, and accurately to the length established at the site and work into place without springing or forcing. Pipe shall be cut by means of pipe cutters, sharp diamond chisels or other approved means which will produce a clean and true cut, free from irregularities. Replace a pipe or fitting that does not allow sufficient space for installation of jointing material. Blocking or wedging between bells and spigots is not permitted. Lay bell-and-spigot pipe with the bell end pointing in the direction of laying. Grade the pipeline in straight lines; avoid the formation of dips and low points. Support pipe at the design elevation and grade. Secure firm, uniform support. Wood support blocking is not permitted. Lay pipe so that the full length of each section of pipe and each fitting rests solidly on the pipe bedding; excavate recesses to accommodate bells, joints, and couplings. Provide anchors and supports for fastening work into place. Make provision for expansion and contraction of pipelines. Keep trenches free of water until joints have been assembled. At the end of each workday, close open ends of pipe temporarily with wood blocks or bulkheads. Do not lay pipe when conditions of trench or weather prevent installation. Provide a minimum of 2 1/2 feet depth of cover over top of pipe.

3.2.1.1.3 Sewer Manholes

No water piping is to pass through or come in contact with any part of a sewer manhole.

3.2.1.1.4 Water Piping Crossing Sewer Piping

Provide at least 18 inches above the top (crown) of the sewer piping and the bottom (invert) of the water piping whenever possible. Measure the distance edge-to-edge. Where water lines cross under gravity sewer lines, construct sewer line of AWWA compliant ductile iron water piping with rubber-gasketed joints and no joint located within 10 feet, horizontally, of the crossing. Lay water lines which cross sewer force mains and inverted siphons at least 2 feet above these sewer lines; when joints in the sewer line are closer than 3 feet horizontally from the water line relay the sewer line to ensure no joint closer than 3 feet.

Normal Conditions: Provide a separation of at least 18 inches between the bottom of the water piping and the top of the sewer piping in cases where water piping crosses above sewer piping.

Unusual Conditions: When local conditions prevent a vertical separation

described above, construct sewer piping passing over or under water piping of AWWA compliant ductile iron water piping and perform hydrostatic sewer test, without leakage, prior to backfilling. Construct sewer crossing with a minimum 20 feet length of the AWWA compliant ductile iron water piping, centered at the point of the crossing so that joints are equidistant and as far as possible from the water piping. Protect water piping passing under sewer piping by providing a vertical separation of at least 18 inches between the bottom of the sewer piping and the top of the water piping; adequate structural support for the sewer piping to prevent excessive deflection of the joints and the settling on or damage to the water piping.

3.2.1.1.5 Penetrations

Provide a watertight mechanical elastomeric modular seal with steel wall sleeves with an integrated 1/2 inch water stop-ring.

3.2.1.1.6 Flanged Pipe

Only install flanged pipe aboveground or with the flanges in valve pits.

3.2.1.2 Ductile-Iron Piping

Unless otherwise specified, install pipe and fittings in accordance with the paragraph GENERAL REQUIREMENTS and with the requirements of AWWA C600 for pipe installation, joint assembly, valve-and-fitting installation, and thrust restraint.

- a. Jointing: Make push-on joints with the gaskets and lubricant specified for this type of joint; assemble in accordance with the applicable requirements of AWWA C600 and AWWA M41 for joint assembly.

Make mechanical joints with the gaskets, glands, bolts, and nuts specified for this type of joint; assemble in accordance with the applicable requirements of AWWA C600 and AWWA M41 for joint assembly and the recommendations of Appendix A to AWWA C111/A21.11. Make flanged joints with the gaskets, bolts, and nuts specified for this type of joint.

Make flanged joints up tight; avoid undue strain on flanges, fittings, valves, and other equipment and accessories. Align bolt holes for each flanged joint. Use full size bolts for the bolt holes; use of undersized bolts will not be permitted. Do not allow adjoining flange faces to be out of parallel to such degree that the flanged joint cannot be made watertight without overstraining the flange. When flanged pipe or fitting has dimensions that do not allow the making of a flanged joint as specified, replace it.

- b. Allowable Deflection: Follow AWWA C600 and AWWA M41 for the maximum allowable deflection. If the alignment requires deflection in excess of the above limitations, provide special bends or a sufficient number of shorter lengths of pipe to achieve angular deflections within the limit set forth.

3.2.1.2.1 Buried Piping

- a. The contractor shall furnish, deliver, haul, store, protect and perform all work necessary for installing, jointing and testing the

ductile cast iron pipes, fittings and accessories required. Including everything necessary to complete the work in readiness for operation, as shown on plans.

- b. Contractor shall excavate all trenches and lay pipes as specified in Section 31 00 00 EARTHWORK.
- c. Placing pipe in the trench. Proper implements, tools, and facilities satisfactory to the Contracting Officer shall be provided and used by the contractor for the safe and convenient prosecution of the work. All pipe, fittings, valves, and similar pieces will be carefully lowered into the trench. Piece by piece, by means of derricks, ropes, or other suitable tools or equipment; in such manner as to prevent damage to the pipe. Under no circumstances shall pipe or accessories be dropped or dumped into trench.

In lowering the pipe into the trench, care shall be exercised to avoid bumping the pipe already in the trench. Timbers shall be used in front of the end of the pipe previously laid. To serve as a bumper to prevent damage from a swinging pipe, as it is lowered into the trench.

- d. Pipe Kept Clean. All foreign matter or dirt shall be removed from the inside of the pipe before it is lowered into its position in the trench and it shall be kept clean by approved means during and after laying.
- e. Blocking of pipe. Only if approved by the Contracting Officer, shall pipes be laid on wood blocking and hold in correct alignment by wood wedges, placed transversely to the line of pipe. Blocking and wedges shall be out from sound lumber.

Blocks shall be bedded firmly with uniform bearing, and shall be level across the bottom of the trench. The trench shall be excavated at least 2.5 cm. below grade, and the block shall project at least 2.5 cm. above the trench bottom. They shall be aligned to the proper grade.

- f. Making the Joints. All joints shall be made in strict accordance with the instructions of the pipe manufacturer.

The contractor shall assume full responsibility for satisfactory performance of the pipe in the line, as far as meeting the leakage test specified on the priory mentioned correspondent specification.

- g. Prevention of trench water from entering pipe. At times when pipe laying is not in progress, the open ends of the pipe shall be tightly closed by approved means, and no trench water shall be permitted to enter the pipe. No pipe shall be laid in water, or when the trench conditions or the weather are unsuitable for such work, except by permission of the Contracting Officer.
- h. Cutting pipe. Pipe shall be cut by means of pipe cutters, sharp diamond point chisels or other approved means which will produce a clean true cut, free from irregularities, and leave a smooth end at right angles to the axis of the pipe. In planning his work, the contractor shall exercise care to minimize the number of cuts required and to avoid wasting pipe.
- i. Installation and joints assembly of pipe, fittings, and appurtenances

shall be in accordance with AWWA C600, Sections 3.3 and 3.4 except as specified otherwise.

- j. Longitudinal joint deflection shall not exceed 75 percent of the pipe manufacturer's maximum recommended deflection. In no case longitudinal joint deflection for push-on-type-joint pipe shall exceed the following values:

Nominal Pipe Size	Deflection Angle (degrees)	Maximum Offset (Inches) Laying Length = 18 ft	Maximum Offset in Inches Laying Length = 20 ft
3	5	19	21
4	5	19	21
6	5	19	21
10	5 / 3(1)	19 / 11(1)	21 / 12(1)
12	5	19	21
14	3	11	12
16	3	11	12
18	3	11	12
20	3	11	12
24	3	11	12
30	3 / 1.5(1)	11 / 6(1)	12 / 6(1)
36	3	11	12
42	3	11	12
48	3	11	12
54	3	11	12
60	3	11	12
64	3	11	12

(1) When using restraint joints, all joints for nominal pipe size of 10 inches shall not exceed 3 degrees deflection angle and all joints with nominal pipe size of 30 inches shall not exceed 1.5 degrees deflection angle.

Attachment: Maximum Deflection Diagram

- k. Valves shall be installed so that adjacent pipe will not be required to support the weight of the valve. Pipe shall be supported in such a

manner to prevent stress on the valve. In no case shall valves be used to bring misaligned pipe into alignment. Valves shall be installed in the closed position.

- l. Dead ends shall be closed with plugs or caps that are suitably restrained to prevent blowing off under test pressure. If a valve precedes the plug or cap it too shall be restrained against blowing off.
- m. All pipe and fittings shall be polyethylene encased with 4-mil high-density cross-laminated polyethylene film in accordance with [AWWA C105/A21.5](#). Provide polyethylene encasement as a flat tube meeting the dimensions of Table 1 in [AWWA C105/A21.5](#)AWWA C105.

3.2.1.2.2 Detectable Marking Tape

The Contractor furnish and lay all detectable marking tape, to be used above all water pipe and all sanitary force main, and/or cross country as shown in the Construction Drawings or as approved by the Contracting Officer including all other materials, labor, tools and appliances necessary to complete the work.

Detectable marking tape shall be installed directly above and parallel to the pipelines and about 12 inches (0.30M) below the surface.

3.2.1.2.3 Installation Above Grade

- a. Install piping and fittings true to grade and alignment with a tolerance of 1/4 -inch in the vertical or horizontal plan. Provide rigidly supported thrust anchors and expansion joints where required.
- b. Install proper size sleeves for all pipes passing through floors or walls as shown in the Construction Drawings. Where indicated in the Construction Drawings or required for liquid or gas-tightness seal the pipe with a mechanical seal.
- c. Furnish and install concrete inserts the hangers and supports in the concrete as it is placed. Set in compliance with the requirements of the piping layout and jointing method. Verify locations from approved piping layout drawings and the structural drawings.
- d. Flanged joints shall be made with bolt studs with a nut on each end, or stud with nuts where the flange is tapped. The number and size of bolts shall comply with AWWA C 110.
- e. Prior to assembly of grooved couplings, the grooves as well as other parts shall be thoroughly cleaned. The ends of the pipes and outside of the gaskets shall be moderately coated with petroleum jelly, cup grease, soft soap, or graphite paste, and the gasket shall be slipped over one pipe end. After the other pipe has been brought to the correct position, the gasket shall be centered properly over the pipe ends with the lips against the pipes. The housing sections then shall be placed. After the bolts have been inserted, the nuts shall be tightened until the housing sections are firmly in contact, metal-to-metal, without excessive bolt tension.
- f. Valves, fittings, equipment, and appurtenances shall be set and jointed as indicated in the Construction Drawings or as required. Pipe

and appurtenance connected to equipment shall be supported in such a manner as to prevent any strain being imposed on the equipment. When manufacturers have indicated requirements that piping loads shall not be transmitted to their equipment, a certification shall be submitted stating that such requirements have been complied with.

- g. All flanges and coupling bolts shall be tightened in accordance with the best practice identified in the governing standards or according to the manufacturer's written recommendation.

3.2.1.2.4 Painting

All exposed piping shall be painted as specified in PRASA Specifications Section 099100, Painting.

3.2.2 Valves

- a. The valves and appurtenances shall be installed by the Contractor in accordance with the details shown on the Drawings, and in accordance with approved submittals and manufacturer's instructions. Installation shall include furnishing all required oil and grease for initial operation. Grade and types shall conform to the manufacturer's recommendations.
- b. Provide valves in quantity, size and type with all required accessories as shown on the Drawings. Install all valves and appurtenances in accordance with manufacturer's instructions.
- c. Install pressure vacuum assemblies of type, size, and capacity indicated. Include valves and test cocks. Install according to the requirements of plumbing and health department and authorities having jurisdiction. Do not install pressure vacuum breaker assemblies in vault or other space subject to flooding.
- d. Install suitable corporation stops at all points shown and required where air binding of pipelines might occur.
- e. Install valves so that operating handwheels or wrenches may be conveniently turned from operating floor but without interfering with access, and as approved by Contracting Officer.
- f. Unless otherwise approved install all valves plumb and level. Valves shall be installed free from distortion and strain caused by misaligned piping, equipment, and other causes.
- g. Valve boxes shall be set plumb, and centered with the bodies directly over the valves so that traffic loads are not transmitted to the valve. Earth fill shall be carefully tamped around each valve box to a distance of 4 feet (1.22.m) on all sides of the box, or to the undisturbed trench face, if less than 4 feet(1.22 m).
- h. Hose valves and connecting pipe shall have at least the same depth of cover as the distributing pipe. The hose valves shall be set upon a concrete support not less than 12-inches square as shown on the drawings and 15-inches (0.381. m) square.

3.2.2.1 Field Test and Adjustments

Notify the Contracting Officer a minimum of five days in advance of hydrostatic testing. Coordinate the proposed method for disposal of wastewater from hydrostatic testing. Perform field tests, and provide labor, equipment, and incidentals required for testing, except that water needed for field tests will be furnished as set forth in paragraph AVAILABILITY AND USE OF UTILITY SERVICES in Section 01 50 02 TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS. Provide documentation that all items of work have been constructed in accordance with the Contract documents.

- a. Adjust all parts and components as required to provide correct operation.
- b. A manufacturer's representative for each type of valve shall check and approve the installation before operation. He shall test operate the valves in the presence of the Contracting Officer and verify that all equipment and controls conform to requirements. He shall revisit the job site as often as necessary until all trouble is corrected and the installation is entirely unsatisfactory.
- c. Conduct functional field test of each valve in presence of the Contracting Officer to demonstrate that each part and all components together function correctly.
- d. Test all installed valves hydrostatically with the corresponding piping at test pressures specified in Section 3.10.1. Valves shall withstand full test pressures without any visible leakage. All necessary testing equipment and labor shall be provided by the Contractor. Valves found defective shall be repaired or replaced at the Contractor's expense.

3.3 FIELD TESTING AND DISINFECTION

Perform tests of piping in the valve manholes as part of the testing of the direct buried conduit system. Include the piping in the valve manhole in these tests and perform in accordance with the system supplier's Approved Brochure or the contract specifications.

3.3.1 General

- a. Test all piping as specified below except as otherwise authorized by Contracting Officer.
- b. Piping shall be tested in sections. The maximum section size is the distance between two consecutive valves in the system.
- c. Notify the Contracting Officer and the PRASA 48 hours in advance of testing.
- d. Provide all testing apparatus, including pumps, hoses, gates and fittings.
- e. Pipelines shall hold the specified test pressure for a period of 2 hours.
- f. Pipelines which fail to hold specified test pressure or which exceed the allowable leakage rate shall be repaired and retested, at no

additional cost to the Government.

- g. Test pressures required are at the lowest elevation of the pipeline section being tested unless otherwise specified.
- h. Unless otherwise approved, conduct all tests in the presence of the Contracting Officer and the PRASA.
- i. Clean of all dirt, dust, oil, grease and other foreign material, before pressure and leakage tests.

3.3.2 Flushing

Prior to testing, pipes shall be thoroughly flushed to remove all foreign material. Material flushed from the pipes shall be removed from the site.

3.3.3 Testing of Pressurized Piping System

- a. Conduct test on buried piping that is to be hydrostatically tested after the trench has been completely backfilled. The Contractor may, if field conditions permit and if approved by the Contracting Officer, partially backfill the trench and leave the joints open for inspection and conduct an initial service leak test. The acceptance test shall not, however, be conducted until backfilling has been completed.
- b. Conduct the test on exposed piping after the piping has been completely installed. Installation includes supports, hangers, and anchors.
- c. Testing of pipe with concrete thrust blocking shall not be performed until the concrete has cured at least five days.
- d. If any pipe fails to meet the specified requirements for water tightness, the cause of the excessive leakage shall be determined and remedied and the pipe retested until requirements are satisfied.
- e. No pipe installation will be accepted until known leaks have been repaired. Locating and repairing of leaks shall be performed by the Contractor at no additional cost to the Government.
- f. Submit written reports on pipe tests and certify that required tests have been successfully completed before the piping is accepted. Reports shall be submitted in forms accepted by the Contracting Officer.

g. Pressure and Leakage Tests of Underground Pressure Piping

- 1. Hydrostatic pressure and **leakage test** shall conform to Section 4 of **AWWA C600** with the exception that the Contractor shall furnish all gauges, meters, pressure pumps and other equipment needed to test the pipeline.
- 2. The pressure required for the field **hydrostatic pressure test** shall be 1.5 times the normal working pressure (50 psi) of the piping system not to exceed the pipe class specified in the Construction Drawings. The pressure shall be maintained at the point for a period of not less than two (2) hours, during this time the actual amount of water force in the pipeline shall be determined by a measurement in the calibrated tank or barrel. This

amount shall be taken as a basic for computing the leakage per twenty-four (24) hours.

3. The Contractor shall provide the necessary water for filling the pipelines for testing. The Contractors shall provide temporary plugs and blocking necessary to maintain the required test pressures. Corporation cocks at least $\frac{3}{4}$ inch diameter, pipe riser and angle globe valves shall be provided at each pipe dead-end in order to bleed air from the line.
4. All visible leaks shall be repaired and leakage eliminated regardless of total leakage as shown by test. Pipelines which fail to meet tests shall be repaired and retested as necessary until test requirements are complied with. Defective materials, pipes, valves, and accessories shall be removed and replaced. Pipelines shall be tested in such sections by shutting valves or installing temporary plugs as required. The pipeline shall be filled with water, air removed, and the test pressure maintained in the pipe for the entire test period by means of a force pump to be furnished by the Contractor. Accurate means shall be provided for measuring the water required to maintain the test pressure. Test pressure shall not vary by more than +5 psi for the duration of the test.
5. The amount of leakage permitted for underground piping shall be as follows: Ductile Iron Pipes: As per latest [AWWA C600](#).
6. Remove and adequately dispose of all blocking material and equipment after completion and acceptance of the field hydrostatic test. Any damage to the pipe coating shall be repaired by the Contractor at no additional expense to the Government. Pipelines shall be totally free of material and clean prior to final acceptance.
7. Test to comply with specifications requirements.
8. Test pipelines in excavation or embedded in concrete before backfilling or placing of concrete and test exposed piping before field painting.
9. The section to be tested shall be closed by valves, temporary flanges, plugs or bulkheads, as required. The pipeline shall then be filled with water. A force pump with calibrated tank or barrel and pressure gauge shall be attached to the pipeline in a satisfactory and approved manner, using small outlets in the pipe or taps made in the plugs, caps or fittings for this purpose.
10. While under test pressure, the leakage from all pipes, valves and fittings under test shall be in accordance with AWWA Standard C600 (latest revision).

$$L = S \cdot D \cdot P^{1/2} / (133,200)$$

L = leakage allowed, in gallons per hour

S = Length of pipe tested in feet

D = nominal diameter of the pipe, in inches

P = average test pressure during the hydrostatic test, in pounds per square inch (gauge)

11. When testing against closed metal-seated valves, an additional testing allowance per closed valve of .0078 gal/h/in of nominal valve size shall be allowed. When hydrants are in the test section, the test shall be made against the main valve in the hydrant.
 - a. If the leakage discovered on test exceeds the amount specified the Contractor shall replace defective pipe, fittings, repair joints, tighten bolts, or do any other work necessary to reduce leakage to the specified amount and shall then re-test.
 - b. The Contractor shall furnish all labor, materials and equipment required for tests, including force pumps, calibrated tank or barrel, screw plugs for holes tapped in the pipes, connections and bracing the uncompleted work. If unable to achieve and maintain specified pressure for one hour with no additional pumping, section failed to pass test.
 - c. If section fails pressure and leakage test, locate, uncover, and repair or replace defective pipe, fitting, or joint, at no additional expense and without time extension. Conduct additional tests and repairs until section passes test, at no additional cost to the Government.

3.3.4 Disinfection

Disinfection of systems supplying non-potable water is not required.

Prior to disinfection, provide [disinfection methods](#), proposed neutralization and disposal methods of wastewater from disinfection as part of the disinfection submittal. Disinfect new water piping and existing water piping affected by Contractor's operations in accordance with [AWWA C651](#). Disinfect new water piping using the [AWWA C601](#) Standard for Disinfection Water Mains and [AWWA C651](#). Ensure a free chlorine residual of not less than 10 parts per million after 24 hour holding period and prior to performing bacteriological tests. Disinfection shall comply with applicable requirements of all governing authorities. The contractor shall obtain and pay for all required permits, fees, and inspections by authorities having jurisdiction, including but not limited to PRASA. The contractor shall make all necessary provisions to get and dispose all water required the for disinfection process.

- a. Perform bacteriological tests as required by PRASA and Puerto Rico Department of Health, and in accordance with latest [AWWA C601](#) standards.
- b. Dosage to produce minimum 10 ppm after minimum of 24 hour contact period.
- c. After treatment, flush with clean water until residual chlorine content less than 0.2 ppm.
- d. Prevent contamination of water in existing water mains. Dispose of water used in disinfecting and flushing in acceptable manner.

3.4 MANUFACTURER'S SERVICES

For each type of valve furnished, the Contractor shall furnish the services of a manufacturer's a qualified factory-trained servicemen who

shall assist as necessary, in the installation of the equipment, check the installation before it is placed into operation, supervise testing, and supervise initial operations.

3.5 SYSTEM STARTUP

For each type of valve furnished, the Contractor shall furnish the services of a manufacturer's qualified factory-trained servicemen who shall assist as necessary, in the installation of the equipment, check the installation before it is placed into operation, supervise testing, and supervise initial operations.

3.6 CLEANUP

Upon completion of the installation of water lines and appurtenances, remove all debris and surplus materials resulting from the work.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 30 00

SANITARY SEWERAGE

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY CONTROL
 - 1.3.1 Installer Qualifications
- 1.4 DELIVERY, STORAGE, AND HANDLING
 - 1.4.1 Delivery and Storage
 - 1.4.1.1 Piping
 - 1.4.1.2 Cement, Aggregate, and Reinforcement
 - 1.4.2 Handling
- 1.5 BYPASS PUMPING

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
 - 2.1.1 Sanitary Sewer Gravity Pipeline
- 2.2 MATERIALS
 - 2.2.1 Gravity Pipe
 - 2.2.1.1 Ductile Iron Gravity Sewer Pipe and Associated Fittings
 - 2.2.1.1.1 Ductile Iron Gravity Pipe and Fittings
 - 2.2.2 Ceramic Epoxy Lining
 - 2.2.3 Pipe Casing
 - 2.2.4 Polyethylene Encasement
 - 2.2.5 Portland Cement
 - 2.2.6 Portland Cement Concrete
 - 2.2.7 Precast Concrete Manholes
 - 2.2.8 Gaskets and Resilient Manhole Connectors
 - 2.2.9 External Preformed Rubber Joint Seals
 - 2.2.10 Frames, Covers, and Gratings for Manholes
 - 2.2.11 Manhole Steps
 - 2.2.12 Sluice Gates
 - 2.2.12.1 Frame
 - 2.2.12.2 Slide
 - 2.2.12.3 Seals
 - 2.2.12.4 Operating Stem
 - 2.2.12.5 Stem Guides
 - 2.2.12.6 Manual Operator Mechanism
 - 2.2.12.7 Welds
 - 2.2.12.8 Anchor Bolts
 - 2.2.13 HDPE Lining for Concrete Structures
- 2.3 Concrete Box Penetrations

PART 3 EXECUTION

- 3.1 PREPARATION

- 3.1.1 Installation Drawings
- 3.2 INSTALLATION
 - 3.2.1 Connections to Existing Lines
 - 3.2.2 General Requirements for Installation of Pipelines
 - 3.2.2.1 Earthwork
 - 3.2.2.2 Pipe Laying and Jointing
 - 3.2.3 Special Requirements
 - 3.2.3.1 Installation of Ductile Iron Gravity Sewer Pipe
 - 3.2.4 Manhole Construction
 - 3.2.5 Sluice Gates
 - 3.2.5.1 Sluice Gate Operation
 - 3.2.6 Miscellaneous Construction and Installation
 - 3.2.6.1 Connecting to Existing Manholes
 - 3.2.6.2 Metal Work
 - 3.2.6.2.1 Workmanship and Finish
 - 3.2.6.2.2 Field Painting
- 3.3 FIELD QUALITY CONTROL
 - 3.3.1 Pipe Leakage Tests
 - 3.3.1.1 Infiltration Test
 - 3.3.1.2 Exfiltration Test
 - 3.3.2 Low-Pressure Air Test
 - 3.3.3 Inspection
- 3.4 SLUICE GATE TESTING AND ACCEPTANCE
 - 3.4.1 Sluice Gate Leakage Field Test
 - 3.4.2 Acceptance Trial Operation And Test

-- End of Section Table of Contents --

SECTION 33 30 00

SANITARY SEWERAGE

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C104/A21.4	(2016) Cement-Mortar Lining for Ductile-Iron Pipe and Fittings for Water
AWWA C105/A21.5	(2018) Polyethylene Encasement for Ductile-Iron Pipe Systems
AWWA C111/A21.11	(2017) Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings
AWWA C561	(2014) Fabricated Stainless-Steel Slide Gates
AWWA C600	(2017) Installation of Ductile-Iron Mains and Their Appurtenances

AMERICAN WELDING SOCIETY (AWS)

AWS D1.2/D1.2M	(2014; Errata 1 2014; Errata 2 2020) Structural Welding Code - Aluminum
----------------	---

ASTM INTERNATIONAL (ASTM)

ASTM A48/A48M	(2003; R 2016) Standard Specification for Gray Iron Castings
ASTM A240/A240M	(2020) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A276/A276M	(2017) Standard Specification for Stainless Steel Bars and Shapes
ASTM A536	(1984; R 2019; E 2019) Standard Specification for Ductile Iron Castings
ASTM A746	(2018) Standard Specification for Ductile Iron Gravity Sewer Pipe
ASTM B117	(2019) Standard Practice for Operating Salt Spray (Fog) Apparatus

ASTM C33/C33M	(2018) Standard Specification for Concrete Aggregates
ASTM C94/C94M	(2020) Standard Specification for Ready-Mixed Concrete
ASTM C150/C150M	(2020) Standard Specification for Portland Cement
ASTM C443	(2020) Standard Specification for Joints for Concrete Pipe and Manholes, Using Rubber Gaskets
ASTM C478	(2018) Standard Specification for Circular Precast Reinforced Concrete Manhole Sections
ASTM C478M	(2018) Standard Specification for Precast Reinforced Concrete Manhole Sections (Metric)
ASTM C923	(2008; R 2013; E 2016) Standard Specification for Resilient Connectors Between Reinforced Concrete Manhole Structures, Pipes and Laterals
ASTM C972	(2000; R 2011) Compression-Recovery of Tape Sealant
ASTM D412	(2016) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension
ASTM D624	(2000; R 2020) Standard Test Method for Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers
ASTM D714	(2002; R 2017) Standard Test Method for Evaluating Degree of Blistering of Paints
ASTM D2000	(2018) Standard Classification System for Rubber Products in Automotive Applications
ASTM D4020	(2011) Ultra-High-Molecular-Weight Polyethylene Molding and Extrusion Materials
ASTM D4101	(2017) Standard Classification System and Basis for Specification for Polypropylene Injection and Extrusion Materials
ASTM E96/E96M	(2016) Standard Test Methods for Water Vapor Transmission of Materials
ASTM G21	(2015; R 2021; E 2021) Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

ASTM G95 (2013) Standard Test Method for Cathodic
Disbondment Test of Pipeline Coatings
(Attached Cell Method)

EUROPEAN COMMITTEE FOR STANDARDIZATION (CEN/CENELEC)

EN 598 (2009) Ductile Iron Pipes, Fittings,
Accessories and Their Joints for Sewerage
Applications. Requirements and Test Methods

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.27 (Nov 2016) Scaffolds and Rope Descent
Systems

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. **Submittals with a "PR" designation shall also be submitted to the PRASA for review and approval.** Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contractor's License; G, DO, PR

Bypass Pumping Plan; G, DO, PR

SD-02 Shop Drawings

Installation Drawings; G, DO, PR

SD-03 Product Data

Precast Concrete Manholes

Frames, Covers, and Gratings

Gravity Pipe

Resilient Manhole Connector

HDPE Lining For Concrete Structures; G, DO, PR

Sluice Gates; G, DO, PR

Polyethylene Encasement

Ceramic Epoxy Lining; G, DO, PR

SD-06 Test Reports

Infiltration Test; G, DO, PR

Exfiltration Test; G, DO, PR

Low-Pressure Air Test; G, DO, PR

Sluice Gate Operation; G, D0, PR

Acceptance Trial Operation And Test

SD-07 Certificates

Portland Cement

Welder Certification; G, D0, PR

1.3 QUALITY CONTROL

1.3.1 Installer Qualifications

Install specified materials by a licensed underground utility Contractor licensed for such work in the state where the work is to be performed. Verify installing [Contractor's License](#) is current and state certified or state registered.

1.4 DELIVERY, STORAGE, AND HANDLING

1.4.1 Delivery and Storage

Check upon arrival; identify and segregate as to types, functions, and sizes. Store off the ground in a manner affording easy accessibility and not causing excessive rusting or coating with grease or other objectionable materials.

1.4.1.1 Piping

Inspect materials delivered to site for damage; store with minimum of handling. Store materials on site in enclosures or under protective coverings. Store plastic piping and jointing materials and rubber gaskets under cover out of direct sunlight. Do not store materials directly on the ground. Keep inside of pipes and fittings free of dirt and debris.

1.4.1.2 Cement, Aggregate, and Reinforcement

As specified in Section [03 31 01](#) CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS.

1.4.2 Handling

Handle pipe, fittings, and other accessories in such manner as to ensure delivery to the trench in sound undamaged condition. Carry, do not drag, pipe to trench. Store solvents, solvent compounds, lubricants, elastomeric gaskets, and any similar materials required to install the plastic pipe in accordance with the manufacturer's recommendation and discard those materials if the storage period exceeds the recommended shelf life. Discard solvents in use when the recommended pot life is exceeded.

1.5 BYPASS PUMPING

Provide a temporary pumping system when required to maintain existing sanitary sewerage flows within project limits. The temporary pumping system shall be designed with sufficient capacity to pump peak flow of 2200 gpm for the duration of the work required to restore the flows

through the gravity system. The temporary pumping system includes the pumps, suction lines, discharge lines, pipe plugs, and ancillary components.

Submit a [bypass pumping plan](#) with a description of the proposed temporary pumping system which includes the following:

- a. Bypass pump sizes, pump sizing calculations, capacity, and power requirements
- b. Static lift, friction losses, and flow velocity calculations
- c. Pump curves showing operating range
- d. Sewer plug method and type
- e. Suction/Discharge pipe size, material, thrust restraint (if required), and trenching and backfill requirements (if buried)
- f. Method for protecting discharge manholes or structures from erosion and damage
- g. Emergency Plans and procedures in the event of a pump failure, sewerage overflow, service backups, and sewerage spillage.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

2.1.1 Sanitary Sewer Gravity Pipeline

Provide gravity sanitary sewer lines of ductile-iron pipe.

2.2 MATERIALS

Provide materials conforming to the respective specifications and other requirements specified below. Submit manufacturer's product specification, standard drawings or catalog cuts.

2.2.1 [Gravity Pipe](#)

2.2.1.1 Ductile Iron Gravity Sewer Pipe and Associated Fittings

2.2.1.1.1 Ductile Iron Gravity Pipe and Fittings

Provide ductile iron pipe conforming to [ASTM A746](#) with ceramic epoxy lining in conforming to [AWWA C104/A21.4](#), Pressure Class 250. Provide push-on joints conforming to [AWWA C111/A21.11](#).

2.2.2 [Ceramic Epoxy Lining](#)

Provide ceramic epoxy lining which shall be amine cured novalax epoxy containing at least 20 percent by volume ceramic quartz pigment. Pipe lining shall be installed per the manufacturer's requirements. Submit manufacturer's product data and installation requirements. Lining shall meet or exceed the following test requirements:

Test	Required Value	Method
Permeability	0.00 for Procedure A with a duration of 30-days	ASTM E96/E96M

Test	Required Value	Method
Salt Spray (scribed panel)	0.0 undercutting after 2-years	ASTM B117
Cathodic Disbondment	0.5-mm (max) undercutting after 30-days	ASTM G95 (1.5 volts @ 77 degrees F)
Immersion Testing 20% Sulfuric Acid 140°F 25% Sodium Hydroxide 160°F Distilled Water 120°F Tap Water (scribed panel)	No effect after 2-years No effect after 2-years No effect after 2-years 0.0 undercutting after 2-years	ASTM D714
Bacterial Resistance	Trace amounts after 7 days at 30°C	ASTM G21
Abrasion Resistance	3 mils (max) loss after 1-million cycles	EN 598

2.2.3 Pipe Casing

See specification section 33 05 07 PILOT TUBE GUIDED BORING AND JACKING for requirements of the steel pipe casing.

2.2.4 Polyethylene Encasement

Provide a polyethylene encasement for all pipe and fittings which are buried and exposed to soil. Encasement shall be a minimum of 4-mil thick consisting of high density cross laminated polyethylene film in accordance with AWWA C105/A21.5. Provide polyethylene encasement as a flat tube meeting the dimensions of Table 1 in AWWA C105/A21.5. Submit manufacturer's product data.

2.2.5 Portland Cement

Submit certificates of compliance stating the type of cement used in manufacture of precast manholes. Provide portland cement conforming to ASTM C150/C150M, Type II for concrete used in manholes and type optional for cement used in concrete cradle. Use a cement containing less than 0.60 percent alkalis where aggregates are alkali reactive, as determined by Appendix XI of ASTM C33/C33M.

2.2.6 Portland Cement Concrete

Provide portland cement concrete conforming to ASTM C94/C94M, compressive strength of 4000 psi at 28 days, except for concrete cradle. Concrete used for cradle is to have a compressive strength of 2500 psi minimum at 28 days. Protect concrete in place from freezing and moisture loss for 7 days.

2.2.7 Precast Concrete Manholes

Provide precast concrete manholes, risers, base sections, and tops

conforming to [ASTM C478](#).

2.2.8 Gaskets and Resilient Manhole Connectors

Provide gaskets for joints between manhole sections conforming to [ASTM C443](#). Resilient connectors for making joints between manhole and pipes entering manhole are to conform to [ASTM C923](#) with stainless clamps, bolts, or other hardware. Submit [resilient manhole connector](#) manufacturer's product data and installation instructions.

2.2.9 External Preformed Rubber Joint Seals

An external preformed rubber joint seal is an accepted method of sealing cast iron covers to precast concrete sections to prevent ground water infiltration into sewer systems. All finished and sealed manholes constructed in accordance with paragraph entitled "Manhole Construction" are to be tested for leakage in the same manner as pipelines as described in paragraph entitled "Leakage Tests." The seal is to be multi-section with a neoprene rubber top section and all lower sections made of Ethylene Propylene Diene Monomer (EPDM) rubber with a minimum thickness of [60 mils](#). Each unit is to consist of a top and bottom section and have mastic on the bottom of the bottom section and mastic on the top and bottom of the top section. The mastic is to be a non-hardening butyl rubber sealant and seal to the cone/top slab of the manhole/catch basin and over the lip of the casting. Extension sections are to cover up to two more adjusting rings. Properties and values are listed in the following table:

Properties, Test Methods and Minimum Values for Rubber used in Preformed Joint Seals				
Physical Properties	Test Methods	EPDM	Neoprene	Butyl Mastic
Tensile, psi	ASTM D412	1840	2195	--
Elongation, percent	ASTM D412	553	295	350
Tear Resistance, ppi	ASTM D624 (Die B)	280	160	--
Rebound, percent, 5 minutes	ASTM C972 (mod.)	--	--	11
Rebound, percent, 2 hours	ASTM C972	--	--	12

2.2.10 [Frames, Covers, and Gratings](#) for Manholes

Submit certification on the ability of frame and cover to carry the imposed live load. Frame and cover are to be cast gray iron, [ASTM A48/A48M](#), Class 35B, cast ductile iron, [ASTM A536](#), Grade 65-45-12, or reinforced concrete, [ASTM C478](#) [ASTM C478M](#). Frames and covers are to be circular without vent holes. Size are to be as indicated on the plans. Stamp or cast the words "Sanitary Sewer" into covers so that it is plainly visible.

2.2.11 Manhole Steps

Stainless steel conforming to 29 CFR 1910.27 with a plastic coating pressure-molded to the steel is to be used. Core of step shall be 316 stainless steel conforming to ASTM A240/A240M. Provide plastic coating conforming to ASTM D4101, copolymer polypropylene. Steps are not required in manholes less than 4 feet deep.

2.2.12 Sluice Gates

Provide sluice gates of the self-contained type with yoke and bench stand operators compliant with AWWA C561. Sluice gates shall be of the non-rising stem type and actuated manually. Sluice gate shall be rated for a maximum operating head of 40-ft. Gate leaf, frame, and appurtenances shall be of a single manufacturer and be factory assembled to the maximum extent possible. Sluice gates shall provide perimeter seals which meet or exceed the leakage rate requirements in AWWA C561. Submit manufacturer's product data and installation requirements.

2.2.12.1 Frame

Provide stainless steel frames meeting ASTM A240/A240M, Type 316L constructed of members with a minimum 1/4-inch thickness. Frame design shall allow for mounting directly to a wall. The frame shall extend to accommodate the entire height of the sluice gate in the fully opened position. The yoke shall be formed by a structural member(s) and secured to provide a on-piece rigid assembly.

2.2.12.2 Slide

Provide slide meeting ASTM A240/A240M, Type 316L stainless steel constructed of members with a minimum 1/4-inch thickness. Slide shall not deflect more than 1/360 of the span or 1/16-inch, whichever is greater, under the maximum design head. Stiffeners shall be stitch welded to slide.

2.2.12.3 Seals

Provide gates with a self-adjusting seal system to restrict leakage. Invert seals shall be EPDM conforming to ASTM D2000. Stem and seat seals shall be ultra-high molecular weight polyethylene (UHMWPE) conforming to ASTM D4020 to prevent metal to metal contact between the frame and slide.

2.2.12.4 Operating Stem

Provide non-rising type operating stems constructed of 316 stainless steel conforming to ASTM A276/A276M. Provide ACME or stub threading to allow for full operation of the sluice gate plus or minus 2-inches.

2.2.12.5 Stem Guides

Provide stainless steel guides conforming to ASTM A240/A240M Type 316L with UHMW bushings. Stem guides shall be adjustable in two directions, providing at least 1/2-inch of adjustment in either direction.

2.2.12.6 Manual Operator Mechanism

Gates shall be operated by manually by a square nut mounted in a floor box set in the lid of the concrete structure. Manual operation of less than 40-pound effort is required when the gate is in the closed position and

experiencing the maximum operating head. Provide an aluminum or stainless steel T-handle wrench for operation.

2.2.12.7 Welds

Shop welding, welding procedures, and welders shall be qualified and certified in accordance with ASME Section IX or [AWS D1.2/D1.2M](#). Submit [welder certification](#).

2.2.12.8 Anchor Bolts

Manufacturer to provide anchor bolts for mounting the gate and/or appurtenances. Anchor bolts shall be 316 stainless steel conforming to [ASTM A276/A276M](#) of at least 1/2-inch diameter. If epoxy type anchors are provided, the gate manufacturer shall provide the studs and nuts.

2.2.13 [HDPE Lining for Concrete Structures](#)

Provide HDPE coating conforming to PRASA Specification Section 029200 on all exposed interior concrete surfaces of sanitary sewer manholes, influent box, and effluent box. PRASA Specification 029200 is included as a reference within Appendix B. Submit manufacturer's product data and installation requirements.

2.3 Concrete Box Penetrations

Penetrations through concrete box walls and bottom slabs shall be of the sizes and type as shown in the Construction Drawings.

Wall penetration shall be provided with a watertight mechanical elastomeric modular seal with stainless steel wall sleeves with an integrated 1/2" water stop-ring.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Installation Drawings

Submit [Installation Drawings](#) showing complete detail, both plan and side view details with proper layout and elevations. Include the pipe lengths, fittings, and grades accounting for any adjustments made to fit actual verified field conditions.

3.2 INSTALLATION

Backfill after inspection by the Contracting Officer. Before, during, and after installation, protect pipe and fittings from any environment that would result in damage or deterioration to the material. Keep a copy of the manufacturer's instructions available at the construction site at all times and follow these instructions unless approved otherwise by the Contracting Officer.

3.2.1 Connections to Existing Lines

Obtain approval from the Contracting Officer before making connection to existing line. Conduct work so that there is minimum interruption of service on existing line.

3.2.2 General Requirements for Installation of Pipelines

These general requirements apply except where specific exception is made in the following paragraphs entitled "Special Requirements."

3.2.2.1 Earthwork

Perform earthwork operations in accordance with Section 31 00 00 EARTHWORK.

3.2.2.2 Pipe Laying and Jointing

Inspect each pipe and fitting before and after installation; replace those found defective and remove from site. Provide proper facilities for lowering sections of pipe into trenches. Lay nonpressure pipe with the bell ends in the upgrade direction. Adjust spigots in bells to give a uniform space all around. Blocking or wedging between bells and spigots will not be permitted. Replace by one of the proper dimensions, pipe or fittings that do not allow sufficient space for installation of joint material. At the end of each work day, close open ends of pipe temporarily with wood blocks or bulkheads. Provide batterboards not more than 25 feet apart in trenches for checking and ensuring that pipe invert elevations are as indicated. Laser beam method may be used in lieu of batterboards for the same purpose.

3.2.3 Special Requirements

3.2.3.1 Installation of Ductile Iron Gravity Sewer Pipe

Unless otherwise specified, install pipe and associated fittings in accordance with paragraph entitled "General Requirements for Installation of Pipelines" of this section and with the requirements of AWWA C600 for pipe installation and joint assembly.

- a. Make push-on joints with the gaskets and lubricant specified for this type joint and assemble in accordance with the applicable requirements of AWWA C600 for joint assembly. Make mechanical-joints with the gaskets, glands, bolts, and nuts specified for this type joint and assemble in accordance with the applicable requirements of AWWA C600 for joint assembly and the recommendations of Appendix A to AWWA C111/A21.11.
- b. Exterior protection: Completely encase buried ductile iron pipelines with polyethylene tube or sheet in accordance with AWWA C105/A21.5, using Class A polyethylene film.

3.2.4 Manhole Construction

Construct base slab of precast concrete base sections. Make inverts in precast concrete bases with a smooth-surfaced semi-circular bottom conforming to the inside contour of the adjacent sewer sections. For changes in direction of the sewer and entering branches into the manhole, make a circular curve in the manhole invert of as large a radius as manhole size will permit. For precast concrete construction, make joints between manhole sections with the gaskets specified for this purpose; install in the manner specified for installing joints in concrete piping. Parging will not be required for precast concrete manholes. Make joints between concrete manholes and pipes entering manholes with the resilient connectors specified for this purpose; install in accordance with the recommendations of the connector manufacturer. Where a new manhole is

constructed on an existing line, remove existing pipe as necessary to construct the manhole. Cut existing pipe so that pipe ends are approximately flush with the interior face of manhole wall, but not protruding into the manhole. Use resilient connectors as previously specified for pipe connectors to concrete manholes.

3.2.5 Sluice Gates

Install sluice gates per the manufacturer's instructions. Gates and assemblies shall be installed in a true vertical plane, square, and plumb. Fill voids between the gate frame and wall with non-shrink grout in accordance with the manufacturer's recommendations.

3.2.5.1 Sluice Gate Operation

After installation is complete, cycle each gate in the presence of the Contracting Officer to confirm operation without binding, scraping, or distorting. Measure and submit the required effort to operate each gate and confirm it is equal to or less than the value specified in Section "Manual Operator Mechanism" within this specification.

3.2.6 Miscellaneous Construction and Installation

3.2.6.1 Connecting to Existing Manholes

Connect pipe to existing manholes such that finish work will conform as nearly as practicable to the applicable requirements specified for new manholes, including all necessary concrete work, cutting, and shaping. Center the connection on the manhole. Holes for the new pipe are to be of sufficient diameter to allow packing cement mortar around the entire periphery of the pipe but no larger than 1.5 times the diameter of the pipe. Cut the manhole in a manner that will cause the least damage to the walls.

3.2.6.2 Metal Work

3.2.6.2.1 Workmanship and Finish

Perform metal work so that workmanship and finish will be equal to the best practice in modern structural shops and foundries. Form iron to shape and size with sharp lines and angles. Do shearing and punching so that clean true lines and surfaces are produced. Make castings sound and free from warp, cold shuts, and blow holes that may impair their strength or appearance. Give exposed surfaces a smooth finish with sharp well-defined lines and arises. Provide necessary rabbets, lugs, and brackets wherever necessary for fitting and support.

3.2.6.2.2 Field Painting

After installation, clean cast-iron frames, covers, gratings, and steps not buried in concrete to bare metal, remove mortar, rust, grease, dirt, and other deleterious materials and apply a coat of bituminous paint. Do not paint surfaces subject to abrasion.

3.3 FIELD QUALITY CONTROL

The Contractor is responsible for quality control, however, the Contracting Officer will conduct field inspections and witness field tests specified in this section. Contractor is responsible for performing all

field tests and providing test reports. Be able to produce evidence, when required, that each item of work has been constructed in accordance with the drawings and specifications.

3.3.1 Pipe Leakage Tests

Perform leakage testing after the pipe has been laid and backfilled. Testing shall be done in short sections between manholes and between a manhole and influent/effluent box. Perform infiltration test when the groundwater is at least one foot above the top of the pipe to be tested. Perform exfiltration tests when the groundwater is less than one foot above the pipe crown. The maximum allowable amount of infiltration and exfiltration measured by test shall be at a rate of not greater than 50-gallons per inch of pipe diameter per mile per 24 hours. Repairs required due to a failed test shall be completed at the Contractor's expense.

3.3.1.1 Infiltration Test

Infiltration test shall be completed as follows:

1. Install plugs at both ends of pipe to be tested.
2. Install suitable measuring device at the lower end
3. Perform test at the lower end of the pipe by means of a measurement device such as a V-notch weir, Parshall flume, or by direct measurement of volume and time. After measuring device is installed allow sufficient time for the infiltration water to build up and level off behind the weir until a steady uniform flow is observed.
4. Measure the amount of water flowing through the outlet over a specified period of time.

3.3.1.2 Exfiltration Test

Exfiltration test shall be completed as follows:

1. Install plugs at both ends of pipe to be tested, providing suitable means for total air removal.
2. Pipe segment to be tested shall be filled with water to a minimum height, not less than 3 feet, over the top of the pipeline at the upstream manhole.
3. Allow water to stand for at least 4 hours or overnight.
4. Refill manhole to original elevation
5. Observe either the drop in elevation in 30 minutes or the quantity of water required to maintain the reach in full condition for 30 minutes.
6. Subtract manhole loss to get actual line loss.

3.3.2 Low-Pressure Air Test

Perform separate low-pressure air tests on the portions of the pipe between the drilled shafts within the casing, between the drilled shaft

wall and influent box, and between the drilled shaft wall and effluent box after the pipe has been laid and backfilled. Low-pressure air test shall be performed as follows:

1. Plug the test section at each end. One of the plugs used at the manhole must be tapped and equipped for the air inlet connection for filling the lines from the air compressor.
2. All openings within the sewer test section should be properly capped or plugged, and carefully braced against the internal pressure to prevent air leakage by slippage and blowouts.
3. Connect air hose to tapped plug selected for the air inlet. Then connect the other end of air hose to the portable air control equipment which consists of valves and pressure gauges used to control:
 - a. The air entry rate to the sewer test section
 - b. To monitor the air pressure in the pipe line. More specifically, the air control equipment includes a shut-off valve, pressure reduction valve and a monitoring pressure gauge having a pressure range from 0 to 5 psi. the gauge should have minimum divisions of 1.0 psi and an accuracy of ± 0.04 psi.
4. Connect another air hose between the air compressor (or other source of compressed air) and the air control equipment. This completes the test equipment set-up. Test operations may commence.
5. Supply air to the test section slowly, filling the pipe line until a constant pressure of 3.5 psi is maintained. The air pressure must be regulated to prevent the pressure inside the pipe from exceeding 5.0 psig.
6. When the constant pressure of 5.0 psig is reached, throttle the air supply to maintain the internal pressure above 3.0 psig for at least 5 minutes. This time permits the temperature of the pipe wall. During this stabilization period it is advisable to check all capped and plugged fittings with soap solution to detect any leakage at these connections. If leakage is detected at any cap or plug, release the pressure in the pressure in the line and tighten all leaky caps and plugs. Then start the test operation again by supplying air. When it is necessary to bleed off the air to tighten or repair a faulty plug, a new five-minute interval must be allowed after the pipeline has been refilled.
7. After the stabilization period, adjust the air pressure to 3.5 psig and shut off or disconnect the air supply. Observe the gauge until the air pressure reaches 3.0 psig. At 3.0 psig commence timing with a stop watch which is allowed to run until the line pressure drops to 2.5 psig at which time the stop watch is stopped. The time required, as shown on the stopwatch, for pressure loss of 0.5 psig is used to compute the air loss. Most authorities consider it necessary to determine the air temperature inside the pipe and the barometric pressure at the time of the test.
8. If the time, in minutes and seconds, for the air pressure to drop from 3.0 to 2.5 psig is greater than 12 minutes and 45 seconds, the section undergoing test shall have passed and shall be presumed to be free of defects. The test may be discontinued at the time.

9. If the time in minutes and seconds, for the 0.5 psig drop is less than 12 minutes and 45 seconds, the section of the pipe shall not have passed the test; therefore, adequate repairs must be made and the line retested.

3.3.3 Inspection

Check each straight run of pipeline for gross deficiencies by holding a light in a manhole; the light must show a practically full circle of light through the pipeline when viewed from the adjoining end of line.

3.4 SLUICE GATE TESTING AND ACCEPTANCE

3.4.1 Sluice Gate Leakage Field Test

Perform a field leakage test after installation is complete. Allowable leakage rates for seating head conditions shall be less than 0.05 gallons per minute per foot of perimeter seal. Allowable leakage rates for unseating head conditions shall be less than 0.10 gallons per minute per linear foot of perimeter seal.

3.4.2 [Acceptance Trial Operation And Test](#)

After the gate assembly has been installed, including operating machinery, the Contracting Officer will examine the complete system for final acceptance. Operation and test results shall be furnished to the Contracting Officer. The assembly will be examined first to determine whether or not the workmanship conforms to the specification requirements. Operate the gate throughout its full operating range a sufficient number of times to demonstrate proper operation. The initial operation of the gate assembly shall be conducted in the dry. The second trial operation and testing of the gate assembly shall be conducted in the wet. The workmanship in the fabrication and installation of the gate assembly shall be such that the gate leaf shall form a watertight barrier when lowered to the seated position. Adjustments shall be made to the operation and control apparatus until all components function as required. Required repairs or replacements to correct defects, as determined by the Contracting Officer, shall be made at no additional cost to the Government. The trial operation and testing shall be repeated after defects are corrected.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 40 00

STORM DRAINAGE UTILITIES

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 DELIVERY, STORAGE, AND HANDLING
 - 1.3.1 Delivery and Storage
 - 1.3.2 Handling

PART 2 PRODUCTS

- 2.1 PIPE FOR CULVERTS AND STORM DRAINS
 - 2.1.1 Concrete Pipe
 - 2.1.2 PVC Pipe for Temporary Storm Drains
- 2.2 DRAINAGE STRUCTURES
 - 2.2.1 Flared End Sections
 - 2.2.2 Precast Reinforced Concrete Box
- 2.3 MISCELLANEOUS MATERIALS
 - 2.3.1 Concrete
 - 2.3.2 Mortar
 - 2.3.3 Precast Concrete Segmental Blocks
 - 2.3.4 Brick
 - 2.3.5 Precast Reinforced Concrete Manholes
 - 2.3.6 Precast Reinforced Drop Inlet
 - 2.3.7 Frame and Cover for Gratings
 - 2.3.8 Drainage Inlet
 - 2.3.9 Flap Gates
 - 2.3.10 Joints
 - 2.3.10.1 Flexible Watertight Joints
 - 2.3.10.2 External Sealing Bands
 - 2.3.10.3 Flexible Watertight, Gasketed Joints
- 2.4 LADDER RUNGS
- 2.5 RESILIENT CONNECTORS
- 2.6 EROSION CONTROL RIP RAP

PART 3 EXECUTION

- 3.1 INSTALLATION OF PIPE CULVERTS, STORM DRAINS, AND DRAINAGE STRUCTURES
 - 3.1.1 Trenching
 - 3.1.2 Removal of Rock
 - 3.1.3 Removal of Unstable Material
 - 3.1.4 Dewatering
 - 3.1.5 Stormwater Pump Station Outfall Diversion Plan
- 3.2 BEDDING
 - 3.2.1 Concrete Pipe Requirements
- 3.3 PLACING PIPE

- 3.3.1 Concrete Pipe
- 3.3.2 Elliptical and Elliptical Reinforced Concrete Pipe
- 3.3.3 Multiple Culverts
- 3.3.4 Jacking Pipe Through Fills
- 3.4 JOINTING
 - 3.4.1 Concrete and Clay Pipe
 - 3.4.1.1 Cement-Mortar Bell-and-Spigot Joint
 - 3.4.1.2 Cement-Mortar Oakum Joint for Bell-and-Spigot Pipe
 - 3.4.1.3 Cement-Mortar Diaper Joint for Bell-and-Spigot Pipe
 - 3.4.1.4 Flexible Watertight Joints
 - 3.4.1.5 External Sealing Band Joint for Noncircular Pipe
- 3.5 DRAINAGE STRUCTURES
 - 3.5.1 Manholes and Inlets
 - 3.5.2 Walls and Headwalls
- 3.6 BACKFILLING
 - 3.6.1 Backfilling Pipe in Trenches
 - 3.6.2 Backfilling Pipe in Fill Sections
 - 3.6.3 Movement of Construction Machinery
 - 3.6.4 Compaction
 - 3.6.4.1 General Requirements
 - 3.6.4.2 Minimum Density
- 3.7 FIELD PAINTING
 - 3.7.1 Cast-Iron Covers, Frames, Gratings, And Steps
- 3.8 FIELD QUALITY CONTROL
 - 3.8.1 Tests
 - 3.8.1.1 HYDROSTATIC TEST ON WATERTIGHT JOINTS
 - 3.8.1.2 Leakage Test
 - 3.8.1.3 Determination of Density
 - 3.8.2 Inspection
 - 3.8.2.1 Post-Installation Inspection
 - 3.8.2.1.1 Concrete
 - 3.8.2.1.2 Post-Installation Inspection Report
 - 3.8.3 Repair Of Defects
 - 3.8.3.1 Leakage Test
 - 3.8.3.2 Deflection Testing
 - 3.8.3.3 Inspection
 - 3.8.3.3.1 Concrete
- 3.9 PROTECTION
- 3.10 WARRANTY PERIOD

-- End of Section Table of Contents --

SECTION 33 40 00
STORM DRAINAGE UTILITIES

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN RAILWAY ENGINEERING AND MAINTENANCE-OF-WAY ASSOCIATION
(AREMA)

AREMA Eng Man (2017) Manual for Railway Engineering

ASTM INTERNATIONAL (ASTM)

ASTM A48/A48M (2003; R 2016) Standard Specification for
Gray Iron Castings

ASTM A240/A240M (2020) Standard Specification for Chromium
and Chromium-Nickel Stainless Steel Plate,
Sheet, and Strip for Pressure Vessels and
for General Applications

ASTM A536 (1984; R 2019; E 2019) Standard
Specification for Ductile Iron Castings

ASTM A929/A929M (2018) Standard Specification for Steel
Sheet, Metallic-Coated by the Hot-Dip
Process for Corrugated Steel Pipe

ASTM B26/B26M (2014; E 2015) Standard Specification for
Aluminum-Alloy Sand Castings

ASTM C32 (2013; R 2017) Standard Specification for
Sewer and Manhole Brick (Made from Clay or
Shale)

ASTM C55 (2017) Standard Specification for Concrete
Building Brick

ASTM C62 (2017) Standard Specification for Building
Brick (Solid Masonry Units Made from Clay
or Shale)

ASTM C76 (2020) Standard Specification for
Reinforced Concrete Culvert, Storm Drain,
and Sewer Pipe

ASTM C139 (2017) Standard Specification for Concrete
Masonry Units for Construction of Catch
Basins and Manholes

ASTM C270	(2019) Standard Specification for Mortar for Unit Masonry
ASTM C425	(2004; R 2013) Standard Specification for Compression Joints for Vitrified Clay Pipe and Fittings
ASTM C443	(2020) Standard Specification for Joints for Concrete Pipe and Manholes, Using Rubber Gaskets
ASTM C478	(2018) Standard Specification for Circular Precast Reinforced Concrete Manhole Sections
ASTM C877	(2008) External Sealing Bands for Concrete Pipe, Manholes, and Precast Box Sections
ASTM C923	(2008; R 2013; E 2016) Standard Specification for Resilient Connectors Between Reinforced Concrete Manhole Structures, Pipes and Laterals
ASTM C969	(2019) Standard Practice for Infiltration and Exfiltration Acceptance Testing of Installed Precast Concrete Pipe Sewer Lines
ASTM C990	(2009; R 2014) Standard Specification for Joints for Concrete Pipe, Manholes and Precast Box Sections Using Preformed Flexible Joint Sealants
ASTM C1103	(2019) Standard Practice for Joint Acceptance Testing of Installed Precast Concrete Pipe Sewer Lines
ASTM C1433	(2020) Standard Specification for Precast Reinforced Concrete Monolithic Box Sections for Culverts, Storm Drains, and Sewers
ASTM D1056	(2014) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber
ASTM D1171	(2016; E 2016) Standard Test Method for Rubber Deterioration - Surface Ozone Cracking Outdoors (Triangular Specimens)
ASTM D1557	(2012; E 2015) Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft ³) (2700 kN-m/m ³)
ASTM D2167	(2015) Density and Unit Weight of Soil in Place by the Rubber Balloon Method
ASTM D6938	(2017a) Standard Test Method for In-Place

Density and Water Content of Soil and
Soil-Aggregate by Nuclear Methods (Shallow
Depth)

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Outfall Diversion Plan; G, R0

SD-04 Samples

Pipe for Culverts and Storm Drains; G, R0

SD-07 Certificates

Oil Resistant Gasket

Leakage Test; G, R0

Hydrostatic Test on Watertight Joints; G, R0

Flap Gates; G, R0

Determination of Density

Frame and Cover for Gratings; G, R0

Post-Installation Inspection Report; G, R0

SD-08 Manufacturer's Instructions

Placing Pipe

1.3 DELIVERY, STORAGE, AND HANDLING

1.3.1 Delivery and Storage

Materials delivered to site shall be inspected for damage, unloaded, and stored with a minimum of handling. Materials shall not be stored directly on the ground. The inside of pipes and fittings shall be kept free of dirt and debris. Before, during, and after installation, plastic pipe and fittings shall be protected from any environment that would result in damage or deterioration to the material. Keep a copy of the manufacturer's instructions available at the construction site at all times and follow these instructions unless approved otherwise by the Contracting Officer. Solvents, solvent compounds, lubricants, elastomeric gaskets, and any similar materials required to install plastic pipe shall be stored in accordance with the manufacturer's recommendations and shall be discarded if the storage period exceeds the recommended shelf life. Solvents in use shall be discarded when the recommended pot life is exceeded.

1.3.2 Handling

Materials shall be handled in a manner that ensures delivery to the trench in sound, undamaged condition. Pipe shall be carried to the trench, not dragged.

PART 2 PRODUCTS

2.1 PIPE FOR CULVERTS AND STORM DRAINS

Pipe for culverts and storm drains shall be of the sizes indicated and shall conform to the requirements specified.

2.1.1 Concrete Pipe

Manufactured in accordance with and conforming to ASTM C76, Class III unless otherwise specified.

2.1.2 PVC Pipe for Temporary Storm Drains

Manufactured in accordance with and conforming to ASTM F-949 unless otherwise specified.

2.2 DRAINAGE STRUCTURES

2.2.1 Flared End Sections

Sections shall be of a standard design fabricated from zinc coated steel sheets meeting requirements of ASTM A929/A929M.

2.2.2 Precast Reinforced Concrete Box

Manufactured in accordance with and conforming to ASTM C1433.

2.3 MISCELLANEOUS MATERIALS

2.3.1 Concrete

Unless otherwise specified, concrete shall conform to the requirements for Class A concrete under section 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS.

2.3.2 Mortar

Mortar for pipe joints, connections to other drainage structures, and brick or block construction shall conform to ASTM C270, Type M, except that the maximum placement time shall be 1 hour. The quantity of water in the mixture shall be sufficient to produce a stiff workable mortar. Water shall be clean and free of harmful acids, alkalis, and organic impurities. The mortar shall be used within 30 minutes after the ingredients are mixed with water. The inside of the joint shall be wiped clean and finished smooth. The mortar head on the outside shall be protected from air and sun with a proper covering until satisfactorily cured.

2.3.3 Precast Concrete Segmental Blocks

Precast concrete segmental block shall conform to ASTM C139, not more than 8 inches thick, not less than 8 inches long, and of such shape that joints

can be sealed effectively and bonded with cement mortar.

2.3.4 Brick

Brick shall conform to [ASTM C62](#), Grade SW; [ASTM C55](#), Grade S-I or S-II; or [ASTM C32](#), Grade MS. Mortar for jointing and plastering shall consist of one part portland cement and two parts fine sand. Lime may be added to the mortar in a quantity not more than 25 percent of the volume of cement. The joints shall be filled completely and shall be smooth and free from surplus mortar on the inside of the structure. Brick structures shall be plastered with [1/2 inch](#) of mortar over the entire outside surface of the walls. For square or rectangular structures, brick shall be laid in stretcher courses with a header course every sixth course. For round structures, brick shall be laid radially with every sixth course a stretcher course.

2.3.5 Precast Reinforced Concrete Manholes

Conform to [ASTM C478](#). Joints between precast concrete risers and tops shall be full-bedded in cement mortar and shall be smoothed to a uniform surface on both interior and exterior of the structure.

2.3.6 Precast Reinforced Drop Inlet

Precast reinforced drop inlet shall be of the type shown on the plans.

2.3.7 Frame and Cover for Gratings

Submit certification on the ability of frame and cover or gratings to carry the imposed live load. Frame and cover for gratings shall be of the type shown on the plans and cast gray iron, [ASTM A48/A48M](#), Class 35B; cast ductile iron, [ASTM A536](#), Grade 65-45-12; or cast aluminum, [ASTM B26/B26M](#), Alloy 356.0-T6. Weight, shape, size, and waterway openings for grates and curb inlets shall be as indicated on the plans. The word "Storm Sewer" shall be stamped or cast into covers so that it is plainly visible.

2.3.8 Drainage Inlet

Drainage inlet shall be of the type shown on the plans.

2.3.9 Flap Gates

a. Flap Gates shall be of the type shown on the plans with an engineered composite fiberglass reinforced polyester (FRP) gate body. The flap shall be one-piece hinge/seal assembly of molded neoprene rubber for 4 inch to 18 inch size. The neoprene flap shall have a steel plate vulcanized into it and a raised rounded 1/8" (3mm) wide seating surface. Flap material for 20 inch to 30 inch size shall be FRP plate bonded and fastened to neoprene rubber. Hinge material shall be one-piece hinge/seal assembly of molded neoprene rubber.

b. Composition of the laminate shall be in accordance with the recommendations shown in the Quality Assurance Report for Reinforced Thermoplastic (RTP) Corrosion Resistance Equipment prepared under the sponsorship of the Society of the Plastics Industry, Inc. (SPI) and the Material Technology Institute (MTI) of the Chemical Process Industry for "Hand Lay-up Laminates" and shall meet the specifications for Type I, Grade 10 Laminates shown in Appendix. M-1 of said report. Visual inspection for defects shall be made without the aid of magnification, and

defects shall be classified as shown in Table 1 Level II of ANSI/ASTM D2563-0, approved 1977, (or any subsequent revision). Deflection across the gate width shall be limited to: L/360 or 1/4" (6mm), whichever is less, at the maximum operating head. Gate diameter as shown on the plans and/or gate schedule.

c. Each gate shall be molded individually to the exact dimensions specified. Flap gate bodies shall be manufactured of gray fiberglass reinforced polyester (FRP) containing ultraviolet absorbers. The surfaces shall be resin rich to a depth of 0.010-0.020 inches (0.3-.05mm) and reinforced with C-glass or polymeric fiber surfacing material. The surface shall be free of exposed reinforcing fibers. The composition of these layers shall be approximately 95% (by weight) resin. The remaining laminate shall be made of resin and reinforcing fibers in a form, orientation, and position in the laminate to meet the mechanical requirements. The body flange shall be drilled for mounting directly using 1/2" (13mm) diameter mounting bolts. Elastomeric flap/seal/hinge shall be made of molded neoprene having a hardness range of 45 - 65 shore A durometer.

d. Structural characteristics for a 1/8" (3mm) glass mat laminate shall meet the following minimum physical properties.

Tensile Strength	11,000 psi (1034 ksc)
Flexural Modulus	900,000 psi (70307 ksc)
Flexural Strength	20,000 psi (1406 ksc)
Compressive Strength	20,000 psi (1547 ksc)
Impact Strength	8.5 ft-lbs./in/ (1.24 kgf*m/25 mm)
Water Absorption	0.40% (in 24 hours)

e. Flap/seal/hinge: Neoprene shall have the following physical characteristics.

Specific Gravity	1.25
Hardness	55 - 65 Shore A
Tensile Strength	1500 psi min. (0.07 ksc)
Elongation	300% min.
Low temperature	- 40 deg

f. Contractor shall provide shop drawings showing all critical flap gate dimensions and showing principal parts and materials to the Contracting Officer for approval prior to construction.

g. Install gates per manufacturer's recommendations. Consult operation

and maintenance manual.

2.3.10 Joints

2.3.10.1 Flexible Watertight Joints

- a. Flexible watertight joints shall be made with plastic or rubber-type gaskets for concrete pipe and with factory-fabricated resilient materials for clay pipe. The design of joints and the physical requirements for preformed flexible joint sealants shall conform to [ASTM C990](#), and rubber-type gaskets shall conform to [ASTM C443](#). Factory-fabricated resilient joint materials shall conform to [ASTM C425](#). Gaskets shall have not more than one factory-fabricated splice, except that two factory-fabricated splices of the rubber-type gasket are permitted if the nominal diameter of the pipe being gasketed exceeds [54 inches](#).
- b. Rubber gaskets shall comply with the [oil resistant gasket](#) requirements of [ASTM C443](#). Certified copies of test results shall be delivered to the Contracting Officer before gaskets or jointing materials are installed. Alternate types of watertight joint may be furnished, if specifically approved.

2.3.10.2 External Sealing Bands

Requirements for external sealing bands shall conform to [ASTM C877](#).

2.3.10.3 Flexible Watertight, Gasketed Joints

- a. Gaskets: When infiltration or exfiltration is a concern for pipe lines, the couplings may be required to have gaskets. The closed-cell expanded rubber gaskets shall be a continuous band approximately [7 inches](#) wide and approximately [3/8 inch](#) thick, meeting the requirements of [ASTM D1056](#), Type 2, and shall have a quality retention rating of not less than 70 percent when tested for weather resistance by ozone chamber exposure, Method B of [ASTM D1171](#). Rubber O-ring gaskets shall be [13/16 inch](#) in diameter for pipe diameters of [36 inches](#) or smaller and [7/8 inch](#) in diameter for larger pipe having [1/2 inch](#) deep end corrugation. Rubber O-ring gaskets shall be [1-3/8 inches](#) in diameter for pipe having [1 inch](#) deep end corrugations. O-rings shall meet the requirements of [ASTM C990](#) or [ASTM C443](#). Preformed flexible joint sealants shall conform to [ASTM C990](#), Type B.
- b. Connecting Bands: Connecting bands shall be of the type, size and sheet thickness of band, and the size of angles, bolts, rods and lugs as indicated or where not indicated as specified in the applicable standards or specifications for the pipe. Exterior rivet heads in the longitudinal seam under the connecting band shall be countersunk or the rivets shall be omitted and the seam welded. Watertight joints shall be tested and shall meet the test requirements of paragraph HYDROSTATIC TEST ON WATERTIGHT JOINTS.

2.4 LADDER RUNGS

Ladder rungs shall be provided where the depth of the storm drainage structure exceeds [4 feet](#) and shall be cast-in-place in structures. These ladders shall be not less than [14 inches](#) in width, with [1/2 inch](#) minimum diameter rungs spaced [14 inches](#) (maximum) apart. The two stringers (if needed) shall be a minimum [3/8 inch](#) thick and [2-1/2 inches](#) wide. Ladder

rung reinforcement shall be Grade 316 stainless steel conformance with
ASTM A240/A240M.

2.5 RESILIENT CONNECTORS

Flexible, watertight connectors used for connecting pipe to manholes and inlets shall conform to ASTM C923.

2.6 EROSION CONTROL RIP RAP

Unless otherwise specified, rip rap shall conform to the applicable portions of Section 35 31 19 STONE, CHANNEL, SHORELINE/COASTAL PROTECTION FOR STRUCTURES.

PART 3 EXECUTION

3.1 INSTALLATION OF PIPE CULVERTS, STORM DRAINS, AND DRAINAGE STRUCTURES

Excavation of trenches, and for appurtenances and backfilling for culverts and storm drains, shall be in accordance with the applicable portions of Section 31 00 00 EARTHWORK and the requirements specified below.

3.1.1 Trenching

The width of trenches at any point below the top of the pipe shall be not greater than as provided in the plans to permit satisfactory jointing and thorough tamping of the bedding material under and around the pipe. Sheeting and bracing, where required, shall be placed within the trench width as specified, without any overexcavation. Where trench widths are exceeded, redesign with a resultant increase in cost of stronger pipe or special installation procedures will be necessary. Cost of this redesign and increased cost of pipe or installation shall be borne by the Contractor without additional cost to the Government.

3.1.2 Removal of Rock

Rock in either ledge or boulder formation shall be replaced with suitable materials to provide a compacted earth cushion having a thickness between unremoved rock and the pipe of at least 8 inches or 1/2 inch for each foot of fill over the top of the pipe, whichever is greater, but not more than three-fourths the nominal diameter of the pipe. Where bell-and-spigot pipe is used, the cushion shall be maintained under the bell as well as under the straight portion of the pipe. Rock excavation shall be as specified and defined in Section 31 00 00 EARTHWORK.

3.1.3 Removal of Unstable Material

Removal of unstable material requirements are provided in Section 31 00 00 EARTHWORK.

3.1.4 Dewatering

Dewatering requirements are provided in Section 31 00 00 EARTHWORK.

3.1.5 Stormwater Pump Station Outfall Diversion Plan

Provide a work plan describing how the existing discharge will be maintained during the installation of the drilled shafts. The rated capacity of the two existing pumps is 234 cfs. Bypass sheets at the outlet

shall protect to elevation +3.0 at minimum. The existing discharge conduits and flap gates are to remain. The plan shall include, at minimum, temporary pumping capacity, temporary diversion structures, and provisions for preparing the site during large storm events or evacuation plan. See the contract drawings for additional information.

3.2 BEDDING

The bedding surface for the pipe shall provide a firm foundation of uniform density throughout the entire length of the pipe. Pipe bedding shall be as specified and defined in Section 31 00 00 EARTHWORK.

3.2.1 Concrete Pipe Requirements

Bell holes and depressions for joints shall be removed and formed so entire barrel of pipe is uniformly supported. The bell hole and depressions for the joints shall be not more than the length, depth, and width required for properly making the particular type of joint.

3.3 PLACING PIPE

Each pipe shall be thoroughly examined before being laid; defective or damaged pipe shall not be used. Pipelines shall be laid to the grades and alignment indicated. Proper facilities shall be provided for lowering sections of pipe into trenches. Lifting lugs in vertically elongated pipe shall be placed in the same vertical plane as the major axis of the pipe. Pipe shall not be laid in water, and pipe shall not be laid when trench conditions or weather are unsuitable for such work. Diversion of drainage during construction shall be provided as necessary.

3.3.1 Concrete Pipe

Laying shall proceed upgrade with spigot ends of bell-and-spigot pipe and tongue ends of tongue-and-groove pipe pointing in the direction of the flow.

3.3.2 Elliptical and Elliptical Reinforced Concrete Pipe

The manufacturer's reference lines, designating the top of the pipe, shall be within 5 degrees of a vertical plane through the longitudinal axis of the pipe, during placement. Damage to or misalignment of the pipe shall be prevented in all backfilling operations.

3.3.3 Multiple Culverts

Where multiple lines of pipe are installed, adjacent sides of pipe shall be at least half the nominal pipe diameter or 3 feet apart, whichever is less.

3.3.4 Jacking Pipe Through Fills

Methods of operation and installation for jacking pipe through fills shall conform to requirements specified in Volume 1, Chapter 1, Part 4 of AREMA Eng Man.

3.4 JOINTING

3.4.1 Concrete and Clay Pipe

3.4.1.1 Cement-Mortar Bell-and-Spigot Joint

The first pipe shall be bedded to the established grade line, with the bell end placed upstream. The interior surface of the bell shall be thoroughly cleaned with a wet brush and the lower portion of the bell filled with mortar as required to bring inner surfaces of abutting pipes flush and even. The spigot end of each subsequent pipe shall be cleaned with a wet brush and uniformly matched into a bell so that sections are closely fitted. After each section is laid, the remainder of the joint shall be filled with mortar, and a bead shall be formed around the outside of the joint with sufficient additional mortar. If mortar is not sufficiently stiff to prevent appreciable slump before setting, the outside of the joint shall be wrapped or bandaged with cheesecloth to hold mortar in place.

3.4.1.2 Cement-Mortar Oakum Joint for Bell-and-Spigot Pipe

A closely twisted gasket shall be made of jute or oakum of the diameter required to support the spigot end of the pipe at the proper grade and to make the joint concentric. Joint packing shall be in one piece of sufficient length to pass around the pipe and lap at top. This gasket shall be thoroughly saturated with neat cement grout. The bell of the pipe shall be thoroughly cleaned with a wet brush, and the gasket shall be laid in the bell for the lower third of the circumference and covered with mortar. The spigot of the pipe shall be thoroughly cleaned with a wet brush, inserted in the bell, and carefully driven home. A small amount of mortar shall be inserted in the annular space for the upper two-thirds of the circumference. The gasket shall be lapped at the top of the pipe and driven home in the annular space with a caulking tool. The remainder of the annular space shall be filled completely with mortar and beveled at an angle of approximately 45 degrees with the outside of the bell. If mortar is not sufficiently stiff to prevent appreciable slump before setting, the outside of the joint thus made shall be wrapped with cheesecloth. Placing of this type of joint shall be kept at least five joints behind laying operations.

3.4.1.3 Cement-Mortar Diaper Joint for Bell-and-Spigot Pipe

The pipe shall be centered so that the annular space is uniform. The annular space shall be caulked with jute or oakum. Before caulking, the inside of the bell and the outside of the spigot shall be cleaned.

- a. Diaper Bands: Diaper bands shall consist of heavy cloth fabric to hold grout in place at joints and shall be cut in lengths that extend one-eighth of the circumference of pipe above the spring line on one side of the pipe and up to the spring line on the other side of the pipe. Longitudinal edges of fabric bands shall be rolled and stitched around two pieces of wire. Width of fabric bands shall be such that after fabric has been securely stitched around both edges on wires, the wires will be uniformly spaced not less than 8 inches apart. Wires shall be cut into lengths to pass around pipe with sufficient extra length for the ends to be twisted at top of pipe to hold the band securely in place; bands shall be accurately centered around lower portion of joint.

- b. Grout: Grout shall be poured between band and pipe from the high side of band only, until grout rises to the top of band at the spring line of pipe, or as nearly so as possible, on the opposite side of pipe, to ensure a thorough sealing of joint around the portion of pipe covered by the band. Silt, slush, water, or polluted mortar grout forced up on the lower side shall be forced out by pouring, and removed.
- c. Remainder of Joint: The remaining unfilled upper portion of the joint shall be filled with mortar and a bead formed around the outside of this upper portion of the joint with a sufficient amount of additional mortar. The diaper shall be left in place. Placing of this type of joint shall be kept at least five joints behind actual laying of pipe. No backfilling around joints shall be done until joints have been fully inspected and approved.

3.4.1.4 Flexible Watertight Joints

Gaskets and jointing materials shall be as recommended by the particular manufacturer in regard to use of lubricants, cements, adhesives, and other special installation requirements. Surfaces to receive lubricants, cements, or adhesives shall be clean and dry. Gaskets and jointing materials shall be affixed to the pipe not more than 24 hours prior to the installation of the pipe, and shall be protected from the sun, blowing dust, and other deleterious agents at all times. Gaskets and jointing materials shall be inspected before installing the pipe; any loose or improperly affixed gaskets and jointing materials shall be removed and replaced. The pipe shall be aligned with the previously installed pipe, and the joint pushed home. If, while the joint is being made the gasket becomes visibly dislocated the pipe shall be removed and the joint remade.

3.4.1.5 External Sealing Band Joint for Noncircular Pipe

Surfaces to receive sealing bands shall be dry and clean. Bands shall be installed in accordance with manufacturer's recommendations.

3.5 DRAINAGE STRUCTURES

3.5.1 Manholes and Inlets

Construction shall be of precast reinforced concrete, reinforced concrete, plain concrete, or brick; complete with frames and covers or gratings; and with fixed stainless steel ladders where indicated. Pipe studs and junction chambers of prefabricated corrugated metal manholes shall be fully bituminous-coated and paved when the connecting branch lines are so treated. Pipe connections to concrete manholes and inlets shall be made with flexible, watertight connectors.

3.5.2 Walls and Headwalls

Construction shall be as indicated.

3.6 BACKFILLING

3.6.1 Backfilling Pipe in Trenches

After the pipe has been properly bedded, selected material from excavation or borrow, at a moisture content that will facilitate compaction, shall be placed along both sides of pipe in layers not exceeding 6 inches in compacted depth. Initial and final backfill materials to consist of

material per Section 31 00 00 EARTHWORK specification. The backfill shall be brought up evenly on both sides of pipe for the full length of pipe. The fill shall be thoroughly compacted under the haunches of the pipe. Each layer shall be thoroughly compacted with mechanical tampers or rammers. This method of filling and compacting shall continue until the fill has reached an elevation equal to the midpoint (spring line) of concrete pipe. The remainder of the trench shall be backfilled and compacted by spreading and rolling or compacted by mechanical rammers or tampers in layers not exceeding 6 inches. Tests for density shall be made as necessary to ensure conformance to the compaction requirements specified below. All sheeting and portions of bracing shall be removed as the work is completed. Untreated sheeting shall not be left in place beneath structures or pavements.

3.6.2 Backfilling Pipe in Fill Sections

For pipe placed in fill sections, backfill material and the placement and compaction procedures shall be as specified below. Initial and final backfill materials to consist of material per Section 31 00 00 EARTHWORK specification. The fill material shall be uniformly spread in layers longitudinally on both sides of the pipe, not exceeding 6 inches in compacted depth, and shall be compacted by rolling parallel with pipe or by mechanical tamping or ramming.

3.6.3 Movement of Construction Machinery

When compacting by rolling or operating heavy equipment parallel with the pipe, displacement of or injury to the pipe shall be avoided. Movement of construction machinery over a culvert or storm drain at any stage of construction shall be at the Contractor's risk. Any damaged pipe shall be repaired or replaced at the Contractor's expense.

3.6.4 Compaction

3.6.4.1 General Requirements

Cohesionless materials include gravels, gravel-sand mixtures, sands, and gravelly sands. Cohesive materials include clayey and silty gravels, gravel-silt mixtures, clayey and silty sands, sand-clay mixtures, clays, silts, and very fine sands. When results of compaction tests for moisture-density relations are recorded on graphs, cohesionless soils will show straight lines or reverse-shaped moisture-density curves, and cohesive soils will show normal moisture-density curves.

3.6.4.2 Minimum Density

Backfill over and around the pipe and backfill around and adjacent to drainage structures shall be compacted at the approved moisture content. The density shall be as stated on the plans but not less than 95 percent of the maximum density.

3.7 FIELD PAINTING

3.7.1 Cast-Iron Covers, Frames, Gratings, And Steps

After installation, clean cast-iron, not buried in masonry or concrete, of mortar, rust, grease, dirt, and other deleterious materials to bare metal

and apply a coat of bituminous paint.

3.8 FIELD QUALITY CONTROL

3.8.1 Tests

Testing is the responsibility of the Contractor. Perform all testing and retesting at no additional cost to the Government.

3.8.1.1 HYDROSTATIC TEST ON WATERTIGHT JOINTS

Watertight joints shall be tested and shall meet test requirements of paragraph HYDROSTATIC TEST ON WATERTIGHT JOINTS. Rubber gaskets shall comply with the oil resistant gasket requirements of [ASTM C443](#). Certified copies of test results shall be delivered to the Contracting Officer before gaskets or jointing materials are installed.

3.8.1.2 Leakage Test

Lines shall be tested for leakage by low pressure air or water testing or exfiltration tests, as appropriate, prior to completing backfill. [Low pressure air testing for concrete pipes shall conform to ASTM C969](#). Testing of individual joints for leakage by low pressure air or water shall conform to [ASTM C1103](#). Prior to exfiltration tests, the trench shall be backfilled up to at least the lower half of the pipe. If required, sufficient additional backfill shall be placed to prevent pipe movement during testing, leaving the joints uncovered to permit inspection. Visible leaks encountered shall be corrected regardless of leakage test results. When the water table is [2 feet](#) or more above the top of the pipe at the upper end of the pipeline section to be tested, infiltration shall be measured using a suitable weir or other device acceptable to the Contracting Officer. An exfiltration test shall be made by filling the line to be tested with water so that a head of at least [2 feet](#) is provided above both the water table and the top of the pipe at the upper end of the pipeline to be tested. The filled line shall be allowed to stand until the pipe has reached its maximum absorption, but not less than 4 hours. After absorption, the head shall be reestablished. The amount of water required to maintain this water level during a 2-hour test period shall be measured. Leakage as measured by the exfiltration test shall not exceed [0.2 gallons per inch in diameter per 100 feet](#) of pipeline per hour.

3.8.1.3 Determination of Density

Testing shall be performed by an approved commercial testing laboratory or by the Contractor subject to approval. Tests shall be performed in sufficient number to ensure that specified density is being obtained. All density tests are to be reported in the next CQC report within 24 hours of the test. Laboratory tests for moisture-density relations shall be made in accordance with [ASTM D1557](#) except that mechanical tampers may be used provided the results are correlated with those obtained with the specified hand tamper. Field density tests shall be determined in accordance with [ASTM D2167](#) or [ASTM D6938](#). When [ASTM D6938](#) is used, the calibration curves shall be checked and adjusted, if necessary, using the sand cone method as described in paragraph Calibration of the referenced publications. [ASTM D6938](#) results in a wet unit weight of soil and [ASTM D6938](#) shall be used to determine the moisture content of the soil. The calibration curves furnished with the moisture gauges shall be checked along with

density calibration checks as described in [ASTM D6938](#). Test results shall be furnished the Contracting Officer. The calibration checks of both the density and moisture gauges shall be made at the beginning of a job on each different type of material encountered and at intervals as required to comply with contract requirements.

3.8.2 Inspection

3.8.2.1 Post-Installation Inspection

Visually inspect each segment of concrete pipe for alignment, settlement, joint separations, soil migration through the joint, cracks, buckling, bulging and deflection. An engineer must evaluate all defects to determine if any remediation or repair is required.

3.8.2.1.1 Concrete

Cracks with a width greater than [0.01 inches](#). An engineer must evaluate all pipes with cracks with a width greater than [0.01 inches](#) but less than [0.10 inches](#) to determine if any remediation or repair is required.

3.8.2.1.2 Post-Installation Inspection Report

The deflection results and final post installation inspection report must include: pipe location identification, equipment used for inspection, inspector name, deviation from design, grade, deviation from line, deflection and deformation of flexible pipe, inspector notes, condition of joints, condition of pipe wall (e.g. distress, cracking, wall damage dents, bulges, creases, tears, holes, etc.).

3.8.3 Repair Of Defects

3.8.3.1 Leakage Test

When leakage exceeds the maximum amount specified, correct source of excess leakage by replacing damaged pipe and gaskets and retest at no additional cost to the Government.

3.8.3.2 Deflection Testing

When deflection readings are in excess of the allowable deflection of average inside diameter of pipe are obtained, remove pipe which has excessive deflection and replace with new pipe. Retest 30 days after completing backfill, leakage testing and compaction testing at no additional cost to the Government.

3.8.3.3 Inspection

Replace pipe or repair defects indicated in the Post-Installation Inspection Report at no additional cost to the Government.

3.8.3.3.1 Concrete

Replace pipes having cracks with a width greater than [0.1 inches](#) at no additional cost to the Government.

3.9 PROTECTION

Protect storm drainage piping and adjacent areas from superimposed and

external loads during construction.

3.10 WARRANTY PERIOD

Pipe segments found to have defects during the warranty period must be replaced with new pipe and retested at no additional cost to the Government.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 71 01

OVERHEAD TRANSMISSION AND DISTRIBUTION

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
- 1.3 ADMINISTRATIVE REQUIREMENTS
 - 1.3.1 Pre-Installation Meetings
- 1.4 SUBMITTALS
- 1.5 MAINTENANCE MATERIAL SUBMITTALS
 - 1.5.1 Additions to Operations and Maintenance Data
- 1.6 QUALITY CONTROL
 - 1.6.1 Regulatory Requirements
 - 1.6.2 Standard Products
 - 1.6.2.1 Material and Equipment Manufacturing Date
 - 1.6.3 Ground Resistance Test Reports
 - 1.6.4 Qualifications
- 1.7 QUALITY ASSURANCE
 - 1.7.1 Submittal Review and Inspection Reports
- 1.8 DELIVERY, STORAGE, AND HANDLING
- 1.9 WARRANTY
- 1.10 PLSCADD DATA

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
 - 2.1.1 Design Requirements
- 2.2 EQUIPMENT
 - 2.2.1 Hardware
 - 2.2.1.1 Pins
 - 2.2.1.2 Hot-Line Clamps
 - 2.2.1.3 Secondary Racks
 - 2.2.2 Grounding and Bonding
 - 2.2.2.1 Driven Ground Rods
 - 2.2.2.2 Grounding Conductors
 - 2.2.2.3 Grounding Connections
 - 2.2.3 Conduit Risers and Conductors
 - 2.2.4 Group-Operated Load Interrupter Switches
 - 2.2.4.1 Manually Operated Type (Switch Handle Operated)
- 2.3 COMPONENTS
 - 2.3.1 Steel Poles
 - 2.3.1.1 Standard Embedded Steel Poles
 - 2.3.1.2 Custom Steel Poles
 - 2.3.1.3 Custom Pole Foundations
 - 2.3.2 Crossarms and Brackets
 - 2.3.2.1 Armless Construction
 - 2.3.3 Insulators
 - 2.3.4 Secondary Cables

- 2.3.5 Surge Arresters
- 2.3.6 Fused Cutouts
- 2.3.7 Transformer (Overhead-Type Distribution)
- 2.3.8 Nameplates
 - 2.3.8.1 Manufacturer's Nameplate
- 2.4 MATERIALS
 - 2.4.1 Overhead Conductors, Connectors and Splices
 - 2.4.1.1 Aluminum Conductor Steel Reinforced (ACSR)
 - 2.4.1.2 Connectors and Splices
 - 2.4.2 Electrical Tapes
 - 2.4.3 Caulking Compound
- 2.5 TESTS, INSPECTIONS, AND VERIFICATIONS
 - 2.5.1 Transformer Test Schedule
 - 2.5.2 Routine and Other Tests

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Overhead Service
 - 3.1.2 Tree Trimming
 - 3.1.3 Steel Pole Setting
 - 3.1.3.1 Cast-In-Place Foundations
 - 3.1.3.2 Embedded Poles
 - 3.1.4 Hardware
 - 3.1.5 Grounding
 - 3.1.5.1 Grounding Electrode Installation
 - 3.1.5.2 Grounding Electrode Conductors
 - 3.1.5.3 Grounding Electrode Connections
 - 3.1.5.4 Grounding and Grounded Connections
 - 3.1.5.5 Protective Molding
 - 3.1.6 Conductor Installation
 - 3.1.6.1 Line Conductors
 - 3.1.6.2 Connectors and Splices
 - 3.1.6.3 Conductor-To-Insulator Attachments
 - 3.1.6.4 Armor Rods
 - 3.1.6.5 Ties
 - 3.1.6.6 Low-Voltage Insulated Cables
 - 3.1.6.7 Reinstalling Conductors
 - 3.1.6.8 New Conductor Installation
 - 3.1.6.9 Fittings
 - 3.1.6.10 Aluminum Connections
 - 3.1.7 Pole Top Switch Installation
 - 3.1.7.1 Operating Handle
 - 3.1.8 Risers
 - 3.1.9 Transformer Installation
 - 3.1.10 Field Applied Painting
 - 3.1.11 Field Fabricated Nameplate Mounting
- 3.2 FIELD QUALITY CONTROL
 - 3.2.1 General
 - 3.2.2 Safety
 - 3.2.3 Medium-Voltage Preassembled Cable Test
 - 3.2.4 Sag and Tension Test
 - 3.2.5 Low-Voltage Cable Test
 - 3.2.6 Pre-Energization Services
 - 3.2.7 Performance of Acceptance Checks and Tests
 - 3.2.7.1 Overhead-Type Distribution Transformers
 - 3.2.7.2 Pole Top Interrupter Switch
 - 3.2.7.3 Grounding System
 - 3.2.8 Devices Subject to Manual Operation

3.2.9 Follow-Up Verification

-- End of Section Table of Contents --

SECTION 33 71 01

OVERHEAD TRANSMISSION AND DISTRIBUTION

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C135.14 (1979) Staples with Rolled or Slash Points
for Overhead Line Construction

ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M (2017) Standard Specification for Zinc
(Hot-Dip Galvanized) Coatings on Iron and
Steel Products

ASTM A153/A153M (2016a) Standard Specification for Zinc
Coating (Hot-Dip) on Iron and Steel
Hardware

ASTM A572/A572M (2021; E 2021) Standard Specification for
High-Strength Low-Alloy Columbium-Vanadium
Structural Steel

ASTM A575 (2020) Standard Specification for Steel
Bars, Carbon, Merchant Quality, M-Grades

ASTM A576 (2017) Standard Specification for Steel
Bars, Carbon, Hot-Wrought, Special Quality

ASTM B8 (2011; R 2017) Standard Specification for
Concentric-Lay-Stranded Copper Conductors,
Hard, Medium-Hard, or Soft

ASTM B117 (2019) Standard Practice for Operating
Salt Spray (Fog) Apparatus

ASTM B232/B232M (2017) Standard Specification for
Concentric-Lay-Stranded Aluminum
Conductors, Coated-Steel Reinforced (ACSR)

ASTM D1654 (2008; R 2016; E 2017) Standard Test
Method for Evaluation of Painted or Coated
Specimens Subjected to Corrosive
Environments

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 404 (2012) Standard for Extruded and Laminated

Dielectric Shielded Cable Joints Rated
2500 V to 500,000 V

IEEE C2	(2023) National Electrical Safety Code
IEEE C57.12.00	(2021) General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers
IEEE C57.12.90	(2021) Test Code for Liquid-Immersed Distribution, Power, and Regulating Transformers
IEEE C135.1	(1999) Standard for Zinc-Coated Steel Bolts and Nuts for Overhead Line Construction
IEEE C135.2	(1999) Threaded Zinc-Coated Ferrous Strand-Eye Anchor Rods and Nuts for Overhead Line Construction
IEEE C135.22	(1988) Standard for Zinc-Coated Ferrous Pole-Top Insulator Pins with Lead Threads for Overhead Line Construction
IEEE Stds Dictionary	(2009) IEEE Standards Dictionary: Glossary of Terms & Definitions

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS	(2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems
----------	--

LUMA PUERTO RICO (LUMA)

LUMA 4751.001	(2022) Transmission Line Design Criteria Document
LUMA 4751.002	(2021) LUMA Transmission Design Manual

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA C135.4	(1987) Zinc-Coated Ferrous Eyebolts and Nuts for Overhead Line Construction
NEMA WC 74/ICEA S-93-639	(2012) 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 4 2023; ERTA 5 2023; ERTA 6 2023) National Electrical Code
---------	---

U.S. DEPARTMENT OF AGRICULTURE (USDA)

RUS 202-1	(2004) List of Materials Acceptable for Use on Systems of RUS Electrification
-----------	--

Borrowers

UNDERWRITERS LABORATORIES (UL)

UL 6	(2007; Reprint Sep 2019) UL Standard for Safety Electrical Rigid Metal Conduit-Steel
UL 467	(2013; Reprint Jun 2017) UL Standard for Safety Grounding and Bonding Equipment
UL 486A-486B	(2018; Reprint May 2021) UL Standard for Safety Wire Connectors

1.2 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE Stds Dictionary.

1.3 ADMINISTRATIVE REQUIREMENTS

Section 26 08 00 APPARATUS INSPECTION AND TESTING applies to this section with additions and modifications specified herein.

1.3.1 Pre-Installation Meetings

Within 30 calendar days after date of award, submit for the approval of the Contracting Officer six (6) copies of specified drawings of all equipment to be furnished under this contract, together with weights and overall dimensions. Submit the following data and drawings:

- a. Connection Diagrams
- b. Fabrication Drawings
- c. Installation Drawings

Submit certification from the manufacturer indicating conformance with the specified poles and transformer losses:

- a. Steel Poles
- b. Transformer Losses

Submit operation and maintenance data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Connection Diagrams; G, D0

Fabrication Drawings; G, D0

Installation Drawings; G, D0

SD-03 Product Data

Conductors; G, D0

Insulators; G, D0

Steel Poles; G, D0

Nameplates; G, D0

Pole Top Switch; G, D0

Cutouts; G, D0

Transformer; G, D0

Surge Arresters; G, D0

SD-05 Design Data

Steel Pole Design; G, D0

Drilled Shaft Foundations; G, D0

Direct Embedment Foundation; G, D0

SD-06 Test Reports

Field Quality Control; G, D0

Ground Resistance Test Reports; G, D0

SD-07 Certificates

Transformer Losses; G, D0

Qualifications of Licensed Engineer in Puerto Rico; G, D0

Custom Steel Pole Certification; G, D0

SD-09 Manufacturer's Field Reports

Routine and Other Tests; G, D0

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G, D0

SD-11 Closeout Submittals

Transformer Test Schedule; G, D0

1.5 MAINTENANCE MATERIAL SUBMITTALS

1.5.1 Additions to Operations and Maintenance Data

In addition to requirements of Data Package 5, include the following in the [operation and maintenance manuals](#) provided:

- a. Assembly and installation drawings
- b. Prices for spare parts and supply list
- c. Date of purchase

1.6 QUALITY CONTROL

1.6.1 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Provide equipment, materials, installation, and workmanship in accordance with the mandatory and advisory provisions of [NFPA 70](#) and [IEEE C2](#) unless more stringent requirements are specified or indicated.

1.6.2 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Provide products that have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period includes applications of equipment and materials under similar circumstances and of similar size. Provide a product that has been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, provide items that are products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.6.2.1 Material and Equipment Manufacturing Date

Do not use products manufactured more than 3 years prior to date of delivery to site, unless specified otherwise.

1.6.3 [Ground Resistance Test Reports](#)

Submit the measured ground resistance of grounding system. When testing grounding electrodes and grounding systems, identify each grounding electrode and each grounding system for testing. Include the test method and test setup (i.e. pin location) used to determine ground resistance and soil conditions at the time the measurements were made.

1.6.4 Qualifications

Contractor shall engage the services of a qualified Puerto Rico Professional Engineer to provide inspection and review of the electrical distribution system throughout Construction.

- a. Submit name and [qualifications of licensed engineer](#). Professional Engineer shall have been regularly engaged in the inspection of electrical materials, devices, installations, and systems associated with transmission/distribution systems for a minimum of 5 years. The Professional Engineer shall be a resident of Puerto Rico.
- b. Include a list of three comparable jobs performed by the Professional Engineer with specific names and telephone numbers for reference.

1.7 QUALITY ASSURANCE

1.7.1 Submittal Review and Inspection Reports

Professional Engineer shall provide submittal reviews and ensure compliance with specified requirements, identify deficiencies, and recommend corrective action when appropriate.

1.8 DELIVERY, STORAGE, AND HANDLING

Visually inspect devices and equipment when received and prior to acceptance from conveyance. Protect stored items from the environment in accordance with the manufacturer's published instructions. Replace damaged items. Store oil filled transformers and switches in accordance with the manufacturer's requirements. [Handle and store metal poles in accordance with the manufacturer's instructions.](#)

1.9 WARRANTY

Support the equipment items by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.10 PLSCADD DATA

PLSCADD data files are available from the Government upon request.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Consider materials specified herein or shown on contract drawings which are identical to materials listed in [RUS 202-1](#) as conforming to requirements. Provide equipment and component items, not hot-dip galvanized or porcelain enamel finished, with corrosion-resistant finishes which withstand 480 hours of exposure to the salt spray test specified in [ASTM B117](#) without loss of paint or release of adhesion of the paint primer coat to the metal surface in excess of [1/16 inch](#) from the test mark. Provide the described test mark and test evaluation in accordance with [ASTM D1654](#) with a rating of not less than 7 in accordance with TABLE 1, (Procedure A). Coat cut edges or otherwise damaged surfaces of hot-dip galvanized sheet steel or mill galvanized sheet steel with a zinc rich paint conforming to the manufacturer's standard.

2.1.1 Design Requirements

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Provide products that have been in

satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period includes applications of equipment and materials under similar circumstances and of similar size. Provide a product that has been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, provide items that are products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

2.2 EQUIPMENT

2.2.1 Hardware

Provide hot-dip galvanized hardware in accordance with [ASTM A153/A153M](#) and [ASTM A123/A123M](#).

Provide zinc-coated hardware that complies with [IEEE C135.1](#), [IEEE C135.2](#), [NEMA C135.4](#), [ANSI C135.14](#), [IEEE C135.22](#). Provide steel hardware that complies with [ASTM A575](#) and [ASTM A576](#). Provide pole-line hardware that is hot-dip galvanized steel. Install washers under boltheads and nuts on wood surfaces and elsewhere as required. Provide washers used on through-bolts and double-arming bolts that are approximately [2-1/4 inches square](#) and [3/16-inch](#) thick. Make the diameter of holes in washers the correct standard size for the bolt on which a washer is used. Provide washers for use under heads of carriage-bolts, of the proper size to fit over square shanks of bolts. Use eye bolts, bolt eyes, eyenuts, strain-load plates, lag screws, fasteners, hooks, shims, and clevises wherever required to support and to protect poles, brackets, crossarms, and insulators.

2.2.1.1 Pins

Provide pins that are zinc-coated forged steel with lead-thread height to suit the insulator to be installed, but not less than [4-1/2-inches high by 5/8-inch diameter](#). Provide shoulder that is not less than [2-inch](#) diameter and that is designed to distribute the load uniformly to the crossarm. Provide shank that is not less than [5/8-inch diameter by 5-3/4-inch length](#), equipped with a [2-inch](#) square washer, nut, and locknut, and that projects not less than [1/8-inch](#) nor more than [2-inches](#) beyond the locknut. Use broad-based corner pins of drop-forged welded steel or malleable iron for turning small angles, as indicated.

2.2.1.2 Hot-Line Clamps

Provide hot line clamps as shown in the plans and in compliance with LUMA Equipment Specification Document No. 4350.097.

2.2.1.3 Secondary Racks

Provide secondary racks that are the 2-, 3-, or 4-wire type as required and are furnished complete with spool insulators.

Provide racks that meet industry requirements for the strength and deflection of heavy-duty steel racks and that are either galvanized steel or aluminum alloy.

Provide top of insulator points that are rounded and smooth. Hold insulators in place with a [5/8-inch](#) buttonhead bolt equipped with a

nonferrous cotter pin, or equivalent, at the bottom.

2.2.2 Grounding and Bonding

2.2.2.1 Driven Ground Rods

Provide copper-clad steel ground rods conforming to [UL 467](#) not less than [3/4-inch](#) in diameter by [10-feet](#) in length. Sectional type rods are acceptable for rods [20-feet](#) or longer.

2.2.2.2 Grounding Conductors

[ASTM B8](#). Provide soft drawn copper wire ground conductors a minimum No. 4 AWG. Provide PVC ground wire protectors.

2.2.2.3 Grounding Connections

[UL 467](#). Exothermic weld or compression connector.

2.2.3 Conduit Risers and Conductors

Provide PVC riser shield containing a PVC back plate and PVC extension shield or a rigid galvanized steel conduit, as indicated, and conforming to [UL 6](#). Provide conductors and terminations as specified in the plans.

2.2.4 Group-Operated Load Interrupter Switches

2.2.4.1 Manually Operated Type (Switch Handle Operated)

Provide group operated air break switch as shown in the plans and in compliance with LUMA Equipment Standard Document No. 4751.067.

2.3 COMPONENTS

2.3.1 Steel Poles

2.3.1.1 Standard Embedded Steel Poles

Provide poles of lengths indicated and strengths in accordance with [LUMA 4751.001](#) and [LUMA 4751.002](#). LUMA design standards will be furnished by the USACE. Provide material for poles in accordance with [ASTM A572/A572M](#), Grade 65.

Provide a [steel pole design](#), including [direct embedment foundation design](#), for withstanding the loads specified in Contract drawings, which include the appropriate overload capacity factors. The poles are hot-dip galvanized in accordance with [ASTM A123/A123M](#) and are not painted. Provide poles that have tapered tubular members, either round in cross-section or polygonal, and that comply with strength calculations performed by a registered professional engineer. Provide certification, from the manufacturer, that the technical requirements of this specification are met. Provide welded construction poles with no bolts, rivets, or other means of fastening except as specifically approved. Welding shall follow requirements specified in [Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS](#). Provide slip joints as required by LUMA Standards. Submit calculations in accordance with the design data portion of paragraph SUBMITTALS. Provide pole markings as per LUMA standards. Provide attachment requirements as indicated, including grounding provisions. Climbing access is required. Provide bases of the anchor-bolt-mounted type.

2.3.1.2 Custom Steel Poles

Provide poles as indicated in accordance with Contract Drawings. Provide material for poles in accordance with [ASTM A572/A572M](#), Grade 65.

Provide steel pole fabrication and installation drawings. The poles are hot-dip galvanized in accordance with [ASTM A123/A123M](#) and are not painted. Provide poles that have tapered tubular members, either round in cross-section or polygonal. Provide welded construction poles with no bolts, rivets, or other means of fastening except as specifically approved. Welding shall follow requirements specified in Section [05 50 13 MISCELLANEOUS METAL FABRICATIONS](#). Provide slip joints as required by LUMA Standards. Provide [certification](#) from the manufacturer that the technical requirements of this specification are met including calculations for any deviations from the final design provided on the Contract Drawings. Provide pole markings as per LUMA standards. Provide attachment requirements as indicated, including grounding provisions. Climbing access is required. Provide bases of the anchor-bolt-mounted type per the Contract Drawings.

2.3.1.3 Custom Pole Foundations

The final design for [drilled shaft foundations](#) shall be designed by a qualified Professional Engineer that is licensed in Puerto Rico. Design shall follow LUMA standards as well as USACE requirements. Ground reaction loads are provided on the Contract Drawings.

Top of shaft deflections shall be within 10% of the shaft diameter based on factored loads.

2.3.2 Crossarms and Brackets

2.3.2.1 Armless Construction

Provide pole mounting brackets for line-post or pin insulators and eye bolts for suspension insulators as shown. Attach brackets to poles with a minimum of two bolts. Provide brackets either integrally as part of an insulator or attached to an insulator with a suitable stud. Provide bracket mounting surface suitable for the shape of the pole. Provide brackets for wood poles that have wood gripping members. Provide horizontal offset brackets that have a 5-degree uplift angle. Provide pole top brackets that conform to [IEEE C135.22](#), except for modifications necessary to provide support for a line-post insulator. Provide brackets that have a strength exceeding that of the required insulator strength, but in no case less than a [2800 pound](#) cantilever strength.

2.3.3 Insulators

Provide insulators as shown in the plans and in compliance with the latest LUMA Standards.

2.3.4 Secondary Cables

Provide secondary cables as shown in the plans and in compliance with LUMA Overhead Electrical Distribution Manual, Table 10 - Conductors for the Secondary Bushings of Distribution Transformers (Line Voltage 240 V).

2.3.5 Surge Arresters

Provide distribution class surge arresters as shown in the plans and in compliance with LUMA Equipment Specification Document No. 4350.010.

2.3.6 Fused Cutouts

Provide fused cutouts as shown in the plans and in compliance with LUMA Equipment Specification Document No. 4350.011.

2.3.7 Transformer (Overhead-Type Distribution)

Provide single phase, oil immersed distribution transformer as shown in the plans and in compliance with LUMA Equipment Specification Document No. 4350.016.

2.3.8 Nameplates

2.3.8.1 Manufacturer's Nameplate

Provide each item of equipment with a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent is not acceptable. Provide equipment containing liquid-dielectrics with the type of dielectric on the nameplate. Nameplate shall contain customer's name, month and year of manufacturing, length, ultimate moment in FT-KIPS, approximate weight, and type/class. Nameplate shall be located at $0.14L + 6$ feet from butt, where "L" is the length of pole and $0.14L$ is the ground line level. Pole shall have clear marking showing ground line (NT) and center of gravity (CG). These marks shall, in no way, affect the structural capacity of the pole.

2.4 MATERIALS

2.4.1 Overhead Conductors, Connectors and Splices

Provide bare aluminum conductor steel reinforced (ACSR) Conductors of sizes and types indicated. Where aluminum conductors are connected to dissimilar metal, use fittings conforming to UL 486A-486B.

2.4.1.1 Aluminum Conductor Steel Reinforced (ACSR)

ASTM B232/B232M, aluminum. Provide Transmission line conductors as shown in the plans and in compliance with LUMA Equipment Standard Document No. 4350.062. Provide Distribution line conductors as shown in the plans and in compliance with LUMA Equipment Standard Document No. 4350.063.

2.4.1.2 Connectors and Splices

Provide connectors and splices of copper alloys for copper conductors, aluminum alloys for aluminum-composition conductors, and a type designed to minimize galvanic corrosion for copper to aluminum-composition conductors. Provide aluminum-composition, aluminum-composition to copper, and copper-to-copper that complies with UL 486A-486B.

2.4.2 Electrical Tapes

Provide electrical tape as shown in the plans and in compliance with LUMA Equipment Specification Document No. 4350.130.

2.4.3 Caulking Compound

Provide compound for sealing of conduit risers that is of a puttylike consistency workable with hands at temperatures as low as 35 degrees F, that does not slump at a temperature of 300 degrees F, and that does not harden materially when exposed to air. Provide compound that readily caulks or adheres to clean surfaces of the materials with which it is designed to be used. Provide compound that has no injurious effects upon the workmen or upon the materials.

2.5 TESTS, INSPECTIONS, AND VERIFICATIONS

2.5.1 Transformer Test Schedule

The Government reserves the right to witness tests. Provide transformer test schedule for tests to be performed at the manufacturer's test facility. Submit required test schedule and location, and notify the Contracting Officer 30 calendar days before scheduled test date. Notify Contracting Officer 15 calendar days in advance of changes to scheduled date.

a. Test Instrument Calibration

- (1) Provide a manufacturer that has a calibration program which assures that all applicable test instruments are maintained within rated accuracy.
- (2) Provide an accuracy that is directly traceable to the National Institute of Standards and Technology.
- (3) Provide instrument calibration frequency schedule that does not exceed 12 months for both test floor instruments and leased specialty equipment.
- (4) Provide visible dated calibration labels on all test equipment.
- (5) Provide calibrating standard of higher accuracy than that of the instrument tested.
- (6) Keep up-to-date records that indicate dates and test results of instruments calibrated or tested. For instruments calibrated by the manufacturer on a routine basis, in lieu of third party calibration, include the following:
 - (a) Maintain up-to-date instrument calibration instructions and procedures for each test instrument.
 - (b) Identify the third party/laboratory calibrated instrument to verify that calibrating standard is met.

2.5.2 Routine and Other Tests

IEEE C57.12.00 and IEEE C57.12.90. Perform routine and other tests by the manufacturer on each of the actual transformer(s) prepared for this project to ensure that the design performance is maintained in production. Submit test reports, by serial number and receive approval before delivery of equipment to the project site. Provide required tests as follows:

- a. Polarity
- b. Ratio
- c. No-load losses (NLL) and excitation current
- d. Load losses (LL) and impedance voltage
- e. Dielectric
 - (1) Impulse
 - (2) Applied voltage
 - (3) Induced voltage
- f. Leak

PART 3 EXECUTION

3.1 INSTALLATION

Provide overhead pole line installation conforming to requirements of **IEEE C2** for Grade B construction of overhead lines in medium loading districts and **NFPA 70** for overhead services. Provide material required to make connections into existing system and perform excavating, backfilling, and other incidental labor. Consider street, alleys, roads and drives "public." Provide pole configuration as indicated.

3.1.1 Overhead Service

Terminate overhead service conductors into buildings at service entrance fittings or weatherhead outside building where indicated. Provide drip loops that are formed on conductors at entrances to buildings, cabinets, or conduits.

3.1.2 Tree Trimming

Where lines pass through trees, trim trees at least **15 feet** clear on both sides horizontally and below for medium-voltage lines, and **5-feet** clear on both sides horizontally and below for other lines. Do not allow a branch to overhang horizontal clearances. Where trees are indicated to be removed to provide a clear right-of-way, clearing is specified in Section **31 11 00** CLEARING AND GRUBBING.

3.1.3 Steel Pole Setting

Mount poles on cast-in-place drilled shaft foundations or embedded as shown on the plans. Provide conduit elbows for cable entrances into pole interiors.

3.1.3.1 Cast-In-Place Foundations

Provide concrete foundations with anchor bolts accurately set in foundations using templates supplied by the pole manufacturer. Concrete work and grouting is specified in Section **03 31 01** CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS and **31 63 29** DRILLED CONCRETE PIERS AND SHAFTS. After the concrete has cured, set pole anchor bases on foundations and level by shimming between anchor bases and foundations or

by setting anchor bases on leveling nuts and grouting. Set poles plumb. Provide the manufacturer's standard anchor bolts, and not less than necessary to meet the pole wind loading specified herein and other design requirements.

3.1.3.2 Embedded Poles

Auger holes in accordance with the requirements in 31 63 29 DRILLED CONCRETE PIERS AND SHAFTS. All flowable concrete for embedded pole foundations shall be placed by tremie method and follow requirements specified in Section 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS. Flowable concrete shall be minimum 4,000 psi. The nominal maximum size of aggregate for flowable concrete will be one half (1/2) inches.

3.1.4 Hardware

Install hardware with washer against wood and with nuts and lock nuts applied wrench tight. Provide locknuts on threaded hardware connections. Provide M-F style locknuts and not palnut style.

3.1.5 Grounding

Unless otherwise indicated, install grounding that conforms to IEEE C2 and NFPA 70. Provide pole grounding electrodes with a resistance to ground not exceeding 25 ohms. When work in addition to that indicated or specified is directed in order to obtain specified ground resistance, apply provisions of the contract covering changes.

3.1.5.1 Grounding Electrode Installation

Install grounding electrodes as follows:

- a. Driven rod electrodes - Unless otherwise indicated, locate ground rods approximately 3-feet out from base of the pole and drive into the earth until the tops of the rods are approximately 1-foot below finished grade. Evenly space multiple rods at least 10-feet apart and connect together 2-feet below grade with a minimum No. 6 bare copper conductor.
- b. Plate electrodes - Install plate electrodes in accordance with the manufacturer's instructions and IEEE C2 and NFPA 70.

3.1.5.2 Grounding Electrode Conductors

On multi-grounded circuits, as defined in IEEE C2, provide a single continuous vertical grounding electrode conductor. Bond neutrals, surge arresters, and equipment grounding conductors to this conductor. For single-grounded or ungrounded systems, provide a grounding electrode conductor for the surge arrester and equipment grounding conductors and a separate grounding electrode conductor for the secondary neutrals. On metal poles, use a preformed galvanized steel strap, 5/8-inch wide by 22 gauge minimum by length, secured by a preformed locking method standard with the manufacturer, to support a grounding electrode conductor installation on the pole and space at intervals not exceeding 5-feet with one band not more than 3-inches from each end of the vertical grounding electrode conductor. Size grounding electrode conductors as indicated. Connect secondary system neutral conductors directly to the transformer neutral bushings, then connect with a neutral bonding jumper between the transformer neutral bushing and the vertical grounding electrode conductor

as indicated. Bends greater than 45 degrees in grounding electrode conductor are not permitted.

3.1.5.3 Grounding Electrode Connections

Make above grade grounding connections on pole lines by exothermic weld or by using a compression connector. Make below grade grounding connections by exothermic weld. Make exothermic welds strictly in accordance with manufacturer's written recommendations. Welds which have puffed up or which show convex surfaces indicating improper cleaning, are not acceptable. No mechanical connectors are required at exothermic weldments. Provide compression connectors that are the type that uses a hydraulic compression tool to provide correct pressure. Provide tools and dies recommended by compression connector manufacturer. Provide an embossing die code or similar method as visible indication that a connector has been fully compressed on ground wire.

3.1.5.4 Grounding and Grounded Connections

- a. Where no primary or common neutral exists, bond together surge arresters and frames of equipment operating at over 750 volts and connect to a dedicated primary grounding electrode.
- b. Where no primary or common neutral exists, transformer secondary neutral bushing, secondary neutral conductor, and bond together frames of equipment operating at under 750 volts and connect to a dedicated secondary grounding electrode.
- c. When a primary or common neutral exists, connect all grounding and grounded conductors to a common grounding electrode.

3.1.5.5 Protective Molding

Protect grounding conductors which are run on surface of wood poles by PVC molding extending from ground line throughout communication and transformer spaces.

3.1.6 Conductor Installation

3.1.6.1 Line Conductors

Handle conductors with care necessary to prevent nicking, kinking, gouging, abrasions, sharp bends, cuts, flattening, or otherwise deforming or weakening conductor or any damage to insulation or impairing its conductivity. Remove damaged sections of conductor and splice conductor. Provide conductors that are paid out with the free end of conductors fixed and cable reels portable, except where terrain or obstructions make this method unfeasible. Make the bend radius for any insulated conductor not less than the applicable NEMA specification recommendation. Do not draw conductors over rough or rocky ground, nor around sharp bends. When installed by machine power, provide conductors that are drawn from a mounted reel through stringing sheaves in straight lines clear of obstructions. Check the initial sag and tension, in accordance with the manufacturer's approved sag and tension charts, within an elapsed time after installation as recommended by the manufacturer.

3.1.6.2 Connectors and Splices

Provide conductor splices, as installed, that exceed ultimate rated

strength of conductor and are of the type recommended by conductor manufacturer. No splices are permitted within 10-feet of a support. Provide connectors and splices that are mechanically and electrically secure under tension and are of the nonbolted compression type. Make splices have a tensile strength of not less than the rated breaking strength of the conductor. Provide splice materials, sleeves, fittings, and connectors that are noncorrosive and that do not adversely affect conductors. Wire brush and apply an oxide inhibitor to aluminum-composition conductors before making a compression connection. Connectors which are factory-filled with an inhibitor are acceptable. Provide types of inhibitors and compression tools recommended by the connector manufacturer. Provide primary line apparatus taps by means of hot line clamps attached to compression type bail clamps (stirrups). Provide solderless pressure type low-voltage connectors for copper conductors. Smoothly tape noninsulated connectors to provide a waterproof insulation equivalent to the original insulation, when installed on insulated conductors. On overhead connections of aluminum and copper, install the aluminum above the copper.

3.1.6.3 Conductor-To-Insulator Attachments

Attach conductors to insulators by means of clamps, shoes or tie wires, in accordance with the type of insulator. For insulators requiring conductor tie-wire attachments, provide tie-wire sizes as specified in TABLE I.

TABLE I	
TIE-WIRE REQUIREMENTS	
CONDUCTOR	TIE WIRE
Copper (AWG)	Soft-Drawn Copper (AWG)
6	8
4 and 2	6
1 through 3/0	4
4/0 and larger	2
AAC, AAAC, or ACSR (AWG)	AAAC OR AAC (AWG)
Any size	6 or 4

3.1.6.4 Armor Rods

Provide armor rods for AAC, AAAC, and ACSR conductors. Install armor rods at supports, except armor rods are not required at primary dead-end assemblies if aluminum or aluminum-lined zinc-coated steel clamps are used. Provide lengths and methods of fastening armor rods in accordance with the manufacturer's recommendations. For span lengths of less than 200-feet, use of flat aluminum armor rods is allowed. Use flat armor rods, not less than 0.03 by 0.25 inch on No. 1 AWG AAC and AAAC and smaller conductors and on No. 5 AWG ACSR and smaller conductors. On larger sizes, provide flat armor rods that are not less than 0.05 by 0.30 inches. For span lengths of 200-feet or more, use preformed round armor

rods.

3.1.6.5 Ties

Provide ties on pin insulators tight against conductor and insulator and ends turned down flat against conductor so that no wire ends project.

3.1.6.6 Low-Voltage Insulated Cables

Support low-voltage cables on clevis fittings using spool insulators. Provide dead-end clevis fittings and suspensions insulators where required for adequate strength. Provide dead-end construction that has a strength exceeding the rated breaking strength of the neutral messenger. Provide clevis attachments with not less than **5/8-inch** through-bolts. Use secondary racks when installed on wood poles and where the span length does not exceed **200-feet**. Provide two-, three-, or four-wire secondary racks, complete with spool insulators. Provide racks that meet strength and deflection requirements for heavy-duty steel racks, and are rounded and smooth to avoid damage to conductor insulation. Hold each insulator in place with a **5/8-inch** button-head bolt equipped with a nonferrous cotter pin, or equivalent, at the bottom. Provide racks for dead-ending four No. 4/0 AWG or four larger conductors that are attached to poles with three **5/8-inch** through-bolts. Attach other secondary racks to poles with at least two **5/8-inch** through-bolts. Provide minimum vertical spacing between conductors of not less than **8-inches**.

3.1.6.7 Reinstalling Conductors

String existing conductors to be reinstalled or resagged to "final" sag table values indicated for the particular conductor type and size involved.

3.1.6.8 New Conductor Installation

String new conductors to "initial" sag table values indicated for conductor type and size of conductor and ruling span indicated.

3.1.6.9 Fittings

Provide dead end fittings, clamp or compression type, that conform to written recommendations of conductor manufacturer and that develop full ultimate strength of conductor.

3.1.6.10 Aluminum Connections

Make aluminum connections to copper or other material using only splices, connectors, lugs, or fittings designed for that specific purpose. Keep a copy of manufacturer's instructions for applying these fittings at job site for use of the inspector.

3.1.7 Pole Top Switch Installation

Install pole top switch strictly according to manufacturer's installation drawings and information.

3.1.7.1 Operating Handle

Locate approximately **5 feet** above ground on field side of pole.

3.1.8 Risers

Secure galvanized steel conduits on poles by two hole galvanized steel pipe straps spaced as indicated and within 3-feet of any outlet or termination. Ground metallic conduits. Secure PVC riser shields on poles as indicated.

3.1.9 Transformer Installation

Carefully install transformers so as not to scratch finishes or damage bushings. Install transformers in accordance with the manufacturer's instructions. After installation, inspect surfaces and touch up scratches with a finish provided by the transformer manufacturer for this purpose.

3.1.10 Field Applied Painting

Paint electrical equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Corrosion resistance of field applied painting shall match that of manufacturer's factory applied painting.

3.1.11 Field Fabricated Nameplate Mounting

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

3.2 FIELD QUALITY CONTROL

3.2.1 General

Perform field testing in the presence of the Contracting Officer and/or LUMA. Notify the Contracting Officer 20 days prior to conducting tests. Furnish materials, labor, and equipment necessary to conduct field tests. Perform tests and inspections recommended by the manufacturer unless specifically waived by the Contracting Officer. Maintain a written record of tests which includes date, test performed, personnel involved, devices tested, serial number and name of test equipment, and test results. Sign and date field reports.

3.2.2 Safety

Provide and use safety devices such as rubber gloves, protective barriers, and danger signs to protect and warn personnel in the test vicinity. Replace any devices or equipment which are damaged due to improper test procedures or handling.

3.2.3 Medium-Voltage Preassembled Cable Test

After installation, prior to connection to an existing system, and before the operating test, give the medium-voltage preassembled cable system a high potential test. Apply direct-current voltage on each phase conductor of the system by connecting conductors at one terminal and connecting grounds or metallic shieldings or sheaths of the cable at the other terminal for each test. Prior to the test, isolate the cables by opening applicable protective devices and disconnecting equipment. Provide the method, voltage, length of time, and other characteristics of the test for initial installation in accordance with NEMA WC 74/ICEA S-93-639 for the particular type of cable installed, and do not exceed the recommendations

of IEEE 404 for cable joints unless the cable and accessory manufacturers indicate higher voltages are acceptable for testing. For any cable that fails due to a weakness of conductor insulation or due to defects or injuries incidental to the installation or because of improper installation of cable, cable joints, terminations, or other connections, make necessary repairs or replace cables as directed. Retest repaired or replaced cables.

3.2.4 Sag and Tension Test

Give the Contracting Officer prior notice of the time schedule for stringing conductors or cables serving overhead medium-voltage circuits. The Contracting Officer reserves the right to witness the procedures used for ascertaining that initial stringing sags and tensions are in compliance with requirements for the applicable loading district and cable weight.

3.2.5 Low-Voltage Cable Test

For underground secondary or service laterals from overhead lines, provide the low-voltage cable, complete with splices, that is tested for insulation resistance after the cables are installed, in their final configuration, ready for connection to the equipment, and prior to energization. The 500 volts dc test voltage, applied for one minute between each conductor and ground and between all possible combinations of conductors in the same trench, duct, or cable, with other conductors in the same trench, duct, or conduit. Provide insulation with a minimum value of:

$$R \text{ in megohms} = (\text{rated voltage in kV} + 1) \times 1000 / (\text{length of cable in feet})$$

Repair or replace each cable failing this test. Retest the repaired cable then until failures have been eliminated.

3.2.6 Pre-Energization Services

Perform the following services on the equipment listed below. Perform these services subsequent to testing but prior to the initial energization. Inspect the equipment to ensure that installation is in compliance with the recommendations of the manufacturer and as shown on the detail drawings. Inspect terminations of conductors at major equipment to ensure the adequacy of connections. Inspect bare and insulated conductors between such terminations to detect possible damage during installation. If factory tests were not performed on completed assemblies, perform tests after the installation of completed assemblies. Inspect components for damage caused during installation or shipment and to ensure that packaging materials have been removed. Provide components capable of being both manually and electrically operated that are operated manually prior to the first electrical operation. Provide components capable of being calibrated, adjusted, and tested and calibrate, adjust and test in accordance with the instructions of the equipment manufacturer. Items for which such services are provided, but are not limited to, are the following:

- a. Switches

3.2.7 Performance of Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations and include

the following visual and mechanical inspections and electrical tests, performed in accordance with **NETA ATS**.

3.2.7.1 Overhead-Type Distribution Transformers

a. Visual and mechanical inspection

- (1) Compare equipment nameplate information with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method. Thermographic survey is not required.
- (4) Perform specific inspections and mechanical tests as recommended by manufacturer.
- (5) Verify correct equipment grounding.

b. Electrical tests

- (1) Ensure that the series-multiple voltage-changing switch is in the correct position. Transformers are normally shipped in the series position.
- (2) Perform insulation-resistance tests.
- (3) Perform continuity test.
- (4) Set tap changer to provide a secondary voltage of 120/240.

3.2.7.2 Pole Top Interrupter Switch

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate information with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Verify appropriate equipment grounding.
- (4) Perform mechanical operator tests in accordance with manufacturer's instructions.
- (5) Verify correct blade alignment, blade penetration, travel stops, arc interrupter operation, and mechanical operation.

b. Electrical Tests

- (1) Perform insulation-resistance tests.
- (2) Perform dc over-potential tests.
- (3) Perform contact-resistance tests across each switch blade.

3.2.7.3 Grounding System

a. Visual and mechanical inspection

- (1) Inspect ground system for compliance with contract plans and specifications.

b. Electrical tests

- (1) Perform ground-impedance measurements utilizing the fall-of-potential method. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground testing megger in accordance with manufacturer's instructions to test each ground or group of grounds. Provide an instrument that is equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test.

3.2.8 Devices Subject to Manual Operation

Operate each device subject to manual operation at least three times, demonstrating satisfactory operation each time.

3.2.9 Follow-Up Verification

Upon completion of acceptance checks and tests, show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function. As an exception to requirements stated elsewhere in the contract, give the Contracting Officer 5 working days advance notice of the dates and times of checking and testing.

-- End of Section --

SECTION TABLE OF CONTENTS
DIVISION 34 - TRANSPORTATION
SECTION 34 71 13
VEHICLE CRASH BARRIERS

PART 1 GENERAL

1.1 REFERENCES

PART 2 PRODUCTS

2.1 GUARDRAIL POSTS, W-BEAMS, END SECTIONS, AND HARDWARE

PART 3 EXECUTION

3.1 GUARDRAIL POSTS AND W-BEAMS

-- End of Section Table of Contents --

SECTION 34 71 13

VEHICLE CRASH BARRIERS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

COMMONWEALTH OF PUERTO RICO, DEPARTMENT OF TRANSPORTATION AND
PUBLIC WORKS (PRHTA)

PRHTA Specifications (2005) Standard Specifications for Road
and Bridge Construction

Special Provisions, when applicable, are appended and included as an attachment to this specification section. Where PRHTA Specifications Section, and/or Special Provision(s) is cited that contains references to other PRHTA sections, they shall also form a part of this specification to the extent referenced. Where reference is made to "Highway Authority" or "Authority's Representative" in PRHTA Specifications shall mean "Government" or "Contracting Officer" as defined in the Contract Documents. Where PRHTA Specifications refer to the "Engineer", "Engineer of Tests", or "Division of Tests" it shall be understood to mean the "Contracting Officer" as stated in the Contract Documents. In case of conflict between the PRHTA specifications and the Specifications in this Section that follows, the Specifications in this Section shall govern.

Paragraphs title "Method of Measurement" and "Basis of Payment" of referenced PRHTA Specifications shall not apply.

PART 2 PRODUCTS

2.1 GUARDRAIL POSTS, W-BEAMS, END SECTIONS, AND HARDWARE

Guardrail posts, w-beam sections, end sections, and hardware shall be in accordance with **PRHTA Specifications** Section 606.

PART 3 EXECUTION

3.1 GUARDRAIL POSTS AND W-BEAMS

Install guardrail posts and w-beam sections as indicated in **PRHTA Specifications** Section 606.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

SECTION 35 20 23

DREDGING

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SCOPE
- 1.3 DEFINITION
- 1.4 SUBMITTALS
- 1.5 QUALITY CONTROL
 - 1.5.1 Notice of Intent To Dredge
 - 1.5.2 Relocation of Navigation Aids
 - 1.5.3 Drag Bar
 - 1.5.4 Daily/Monthly Report of Operations
- 1.6 DREDGING PLAN
- 1.7 DREDGING RESTRICTIONS
 - 1.7.1 Order of Work
 - 1.7.2 Transportation of Material
- 1.8 SPECIAL INSTRUCTIONS
 - 1.8.1 Historical Period Shipwreck Sites
- 1.9 UTILITY CROSSINGS
 - 1.9.1 General
- 1.10 PERMITS
- 1.11 FUEL OIL TRANSFER OPERATIONS
- 1.12 SIGNAL LIGHTS
- 1.13 NOTICE TO MARINERS -- DREDGING CONTRACTS
- 1.14 FINAL CLEANUP
- 1.15 WORK VIOLATIONS
- 1.16 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM (DREDGES AND DUMP SCOWS)
 - 1.16.1 System Requirements
- 1.17 ENVIRONMENTAL PROTECTION REQUIREMENTS

PART 2 PRODUCTS (NOT APPLICABLE)

PART 3 EXECUTION

- 3.1 NOTIFICATION OF COAST GUARD
 - 3.1.1 Navigation Aids
 - 3.1.2 Dredging Aids
 - 3.1.3 Surveillance of Ocean Disposal
- 3.2 WORK AREA
 - 3.2.1 Access
 - 3.2.2 Protection of Existing Waterways
 - 3.2.3 Adjacent Property and Structures
 - 3.2.4 Subaqueous Utility Crossings
- 3.3 DISPOSAL OF EXCAVATED MATERIAL
 - 3.3.1 General

- 3.3.2 Ocean Dredged Material Disposal Site (ODMDS)
- 3.3.3 Dredge Location Control
 - 3.3.3.1 Dredge Control
- 3.3.4 Misplaced Materials
- 3.3.5 Dredge Pipelines
 - 3.3.5.1 Dredge Discharge Pipeline
 - 3.3.5.2 Floating Pipeline
- 3.3.6 Booster Pumps
- 3.4 REQUIRED DEPTH AND STRUCTURAL OFFSETS
 - 3.4.1 Required Depth
 - 3.4.2 Structural Offsets
 - 3.4.3 Excessive Dredging
 - 3.4.4 Acceptance Sections Plans
- 3.5 SURVEYS
 - 3.5.1 General
 - 3.5.2 Contractor Representative
 - 3.5.3 Tide Data
 - 3.5.3.1 Real Time Kinematic (RTK) GPS
 - 3.5.3.2 Kinematic Tidal Datum
- 3.6 INSPECTION
 - 3.6.1 Quality Assurance Representative (QAR)
 - 3.6.2 Failure to Comply
- 3.7 FINAL EXAMINATION AND ACCEPTANCE
 - 3.7.1 Final Examination of Work
 - 3.7.2 Bed Leveling
 - 3.7.3 Bed Leveling - Reporting and Documenting
 - 3.7.4 Final Acceptance
- 3.8 SHOALING
- 3.9 CONTINUITY OF WORK
- 3.10 NOISE CONTROL
- 3.11 DREDGE CERTIFICATION
- 3.12 GENERAL DREDGING SAFETY REQUIREMENTS
 - 3.12.1 Dredge Plant Restriction Definitions
 - 3.12.1.1 Certification Required Line - Coast Guard Inspected (CRL-C)
 - 3.12.1.2 Certification Required Line - Load Line (CRL-L)
 - 3.12.2 Certification of Attendant and Auxiliary Plant
 - 3.12.3 Minimum Tow-Boat Requirements
- 3.13 CONSTRUCTION FORMS AND DETAILS

-- End of Section Table of Contents --

SECTION 35 20 23

DREDGING

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety and Occupational Health (SOH) Requirements
EM 1110-1-1000	Photogrammetric and LiDAR Mapping
EM 1110-1-1002	(2012) Survey Markers and Monumentation
EM 1110-1-1003	(2011) NAVSTAR Global Positioning System Surveying
EM 1110-1-2909	(2012) Geospatial Data and Systems
EM 1110-2-1003	(2013) Hydrographic Surveying
TR-12-6	(2015) A/E/C Standard Release 6.0

FLORIDA BOARD OF PROFESSIONAL SURVEYORS AND MAPPERS (FBPSM)

FBPSM	Minimum Technical Standards, Chapters 177, 472, 61G17
-------	--

1.2 SCOPE

The work covered by this section consists of furnishing all labor, materials, and equipment, and performing excavation of material removed from within the project limits by hydraulic or barge-based mechanical dredging if the Contractor elects to utilize these methods. Disposal requirements of all material designated to the Ocean Dredged Material Disposal Site (ODMDS) is also covered by this specification. For additional requirements related to excavation and disposal within the La Chuleta upland disposal site, see specification section 31 00 00 EARTHWORK.

This scope also includes all necessary measures for protection of the environment. Environmental protection requirements under this contract are as important to overall completion of the work as other technical aspects. Failure to meet the requirements of these specifications for environmental protection may result in work stoppages or termination for default. No part of the time lost due to any such work stoppages shall be made the subject of claims for extensions of time or for excess costs or damages by the Contractor. If the Contractor fails or refuses to promptly repair any damage caused by violation of the provisions of these specifications, the Contracting Officer may have the necessary work

performed and charge the cost thereof to the Contractor.

Refer to specification section 01 22 00 MEASUREMENT AND PAYMENT and the Contract Drawings for pay limit lines.

1.3 DEFINITION

Hard material is defined as material requiring the use of special equipment for economical removal, and includes boulders or fragments too large to be removed in one piece by the dredge.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Notice Of Intent To Dredge

Relocation Of Navigation Aids

Drag Bar

Dredging Plan; G, D0

SD-07 Certificates

Acceptance Sections Plans; G, D0

Historical Period Shipwreck Sites

Surveys

Daily/Monthly Report Of Operations

Misplaced Materials

Declaration Of Inspection

Surveillance Of Ocean Disposal

1.5 QUALITY CONTROL

1.5.1 Notice of Intent To Dredge

Prior to commencement of work on this contract, the Contractor shall notify the Commander, Seventh Coast Guard District of his intended operations to dredge and request that it be published in the Local Notice to Mariners. This notification must be given in sufficient time so that it appears in the Notice to Mariners at least two weeks prior to the commencement of this dredging operation. A copy of the notification shall be provided to the Contracting Officer.

1.5.2 Relocation of Navigation Aids

The Contractor shall not remove, change the location of, obstruct, willfully damage, make fast to, or interfere with any aid to navigation. Within 7 calendar days following receipt of Notice of Award, the Contractor shall notify the Commander, Seventh Coast Guard District, Miami, Florida, of his plan to dredge adjacent to any aids which require relocation to facilitate dredging. The notification shall be sent via Fax to 305-415-6757, ATTN: Mr. Gene Stratten. This notification shall be immediately followed by a formal written request with a copy to the Contracting Officer. The Contractor shall also contact the U.S. Coast Guard, 305-415-6750, for information concerning the position to which the aids will be relocated.

1.5.3 Drag Bar

Submit drawings and one photograph showing drag bar equipment used for final leveling work indicated in the paragraph FINAL EXAMINATION AND ACCEPTANCE below.

1.5.4 Daily/Monthly Report of Operations

The Contractor shall prepare on RMS and submit one (1) hard copy of the Daily Report of Operations, using either ENG Form No. 27A or ENG Form No. 4267, for each dredge and/or unloader working. This report shall be submitted on a daily basis and not in groups (groups = multi-days reports packaged together at one time). In addition to the daily report, the Contractor shall prepare a Monthly Report of Operations for each month or partial month's work on either ENG Form No. 27A or ENG Form No. 4267. The monthly report shall be submitted on or before the 7th of each month, consolidating the previous month's work. Upon completion of the job, the Contractor shall submit a consolidated job report, combining the monthly reports. The Contractor shall distribute one copy of each report to the District Engineer; ATTN: CESAJ-EN-C; U.S. Army Engineer District, Jacksonville, P.O. Box 4970; Jacksonville, Florida 32232-0019. Reports shall be submitted on a monthly basis with daily reports accompanying the monthly report and job report.

Additionally, one copy of the form(s) shall be maintained by the Contractor on the dredge(s) for the Contracting Officer's inspection purpose. Further instructions on the preparation of the reports will be furnished at the Preconstruction Conference.

1.6 DREDGING PLAN

Provide a [dredging plan](#) which includes a summary of dredging methods and a listing of dredging equipment utilized for channel excavation. The plan shall include, at minimum, dredge equipment specifications, barge capacities, vessel names, and methods for horizontal and vertical dredge position control. Include any mooring devices, the planned access route for offshore disposal and procedures for protecting existing structures along the access route and in-channel staging areas. Provide a method for tracking offshore disposal materials quantities. Tracking of offshore disposal quantity is required for ensuring ODMDS permit compliance and not for measurement and payment procedures.

1.7 DREDGING RESTRICTIONS

1.7.1 Order of Work

Final dredging of the channel shall be completed from downstream to upstream (North to South) within the project limits, except for material removed for establishment of the required Bypass Channel as shown on the drawings. If necessary for coordination purposes with other contracts or project schedule requirements, exceptions will be approved by the Contracting Officer.

1.7.2 Transportation of Material

Water and dredge material shall not be permitted to overflow or spill out of barges during transport to the disposal site.

1.8 SPECIAL INSTRUCTIONS

Dredging of in-situ rock is not required; but if encountered, the location and extent of any in-situ rock above grade or required depth shall be reported to the Contracting Officer.

Except as indicated, the Government has no knowledge of cables, pipes, or other artificial obstructions or of any wrecks, wreckage, or other material that would necessitate the use of explosives or the employment of additional equipment for economical removal. If actual conditions differ from those stated or shown, or both, an adjustment in contract price or time for completion, or both, will be made in accordance with FAR 52.236-2 Differing Site Conditions.

1.8.1 [Historical Period Shipwreck Sites](#)

If any shipwreck, artifact, or other objects of antiquity that have scientific or historical value, or are of interest to the public, are discovered, located, and/or recovered, the Contractor acknowledges that:

- a. The site(s), articles, or other materials are the property of the Commonwealth of Puerto Rico, with title vested in the Department of State, Division of Historical Resource; and that,
- b. He will immediately notify the Contracting Officer.

1.9 UTILITY CROSSINGS

1.9.1 General

It is the Contractor's responsibility to investigate the location of all utility crossings. The Contractor shall take precautions against damages which might result from his operations in the vicinity of the utility crossings. If any damage occurs as a result of his operations, the Contractor will be required to suspend dredging until the damage is repaired and approved by the Contracting Officer. Costs of such repairs and downtime of the dredge and attendant plant shall be at the Contractor's expense.

1.10 PERMITS

The Contractor's attention is directed to the Clause PERMITS AND RESPONSIBILITIES of Section 00700 CONTRACT CLAUSES in Volume 1 and the

paragraph PERMITS AND AUTHORIZATIONS of Section 01 57 20 ENVIRONMENTAL PROTECTION.

The Contractor shall comply with conditions and requirements of the Corps of Engineers Permit and other Commonwealth or Federal permits. The Contracting Officer will secure the permit for dredging and disposal of material within the ODMDS as indicated.

1.11 FUEL OIL TRANSFER OPERATIONS

In accordance with U.S. Coast Guard regulations (33 CFR 156.120), couplings used in fuel oil transfer operations on any vessel with a capacity of 250 or more barrels of oil shall be either a bolted or full-threaded connection; or a quick-connect coupling approved by the Commandant; or an automatic back-pressure shutoff nozzle used to fuel the vessel. An executed fuel oil transfer (Declaration) form signed by the tanker operator shall be submitted to the Contracting Officer for each refueling operation. The U.S. Coast Guard shall also be notified prior to any refueling. A copy of the [Declaration of Inspection](#) for Refueling is on the web site indicated in the paragraph CONSTRUCTION FORMS AND DETAILS below.

1.12 SIGNAL LIGHTS

The Contractor shall display signal lights and conduct operations in accordance with the General Regulations of the Department of the Army and of the Coast Guard governing lights and day signals to be displayed by towing vessels with tows on which no signals can be displayed, vessels working on wrecks, dredges, and vessels engaged in laying cables or pipe or in submarine or bank protection operations, lights to be displayed on dredge pipe lines, and day signals to be displayed by vessels of more than 65 feet in length moored or anchored in a fairway or channel, and the passing by other vessels of floating plant working in navigable channels, as set forth in Commandant U.S. Coast Guard Instruction M16672.2, Navigation Rules: International-Inland (COMDTINST M16672.2), or 33 CFR 81 Appendix A (International) and 33 CFR 84 through 33 CFR 89 (Inland) as applicable.

1.13 NOTICE TO MARINERS -- DREDGING CONTRACTS

Should the Contractor, during operations, encounter any objects on the channel bottom which could be a hazard to navigation, the Contractor shall immediately notify the Contracting Officer as to the location of said object and shall provide any other pertinent information necessary for the Contracting Officer to prepare and issue a Notice to Mariners.

1.14 FINAL CLEANUP

Final cleanup, as stated in the paragraph COMMENCEMENT, PROSECUTION, AND COMPLETION OF WORK of Section 00700 CONTRACT CLAUSES in Volume 1, shall include the removal of all the Contractor's plant and equipment either for disposal or reuse. Plant and/or equipment and/or materials to be disposed of shall only be disposed in a manner and at locations approved by the Contracting Officer. Unless otherwise approved by the Contracting Officer, the Contractor will not be permitted to abandon any equipment in the disposal area or other areas adjacent to the worksite. Failure to promptly remove all plant, pipeline, equipment, and materials upon completion of the dredging will be considered a delay in the completion of the final cleanup and demobilization work. In such case, the Government

will exercise its right as stated in Clause DEFAULT (FIXED-PRICE CONSTRUCTION) of Section 00700 CONTRACT CLAUSES in Volume 1 to remove any plant and/or equipment and/or materials at the Contractor's expense.

1.15 WORK VIOLATIONS

Work done in violation of these specifications or a verbal or written stop order of the Contracting Officer will be considered as unsatisfactory progress for purposes of progress payments in accordance with Clause PAYMENTS UNDER FIXED-PRICE CONSTRUCTION CONTRACTS of Section 00700 CONTRACT CLAUSES in Volume 1.

1.16 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM (DREDGES AND DUMP SCOWS)

1.16.1 System Requirements

See Section 35 20 26 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SCOW - ULLAGE PROFILE and Section 35 20 27 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM PIPELINE HYDRAULIC DREDGE. However, in the event of NDQMP System failure (not fully operational), the Contractor shall notify the Contracting Officer. Any delays resulting from NDQMP System failure shall be at the Contractor's expense. When required to calculate displacement, the density of water shall be 64.0 pounds per cubic foot or its metric equivalent of 1025 kilograms per cubic meter.

1.17 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain during the life of the contract, environmental protective measures. Also, provide environmental protective measures required to correct conditions, such as oil spills or debris, that occur during the dredging operations. Comply with Federal, State, and local regulations pertaining to water, air, and noise pollution. Refer to specification section 01 57 25 TURBIDITY AND DISPOSAL MONITORING for additional requirements.

PART 2 PRODUCTS (NOT APPLICABLE)

PART 3 EXECUTION

3.1 NOTIFICATION OF COAST GUARD

3.1.1 Navigation Aids

Navigation aids located within or near the areas required to be dredged will be removed, if necessary, by the U.S. Coast Guard in advance of dredging operations. Notify the U.S. Coast Guard as indicated in paragraph SUBMITTALS above for [relocation of navigation aids](#). The Contractor shall not remove, change the location of, obstruct, willfully damage, make fast to, or interfere with any aid of navigation.

3.1.2 Dredging Aids

The Contractor shall obtain approval from the U.S. Coast Guard for all buoys, dredging aid markers to be placed in the water, and dredging aid markers affixed with a light prior to the installation. Dredging aid markers and lights shall not be colored or placed in a manner that they will obstruct or be confused with navigation aids.

3.1.3 Surveillance of Ocean Disposal

The Contractor shall notify the local Coastal Guard Captain of the Port at least 5 calendar days prior to the first ocean disposal. The notification will be by certified mail with a copy to the Contractor Officer. The following information shall be included in the notification:

- (1) Project designation; Corps of Engineers' Contracting Officer's name and contract number; and, the Contractor's name, address, and telephone number.
- (2) Port of departure.
- (3) Location of ocean disposal area.
- (4) Quantity of material to be deposited in ocean.
- (5) Schedule for ocean disposal, give date and time proposed for first ocean disposal.

3.2 WORK AREA

The Contractor will be permitted to exclude the public from the work areas in the immediate vicinity of his dredging, transporting, and disposal operations. Enforcement shall be the Contractor's responsibility at no additional cost to the Government. The enforcement shall be coordinated with local enforcement agencies and will be subject to approval of the Contracting Officer.

3.2.1 Access

The Contractor shall be responsible for providing and maintaining access necessary for his equipment and plant to and from the work site, mooring area, staging area and disposal area. The Contractor shall ascertain the environmental conditions which can affect the access such as climate, winds, currents, waves, depths, shoaling, and scouring tendencies.

3.2.2 Protection of Existing Waterways

The Contractor shall conduct his operations in such a manner that material or other debris are not pushed outside of dredging limits or otherwise deposited in existing side channels, basins, docking areas, or other areas being utilized by vessels. The Contractor will be required to change his method of operations as may be required to comply with the above requirements. Should any bottom material or other debris be pushed into areas described above, as a result of the Contractor's operations, the same must be promptly removed by and at the expense of the Contractor to the satisfaction of the Contracting Officer.

3.2.3 Adjacent Property and Structures

No dredging will be permitted within 25 feet of any private or public structure unless otherwise shown in the drawings. Any damage to private or public property or structures resulting from the disposal or dredging operations shall be repaired promptly by the Contractor at his expense. Any damage to structures as a result of Contractor's negligence will result in suspension of dredging and require prompt repair at the Contractor's expense as a prerequisite to the resumption of dredging.

3.2.4 Subaqueous Utility Crossings

The Contractor shall be responsible for verifying the locations and depths of all utility crossings and take precautions against damages which might

result from his operations, especially the sinking of dredge spuds and/or anchors into the channel bottom, in the vicinity of utility crossings. If any damage occurs as a result of his operations, the Contractor will be required to suspend dredging until the damage is repaired and approved by the Contracting Officer. Costs of such repairs and downtime of the dredge and attendant plant shall be at the Contractor's expense.

3.3 DISPOSAL OF EXCAVATED MATERIAL

3.3.1 General

Material excavated shall be transported to and deposited in the disposal areas designated on the drawings.

3.3.2 Ocean Dredged Material Disposal Site (ODMDS)

The excavated material meeting the permit requirements for offshore disposal shall be transported to and deposited in the appropriate ODMDS location shown on the drawings. Only material within the existing Rio Piedras banks is permitted for disposal at the ODMDS. This limit is defined as a vertical plane where the mean higher high water (MHHW) meets the ascending bank. The MHHW is defined as El. 4.5 from station 88+33 to 118+50 and El. 6.0 from station 123+50 to 147+40 for project purposes. Cross sections within the contract drawings show the limits at each project station. The average distance to which the material will have to be transported to each ODMDS is also shown on the drawings.

3.3.3 Dredge Location Control

The Contractor shall have, in continuous operation on the dredge, electronic positioning equipment that will accurately and continuously compute and plot the position and the deepest penetration of the dredge.

3.3.3.1 Dredge Control

Other dredge types shall meet all of the following requirements, and this information shall be compiled and submitted to the Contracting Officer bi-weekly:

- a. The Contractor is required to have electronic positioning equipment that shall horizontally locate the dredge when operating on the project, as well as determine the depth of dredging. Continuous locations of the dredges shall be made at all times during dredging.
- b. A color-coded plot, in the project reference datum, of the draghead, cutterhead, or other hydraulic or mechanical dredging device shall also be recorded daily depicting the vertical and horizontal limits of the material dredged that day. Any horizontal or vertical dredge violations shall be clearly defined.
- c. The Contractor is required to calibrate the equipment as required by the manufacturer or as required by the Contracting Officer. Proof of calibration shall be submitted to the Contracting Officer.
- d. For Horizontal control, the location of the dredge is to be computed by coordinates with a probable range error not to exceed +/- 3 feet. The Contractor's method of horizontal location of the dredge shall be submitted for review. LORAN-C shall not be permitted for location control.

e. The Contractor is also required to have a depth of dredging indicator accurate to within one foot. The instrument used shall indicate the depth of dredging at all times. This equipment shall include real-time measurement of the water (tide) level. The reported elevation of dredging shall be adjusted by the measured water level elevation and shall have a probable range error not to exceed 1-foot vertical.

f. If the dredge is outside the dredge area limits, the Contractor shall provide reason the dredge is outside the dredge area limits such as turning around annotated on the position chart and the dredge depth within 1-foot accuracy. Add notation in the Contractor's Quality Control Report for each occurrence.

3.3.4 Misplaced Materials

If any material is deposited other than in places designated or approved, the Contractor may be required to remove such misplaced material and redeposit it where directed at their expense.

3.3.5 Dredge Pipelines

3.3.5.1 Dredge Discharge Pipeline

The Contractor shall plainly mark the pipeline access routes with conspicuous stakes, targets and/or buoys to be maintained throughout the contract operations. A tight dredge discharge pipeline shall be maintained to prevent spilling of dredged material or dredge water. The Contractor shall provide and maintain radio communication between the dredge and the disposal vessels and the dredge and the Contracting Officer. The pipeline shall be inspected at least twice daily for leaks. Failure to immediately repair leaks in the discharge pipeline will result in suspension of dredging operations and require prompt repair of pipeline as a prerequisite to the resumption of dredging. Any damage to private or public property resulting from the Contractor's operations shall be repaired by the Contractor at his expense.

3.3.5.2 Floating Pipeline

Should the Contractor's pipeline not rest on the bottom, it will be considered a floating pipeline and shall be visible on the surface and clearly marked. In no case will the Contractor's pipeline be allowed to fluctuate between the surface and the bottom, or lie partly submerged. Lights shall be installed on the floating pipeline as required in paragraph SIGNAL LIGHTS above. The lights shall be supported either by buoys or by temporary piling, provided by the Contractor and approved by the Contracting Officer. Where the pipeline does not cross a navigable channel, the flashing yellow all-around lights shall be spaced not over 200 feet apart, unless closer spacing is required by U.S. Coast Guard personnel, in which case the requirements of the U.S. Coast Guard shall govern, at no additional cost to the Government.

3.3.6 Booster Pumps

Any booster pumps installed by the Contractor shall be located at least 300 feet from any residential-type building or house. Booster pumps, their prime movers, and any auxiliary equipment shall be fitted or equipped with mufflers, noise control enclosures, or other engineering noise control methods, measures, and features such that steady noise

emanating from this equipment does not exceed 85 decibels on the A scale at slow response, and impulsive noise does not exceed 140 decibels. Such items shall be maintained throughout the course of the work.

3.4 REQUIRED DEPTH AND STRUCTURAL OFFSETS

3.4.1 Required Depth

To a depth of not more than the final channel invert as shown on the Contract Drawings, will be estimated and paid for in accordance with the provisions contained in the subparagraphs "Measurement" and "Payment" of Section 01 22 00 MEASUREMENT AND PAYMENT.

3.4.2 Structural Offsets

See Section 31 00 00 EARTHWORK, paragraph "Structural Offsets" for requirements for working near existing and new structures constructed as part of the project.

3.4.3 Excessive Dredging

It is recognized that the limits of dredging need to extend beyond the pay prism in order to remove material from within the pay prism. Therefore, the limits of active dredging may exceed the limits of the pay prism in vertical dimension to the extent necessary to remove material from the pay prism. Nevertheless, material taken from beyond the limits of the pay prism will be deducted from the total amount dredged as excessive over-depth dredging for which payment will not be made. Nothing herein shall be construed to prevent payment for the removal of shoals performed in accordance with the applicable provisions of the paragraphs FINAL EXAMINATION AND ACCEPTANCE or SHOALING of this Section.

3.4.4 Acceptance Sections Plans

For purposes of payment, the work to be done will be divided into acceptance sections. Prior to any dredging, the Contractor shall prepare and provide a proposed acceptance sections plan for approval by the Contracting Officer. Generally, the work should be divided into areas that equate to approximately 30 days of dredging or that constitute separable or definable features of work.

3.5 SURVEYS

3.5.1 General

The Contracting Officer shall be notified, in writing, 10 days in advance of the need for pre-dredging and after-dredging surveys. Surveys will be performed in accordance with the paragraph QUANTITY SURVEYS of Section 00700 CONTRACT CLAUSES in Volume 1; paragraph LAYOUT OF WORK of Section 01 11 00 SUMMARY OF WORK; Section 01 45 04 CONTRACTOR QUALITY CONTROL; EM 1110-1-1000, EM 1110-1-1002, EM 1110-1-1003, EM 1110-1-2909, and EM 1110-2-1003; FBPSM; and, TR-12-6. A copy of the EM's can be downloaded from the following web site:
<http://www.publications.usace.army.mil/USACEPublications/EngineerManuals.aspx>
A copy of the TSS can be downloaded from the following web site:
<https://cadbimcenter.erdc.dren.mil/default.aspx?p=a&t=1&i=2>.

3.5.2 Contractor Representative

The Contractor or his authorized representative will be notified when soundings and/or sweepings are to be made, and will be offered the opportunity to be on board the Government survey vessel, but may decline the offer. All in-place measurement surveys and final acceptance sweep surveys will be performed by the Government. The Contractor's representative shall be fully knowledgeable in offshore construction subsurface surveying procedures, techniques, equipment, and horizontal and vertical calibration methods, and state-of-the-art horizontal and vertical accuracy limitations. The Contractor's representative may observe and review, in progress, the adequacy and accuracy of the survey for in-place payment purposes, and for the potential existence of collusion, fraud, or obvious error in the data.

3.5.3 Tide Data

3.5.3.1 Real Time Kinematic (RTK) GPS

The government will utilize RTK GPS for determining Real Time water levels (tide corrections) for measurement and payment purposes. The Contractor is responsible for providing all applicable RTK equipment for surveying and dredging operations.

3.5.3.2 Kinematic Tidal Datum

A file listing the separations between NAVD 88 and the Chart Datum (Mean Lower Low Water) will be provided to the Contractor for entry into the hydrographic survey software. The Kinematic Tidal Datum file shall be obtained from the Construction Representative prior to commencing surveying and dredging operations.

3.6 INSPECTION

3.6.1 Quality Assurance Representative (QAR)

The QAR shall be notified prior to the establishment of horizontal control work (baseline layout, ranges, station flags, shore-based control for EPS/RPS, etc.) and vertical control work (tide staff(s), upland cross sections, construction elevations top/invert, maximum/minimum elevations of dredged materials within disposal area(s), etc.), but the presence or absence of the QAR shall not relieve the Contractor of his responsibility for proper execution of the work in accordance with the specifications. The Contractor will be required:

- a. To furnish, on the request of the Contracting Officer or any QAR, the use of such boats, boatmen, laborers, and material forming a part of the ordinary and usual equipment and crew of the dredging plant as may be reasonably necessary in inspecting and supervising the work.
- b. To furnish, on the request of the Contracting Officer or any QAR, suitable transportation from all points on shore designated by the Contracting Officer to and from the various pieces of plant.

3.6.2 Failure to Comply

In conjunction with the Clause INSPECTION OF CONSTRUCTION of Section 00700 CONTRACT CLAUSES of Volume 1, should the Contractor refuse, neglect, or

delay compliance with these requirements, the specific facilities may be furnished and maintained by the Contracting Officer and the cost thereof will be deducted from any amounts due or to become due the Contractor.

3.7 FINAL EXAMINATION AND ACCEPTANCE

3.7.1 Final Examination of Work

As soon as practicable and no later than three (3) weeks after the completion of the entire work or any section thereof (if the work is divided into sections) as in the opinion of the Contracting Officer will not be subject to damage by further operations under the contract, such work will be thoroughly examined at the cost and expense of the Government by sounding or by sweeping, or both, as determined by the Contracting Officer. Should any shoals, lumps, or other lack of contract depth be disclosed by this examination, the Contractor will be required to remove same by dragging the bottom in accordance with the subparagraph "Bed Leveling" below or by dredging at the contract rate of dredging. The Contractor or his authorized representative will be notified when soundings and/or sweepings are to be made and will be permitted to accompany the survey party. When the area is found to be in a satisfactory condition, it will be accepted finally. Should more than two sounding or sweeping operations by the Government over an area be necessary by reason of work for the removal of shoals disclosed at a prior sounding or sweeping, the cost of such third and any subsequent soundings or sweeping operations will be charged against the Contractor at the rate of \$5,500 per day for each day in which the Government plant is engaged in sounding or sweeping and/or is enroute to or from the site or held at or near the said site for such operation.

3.7.2 Bed Leveling

Bed leveling by dragging the bottom with a drag bar or other apparatus shall be allowed only in the designated dredging areas shown on the drawings. Dragging in areas outside of the designated dredging areas shown on the drawings is specifically prohibited without written approval of the Contracting Officer.

3.7.3 Bed Leveling - Reporting and Documenting

The contractor shall fully document all bed leveling activity including date and time for initiation and completion of bed leveling. All bed leveling activity shall be documented on the Contractor's Quality Control Report (QCR).

3.7.4 Final Acceptance

Final acceptance of the whole or a part of the work and the deductions or corrections of deductions made thereon will not be reopened after having once been made, except on evidence of collusion, fraud or obvious error, and the acceptance of a completed section shall not change the time of payment of the retained percentages of the whole or any part of the work.

3.8 SHOALING

If, before the contract is completed, shoaling occurs in any section previously accepted, including shoaling in the finished channel because of the natural lowering of the side slopes, redredging at contract price, within the limits of available funds may be done if agreeable to both the

Contractor and the Contracting Officer.

3.9 CONTINUITY OF WORK

Progress payments will be made as work progresses in accordance with 52.232-5 PAYMENTS UNDER FIXED-PRICE CONSTRUCTION CONTRACTS (MAY 2014) and subparagraph "Excavation, Unclassified" of Section 01 22 00 MEASUREMENT AND PAYMENT. Full payment for and formal acceptance of acceptance sections will not be rendered until the acceptance section and the first 100 feet of area of all adjacent acceptance section bordering subject section is excavated to the full depth and extent required by this contract. Nor will full payment be made for excavation in any area not adjacent to and in prolongation of areas where full depth has been secured, except by decision of the Contracting Officer.

3.10 NOISE CONTROL

All hauling and excavating equipment and dredge/barges, boats, and tugs used on this work shall be equipped with satisfactory mufflers or other noise abatement devices. The Contractor shall conduct his operations so as to comply with all Federal, State and local laws pertaining to noise. The use of horns and whistle signals shall be held to the minimum necessary in order to ensure as quiet an operation as possible.

3.11 DREDGE CERTIFICATION

During dredging operations the Contractor's dredge shall have a current Certificate of Inspection issued by the U.S. Coast Guard.

3.12 GENERAL DREDGING SAFETY REQUIREMENTS

3.12.1 Dredge Plant Restriction Definitions

To increase the safety of dredges on Federal contracts, the contract plans and specifications may identify a location within the Federal channel (usually indicated by a channel station or navigation aid) requiring minimum equipment standards. This is a Certification Required Line (CRL). The following are specific designations that may apply to this contract.

3.12.1.1 Certification Required Line - Coast Guard Inspected (CRL-C)

A designation of Certification Required Line - Coast Guard inspected (CRL-C) means only dredges with a current U.S. Coast Guard Certificate of Inspection for the intended use of the vessel will be allowed to work seaward of this line under this contract. The inspections shall include both topside and drydock. The topside inspection shall have occurred within 15 months prior to the date that dredging commences. The drydock inspection shall have occurred within 3 years prior to the date that dredging commences.

3.12.1.2 Certification Required Line - Load Line (CRL-L)

A designation of Certification Required Line - Load Line (CRL-L) means only dredges having a current Load Line Certification issued pursuant to 46 CFR 41-47 will be allowed to work seaward of this line under this contract. Load Line inspections may be performed by either the American Bureau of Shipping (ABS) or Det Norske Veritas (DNV) on behalf of the U.S. Coast Guard. The statutory requirement for Load Line Certification is 46

U.S.C.
5101-5116.

3.12.2 Certification of Attendant and Auxiliary Plant

In addition to the requirements of EM 385-1-1, Section 19, Subparagraph 19.A.01.b., all supporting floating plant for dredging operations (to include, but not limited to, anchor barges, deck barges, deck/work barges, tug boats, tenders, workboats, pump-out barges, derrick barges, and spider barges) shall have a current marine survey inspection certificate. The only exceptions to this requirement include pipeline support pontoons or floats and outboard skiffs under 16 feet in length. The certification shall have been issued by a licensed and accredited marine surveyor within the previous 12 months. The marine surveyor must be accredited by either the National Association of Marine Surveyors or the Society of Accredited Marine Surveyors. No attendant plant that does not have a valid certification shall be used on this contract. The certifications and inspection shall be appropriate for the intended use of the plant in all locations specified in this contract. Unless the Contracting Officer decides otherwise, documentation of the inspection for each piece of plant shall be provided prior to issuance of Notice to Proceed.

3.12.3 Minimum Tow-Boat Requirements

In addition to the requirements of EM 385-1-1, Section 19, Subparagraph 19.A.03.e., when dredging with non self-propelled dredges seaward of the CRL-L or CRL-C, the contractor shall provide the following.

- a. One primary towboat of adequate size and horsepower to move any and all attendant and support plant against prevailing tides, currents, and winds during normal and severe weather conditions. The towboat shall be available within the project work area at all times.
- b. A towboat(s) of adequate size and horsepower to move the dredge(s) against prevailing tides, currents, and winds during normal and severe weather conditions shall be able to mobilize to the work site, and move the dredge(s) to safe harbor with 4 to 6 hours of first notification.

3.13 CONSTRUCTION FORMS AND DETAILS

See the "Construction Forms & Details" module at the following web address:

<URL

HREF="http://www.saj.usace.army.mil/About/DivisionsOffices/Engineering/DesignBranch/SpecificationsSection.aspx.">http://www.saj.usace.army.mil/About/DivisionsOffices/Eng

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

SECTION 35 20 26

NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SCOW - ULLAGE PROFILE

PART 1 GENERAL

- 1.1 DESCRIPTION
- 1.2 SUBMITTALS
- 1.3 PAYMENT
- 1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION
- 1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

PART 2 PRODUCTS (Not Applicable)

PART 3 EXECUTION

- 3.1 REQUIREMENTS FOR REPORTED DATA
 - 3.1.1 Scow Name
 - 3.1.2 Contract Number
 - 3.1.3 Load Number
 - 3.1.4 Horizontal Positioning
 - 3.1.5 Date and Time
 - 3.1.6 Hull Status
 - 3.1.6.1 Open-Water Disposal
 - 3.1.6.2 Offloading
 - 3.1.7 Course
 - 3.1.8 Speed
 - 3.1.9 Heading
 - 3.1.10 Draft
 - 3.1.11 Displacement
 - 3.1.12 Bin Ullage Sounding
 - 3.1.13 Bin Volume
- 3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS
 - 3.2.1 Telemetry
 - 3.2.2 Data Reporting Frequency
 - 3.2.3 Data Transmission to the Web Service
 - 3.2.4 XML-Formatted Sensor Data String
 - 3.2.5 Contractor Data Backup
- 3.3 PERFORMANCE REQUIREMENTS
- 3.4 LIST OF ITEMS TO BE PROVIDED BY THE CONTRACTOR

-- End of Section Table of Contents --

SECTION 35 20 26

NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM
SCOW - ULLAGE PROFILE

PART 1 GENERAL

1.1 DESCRIPTION

The work under this contract requires use of the National Dredging Quality Management Program (DQM) to monitor the scow's status at all times during the contract and to manage data history. For the purpose of these specifications, a scow is defined as any non-self-propelled vessel used to transport dredged material. This includes, but is not limited to, split-hull scows, pocket scows, hopper barges, and deck barges.

This performance-based specification section identifies the minimum required output and the precision and instrumentation requirements. The requirements may be satisfied using equipment and technical procedures selected by the Contractor.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office responsible for review of the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00, "SUBMITTAL PROCEDURES":

[SD-01 Preconstruction Submittals](#)

[Revisions Or Addendum To The Dredge Plant Instrumentation Plan; G, SAM-OP-J](#)

[SD-07 Certificates](#)

[Letter of National Dredging Quality Management Program Certification; G, DO](#)

1.3 PAYMENT

No separate payment shall be made for installation, operation, and maintenance of the DQM-certified system as specified herein for the duration of the dredging operations; all costs in connection therewith will be considered a subsidiary obligation of the Contractor and covered under the contract unit price for dredging in the bidding schedule.

1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION

The Contractor is required to have a current certification from DQM for the scow instrumentation system to be used under this contract. Provide a [letter of National Dredging Quality Management Program Certification](#) to the Contracting Officer. Criteria for certification shall be based on the most recent specification posted on the DQM

website(<http://dqm.usace.army.mil/Specifications/Index.aspx>), Compliance with these criteria shall be verified by onsite quality assurance (QA) checks conducted by the DQM Support Center Data Acquisition and Analysis Team and by periodic review of the transmitted data. If a system is installed specifically for this contract, in order to ensure that it is capable of transmitting quality data to the DQM database, the QA checks should take place either prior to the start of the contract or, with prior approval of the local USACE District, as soon as practical after dredging commences. DQM Certification is valid for one year from the date of certification and is contingent upon the system's ability to meet the performance requirements as outlined in Paragraph 3.3, "Performance Requirements." If issues with data quality are not corrected within 48 hours, the system certification shall be revoked and additional QA checks by the Data Acquisition and Analysis Team may be necessary.

Annual DQM Certification shall be based on the following:

- A series of QA checks as outlined on the DQM website (<https://dqm.usace.army.mil/Certifications/Index.aspx>)

- Verification of data acquisition and transfer as described in Paragraph 3.3, "Performance Requirements"

- Review of the Dredge Plant Instrumentation Plan (DPIP) as described in Paragraph 1.5, "Dredge Plant Instrumentation Plan (DPIP)"

The Dredging Contractor shall have personnel who are familiar with the system instrumentation and who have the ability to recalibrate the sensors on site during the QA process. The Dredging Contractor shall coordinate pickup times and locations and provide transportation to and from any platform with a DQM system to team personnel in a timely manner. The Dredging Contractor shall also have on site for the QA checks a tug capable of towing the scow. As a general rule, DQM Data Acquisition and Analysis Team personnel will come with personal protective equipment (PPE) consisting of hardhats, steel toe boots, and life jackets. If additional safety equipment is needed - such as eye protection, safety harnesses, work gloves, or personal location beacons - these items shall be provided to the team while on site. The Contractor shall submit a test data package to the DQM database from the system on each scow and have it accepted by the DQM Support Center prior to scow compliance checks. The Contractor shall also submit data collected during the QA Checks from the scow monitoring system to the DQM database and the Data Acquisition and Analysis Team personnel while on site. It is the Dredging Contractor's obligation to inform the QA team if the location designated for the QA checks has any site-specific safety concerns prior to their arrival on site.

The owner or operator of the scow shall contact DQM at DQM-AnnualQA@rpsgroup.com on an annual basis, or at least three weeks prior to the proposed beginning of dredging, to schedule QA checks. This notification is meant to make the Data Acquisition and Analysis Team aware of a target date and the contract on which the plant will be used. At least one week prior to the target date, the Dredging Contractor shall contact the Data Acquisition and Analysis Team and verbally coordinate a specific date and location. The Contractor shall then follow up this conversation with a written email confirmation. The owner/operator shall coordinate the QA checks with all local authorities including, but not limited to, the local USACE Contracting Officer's Representative (COR).

Recertification is required for any yard work which produces modification to displacement (for example, a change in scow lines, or repositioning or repainting hull marks), modification to bin volume (change in bin dimensions or addition or subtraction of structure), or changes in sensor type or location; these changes shall be reported in the sensor log section of the DPIIP. A system does not have to be transmitting data between jobs; however, in order to retain certification during this period, the system sensors or hardware should not be disconnected or removed from the scow. If the system is powered down, calibration coefficients shall be retained.

1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIIP)

The Contractor shall have a digital copy of the DPIIP on file with the DQM Support Center. While working on site, the Contractor shall also maintain on the dredge a copy of the DPIIP which is easily accessible to Government personnel at all times. This document shall describe the sensors used, configuration of the system, how sensor data will be collected, how quality control on the data will be performed, and how sensors/data reporting equipment will be calibrated and repaired if they fail. A description of computed scow-specific data and how the sensor data will be transmitted to the DQM database shall also be included. The Contractor shall submit to the DQM Support Center any addendum or modifications made to the plan, subsequent to its original submission, prior to start of work.

A complete list of the required DPIIP contents is provided on the DQM website (<https://dqm.usace.army.mil/Certifications/Index.aspx>). The Contractor shall submit to the DQM Support Center any [revisions or addendum to the dredge plant instrumentation plan](#), subsequent to its original submission, prior to the start of work. Any changes to the computation methods shall be approved by the DQM Support Center prior to their implementation.

PART 2 PRODUCTS (Not Applicable)

PART 3 EXECUTION

3.1 REQUIREMENTS FOR REPORTED DATA

The Contractor shall provide, operate, and maintain all hardware and software to meet these specifications. The Contractor shall be responsible for replacement, repair, and calibration of sensors and other necessary data acquisition equipment needed to supply the required data.

Repairs shall be completed within 48 hours of any sensor failure. Upon completion of a repair, replacement, installation, modification, or calibration the Contractor shall notify the Contracting Officer's Representative (COR). The COR may request recalibration of sensors or other hardware components at any time during the contract as deemed necessary.

The Contractor shall keep a log of sensor repair, replacement, installation, modification, and calibration in the onsite copy of the DPIIP. The log shall contain a three-year history of sensor maintenance, including the time of sensor failures (and subsequent repairs), the time and results of sensor calibrations, the time of sensor replacements, and the time that backup sensor systems were initiated to provide the required data. It shall also contain the name of the person responsible for the

sensor work.

Sensors installed shall be capable of collecting parameters within specified accuracies and resolutions indicated in the following subparagraphs.

With the exception of position and any value calculated, reported sensor values should represent a weighted average with the highest and lowest values not included in the calculated average for the given interval. The averaging routine used should be consistent across all event triggers. This information should be documented in the DPIP sections that say "Calculations done external to the instrumentation."

These data-reporting requirements cover the collection of electronic data on a scow through the entire dredging cycle. Disposal events can consist of both open-water disposal and offloading. Open-water disposal is the placement of material via bottom doors or split hull. Offloading is the placement of material via either hydraulic or mechanical means.

3.1.1 Scow Name

Each scow shall be assigned a unique name that will remain constant from one dredging operation to the next.

3.1.2 Contract Number

The USACE-assigned contract number for the project will be reported.

3.1.3 Load Number

A DQM load number shall document the end of a disposal event for a given scow.

3.1.4 Horizontal Positioning

Horizontal positioning shall be recorded as the geographic coordinates of the vessel as indicated by the location of the Global Positioning System (GPS) antenna. All locations shall be obtained using a positioning system operating with a minimum accuracy level of 1 to 3 meters horizontal Circular Error Probable (CEP). Positions shall be reported as Latitude/Longitude WGS 84 in decimal degrees. West Longitude and South Latitude values are reported as negative.

3.1.5 Date and Time

The date and time shall be reported to the nearest second and referenced to Universal Time Coordinated (UTC) based on a 24-hour format: yyyy-mm-dd hh:mm:ss.

3.1.6 Hull Status

Hull status is meant to reflect a condition when material could be removed or released from the scow.

For this contract, hull status shall register closed prior to leaving the disposal area.

3.1.6.1 Open-Water Disposal

An open split hull or open bottom door of a scow shall be indicated by

reporting an "OPEN" value. A closed split hull or closed bottom door of a scow shall be indicated by reporting a "CLOSED" value. An open status shall be indicated as the bin starts to open, and a closed status shall be indicated only once the bin is fully closed. For pocket scows, the open/closed status shall correspond to the compartment which is first to open and last to close.

3.1.6.2 Offloading

For non-dumping scows, an "OPEN" value shall indicate that the bin is in the process of being unloaded, either by pumping or mechanical means.

3.1.7 Course

Scow course-over-ground (COG) shall be provided using industry-standard equipment. The Contractor shall provide scow course-over-ground (to the nearest whole degree) with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention.

3.1.8 Speed

Scow speed-over-ground shall be provided in knots using industry-standard equipment with a minimum accuracy of 1.0 knot and resolution to the nearest 0.1 knot.

3.1.9 Heading

Scow heading shall be provided using industry-standard equipment. The scow heading shall be accurate to within 5 degrees and reported to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention.

3.1.10 Draft

All reported draft measurements shall be in feet, tenths, and hundredths with an accuracy of +/- 0.1 foot relative to observed physical draft readings. The measurements shall be reported at a resolution of two decimal places (hundredths of a foot). The reported forward draft value shall be equal to the sum of the visual forward port and starboard draft mark readings divided by two. The reported aft draft value shall be equal to the sum of the visual aft port and starboard draft mark readings divided by two. Forward draft, aft draft, and average draft will be reported. Sensors shall be placed at an optimum location on the scow to be reflective of observed physical draft mark readings at any trim or list. Minimum accuracies are conditional to relatively calm water. The sensor value reported shall be an average of at least ten samples per event, with at least one maximum value and one minimum value removed, and the minimum eight remaining values averaged. When the average draft is calculated for the purpose of determining displacement, significant digits for average draft shall be maintained such that if forward draft were 0.15 and aft draft were 0.1, then the average draft would be 0.125.

3.1.11 Displacement

Scow displacement shall be reported in long tons, based on the most accurate method available for the scow. The minimum standard of accuracy for displacement is interpolation from the displacement table, based on the average draft. For this contract the density of water used to calculate displacement shall be _____ * kg/cubic meter and shall be used

for an additional interpolation between the fresh and salt water tables.

** The water density used is project-/location-specific. Enter the appropriate water density in the blank:*

- Fresh Water: 1000 kg/m³ (1g/cm³)
- Salt Water: 1027-1030 kg/m³ (1.027-1.03g/cm³)

3.1.12 Bin Ullage Sounding

All reported ullage soundings shall be in feet, tenths, and hundredths with an accuracy of +/- 0.1 foot with respect to the combing and be representative of the forward and aft extents of the hopper as close to the centerline as is possible. The measurements shall be reported at a resolution of two decimal places (hundredths of a foot). If sensors must be offset from the centerline of the bin, they should be offset to opposite sides of the vessel. Forward ullage, aft ullage, and average ullage soundings will be reported. The sensor value reported shall be an average of at least ten samples per event, with at least one maximum value and one minimum value removed, and the minimum eight remaining values averaged. When the average ullage is calculated for the purpose of determining the hopper volume, significant digits for the average ullage shall be maintained such that if the forward ullage were 0.15 and aft ullage were 0.1, then the average ullage would be 0.125. Special arrangements for pocket scows may be made in consultation with the DQM Support Center.

3.1.13 Bin Volume

Scow bin volume shall be reported in cubic yards based on the most accurate method available for the scow. The minimum standard of accuracy for bin volume is interpolation from the bin volume table based on the average ullage soundings.

3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS

The Contractor's DQM system shall be capable of collecting, displaying, and transmitting information to the DQM database. The parameters which shall be reported to the DQM database include trip number, date and time, hull status, scow course, scow speed, scow heading, draft, displacement, ullage, and bin volume. An easily accessible, permanent visual display on the scow shall show in real time the parameters collected by the system in the same units as are used for data submitted to the DQM database. In the event a reported parameter is calculated based on multiple sensors, the sensor values as used in the equation shall be able to be viewed in addition to the required parameter. If a hardware problem occurs, or if a part of the system is physically damaged, then the Contractor shall be responsible for repairing it within 48 hours of determination of the condition.

3.2.1 Telemetry

The Contractor may select any commercial satellite, cellular phone, or other data communications systems available, as long as it is capable of transmitting real-time data as well as enough additional bandwidth to clear historically queued data when a connection is reobtained. If connectivity is lost, unsent data shall be queued and transmitted upon restoration of connectivity. Delays in pushing real-time data to the DQM database should not exceed four hours. Exceptions to these requirements

may be granted by the DQM Center on a case-by-case basis with consideration for contract-specific requirements, site-specific conditions, and extreme weather events.

The data transmission process from the scow to the DQM database must be automated. The data may be sent from the scow directly to the DQM database or to a shore-based system. Data transmitted to the DQM database should be raw data; any processing of the data conducted shoreside shall be done using repeatable automated software or programming routine. A description of this process shall be included in the DPIIP.

3.2.2 Data Reporting Frequency

Disposal activities shall be logged with high temporal and spatial resolution. Data shall be logged as a series of events. Each set of measurements (time, position, etc.) will be considered an event. Any required information in Paragraph 3.1, "Requirements for Reported Data," that is not an averaged variable (that is, draft and ullage) shall be collected within 1 second of the reported time. Data shall be measured with sufficient frequency by the scow system to resolve the events to the accuracy specified in the following table. Any averaged variable must be collected and computed within this sampling interval. Event types "Sailing," "Loading/Stationary," "Offloading," and "Open Water Disposal" are triggered by a time criterion; the criterion should be consistent across the "Sailing" and "Open Water Disposal" event types and should not change for the data collected on a given scow. This criterion should be documented by the Contractor in the DPIIP.

Event Type	Event Trigger Descriptions	Event Time Resolution	Event Position
Loading/ Stationary	No change in position with hull status closed An elapsed time of 1 hour since the last event. No change in position with hull status open -----NONCLOSURE----- In the event a scow has completed an open water disposal and transited back to a holding station without closing the hull, the sampling shall be changed to once per hour.	1 minute	N/A
Sailing	Change in position with hull status closed Time from the last sample equals 1 minute.	1 second	+/-10 ft
Open Water Disposal	Hull status open A position must be recorded within 1 second of the hull status going from closed to open and again within 1 second of the hull status going from open to closed. The position shall be reported at any equal interval from 6 to 12 seconds. This interval shall always remain consistent for the dredge plant.	1 second	+/-10 ft
Offloading	Offloading material, hull status reported as open A position must be recorded within 1 minute arrival at the offload location and within one second of the material starting to be	1 second	+/-10 ft

removed from scow. The time from the last sample equals 1 minute.
-----STANDBY OFFLOADING-----
In the event a scow is not being actively offloaded at the offload location for a time equal to one hour, the sampling interval shall be equal to once an hour. 1 minute

Example: The scow is stationary for 1 hour and 15 minutes, and then it sails to the disposal area. You should have a "Loading/Stationary" event at time 0, time 1 hour, and time 1 hour and 15 minutes. Then, for "Sailing," within 1 second of an elapsed time of 1 minute from the 1 hour and 15 minutes event, another event occurs.

3.2.3 Data Transmission to the Web Service

A Simple Object Access Protocol (SOAP) web service shall be used to report sensor data to the DQM database. Data shall be transmitted as it is collected in real time and pushed to the DQM web service. If the web service is not available or returns an error message, the data shall be stored in a queue and transmitted upon re-establishment of the connection, starting with the oldest data in the queue and continuing until real-time transmission is restored. Delays in pushing real-time data to the DQM database should not exceed four hours. Exceptions to these requirements may be granted by the DQM Support Center on a case-by-case basis with consideration for contract-specific requirements, site-specific conditions, and extreme weather events.

Contact dqm-support@usace.army.mil to obtain the web service URL and the appropriate key credentials and communication protocol.

The data transmission method call takes two arguments: a string containing the plant identifier assigned by the DQM Support Center and a second string containing the JSON-formatted sensor data. The method returns the string "OK" if the data is received. If the data is not received, either the web service or the client application throws an error.

3.2.4 XML-Formatted Sensor Data String

Each scow event shall be passed as a string on one continuous line of data. The example below is broken up by variable for ease of reading:

```
<?xml version="1.0"?>
<SCOW_DREDGING_DATA version="2.5">
  <SCOW_NAME>AU1994</SCOW_NAME>
  <CONTRACT>W123BA-09-D-0087_RL01</CONTRACT>
  <TRIP_NUMBER>34</TRIP_NUMBER>
  <X_POSITION>-81.670632</X_POSITION>
  <Y_POSITION>41.528987</Y_POSITION>
  <DATE_TIME>2010-08-14 10:50:15</DATE_TIME>
  <SCOW_SPEED>0.0</SCOW_SPEED>
  <SCOW_COURSE>0.0</SCOW_COURSE>
  <HULL_STATUS>OPEN</HULL_STATUS>
  <SCOW_HEADING></SCOW_HEADING>
  <SCOW_FWD_DRAFT></SCOW_FWD_DRAFT>
  <SCOW_AFT_DRAFT></SCOW_AFT_DRAFT>
  <SCOW_AVG_DRAFT></SCOW_AVG_DRAFT>
  <ULLAGE_FWD></ULLAGE_FWD>
```

```
<ULLAGE_AFT></ULLAGE_AFT>
<ULLAGE_AVG></ULLAGE_AVG>
<SCOW_BIN_VOLUME></SCOW_BIN_VOLUME>
<SCOW_DISPLACEMENT></SCOW_DISPLACEMENT>
<SCOW_LIGHTSHIP></SCOW_LIGHTSHIP>
<SCOW_TDS></SCOW_TDS>
<ADDITIONAL_DATA>Some more scow info, if needed</ADDITIONAL_DATA>
</SCOW_DREDGING_DATA>
```

DATE_TIME values shall be formatted as YYYY-MM-DD HH:MM:SS, as shown above. If, for any reason, a field has no value, the enclosing XML tags should be sent with nothing between them (for example, <DRAFT_AFT></DRAFT_AFT>). The web service cannot handle a "null" value or any other indicators of no value collected.

3.2.5 Contractor Data Backup

The Contractor shall maintain an archive of all data sent to the DQM database during the dredging contract. The Contracting Officer's Representative (COR) may require, at no increase in the contract price, that the Contractor provide a copy of these data covering specified time periods. The data shall be provided in the HTML format which would have been transmitted to the DQM database. Data submission shall be via storage medium acceptable to the COR.

At the end of the dredging contract, the Contractor shall contact the DQM Support Center prior to discarding the data. The DQM Support Center will verify that all data has been received and appropriately archived before giving the Contractor discard permission. The Contractor shall record in a separate section at the end of the scow's onsite copy of the DPIIP the following information:

- Person who made the call
- Date of the call
- DQM representative who gave permission to discard

3.3 PERFORMANCE REQUIREMENTS

The Contractor's DQM system shall be fully operational at the start of dredging operations and fully certified prior to moving dredge material on the contract (see Paragraph 1.4, "National Dredging Quality Management Program Certification"). To meet contract requirements for operability, in addition to certification, the Contractor's system shall provide a data string with values for all parameters while operating, as described within the specifications. Additionally, all hardware shall be compliant with DPIIP requirements (Paragraph 1.5, "Dredge Plant Instrumentation Plan (DPIIP)"). Quality data strings are considered to be those providing values for all parameters reported when operating according to the specification. Repairs necessary to restore data return compliance shall be made within 48 hours. Failure by the Contractor to report the required data within the specified time window for scow measurements (see Paragraph 3.2.2, "Data Reporting Frequency," and Paragraph 3.2.3, "Data Transmission to the Web Service") and failure to receive DQM certification prior to dredging will result in withholding of up to 10% of the contract progress payment per clause 52.232-5.

3.4 LIST OF ITEMS TO BE PROVIDED BY THE CONTRACTOR

DPIP	Paragraph 1.5, "Dredge Plant Instrumentation Plan (DPIP)"
DQM SYSTEM	
Sensor instrumentation	Paragraph 3.1, Requirements for Reported Data"
SCOW DATA	
Event documentation	Paragraph 3.2.2, "Data Reporting Frequency"
Data reports	Paragraph 3.2.3, "Data Transmission to the Web Service"
QA EQUIPMENT ON THE DREDGE	
Clear and accurate draft marks	
Ullage tape	
-- End of Section --	

SECTION TABLE OF CONTENTS

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

SECTION 35 20 27

NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM PIPELINE HYDRAULIC DREDGE

PART 1 GENERAL

- 1.1 DESCRIPTION
- 1.2 SUBMITTALS
- 1.3 PAYMENT
- 1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION
- 1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

PART 2 PRODUCTS (Not Applicable)

PART 3 EXECUTION

- 3.1 REQUIREMENTS FOR REPORTED DATA
 - 3.1.1 Message Bundle Data
 - 3.1.1.1 Messages
 - 3.1.1.1.1 Message Time
 - 3.1.1.1.2 Comment
 - 3.1.1.2 Dredge Events - Work Event
 - 3.1.1.2.1 Vertical Correction
 - 3.1.1.2.2 Cutter/Suction Head Location and Movement
 - 3.1.1.2.2.1 Cutter/Suction Head Horizontal Position
 - 3.1.1.2.2.2 Cutter/Suction Invert Depth
 - 3.1.1.2.2.3 Cutter/Suction Head Heading
 - 3.1.1.2.3 Dredge Activity
 - 3.1.1.2.3.1 Slurry Velocity
 - 3.1.1.2.3.2 Slurry Density
 - 3.1.1.2.3.3 Pump RPM
 - 3.1.1.2.3.4 Pump Vacuum
 - 3.1.1.2.3.5 Pump Outlet Pressure
 - 3.1.1.2.4 Outfall Information (Open Water/Spill Barge Disposal)
 - 3.1.1.2.4.1 Discharge Horizontal Position
 - 3.1.1.3 Dredge Events - State Event
 - 3.1.1.3.1 Message Time
 - 3.1.1.3.2 Contract Event
 - 3.1.1.3.2.1 Contract Number
 - 3.1.1.3.2.2 Contract Start and End
 - 3.1.1.3.3 Tide Station/River Stage Gage Event
 - 3.1.1.3.3.1 Station Name
 - 3.1.1.3.4 Length of Pipe Event
 - 3.1.1.3.4.1 Floating Pipe
 - 3.1.1.3.4.2 Submerged Pipe
 - 3.1.1.3.4.3 Shore Pipe
 - 3.1.1.3.5 Booster Pump Event
 - 3.1.1.3.5.1 Number of Booster Pumps
 - 3.1.1.3.6 Dredge Advance
 - 3.1.1.3.7 Outfall Information
 - 3.1.1.3.7.1 Discharge Location

- 3.1.1.3.7.2 Discharge Horizontal Position
- 3.1.1.3.7.3 Discharge Outfall Heading
- 3.1.1.3.7.4 Discharge Pipe Elevation
- 3.1.1.3.8 Non-effective Work Event
 - 3.1.1.3.8.1 Non-effective Work Interval
 - 3.1.1.3.8.2 Dredge Function Code
 - 3.1.1.3.8.3 Additional Comments
- 3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS
 - 3.2.1 Computer Requirements
 - 3.2.2 Software
 - 3.2.3 UPS
 - 3.2.4 Internet Access
 - 3.2.5 Data Routing Requirements
- 3.3 DREDGE MONITORING DATA
 - 3.3.1 General
 - 3.3.2 Data Measurement Frequency
 - 3.3.2.1 Work Event Messages
 - 3.3.2.2 State Event Messages
 - 3.3.3 Parameter Transmission to the Web Service
 - 3.3.4 Contractor Data Backup
- 3.4 PERFORMANCE REQUIREMENTS
- 3.5 LIST OF ITEMS PROVIDED BY THE CONTRACTOR

-- End of Section Table of Contents --

SECTION 35 20 27

NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM
PIPELINE HYDRAULIC DREDGE

PART 1 GENERAL

1.1 DESCRIPTION

The work under this contract requires use of the US Army Corps of Engineers (USACE) National Dredging Quality Management Program (DQM) to monitor the dredge's status at all times during the contract and manage data history.

This performance-based specification section identifies the minimum required output as well as the precision and instrumentation requirements. The requirements may be satisfied using equipment and technical procedures selected by the Contractor.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office responsible for review of the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00, "Submittal Procedures":

[SD-01 Preconstruction Submittals](#)

[Dredge Plant Instrumentation Plan](#) Revisions or Addendum; G

[SD-07 Certificates](#)

[National Dredging Quality Management Program Certification](#); G, R0

1.3 PAYMENT

No separate payment shall be made for the installation, operation, and maintenance of the DQM-certified system as specified herein for the duration of the dredging operations; all costs in connection therewith shall be considered a subsidiary obligation of the Contractor and covered under the contract unit price for dredging in the bidding schedule.

1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION

The Contractor is required to have a current [National Dredging Quality Management Program Certification](#) from the DQM Program for the cutter/suction head hydraulic dredge instrumentation system to be used under this contract. Standard Operating Procedures (SOP) and criteria for certification are presented on the DQM website at <https://dqm.usace.army.mil>.

1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

The Contractor shall have a digital copy of the [Dredge Plant](#)

Instrumentation Plan (DPIP) on file with the DQM Support Center. While working on site, the Contractor shall also maintain on the dredge a copy of the DPIP, which is easily accessible to Government personnel at all times. This document shall accurately describe the sensors used, the configuration of the system, how sensor data will be collected, how quality control on the data will be performed, and how the sensors/data-reporting equipment will be calibrated and repaired if it fails. A description of the computed dredge-specific data and how the sensor data will be transmitted to the DQM database shall also be included. Prior to the start of work, the Contractor shall submit to the DQM Support Center any addendum or modifications made to the plan subsequent to its original submission. Requirements and a template for the DPIP are available on the DQM website at <https://dqm.usace.army.mil>.

PART 2 PRODUCTS (Not Applicable)

PART 3 EXECUTION

3.1 REQUIREMENTS FOR REPORTED DATA

The Contractor shall provide, operate, and maintain all hardware and software to meet these specifications. The Contractor shall also be responsible for the replacement, repair, and calibration of the sensors and other necessary data acquisition equipment needed to supply the required data.

The procedure to complete a repair shall be documented and completed as soon as practical. If repair is not possible within two business days of any sensor failure, a plan and timeline to complete the repair shall be submitted. Upon completion of a repair, replacement, installation, modification, or calibration, the Contractor shall notify the Contracting Officer's Representative (COR). The COR may request recalibration of the sensors or other hardware components at any time during the contract as deemed necessary.

The Contractor shall keep a log of sensor repair, replacement, installation, modification, and calibration in the dredge's onboard copy of the DPIP. The log shall contain a three-year history of sensor maintenance, including the time of the sensor failures (and subsequent repairs), the time and results of sensor calibrations, the time of sensor replacements, and the time that backup sensor systems were initiated to provide the required data. It shall also contain the name of the person responsible for the sensor work.

Sensors installed shall be capable of collecting parameters within the specified accuracies and resolutions indicated in the following subparagraphs and transmit these parameters to the DQM database. All data shall be transmitted in JSON message bundles. Each bundle can contain multiple message types. Sensor data shall be transmitted as work event messages, and data which relates to the operational state of the dredge or its sensors shall be transmitted as state event messages. (See Paragraph 3.3.3, "Parameter Transmission to the Web Service.")

3.1.1 Message Bundle Data

Every message bundle shall contain descriptive data that relates the message to a given dredge plant and date/time. The start of a message bundle shall be identified by the tag "DQM_data".

3.1.1.1 Messages

Messages contain operational data that populates the DQM database for a dredge plant. A message shall consist of an event type and its associated data (as defined in Paragraph 3.1.1.1.3, "Dredge Events"), a date/time stamp indicating when the event occurred or started, and a comment providing clarification or metadata about the situation. There are multiple event types, but they all fall into one of two categories - work events and state events.

3.1.1.1.1 Message Time

In a work event message, message time is the date and time that the data is collected from the sensors; in a state event message, message time is the date and time that the state event begins. The message time shall be reported to the nearest second and referenced to Coordinated Universal Time (UTC) time based on a 24-hour format (*YYYY-MM-DD HH:MM:SS*). In order to ensure accuracy and reliability, the time stamp shall be synchronized to UTC format from an accurate, unchangeable source (for example, a GPS National Marine Electronics Association (NMEA) datastring). Message time shall be identified by the tag "msg_time".

3.1.1.1.2 Comment

Comments concerning the work event or state event messages being transmitted provide descriptive information that relates to the data. An example of a comment for work event data is information about a sensor issue; an example of a comment for state event data is a description of operations. A comment shall be identified by the introductory tag "comment", and the comment shall consist of no more than 250 characters.

3.1.1.2 Dredge Events - Work Event

There are two types of dredge event messages - work event messages and state event messages. Work event messages contain data that are instantaneously collected or calculated from sensors and are logged as a series of events. Work events are triggered by a time interval change (as described in Paragraph 3.3.2.1, "Work Event Messages"). All work event messages shall be initiated by the header tag "work_event".

3.1.1.2.1 Vertical Correction

The variation of the water level from the vertical datum for the river stage or tidal gage described in the state events shall be obtained using appropriate equipment to give the water level with an accuracy of +/- 0.1 ft. Vertical correction values above project datum described in the dredging specification shall be entered with a positive sign and those below with a negative sign. The tag for vertical correction shall be "vert_correction".

3.1.1.2.2 Cutter/Suction Head Location and Movement

The X, Y, and Z components of the cutter/suction head location shall be monitored. Additional calculations made from the observed values determine the rates of movement to track the progress of the dredge.

3.1.1.2.2.1 Cutter/Suction Head Horizontal Position

The forwardmost point of the cutter/suction head shall be obtained using a

positioning system operating with a minimum accuracy level of 3-10 feet horizontal Circular Error Probable (CEP). It shall be reported as Latitude/Longitude WGS 84 in decimal degrees with West Longitude and South Latitude values reported as negative. Position values shall be identified by the tags "ch_latitude" and "ch_longitude".

3.1.1.2.2.2 Cutter/Suction Invert Depth

Cutter/suction invert depth is the depth of the invert of the suction mouth relative to the surface of the water. Instrumentation shall be capable of reporting to an accuracy of +/- 0.5 foot and a resolution to the nearest 0.1 foot with no tidal adjustments. Minimum accuracies are conditional to relatively calm water. The tag "ch_depth" shall be used to identify the cutter/suction head depth.

3.1.1.2.2.3 Cutter/Suction Head Heading

The cutter/suction head heading is the angle of the centerline of the cutter/suction head and dredge ladder measured relative to true north. All headings shall be provided using industry-standard equipment. The heading shall be accurate to within 5 degrees and reported to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention. The tag "ch_heading" shall be used to identify the cutter/suction head heading.

3.1.1.2.3 Dredge Activity

Dredge activity shall be monitored using a combination of the following parameters.

3.1.1.2.3.1 Slurry Velocity

A flow-metering device, calibrated according to the manufacturer's specifications, shall be used to record the slurry velocity to the nearest 0.01 fps with an accuracy of plus 0.5 fps. If the manufacturer does not specify a frequency of recalibration, calibration shall be conducted prior to the commencement of work. The slurry velocity shall be measured for the same pipeline inside diameter as that used for the slurry density measurement. The tag "slurry_velocity" shall be associated with this value.

3.1.1.2.3.2 Slurry Density

A density-metering device, calibrated according to the manufacturer's specifications, shall be used to record the slurry density to the nearest 0.01 g/cc. It is understood that the accuracy of this sensor can vary based on several factors, including the type of material, the magnitude of the cut, and the length of time since calibration. If the manufacturer does not specify a frequency of recalibration, calibration shall be conducted prior to the commencement of work. Continuous monitoring of this sensor ensures that drift and other factors inherent in the dredging process can be accounted for in monitoring dredge activity. The tag "slurry_density" shall be associated with this value.

3.1.1.2.3.3 Pump RPM

The pump rpm is the number of revolutions per minute measured for the slurry pump shaft. The shaft revolution rate (rev/min) shall be measured with the highest level of accuracy that is standard on the vessel's operational displays either at the bridge or in the engine room. This

value shall be identified by the tag "rpm".

3.1.1.2.3.4 Pump Vacuum

The vacuum pressure of the dredge pump(s) (inches of mercury) shall be measured as near to the eye as practicable in the pump's suction pipe with the highest level of accuracy that is standard on the vessel's operational displays either at the leverman's controls or in the engine room. Vacuum pressure shall be identified by the tag "vacuum".

3.1.1.2.3.5 Pump Outlet Pressure

The pump outlet pressure shall be measured in the discharge line on the pump side of the flap valve in terms of pounds per square inch (psi) on a gauge. Pump outlet pressure shall be identified by the tag "outlet_psi".

3.1.1.2.4 Outfall Information (Open Water/Spill Barge Disposal)

The X and Y position of the terminal end of the outfall pipe shall be monitored continuously and the position reported as part of the work event string.

3.1.1.2.4.1 Discharge Horizontal Position

The horizontal position of the outfall end of the discharge pipe shall be obtained using a positioning system operating with a minimum accuracy level of 3-10 feet horizontal Circular Error Probable (CEP). It shall be reported as Latitude/Longitude WGS 84 in decimal degrees with West Longitude and South Latitude values being reported as negative. Position values shall be identified by the tags "outfall_latitude" and "outfall_longitude".

3.1.1.3 Dredge Events - State Event

There are two types of dredge event messages - work event messages and state event messages. State event messages provide information about the current state of the dredge equipment or operations. They are created and sent only when a state changes. Since state events often cannot be collected in real time, state events are tagged with a date time stamp (referenced to Coordinated Universal Time (UTC)) that indicates when the state change happened relative to the work event message tag. This data is considered to be "true" until another state event tag of the same type is received. Each type of state event message shall be indicated by a specific header tag as enumerated in the following subparagraphs. State events can be transmitted along with work event message bundles directly by the contractor using the indicated format, or they can be entered on the "State" tab in the DQM-provided software. However, they should be sent only if the state value changes.

3.1.1.3.1 Message Time

The state event time is the date and time that the event starts. The leverman's time shall be entered to the nearest second as local time and automatically converted to and reported in UTC based on a 24-hour format (YYYY-MM-DD HH:MM:SS). Message time shall be identified by the tag "msg_time".

3.1.1.3.2 Contract Event

Information concerning the contract under which dredging is being performed shall be reported at the start and completion of each contract using the header tag "contract_event".

3.1.1.3.2.1 Contract Number

The USACE-assigned contract number for the project shall be reported using the tag "contract_number".

3.1.1.3.2.2 Contract Start and End

The start and end of a contract shall be reported using the tag "event_type" with the appropriate value of "start" or "end".

3.1.1.3.3 Tide Station/River Stage Gage Event

Properties associated with the vertical correction (see Paragraph 3.1.1.1.3.1.1, "Vertical Correction") for the tide station/river stage gage shall be grouped together under the header tag "station_event". This information shall be sent at the start of the contract and each time the dredge has moved enough to change the station being used.

3.1.1.3.3.1 Station Name

The station name is a concise name defining the tide station/river stage gage begin referred to. It shall be introduced by the tag "station_name", and it shall consist of a descriptor of no more than 25 characters.

3.1.1.3.4 Length of Pipe Event

The leverman's estimate of the length of pipe downflow from the dredge pump, measured to the nearest whole foot, shall be reported under the header tag "pipe_length_event". This information shall be sent at the start of the contract and at the completion of each 24-hour period ending at midnight local time.

3.1.1.3.4.1 Floating Pipe

The total length of floating pipe shall be reported with the tag "length_floating".

3.1.1.3.4.2 Submerged Pipe

The total length of floating pipe shall be reported with the tag "length_submerged".

3.1.1.3.4.3 Shore Pipe

The total length of shore pipe shall be reported with the tag "length_land".

3.1.1.3.5 Booster Pump Event

Information concerning the booster pumps being used shall be included under the header tag "booster_pump_event". A message shall be sent to indicate any change in the status of the booster pumps being used.

3.1.1.3.5.1 Number of Booster Pumps

Upon the addition or removal of a booster pump, the total number of booster pumps being used shall be reported with the tag "booster_total".

3.1.1.3.6 Dredge Advance

The dredge advance, the total forward progress of the dredge relative to the centerline of the cut, shall be measured to the nearest whole foot and cumulatively calculated over a 24-hour period from midnight to midnight local time. It shall be identified by the tag "advance_daily". The msg_time associated with this tag shall be reported as the first timestamp of the following 24-hour period (based on the local time) rather than as midnight of the day for which the value was calculated, and it shall be reported in Greenwich Mean Time (GMT).

3.1.1.3.7 Outfall Information

The X and Y position of the terminal end of the outfall pipe shall be monitored and sent at the start of the contract and thereafter according to the following table. Discharge Heading and Pipe Elevation may be omitted if the dredge is not discharging into an upland disposal site. For beach nourishment, the horizontal X and Y position of the outfall shall be sent at the start of the contract and at the completion of each 24-hour period ending at midnight local time.

Discharge Location	Horizontal Position	Discharge Pipe Elevation	Discharge Outfall Heading
Open Water	Continuous Work Event	N/A	N/A
Scow	Upon Change	N/A	N/A
Beach	Every 24 Hours	N/A	N/A
Upland	Upon Change	Upon Change	Upon Change

3.1.1.3.7.1 Discharge Location

Information on where the slurry is being discharged shall be reported with the tag "outfall_location". Acceptable values include "upland", "open water", "beach", and "scow".

3.1.1.3.7.2 Discharge Horizontal Position

The horizontal position of the outfall end of the discharge pipe shall be obtained using a positioning system operating with a minimum accuracy level of 3-10 feet horizontal Circular Error Probable (CEP). It shall be reported as Latitude/Longitude WGS 84 in decimal degrees with West Longitude and South Latitude values being reported as negative. Position values shall be identified by the tags "outfall_latitude" and "outfall_longitude".

3.1.1.3.7.3 Discharge Outfall Heading

The discharge outfall heading is the angle relative to true north measured from the centerline of the pipe in the direction of discharge. All headings shall be provided using industry-standard equipment. They shall

be accurate to within 5 degrees and reported to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention. The discharge heading shall be identified by the tag "outfall_heading".

3.1.1.3.7.4 Discharge Pipe Elevation

The discharge pipe elevation is the height of the outfall measured in feet and tenths of a foot relative to the project datum. The required accuracy is contingent upon contract requirements. The tag "outfall_elevation" shall be used to identify this elevation.

3.1.1.3.8 Non-effective Work Event

Delays and dredge downtime shall be reported at the conclusion of the event. The reason for the non-effective work time shall be submitted under the header tag "non_eff_event" within 24 hours of the event.

3.1.1.3.8.1 Non-effective Work Interval

The start and end times for the non-effective work event shall be reported using the tags "msg_start_time" and "msg_end_time".

3.1.1.3.8.2 Dredge Function Code

The dredge operator indication of production delays, as listed on Form 4267, shall be transmitted at the end of the non-effective interval. Dredge function event messages shall be identified by the tag "function_code" and shall consist of one of the following standardized entries to indicate the operation:

AGV	Assisting Grounded Vessels
CCH	Change Cutterhead
CCSH	Clear Cutter Suction
CLPJ	Change Location Bar
COLL	Collision
CPPL	Clear Pump Pipeline
CPR	Change Impeller
DR	Dike Repair
FBD	Fire Boat Drills
HPL	Handling Pipe Line
HSL	Handling Swing Line
HSP	Handling Shore Pipe
LDNE	Loss Due to Natural Elements
LDPV	Loss Due to Passing Vessel
LNL	Transfer to New Location
MISC	Miscellaneous
MOB	Mobilization & Demobilization
MSC	Miscellaneous/Non-pay
OC	Out of Commission
OR	Operating Repairs
P	Preparation
PREP	Preparation & Making Up Tow
RPL	Repair Pipeline
SB	Sounding & Buoying
SBT	Stand-By Time as Directed
SH	Sundays-Holidays
TFS	Taking on Fuel & Supplies
TOW	Time on Tow

WAP Waiting Attendant Plant

3.1.1.3.8.3 Additional Comments

The "comment" tag shall be used to provide additional explanation for the noted delays or downtimes. For example, when the code "LDPV" (Loss Due to Passing Vessel) is indicated, the name of the vessel and the number of tows shall be listed with the "comment" tag.

3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS

The Contractor's DQM system shall be capable of collecting and transmitting information to the DQM onboard computer. The applicable parameters from Paragraph 3.1, "Requirements for Reported Data," shall be recorded as events locally and continuously transmitted to the DQM database anytime an Internet connection is available. The dredge shall be equipped with a DQM computer system consisting of a computer, monitor, keyboard, mouse, data modem, Universal Power Supply (UPS), and network hub. The computer system shall be a standalone system, exclusive to the DQM monitoring system, and shall have USACE DQM software installed on it. If a hardware problem occurs, or if a part of the system is physically damaged, then the Contractor shall be responsible for repairing it within two business days of the determination of the condition or submitting a plan and timeline for repair if the repair will take more than two business days.

3.2.1 Computer Requirements

The Contractor shall provide a dedicated onboard computer for use by the Dredging Quality Management system. This computer shall run the USACE DQM software and receive data from the Contractor's data-reporting interface. This computer must meet or exceed the following performance specifications:

CPU	Intel or AMD processor with a (non-overclocked) clock speed of at least 1.6 gigahertz (GHz)
Hard drive	250 gigabytes (GB); internal
RAM	4 gigabytes (GB)
Ethernet adapter	Internal network card with an RJ 45 connector
Ports	1 free serial port with standard 9-pin connectors; 1 free USB port
Other hardware	Keyboard, mouse, monitor

The Contractor shall install a fully licensed copy of Windows 7 Professional Operating System or later on the computer specified above. The Contractor shall also install any necessary manufacturer-provided drivers for the installed hardware.

This computer shall be located and oriented to allow data entry and data viewing as well as to provide access to data ports for connection of external hardware.

3.2.2 Software

The DQM computer's primary function is to transmit data to the DQM shoreside database. No other software which conflicts with this function shall be installed on it. The DQM computer shall also have the USACE-provided Dredging Quality Management Onboard Software (DQMOBS) installed on it by DQM personnel.

3.2.3 UPS

The Contractor shall supply an Uninterruptible Power Supply (UPS) for the computer and networking equipment. It shall interface with the DQM computer to communicate UPS status, and it shall provide backup power at 1 kVA for a minimum of 10 minutes. The Contractor shall ensure that sufficient power outlets are available to run all specified equipment.

3.2.4 Internet Access

The Contractor shall maintain an Internet connection capable of transmitting real-time data to the DQM server as well as enough additional bandwidth to clear historically queued data when a connection is re-established. If connectivity is lost, unsent data shall be queued and transmitted upon restoration of connectivity. Delays in pushing real-time data to the DQM database should not exceed four hours. Exceptions to these requirements may be granted by the DQM Support Center on a case-by-case basis with consideration for contract-specific requirements, site-specific conditions, and extreme weather events.

The Contractor shall acquire and install all necessary hardware and software to make the Internet connection available for data transmission to the DQM web service. The hardware and software shall be configured to allow the DQM Support Center remote access to this computer, and the telemetry system shall be capable of meeting these minimum reporting requirements in all operating conditions.

In areas with poor cellular service and at the local District's discretion, it may be required to manually download the data on a daily basis using the protocol for retrieving and submitting backup files provided by the DQM Support Center. This method of data transmission should be used only if Internet connectivity is unavailable at the dredging site, and it should be considered a temporary measure.

3.2.5 Data Routing Requirements

Onboard sensors continually monitor dredge conditions, operations, and efficiency and route this information to the shipboard dredge-specific system (DSS) computer to assist in guiding dredge operations. Portions of this Contractor-collected information, as described in this specification, shall be routed to the DQM computer on a real-time basis. Standard sensor data shall be sent to the DQM computer via an RS-232 serial interface with a baud rate of 9600 or 19200 bps. The serial interface shall be configured as 8 bits, no parity, and no flow control.

Information regarding changes in the state of the dredge shall be digitally logged and transmitted as close to the time of the occurrence as possible. These events can either be included in a separate message bundle going to the DQM onboard computer, or they can be entered on the "State" tab in the DQM Pipeline Software.

3.3 DREDGE MONITORING DATA

3.3.1 General

Onboard sensors continuously collect dredging data in support of the dredge Contractor's operations. Portions of this Contractor-collected information, as described in this specification, and calculations based on

them shall be stored and transmitted to the DQM database on a near real-time basis. Additionally, information regarding the state of the dredge shall be digitally logged and transmitted.

3.3.2 Data Measurement Frequency

The frequency of data transmission is dependent on the type of message being sent. Work Event messages contain data that are instantaneously collected or calculated from sensors and are logged as a series of events. State event messages are activated by a change in the dredge state.

3.3.2.1 Work Event Messages

Data shall be logged as a series of events. Each event shall consist of a dataset containing dredge information (as defined in Paragraph 3.1, "Requirements for Reported Data"). Each set of measurements (for example, time and position) shall be considered an event, and there shall be a 6-12 second interval between work events. This interval shall remain consistent across event types for the dredge plant.

A standard data string shall be recorded within one second of an event trigger with the time stamp and all parameters reflecting when the event happened.

3.3.2.2 State Event Messages

A set of descriptive information (event name, time, description, comment) shall be considered a state event. These events shall be recorded within 24 hours of a change in state with the time stamp reflecting when the event happened.

3.3.3 Parameter Transmission to the Web Service

The data shall be formatted as JSON (JavaScript Object Notation, as defined at <http://www.json.org>) strings of arbitrary length. These JSON strings represent a hierarchical data structure consisting of a message bundle which may contain 0-3 automatic data messages and any number of manual data messages.

A tag/parameter is reported only when it contains a value. No "Null" value strings shall be included in a message bundle.

Message bundle

```
{
  "DQM_Data": {
    "messages": [
      {
        "work_event": {
          "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
          "vert_correction": <floating point 100th decimal place>,
          "ch_latitude":    <decimal to 6 decimal places>,
          "ch_longitude":   <decimal to 6 decimal places>,
          "ch_depth":       <floating point 100th decimal
place>,
          "ch_heading":     <integer value 000-359>,
          "slurry_velocity": <floating point 100th decimal place>,

```

```

        "slurry_density":    <floating point 100th decimal place>,
        "pump_rpm":          <integer>,
        "vacuum":            <floating point 100th decimal place>,
        "outlet_psi":        <floating point 100th decimal place>,
        "comment":           <string>},
    },
    {
        "contract_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "contract_number": <string>,
            "event_type":    <string - "start" or "end">,
            "comment":       <string>
        }
    },
    {
        "station_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "station_name":  <string>,
            "comment":       <string>
        }
    },
    {
        "pipe_length_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "length_floating": <integer>,
            "length_submerged": <integer>,
            "length_land":    <integer>,
            "comment":       <string>
        }
    },
    {
        "booster_pump_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DDHH:MM:SS>,
            "booster_total": <integer>,
            "comment":       <string>
        }
    },
    {
        "advance_Event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "advance_daily": <integer>,
            "comment":       <string>
        }
    },
    {
        "outfall_position": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "outfall_location": <string-"upland", "beach", "scow",
"open water">
            "outfall_latitude": <decimal to 6 decimal places>,
            "outfall_longitude": <decimal to 6 decimal places>,
            "outfall_heading": <integer value 000-359>,
            "outfall_elevation": <floating point 10th decimal place>,
            "comment":       <string>
        }
    },
    {
        "non_eff_event": {

```

```
        "msg_start_time":    <24-hour UTC time YYYY-MM-DD HH:MM:SS>,  
        "msg_end_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,  
        "function_code":     <string - 1 to 4 characters>,  
        "comment":           <string>  
    }  
}  
]  
}  
}
```

3.3.4 Contractor Data Backup

The Contractor shall maintain an archive of all data sent to the DQM computer during the dredging contract. The COR may require, at no increase in the contract price, that the Contractor provide a copy of these data covering specified time periods. The data shall be provided in the same JSON format as would have been transmitted to the DQM computer. There shall be no line breaks between the parameters, and each record string shall be on separate line. The naming convention for the files shall be < dredgename>_<StartYYYYMMddhhmmss>_<EndYYYYMMddhhmmss>.txt. Data submission shall be via a storage medium acceptable to the COR.

At the end of the dredging contract, the Contractor shall call the National DQM Support Center prior to discarding the data. The DQM Support Center will verify that all data has been received and appropriately archived before giving the Contractor discard permission. The Contractor shall then record the following information in a separate section at the end of the dredge's onboard copy of the DPIP:

- Person who called the National DQM Support Center
- Date of the call
- DQM representative who gave permission to discard the data

3.4 PERFORMANCE REQUIREMENTS

The Contractor's National Dredging Quality Management Program's data transmission shall be fully operational at the start of dredging operations. To meet contract requirements for operability, the Contractor's system shall provide an accurate data string return and be compliant with hardware requirements. Data string return is defined as the number of quality records within an event or state tag sent by the contractor's system to the DQM database. Quality data strings are considered to be those providing accurate values for all parameters reported when operating according to the specification. Repairs necessary to restore data return compliance shall be made within two business days, or a plan and timeline for repair shall be submitted if the repair will take more than two business days. Failure by the Contractor to report quality data within the specified time window for dredge measurements as stated in the specifications (see Paragraph 3.2.4, "Internet Access"; Paragraph 3.3.2, "Data Measurement Frequency"; and Paragraph 3.3.3, "Parameter Transmission to the Web Service"), will result in withholding of up to 10% of the contract progress payment per clause 52.232-5.

3.5 LIST OF ITEMS PROVIDED BY THE CONTRACTOR

- DPIP <https://dqm.usace.army.mil>
- DQM System Paragraph 3.2, "National Dredging Quality Management

Program System Requirements," including all subparagraphs

- Dredge Data Paragraph 3.3, "Dredging Monitoring Data"

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

SECTION 35 31 19.20

ARTICULATING CONCRETE BLOCK REVETMENTS

PART 1 GENERAL

- 1.1 RELATED SECTIONS
- 1.2 REFERENCES
- 1.3 DEFINITIONS
 - 1.3.1 Articulating Concrete Block (ACB) Revetment System
 - 1.3.2 Blocks
- 1.4 SUBMITTALS
- 1.5 DELIVERY, STORAGE, AND HANDLING
 - 1.5.1 Blocks
 - 1.5.2 Microgrid
- 1.6 SCHEDULING

PART 2 PRODUCTS

- 2.1 ARTICULATING CONCRETE BLOCK
 - 2.1.1 Hydraulic Stability
 - 2.1.1.1 Flume Test
 - 2.1.1.2 Extrapolation of Hydraulic Stability
 - 2.1.2 Matrix Assembly - Cabled Systems
 - 2.1.3 Structural requirements
 - 2.1.3.1 Compressive Strength
 - 2.1.3.2 Water Absorption for Dry Cast Units
 - 2.1.3.3 Saturated Surface-Dry Density
 - 2.1.3.4 Air Entraining
- 2.2 GEOTEXTILE
- 2.3 MICROGRID
- 2.4 CABLE
 - 2.4.1 Installation Requirements for Cable
 - 2.4.2 Design Requirements for Cable
- 2.5 VOID FILLER
 - 2.5.1 Aggregate
 - 2.5.2 Topsoil and Seed
- 2.6 STABILIZED STONE DRAINAGE LAYER

PART 3 EXECUTION

- 3.1 SUBGRADE PREPARATION
 - 3.1.1 Clearing
 - 3.1.2 Bank Grading
 - 3.1.3 Compaction and Subgrade Finishing
 - 3.1.4 Grade Tolerances
 - 3.1.5 Subgrade Surface Tolerances
- 3.2 GEOTEXTILE INSTALLATION
 - 3.2.1 General
- 3.3 MICROGRID INSTALLATION

- 3.3.1 Placement
- 3.3.2 Overlaps and Fasteners
- 3.4 BLOCK INSTALLATION
 - 3.4.1 Placement of Pre-Assembled Mattresses
 - 3.4.2 Tolerances
- 3.5 CONCRETE JOINTS
 - 3.5.1 General Requirements
 - 3.5.2 Abutments
- 3.6 VOID FILLER AND SEEDING
- 3.7 PROTECTION OF WORK
- 3.8 QUALITY CONTROL TESTING

-- End of Section Table of Contents --

SECTION 35 31 19.20

ARTICULATING CONCRETE BLOCK REVETMENTS

PART 1 GENERAL

1.1 RELATED SECTIONS

03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS
31 00 00 EARTHWORK
31 05 19 GEOTEXTILE
31 11 00 CLEARING AND GRUBBING
32 92 19 SEEDING.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO M 288 (2017) Standard Specification for
Geosynthetic Specification for Highway
Applications

ASTM INTERNATIONAL (ASTM)

ASTM C33/C33M (2018) Standard Specification for Concrete
Aggregates

ASTM C42/C42M (2020) Standard Test Method for Obtaining
and Testing Drilled Cores and Sawed Beams
of Concrete

ASTM C140/C140M (2020a) Standard Test Methods for Sampling
and Testing Concrete Masonry Units and
Related Units

ASTM D4873/D4873M (2017) Standard Guide for Identification,
Storage, and Handling of Geosynthetic
Rolls and Samples

ASTM D5262 (2007; R 2016) Standard Test Method for
Evaluating the Unconfined Tension Creep
Behavior of Geosynthetics

ASTM D6637 (2011) Standard Test Method for
Determining Tensile Properties of Geogrids
by the Single or Multi-Rib Tensile Method

NATIONAL CONCRETE MASONRY ASSOCIATION (NCMA)

NCMA TR127B (2010) Design Manual for Segmental

Retaining Walls

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)

FHWA RD-89-199

(1989) Hydraulic Stability of Articulated
Concrete Block Revetment Systems During
Overtopping Flow

1.3 DEFINITIONS

1.3.1 Articulating Concrete Block (ACB) Revetment System

A matrix of interconnected concrete block units for erosion protection. Units are connected by geometric interlock and/or cables, geotextiles, or microgrids, and typically include a geotextile underlayment for subsoil retention.

1.3.2 Blocks

Articulating concrete block revetment units will be referred to as blocks.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Block Installation; G, R0

Microgrid Installation; G, R0

SD-03 Product Data

Articulating Concrete Block; G, R0

Microgrid; G, R0

Stabilized Stone Drainage Layer; G, R0

SD-04 Samples

Articulating Concrete Block

Microgrid

SD-06 Test Reports

Flume Test

SD-07 Certificates

Certificates Of Compliance

1.5 DELIVERY, STORAGE, AND HANDLING

Check products upon delivery to assure that the proper material has been received and is undamaged. For geosynthetics, the guidelines presented in [ASTM D4873/D4873M](#) shall be followed.

1.5.1 Blocks

Provide blocks which are sound and free of defects that would interfere with proper placement or that would impair the strength or longevity of the installation. Discard blocks with the following defects:

- a. Broken appendages.
- b. Chips larger than [2 inches](#) in any dimension.
- c. Cracks wider than [0.02 inches](#) and longer than 33 percent of the nominal height.

Minor cracks, incidental to the usual method of manufacture, or chipping that results from customary methods of handling in shipping, delivery and placement will not be deemed grounds for rejection. Store blocks in a suitable location away from mud, paint, wet cement, and other contamination or disturbance.

1.5.2 Microgrid

The QC inspector shall be present during delivery and unloading of the microgrid. Ensure the material has not been damaged during shipping, storage, or handling. Any material found to be damaged shall be repaired or replaced. Accept delivery of material only after the required submittals have been approved. Each roll shall be labeled with the manufacturer's name, product identification, lot number, roll number, and roll dimensions. Store the rolls in a level and dry area.

1.6 SCHEDULING

To limit ultraviolet light exposure of the geotextile, place the blocks within 7 days after placing the geotextile, and the void filler within 14 days after placing the geotextile.

PART 2 PRODUCTS

2.1 [ARTICULATING CONCRETE BLOCK](#)

Submit descriptive technical data on the blocks, [cables](#), [cable fittings](#), and geotextile. Include all material properties specified under paragraph PRODUCTS. Catalog cuts, technical data sheets, or test data shall be submitted showing that the products meet the specifications. The submittal shall also include a copy of any standard manufacturer's warranties for the products. See below under "geotextile" for more requirements. The ACB shall meet the following criteria:

TABLE 1. ACB Requirements		
Criteria	Required Value	Test Method
Matrix Assembly: Cabled System		

TABLE 1. ACB Requirements		
Criteria	Required Value	Test Method
Thickness, minimum	4.75 inches	N/A
Net Weight/Area, minimum	35 psf	Note a.
Critical Shear Stress, minimum	25.0 psf	FHWA RD-89-199
Critical Velocity, minimum	16.0 ft/sec	FHWA RD-89-199
Curvature Radius, maximum	3 feet	Note b.
Surface Void Area Ratio	25 - 40 percent	Note c.
Drainage Correction Factor (DCF)	20 - 35 percent	Note d.
Block/Geotextile Interface Friction Angle	35 degrees	Note e.
a. Determine the weight of the mattress per unit area with the nominal joint spacing, in a non-submerged condition.		
b. The curvature radius shall be indicative of the ability of the assembled mattress to conform to one dimensional subgrade curves without binding, such as for anchor trenches and swales. The curvature radius shall be demonstrated, if requested by the Contracting Officer.		
c. The surface void area ratio shall be determined at the visible (with filled voids) surface of the blocks, with the joints spaced in a neutral position (50 percent), and shall be expressed as a percentage of the gross mat area. The void area shall include area between the blocks and open cells within the block.		
d. The drainage correction factor shall be the minimum void area ratio (usually taken at the base of the blocks), with the joints spaced in a neutral position (50 percent freeplay in each direction), and shall be expressed as a percentage of the gross mat area.		
e. The concrete surface shall be sufficiently rough to prevent sliding of the blocks on the geotextile. The interface friction must be matched with the selected block and geotextile combination, and shall be included with the ACB and Geotextile Data submittal. The block/geotextile interface friction angle shall be demonstrated, if requested by the Contracting Officer.		

2.1.1 Hydraulic Stability

2.1.1.1 Flume Test

Submit a report of testing for the ACB in substantial conformance with FHWA RD-89-199, except that a drainage layer is not required, at the same time as the ACB and Geotextile Data submittal. The report shall clearly state if the critical shear stress associated with the stability threshold

of the ACB system was derived from laboratory testing that included a sub-block drainage layer as a component of the tested system. The ACB product shall have been tested in a flume chamber in substantial conformance with FHWA RD-89-199. If the product was tested with a drainage layer, the installed product shall incorporate a similar drainage layer with adequate filtration design for the site soils. The flume test shall be based on conservative assumptions for field placement of the blocks (such as block orientation, and joint spacing within construction tolerances). The critical shear stress (and critical velocity) shall be indicated in the test report.

2.1.1.2 Extrapolation of Hydraulic Stability

Extrapolation of critical shear stress for untested blocks within a similar family of ACB shall be subject to limitations. Extrapolation shall only be used for blocks having a similar footprint area and interlock mechanism, but with variable thickness or net weight/area. Extrapolation shall only be accepted if the following conditions are met:

- a. The extrapolation is in strict accordance with hydraulic similitude methods commonly accepted by the industry, and includes quantitative treatment for a block overturning failure mode.
- b. The tested block is the smaller product size in both thickness and net weight/area, and extrapolation does not extend the critical velocity more than 10 feet per second from the tested product size.

2.1.2 Matrix Assembly - Cabled Systems

Cable tied concrete block shall be interconnected by flexible cables running through the blocks. Each block shall be penetrated by a cable that allows articulation of the blocks, but restrains removal of individual blocks. Void filler shall be placed to inhibit lateral movement, cover the geotextile, and increase hydraulic stability. Articulating concrete block, cables, and fittings shall be fabricated into mattresses at the manufacturer's plant.

2.1.3 Structural requirements

Articulating concrete block shall be wet cast using concrete as specified herein, or dry-cast by a vibratory block forming machine. The blocks shall be manufactured to the following requirements:

2.1.3.1 Compressive Strength

The minimum compressive strength shall be 4000 psi for an average of 3 units, and 3500 psi for an individual unit. Compressive strength shall be determined by ASTM C42/C42M for wet cast blocks, or by ASTM C140/C140M for dry cast blocks.

2.1.3.2 Water Absorption for Dry Cast Units

The maximum water absorption for dry cast units shall be 9 pcf for an average of 3 units, and 12 pcf for an individual unit. Water absorption shall be determined by ASTM C140/C140M.

2.1.3.3 Saturated Surface-Dry Density

The minimum saturated surface-dry density shall be 140 for average of 3

units, and 140 for an individual unit.

2.1.3.4 Air Entraining

Wet cast concrete shall be air entrained to contain between 4 and 7 percent total air.

2.2 GEOTEXTILE

The Contractor shall provide Class A Nonwoven Geotextile conforming to the requirements of SECTION 31 05 19 GEOTEXTILE beneath bedding material and ACB's.

2.3 MICROGRID

Submit one properly identified 24 by 24 inches minimum size microgrid sample with the fasteners proposed for use. Provide a microgrid that is a regular network of integrally connected polymer tensile elements.

- a. Submit manufacturer's certified raw and roll material test reports including ultimate strength performed in accordance with ASTM D6637 or ASTM D4595 (modified). Test results not meeting the requirements in Table 2 or in the approved Manufacturer's Quality Control Manual will result in rejection of applicable rolls.
- b. The microgrid structure shall be dimensionally stable and able to retain its geometry under manufacture, transport and installation. The microgrid shall be manufactured with 100 percent virgin resin consisting of polyethylene, polypropylene, polyester, or other approved material and with a maximum of 5 percent in-plant regrind material. Polyester resin shall have a minimum molecular weight of 25,000 and a carboxyl end group number less than 30. Polyethylene and polypropylene shall be stabilized with long term antioxidants.
- c. Submit [Certificates of Compliance](#) for the materials provided and results of conformance testing. Submit an affidavit certifying raw and roll material test results submitted are accurate and that the reinforcement meets the requirements of the project specifications. The affidavit shall be signed by an official authorized to certify on behalf of the manufacturer. documents shall include a statement confirming that all purchased resin used to produce reinforcement is virgin resin.

TABLE 2: Microgrid Properties		
Tensile Properties	Test Method	MARV Values
Ultimate Strength	ASTM D6637	3,600 lbs/ft
Creep Limited Strength	ASTM D5262	2,384 lbs/ft
Long Term Design Strength	NCMA TR127B	2,064 lbs/ft
Aperture Size	-	0.5 x 0.75 inches

2.4 CABLE

2.4.1 Installation Requirements for Cable

Cable used for preassembled mattresses shall be sufficiently sized and fastened for the size/weight of the assembled mattresses such that the assembled mattresses can be placed in compliance with OSHA standards. The manufacturer shall be responsible for determining the minimum cable strength compatible with the mattress size for safe handling. Cable strength shall be based on a minimum factor of safety of 5, and include appropriate reduction factors for mechanically crimped cable, and other fasteners. If applicable, loading conditions shall include the use of a spreader bar for placing the mattresses.

2.4.2 Design Requirements for Cable

ACB's that rely on cables to maintain block to block interconnection shall use ropes manufactured from polyester, stainless steel wire, or galvanized steel wire. The cable shall have a minimum breaking strength of 3,700 pounds. Polyester rope shall be constructed of high tenacity, low elongating, continuous filament polyester fibers; and shall consist of a core construction comprised of parallel fibers contained within an outer jacket or cover.

2.5 VOID FILLER

2.5.1 Aggregate

Aggregate for filling the voids in the block shall meet the requirements of STABILIZED STONE DRAINAGE LAYER below.

2.5.2 Topsoil and Seed

Topsoil for filling the voids in the block and seed for turf establishment shall meet the requirements of Section 32 92 19 SEEDING.

2.6 STABILIZED STONE DRAINAGE LAYER

Aggregate for the stabilized stone drainage layer shall have a minimum specific gravity of 2.24, consist of crushed rock with no fines with a maximum D100 1.5-inch, and meet the physical requirements of ASTM C33/C33M. Drainage stone shall be obtained from a source provided by the Contractor. Stabilize the stone per the manufacturer's recommendations. The D50 of the drainage stone shall be larger than the apertures of the geogrid.

Prior to delivery, submit bulk specific gravity and gradation test results to show that the proposed aggregate materials meet these requirements for drainage stone. Provide the gradation band of the proposed drainage stone that will encompass all possible D50 particle sizes. Submit at the same time as the Microgrid submittal.

PART 3 EXECUTION

3.1 SUBGRADE PREPARATION

Place the geotextile on compacted native soils, or acceptably placed and compacted fill. Do not place the geotextile on surfaces that contain mud, frost, organic soils, embankment that has not met compaction requirements, or where the Contracting Officer determines that unsatisfactory material

remains in or under the subgrade.

3.1.1 Clearing

All vegetation shall be completely removed as specified in Section 31 11 00 CLEARING AND GRUBBING. Remaining roots from trees and brush shall be removed to a depth of 1 foot below the subgrade surface. Loose roots and twigs, turf clods, stones larger than 1/2 inch diameter, and other debris shall be raked and removed from the final surface. Rills and gullies from erosion shall be corrected.

3.1.2 Bank Grading

Grading shall be finished to a smooth surface, typical of that obtainable with a dozer and blade. A rough surface typically obtained with a backhoe or dragline will not be acceptable.

3.1.3 Compaction and Subgrade Finishing

Fill soils shall be compacted to the specified density in Section 31 00 00 EARTHWORK. Incidental grading (where embankment is not otherwise specified) shall be compacted by heavy equipment or by tamping with a bucket to a density characteristic of the surrounding soils. The final surfaces accessible by compaction equipment shall be compacted with a smooth drum roller or vibratory plate tamper until there is no further evidence of consolidation. Where slopes limit operation of compaction equipment, the final surface shall be back-dragged to a dense smooth surface with bladed equipment. Localized loose or soft zones shall be corrected.

3.1.4 Grade Tolerances

The grading tolerance shall be within 2 inches from the prescribed elevations, with no abrupt variations that would cause unacceptable projections of individual blocks.

3.1.5 Subgrade Surface Tolerances

The subgrade shall be maintained in a smooth condition between installation of the geotextile and the blocks. Windrows, stones, clods of cohesive soil, and irregularities shall be raked smooth. Ruts, rills and gullies resulting from traffic, precipitation runoff, groundwater seepage, etc. shall be corrected prior to installation of blocks.

3.2 GEOTEXTILE INSTALLATION

3.2.1 General

Install geotextile in accordance with requirements within 31 05 19 GEOTEXTILE except that seams shall be overlapped a minimum of 18 inches. Seams on slopes and butt end seams shall be shingled so that runoff and channel flow passes over the fabric. Geotextile panels shall be secured before block placement by adequate sandbags, spare blocks, or pins/staples.

3.3 MICROGRID INSTALLATION

Submit Microgrid layout plan along with joint details, sequencing and construction procedures

3.3.1 Placement

Install the microgrid in accordance with the Manufacturer's recommendations. Unroll the microgrid in the direction of reinforcement. After a layer of microgrid has been placed, use suitable means, that do not damage the underlying geosynthetics, to hold the microgrid flat and in place until cover soil can be placed. Microgrid damaged during placement and covering shall be removed and replaced at no additional cost to the Government.

3.3.2 Overlaps and Fasteners

Adjacent rolls of microgrid shall be positioned edge-to-edge and loosely fastened to maintain alignment during fill placement. Adjacent rolls shall not be overlapped. Use fastener type and spacing as recommended by the manufacturer and approved by the Contracting Officer. Metallic fasteners will not be allowed.

3.4 BLOCK INSTALLATION

All placement of blocks shall be in accordance with the manufacturer's recommendations and the Contractor's approved shop drawings. Submit drawings showing details of the ACB and Geotextile Installation, including the block layout patterns in relation to the feature alignment, anticipated locations of cast-in-place concrete joints, mattress junction details, and proposed installation methods for void filling materials.

3.4.1 Placement of Pre-Assembled Mattresses

The mattresses shall be placed directly into position, with a maximum space or gap between mattresses of 3 inches in excess of the nominal joint spacing of blocks within the mattress. Mattresses out of alignment shall be lifted and reset. Mattresses shall not be pushed or pulled laterally after they are in contact with the geotextile. No overlapping of mats will be accepted and no blocks shall project vertically more than 1 inch beyond the adjacent blocks. As adjacent mats are placed, they shall be secured to each other by fastening the protruding horizontal and vertical cable connections and end cable loops together along each side of the mats.

3.4.2 Tolerances

Maximum acceptable block projections (vertical offset from adjacent blocks) for "installation in the dry" shall not exceed 1.0 inch for cabled systems. Typical block projections shall be less than half the maximum projections.

3.5 CONCRETE JOINTS

3.5.1 General Requirements

Use of cast in place concrete joints shall be minimized to the extent practicable. The Contracting Officer shall be informed of all concrete joints not shown on shop drawings prior to field placement. Joints that shall require concrete include:

- a. Joints between cable tied mattresses where the joint is wider than 2 inches.
- b. Termination trenches.

- c. Abutments where the ACB meets headwalls, pipe penetrations, or other constructed features.
- d. Abutments where the ACB's meet sheet piling that will remain after construction.
- e. Abutments where the ACB's meet the proposed drill shaft walls.
- f. Any areas where there are partial blocks (to avoid small blocks with reduced hydraulic stability).

Field placed concrete shall be non-shrink grout meeting the applicable portions of Section 03 31 01 CAST-IN-PLACE STRUCTURAL CONCRETE FOR CIVIL WORKS. All cable ties and anchoring shall be completed prior to placing concrete.

3.5.2 Abutments

The ACB shall abut headwalls, pipe penetrations, sheet piles that will remain after construction, drilled shaft walls, and other constructed features in a neat appearance. Unless a specific detail is indicated on the drawings, voids shall be filled with partial blocks and the gap shall be filled with cast-in-place concrete. The concrete shall be installed flush with the surface of the blocks, and shall be float finished.

3.6 VOID FILLER AND SEEDING

The voids of the articulating concrete block mats shall be filled with topsoil, except that voids below elevation 4.0 shall be filled with aggregate void filler. All cable ties and anchoring shall be completed prior to filling voids. Seeding and maintenance shall be completed in accordance with Section 32 92 19 SEEDING.

3.7 PROTECTION OF WORK

Work shall be protected against damage from subsequent operations. Displaced or broken blocks shall be removed and replaced to conform to all requirements of this section. Damaged material shall not be incorporated. Equipment shall not be allowed on the ACB that could crack, cause abrasion, or otherwise damage the blocks. Vehicles shall not operate directly on geotextile, except that rubber-tired vehicles may operate directly on short reaches of geotextile that meets or exceeds AASHTO M 288 survivability requirements for Class 1 geotextile, if there is no rutting, if the vehicle access is necessary to accomplish the work, and if the Contracting Officer observes the operation and approves. Vehicles shall not operate on the ACB until (during or after) placement of void filler. Vehicle traffic on the ACB shall be restricted to light weight rubber-tired vehicles, and where intermittent access is necessary to accomplish the work. Routine haul routes shall not be established on the ACB. These allowances shall not waive the Contractor's obligation to maintain the installation until acceptance, and verify that vehicle access does not crack, or in any way damage, the ACB.

3.8 QUALITY CONTROL TESTING

The following testing shall be performed independent of the manufacturing process, by an agency other than the manufacturer. The ACB blocks shall be sampled and tested for compressive strength, water absorption and unit

weight. The sample frequency shall be 3 specimens for each 3000 SY. Test methods shall be consistent with those specified in PART 2 PRODUCTS.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

SECTION 35 31 19.26

CONCRETE ARMOR UNITS

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Concrete Armor Unit (CAU) Revetment System
 - 1.2.2 Concrete Armor Unit Bundle
 - 1.2.3 Packing Ratio
- 1.3 SUBMITTALS
- 1.4 DELIVERY, STORAGE, AND HANDLING
 - 1.4.1 Units

PART 2 PRODUCTS

- 2.1 CONCRETE ARMOR UNIT
 - 2.1.1 Matrix Assembly - Interlocking Units
 - 2.1.2 Structural requirements
 - 2.1.2.1 Compressive Strength
 - 2.1.2.2 Water Absorption for Dry Cast Units
 - 2.1.2.3 Air Entraining
- 2.2 VOID FILLING MATERIAL

PART 3 EXECUTION

- 3.1 MARINE MATTRESS
- 3.2 UNIT INSTALLATION AND LAYOUT PLAN
 - 3.2.1 Placement of Pre-Assembled Bundles
- 3.3 VOID FILLER
- 3.4 QUALITY CONTROL TESTING

-- End of Section Table of Contents --

SECTION 35 31 19.26

CONCRETE ARMOR UNITS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C42/C42M (2020) Standard Test Method for Obtaining and Testing Drilled Cores and Sawed Beams of Concrete

ASTM C140/C140M (2020a) Standard Test Methods for Sampling and Testing Concrete Masonry Units and Related Units

1.2 DEFINITIONS

1.2.1 Concrete Armor Unit (CAU) Revetment System

A matrix of concrete units for erosion protection. Units are connected by geometric interlock. Manufactured units may consist of a single complete unit or modules to be assembled to form an individual unit for use in the matrix.

1.2.2 Concrete Armor Unit Bundle

A concrete armor unit bundle consists of multiple units preassembled prior to placement secured by a cable. The cable is intended for holding the bundled units together and not for lifting.

1.2.3 Packing Ratio

The distance between each unit measured center-to-center divided by the overall length of the unit when the units are interlocked.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Unit Installation and Layout Plan; G, R0

SD-03 Product Data

Concrete Armor Unit; G, R0

SD-04 Samples

Concrete Armor Unit

1.4 DELIVERY, STORAGE, AND HANDLING

Check products upon delivery to assure that the proper material has been received and is undamaged.

1.4.1 Units

Provide blocks which are sound and free of defects that would interfere with proper placement or that would impair the strength or longevity of the installation. Discard blocks with the following defects:

- a. Broken appendages.
- b. Chips resulting in a weight loss exceeding 10 percent of the average weight of a unit
- c. Cracks wider than 0.25 inches or 1-inch in depth.

Minor cracks, incidental to the usual method of manufacture, or chipping that results from customary methods of handling in shipping, delivery and placement will not be deemed grounds for rejection. Store units in a suitable location away from mud, paint, wet cement, and other contamination or disturbance.

PART 2 PRODUCTS

2.1 CONCRETE ARMOR UNIT

Submit descriptive technical data on the units. Include all material properties specified under paragraph PRODUCTS. Catalog cuts, technical data sheets, or test data shall be submitted showing that the products meet the specifications. The submittal shall also include a copy of any standard manufacturer's warranties for the products.

The geometry of a concrete armor unit shall consist of six arms extending from a central hub. The unit shall have geometry which exhibits six equally spaced arms, with each arm spaced at 90 degrees from the four adjacent arms. When placed in the most stable configuration, each unit will rest on three of the six arms. Provide a single-layer system which shall be hydraulically stable under the design conditions and requirements shown in Table 1.

TABLE 1. CAU Requirements	
CRITERIA	REQUIRED VALUE
Factor of Safety for Velocity (FOS) See Note (1)	2.0 (min)
Critical Maximum Velocity	32.0 ft/sec

TABLE 1. CAU Requirements	
CRITERIA	REQUIRED VALUE
Critical Shear	12 psf
Friction or Bed Slope	0.00 ft/ft to 0.01 ft/ft
Side Slope	2H:1V
Individual Unit Weight	600 lbs (min)
(1) where FOS is equal to the permissible velocity of the armor unit divided by the design velocity	

2.1.1 Matrix Assembly - Interlocking Units

Interlocking units are assumed to function without the use of cables or similar restraints. Any cables used for assembly and placement shall be assumed as temporary and to provide no benefit in the final installation. Void filler shall be placed to inhibit lateral movement and increase hydraulic stability.

2.1.2 Structural requirements

Concrete armor units shall be wet cast using concrete as specified herein, or dry-cast by a vibratory unit forming machine. The units shall be manufactured to the following requirements:

2.1.2.1 Compressive Strength

The minimum compressive strength shall be 4500 psi for an average of 3 units, and 4000 psi for an individual unit. Compressive strength shall be determined by ASTM C42/C42M for wet cast units, or by ASTM C140/C140M for dry cast units.

2.1.2.2 Water Absorption for Dry Cast Units

The maximum water absorption for dry cast units shall be 9 pcf for an average of 3 units, and 12 pcf for an individual unit. Water absorption shall be determined by ASTM C140/C140M.

2.1.2.3 Air Entraining

Wet cast concrete shall be air entrained to contain between 1 and 6 percent total air.

2.2 VOID FILLING MATERIAL

Stone used for filling voids shall meet the requirement of STONE FILL MATERIAL as specified in Section 35 42 35 MARINE MATTRESS.

PART 3 EXECUTION

3.1 MARINE MATTRESS

Concrete armor units shall be placed on a marine mattress. See the contract drawings for marine mattress layout information and extents. See Section 35 42 35 MARINE MATTRESS for requirements.

3.2 UNIT INSTALLATION AND LAYOUT PLAN

All placement of units shall be in accordance with the manufacturer's recommendations and the Contractor's approved shop drawings. Submit drawings showing details of the CAU, including the unit layout patterns in relation to the feature alignment, bundle assembly methods, lifting details, and proposed installation methods for void filling materials.

3.2.1 Placement of Pre-Assembled Bundles

The bundles shall be placed directly into position as closely as possible, within a maximum of 8 inches between bundles. Bundles out of alignment and tolerance shall be lifted and reset. Cables required for securing units within the bundle shall not be used for lifting. Lifting shall be performed by the use of a spreader bar to distribute the load without crushing interior units. Maximum packing ratio shall be 0.5.

3.3 VOID FILLER

The voids of the concrete armor units in the areas specified on the plans shall be filled with STONE FILL MATERIAL as defined in 35 42 35 MARINE MATTRESS.

3.4 QUALITY CONTROL TESTING

The following testing shall be performed independent of the manufacturing process, by an agency other than the manufacturer. The CAU shall be sampled and tested for compressive strength, water absorption and unit weight. The sample frequency shall be 3 specimens for each 250 units supplied. Test methods shall be consistent with those specified in PART 2 PRODUCTS.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

SECTION 35 31 19

STONE AND CHANNEL PROTECTION FOR STRUCTURES

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Revetments
 - 1.2.2 Standard Drawings
 - 1.2.3 Stone Protection
 - 1.2.4 Riprap
 - 1.2.5 Graded Stone
 - 1.2.6 Channel Protection
- 1.3 SUBMITTALS
- 1.4 DEMONSTRATION STOCKPILES
- 1.5 INSPECTION COORDINATION
- 1.6 WARRANTY
- 1.7 QUALITY CONTROL
 - 1.7.1 Weekly Certificates
 - 1.7.2 Stone Source
 - 1.7.2.1 Quarry Quality Control Supervisor
 - 1.7.3 Initial Stone Test
 - 1.7.4 Gradation
 - 1.7.5 Testing Laboratory
 - 1.7.6 Weight Scale Certificate

PART 2 PRODUCTS

- 2.1 Stone Material Quality
 - 2.1.1 Cracked Stone Criteria
- 2.2 BEDDING MATERIAL
- 2.3 SAND FILL
- 2.4 GEOTEXTILE
- 2.5 RIPRAP
 - 2.5.1 Quarry Operations
 - 2.5.1.1 Curing Stone
 - 2.5.1.2 Temporary Storage at Quarry

PART 3 EXECUTION

- 3.1 CONSTRUCTION METHODS
- 3.2 Field Quality Control
- 3.3 Base Preparation
- 3.4 Geotextile
- 3.5 PLACEMENT OF BEDDING LAYERS
 - 3.5.1 General
 - 3.5.2 Placement of Bedding Material on Prepared Base
- 3.6 PLACEMENT OF RIPRAP
 - 3.6.1 General

- 3.6.2 Placement
 - 3.6.2.1 Above Water
 - 3.6.2.2 Under Water
 - 3.6.2.3 Placement Tolerance
- 3.7 SURVEYS
 - 3.7.1 Placement Surveys
- 3.8 GRADATION TEST DATA SHEETS AND EXAMPLES

-- End of Section Table of Contents --

SECTION 35 31 19

STONE AND CHANNEL PROTECTION FOR STRUCTURES

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO M 43 (2005; R 2009) Standard Specification for
Sizes of Aggregate for Road and Bridge
Construction

ASTM INTERNATIONAL (ASTM)

ASTM C33/C33M (2018) Standard Specification for Concrete
Aggregates

ASTM C88 (2018) Standard Test Method for Soundness
of Aggregates by Use of Sodium Sulfate or
Magnesium Sulfate

ASTM C127 (2015) Standard Test Method for Density,
Relative Density (Specific Gravity), and
Absorption of Coarse Aggregate

ASTM C295/C295M (2019) Standard Guide for Petrographic
Examination of Aggregates for Concrete

ASTM C535 (2016) Standard Test Method for Resistance
to Degradation of Large-Size Coarse
Aggregate by Abrasion and Impact in the
Los Angeles Machine

ASTM D2487 (2017; E 2020) Standard Practice for
Classification of Soils for Engineering
Purposes (Unified Soil Classification
System)

ASTM D5312/D5312M (2012; R 2013) Evaluation of Durability of
Rock for Erosion Control Under Freezing
and Thawing Conditions

ASTM D5313/D5313M (2012; R 2013) Evaluation of Durability of
Rock for Erosion Control Under Wetting and
Drying Conditions

ASTM D5519 (2015) Particle Size Analysis of Natural
and Man-Made Riprap Materials

U.S. ARMY CORPS OF ENGINEERS (USACE)

COE CRD-C 144	(1992) Standard Test Method for Resistance of Rock to Freezing and Thawing
EM 1110-2-1601	(1991; 1994 Change 1) Engineering and Design -- Hydraulic Design of Flood Control Channels
EM 1110-2-2302	(1990) Engineering and Design -- Construction With Large Stone
ERDC/GL TR-81-8	(1981) Evaluation of Quality and Performance of Stone as Riprap or Armor

1.2 DEFINITIONS

1.2.1 Revetments

The term "revetment" applies to various types of stabilization structures that are constructed along the river approximately parallel to the current. The revetments are constructed of stone or piling.

1.2.2 Standard Drawings

Details of various types of structures in general use on the project are shown on standard drawings forming a part of these specifications.

1.2.3 Stone Protection

Stone Protection is defined as a system which includes a layer of bedding material or layers of filter material beneath a layer or layers of riprap. Stone protection is placed around structures in slack water or within a dewatered site. Stone protection may also be used to protect channel banks when it is placed in the dry or in slack water.

1.2.4 Riprap

Riprap is defined as a material having a gradation band similar to those specified in EM 1110-2-1601, Chapter 3, uniform graded material. Riprap is normally produced by mechanical methods, with a jaw crusher and grizzly after the stone has been mined by blasting in a quarry. Riprap gradations have a maximum top size of 3.5 tons.

1.2.5 Graded Stone

Graded Stone is defined as material with gradations that are produced by the mining technique and minimal additional processing other than the use of a skeleton bucket or a bar grizzly. The gradation band have more fines than riprap and have gradations with top size up to 3.5 tons and could be classified as being well graded.

1.2.6 Channel Protection

Channel protection is stone placed in a current as revetment, dikes, or slope paving without the use of a separate layer of bedding or filter material. In this type of environment, bedding sand or geotextiles and materials with gradation bands with a top size of 150 mm (6 inches) will

not stay where placed.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Stone Source; G, D0

Construction Methods; G, D0

SD-04 Samples

Riprap; G, D0

The Contractor shall submit a 10 ton sample of the proposed riprap gradation for initial size analyses and approval at least 30 days prior to use on the project.

SD-05 Design Data

Surveys; G, D0

The Contractor shall submit surveys for both field data and plotted cross sections, as required by paragraph SURVEYS below, for each prepared surface for bedding and riprap placement, and the final bedding and riprap stone surface in accordance with the Survey Section of these specification.

SD-06 Test Reports

Initial Stone Test; G, D0

Gradation; G, D0

SD-07 Certificates

Testing Laboratory; G, D0

Riprap Stone Weekly Certificates; G, D0

Riprap Weekly Certificates; G, D0

The QC Supervisor shall submit weekly certificates for each stone class of new stone material delivered to the project site in accordance with subparagraph "Weekly Certificates" of this section. The Weekly Certificate shall include a statement of compliance attesting that all stone material delivered to the project site has been examined by the QC Supervisor and has been observed to meet all specification requirements

Weight Scale Certificate; G, D0

1.4 DEMONSTRATION STOCKPILES

Following the Contracting Officer's approval of the Stone Source Submittal(s) and all associated data and test results, but prior to approval of the Stone Source (quarry), the Contractor shall make arrangements to provide a pre-production demonstration stockpile for each of the stone gradations (e.g. Riprap and Bedding Stone) at each Stone Source. For each gradation class, the stockpiles shall be segregated by each type of lithology approved by the Contracting Officer after review and approval of the Stone Source Submittal and evaluation testing of stone results. The demonstration stockpiles shall be located at the source of the stone (i.e. within the proposed quarry). The stone material placed in the demonstrative stockpile shall be representative of the overall quality of each lithologic type produced in that source, meet all requirements of the specification, and shall not consist of the most perfect specimens unless it is reasonable to determine that the stone source will produce the required amount of stone of the applicable size range with a degree of quality no less than that existent in the demonstration stockpile. The stone material placed in the demonstrative stockpiles shall also be representative of the stone sizes and shapes intended for delivery to the project site.

1.5 INSPECTION COORDINATION

The Contractor shall provide access to the work on a continual basis as required for inspection and testing by the Government. The Contractor shall provide 48-hour notice of intention to begin new work activities.

1.6 WARRANTY

The Manufacturer shall warrant the EQUIPMENT, MATERIALS and PRODUCTS specified in this section against defective materials and workmanship with the Manufacturer's standard warranty, but for no less than one year from the date of Substantial Completion, and as described in SECTION 00700 - GENERAL TERMS AND CONDITIONS.

The Contractor shall warrant the work against defects for one year from the date of Substantial Completion and as described in SECTION 00700 - GENERAL TERMS AND CONDITIONS.

1.7 QUALITY CONTROL

1.7.1 Weekly Certificates

During construction phases when stone is shipped to the construction site, the Contractor's QC Supervisor shall submit on a weekly basis, by gradation class of material shipped to the project site during that week and certification of the total volume by weight shipped. The weekly certificate shall include weight scale receipts showing tare weight, gross weight and date for each truck, railcar, barge or other transportation method from the stone source that week used to determine that weekly volume. The certificate shall also state that the Contractor's QC Supervisor has examined all stone material delivered to the project site and found that material meets all the requirements of these contract specifications. Weekly Certificates shall be submitted to the Contracting Officer no later than the following Tuesday of that weekly reporting period.

1.7.2 Stone Source

For each individual source of stone (quarry) identified by the Contractor as a potential supplier of material for use in the project structure, the Contractor shall submit in report format, a Stone Source Submittal. The Stone Source Submittal for each source shall contain quarry information, quarry location map, quarry quality control (QC) practices, proposed shipping methods, and the intended products obtained from that designated source. The Stone Source submittal shall also include information on that source including the geological formations of the quarry location, geological formations within the quarry, geologic structure of the formation(s), areas of the quarry producing stone for this project, and quarrying procedures and practices. The submittal shall also include information and discussion on the role and process of all subcontractors (by Company Name) involved in stone material production. Different lithology types within the quarry as well as the lithostratigraphic(s) units proposed as the source of stone for this project shall be discussed.

The Stone Source Submittal shall likewise include both a general map, site specific map, and geological map for each source. The general map shall show the local region of the quarry including major access and shipping routes. The site specific map shall fully define the area of operations within the quarry including the boundaries and working benches of the quarry for the stone to be used on the project. Evaluation Testing of Stone results (SD-06 Test Reports), Bulk Specific Gravity, and Initial Gradation Tests from that designated source shall also be submitted within the Stone Source Submittal. The Stone Source Submittal for each quarry so designated shall state that all stone (i.e. different lithologies) within each stone source proposed as a supply of material for this project has met all requirements stipulated herein. No stone shall be shipped from any proposed stone source(s) until the Contractor has received written approval of that source from the Contracting Officer

It is the Contractor's responsibility to complete all submittals and testing requirements to determine that the Stone Source or combination of sources proposed are capable of providing the specified quality, quantities, and gradation needed and at the rate needed to maintain the scheduled progress of the work.

The Final Stone Source Submittal(s) shall be submitted a minimum of 30 days before scheduling gradation testing for the Contracting Officer's demonstration stockpile and quarry site examinations.

1.7.2.1 Quarry Quality Control Supervisor

The Contractor shall also designate a Quality Control (QC) Supervisor in the Stone Source Submittal who shall be responsible for all QC functions for that proposed stone source. The QC Supervisor shall have at least three (3) years' experience in assessing stone quality and quarry stone production. The QC supervisor shall also be responsible for preparing the Weekly Certificate for each gradation class specified, conducting gradation testing, and answering the Contracting Officer's questions concerning quality control and weekly certificates. The Stone Source Submittal shall include the contact number for the QC Supervisor. The Stone Source Submittal shall be prepared, signed, and stamped by an independent state licensed Professional Geologist, licensed by the Commonwealth of Puerto Rico that issues a Professional Geologist License. The Stone Source Submittal and Test Reports consisting of Bulk Specific

Gravity, Initial Gradation Testing for all gradations, and Evaluation Testing of Stone shall be submitted concurrently. Any of the above submittals submitted individually will be rejected as incomplete.

The Professional Geologist shall also describe all deleterious or detrimental geologic features outlined in subparagraph "Stone Material Quality" that exists in the stone material quarried at that source.

1.7.3 Initial Stone Test

The Contractor shall submit certified test reports prepared by a qualified independent testing laboratory selected and compensated by the Contractor for the tests required herein. These results shall be submitted in the Stone Source Submittal.

1.7.4 Gradation

Stone gradation based on ASTM D5519 Test Method A: Size-Mass Grading. A representative sample shall be tested to achieve a minimum 2% accuracy based on the largest individual piece mass. A Cumulative plot of percent lighter by weight versus stone weight in pounds shall be submitted in the Stone Source Submittal.

1.7.5 Testing Laboratory

The Contractor shall furnish testing certificates from a qualified testing agency, which shall be submitted prior to acceptance of the rock source to verify the conformity to the requirements of the Contract Documents.

All evaluation testing shall be completed by a USACE validated commercial testing laboratory unless otherwise specified herein. The list of USACE-validated laboratories can be found in the Materials Testing Center webpage of the US Army Engineer Research and Development Center:

<https://mtc.erd.c.dren.mil/searchvalidation.aspx>.

1.7.6 Weight Scale Certificate

For all weight scales used to measure weight of stone material including the Stone Source(s), all transfer points, and work site, and prior to the use thereof, the Contractor shall submit pertinent details on the location, type, and construction of the scale, including a copy of the certification of the scale's accuracy from the local weights and measures regulatory agency. Weight Scale Certifications shall be submitted concurrently with the Stone Source Submittal

PART 2 PRODUCTS

2.1 Stone Material Quality

All stone material shall be free of any detrimental geologic features such as, but not limited to, clay seams, shale seams, argillaceous material, any stylolites, schistose seams, detrimental vugs zones or seams, zones of high foliation, and/or other adverse diagenetic features or seams. No schistose stone material or any highly foliated stone material will be acceptable. Sedimentary or metamorphic stone containing carbonaceous, coal, clay seams/lenses/joints or similar deleterious features will not be permitted or accepted. All stone material utilized shall be free of cracks, blast fractures, bedding seams, foliation seams, foliation joints,

joints, any deleterious features such as splits, spalls, delaminations, disaggregations, disjuncture, shelly parting, or combination thereof of such features, and/or other defects that would tend to increase its deterioration from natural causes. An independent, state licensed Professional Geologist, licensed by the Commonwealth of Puerto Rico shall determine if the aforementioned features are present within stone material from any of the proposed Stone Source(s). Discussion of Stone Material Quality and any detrimental or deleterious features present within any source shall be included in Evaluation Testing of Stone and the Stone Source Submittal for that Stone Source. Examples of deleterious and detrimental features in stone can be viewed in [ERDC/GL TR-81-8](#). [EM 1110-2-1601](#) and [EM 1110-2-2302](#) provide guidance on the stone properties, characteristics, and gradations specified herein.

2.1.1 Cracked Stone Criteria

Criteria used by the Contracting Officer for a "fractured" or "cracked" stone is defined as "any stone which contains one visible and continuous crack or fracture and exposed on two or more faces of the stone, or any stone which contains two or more visible non-continuous cracks or one continuous fracture on any one face of the stone." A continuous crack or fracture is defined as "an exposed unbroken and uninterrupted visible crack or fracture with a length equal to or greater than one-half of the dimension of the face on which it is exposed". Evaluation of a crack or fracture along a stone face shall only be based upon length and not width (openness) of that crack or fracture

2.2 BEDDING MATERIAL

The CONTRACTOR shall place a blanket of bedding material beneath permanent riprap materials. Some bedding material shall be placed in gabions as shown on the drawings and as specified herein. Stone for use in granular bedding shall weigh not less than 155 lbs/cf (saturated surface dry). The material shall be composed of tough, durable particles, shall be reasonably free from thin, flat, and elongated pieces, and shall contain neither organic matter nor soft, friable particles in quantities considered objectionable by the Contracting Officer. Bedding stone shall be placed within the limits shown on the drawings and shall be reasonable well graded within the following limits:

U.S. STANDARD SIEVE	PERMISSIBLE LIMITS PERCENT BY WEIGHT, PASSING
No. 1 (AASHTO M 43) For Use Below Class III and IV Riprap	
4 in.	100
3-1/2 in.	90 to 100
3 in.	-
2-1/2 in.	25 to 60
2 in.	-

U.S. STANDARD SIEVE	PERMISSIBLE LIMITS PERCENT BY WEIGHT, PASSING
No. 1 (AASHTO M 43) For Use Below Class III and IV Riprap	
1-1/2 in.	0 to 15
3/4 in.	0 -5

Minimum and Maximum Allowable Particle Size for Use Below Class VII and VIII Riprap					
Class	Median Particle Diameter	d15 Min d15 Max	d50 Min d50 Max	d85 Min d85 Max	d100 Max
I	6-inch	3.7-inch 5.2-inch	5.7-inch 6.9-inch	7.8-inch 9.2-inch	12.0-inch

2.3 SAND FILL

Sand shall be a clean, free draining sand in accordance with classification SP in ASTM D2487, except that no more than 5 percent by weight of the material smaller than a No. 4 sieve, shall pass a No. 200 sieve. If sand meeting this criterion is available on site it may be used. Sand shall meet ASTM C33/C33M gradation requirements for fine aggregate.

2.4 GEOTEXTILE

The Contractor shall provide Class A Nonwoven Geotextile conforming to the requirements of SECTION 31 05 19 GEOTEXTILE beneath bedding material and riprap.

2.5 RIPRAP

The Contractor shall furnish stone for riprap that is sound, durable and angular in shape. No more than 10% of the stone for any gradation shall have an elongation (ratio of greatest dimension to least dimension) greater than 3:1, and no stone shall have an elongation greater than 4:1. The riprap material shall conform to the following additional requirements.

Native stones for riprap shall be graded in size to produce a dense mass. Riprap shall consist of dense, natural rock fragments. Stones shall be resistant to weathering and to water action; free from quarry fines, soil, shell and organic material; and shall meet the gradation requirements below. Shale and stones with shale seams are not acceptable.

Riprap shall conform to the size and weights as follows:

Minimum and Maximum Allowable Particle Size					
Class	Median Particle Diameter	d15 Min d15 Max	d50 Min d50 Max	d85 Min d85 Max	d100 Max
III	12-Inch	7.3-Inch 10.5-Inch	11.5-Inch 14.0-Inch	15.5-Inch 18.5-Inch	24.0-Inch
IV	20-Inch	9.0-Inch 12.0-Inch	14.0-Inch 17.0-Inch	19.0-Inch 23.0-Inch	30.0-Inch
VII	24-Inch	14.5-Inch 21.0-Inch	23.0-Inch 27.5-Inch	31.0-Inch 37.0-Inch	48.0-Inch
VIII	30-Inch	18.5-Inch 26.0-Inch	28.5-Inch 34.5-Inch	39.0-Inch 46.0-Inch	60.0-Inch

Minimum and Maximum Allowable Particle Weights					
Class	Median Particle Weight	w15 Min w15 Max	w50 Min w50 Max	w85 Min w85 Max	w100 Max
III	150-lb	32-lb 93-lb	120-lb 210-lb	310-lb 510-lb	1100-lb
IV	300-lb	62-lb 180-lb	240-lb 420-lb	600-lb 1000-lb	2200-lb
VII	1000-lb	260-lb 740-lb	950-lb 1700-lb	2500-lb 4100-lb	9000-lb
VIII	2000-lb	500-lb 1450-lb	1900-lb 3300-lb	4800-lb 8000-lb	17600-lb

The greatest dimension of 50 percent of the stones shall be at least two-thirds but not more than 1 1/2 times the diameter of the average size. Material shall be of shapes that shall form a stable protection structure of required depth. Rounded boulders or cobbles shall not be used.

Riprap shall have a minimum bulk specific gravity, saturated surface dry (SSD) of 2.65 based upon water having a unit weight of 62.4 pounds per cubic foot. The method of bulk specific gravity (SSD) shall be [ASTM C127](#)

Test	Test Method	Test Value
Unit Weight	ASTM C127	165 pcf(saturated surface dry)
Specific Gravity	ASTM C127	greater than or equal to 2.65 (saturated surface dry)
Absorption	ASTM C127	Less than 1 percent

Test	Test Method	Test Value
Sulfate Soundness	ASTM C88	Less than 5 percent loss
LA Abrasion	ASTM C535	Less than 20 percent loss for 500 revolutions
Freeze and Thaw	ASTM D5312/D5312M	Less than 10 percent loss for 20 cycles
Wetting and Drying	COE CRD-C 144 ASTM D5313/D5313M	Less than 8 percent loss for 30 cycles with no progressive cracking
Petrography	ASTM C295/C295M	Fresh interlocking crystalline, no clay minerals, and no soluble materials

Control of Gradation shall be by visual inspection. The Contractor shall furnish a sample of the proposed gradation of at least five tons or 10 percent of the total riprap weight, whichever is less. If approved, the sample may be incorporated into the finished riprap at a location where it can be used as a frequent reference for judging the gradation of the remainder of riprap. Any difference of opinion between the Government and the Contractor shall be resolved by dumping and checking the gradation of two random truckloads of stones. Arranging for and the costs of mechanical equipment, a sorting site, and labor needed in checking gradation shall be the Contractor's responsibility.

The acceptability of the stones shall be determined by the Government prior to placement. All stone shall be approved by the Contracting Officer prior to placement.

2.5.1 Quarry Operations

Conduct quarry operations in a manner to produce stone conforming to the requirements specified, this may involve selective quarrying, handling, processing, blending, and loading as necessary, all of which shall be as specified in SECTION 01 45 00 QUALITY CONTROL. Control blasting and handling of rock to produce rock of the size ranges and quality specified. Techniques such as the use of proper hole diameter, hole depth, hole angle, burden and spacing distances, types and distribution of explosives. Delay intervals and sequence, removal of muck piles between each shot, and special handling techniques are required as necessary to produce the specified materials. All aspects of blasting operations shall be specifically designed so that the end product is not damaged from the blasting technique and that the stone is suitable for the intended purpose.

2.5.1.1 Curing Stone

Conduct curing operations on freshly quarried stone to allow it to release stored energy and moisture and to allow the stone to demonstrate that it will not fracture during the energy release and drying-out phase. Stones of sizes which are individually picked shall be temporarily stockpiled at the quarry site before being shipped to the project site, unless this requirement is waived by the Contracting Officer. Such waiver shall be granted only if the stone has characteristics that make curing unnecessary.

2.5.1.2 Temporary Storage at Quarry

Storage of stone materials subsequent to shipment from the quarry and prior to permanent placement in the required work shall be subject to approval of the Contracting Officer. Underwater storage of stone materials is prohibited.

PART 3 EXECUTION

3.1 CONSTRUCTION METHODS

Submit the proposed method of construction, to include the sequence of Geotextile, Bedding Stone and Riprap. Construction Methods shall also include the planned method of material placement and equipment to be used during each construction phase. The Construction Methods shall be submitted to the Contracting Officer 30 calendar days prior to the scheduled start of work.

3.2 Field Quality Control

The Contractor shall recombine the riprap stone sample used for gradation analysis, transport to the project site, and place in a location acceptable to Government. Field control of riprap gradation shall be by visual comparison of the representative sample to arriving loads. Arriving loads not bearing reasonable similarity to the sample will be rejected. All rejected stone shall be replaced at no additional cost to the Government.

3.3 Base Preparation

The Contractor shall prepare the subgrade to the lines, slopes and elevations indicated. The Contractor shall clear the subgrade of sticks, stones, debris and other materials that could puncture the overlying geotextile. The finished subgrade shall not vary from design grade by more than plus 2 inches and minus 4 inches at any location above water. The prepared base shall be approved by the Contracting Officer. Where such areas are below the allowable tolerance limit they shall be brought to grade by fill with earth similar to the adjacent material or with sand fill and then compacted to a density equal to the adjacent in place material. Subaqueous areas on which bedding material and riprap are to be placed shall be graded and/or dressed to conform to cross sections shown on the contract drawings within an allowable tolerance of plus 1 foot and minus 2 feet from the specified design grades. Where such areas are below the allowable minus tolerance limit they shall be filled with sand fill. As an alternative, these areas may be filled with bedding material. No payment will be made for any material thus required.

3.4 Geotextile

The Contractor shall place Class A Non-woven Geotextile in accordance with the requirements of SECTION 31 05 19 GEOTEXTILE, and as shown on the contract drawings. The geotextile must be in intimate contact with the subgrade.

3.5 PLACEMENT OF BEDDING LAYERS

3.5.1 General

A bedding layer conforming of bedding material specified in BEDDING MATERIAL shall be placed on the prepared base as described below, in accordance with the details shown on the contract drawings, and within the limits shown on the contract drawings or staked in the field. A tolerance of plus 2 inches and minus 1 inch from the slope lines and grades shown on the contract drawings shall be allowed in the finished surface of the bedding, except that the extreme of this tolerance shall not be continuous over an area greater than 200 square feet.

3.5.2 Placement of Bedding Material on Prepared Base

Bedding material shall be either placed in gabion units and set or spread uniformly on the prepared base to the grades as indicated on the contract drawings and in such manner as to avoid damage to the prepared base. Geotextile as shown in the contract drawings and as specified in SECTION 31 05 19 GEOTEXTILE shall be placed under the gabion units. Placing of bedding material by methods which tend to segregate the particle sizes within the bedding layer shall not be permitted. Placement shall begin at the bottom of the area to be covered and continue up slope. Subsequent loads of material shall be placed against previously placed material in such a manner as to ensure a relatively homogenous mass. Any damage to the surface of the prepared base during placing of the material shall be repaired before proceeding with the work at the Contractor's expense. Compaction of material placed on the prepared base shall not be required, but the material surface shall be finished to present an adequately even surface, free from mounds or windrows.

3.6 PLACEMENT OF RIPRAP

3.6.1 General

Riprap shall be placed on the bedding layers specified in paragraph(s) BEDDING MATERIAL within the limits shown on the contract drawings.

3.6.2 Placement

Under water placement rates shall be used when the top of the layer to be placed is covered by more than 3 feet of water.

3.6.2.1 Above Water

Riprap shall be placed in a manner which shall produce a well-graded mass of rock with the minimum practicable percentage of voids, and shall be constructed, within the specified tolerances, to the lines and grades shown on the contract drawings or staked in the field. The average tolerance of the entire job shall have no more than 50 percent of the tolerances specified above. Riprap shall be placed to its full course thickness in one operation and in such manner as to avoid displacing the bedding material. The large stones shall be well distributed. Placement shall begin at the bottom of the area to be covered and continue up slope. Subsequent loads of material shall be placed against previously placed material in such a manner as to ensure a relatively homogenous mass. The finished riprap shall be free from objectionable pockets of small stones and clusters of larger stones. Placing riprap in layers

shall not be permitted. Placing riprap by dumping it into chutes, or by similar methods likely to cause segregation of the various sizes, shall not be permitted. Placing riprap by dumping it at the top of the slope and pushing it down the slope shall not be permitted. No equipment shall be operated directly on the completed stone protection system. The desired distribution of the various sizes of stones throughout the mass shall be obtained by selective loading of the material at the quarry or other source; by controlled dumping of successive loads during final placing; or by other methods of placement which shall produce the specified results. Each truckload shall be representative of the gradation requirements. Rearranging of individual stones shall be required to the extent necessary to obtain a well-graded distribution of stone sizes as specified above. Maintain the stone protection until accepted by the Contracting Officer; any material displaced prior to acceptance due to the Contractor's negligence shall be replaced with no additional payment and to the lines and grades shown on the contract drawings.

3.6.2.2 Under Water

Riprap placed in the wet shall be placed evenly at a rate of 3 tons per square 100 square feet per pass for Class I riprap, 12 tons per square 100 square feet per pass for Class III riprap, 24 tons per square 100 square feet per pass for Class VII riprap, and 30 tons per square 100 square feet per pass for Class VIII riprap. Prior to starting work, submit the proposed method of placing riprap under water to the Contracting Officer for approval. Riprap to be placed in the wet shall be done during periods of low water levels during the months of June through November. The riprap shall be placed in two passes, with the second pass perpendicular to the first pass.

3.6.2.3 Placement Tolerance

Finished grade of the riprap shall be within (+\-) 4 inches of the grades shown on the contract drawings. Riprap placed above water shall be verified by topographic survey.

The Contractor shall document the correct placement of riprap placed below water to within the tolerances specified by using one of the following methods: sounding surveys using a sounding pole or sounding basket on a lead line, divers, or sonar bottom profiles.

3.7 SURVEYS

The Contractor shall establish and maintain quality control for all work performed at the job site under this section to assure compliance with contract requirements. Maintain records of his quality control tests, inspections and corrective actions. Quality control measures shall cover all construction operations including, but not limited to, the placement of all materials to the slope and grade lines shown and in accordance with this section.

3.7.1 Placement Surveys

Surveys made by the Contractor are required on each material placed for determining that the material layers are acceptably placed in the work. The Contractor shall make survey checks as the work progresses to verify lines, grade thicknesses, and tolerances are established for completed work. At least one (1) placement-control survey every 50 linear feet shall be made for stone protection placed, and shall be measured along the

revetment alignment. The cross sections determined by the pre-placement and post-placement survey shall be approved and field inspection performed by the Contracting Officer before proceeding with the next 50-foot length of stone protection placement. Cross sections shall be taken on lines 50-feet apart measured along the revetment alignment. Elevations shall be taken on 5-foot intervals and at breaks along each cross section. Approval of cross sections based upon placement surveys shall not constitute final acceptance of the work. However, other cross section spacing and reading intervals may be used if determined appropriate and approved by the Contracting Officer. Additional elevations shall be taken as the Contracting Officer may deem necessary or advisable. The surveys shall be conducted in the presence of an authorized representative of the Contracting Officer, unless this requirement is waived by the Contracting Officer.

3.8 GRADATION TEST DATA SHEETS AND EXAMPLES

See next page.

Quarry _____ Type of Stone Tested _____

T E S T R E P R E S E N T S		
Contract No.	District	Tons
	TOTAL	

			Cumulative		
Stone Size (lbs)	Weight Retained	Individual % Retained	Percent Retained	Percent Passing	Specification % Finer by Wt
Total Weight					
Max Size Stone					

I certify that the above stone sample is representative of the total tonnage covered by this test report.

Contractor Representative	_____
Government Representative	_____

F O R

E X A M P L E

O N L Y

EXAMPLE GRADATION SPECIFICATIONS	
PERCENT LIGHTER BY WEIGHT	STONE WEIGHT IN LBS
100	400 - 160
50	160 - 80
15	80-30

EXAMPLE WORKSHEET				
STONE SIZE LBS	INDIVIDUAL WT. RETAINED	INDIVIDUAL PERCENT RETAINED	CUMULATIVE RETAINED	PERCENT PASSING
400	0	0	0	100
160	9600	30	30	70
80	11,200	35	65	35
30	8000	25	90	10
<30	3200	10	100	-
TOTAL	32,000 pounds			
NOTE: Largest stone 251 pounds				

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION

SECTION 35 42 35

MARINE MATTRESS

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 REFERENCES
- 1.3 SUBMITTALS
- 1.4 DEFINITIONS
 - 1.4.1 Polymeric Marine Mattress
 - 1.4.2 Geogrid
 - 1.4.3 Minimum Average Roll Values
 - 1.4.4 True Tensile Modulus in Use
 - 1.4.5 Junction Strength
 - 1.4.6 Flexural Stiffness (Also Known as Flexural Rigidity)
 - 1.4.7 Resistance to Installation Damage
 - 1.4.8 Resistance to Long Term Degradation
 - 1.4.9 Ultraviolet Stability
- 1.5 STORAGE
- 1.6 ON-SITE TECHNICAL ASSISTANCE
- 1.7 QUALITY ASSURANCE

PART 2 PRODUCTS

- 2.1 STRUCTURAL GEOGRID
- 2.2 STONE FILL MATERIALS
- 2.3 MECHANICAL CONNECTION ELEMENTS
- 2.4 UV STABILIZED BRAID

PART 3 EXECUTION

- 3.1 EXAMINATION
- 3.2 MANUFACTURER'S INSTRUCTIONS
- 3.3 TEST MATTRESS
 - 3.3.1 General
 - 3.3.2 Coordination
- 3.4 FINAL FABRICATION AND FILLING OF MARINE MATTRESSES
 - 3.4.1 Marine Mattress Fabrication Plan
 - 3.4.2 Mechanical Connections
 - 3.4.3 Seaming
 - 3.4.4 Extension
 - 3.4.5 Geotextile
 - 3.4.6 Filling
 - 3.4.7 Compaction of Stone
 - 3.4.8 Lifting Hoops
 - 3.4.9 Final Assembly
- 3.5 PROTECTION OF MARINE MATTRESSES DURING TRANSPORTATION
- 3.6 SITE PREPARATION AND MARINE MATTRESS INSTALLATION
- 3.7 REPAIRS

3.8 CONTRACTOR QUALITY CONTROL

-- End of Section Table of Contents --

SECTION 35 42 35

MARINE MATTRESS

PART 1 GENERAL

1.1 SUMMARY

The work covered by this section consists of furnishing all work, labor, materials, plant and equipment and performing all operations required to provide polymeric Marine Mattress to close conformity with the lines, grades and dimensions shown on the contract drawings or approved by the Contracting Officer.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publication is referred to in the text by basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO HB-17 (2002; Errata 2003; Errata 2005, 17th Edition) Standard Specifications for Highway Bridges

ASTM INTERNATIONAL (ASTM)

ASTM D1241 (2015) Materials for Soil-Aggregate Subbase, Base, and Surface Courses

ASTM D1388 (2018) Standard Test Method for Stiffness of Fabrics

ASTM D4355/D4355M (2014) Deterioration of Geotextiles from Exposure to Light, Moisture and Heat in a Xenon-Arc Type Apparatus

ASTM D4759 (2011; R 2018) Standard Practice for Determining the Specification Conformance of Geosynthetics

ASTM D5818 (2011; R 2018) Standard Practice for Exposure and Retrieval of Samples to Evaluate Installation Damage of Geosynthetics

GEOSYNTHETIC INSTITUTE (GSI)

GRI GG1-87 Standard Test Method for Geogrid Rib Tensile Strength

GRI GG2-87 Standard Test for Geogrid Junction Strength

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 9090

Compatibility Test for Wastes and Membrane
Liners

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Mechanical Connection; G, D0

Marine Mattress Fabrication Plan; G, D0

Marine Mattress Installation; G, D0

SD-04 Samples

Structural Geogrid; G, D0

Stone Fill Materials; G, D0

Mechanical Connection; G, D0

SD-07 Certificates

Geogrid; G, D0

SD-08 Manufacturer's Instructions

Manufacturer's Instructions; G, D0

SD-09 Manufacturer's Field Reports

Roll Values; G, D0

1.4 DEFINITIONS

1.4.1 Polymeric Marine Mattress

A non-metallic compartmental structure filled densely and tightly with stone prior to installation. Filling is achieved while each unit is positioned on edge prior to installation. Units are comprised of structural geogrid, braid, and mechanical connection elements fabricated to allow placement and provide containment of aggregate fill.

1.4.2 Geogrid

An integrally formed grid structure manufactured of a stress resistance high density polyethylene (HDPE) and/or polypropylene (PP) material with molecular weight and molecular characteristics which impart high resistance to:

- a. Loss of load capacity or structural integrity when the geogrid

is subjected to mechanical stress in installation.

b. Deformation when the geogrid is subjected to applied force in use.

c. Loss of load capacity or structural integrity when the geogrid is subjected to long-term environmental stress.

1.4.3 Minimum Average Roll Values

Value based on testing and determined in accordance with ASTM D4759. All values given in the Table in this section are minimum average roll values.

1.4.4 True Tensile Modulus in Use

The ratio of tensile strength to corresponding strain (e.g., 1 percent). The tensile strength is measured via GRI GG1-87 as modified by AASHTO HB-17 using a single rib having the greater of 3 junctions or 8 inches and tested at a strain rate of 10 percent per minute based on this gauge length without deforming test materials under load before measuring such resistance or employing "secant" or "offset" tangent methods of measurement so as to overstate tensile properties. Values shown are minimum average roll values.

1.4.5 Junction Strength

Breaking tensile strength of junctions when tested in accordance with GRI GG2-87 as modified by AASHTO HB-17 using a single rib having the greater of 3 junctions or 8 inches and tested at a strain rate of 10 percent per minute based on this gauge length. Values shown are minimum average roll values.

1.4.6 Flexural Stiffness (Also Known as Flexural Rigidity)

Resistance to bending force measured via ASTM D1388. Values shown are minimum average roll values.

1.4.7 Resistance to Installation Damage

Resistance to loss of load capacity or structural integrity when subjected to mechanical stress in installation measured via ASTM D5818 in a crushed stone classified as a poorly graded gravel with a maximum 2 inch particle size (GP). Values shown are typical values.

1.4.8 Resistance to Long Term Degradation

Resistance to loss of load capacity or structural integrity when subjected to chemically aggressive environments measured via EPA 9090 immersion testing. Values shown are typical values.

1.4.9 Ultraviolet Stability

The ratio of tensile strength after exposure to the tensile strength prior to exposure with exposure per ASTM D4355/D4355M and tensile strengths measured via GRI GG1-87 as specified in subparagraph "True Tensile Modulus in Use" of paragraph DEFINITIONS above.

1.5 STORAGE

Store mattress materials in conditions as recommended by the manufacturer. Prevent excessive mud, wet cement, epoxy, and like materials from affixing to the mattress materials. Cover stockpiles of mattresses, at the point of fabrication and at the project site, and protect if not installed within one month of production.

1.6 ON-SITE TECHNICAL ASSISTANCE

Coordinate for a qualified representative of the polymeric marine mattress supplier to be present at the job site during the entire first week of installation to provide technical assistance as needed. The Contractor shall remain solely responsible for the quality of installation.

1.7 QUALITY ASSURANCE

Prior to the installation of the units, the Contractor shall arrange a meeting at the site with the system supplier and, where applicable, the system installer. The Contracting Officer shall be notified at least 5 days in advance of the time of the meeting.

PART 2 PRODUCTS

2.1 STRUCTURAL GEOGRID

The Contractor shall submit geogrid product data sheet and a manufacturer's certificate or affidavit signed by a legally authorized official from the company manufacturing the marine mattresses, that all materials contained within that shipment meet the composition, physical, and manufacturing requirements stated in this specification. Certificate of compliance shall be submitted for the mattress prior to delivery to the work site.

a. Unless otherwise specified on the contract drawings, shop drawings, or approved by the Contracting Officer, the structural geogrid type shall be:

- (1) Type 1 for the internal diaphragms of the units.
- (2) Type 2 for the top, bottom, sides.
- (3) Type 1 or Type 2 for extensions of the units.

b. The structural geogrid shall be produced from virgin resin and classified as HDPE or PP and shall possess complete continuity of all properties throughout its structure.

c. The structural geogrid shall accept applied force in use by positive mechanical interlock (i.e., direct mechanical keying) with:

- (1) Compacted soil or construction fill materials;
- (2) Contiguous sections of itself when overlapped and embedded in compacted soil or construction fill materials; and,
- (3) Rigid mechanical connection elements such as bodkins, pins or hooks.

d. The structural geogrid shall have the following characteristics:

PROPERTY	UNITS	TYPE 1	TYPE 2
True 1 percent Tensile Modulus in Use (MD)	kN/m (lb/ft)	350 (24,000)	1,650 (113,090)
Junction Strength (MD)	kN/m (lb/ft)	24.30 (1,665)	100.8(6,908)
Flexural Stiffness	mg-cm	6,500,000	670,000
Resistance to Installation Damage	percent GP	85	85
Resistance to Long Term Degradation	percent	100	100
Ultraviolet Stability (Retained Strength @ 500 hours)	percent	98	98

2.2 STONE FILL MATERIALS

Aggregate for filling the marine mattresses shall meet the requirements of [ASTM D1241](#), Gradation B.

2.3 MECHANICAL CONNECTION ELEMENTS

a. The mechanical connection elements shall be as shown on the contract drawings and shop drawings and shall be composed of HDPE and/or, unless otherwise approved by the Contracting Officer.

b. The mechanical connection used shall be bodkin type, unless otherwise approved by the Contracting Officer.

2.4 UV STABILIZED BRAID

a. The braid used for tying and lacing in the fabrication of the units shall be 8-strand hollow-core braid composed of HDPE or PP. Each strand shall consist of a bundle of monofilament HDPE or PP.

b. The braid shall have a nominal diameter of not less than 3/16 inch and a breaking strength of not less than 400 pounds on a test specimen 36 inches in length.

c. The braid shall be UV stabilized with a minimum carbon black content of 2.0 percent by weight.

PART 3 EXECUTION

3.1 EXAMINATION

The Contractor shall check the geogrid, braid and mechanical connection elements upon delivery to verify that the proper material has been received. These materials shall be inspected by the Contractor to be free of flaws or damage occurring during manufacturing, shipping, or handling.

3.2 MANUFACTURER'S INSTRUCTIONS

The Contractor shall submit manufacturer's fabrication instructions, installation instructions, and general recommendations.

3.3 TEST MATTRESS

3.3.1 General

Fabricate a test polymeric marine mattress (geogrid, geotextile, and stone). The Contracting Officer and/or his designated representative shall be present during the assembly of the test mattress. The test mattress shall be fabricated at the facility where all the mattresses are to be fabricated. Utilize and demonstrate the personnel, equipment, materials, and methods proposed in the approved work plan for fabricating the mattresses. The Contractor shall not fabricate any other mattresses until the test mattress is approved. Photographs of the fabrication operation shall be taken and provided to the Contracting Officer.

3.3.2 Coordination

Provide 10 days advance notification to the Contracting Officer Representative (COR) prior to commencing the construction of the test polymeric marine mattress. This advance notification to the COR will serve to facilitate coordination of the site visit with the Government's various technical and project personnel who will conduct the inspection.

3.4 FINAL FABRICATION AND FILLING OF MARINE MATTRESSES

3.4.1 Marine Mattress Fabrication Plan

Submit shop drawings, depicting the fabrication of the polymetric marine mattresses including all connections. The equipment, labor, materials, and methods, shall be detailed for the fabrication of the marine mattresses.

Empty mattress units shall be assembled in accordance with the directions by the manufacturer and as indicated on the Plans. The joints where the ends and baffles of each unit join the top of bottom of the unit shall be closed with a mechanical connection.

3.4.2 Mechanical Connections

The joints where the ends and baffles of each unit join the top or bottom of the unit shall be made with a mechanical connection between geogrid elements.

3.4.3 Seaming

Unless otherwise shown on the contract drawings and shop drawings or approved by the Contracting Officer:

a. The joints along the sides of each mattress unit shall be secured by seaming with braid using a lock-stitch configuration to provide complete closure of each unit.

b. Stitches shall be spaced evenly along each seam, with a minimum of 6 stitches per foot of seam. The braid material shall be stitched

through each pair of apertures along each seam at least once. The braid material shall be securely knotted to the geogrid material at each end of each seam and at a spacing not to exceed 3 feet along any stitched seam. The ends of each piece of braid used shall be knotted to prevent raveling of the braid.

c. The braiding shall be sufficiently tight to prevent openings greater than 1 inch along the seam, but shall not be cinched so tightly that overlaps and binding result.

d. Seaming to connect adjacent units is not required.

e. All cut ends of braid material shall be knotted within 1 inch to 2 inches of the end to prevent unraveling of the braid material.

3.4.4 Extension

A 2-foot geogrid extension shall be attached to the underside of the marine mattress as indicated on the drawings. The extension shall be attached in the same manner as specified herein for connecting baffles and sides to the primary panel of the mattress.

3.4.5 Geotextile

Prior to filling, geotextile shall be attached to the top, bottom and 2-foot extension of the marine mattress as recommended by the manufacturer.

3.4.6 Filling

All requirements herein apply unless otherwise shown on the contract drawings and shop drawings or approved by the Contracting Officer:

a. Each unit shall be filled and the fill densified while the unit is supported in an upright position resting on its side with the open side facing upward and the long direction of the compartments running vertical. Each compartment shall be filled in lifts and each lift shall be tightly packed, except the final lift.

b. Densification of the stone fill material and complete filling of each compartment shall be accomplished by rodding and/or vibration.

c. The filling sequence of the compartments within each unit shall be appropriate to prevent excess deformation or displacement of the interior diaphragms.

d. When filling and fabrication of a unit are complete, the unit shall be rotated to a horizontal position resting on its bottom in order to facilitate subsequent lifting.

e. The typical lift height shall not exceed 3 feet (loose) or 2.5 feet (packed). The final lift height shall not exceed 10 inches in height and should overfill each compartment by approximately 2 inches.

f. Filling shall not occur on the beach.

3.4.7 Compaction of Stone

Packing of the stone fill material and complete filling of each compartment shall be accomplished by rodding and/or vibration. The degree

of filling and packing shall be adequate to achieve complete filling as evidenced by tightly confined stone particles, tensioned interior diaphragms, snug bodkin connections, slight bulging of each compartment, and no evidence of air space between compartment during lifting. Excessive bulging of the unit or displacement on the interior diaphragms, such as caused by over packing or inadequate support, shall be rejected. Filling shall be accomplished in a manner that does not cause excessive damage to the geotextile, geogrid, mechanical connection elements, or the braid.

3.4.8 Lifting Hoops

Lifting hoops shall be formed by joining the top and bottom layers of grid from each unit by means of approved mechanical connections. When filling and fabrication of a unit are complete, the unit shall be rotated to a horizontal position resting on its bottom in order to facilitate subsequent lifting.

3.4.9 Final Assembly

The Contractor shall notify the Contracting Officer of the time and location that the mattresses will be assembled and filled. The mattresses may not be assembled and filled on the beach. It is preferred that the mattresses be assembled offsite, in the staging area, or on a barge. Regardless of the assembly location, all mattresses delivered to the site shall meet the contract requirements at the time of installation. The filled mattresses shall be free from debris, mud, dirt and other refuse prior to installation. If necessary, the filled mattresses shall be washed prior to delivery to the beach.

3.5 PROTECTION OF MARINE MATTRESSES DURING TRANSPORTATION

Utilize typical industry methods to protect the edge and corners of the mattresses during transportation.

3.6 SITE PREPARATION AND MARINE MATTRESS INSTALLATION

The Contractor shall submit installation drawings showing the mattress layout patterns for the project in relation to the feature alignment. The layout shall meet all requirements as shown on the drawings and specifications.

3.7 REPAIRS

Any units damaged during installation shall be replaced or repaired in a manner approved by the Contracting Officer. Any such measures required shall be at no additional cost to the Government. Damaged geogrid shall be repaired by placing a geogrid patch over the damaged area such that it overlaps onto the acceptable geogrid material by at least one foot in all directions; the perimeter edges of the patch shall be attached to the mattress in accordance with the seaming requirements, above. Damaged braid shall be repaired by installing a new braided seam in accordance with above, and extending at least one foot in either direction beyond the location of the damaged braid. Failed units shall be replaced at the direction of the Contracting officer at no cost to the Government.

3.8 CONTRACTOR QUALITY CONTROL

Conduct an inspection of each fabricated mattress prior to and after filling with stone. The results of these inspections shall be documented

in the daily Quality Control reports required under Section 01 45 04
CONTRACTOR QUALITY CONTROL.

-- End of Section --